

Nonequilibrium Superconductivity : Part II

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Preface

Other My Theories

Main experiments

Other Special Theories, Models

Another Explanations of a Typical Theories

Other Experiments

Amazing facts

(I'll reveal it later)

Puzzles for motivation

(I'll reveal it later)

Part VII

Special Nonequilibrium Superconductivity in a Nutshell

Part VIII

Other My Theories

Part IX

Method of Pseudospins and Lax Vector: Other Theory

7 Nonequilibrium Cooper pairing in the nonadiabatic Regime by Yuzbashyan et al.

Abstract

We obtain a complete solution for the mean-field dynamics of the BCS paired state with a large, but finite number of Cooper pairs in the nonadiabatic regime. We show that the problem reduces to a classical integrable Hamiltonian system and derive a complete set of its integrals of motion. The condensate exhibits irregular multi-frequency oscillations ergodically exploring the part of the phase-space allowed by the conservation laws. In the thermodynamic limit however the system can asymptotically reach a steady state.

The study of the dynamics of the BCS superconductors has a long history[1]. Early attempts to describe nonstationary superconductivity were based on the time-dependent Ginzburg-Landau (TDGL) equation [2, 3, 4], which reduces the problem to the time evolution of a single collective order parameter $\Delta(t)$. The TDGL approach is valid only provided the system quickly reaches an equilibrium with the instantaneous value of $\Delta(t)$, i.e. a local equilibrium is established faster than the time scale of the order parameter variation, $\tau_\Delta \simeq 1/\Delta$. This requirement limits the applicability of the TDGL to special situations where pair breaking dominates, e.g. due to a large concentration of magnetic impurities. An alternative to TDGL is the Boltzmann kinetic equation[5, 6] for the quasiparticle distribution function coupled to a self-consistent equation for $\Delta(t)$. This approach is justified only when external parameters change slowly on the τ_Δ time scale, so that the system can be characterized by a quasiparticle distribution.

Is it possible to describe theoretically the dynamics of a BCS paired state in the nonadiabatic regime when external parameters change substantially on the τ_Δ time scale? In particular, an important question is whether, following a sudden perturbation, the condensate reaches a steady state on a τ_Δ time scale or on a much longer quasiparticle energy relaxation time scale τ_ϵ . In the nonadiabatic regime both TDGL and the Boltzmann kinetic equations fail and one has to deal with the coupled coherent dynamics of individual Cooper pairs. Recent studies[15, 16, 17, 18] of this outstanding problem were motivated by experiments on fermionic pairing in cold atomic alkali gases[8, 7]. The strength of pairing interactions in these systems can be fine tuned rapidly by a magnetic field, making it easier than in metals to access the nonadiabatic regime experimentally.

The main result of the present paper is an explicit general solution for the *dynamics* of the BCS model, which describes a spatially homogenous condensate at times $t \ll \tau_\epsilon$. We employ the usual BCS mean-field approximation, which is accurate when the number of Cooper pairs is large[10, 11]. It turns out that the mean-field BCS dynamics can be formulated as a nonlinear *classical Hamiltonian* problem. We obtain the exact solution for all initial conditions and a complete set of integrals of motion for the mean-field BCS dynamics.

In this paper we assume that the number of Cooper pairs in the system is arbitrary large, but *finite*. In this case the typical evolution at times $t \ll \tau_\epsilon$ is *quasi*-periodic with a large number of incommensurate frequencies. The condensate exhibits irregular multi-frequency oscillations ergodically exploring the part of the phase-space allowed by the conservation laws. The system returns arbitrarily close to its initial state at irregular time intervals. However, the return time

diverges in the thermodynamic limit for most physical initial conditions, while the solution asymptotically reaches a *steady* state on the τ_Δ time scale. The system thermalizes on a much larger energy relaxation time scale τ_ϵ [9].

The dynamics of the BCS condensate following a sudden change of external parameters has been previously discussed by a number of authors[11, 12, 13, 14, 15, 16, 17, 18, 19]. Most notably, a linear analysis around the BCS ground state has been performed[11, 12] and some simple particular solutions for the nonlinear mean-field dynamics in the context of superconductivity have been reported[14, 15]. We discuss below how these results fit into the general picture.

We begin our description of the nonequilibrium Cooper pairing in the non-dissipative regime, $t \ll \tau_\epsilon$, with the BCS model[20, 21, 22].

$$\hat{H}_{BCS} = \sum_{j,\sigma} \epsilon_j \hat{c}_{j\sigma}^\dagger \hat{c}_{j\sigma} - g \sum_{j,q} \hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow}^\dagger \hat{c}_{q\downarrow} \hat{c}_{q\uparrow} \quad (7.1)$$

where ϵ_j are single-particle energies. The pairing is between time reversed states $|j \uparrow\rangle$ and $|j \downarrow\rangle$ [23]. Our goal is to determine the evolution of a state that was driven out of equilibrium at, say, $t = 0$.

There are several equivalent ways to derive mean-field equations of motion. One can start with the BCS product state, $\prod_j \left(U_j(t) + V_j(t) \hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow}^\dagger \right) |0\rangle$, and use Bogoliubov-de Gennes equations for the time-dependent amplitudes $U_j(t)$ and $V_j(t)$. Alternatively, one can study the evolution of the normal, $G_j(t) = -i\langle [\hat{c}_{j\uparrow}(t), \hat{c}_{j\uparrow}^\dagger(t)] \rangle$, and anomalous, $F_j(t) = -i\langle [\hat{c}_{j\uparrow}(t), \hat{c}_{j\downarrow}(t)] \rangle$, Green's functions at coinciding times[12].

The most convenient for us approach to the BCS mean-field dynamics is based on the Anderson pseudospin representation[11]. Within this approach the mean-field equations are Hamiltonian equations of motion for a classical spin chain. Pseudospin-1/2 operators are related to fermion creation and annihilation operators via $\hat{K}_j^z = (\hat{n}_{j\uparrow} + \hat{n}_{j\downarrow} - 1)/2$ and $\hat{K}_j^- = \hat{c}_{j\downarrow} \hat{c}_{j\uparrow} = (\hat{K}_j^+)^{\dagger}$. Pseudospins are defined on empty and doubly occupied (unblocked) single-particle orbitals ϵ_j . Singly occupied orbitals are decoupled from the dynamics. For n unblocked orbitals the Hamiltonian has the form

$$\hat{H}_{BCS} = \sum_{j=0}^{n-1} 2\epsilon_j \hat{K}_j^z - g \sum_{j,q} \hat{K}_j^+ \hat{K}_q^- \quad (7.2)$$

The mean field approximation is accurate[11, 10] in the thermodynamic limit due to the infinite range of interactions between spins in the Hamiltonian (11.19). Therefore, the effective field seen by each pseudospin in (11.19) can be replaced with its quantum mechanical average, $\mathbf{b}_j(t) = (-2\Delta_x(t), -2\Delta_y(t), 2\epsilon_j)$, where $\Delta(t) \equiv \Delta_x(t) - i\Delta_y(t) \equiv g \sum_j \langle \hat{K}_j^-(t) \rangle$ is the BCS gap function. In this approximation, each spin evolves in the self-consistent field: $\dot{\hat{\mathbf{K}}}_j = i[\hat{\mathbf{H}}_{BCS}, \hat{\mathbf{K}}_j] \approx \mathbf{b}_j \times \hat{\mathbf{K}}_j$. Taking the quantum mechanical average of these equations with respect to the time-dependent state of the system, we obtain for $\mathbf{s}_j(t) = \langle \hat{\mathbf{K}}_j(t) \rangle$

$$\dot{\mathbf{s}}_j = \mathbf{b}_j \times \mathbf{s}_j \quad \mathbf{b}_j = (-2gJ_x, -2gJ_y, 2\epsilon_j) \quad \mathbf{J} = \sum_{q=0}^{n-1} \mathbf{s}_q \quad (7.3)$$

The components of the classical spins $s_j^z(t)$ and $s_j^\pm = s_j^x \pm is_j^y$ are related to Bogoliubov amplitudes and equal times Green's functions as $2s_j^z = |V_j|^2 - |U_j|^2$, $s_j^- = \bar{U}_j V_j$ and $G_j(t) = is_j^z(t)$, $F_j(t) = is_j^-(t)$, respectively. Evolution equations (10.8) conserve the square of the average for each spin: $ds_j^2/dt = 0$. If the spins initially were in a product state, $\mathbf{s}_j^2 = 1/4$. Note also that $\Delta(t) = gJ_-(t)$.

One can check that Eqs. (10.8) are equations of motion for a *classical* spin Hamiltonian

$$H_{BCS} = \sum_{j=0}^{n-1} 2\epsilon_j s_j^z - g \sum_{j,q} s_j^+ s_k^- \quad (7.4)$$

It means that Eqs. (10.8) are Hamilton equations $\dot{\mathbf{s}}_j = \{H_{BCS}, \mathbf{s}_j\}$ derived from Hamiltonian (10.11) using the usual angular momentum Poisson brackets

$$\{s_j^a, s_k^b\} = -\varepsilon_{abc} \delta_{jk} s_j^c \quad (7.5)$$

where a, b , and c stand for spatial indexes x, y , and z . The classical model (10.11) can be obtained from its quantum counterpart (11.19) by replacing operators with classical dynamical variables and commutators with Poisson brackets.

Both the classical (10.11) and quantum models (11.18) and (11.19) are integrable[24, 25, 26]. To show this, one can introduce a vector function (Lax-vector) of an auxiliary parameter u

$$\mathbf{L}(u) = -\frac{\hat{\mathbf{z}}}{g} + \sum_j \frac{\mathbf{s}_j}{u - \epsilon_j}, \quad (7.6)$$

where $\hat{\mathbf{z}}$ is a unit vector along the z axis. Poisson brackets between components of $\mathbf{L}(u)$ at different values of u can be evaluated using Eq. (11.8).

$$\{L^a(v), L^b(w)\} = \varepsilon_{abc} \frac{L^c(v) - L^c(w)}{v - w} \quad (7.7)$$

(Relations (11.11) hold for each term in (11.12) separately; all terms Poisson-commute with each other). It follows from Eq. (11.11) that the lengths of the Lax vector at different values of u Poisson-commute:

$$\{\mathbf{L}^2(v), \mathbf{L}^2(w)\} = 0 \quad (7.8)$$

The scalar function $\mathbf{L}^2(u)$ can be represented in the form

$$\mathbf{L}^2(u) = \frac{1}{g^2} + \sum_{j=0}^{n-1} \left(\frac{2H_j}{u - \epsilon_j} + \frac{s_j^2}{(u - \epsilon_j)^2} \right) \quad (7.9)$$

where

$$H_j = \sum_{k=0}^{n-1} \left(\frac{\mathbf{s}_j \cdot \mathbf{s}_k}{\epsilon_j - \epsilon_k} - \frac{s_j^z}{g} \right) \quad (7.10)$$

Since Eq. (11.13) holds for any v and w , all H_j Poisson-commute with each other. Therefore, each H_j , as well as any algebraic combination of H_j , defines a classical model[27] that has n degrees of freedom (n classical spins) and n integrals of motion (including itself) and thus is integrable in the usual sense[28]. Note that the sum of H_j is proportional to the z -component of the total spin $\mathbf{J}$, therefore J_z is conserved by all H_j and their combinations. Moreover, the following identity follows from Eqs. (10.11, 7.10)

$$H_{BCS} = -g \sum_j \epsilon_j H_j + \text{const} \quad (7.11)$$

This implies that the classical BCS model (10.11) Poisson-commutes with all H_j 's and thus is also integrable. Equations (7.10) and (7.11) can be straightforwardly quantized by replacing $\mathbf{s}_j \rightarrow \hat{\mathbf{K}}_j$. The resulting operators $\hat{H}_j$ all pairwise commute, thus showing the integrability of quantum models (11.18) and (11.19).

To obtain the general solution for the mean-field dynamics of the BCS model (11.18), we follow the method of Ref. [29] and introduce $n - 1$ *separation* variables u_k as zeros of $L_-(u) = L_x(u) - iL_y(u)$, i.e. $\sum_j s_j^-/(u_k - \epsilon_j) = 0$.

Equations of motion for the variables u_k are[30]

$$\dot{u}_k = 2i\sqrt{Q_{2n}(u_k)} \prod_{m \neq k} (u_k - u_m)^{-1} \quad (7.12)$$

where $Q_{2n}(u)$ is the *spectral polynomial* defined as

$$Q_{2n}(u) = g^2 \mathbf{L}^2(u) \prod_j (u - \epsilon_j)^2 \quad (7.13)$$

By Eq. (10.16), the coefficients of $Q_{2n}(u)$ depend only on the integrals of motion H_j .

Eqs. (7.12) constitute the well-known Jacobi's inversion problem solvable in terms of hyperelliptic theta functions[31]. Here we outline the final answer, the details will be reported elsewhere[30]. Klenian σ - and ζ -functions of *genus* G (in our case $G = n - 1$) are defined as

$$\begin{aligned} \zeta_l(\mathbf{x}) &= \frac{\partial \ln \sigma(\mathbf{x})}{\partial x_l} \quad \sigma(\mathbf{x}) = \sum_{\mathbf{m} \in Z^G} \exp [S_{\mathbf{m}}(x)/2] \\ S_{\mathbf{m}}(x) &= \mathbf{x} \cdot \eta \omega^{-1} \mathbf{x} + 2i\pi(\mathbf{m} \cdot \tau \mathbf{m} + \omega^{-1} \mathbf{x} \cdot \mathbf{m}) \end{aligned} \quad (7.14)$$

where the sum is over all G -dimensional integer vectors $\mathbf{m}$, $\tau = \omega' \omega^{-1}$, and ω , ω' , and η are $G \times G$ matrices of periods (see below). The solution is

$$s_j^-(t) = \langle \hat{c}_{j\downarrow}(t) \hat{c}_{j\uparrow}(t) \rangle = J_-(t) r(\epsilon_j, t) \prod_{k \neq j} \frac{\epsilon_j}{\epsilon_j - \epsilon_k} \quad (7.15)$$

$$\Delta(t) = g J_-(t) = g \sum_j s_j^-(t) = c_n e^{-i\beta t} \frac{\sigma(\mathbf{x} + \mathbf{d})}{\sigma(\mathbf{x} - \mathbf{d})} \quad (7.16)$$

Here $\mathbf{x}^T = i(c_1, \dots, c_{n-2}, 2t + c_{n-1})$; $\mathbf{d}$ is a vector of constants; $\beta = gJ_z + \sum_j \epsilon_j$; $c_1, \dots, c_n$ are constants fixed by the initial conditions, and

$$r(u, t) = 1 - \sum_{k=1}^{n-1} (\zeta_k(\mathbf{x} + \mathbf{d}) - \zeta_k(\mathbf{x} - \mathbf{d}) + a_k) u^{k-n} \quad (7.17)$$

Constants a_k , the matrices of periods, constant vector $\mathbf{d}$, are all uniquely determined[32] by the spectral polynomial $Q_{2n}(u)$, i.e. by integrals of motion.

The evolution of $\mathbf{s}_j(t) = \langle \hat{\mathbf{K}}_j(t) \rangle$ described by the general solution is typical of an integrable system[28]. It is characterized by n frequencies, which in our case can be determined[30] in terms of integrals of motion, and are typically incommensurate. Note that $|\Delta(t)|$ contains only $n - 1$ frequencies. The typical dynamics is stable against perturbations destroying integrability[28].

Now let us discuss some particular solutions. There are two types of equilibrium states that play an important role in the dynamics. In *normal* states all spins are parallel to the z -axis, $2\mathbf{s}_j^z = \pm 1$. Since $2\mathbf{s}_j = \langle \hat{n}_j \rangle - 1$, these states correspond to the ground state and excitations of the single-particle part of the Hamiltonian (11.18) (Fermi gas). They are stationary within the mean-field dynamics (10.8). For a finite system, they are non-stationary for the quantum Hamiltonian (11.18) and their short time dynamics is entirely driven by quantum corrections (cf. Refs. [33, 34]).

The second type of equilibrium states are *anomalous* ones, which correspond to the BCS ground state and excitations. These states are obtained by aligning each spin in (10.11) self-consistently along the effective magnetic field acting on it. The self-consistency condition is the

BCS gap equation. As $\mathbf{s}_j = \langle \hat{\mathbf{K}}_j \rangle$, one can obtain the BCS wave-function and energy spectrum from anomalous equilibrium configurations of classical spins $\mathbf{s}_j$.

It turns out that equilibrium states are a part of a more general scheme when the dynamics of n spins degenerates to that of $m < n$ *collective* spins (*m-spin solutions*) governed by the same Hamiltonian (10.11) only with m spins and new parameters μ_j instead of ϵ_j . Normal and anomalous states correspond to 0- and 1-spin solutions, respectively. To construct m -spin solutions one has to take the Lax-vector (11.12) to be proportional to that of a system with m spins $\mathbf{t}_k$, $\mathbf{L}(u) = [1 + \sum_j b_j/(u - \epsilon_j)]\mathbf{L}_t(u)$, where b_j are time-independent constants, and $\mathbf{L}_t(u) = -\hat{\mathbf{z}}/g + \sum_k \mathbf{t}_k/(u - \mu_k)$. Then, $2(n - m)$ of $2n$, typically distinct, roots of the spectral polynomial $Q_{2n}(u)$ become doubly degenerate and $n - m$ separation variables u_k are frozen in these double roots, which automatically solves the equations of motion for these $n - m$ variables. The dynamics is obtained by replacing $n \rightarrow m$ and $\epsilon_k \rightarrow \mu_k$ in Eqs. (10.28–10.30) and is characterized by $m < n$ typically incommensurate frequencies. For $m = 2$ the solution is in terms of hyperelliptic functions of genus $G = m - 1 = 1$, i.e. in terms of ordinary elliptic functions.

Now let us discuss the connection of our results with the previous work. The solutions for the mean field BCS dynamics obtained in Ref. [15] are 2-spin solutions in the above classification. They were used in Ref. [15] to describe the evolution beginning from a state infinitesimally close to the normal ground state. In our view, the dynamics in the vicinity of this state can have additional features and deserves further analysis.

The 2-spin solutions resemble the TDGL approach in that they describe the dynamics of all pairs in terms of only two collective degrees of freedom resulting in large amplitude single frequency (periodic) oscillations of the order parameter magnitude $|\Delta(t)|$. Mathematically, they lie on a 1d curve of points in a multi-dimensional (infinite-dimensional in the thermodynamic limit) space of possible values of integrals of motion. The situation with other few spin solutions is similar[35]. In contrast, the general solution we obtained here *typically* has a large (infinite in the thermodynamic limit) number of incommensurate frequencies and a substantially reduced amplitude. The difference between the general and few spin solutions is clear in a linear analysis[11, 12] around the BCS ground state that displays normal modes with frequencies $\omega_k = 2\sqrt{\epsilon_k^2 + \Delta_0^2}$, where ϵ_k are single-particle energies and Δ_0 is the equilibrium order parameter. In the linear regime, the general solution becomes an arbitrary superposition of all normal modes, while few spin solutions single out all, but few modes. For example, 2-spin solutions of Ref. [15] correspond to a single normal mode with a frequency $2\Delta_0$.

In conclusion, we have obtained the explicit general solution for the mean-field dynamics of the BCS paired state and discussed a number of special cases including two types of equilibrium states and few spin solutions. A still open problem is to fully analyze the solution in the thermodynamic limit. It is also desirable to better understand the dynamics in the vicinity of normal states where quantum effects become important. Finally, it is interesting to identify experimental setups where peculiar features of the nonequilibrium Cooper pairing in the nonadiabatic regime can be observed such as e.g. cold Fermi gases.

Note Added. Recently, we became aware of a publication[36] that is in agreement with some of our conclusions – that the initial dynamics of the normal ground state is driven by quantum corrections and that the system can reach a steady state at large times.

8 Quantum quench in a $p+ip$ superfluid: Winding numbers and topological states far from equilibrium by Foster, Dzero, Gurarie, Yuzbashyan

Abstract

We study the non-adiabatic dynamics of a 2D $p + ip$ superfluid following an instantaneous quantum quench of the BCS coupling constant. The model describes a topological superconductor with a non-trivial BCS (trivial BEC) phase appearing at weak (strong) coupling strengths. We extract the exact long-time asymptotics of the order parameter $\Delta(t)$ by exploiting the integrability of the classical p-wave Hamiltonian, which we establish via a Lax construction. Three different types of asymptotic behavior can occur depending upon the strength and direction of the interaction quench. We refer to these as the non-equilibrium phases **I**, **II**, **III**, characterized as follows. In phase **I**, the order parameter asymptotes to zero due to dephasing. In phase **II**, $\Delta \rightarrow \Delta_\infty$, a non-zero constant. Phase **III** is characterized by persistent oscillations of $\Delta(t)$. For quenches within phases **I** and **II**, we determine the topological character of the asymptotic states. We show that two different formulations of the bulk topological winding number, although equivalent in the BCS or BEC ground states, must be regarded as independent out of equilibrium. The first winding number Q characterizes the Anderson pseudospin texture of the initial state; we show that Q is generically conserved. For $Q \neq 0$, this leads to the prediction of a “gapless topological” state when Δ asymptotes to zero. The presence or absence of Majorana edge modes in a sample with a boundary is encoded in the second winding number W , which is formulated in terms of the retarded Green’s function. We establish that W can change following a quench across the quantum critical point. When the order parameter asymptotes to a non-zero constant, the final value of W is well-defined and quantized. We discuss the implications for the (dis)appearance of Majorana edge modes. Finally, we show that the parity of zeros in the *bulk* out-of-equilibrium Cooper pair distribution function constitutes a $\mathbb{Z}_2$ -valued quantum number, which is non-zero whenever $W \neq Q$. The pair distribution can in principle be measured using RF spectroscopy in an ultracold atom realization, allowing direct experimental detection of the $\mathbb{Z}_2$ number. This has the following interesting implication: topological information that is experimentally inaccessible in the bulk ground state can be transferred to an observable distribution function when the system is driven far from equilibrium.

8.1 Introduction

Topology has emerged as a key tool to characterize phases of quantum many-body particle systems. A recent application is the classification of topological insulators and superconductors.[1, 2] These are distinguished by a topological winding number in the bulk; when this number is quantized to a non-zero integer value, it implies the presence of gapless, delocalized states at the sample surface. Both the bulk topological invariant and the gapless surface states are argued to be protected against generic local perturbations.

A natural means to generate non-trivial dynamics in a topological system is via a *global* deformation of the system Hamiltonian, otherwise known as a quantum quench.[3, 4, 5, 6, 19, 18, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20] Through the evolution induced by a quench, one can probe the stability of the system topology—when and how can it change? Under what circumstances does it remain well-defined when the system is coherently driven far away from its ground state? Finally, can the quench be employed as an experimental tool to reveal the bulk topology?

In the setup for a quench, a many-particle system is initially prepared in a particular pure state; this can be taken as the ground state of some initial Hamiltonian. In addition, we assume that there is a gap to excitations. Performing the quench, a parameter of the Hamiltonian (such as the interparticle interaction strength) is changed over a time interval much shorter than the inverse excitation gap. The system subsequently evolves as a highly excited, coherent admixture of many-body eigenstates of the final, post-quench Hamiltonian. Quantum quenches have become a standard protocol to investigate ultracold atomic systems.[21, 3, 4, 5] Ultracold gases are engineered to be well-isolated from any outside environment or heat bath, and typically

exhibit a high degree of external tunability. The long-time out-of-equilibrium dynamics induced by a quench in an isolated many particle system can show different *dynamical phases* as function of the quench parameters.[6, 19, 18, 9, 10, 11, 12, 13, 14, 13, 14]

In this work, we probe the response of a 2D topological $p + ip$ superfluid[24, 25, 1] to an instantaneous quantum quench. We envisage an ultracold fermionic atom[26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37] or molecule[38] realization of the system, such that the effective pairing interaction strength can be tuned externally, e.g. by manipulating a Feshbach resonance.[26, 27, 28, 29, 30, 31, 32] Initially, the system occupies the ground state of the pre-quench Hamiltonian, residing within either the topologically non-trivial BCS or trivial BEC phase (see below). Subsequently, the BCS interaction coupling is deformed instantaneously to stronger or weaker pairing. We consider quenches both within and between the BCS and BEC phases. We calculate the asymptotic time evolution[6, 19, 18, 9, 10, 11, 12, 13, 14, 15] using an integrable version of the p-wave BCS Hamiltonian.[39, 40, 41, 42, 43] Our treatment is exact in the thermodynamic limit when pair-breaking can be neglected.[6, 11, 10]

An overview of our main results is provided in Sec. 8.2; these include the following: First, we compute the out-of-equilibrium phase diagram shown in Fig. 8.4, as determined by the exact long-time dynamics of the order parameter Δ (which is *not* the same as the quasiparticle gap for p-wave—see Secs. 8.1.1, 8.3.2, and Appendix 8.7 for a brief review). Similar to the s-wave case,[6, 19, 18, 9, 10, 11, 12, 13, 14] we find that Δ exhibits one of three behaviors in the long time limit $t \rightarrow \infty$: for strong-to-weak pairing quenches within phase **I** of Fig. 8.4, $\Delta(t) \rightarrow 0$ due to dephasing.[13, 14] In phase **II**, which includes the case of zero quench, $\Delta(t) \rightarrow \Delta_\infty$, a non-zero constant.[9, 10, 12] Finally, for weak-to-strong pairing quenches within phase **III**, $\Delta(t)$ exhibits coherent, undamped oscillations[6, 12, 13] as $t \rightarrow \infty$, see Fig. 8.7.

In the ground state, the bulk topology and edge states are encoded in a $\mathbb{Z}$ -valued winding number. We consider two formulations: the winding of the Anderson pseudospin texture Q , [24, 25] and of the retarded single-particle Green's function W . [24, 44, 45] In the ground state $W = Q$, and $Q = 1$ ($Q = 0$) in the weak pairing BCS (strong pairing BEC) phase. We show that Q does not evolve from its initial value following a quench, Fig. 8.8. We identify a “gapless topological” phase, characterized by $Q = 1$ and $\lim_{t \rightarrow \infty} \Delta(t) = 0$. Although equivalent in equilibrium, we find that Q and W must be regarded as independent following a quench. In particular, the presence or absence of Majorana edge modes in a sample with a boundary is encoded in W , not Q . Moreover, a quench across the topological quantum phase transition (e.g., from BCS to BEC) induces a change in W . Whenever $\lim_{t \rightarrow \infty} \Delta(t) = \Delta_\infty \neq 0$, W nevertheless assumes a quantized value in the asymptotic steady-state, Fig. 8.9. We discuss implications for the (dis)appearance of Majorana edge modes.

Some of the results discussed in this paper also appear in an abbreviated form in Ref. [46]. The quench phase diagram and the asymptotic values of the winding number W for quenches in phase **II** are presented in that work, as well as the link between W and Majorana edge modes. In Ref. [46], we show that a topologically non-trivial cold atomic p-wave superfluid could be induced by quenching from very weak initial coupling to strong pairing, using a Feshbach

Figure 8.1: Anderson pseudospin description of (a) a normal Fermi liquid (b) an s-wave superconductor, in any number of spatial dimensions at zero temperature. In this figure, k measures the radial coordinate along any direction in momentum space.

resonance. In the non-trivial case, the order parameter oscillates periodically in time (phase **III**), and the presence of edge modes is established using a Floquet analysis.[47, 48, 49, 50] We do not discuss Floquet in this paper. Instead, we provide the detailed derivation of the phase diagram and the topological characterization of phases **I** and **II**. We consider both formulations Q and W of the bulk winding number. We compute the long-time dynamics of the order parameter exactly, using a variant of the Lax construction employed in the s-wave case.[11, 10, 12, 13, 14] Finally, in this paper we search for *bulk* signatures of the system topology.

Because the topology resides in a quantum mechanical Berry phase, it is typically difficult to measure a bulk invariant directly. We show that when Δ asymptotes to a non-zero constant and $Q \neq W$, as occurs for a quench across the quantum critical point, the number of zeroes in the out-of-equilibrium *Cooper pair distribution function* is odd, as demonstrated in Figs. 8.10 and 8.13. The parity of these zeroes constitutes a non-equilibrium $\mathbb{Z}_2$ -valued bulk winding number. We show that this number can in principle be detected through the modulation of the absorption amplitude in RF spectroscopy. This is unique to the non-equilibrium preparation; the winding number distinguishing the BCS and BEC ground states cannot be ascertained via a bulk RF measurement.

In the remainder of this Introduction, we briefly review the topological character of 2D $p+ip$ superfluids. We close with an outline for the organization of this paper.

8.1.1 Topological superfluidity in 2D

The topological properties of 2D $p+ip$ superconductors were originally obtained by Volovik[24] in the context of $^3\text{He-A}$, and subsequently discussed in the context of the fractional quantum Hall effect by Read and Green.[25] The simplest p-wave channel BCS Hamiltonian for spinless (or spin-polarized) fermions is[24]

$$H = \sum_{\mathbf{k}}' \frac{k^2}{m} s_{\mathbf{k}}^z - \frac{2G}{m} \sum_{\mathbf{k}, \mathbf{q}}' \mathbf{k} \cdot \mathbf{q} s_{\mathbf{k}}^+ s_{\mathbf{q}}^-, \quad (8.1)$$

where $G > 0$ is a dimensionless, attractive BCS interaction strength. Eq. (8.1) is expressed in terms of $SU(2)$ Anderson pseudospin[51] operators, defined as follows:

$$\begin{aligned} s_{\mathbf{k}}^z &\equiv \frac{1}{2}(c_{\mathbf{k}}^\dagger c_{\mathbf{k}} + c_{-\mathbf{k}}^\dagger c_{-\mathbf{k}} - 1), \\ s_{\mathbf{k}}^+ &\equiv c_{\mathbf{k}}^\dagger c_{-\mathbf{k}}, \\ s_{\mathbf{k}}^- &\equiv c_{-\mathbf{k}} c_{\mathbf{k}}, \end{aligned} \quad (8.2)$$

where $c_{\mathbf{k}} c_{\mathbf{q}}^\dagger + c_{\mathbf{q}}^\dagger c_{\mathbf{k}} = \delta_{\mathbf{k}, \mathbf{q}}$. The primed sums in Eq. (8.1) run over 2D momenta in the half plane $\mathbf{k} = \{k^x \in \mathbb{R}, k^y \geq 0\}$; with $\{\mathbf{k}, \mathbf{q}\}$ restricted to this range, the pseudospins satisfy $[s_{\mathbf{k}}^a, s_{\mathbf{q}}^b] = i\epsilon^{abc} \delta_{\mathbf{k}, \mathbf{q}} s_{\mathbf{k}}^c$. In Eq. (8.1), we have assumed that Cooper pairs are created and destroyed with zero center-of-mass momentum only (“reduced BCS” theory).[Schrieffer(1989)] This neglects pair-breaking processes; we address the limitations of this approximation in the conclusion Sec. 8.6.

The Anderson pseudospins provide a simple way to visualize the ground state of a BCS superconductor. The expectation $\langle s_{\mathbf{k}}^z \rangle$ measures the double-occupancy of a pair of states related by time-reversal symmetry; in Eq. (8.2), $\langle s_{\mathbf{k}}^z \rangle = 1/2$ ($-1/2$) implies that the states $\{\mathbf{k}, -\mathbf{k}\}$ are occupied (vacant). The Fermi liquid ground state is a discontinuous domain wall, as depicted in Fig. 8.1(a). By contrast, an s-wave paired state exhibits smooth pseudospin canting near $k = k_F$, Fig. 8.1(b). In the thermodynamic limit, H in Eq. (8.1) has a $p+ip$ ground state, defined in terms of the order parameter

$$\Delta(\mathbf{k}) \equiv -2G \sum_{\mathbf{q}}' \mathbf{k} \cdot \mathbf{q} \langle s_{\mathbf{q}}^- \rangle = \Delta_0 (k^x - ik^y). \quad (8.3)$$

Figure 8.2: Momentum space pseudospin texture of the $p+ip$ ground state (a) in the topological BCS phase, $Q = 1$ (b) at the BCS-BEC quantum phase transition, and (c) in the topologically-trivial BEC phase, $Q = 0$. Here Q denotes the pseudospin winding number (skyrmion charge), defined via Eq. (10.17). For a $p+ip$ state, $Q = 1$ ($Q = 0$) when the $\mathbf{k} = 0$ spin $\vec{s}_0$ is up (down). By contrast, Q is ill-defined at the quantum phase transition.

The amplitude Δ_0 is non-zero for any $G > 0$. The pseudospin texture for the weak-pairing BCS ground state is shown in Fig. 8.2(a). The $p + ip$ texture differs from the s-wave one in that the canting angle in spin space is correlated to the polar angle ϕ_k in momentum space; the result is a topologically non-trivial (skyrmion) configuration,[24, 25] discussed in more detail below.

The quasiparticle energy in the $p + ip$ paired state is given by

$$E_k = \sqrt{\left(\frac{k^2}{2} - \mu\right)^2 + k^2 \Delta_0^2}, \quad (8.4)$$

where μ is the chemical potential. Here we have set $m = 1$, since the mass can be factored from H in Eq. (8.1); μ and $(\Delta_0)^2$ both carry units of density. The spectrum is fully-gapped for any $\mu \neq 0$. In a system with fixed density n , the chemical potential is a monotonically decreasing function of the pairing amplitude Δ_0 :

$$\begin{aligned} \mu = & \left[2\pi n - \frac{\Delta_0^2}{2} \log\left(\frac{2\Lambda}{e\Delta_0^2}\right) \right] \theta(\Delta_{\text{QCP}} - \Delta) \\ & + \left[\frac{\Delta_0^2}{2} - \Lambda \exp\left(-\frac{4\pi n}{\Delta_0^2} - 1\right) \right] \theta(\Delta - \Delta_{\text{QCP}}), \end{aligned} \quad (8.5)$$

where Λ is a high-energy cutoff (see Appendix 8.7 for details), and $\theta(\Delta)$ denotes the unit step function. A plot of μ versus Δ_0 is shown in Fig. 8.3. At the special value $\Delta_0 = \Delta_{\text{QCP}}$ [defined via Eq. (8.164)], $\mu = 0$. Here, the bulk quasiparticle spectrum develops a massless Dirac node at $\mathbf{k} = 0$.

Δ_{QCP} marks a *topological* quantum phase transition between the topologically non-trivial, weak-pairing BCS phase ($0 < \Delta_0 < \Delta_{\text{QCP}}$) and the trivial, strong-pairing BEC phase ($\Delta_0 > \Delta_{\text{QCP}}$). [25] These can be distinguished by a bulk topological invariant. There are several equivalent formulations of the invariant in equilibrium. We will employ two different definitions. The first measures the winding of the pseudospin texture,[24, 25]

$$Q \equiv 8\pi \epsilon_{abc} \int \frac{d^2 \mathbf{k}}{(2\pi)^2} \frac{1}{k} \langle s_{\mathbf{k}}^a \rangle \partial_k \langle s_{\mathbf{k}}^b \rangle \partial_{\phi_k} \langle s_{\mathbf{k}}^c \rangle, \quad (8.6)$$

where ϕ_k denotes the polar angle in momentum space. A generalized $p + ip$ state can be defined by the pseudospin configuration

$$\begin{aligned} \langle s_{\mathbf{k}}^- \rangle & \equiv \frac{1}{2} \sqrt{1 - \varrho^2(k)} \exp[-i\phi_k + i\Theta(k)], \\ \langle s_{\mathbf{k}}^z \rangle & \equiv \frac{1}{2} \varrho(k), \end{aligned} \quad (8.7)$$

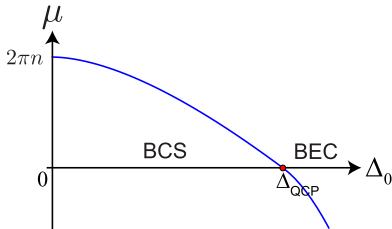


Figure 8.3: Zero temperature chemical potential μ as a function of the ground state order parameter amplitude Δ_0 in the $p + ip$ ground state, for fixed particle density n . The point $\{\Delta_0, \mu\} = \{\Delta_{\text{QCP}}, 0\}$ is a quantum phase transition between the topologically non-trivial BCS and trivial BEC phases. The quasiparticle spectrum has a gapless Dirac node in the bulk at $\mathbf{k} = 0$ when $\Delta_0 = \Delta_{\text{QCP}}$.

where $\varrho(k)$ and $\Theta(k)$ are real, continuous functions of k , independent of ϕ_k , $|\varrho(k)| \leq 1$, and $|\varrho(0)| = -\varrho(k \rightarrow \infty) = 1$. Then the integrand reduces to a total derivative, leading to $Q = \frac{1}{2} \{ \text{sgn}[\varrho(0)] + 1 \}$. In the $p + ip$ ground state,

$$\varrho(k) = \frac{2\mu - k^2}{2E_k}, \quad \Theta(k) = 0,$$

so that

$$Q = \begin{cases} 1, & \mu > 0 \text{ (BCS)}, \\ 0, & \mu < 0 \text{ (BEC)}. \end{cases} \quad (8.8)$$

At $\Delta_0 = \Delta_{\text{QCP}}$ ($\mu = 0$), both $\langle \vec{s}_{\mathbf{k}=0} \rangle$ and Q are undefined. Pseudospin textures at the critical point and in the BEC phase are depicted in Figs. 8.2(b,c).

A different formulation of the invariant based upon the TKNN formula[53] was derived by Volovik,[24] and utilizes the retarded single particle Green's function

$$\begin{aligned} \mathcal{G}_{\mathbf{k}}(t, t') \equiv & -i \begin{bmatrix} \langle \{c_{-\mathbf{k}}^\dagger(t), c_{-\mathbf{k}}(t')\} \rangle & \langle \{c_{-\mathbf{k}}^\dagger(t), c_{\mathbf{k}}^\dagger(t')\} \rangle \\ \langle \{c_{\mathbf{k}}(t), c_{-\mathbf{k}}(t')\} \rangle & \langle \{c_{\mathbf{k}}(t), c_{\mathbf{k}}^\dagger(t')\} \rangle \end{bmatrix} \\ & \times \theta(t - t'). \end{aligned} \quad (8.9)$$

The winding number W is[24, 44, 45]

$$W \equiv \frac{\epsilon_{\alpha\beta\gamma}}{3!} \int_{-\infty}^{\infty} d\omega \int \frac{d^2\mathbf{k}}{(2\pi)^2} \text{Tr} \begin{bmatrix} \mathcal{G}^{-1}(\partial_\alpha \mathcal{G}) \\ \times \mathcal{G}^{-1}(\partial_\beta \mathcal{G}) \\ \times \mathcal{G}^{-1}(\partial_\gamma \mathcal{G}) \end{bmatrix}, \quad (8.10)$$

where Tr denotes the trace in Nambu (particle-hole) space, and $\mathcal{G} \equiv \mathcal{G}_{\mathbf{k}}(i\omega)$ is the Fourier transform of $\mathcal{G}_{\mathbf{k}}(t, 0)$, analytically-continued to imaginary frequency. In both the BCS and BEC ground states, $W = Q$.

In the BCS phase, the advent of a non-zero bulk winding number implies the presence of 1D chiral Majorana edge states at the boundary of a superfluid droplet.[24, 25] Gapless Majorana edge channels are the hallmark of a topological superconductor.[1] When a temperature gradient is applied across the droplet, these states carry a perpendicular, dissipationless energy current, with a quantized thermal Hall conductance[54, 25, 55]

$$\kappa_{xy} = \frac{\pi^2 k_B T}{12\pi\hbar}.$$

Here T is the average temperature of the bulk. Additional ‘‘Majorana’’ signatures include isolated zero modes in type II vortices.[25]

8.1.2 Outline

This paper is organized as follows. In Sec. 8.2, we provide an overview of our main results concerning the order parameter dynamics and asymptotic winding numbers following a quench. We discuss implications for Majorana edge states and RF spectroscopy. These results are derived in the remaining sections. In Sec. 8.3, we derive the quench phase diagram from the exact solution to the long-time dynamics. We exploit a new Lax vector construction for the integrable p-wave problem. The solution obtains by classifying the isolated roots of the system's spectral polynomial. In Sec. 8.4, we establish the precise relation between the roots of the spectral polynomial and the steady-state order parameter dynamics. We compute the exact form of the persistent oscillations in $\Delta(t)$ for weak-to-strong quenches, and we present

formulae relating the period and amplitude of these to the isolated roots. In Sec. 8.5, we derive the out-of-equilibrium pseudospin and Cooper pair distribution functions. Using these results, for quenches wherein $\Delta(t)$ asymptotes to a finite constant (which may be zero), we derive the power-law approach to this value. We also compute the asymptotic values of the winding numbers W and Q , defined above. We conclude with open questions in Sec. 8.6.

Various technical details are relegated to the appendices. Ground state properties, including the tunneling density of states, are reviewed in Appendix 8.7. Appendix 8.9 supplies additional results on persistent order parameter oscillations in a narrow sliver of the quench phase diagram. Finally, in Appendix 8.10 we compute the coherence factors and single particle Green's functions following the quench.

8.2 P+ip superfluid quench: Key results

8.2.1 Chiral p-wave BCS model

To study quench dynamics in a 2D $p + ip$ superfluid, we consider a “chiral” variant[41] of the model in Eq. (8.1), defined via

$$H = \sum_{\mathbf{k}} k^2 s_{\mathbf{k}}^z - G \sum_{\mathbf{k}, \mathbf{q}} (k^x - ik^y)(q^x + iq^y) s_{\mathbf{k}}^+ s_{\mathbf{q}}^-, \quad (8.11)$$

where the mass $m = 1$. The relation between Eqs. (8.1) and (8.11) follows from

$$\mathbf{k} \cdot \mathbf{q} = \frac{1}{2} [(k^x - ik^y)(q^x + iq^y) + (k^x + ik^y)(q^x - iq^y)],$$

discarding the second term. In the thermodynamic limit, H in Eq. (8.11) possesses the same $p + ip$ ground state (BCS product wavefunction)[Schrieffer(1989)] as Eq. (8.1), in both the topologically non-trivial BCS and trivial BEC phases. However, the model in Eq. (8.11) breaks time-reversal symmetry explicitly, and preferentially selects $k^x - ik^y$ over $k^x + ik^y$ pairing. These are degenerate in the time-reversal invariant Hamiltonian of Eq. (8.1).

We work with Eq. (8.11) instead of Eq. (8.1) because the former is of Richardson-Gaudin[56, 57, 58] type and therefore integrable;[39, 40, 41, 42, 43] e.g., equilibrium properties can be extracted exactly via the Bethe ansatz. We can absorb the polar phase into the pseudospins $s_{\mathbf{k}}^- \rightarrow \exp(-i\phi_{\mathbf{k}}) s_{\mathbf{k}}^-$, and sum spins along arcs in momentum space:

$$\vec{s}_{\mathbf{k}} \equiv \frac{1}{\pi} \int_0^\pi d\phi_{\mathbf{k}} \vec{s}_{\mathbf{k}}. \quad (8.12)$$

As a result Eq. (8.11) reduces to an effective “1D” model

$$H = \sum_i \varepsilon_i s_i^z - G \sum_i \sqrt{\varepsilon_i} s_i^+ \sum_j \sqrt{\varepsilon_j} s_j^-, \quad (8.13)$$

where $\varepsilon_i \equiv k_i^2$. The Heisenberg equations of motion for the pseudospins are

$$\begin{aligned} \frac{d}{dt} \langle \vec{s}_i \rangle &= - \langle \vec{B}_i \times \vec{s}_i \rangle, \\ \vec{B}_i &\equiv -\varepsilon_i \hat{z} - 2\sqrt{\varepsilon_i} (\Delta_x \hat{x} + \Delta_y \hat{y}), \end{aligned} \quad (8.14)$$

$$\Delta \equiv \Delta_x - i\Delta_y \equiv -G \sum_i \sqrt{\varepsilon_i} s_i^-. \quad (8.15)$$

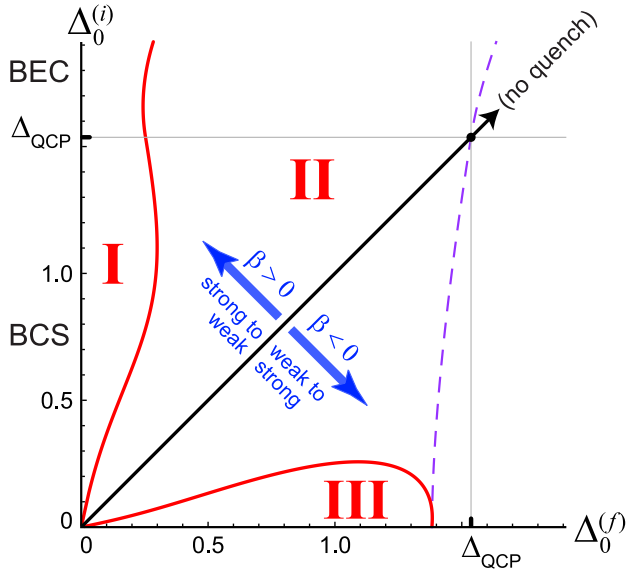


Figure 8.4: Exact interaction strength quench phase diagram, extracted from the isolated roots of the spectral polynomial. The vertical axis measures the initial pairing amplitude $\Delta_0^{(i)}$. Δ_{QCP} marks the equilibrium BCS-to-BEC topological quantum phase transition. The horizontal axis specifies the post-quench Hamiltonian through $\Delta_0^{(f)}$, which is the order parameter the system would exhibit in its ground state. The diagonal line $\Delta_0^{(i)} = \Delta_0^{(f)}$ is the case of no quench. Off-diagonal points to the left (right) describe strong-to-weak (weak-to-strong) pairing quenches, wherein the dynamic variable $\Delta(t)$ evolves away from its initial value. Similar to the s-wave case,[6, 19, 18, 9, 11, 10, 12, 13, 14] the $p + ip$ system exhibits three different dynamical phases defined by the long-time asymptotics ($t \rightarrow \infty$) of Δ . In phase **I**, $\Delta(t) \rightarrow 0$ due to dephasing,[13, 14] in **II**, $\Delta(t) \rightarrow \Delta_\infty$, a non-zero constant,[10, 12] and in phase **III**, $\Delta(t)$ shows persistent oscillations.[6, 12, 13] The dashed purple line is the non-equilibrium continuation of the quantum phase transition, in the sense that the asymptotic value of the chemical potential $\mu_\infty \equiv \lim_{t \rightarrow \infty} \mu(t)$ vanishes. This leads to a change in the retarded Green's function winding number W (Fig. 8.9, below). For this plot and all subsequent figures, we choose the Fermi energy $\mathcal{E}_F = 2\pi n = 5.18$ and the energy cutoff $\Lambda = 50\mathcal{E}_F$, so that $\Delta_{\text{QCP}} = 1.54$.

In the first equation, we take the expectation $\langle \cdots \rangle$ with respect to the initial state. Due to the infinite-ranged nature of the interactions in the BCS Hamiltonian, self-consistent mean field theory becomes *exact* in the thermodynamic limit.[51, 56] This is because Δ is an extensive variable depending upon all N of the spins in the system, and can be replaced by its expectation value in the limit $N \rightarrow \infty$. For a global quench, the instantaneous state of the system is described by a BCS product wavefunction at all times, albeit one parameterized by time-dependent coherence factors.[6] This implies that the problem reduces to solving Eq. (8.14), treating the spins and Δ as classical variables.[6, 9, 11, 10]

In Appendix 8.8, we demonstrate that the classical dynamics following from a $p + ip$ initial pseudospin configuration are in fact *identical* when generated by either Eq. (8.1) or Eq. (8.11); we therefore expect our predictions hold for the full quantum dynamics of Eq. (8.1) as well, in the thermodynamic limit. The main approximation employed in the present work is not tied to the distinction between Eqs. (8.1) and (8.11), but rather the neglect of pair-breaking processes. These are always present, break the integrability of the BCS Hamiltonian, and should ultimately induce thermalization. We will discuss time scales relevant to pair-breaking in Sec. 8.6.

8.2.2 Non-equilibrium phase diagram and asymptotic order parameter dynamics

We consider quenches in the model of Eq. (8.11) [or equivalently, Eq. (8.13)]. The system is initially prepared in the $p + ip$ ground state of the pre-quench Hamiltonian H_i , which has interaction strength G_i and amplitude $\Delta_0^{(i)}$. At time $t = 0$, the coupling is instantaneously deformed $G_i \rightarrow G_f$. We can label the quench by the initial pairing amplitude $\Delta_0^{(i)}$ and the quench parameter

$$\beta \equiv 2\pi \left(\frac{1}{g_f} - \frac{1}{g_i} \right), \quad (8.16)$$

where

$$g \equiv G\mathfrak{L}^2/4 \quad (8.17)$$

is the interaction strength that remains well-defined in the thermodynamic limit; $\mathfrak{L}$ is the linear system size. The case of zero quench has $\beta = 0$; $\beta > 0$ ($\beta < 0$) signifies a quench towards weaker (stronger) pairing in the post-quench Hamiltonian H_f . Although g carries units of length-squared and is therefore formally irrelevant in an RG sense, the integrals necessary to compute the quench dynamics are at most logarithmically divergent in the high energy ultraviolet cutoff Λ . These can be evaluated to logarithmic accuracy. Parameters $(\Delta_0)^2$, μ , β , and the fixed particle density n carry the same units; the latter sets the natural scale.

A quench $\{\Delta_0^{(i)}, \beta\}$ can also be specified via “quench coordinates” $\{\Delta_0^{(i)}, \Delta_0^{(f)}\}$, where $\Delta_0^{(f)}$ denotes the pairing amplitude associated to the *ground state* of H_f . This is not to be confused with the dynamic variable $\Delta(t)$, which evolves away from its initial value for any $\beta \neq 0$. Using the BCS Eq. (8.162) in Appendix 8.7, we can express $\beta = \beta(\Delta_0^{(i)}, \Delta_0^{(f)})$, a function of the pre- and post-quench Hamiltonian ground state order parameter amplitudes, with $\beta(\Delta_0, \Delta_0) = 0$. An explicit formula appears in Eq. (8.69).

In this work, we employ a generalized Lax construction[10, 11] to solve the integrable dynamics of the classical pseudospins governed by Eq. (8.13), given a $p + ip$ paired initial state. The key to the solution is the so-called “spectral polynomial” $\mathcal{Q}_{2N}(u; \beta)$, defined via a suitable Lax vector norm (see Sec. 8.3.1 for details). For a system of N spins, $\mathcal{Q}_{2N}(u; \beta)$ is a rank $2N$ polynomial in an auxiliary parameter u ; it is also a conserved integral of motion for any value of u . The polynomial coefficients (which are also integrals of motion) are complicated functions of the pseudospins $\{\vec{s}_j\}$. The spectral polynomial encodes all essential aspects of the quench.[10, 11, 12, 13] It is a function of the post-quench coupling strength G_f , and of the pre-quench state; the coefficients can be evaluated in terms of the spin configuration at $t = 0$.

There is a separation of global versus local dynamics in the BCS quench problem. The long-time evolution of the order parameter is determined by the isolated roots[11, 10, 12] of $\mathcal{Q}_{2N}(u; \beta)$. These always appear in pairs and are few in number for a quench. In a quench with M isolated pairs, $\Delta(t \rightarrow \infty)$ is governed by an effective M -spin problem, with parameters specified by the roots.[12] Once the asymptotic dynamics of $\Delta(t)$ are determined, more detailed information can be extracted. In particular, the pseudospin distribution required to compute winding numbers and Green’s functions in the limit $t \rightarrow \infty$ follows from the conservation of the spectral polynomial and the behavior of $\Delta(t)$.

Our results for the order parameter dynamics are summarized in Fig. 8.4, which shows the non-equilibrium phase diagram. The initial pre-quench state is labeled by $\Delta_0^{(i)}$ on the vertical axis; the post-quench Hamiltonian is identified by $\Delta_0^{(f)}$ (the *ground state* pairing amplitude of H_f) on the horizontal. The diagonal line $\Delta_0^{(i)} = \Delta_0^{(f)}$ corresponds to the ground state (no quench), while points to the left (right) of this line indicate strong-to-weak (weak-to-strong) quenches. Each point in this diagram represents a specific quench. As in previous studies of

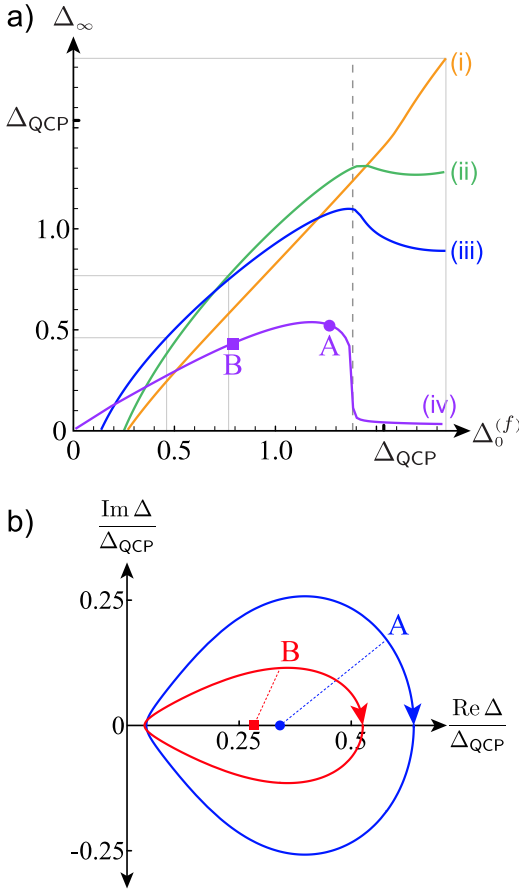


Figure 8.5: Asymptotic values of the non-equilibrium order parameter induced by various quenches. In (a), we plot $\lim_{t \rightarrow \infty} \Delta(t) = \Delta_\infty$ as a function of the post-quench ground state amplitude $\Delta_0^{(f)}$, for fixed values of the initial $\Delta_0^{(i)}$. Curves (i)–(iv) respectively correspond to $\Delta_0^{(i)}/\Delta_{\text{QCP}} = \{1.2, 0.5, 0.3, 0.00651\}$; each gives Δ_∞ along a horizontal cut across the quench phase diagram in Fig. 8.4. The value of Δ_∞ is determined by the isolated roots in Eq. (8.89). For the quench specified by $\{\Delta_0^{(i)}, \Delta_0^{(f)}\}$, the roots are computed from the conserved spectral polynomial [Eq. (8.64)]. The dashed vertical line marks the boundary separating phases **III** and **II** at $\Delta_0^{(i)} = 0$. The portion of curve (iv) to the left of this line in fact represents quenches in phase **III**, while the portion to the right resides in phase **II**. Instead of asymptoting to a constant, $\Delta(t)$ executes periodic amplitude and phase motion in phase **III**. For a given $\Delta_0^{(f)}$ to the left of the dashed line, the value of the curve (iv) specifies the average $(|\Delta_+| + |\Delta_-|)/2$, where $|\Delta_\pm|$ denote the turning points of the orbit in the complex Δ -plane. Phase **III** orbits[59] associated to the quenches marked A and B appear in (b).

the s-wave case,[6, 19, 18, 9, 11, 10, 12, 13, 14] we find that the $p + ip$ order parameter exhibits only three different classes of long-time dynamics, labeled **I**, **II**, **III** in Fig. 8.4. For strong-to-weak pairing quenches in phase **I**, $\Delta(t)$ decays to zero due to dephasing; this is the case of zero isolated pairs in $\mathcal{Q}_{2N}(u; \beta)$. Phase **I** is an example of gapless superconductivity: pair oscillations produce a continuous frequency spectrum with no isolated frequencies separated from the continuum.[13, 14] The Riemann-Lebesgue lemma then implies that $\Delta(t \rightarrow \infty) = 0$. Quenches in phase **II** exhibit $\Delta(t) \rightarrow \Delta_\infty$ as $t \rightarrow \infty$, where Δ_∞ is a non-zero constant. In this case, there is a single pair of isolated roots in $\mathcal{Q}_{2N}(u; \beta)$. Finally, weak-to-strong quenches in phase **III** induce persistent oscillations in $\Delta(t)$; here $\mathcal{Q}_{2N}(u; \beta)$ has two isolated pairs of roots.

A key difference from previous work[6, 9, 11, 10, 12, 13, 14] is that the chemical potential $\mu(t)$ is also a dynamical variable here. This occurs because we consider quenches to and from intermediate and strong pairing, wherein μ deviates from the Fermi energy even in the ground

state [Eq. (8.5) and Fig. 8.3]. In phase **II**, the chemical potential asymptotes to a constant μ_∞ , which is positive (negative) to the left (right) of the dashed purple line shown in Fig. 8.4. This line is the non-equilibrium extension of the topological quantum phase transition at $\Delta_0 = \Delta_{\text{QCP}}$. As discussed below, the asymptotic value ($t \rightarrow \infty$) of the Green's function winding number W [Eq. (8.10)] associated with Majorana edge modes changes across this line.

In phases **II** and **III** wherein the order parameter remains non-zero, the quantitative description of the asymptotic dynamics is entirely encoded in the isolated roots of $\mathcal{Q}_{2N}(u; \beta)$. These solve a particular transcendental equation in the thermodynamic limit, and can be extracted for any quench. Results for the asymptotic order parameter Δ_∞ and chemical potential μ_∞ amplitudes are plotted for horizontal cuts across the quench phase diagram in Figs. 8.5(a) and 8.6. Persistent oscillations in phase **III** are depicted in Figs. 8.5(b) and 8.7. In the latter, the result obtained from the isolated roots is compared to a direct simulation of 5024 coupled spins.

8.2.3 One quench, two winding numbers

Our main purpose is to characterize the dynamics of the system topology following a global quench. We compute the winding numbers Q and W respectively defined by Eqs. (10.17) and (8.10) for quenches in the dynamical phases **I–III** of Fig. 8.4.

In the initial BCS or BEC ground state, $Q = W$. We find that the pseudospin winding number Q does not change following a quench, as indicated in Fig. 8.8. By contrast, the retarded Green's function winding W undergoes a dynamical transition for a quench across the quantum critical point. We argue below that W determines the presence or absence of chiral Majorana edge modes in the post-quench asymptotic state for quenches in phase **II**. A quench in which $W \neq Q$ as $t \rightarrow \infty$ incurs a non-equilibrium topological quantum phase transition.

In phase **II** of Fig. 8.4 wherein $\{\Delta(t), \mu(t)\} \rightarrow \{\Delta_\infty, \mu_\infty\}$ as $t \rightarrow \infty$, $W = 1$ ($W = 0$) when $\mu_\infty > 0$ ($\mu_\infty < 0$). These regions are indicated in Fig. 8.9. The dashed purple line has $\mu_\infty = 0$, and is the extension of the topological quantum critical point into the non-equilibrium phase diagram.

Phase **II** quenches in which W changes relative to Q occur in two regions. Strong-to-weak pairing quenches across the critical point ($\Delta_0^{(i)} > \Delta_{\text{QCP}}$, $\Delta_0^{(f)}$ to the left of the $\mu_\infty = 0$ line) have $Q = 0$ and $W = 1$, region **C** in Fig. 8.10. Weak-to-strong quenches across the non-equilibrium quantum phase boundary ($\Delta_0^{(i)} < \Delta_{\text{QCP}}$, $\Delta_0^{(f)}$ to the right of the $\mu_\infty = 0$ line) have $Q = 1$ and $W = 0$, region **H** in Fig. 8.10. Our methods allow access to the asymptotic behavior; we do not compute the transient kinetics of the topological transition wherein W changes.

We now discuss implications specific to the particular winding numbers.

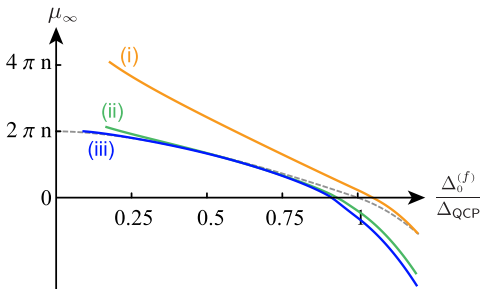


Figure 8.6: Asymptotic value of the non-equilibrium chemical potential $\lim_{t \rightarrow \infty} \mu(t) = \mu_\infty$ for quenches in phase **II**, as a function of the post-quench ground state amplitude $\Delta_0^{(f)}$, for fixed values of the initial $\Delta_0^{(i)}$. Curves (i)–(iii) correspond to the associated quenches in Fig. 8.5(a). The dashed line is the ground state curve [Eq. (8.5)].

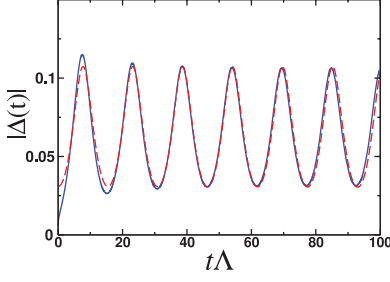


Figure 8.7: Example of persistent order parameter oscillations following a quench. The coordinates $\{\Delta_0^{(i)}, \Delta_0^{(f)}\} = \{.00503, 0.108\}$ place this quench in phase **III** of Fig. 8.4. The blue solid curve is the result of numerical simulation for classical pseudospins (5024 spins). The red dashed curve obtains from the effective two-spin analytical solution, the parameters of which are extracted from the isolated roots of the spectral polynomial $\mathcal{Q}_{2N}(u; \beta)$ (see text).

Non-equilibrium gapless topological phase

The conservation of the pseudospin winding number Q [Eq. (10.17)] is simple to understand. Under the dynamics induced by Eq. (8.11), spins along equal-radius arcs in momentum space evolve collectively. For a $p + ip$ initial state, the relative canting of spins with equal k is determined by the polar phase ϕ_k , and this does not change; the spin texture is chiral at any time $t \geq 0$. The pseudospins can be parameterized as in Eq. (8.7), with $\varrho(k)$ and $\Theta(k)$ now time-dependent parameters. The effective dynamics are captured by the “1D” model in Eq. (8.13). The spin at zero energy is stationary because it is decoupled from Δ , see Eq. (8.14). By continuity, low energy spins remain close to the zero energy spin over a time interval of order the inverse level spacing, beyond which the notion of smooth topology becomes meaningless. Up until this time, Q is conserved.

This has interesting implications in phase **I** of Fig. 8.4, wherein $\Delta(t)$ decays to zero. At sufficiently long times, the effective magnetic field acting upon spin $\vec{s}_i$ reduces to $\vec{B}_i = -\varepsilon_i \hat{z}$. However, the gapless phase is *not* a Fermi liquid ground state, which would have spins aligned along the field,[60] nor can it be understood as a finite temperature normal fluid. Instead, phase **I** is a quench-induced state of gapless superconductivity with a non-zero superfluid density.[14] The spin configuration can be parameterized as

$$\vec{s}_i(t) = \frac{1}{2} \sqrt{1 - \gamma_i^2} \begin{Bmatrix} \cos(\varepsilon_i t + \Theta_i) \hat{x} \\ + \sin(\varepsilon_i t + \Theta_i) \hat{y} \end{Bmatrix} + \frac{1}{2} \gamma_i \hat{z}, \quad (8.18)$$

where Θ_i is some constant phase. The precession frequency of a spin at radius k_i is $\varepsilon_i = k_i^2$, twice the bare energy. The parameter γ_i gives the z-projection of the i^{th} spin in the $t \rightarrow \infty$ limit. This is the “distribution function” for the Anderson pseudospins, equivalent to the fermion mode occupation minus one, which characterizes the out-of-equilibrium state. The zero temperature Fermi liquid would have $\gamma_i = \text{sgn}(\varepsilon_i - 2\mathcal{E}_F)$, with $\mathcal{E}_F$ the Fermi energy. For the quench, we compute γ_i exactly in the thermodynamic limit using the conservation of the Lax vector norm.

In Figs. 8.11 and 8.12, we plot $2s_i^z = \gamma_i$ for representative quenches in regions **A** and **B** of Fig. 8.8. The pseudospin distributions resemble those of the initial, pre-quench ground state with pairing amplitude $\Delta_0^{(i)}$, and the winding Q is the same. Nevertheless, the post-quench state is gapless, due to dephasing of the spins. In particular, a quench in region **B** induces a “gapless topological” state with $Q = 1$ and $\Delta(t \rightarrow \infty) = 0$. The state can be visualized as an undulating (time-evolving) variant of the skyrmion texture shown in Fig. 8.2(a), where the pseudospin $\vec{s}_{\mathbf{k}}$ precesses about $\hat{z}$ at frequency k^2 [Eq. (8.18)].

Knowledge of γ_i allows the self-consistent determination of $\Delta(t)$. For an initial state not at

the quantum critical point $\Delta_0^{(i)} \neq \Delta_{\text{QCP}}$, we find that

$$\Delta(t \rightarrow \infty) \propto \frac{c_1}{\Lambda} \left(\frac{1}{t} \right) \exp(-2i\Lambda t) + c_2 \frac{1}{t^2}, \quad (8.19)$$

where $c_{1,2}$ are constants and Λ is the high-energy cutoff appearing in (e.g.) Eq. (8.5). Ignoring the high-frequency, cutoff-dependent piece, the dominant decay is $1/t^2$. By contrast, for $\Delta_0^{(i)} = \Delta_{\text{QCP}}$ the cutoff-independent decay law is slower:

$$\Delta(t \rightarrow \infty) \propto \frac{\tilde{c}_1}{\Lambda} \left(\frac{1}{t} \right) \exp(-2i\Lambda t) + \tilde{c}_2 \frac{1}{t^{3/2}}. \quad (8.20)$$

Asymptotic Bogoliubov-de Gennes spectrum and edge states

The retarded Green's function winding W in Eq. (8.10) is well-defined as $t \rightarrow \infty$ in phase **II** of Fig. 8.4, whereupon $\{\Delta(t), \mu(t)\} \rightarrow \{\Delta_\infty, \mu_\infty\}$. As discussed above, W changes from its initial value Q for quenches across the topological quantum critical point, regions **C** and **H** in Fig. 8.10. This follows from solving the Bogoliubov-de Gennes equation in the asymptotic steady state:

$$i \frac{d}{dt} \mathcal{G}_{\mathbf{k}}(t, t') = \begin{bmatrix} -\frac{k^2}{2} + \mu_\infty & (k_x + ik_y) \Delta_\infty \\ (k_x - ik_y) \Delta_\infty & \frac{k^2}{2} - \mu_\infty \end{bmatrix} \mathcal{G}_{\mathbf{k}}(t, t'), \quad (8.21)$$

subject to the initial condition

$$\lim_{\delta t \rightarrow 0^+} \mathcal{G}_{\mathbf{k}}(t + \delta t, t) = -i \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}. \quad (8.22)$$

Technically it is the magnitude $|\Delta(t)|$ that asymptotes to a finite constant in phase **II**: Using the definition in Eq. (8.15), the phase of the order parameter winds according to $\Delta(t) \rightarrow \Delta_\infty \exp(-2i\mu_\infty t)$. In Eq. (8.21) and all following equations in this section, we work in the rotating frame that eliminates this phase.[60] The function $\mathcal{G}_{\mathbf{k}}(t, t')$ encodes only the asymptotic pairing amplitude and chemical potential, not the non-equilibrium spin distribution function. The solution to Eq. (8.21) is identical to that in the BCS or BEC ground state, but with Δ_∞ and μ_∞ determined by the quench. Eq. (8.10) then implies that $W = 1$ ($W = 0$) for $\mu_\infty > 0$ ($\mu_\infty < 0$). This is specific to the retarded function; other Green's functions (e.g., Keldysh) do depend upon the asymptotic pseudospin configuration.

The question of Majorana edge modes in the spectrum of a system with a boundary is determined by solving the effective Bogoliubov-de Gennes mean field Hamiltonian H_{BdG} in the appropriate geometry. For the quench, we have

$$H_{\text{BdG}}(t) = \sum_{\mathbf{k}} \{ [k^2 - 2\mu(t)] s_{\mathbf{k}}^z + \mathbf{k} \Delta(t) s_{\mathbf{k}}^+ + \mathbf{k}^* \Delta^*(t) s_{\mathbf{k}}^- \}, \quad (8.23)$$

where $\mathbf{k} \equiv k_x - ik_y$. Since H_{BdG} and $\mathcal{G}_{\mathbf{k}}(t, t')$ encode the same information as $t \rightarrow \infty$, one expects edge modes in the asymptotic spectrum when $W = 1$.

We conclude that Majorana edge modes appear when $W \neq 0$ in the post-quench state. This implies that the edge spectrum can change following a quench across the critical point. In region **C** of Fig. 8.10, the initially trivial BEC state develops edge modes in the H_{BdG} spectrum, while the modes present in the initial BCS state disappear from the spectrum in region **H**. Using a Floquet analysis,[47, 48, 49, 50] we have established that phase **III** also hosts gapless Majorana edge modes. Details appear elsewhere.[46]

Within the bulk integrable theory, we cannot determine the *occupation* of edge states following a quench. This is because a spatial edge breaks the integrability. In future work we will investigate possible experimental signatures of the edge states following a quench, such as a quantized dissipationless energy current.[54, 25, 55] The characterization of Majorana zero modes following a quench in a 1D topological superconductor has been studied numerically,[61] for a parameter change in a non-interacting (non-self-consistent) Bogoliubov-de Gennes model.

8.2.4 A new $\mathbb{Z}_2$: Parity of the non-equilibrium Cooper pair distribution

We have established above that the two different formulations of the bulk topological invariant, although equivalent in equilibrium, can differ following a quench, and we have discussed implications for the presence or absence of chiral Majorana modes at the edge of the sample. Are there any experimentally-accessible bulk signatures of the topological transition that occur when $W \neq Q$? Here we discuss one possibility in the parity of zeroes of the Cooper pair distribution function, defined below.

For a quench in which the order parameter asymptotes to a non-zero constant (phase **II** of Fig. 8.4), the pseudospins precess around the “effective ground state field”

$$\vec{B}_i = -(\varepsilon_i - 2\mu_\infty)\hat{z} - 2\sqrt{\varepsilon_i}\Delta_\infty\hat{x}. \quad (8.24)$$

The solution is

$$\vec{s}_i(t) = \frac{1}{2}\sqrt{1 - \gamma_i^2} \left\{ \cos[2E_\infty(\varepsilon_i)t + \Theta_i] \hat{B}_i \times \hat{y} \right\} - \frac{\gamma_i}{2} \hat{B}_i, \quad (8.25)$$

where $\hat{B}_i \equiv \vec{B}_i/|\vec{B}_i|$, $E_\infty(\varepsilon) \equiv E(\varepsilon; \Delta_\infty, \mu_\infty)$, and

$$E(\varepsilon; \Delta_0, \mu) = \sqrt{\left(\frac{\varepsilon}{2} - \mu\right)^2 + \varepsilon\Delta_0^2}. \quad (8.26)$$

In Eq. (8.25), γ_i is the “Cooper pair distribution,” which measures the projection of the pseudospin onto $-\vec{B}_i$. [Note that this is different from the definition employed in the gapless case, Eq. (8.18).] In the ground state, $\gamma_i = -1$ for all pseudospins, while the configuration with $\gamma_i = +1$ for all i is a metastable negative temperature state. We refer to a spin with $\gamma_i = -1$ as a ground state pair, while $\gamma_i = +1$ is an excited pair.[62]

For the quench, we compute γ_i exactly in the limit $t \rightarrow \infty$ using the conservation of the Lax vector norm. We find that γ_i exhibits an odd number of zeroes whenever $W \neq Q$, whereas the number of zeroes is even (and typically zero) when $W = Q$. Two examples are shown in Fig. 8.13: The first is a quench across the quantum critical point from strong-to-weak pairing, region **C** in Fig. 8.10 with $W = 1$ and $Q = 0$. The second is a strong-to-weak quench within the BCS phase, region **D** in Fig. 8.10 with $W = Q = 1$.

The presence of an odd number of zeroes in γ_i is required by the conservation of the pseudospin winding Q . The effective field in Eq. (8.24) “winds” when $W = 1$ ($\mu_\infty > 0$). When W obtained in the asymptotic steady state differs from its initial, pre-quench value, γ_i must also “wind” so that the pseudospin index Q is conserved. Thus the parity of the number of zeroes in γ_i constitutes a new $\mathbb{Z}_2$ -valued index that encodes the retarded Green’s function invariant W , which can change following a quench.

The Cooper pair distribution appears in the amplitude for photon absorption or emission via RF spectroscopy.[62] In a cold atomic realization, absorption of an RF photon can destroy an Anderson pseudospin by breaking a Cooper pair. The photon is absorbed by one partner,

which is subsequently excited to a different internal state, denoted below as “3.” An atom in state 3 does not participate in pairing.

The RF-induced tunneling Hamiltonian is

$$H_T = T \sum_{\mathbf{k}} \left[e^{i\omega_L t} c_{\mathbf{k}}^\dagger d_{\mathbf{k}} + e^{-i\omega_L t} d_{\mathbf{k}}^\dagger c_{\mathbf{k}} \right]. \quad (8.27)$$

Here $d_{\mathbf{k}}$ annihilates a state 3 atom with momentum $\mathbf{k}$ and $\omega_L > 0$ is the frequency of the RF source. We denote the total number of state 3 atoms by

$$N_d \equiv \sum_{\mathbf{k}} d_{\mathbf{k}}^\dagger d_{\mathbf{k}}.$$

In the asymptotic steady-state following a quench in phase **II**, a straight-forward linear response calculation gives the RF current

$$\left\langle \frac{d}{dt} N_d(t) \right\rangle = 2\pi\nu_0 T^2 \frac{\varepsilon_\omega \Delta_\infty^2}{|\omega| |\omega + \Delta_\infty^2|} \theta(\varepsilon_\omega) A(\omega), \quad (8.28)$$

where

$$\begin{aligned} A(\omega) = & \left\{ \begin{aligned} & [1 - \gamma(\varepsilon_\omega)][1 - n^{(d)}(\varepsilon_\omega)] \\ & - [1 + \gamma(\varepsilon_\omega)]n^{(d)}(\varepsilon_\omega) \end{aligned} \right\} \theta(\omega) \\ & + \left\{ \begin{aligned} & [1 + \gamma(\varepsilon_\omega)][1 - n^{(d)}(\varepsilon_\omega)] \\ & - [1 - \gamma(\varepsilon_\omega)]n^{(d)}(\varepsilon_\omega) \end{aligned} \right\} \theta(-\omega). \end{aligned} \quad (8.29)$$

In these equations, $\theta(\varepsilon)$ denotes the Heaviside unit step function, and $n^{(d)}(\varepsilon = k^2)$ is the initial occupation of atoms in state 3, equal to $\langle d_{\mathbf{k}}^\dagger d_{\mathbf{k}} \rangle_0$. The frequency ω is defined by

$$\omega \equiv \omega_L - \mathcal{E}_{3,2},$$

where $\mathcal{E}_{3,2} > 0$ denotes the atomic transition energy between states 3 and 2. We ignore high frequency processes that involve counter-rotating terms with $\omega_L \rightarrow -\omega_L$. In Eq. (8.28), ν_0 denotes the bare density of states [Eq. (8.33)] and the mode energy $\varepsilon_\omega/2$ is defined below in Eq. (8.30).

Eq. (8.29) follows from simple kinematics. The first term proportional to $[1 - n^{(d)}]$ describes the process wherein a photon with energy $\omega_L > \mathcal{E}_{3,2}$ is absorbed by a ground state Cooper pair with initial energy $-E(\varepsilon_\omega)$, exciting one partner to state 3 with energy $\varepsilon_\omega/2 + \mathcal{E}_{3,2} - \mu_\infty$. The remaining unpaired fermion carries energy zero, since $s_{\mathbf{k}}^a c_{\mathbf{k}}^\dagger |0\rangle = s_{\mathbf{k}}^a c_{-\mathbf{k}}^\dagger |0\rangle = 0$, where modes $\{\mathbf{k}, -\mathbf{k}\}$ are vacant in $|0\rangle$. The conservation of energy gives

$$\omega - E_\infty(\varepsilon_\omega) = \frac{\varepsilon_\omega}{2} - \mu_\infty,$$

which has the unique solution

$$\varepsilon_\omega = \frac{\omega(\omega + 2\mu_\infty)}{\Delta_\infty^2 + \omega}. \quad (8.30)$$

The second term in Eq. (8.29) proportional to $n^{(d)}$ is the inverse stimulated emission process. The third term describes the destruction of an excited state Cooper pair due to a photon absorption with $\omega_L < \mathcal{E}_{3,2}$, again creating a state 3 atom with energy $\varepsilon_\omega/2 + \mathcal{E}_{3,2} - \mu_\infty$ and an unpaired particle with zero energy. Energy balance is

$$\omega + E_\infty(\varepsilon_\omega) = \frac{\varepsilon_\omega}{2} - \mu_\infty,$$

with ε_ω again given by Eq. (8.30) and $\omega < 0$. The fourth term is the inverse emission process. The factor $\theta(\varepsilon_\omega)$ in Eq. (8.28) disallows unphysical processes requiring negative mode energies.

Conceptually, the simplest situation has an initially empty state 3 band; then $n^{(d)}(\varepsilon) = 0$ for all ε . Eq. (8.29) implies that the Cooper pair distribution function $\gamma(\varepsilon)$ can in principle be extracted from the RF spectroscopy current. However, this result ignores complications involving “off-diagonal” processes[62] that can become important for transitions involving states far from the Fermi energy. We defer a full treatment to future work.

8.3 Quench phase diagram

In this section, we derive the quench phase diagram in Fig. 8.4. Ground state properties of the model are reviewed in Appendix 8.7.

8.3.1 Lax construction, spectral polynomial, separation variables

Our starting point is the “1D” Hamiltonian in Eq. (8.13). This can be derived from the 2D chiral p-wave model[39, 41] in Eq. (8.11) by a canonical rescaling of the pseudospins,

$$s_{\mathbf{k}}^- \rightarrow \exp(-i\phi_k) s_{\mathbf{k}}^-, \quad s_{\mathbf{k}}^+ \rightarrow \exp(i\phi_k) s_{\mathbf{k}}^+, \quad (8.31)$$

where ϕ_k is the polar angle of $\mathbf{k}$. Applying Eq. (8.31) to H in Eq. (8.11) eliminates the phases of the complex momenta appearing in the pairing term. Pseudospins with the same momentum radius evolve collectively. For each k , we sum spins along the arc in Eq. (8.12) to obtain a single radial pseudospin $\vec{s}_k$. The Hamiltonian reduces to Eq. (8.13), where $\varepsilon_i \equiv k_i^2$ and $\vec{s}_i \equiv \vec{s}_{k_i}$. At any time t following a quench, the full 2D spin configuration is easily reconstructed.

In what follows, we switch frequently between discrete and continuum formulations of the problem. The connection is given by

$$\sum_i \Leftrightarrow \nu_0 \int_0^{2(\Lambda+\mu)} d\varepsilon, \quad (8.32)$$

where

$$\nu_0 \equiv \frac{\mathfrak{L}^2}{8\pi} \quad (8.33)$$

is the (bare) density of states and $\mathfrak{L}$ denotes the linear system size. On the right-hand side of Eq. (8.32), Λ is the high-energy cutoff; the chemical potential is incorporated here as a convenience, see Appendix 8.7. Using these conventions, all spins have $(\vec{s}_i)^2 = 1/4$.

Although the model in Eq. (8.13) is classically integrable as we demonstrate below, the spin equations of motion in Eq. (8.14) are not directly useful. Instead, we introduce a new Lax vector construction inspired by the s-wave case[56, 57, 11, 10, 12] and the Bethe ansatz formulation[41, 42, 43] of the p-wave model.

For a system of N pseudospins, we define the Lax vector components

$$\begin{aligned} L^+(u) &\equiv \sum_{i=1}^N \frac{\sqrt{\varepsilon_i} s_i^+}{\varepsilon_i - u}, \\ L^-(u) &\equiv \sum_{i=1}^N \frac{\sqrt{\varepsilon_i} s_i^-}{\varepsilon_i - u}, \\ L^z(u) &\equiv \sum_{i=1}^N \frac{\varepsilon_i s_i^z}{\varepsilon_i - u} + \frac{1}{2G}, \end{aligned} \quad (8.34)$$

where G is the interaction strength in Eq. (8.13). In these equations, u denotes an arbitrary complex-valued parameter. We also introduce a Lax vector norm:

$$L_2(u) \equiv uL^+(u)L^-(u) + [L^z(u)]^2. \quad (8.35)$$

Unlike the s-wave case,[56, 57, 11, 10, 12] the norm is not Euclidean. This is a key distinction that produces a different structure for isolated roots of the spectral polynomial, defined below.

Employing canonical Poisson bracket relations for the spins

$$\{s_i^a, s_j^b\} = \delta_{ij}\epsilon^{abc}s_j^c, \quad (8.36)$$

it is easy to show that

$$\begin{aligned} \{L^+(u), L^-(v)\} &= 2 \left[\frac{L^z(u) - L^z(v)}{u - v} \right], \\ \{L^z(u), L^+(v)\} &= \frac{uL^+(u) - vL^+(v)}{u - v}, \\ \{L^z(u), L^-(v)\} &= - \frac{uL^-(u) - vL^-(v)}{u - v}. \end{aligned} \quad (8.37)$$

These in turn imply that

$$\{L_2(u), L_2(v)\} = 0. \quad (8.38)$$

The Lax norm $L_2(u)$ is a generator for integrals of motion. Explicitly,

$$L_2(u) = \sum_i \frac{H_i}{u - \varepsilon_i} + \sum_i \frac{\varepsilon_i^2}{4(u - \varepsilon_i)^2} + \frac{1}{4G^2}, \quad (8.39)$$

where H_i denotes a central-spin type Hamiltonian,

$$\begin{aligned} H_i &= -\frac{1}{G}\varepsilon_i s_i^z + \varepsilon_i s_i^+ s_i^- \\ &+ \sum_{j \neq i} \left\{ \frac{\varepsilon_i \sqrt{\varepsilon_i \varepsilon_j} [s_i^+ s_j^- + s_j^+ s_i^-] + 2\varepsilon_i \varepsilon_j s_i^z s_j^z}{(\varepsilon_i - \varepsilon_j)} \right\}. \end{aligned} \quad (8.40)$$

There are N independent H_i 's in a system of N spins. Because Eq. (8.38) holds for generic u and v , it implies that the H_i 's are mutually conserved:

$$\{H_i, H_j\} = 0. \quad (8.41)$$

The BCS Hamiltonian in Eq. (8.13) is given by the sum

$$H = -G \sum_i H_i. \quad (8.42)$$

For the spin dynamics generated by H [Eq. (8.14)], the Lax components evolve according to

$$\frac{d\vec{L}(u)}{dt} = \det \begin{bmatrix} \hat{x} & \hat{y} & u\hat{z} \\ 2\Delta^x & 2\Delta^y & u \\ L^x(u) & L^y(u) & L^z(u) \end{bmatrix}, \quad (8.43)$$

where $\vec{L} \equiv \hat{x}L^x + \hat{y}L^y + \hat{z}L^z$ and $L^\pm = L^x \pm iL^y$.

We define the spectral polynomial

$$\mathcal{Q}_{2N}(u) \equiv G^2 \prod_{j=1}^N (u - \varepsilon_j)^2 L_2(u). \quad (8.44)$$

This is a polynomial of degree $2N$ in u , with coefficients that depend upon (a) the coupling strength G and (b) the pseudospin configuration $\{\vec{s}_i\}$. Eq. (8.43) implies that $L_2(u)$ and $\mathcal{Q}_{2N}(u)$ are integrals of motion.

For a quench, the roots of $\mathcal{Q}_{2N}(u)$ provide the key to determine the long-time asymptotic dynamics. Part of the story involves trading the spins for a more convenient set of coordinates. From Eq. (8.34), we write

$$L^-(u) = \frac{\Delta \prod_{\beta=1}^{N-1} (u - u_\beta)}{G \prod_{j=1}^N (u - \varepsilon_j)}. \quad (8.45)$$

In this equation, we have formed a common denominator. The numerator is a polynomial in u of degree $N - 1$ with zeroes $\{u_\beta\}$, which we term *separation variables*. [29, 23, 10] Each u_α is a complicated function of all N $\{s_i^-\}$, the precise form of which we will not need. The separation variables satisfy the Poisson bracket relations $\{u_\alpha, u_\beta\} = 0$. The prefactor in Eq. (8.45) follows by expanding the numerator and matching the coefficient of u^{N-1} with Eq. (8.34), using Eq. (8.15).

The BCS evolution of the Lax vector in Eq. (8.43) implies that

$$\frac{dL^-(u)}{dt} = -i [uL^-(u) - 2\Delta L^z(u)] + \frac{\partial L^-(u)}{\partial u} \left(\frac{du}{dt} \right),$$

allowing for a time-dependent parameter u . Evaluating this equation for a separation variable gives

$$0 = 2i\Delta L^z(u_\alpha) + \frac{\Delta \prod_{\beta \neq \alpha} (u_\alpha - u_\beta)}{G \prod_{j=1}^N (u_\alpha - \varepsilon_j)} \left(\frac{du_\alpha}{dt} \right). \quad (8.46)$$

Using Eqs. (8.35) and (8.44), we obtain the equations of motion

$$\frac{du_\alpha}{dt} = -2i \frac{\sqrt{\mathcal{Q}_{2N}(u_\alpha)}}{\prod_{\beta \neq \alpha} (u_\alpha - u_\beta)}. \quad (8.47)$$

The spins have been entirely eliminated in favor of coupled equations for the separation variables. One can also derive the following equation of motion for $\Delta(t)$, employing Eqs. (8.14), (8.15), and (8.45):

$$\frac{d\Delta}{dt} = i\Delta \left[\sum_{\beta=1}^{N-1} u_\beta - \sum_j \varepsilon_j - 2GH - 2|\Delta|^2 \right]. \quad (8.48)$$

Separation variables are in general complex-valued; to solve the equations of motion in (8.47), one has to choose a proper branch of $\sqrt{\mathcal{Q}_{2N}(u)}$ in the plane of complex u . This can be done by connecting pairs of roots of the polynomial $\mathcal{Q}_{2N}(u)$ with branch cuts. Separation variables cannot cross these cuts in their motion. Note that the number of separation variables ($N - 1$) is one less than the number of the branch cuts (N).

8.3.2 Ground state roots and spectral transitions

In the BCS or BEC ground state, the pseudospins satisfy

$$s_i^- = -\frac{\sqrt{\varepsilon_i}\Delta_0}{2E(\varepsilon_i; \Delta_0, \mu)}, \quad s_i^z = -\frac{(\varepsilon_i - 2\mu)}{4E(\varepsilon_i; \Delta_0, \mu)}, \quad (8.49)$$

where $E(\varepsilon; \Delta_0, \mu)$ is the quasiparticle energy, defined by Eq. (8.26). We take Δ_0 real and positive without loss of generality. Spin $\vec{s}_i$ lies along the field $\vec{B}_i + 2\mu\hat{z}$, which incorporates μ so as to fix the total density n . The pairing amplitude in Eq. (8.15) solves the BCS equation

$$\frac{1}{G} = \sum_i \frac{\varepsilon_i}{2E(\varepsilon_i; \Delta_0, \mu)}. \quad (8.50)$$

Eq. (8.5) determines μ in terms of the density n and Δ_0 . Eq. (8.162) relates these to the interaction strength, see Appendix 8.7.1 for details.

We evaluate the ground state spectral polynomial [Eq. (8.44)] by combining Eqs. (8.49), (8.34), and (8.35). The result is

$$\mathcal{Q}_{2N}(u) = \frac{G^2}{4}(u - u_0^{(+)})(u - u_0^{(-)})[P_{N-1}(u)]^2, \quad (8.51a)$$

$$P_{N-1}(u) \equiv \prod_{j=1}^N (u - \varepsilon_j) F(u; \Delta_0, \mu), \quad (8.51b)$$

$$F(u; \Delta_0, \mu) \equiv \sum_{i=1}^N \frac{\varepsilon_i}{2(u - \varepsilon_i)E(\varepsilon_i; \Delta_0, \mu)}. \quad (8.51c)$$

$P_{N-1}(u)$ denotes a polynomial of degree $N - 1$ in u . The zeroes of $F(u)$ fall between adjacent mode energies $\{\varepsilon_j, \varepsilon_{j+1}\}$, the latter non-negative and non-degenerate. The $N - 1$ distinct, positive real zeroes of $F(u)$ are the roots of $P_{N-1}(u)$. Each of these is a doubly-degenerate root of $\mathcal{Q}_{2N}(u)$. The remaining two roots $u_0^{(\pm)}$ solve $E(u; \Delta_0, \mu) = 0$ [Eq. (8.26)],

$$u_0^{(\pm)} = 2 \left[\mu - (\Delta_0)^2 \pm \Delta_0 \sqrt{(\Delta_0)^2 - 2\mu} \right]. \quad (8.52)$$

We first consider the $2(N - 1)$ positive real roots of $\mathcal{Q}_{2N}(u)$. Eq. (8.35) implies that each such root u_0 also satisfies $L^-(u_0) = 0$, i.e. u_0 is a separation variable [Eq. (8.45)]. Eq. (8.47) shows that this is a stationary solution to the equations of motion. Thus, in the BCS ground state, the $N - 1$ separation variables $\{u_\alpha\}$ are locked to the $N - 1$ distinct, positive real roots of $\mathcal{Q}_{2N}(u)$.

The single pair of isolated roots in Eq. (8.52) encodes key macroscopic features of the superfluid state: the order parameter Δ_0 and the chemical potential μ . These are constrained by Eq. (8.5) for a fixed particle density n , Fig. 8.3. Roots $u_0^{(\pm)}$ take values away from the positive real axis for all $\Delta_0 > 0$ [except at the critical point $\Delta_0 = \Delta_{\text{QCP}}$, Eq. (8.55) below], and cannot serve as stationary solutions for separation variables. Different from the s-wave case, the isolated roots for p-wave can be complex or negative real, due to the non-Euclidean norm in Eq. (8.35).

For the p-wave model, the pattern of isolated roots is tied to the strength of the pairing as measured by Δ_0 . Three special values $\Delta_{\text{Coh}} < \Delta_{\text{MR}} < \Delta_{\text{QCP}}$ separate four domains. These are implicitly defined through Eq. (8.163) in Appendix 8.7.2, which specifies the relation of each to the chemical potential. All three pairing amplitudes $\{\Delta_{\text{Coh}}, \Delta_{\text{MR}}, \Delta_{\text{QCP}}\}$ are of order

$\sqrt{4\pi n / \log(\Lambda/2\pi n)}$, and differ by terms of size $\sqrt{n}[\log(\Lambda/2\pi n)]^{-3/2}$; explicit values are transcribed in Eq. (8.164). In the following, we describe the four pairing domains in terms of the isolated roots, and via spectral features detectable in the tunneling density of states (TDoS). Both are depicted in Fig. 8.14.

For the weakest coupling strengths such that $\Delta_0 < \Delta_{\text{Coh}}$, the minimum of the quasiparticle spectrum $E(\varepsilon; \Delta_0, \mu)$ occurs at a non-zero mode energy $\varepsilon_0 \equiv 2(\mu - \Delta_0^2)$. The quasiparticle gap is

$$E_{\min} \equiv E(\varepsilon_0; \Delta_0, \mu) = \Delta_0 \sqrt{2\mu - \Delta_0^2}, \quad \Delta_0 < \Delta_{\text{Coh}}. \quad (8.53)$$

In this regime, the TDoS $\nu(\omega)$ possesses a coherence peak at the gap edge, Eq. (8.166) in Appendix 8.7.2 and Fig. 8.14(b). The corresponding isolated roots form a complex conjugate pair with positive real part. At $\Delta_0 = \Delta_{\text{Coh}}$ [where $\mu = (\Delta_{\text{Coh}})^2$, Eq. (8.163)], the TDoS coherence peak and the real part of $u_0^{(\pm)}$ both vanish. For stronger pairing, the bulk quasiparticle gap is determined by the chemical potential, and resides at $\varepsilon = 0$:

$$E_{\min} = E(0; \Delta_0, \mu) = |\mu|, \quad \Delta_0 \geq \Delta_{\text{Coh}}. \quad (8.54)$$

Increasing the interaction strength (yet remaining within the topologically non-trivial BCS phase), the roots move into the left-hand complex plane. At $\Delta_0 = \Delta_{\text{MR}}$ [$\mu = (\Delta_{\text{MR}})^2/2$], the roots collapse to a degenerate value on the negative real axis. This is a point (for fixed density) on the “Moore-Read” line discussed in Ref. [41]. Although gapped, the spectrum $E(\varepsilon; \Delta_{\text{MR}}, \mu)$ exhibits zero curvature (Appendix 8.7.2). As Δ_0 is further increased towards the topological phase transition at Δ_{QCP} , the roots split along the negative real axis; $u_0^{(+)}$ ($u_0^{(-)}$) becomes less (more) negative. At the critical point $\Delta_0 = \Delta_{\text{QCP}}$ and $\mu = 0$, the quasiparticle gap vanishes as indicated by the TDoS in Fig. 8.14(b). The roots are

$$u_0^{(+)} = 0, \quad u_0^{(-)} = -4(\Delta_{\text{QCP}})^2. \quad (8.55)$$

Entering the BEC phase with $\Delta_0 > \Delta_{\text{QCP}}$ (and $\mu < 0$), $u_0^{(+)}$ retreats back along the negative axis. Throughout the BEC phase, the TDoS is gapped, and $u_0^{(\pm)}$ remain non-degenerate and negative real.

Thus, the three thresholds $\Delta_0 = \{\Delta_{\text{Coh}}, \Delta_{\text{MR}}, \Delta_{\text{QCP}}\}$ correspond to three special configurations of the isolated roots in the $p + ip$ ground state, marked (2), (3), and (4) in Fig. 8.14(a). Non-equilibrium extensions of all three appear in the quench phase diagram as the lines marked β_{coh} , β_{MR} , and β_{QCP} in Fig. 8.15, discussed below.

8.3.3 Roots of the spectral polynomial and the asymptotic behavior

An instantaneous quench of the BCS coupling strength sends $G_i \rightarrow G_f$ in Eq. (8.13). The initial condition is taken as the (BCS or BEC) $p + ip$ ground state of the pre-quench Hamiltonian. Following the quench, the spins evolve according to Eq. (8.14), and $\Delta(t)$ is self-consistently determined by (8.15), with $G = G_f$.

We label the strength of the quench by β , defined in terms of $\{G_i, G_f\}$ via Eqs. (8.16) and (8.17). We denote the corresponding spectral polynomial as $\mathcal{Q}_{2N}(u; \beta)$ [Eq. (8.44)]. This is a function of the instantaneous spin state $\{\vec{s}_i(t)\}$. Because it is an integral of motion, we can evaluate $\mathcal{Q}_{2N}(u; \beta)$ at $t = 0$ in terms of the pre-quench ground state spin configuration in Eq. (8.49), wherein the initial pairing amplitude $\Delta_0^{(i)}$ is related to G_i via the BCS Eq. (8.50).

In the ground state, all but two of the $2N$ roots of $\mathcal{Q}_{2N}(u; 0)$ reside along the positive real axis; the remaining isolated roots $u_0^{(\pm)}$ in Eq. (8.52) are determined by the pairing amplitude Δ_0

and the chemical potential μ . Various equilibrium spectral transitions (including the topological BCS-BEC transition) are encoded in the isolated root positions, see Fig. 8.14.

Because a quench is a violent perturbation to the many-pair superfluid, one expects to find a different pattern of roots in $\mathcal{Q}_{2N}(u; \beta)$ for any $\beta \neq 0$. In particular, for a *finite* number of spins, all of the real, doubly-degenerate, positive ground state roots split into complex conjugate pairs for an arbitrarily weak quench. However, the splitting for most roots turns out to be small, of the order of the level spacing. In fact, when $\mathcal{Q}_{2N}(u; \beta)$ is evaluated at points $\{\Delta_0^{(i)}, \Delta_0^{(f)}\}$ throughout the quench phase diagram in Fig. 8.4, for even a modest number (e.g. 100) of spins, one finds that all but a few roots always cluster around the positive real axis, even for “large” quenches. As β carries units of density, a large quench has $|\beta| \gg n$.

For all quenches depicted in Fig. 8.4, including those across the topological quantum phase transition (e.g., $\Delta_0^{(i)} > \Delta_{\text{QCP}}$ and $\Delta_0^{(f)} < \Delta_{\text{QCP}}$), we find that $\mathcal{Q}_{2N}(u; \beta)$ exhibits zero, one, or two *isolated* pairs of roots. An isolated pair is well-separated from the positive real u axis, as is the case for the ground state pair $u_0^{(\pm)}$ in Eq. (8.52). The following picture therefore emerges, identical to the s-wave[11, 10, 12, 13, 14] case: The pattern of roots in $\mathcal{Q}_{2N}(u; \beta)$ for a quench is similar to that of the ground state, except that the number M of isolated root pairs can change. There is also a small splitting of the remaining $2(N - M)$ roots away from the positive real u -axis.

The importance of the spectral polynomial roots can be appreciated from the following argument. Suppose $\mathcal{Q}_{2N}(u)$ has a positive real zero u_0 , i.e. $\mathcal{Q}_{2N}(u_0) = 0$. Zeroes of the spectral polynomial coincide with the zeros of $L_2(u)$ and because by definition this quantity is nonnegative when $u > 0$, any real positive root of $L_2(u)$ must also be a double root. Further, since both terms in Eq. (8.35) are nonnegative for $u > 0$, it follows that u_0 must be a root of both $L^z(u)$ and $L^-(u)$. But the roots of $L^-(u)$ are defined to be the separation variables. This implies that one of the separation variables must coincide with the positive real root: $u_\beta(t) = u_0$. It is then “frozen” in time; this is consistent with the equations of motion because both sides of Eq. (8.47) vanish for $u_\beta(t) = u_0 = \text{const}$.

Note that $u_\beta = u_0$ also drops out from the equations of motion for the remaining separation variables. Indeed, we have $\mathcal{Q}_{2N}(u_\alpha) = (u_\alpha - u_0)^2 \mathcal{Q}_{2N-2}(u_\alpha)$ because u_0 is a double root of $\mathcal{Q}_{2N}(u)$; the factor of $(u_\alpha - u_0)$ in the numerator of Eq. (8.47) cancels $(u_\alpha - u_\beta)$ in the denominator. Note also that the order of the spectral polynomial drops by two. It turns out that this kind of reduction occurs for our quench initial conditions in the continuum, $N \rightarrow \infty$, limit. As a result the order of the spectral polynomial drops dramatically to either 0, 2 or 4, and the resulting equations of motion can be explicitly solved.

Suppose there are $N - M$ real double roots $\{u_0^{(\beta)}\}$. The spectral polynomial in Eq. (8.44) then reduces to

$$\mathcal{Q}_{2N}(u) = \prod_{\beta=1}^{N-M} (u - u_0^{(\beta)})^2 \mathcal{Q}_{2M}(u), \quad (8.56)$$

where $\mathcal{Q}_{2M}(u)$ is a polynomial of order $2M$ whose roots are isolated. Now $N - M$ separation variables are equal to the roots, and the remaining nontrivial $M - 1$ variables u_α satisfy, as a consequence of Eq. (8.47),

$$\frac{du_\alpha}{dt} = -2i \frac{\sqrt{\mathcal{Q}_{2M}(u_\alpha)}}{\prod_{\beta \neq \alpha} (u_\alpha - u_\beta)}. \quad (8.57)$$

These are the reduced equations of motion for the remaining $M - 1$ separation variables, which have the same form as the original equations of motion for $N - 1$ variables.

It is possible to reduce the number of degrees of freedom for this problem by using an explicit (“Lax reduction”) procedure, whose outcome allows one to find the asymptotic behavior of the order parameter. We introduce M collective spin variables $\vec{\sigma}_r$, where $r = 1, 2, \dots, M$. They satisfy the same Poisson bracket relations as the original spins, Eq. (8.36). The collective spins

have their own Lax vector $\vec{L}_\sigma$ defined analogously to Eq. (8.34) as

$$\begin{aligned} L_\sigma^\pm(u) &\equiv \sum_{r=1}^M \frac{\sqrt{\chi_r} \sigma_r^\pm}{\chi_r - u}, \\ L_\sigma^z(u) &\equiv \sum_{r=1}^M \frac{\chi_r \sigma_r^z}{\chi_r - u} + \frac{1}{2G}. \end{aligned} \quad (8.58)$$

Here the parameters $\{\chi_r\}$ are chosen in such a way that

$$\vec{L}(u) = \mathcal{A}(u) \vec{L}_\sigma(u), \quad (8.59)$$

where

$$\mathcal{A}(u) \equiv 1 + \sum_{j=1}^N \frac{d_j}{\varepsilon_j - u}. \quad (8.60)$$

Matching the residues of the poles with Eq. (8.34), we require that

$$\sum_{j=1}^N \frac{d_j}{\varepsilon_j - \chi_r} = -1, \quad r \in \{1, \dots, M\}, \quad (8.61a)$$

$$\begin{aligned} s_i^\pm &= \frac{d_i}{\sqrt{\varepsilon_i}} L_\sigma^\pm(\varepsilon_i), \quad s_i^z = \frac{d_i}{\varepsilon_i} L_\sigma^z(\varepsilon_i), \\ i &\in \{1, \dots, N\}. \end{aligned} \quad (8.61b)$$

The first Eq. (8.61a) constrains the parameters $\{\chi_r\}$, while Eq. (8.61b) determines the coefficients

$$d_i = - \frac{\varepsilon_i \zeta_i}{2\sqrt{L_2^{(\sigma)}(\varepsilon_i)}}, \quad (8.62)$$

where $L_2^{(\sigma)}(u) \equiv u L_\sigma^+ L_\sigma^-(u) + (L_\sigma^z)^2(u)$ and $\zeta_i \in \pm 1$. To obtain Eq. (8.62) we have used the fact that $\vec{s}_i^2 = 1/4$. The spectral polynomial indeed takes the form in Eq. (8.56).

The effective spin variables $\{\vec{\sigma}_r\}$ evolve according to the same Hamiltonian in Eq. (8.13) except with energies $\{\chi_r\}$ and with M spins instead of the original N . The order parameter is expressed in terms of $\{\vec{\sigma}_r\}$ in the same way as $\vec{s}_i$ in Eq. (8.15), that is

$$\Delta = -G \sum_{r=1}^M \sqrt{\chi_r} \sigma_r^- \sum_{i=1}^N \frac{d_i}{\chi_r - \varepsilon_i} = -G \sum_{r=1}^M \sqrt{\chi_r} \sigma_r^-. \quad (8.63)$$

As discussed above, the roots of the spectral polynomial for a quench $\mathcal{Q}_{2N}(u; \beta)$ fall into two classes. This equation can be studied numerically or analytically for some finite large value of N . Such a study reveals that most of the roots come in complex conjugate pairs that lie close to the real axis. Their imaginary parts scale as $1/N$ for large N . For each such root pair there is a separation variable that remains close to it (at a distance of order $1/N$) at all times. We call these variables continuum separation variables and the respective zeroes of $\sqrt{L_2(u)}$ continuum roots. [Recall that $L_2(u)$ is proportional to $\mathcal{Q}_{2N}(u; \beta)$ through Eq. (8.44).] In the thermodynamic limit $N \rightarrow \infty$, the continuum roots of $\sqrt{L_2}$ merge with its poles into a cut on the real axis. However, several of the zeros, which we can call the isolated roots, remain far from each other even in the thermodynamic limit.

The contribution of the continuum separation variables to the equations of motion (8.47) for the isolated ones as well as to Eq. (8.48) vanishes as $t \rightarrow \infty$. This can be shown explicitly assuming the joined Fourier spectrum of the continuum separation variables is continuous. Then, for example,

$$\sum_{\beta} u_{\beta} \rightarrow \int A(\omega) e^{i\omega t} d\omega \rightarrow 0$$

as $t \rightarrow \infty$. Here the summation is over continuum separation variables only. Thus at large times $\Delta(t)$ and the isolated separation variables are given by an effective M -spin solution, as outlined above.

With this information it is straightforward to construct the large time asymptotic solutions of the equations of motion. For example, in phase **II** the spectral polynomial has one pair of isolated roots. That means that the reduced problem has only one collective spin ($M = 1$), and the order parameter behaves as $\Delta(t) = \Delta_{\infty} \exp(-2i\mu_{\infty}t)$, as follows from the solution of the equations of motion for just one spin. In turn, it is possible to relate Δ_{∞} and μ_{∞} to the position of that isolated root, by calculating the spectral polynomial for this single-spin problem. Phases **I** and **III** respectively correspond to zero and two-spin problems associated to zero and two pairs of isolated roots. The precise relations between the isolated roots and the order parameter dynamics in phases **II** and **III** are determined in Sec. 8.4.

8.3.4 Spectral polynomial and isolated roots for a quench

The spectral polynomial is defined by Eqs. (8.44), (8.35), and (8.34), where explicit factors of the coupling G take the post-quench value G_f . We evaluate this in terms of the initial pre-quench state in Eq. (8.49) with $\{\Delta_0, \mu\} = \{\Delta_0^{(i)}, \mu_0^{(i)}\}$, where the chemical potential $\mu_0^{(i)}$ is that associated to the initial pre-quench order parameter $\Delta_0^{(i)}$ via Eq. (8.5). The result is

$$\mathcal{Q}_{2N}(u; \beta) = G_f^2 \prod_{j=1}^N (u - \varepsilon_j)^2 L_2(u; \beta), \quad (8.64a)$$

$$L_2(u; \beta) = \left(\frac{\nu_0}{2}\right)^2 \left\{ [E^{(i)}(u) f^{(i)}(u)]^2 + \beta (u - 2\mu_0^{(i)}) f^{(i)}(u) + \beta^2 \right\}, \quad (8.64b)$$

where the “quench parameter” β was defined in Eq. (8.16). In Eq. (8.64), we have introduced the following quantities that characterize the pre-quench state:

$$\begin{aligned} E^{(i)}(u) &= E(u; \Delta_0^{(i)}, \mu_0^{(i)}), \\ f^{(i)}(u) &= \frac{2}{\nu_0} F(u; \Delta_0^{(i)}, \mu_0^{(i)}), \end{aligned} \quad (8.65)$$

where F denotes the function appearing in the ground state spectral polynomial [Eq. (8.51c)], and ν_0 is the density of states in Eq. (8.33).

In the remainder of this section, we demonstrate how to extract isolated roots from Eq. (8.64) in the thermodynamic limit, and we establish the boundaries of the phase diagram in Fig. 8.4. We also determine the non-equilibrium extensions of the special pairing amplitudes $\{\Delta_{\text{Coh}}, \Delta_{\text{MR}}, \Delta_{\text{QCP}}\}$ discussed in Sec. 8.3.2.

The roots of $\mathcal{Q}_{2N}(u; \beta)$ in Eq. (8.64) satisfy

$$f^{(i)}(u) = -\frac{\beta}{\mathcal{E}_{1,2}^{(i)}(u)}, \quad (8.66a)$$

$$\mathcal{E}_{1,2}^{(i)}(u) \equiv \left(\frac{u}{2} - \mu_0^{(i)}\right) \mp i\Delta_0^{(i)} u^{1/2}, \quad (8.66b)$$

where we have solved the quadratic equation for $f^{(i)}$. In the thermodynamic limit, the left-hand side of this equation becomes [via Eq. (8.51c)]

$$f^{(i)}(u) = \int_0^{2[\Lambda + \mu_0^{(i)}]} d\varepsilon \frac{\varepsilon}{(u - \varepsilon)E^{(i)}(\varepsilon)}. \quad (8.67)$$

To logarithmic accuracy in the cutoff Λ ,

$$f^{(i)}(u; \zeta) = -2 \log \left[\frac{2\Lambda}{(\Delta_0^{(i)})^2 + 2|\mu_0^{(i)}|\theta(-\mu_0^{(i)})} \right] + \frac{u}{E^{(i)}(u)} \log \left\{ \frac{\zeta u [u + 2(\Delta_0^{(i)})^2 - 2\mu_0^{(i)} + 2E^{(i)}(u)]}{2 [u(\Delta_0^{(i)})^2 - u\mu_0^{(i)} + 2(\mu_0^{(i)})^2 + 2|\mu_0^{(i)}|E^{(i)}(u)]} \right\} \quad (8.68)$$

The parameter ζ determines the branch cut in the complex u -plane; taking the principal branch for $\log(z)$, the cut lies along the positive (negative) u -axis for $\zeta = -1$ ($\zeta = 1$).

In the thermodynamic limit, isolated roots are solutions to Eqs. (8.66) and (8.68) for u away from the positive real axis (so that we should take $\zeta = -1$). A given quench is defined by the initial pairing amplitude $\Delta_0^{(i)}$ and the quench parameter β [Eq. (8.16)]. Alternatively, one can specify coordinates $\{\Delta_0^{(i)}, \Delta_0^{(f)}\}$ in the quench phase diagram, Fig. 8.4. β is determined through the ground state BCS equation for the initial and final coupling strengths through Eq. (8.162), leading to

$$\begin{aligned} \beta = 2\mu_0^{(f)} \log \left[\frac{2\Lambda e}{(\Delta_0^{(f)})^2 + 2|\mu_0^{(f)}|\theta(-\mu_0^{(f)})} \right] \\ - 2\mu_0^{(i)} \log \left[\frac{2\Lambda e}{(\Delta_0^{(i)})^2 + 2|\mu_0^{(i)}|\theta(-\mu_0^{(i)})} \right]. \end{aligned} \quad (8.69)$$

8.3.5 Threshold roots: Dynamical phase boundaries

A quench located within the dynamical phases marked **I**, **II**, or **III** in Fig. 8.4 respectively exhibits zero, one, or two isolated pairs of spectral polynomial roots. To determine the boundaries of these regions, we look for *threshold* conditions, wherein a complex conjugate pair first separates from or merges with the positive real axis. To that end, we write

$$u \rightarrow u \pm i \operatorname{sgn}(\beta) \eta,$$

where η denotes a positive infinitesimal, and we take u real and positive on the right-hand side of this equation. The real and imaginary parts of Eq. (8.66) then imply that

$$f^{(i)}(u; 1) = -\operatorname{sgn}(\beta) \frac{\pi \sqrt{u}}{\Delta_0^{(i)}} \frac{\left(\frac{u}{2} - \mu_0^{(i)}\right)}{E^{(i)}(u)}, \quad (8.70a)$$

$$|\beta| = \frac{\pi \sqrt{u}}{\Delta_0^{(i)}} E^{(i)}(u). \quad (8.70b)$$

For the generic quench, two pieces of information such as $\{\Delta_0^{(i)}, \beta\}$ must be specified to determine the isolated roots. In Eq. (8.70), $|\beta|$ is no longer a free parameter, as we have constrained the imaginary part of u to be infinitesimal. For a given $\Delta_0^{(i)}$ and $\operatorname{sgn}(\beta)$, the real positive roots of Eq. (8.70) determine $|\beta|$ and $\Delta_0^{(f)}$ as functions of $\Delta_0^{(i)}$, leading to one-parameter curves in the phase diagram shown in Fig. 8.4.

The phase boundaries are labeled $\beta_\zeta^{(\pm)}$ in Fig. 8.15, which depicts a more detailed version of the quench phase diagram. These curves were obtained through the numerical solution of Eq. (8.70), using Eq. (8.69) to determine $\Delta_0^{(f)}$. In the following, we derive analytical results for quenches from a weakly-paired initial BCS state.

Threshold roots at weak initial and final pairing, $\{\Delta_0^{(i)}, \Delta_0^{(f)}\} \ll \Delta_{\text{QCP}}$

For weak initial pairing defined as $\Delta_0^{(i)} \ll \Delta_{\text{QCP}}$, the threshold root

$$u_c^{(1)} \equiv 2\mu_0^{(i)} \simeq 4\pi n + \mathcal{O}(\Delta_0^{(i)})^2 \quad (8.71)$$

solves Eq. (8.70) for both $\text{sgn}(\beta) = \pm 1$. The corresponding quench parameters are

$$\beta_c^{(\pm)} = \pm(2\pi)^2 n. \quad (8.72)$$

Eq. (8.69) reduces to $\Delta_0^{(i)}/\Delta_0^{(f)} \simeq \exp(\beta/8\pi n)$ for $\{\Delta_0^{(i)}, \Delta_0^{(f)}\} \ll \Delta_{\text{QCP}}$. Phase boundaries corresponding to Eq. (8.72) are given by the lines

$$\Delta_0^{(i)} = e^{\pm\pi/2} \Delta_0^{(f)}. \quad (8.73)$$

These are plotted in Fig. 8.15. The weak-pairing thresholds in Eq. (8.73) correspond to the straight-line portions of the curves marked $\beta_c^{(\pm)}$ near the origin $\{\Delta_0^{(i)}, \Delta_0^{(f)}\} = \{0, 0\}$ in Fig. 8.15.

We can go further and determine the roots throughout the weak-pairing BCS-to-BCS region of the quench phase diagram. In phases **II** and **III** with $\{\Delta_0^{(i)}, \Delta_0^{(f)}\} \ll \Delta_{\text{QCP}}$, an isolated root can be parameterized as

$$u \simeq u_c^{(1)} + 2i\delta. \quad (8.74)$$

Employing the branch of Eq. (8.66) with the minus sign $[\mathcal{E}_1^{(i)}(u)]$ and using $\zeta = -1$ in Eq. (8.68), a single complex conjugate isolated pair obtains for quenches satisfying $\beta < \beta_c^{(+)}$. For $\beta \geq \beta_c^{(+)}$, there are no isolated roots (phase **I** in Fig. 8.15). We define

$$\mathbf{B} \equiv \frac{\pi}{2} \left(\frac{\beta}{\beta_c^{(+)}} \right), \quad (8.75a)$$

$$\frac{\delta}{\sqrt{4\pi n} \Delta_0^{(i)}} \equiv \begin{cases} \cos(\theta), & 0 \leq \delta < \sqrt{4\pi n} \Delta_0^{(i)}, \\ \cosh(\theta), & \delta \geq \sqrt{4\pi n} \Delta_0^{(i)}. \end{cases} \quad (8.75b)$$

We note that since $(\Delta_{\text{QCP}})^2$ is of order $4\pi n$ [Eq. (8.164)], we have $\sqrt{4\pi n} \Delta_0^{(i)} \ll u_c^{(1)}$ for $\Delta_0^{(i)} \ll \Delta_{\text{QCP}}$; the imaginary part of the root is always much smaller than the real part in this regime. Using the above definitions, we find that δ is encoded in the transcendental equations

$$\begin{aligned} \theta \tan\left(\frac{\theta}{2}\right) &= \mathbf{B}, \quad 0 \leq \theta \leq \frac{\pi}{2}, \quad 0 \leq \mathbf{B} \leq \frac{\pi}{2}, \\ \theta \tanh\left(\frac{\theta}{2}\right) &= |\mathbf{B}|, \quad 0 < \theta, \quad \mathbf{B} < 0. \end{aligned} \quad (8.76)$$

The first equation applies to strong-to-weak quenches, up to the boundary of phase **II** with phase **I**, $\mathbf{B} = \theta = \pi/2$. The second equation holds for weak-to-strong quenches in phases **II** and **III**. Both equations have exactly one solution in their regions of validity.

The second branch of Eq. (8.66) with the plus sign $[\mathcal{E}_2^{(i)}(u)]$ has zero (one) isolated pair of complex conjugate roots for $\beta > \beta_c^{(-)}$ ($\beta < \beta_c^{(-)}$). Employing Eq. (8.75) to specify the imaginary part the root δ in terms of θ , we find the equations

$$\begin{aligned} \theta \cot\left(\frac{\theta}{2}\right) &= |\mathbf{B}|, \quad 0 \leq \theta \leq \frac{\pi}{2}, \quad -2 < \mathbf{B} \leq -\frac{\pi}{2}, \\ \theta \coth\left(\frac{\theta}{2}\right) &= |\mathbf{B}|, \quad 0 \leq \theta, \quad \mathbf{B} < -2. \end{aligned} \quad (8.77)$$

The two pairs of isolated roots that distinguish phase **III** solve Eqs. (8.76) and (8.77) with $\mathbf{B} < -\pi/2$. Eqs. (8.76) and (8.77) turn out to be identical to the corresponding equations in the s-wave case.[14, 13]

Phase III termination at strong final pairing, $\Delta_0^{(f)} \sim \Delta_{\text{MR}}$

For $\Delta_0^{(i)} \ll \Delta_{\text{QCP}}$, the threshold Eq. (8.70) admits an additional solution for $\text{sgn}(\beta) < 0$ (weak-to-strong quenches), given by

$$u_c^{(2)} = \left\{ \frac{2\Delta_0^{(i)}}{\pi} \log \left[\frac{2\Lambda}{(\Delta_0^{(i)})^2} \right] \right\}^2. \quad (8.78)$$

This threshold root in fact locates the phase **III-II** boundary near the bottom of the phase diagram. Unlike $u_c^{(1)}$ [Eq. (8.71)], this additional isolated root *vanishes* as $\Delta_0^{(i)} \rightarrow 0$. The reason for this is as follows. In the next section we discuss the non-equilibrium extension of the topological quantum critical point, indicated by the dashed curve labeled β_{QCP} in Fig. 8.15. The β_{QCP} line is uniquely defined in phase **II** by the feature that it possesses a vanishing isolated root. Since the phase **III-II** boundary merges with the β_{QCP} line when $\Delta_0^{(i)} \rightarrow 0$, the threshold root $u_c^{(2)}$ also vanishes in that limit.

Via Eq. (8.69), the quench parameter associated with the phase **III** boundary is

$$\beta_c^{(-)}(\Delta_0^{(i)}) = \beta_{\text{QCP}}(\Delta_0^{(i)}) - u_c^{(2)} \log \left(\frac{8\pi^2 n^2}{\Lambda u_c^{(2)}} \right), \quad (8.79)$$

valid in the same limit. In this equation, $\beta_{\text{QCP}}(\Delta_0^{(i)})$ parameterizes the non-equilibrium topological transition line, Eq. (8.80) below. At $\Delta_0^{(i)} = 0$, the threshold root $u_c^{(2)} = 0$ and $\beta_c^{(-)}(0)$ coincides with $\beta_{\text{QCP}}(0)$. The corresponding quench phase diagram coordinates are $\{\Delta_0^{(i)}, \Delta_0^{(f)}\} \simeq \{0, \Delta_{\text{MR}}\}$ as shown in Fig. 8.15, see Eq. (8.81) in the next section.

8.3.6 Non-equilibrium topological and spectral transitions

In the ground state, the topological quantum phase transition at $\{\Delta_0, \mu\} = \{\Delta_{\text{QCP}}, 0\}$ corresponds to the isolated root configuration $u_0^{(\pm)}$ in Eq. (8.55). This is the only pairing amplitude associated to a vanishing isolated root; for any non-zero $\Delta_0 \neq \Delta_{\text{QCP}}$, both $u_0^{(\pm)}$ have finite separation from the positive real axis. For quenches in phase **II** of Figs. 8.4 and 8.15, the chemical potential $\mu(t)$ and the order parameter $\Delta(t)$ respectively asymptote to constants μ_∞ and Δ_∞ , with the latter non-zero. Phase **II** is characterized by a single pair of isolated roots. As discussed below in Sec. 8.4.1, $\{\Delta_\infty, \mu_\infty\}$ have the same relation to the isolated roots $u^{(\pm)}$ of the quench spectral polynomial $\mathcal{Q}_{2N}(u; \beta)$ as $\{\Delta_0, \mu\}$ have to $u_0^{(\pm)}$ in the ground state, c.f. Eqs. (8.52) and (8.89).

We can define a non-equilibrium extension of the topological phase transition through the condition $\mu_\infty = 0$, which corresponds to the vanishing of an isolated root of $\mathcal{Q}_{2N}(u; \beta)$. For a given $\Delta_0^{(i)}$, let us denote the quench parameter $\beta \equiv \beta_{\text{QCP}}$ that yields a vanishing isolated root in phase **II**. At $u = 0$, Eqs. (8.66) and (8.68) yield

$$\beta_{\text{QCP}}(\Delta_0^{(i)}) = -2\mu_0^{(i)} \log \left[\frac{2\Lambda}{(\Delta_0^{(i)})^2 + 2|\mu_0^{(i)}| \theta(-\mu_0^{(i)})} \right]. \quad (8.80)$$

Note that the ground state critical point satisfies this equation, since $\beta_{\text{QCP}} = 0$ for zero quench and $\mu_0^{(i)} = 0$ locates the ground state transition. As discussed in Sec. 8.2.3, the Green's function winding number W defined via Eq. (8.10) changes across this line (Fig. 8.9), indicating the presence or absence of edge states in the spectrum of the asymptotic Bogoliubov-de Gennes Hamiltonian. The topological transition (dashed) line drawn in Figs. 8.4 and 8.15 was obtained by solving Eq. (8.69) numerically to calculate $\Delta_0^{(f)}$ from $\Delta_0^{(i)}$, using Eq. (8.80).

In the limit $\Delta_0^{(i)} \rightarrow 0$ (very weak initial pairing), the topological transition line β_{QCP} terminates at a particular value of $\Delta_0^{(f)}$ in Fig. 8.15. Eqs. (8.80) and (8.69) imply that for

$$\beta(\Delta_0^{(i)}, \Delta_0^{(f)}) = \beta_{\text{QCP}}(\Delta_0^{(i)}),$$

$$\lim_{\Delta_0^{(i)} \rightarrow 0} \Delta_0^{(f)} \simeq \Delta_{\text{MR}}, \quad (8.81)$$

up to terms of size $\sqrt{n}[\log(\Lambda/2\pi n)]^{-5/2}$. By contrast, the ground state pairing amplitudes Δ_{Coh} and Δ_{QCP} differ from Δ_{MR} by terms of order $\sqrt{n}[\log(\Lambda/2\pi n)]^{-3/2}$. [Eq. (8.164) gives explicit formulae for $\{\Delta_{\text{Coh}}, \Delta_{\text{MR}}, \Delta_{\text{QCP}}\}$].

We conclude that the topological transition in the non-equilibrium phase diagram deviates from the equilibrium line $\Delta_0^{(f)} = \Delta_{\text{QCP}}$ for $\Delta_0^{(i)} \ll \Delta_{\text{QCP}}$. At $\Delta_0^{(i)} = 0$, the transition β_{QCP} is such that $\Delta_0^{(f)} \simeq \Delta_{\text{MR}}$, which is of the same order as, but smaller than Δ_{QCP} . The boundary $\beta_c^{(-)}$ separating dynamical phases **II** and **III** also terminates at this point.

In the ground state, the quasiparticle energy gap E_{min} occurs at non-zero (zero) momentum for $\Delta_0 < \Delta_{\text{Coh}}$ ($\Delta_0 \geq \Delta_{\text{Coh}}$) [Eqs. (8.53) and (8.54)]. The isolated ground state roots $u_0^{(\pm)}$ in Eq. (8.52) are purely imaginary at the transition $\Delta_0 = \Delta_{\text{Coh}}$, marked (2) in Fig. 8.14(a). The dotted curve labeled β_{Coh} in Fig. 8.15 is the non-equilibrium extension, obtained via the numerical solution of Eqs. (8.66) and (8.68) ($\zeta = -1$) locating one pair of purely imaginary isolated roots for a given $\Delta_0^{(i)}$. Along this curve, the asymptotic values $\{\Delta_\infty, \mu_\infty\}$ satisfy

$$\mu_\infty = (\Delta_\infty)^2 \quad (8.82)$$

[c.f. Eq. (8.89)].

Throughout phase **II**, the order parameter approaches its asymptotic value via a power-law-damped oscillation. The precession of the pseudospins in the asymptotically constant field $\vec{B}_i$ [Eq. (8.24)] implies that the self-consistent time-evolution of Δ can be expressed as

$$\Delta(t) - \Delta_\infty \propto \int d\varepsilon \alpha(\varepsilon) \exp[-i2E_\infty(\varepsilon)t], \quad (8.83)$$

where $E_\infty(\varepsilon) = E(\varepsilon; \Delta_\infty, \mu_\infty)$ is the asymptotic dispersion relation [Eq. (8.26)]. In Sec. 8.5.3, we show that Eq. (8.83) evaluates to

$$\Delta(t) = \Delta_\infty + ct^{-\alpha} \cos(\Omega t + \phi),$$

where c is a constant and we ignore “non-universal” corrections of order $1/\Lambda$. We will find that $\alpha = 1/2$ and $\Omega = 2E_{\text{min}}$ to the left of the β_{Coh} curve. For quenches to the right of the β_{Coh} line in Fig. 8.15, $\alpha = 2$ and $\Omega = 0$ (excluding certain special cases). The changes in α and Ω are associated to the transition in the asymptotic dispersion relation, as determined by $\{\Delta_\infty, \mu_\infty\}$. For $\Delta_\infty < (\mu_\infty)^2$ (left of β_{Coh}), the minimum in $E(\varepsilon; \Delta_\infty, \mu_\infty) \equiv E_{\text{min}}$ at non-zero ε results in a non-trivial saddle-point for the dynamics. This disappears when E_{min} moves to $\varepsilon = 0$ [$\Delta_\infty \geq (\mu_\infty)^2$, quenches to the right of the β_{Coh} curve].

As discussed in Sec. 8.3.2, the amplitude Δ_{MR} corresponds to a doubly-degenerate, negative real root pair in the ground state polynomial $\mathcal{Q}_{2N}(u)$. Such a pair is marked (3) in Fig. 8.14(a). We look for a non-equilibrium extension in the form of a doubly-degenerate, negative real isolated root $u = -v < 0$ satisfying

$$\mathcal{Q}_{2N}(-v; \beta) = \frac{d}{dv} \mathcal{Q}_{2N}(-v; \beta) = 0.$$

From Eqs. (8.64) and (8.66), these conditions become

$$\beta = \left[\left(\frac{v}{2} + \mu_0^{(i)} \right) - \Delta_0^{(i)} \sqrt{v} \right] f^{(i)}(-v; -1), \quad (8.84a)$$

$$= \frac{\frac{d}{dv} [E^{(i)}(-v) f^{(i)}(-v; -1)]^2}{f^{(i)}(-v; -1) + (v + 2\mu_0^{(i)}) \frac{d}{dv} f^{(i)}(-v; -1)}. \quad (8.84b)$$

On the first line, we have selected the branch of Eq. (8.66) that includes the ground state solution $\{\Delta_0, \mu, \beta\} = \{\Delta_{\text{MR}}, \Delta_{\text{MR}}^2/2, 0\}$; the other branch gives $v \sim \mathcal{O}(1/\Lambda)$, which is beyond the logarithmic accuracy employed here. In the limit of weak initial pairing $\Delta_0^{(i)} \ll \Delta_{\text{QCP}}$, Eq. (8.84) reduces to

$$\sqrt{v} \log \left(\frac{e\Lambda v}{8\pi^2 n^2} \right) \simeq \Delta_0^{(i)} \log \left[\frac{2\Lambda}{(\Delta_0^{(i)})^2} \right].$$

This has the solution

$$\sqrt{v_c} = \frac{\Delta_0^{(i)} \log \left[\frac{2\Lambda}{(\Delta_0^{(i)})^2} \right]}{2\mathcal{W}_0 \left\{ \log \left[\frac{2\Lambda}{(\Delta_0^{(i)})^2} \right] \frac{\Delta_0^{(i)}}{4\pi n} \sqrt{\frac{e\Lambda}{2}} \right\}}, \quad (8.85)$$

valid for $v \geq 8\pi^2 n^2 / e\Lambda$. In this equation, $\mathcal{W}_0(z)$ denotes the $k = 0$ branch of Lambert's W function. The quench parameter is

$$\beta_{\text{MR}}(\Delta_0^{(i)}) = \beta_{\text{QCP}}(\Delta_0^{(i)}) + v_c + \sqrt{v_c} \Delta_0^{(i)} \log \left[\frac{2\Lambda}{(\Delta_0^{(i)})^2} \right]. \quad (8.86)$$

Like the spectral transition line β_{Coh} , β_{MR} converges to β_{QCP} as $\Delta_0^{(i)} \rightarrow 0$; all three curves coalesce at $\Delta_0^{(f)} \simeq \Delta_{\text{MR}}$ [Eq. (8.81)] for vanishing initial interaction strength, as shown in Fig. 8.15. The dot-dashed curve in this figure marked β_{MR} was obtained through the numerical solution to Eq. (8.84), using Eq. (8.69).

Summarizing, in phase **II** of the quench phase diagram in Fig. 8.15, the same patterns of isolated roots that appear in the ground state (diagonal line) also appear for quenches ($\beta \neq 0$). Implications of the lines β_{Coh} and β_{QCP} for asymptotic quench dynamics are discussed in detail in Sec. 8.5, below.

8.4 Steady-state order parameter dynamics

In this section we determine the generalized steady-state behavior of $\Delta(t)$ in the limit $t \rightarrow \infty$. We include the case exhibiting persistent oscillations, phase **III** in Fig. 8.4. Calculations of the *approach* to the steady-state are deferred until Sec. 8.5.

8.4.1 Phase II: Constant $\Delta(t) \rightarrow \Delta_\infty$

In phase **II** of Fig. 8.4, there is a single pair of isolated roots. The reduced problem [Lax reduction, Eqs. (8.58) and (8.59)] has one collective spin $\vec{\sigma}$ and zero separation variables. The order parameter solves a version of Eq. (8.48) with $N = 1$. The solution is

$$\Delta(t) = \Delta_\infty \exp(-2i\mu_\infty t + i\phi_0), \quad (8.87)$$

$$\mu_\infty - \frac{\chi}{2} = G\chi\sigma^z, \quad (8.88)$$

where χ is the mode energy. We relate $\{\Delta_\infty, \mu_\infty\}$ to the roots of the reduced spectral polynomial. Eq. (8.34) implies that

$$L_\sigma^z(u) = \frac{1}{G} \left[\frac{(\frac{\chi}{2} - \mu_\infty)}{u - \chi} + \frac{1}{2} \right].$$

Using Eqs. (8.35), (8.44), and (8.58), the spectral polynomial is

$$\mathcal{Q}_2(u) = \frac{1}{4}(u - u^{(+)})(u - u^{(-)}),$$

with roots

$$u^{(\pm)} = 2 \left[\mu_\infty - (\Delta_\infty)^2 \pm \Delta_\infty \sqrt{(\Delta_\infty)^2 - 2\mu_\infty} \right]. \quad (8.89)$$

We interpret μ_∞ as the out-of-equilibrium chemical potential because the N -spin ground state has $\Delta(t) = \Delta_0 \exp(-2i\mu t)$. This follows from using Eq. (8.49) as the initial condition to Eq. (8.14): Due to the mismatch between $\vec{B}_i$ in the EOM and the ground state field $\vec{B}_i + 2\mu\hat{z}$, the pseudospins uniformly precess $s_i^-(t) = s_i^-(0) \exp(-2i\mu t)$. This can be eliminated by moving to a rotating frame.[60] $\{\Delta_\infty, \mu_\infty\}$ have the same relation to the isolated roots $u^{(\pm)}$ of the quench spectral polynomial $\mathcal{Q}_{2N}(u; \beta)$ as $\{\Delta_0, \mu\}$ have to $u_0^{(\pm)}$ in the ground state, Eq. (8.52).

For a quench, the pairing amplitude and chemical potential evolve from the initial pre-quench state. The isolated root pair for $\mathcal{Q}_{2N}(u; \beta)$ determine $\lim_{t \rightarrow \infty} \{\Delta(t), \mu(t)\} = \{\Delta_\infty, \mu_\infty\}$ via Eq. (8.89). By contrast, the approach to the asymptotic steady-state (typically a power-law-damped oscillation) is governed by the full N -spin distribution function. This is computed exactly in the thermodynamic limit in Sec. 8.5.1, below.

8.4.2 Phase **III**: Oscillating order parameter

In phase **III**, $\mathcal{Q}_{2N}(u; \beta)$ exhibits two isolated pairs of roots. The pair confined to this region nucleates along the boundary marked $\beta_c^{(-)}$ in Fig. 8.15. The second pair persists into phase **II**. At weak initial and final coupling $\{\Delta_0^{(i)}, \Delta_0^{(f)}\} \ll \Delta_{\text{QCP}}$, these solve Eqs. (8.77) and (8.76), respectively.

The isolated roots entirely confined to **III** always appear as a complex conjugate pair, with a positive real part. In what follows, we denote this pair as

$$u_{1,\pm} \equiv u_{1,\text{r}} \pm iu_{1,\text{i}}, \quad u_{1,\{\text{r},\text{i}\}} \geq 0. \quad (8.90)$$

The isolated pair that persists into phase **II** also occurs as a complex conjugate pair throughout the bulk of phase **III**; we denote this pair as

$$u_{2,\pm} \equiv u_{2,\text{r}} \pm iu_{2,\text{i}}, \quad u_{2,\text{i}} \geq 0. \quad (8.91)$$

In Fig. 8.15, there is a very narrow sliver in phase **III** bounded on the left (right) by the β_{MR} ($\beta_c^{(-)}$) curve, of width $\sim \sqrt{n}[\log(\Lambda/2\pi n)]^{-5/2} \ll \{\Delta_{\text{Coh}}, \Delta_{\text{MR}}, \Delta_{\text{QCP}}\}$. [C.f. the text surrounding Eq. (8.81).] Within this sliver, the second pair of roots is negative real (to the right of the β_{MR} line). In this subsection, we consider quenches in the bulk of phase **III**, wherein the isolated roots always occur in two complex conjugate pairs. The sliver with negative real roots is considered in Appendix 8.9.

Pairing energy EOM

As argued in Sec. 8.3.3, the asymptotic dynamics of $\Delta(t)$ for the quench will be the same as in the two-spin solution, which we now derive. We first decompose the complex pairing amplitude into modulus and phase components:

$$\Delta \equiv \sqrt{R} \exp(-i\phi), \quad (8.92)$$

where R is the pairing energy. For the BCS problem with two spins $\vec{\sigma}_{1,2}$, conservation of the total energy E_σ and of the z -angular momentum J_σ imply that

$$\frac{R}{G} + E_\sigma = \chi_1 \sigma_1^z + \chi_2 \sigma_2^z, \quad (8.93a)$$

$$J_\sigma = \sigma_1^z + \sigma_2^z, \quad (8.93b)$$

where $\chi_{1,2}$ are the mode energies [c.f. Eq. (8.58)] and R is the pairing energy defined by Eq. (8.92). Expressing the spins in terms of the latter, we have

$$\begin{aligned}\sigma_p^z &\equiv \mathfrak{a}_p \frac{R}{G} + \mathfrak{b}_p, \\ \mathfrak{a}_{1,2} &= \pm \frac{1}{(\chi_1 - \chi_2)}, \\ \mathfrak{b}_{1,2} &= \pm \left(\frac{E_\sigma - \chi_{2,1} J_\sigma}{\chi_1 - \chi_2} \right).\end{aligned}\tag{8.94}$$

According to the reduction formula Eq. (8.61b), the z -component of the i^{th} spin in the N -spin problem is expressed in terms of $\vec{\sigma}_{1,2}$ via

$$s_i^z = \frac{d_i}{\varepsilon_i} L_\sigma^z(\varepsilon_i) \equiv a_i \frac{R}{G} + b_i.\tag{8.95}$$

Eq. (8.62) implies that the constant a_i is given by

$$a_i = \frac{G \varepsilon_i \zeta_i}{2 \sqrt{\mathcal{Q}_4(\varepsilon_i)}},\tag{8.96}$$

where $\zeta_i = \pm 1$ and we have introduced the fourth order spectral polynomial for the two-spin problem,

$$\mathcal{Q}_4(u) \equiv G^2(u - \chi_1)^2(u - \chi_2)^2 L_2^{(\sigma)}(u).\tag{8.97}$$

In this equation, $L_2^{(\sigma)}(\varepsilon_i)$ denotes the reduced Lax norm defined below Eq. (8.62).

The constants $\{a_i, b_i\}$ must satisfy

$$\begin{aligned}\sum_{i=1}^N a_i &= 0, & \sum_{i=1}^N b_i &= J, \\ \sum_{i=1}^N \varepsilon_i a_i &= 1, & \sum_{i=1}^N \varepsilon_i b_i &= H.\end{aligned}\tag{8.98}$$

The two equations on the first line follow from particle conservation (J is the conserved total z -spin). The remaining equations encode energy conservation; H is the Hamiltonian in Eq. (8.13).

We differentiate Eq. (8.95) to obtain

$$-i \frac{a_i}{\sqrt{\varepsilon_i} R} \dot{R} = \exp(i\phi) s_i^- - \exp(-i\phi) s_i^+,\tag{8.99}$$

where $\dot{x} \equiv dx/dt$ and we have used Eq. (8.14). We also have

$$\begin{aligned}\frac{d}{dt} [\exp(i\phi) s_i^- + \exp(-i\phi) s_i^+] \\ = \frac{a_i}{\sqrt{\varepsilon_i} R} \dot{\phi} \dot{R} - \sqrt{\varepsilon_i} \frac{a_i}{\sqrt{R}} \frac{\dot{R}}{G}.\end{aligned}$$

We define

$$\dot{A} \equiv \frac{1}{2\sqrt{R}} \dot{\phi} \dot{R},\tag{8.100}$$

so that

$$\exp(i\phi)s_i^- = \frac{a_i}{G\sqrt{\varepsilon_i}} \left(-i\frac{1}{2\sqrt{R}}\dot{R} + A - \varepsilon_i\sqrt{R} \right) + c_i. \quad (8.101)$$

Eq. (8.15) implies that

$$\sum_{i=1}^N \sqrt{\varepsilon_i} c_i = 0. \quad (8.102)$$

We compute the modulus-squared of Eq. (8.101):

$$\begin{aligned} -\frac{1}{4R}\dot{R}^2 &= \varepsilon_i R^2 + \varepsilon_i \left(\varepsilon_i + 2G\frac{b_i}{a_i} \right) R \\ &\quad - 2\varepsilon_i \left(\frac{G\sqrt{\varepsilon_i}c_i}{a_i} + A \right) \sqrt{R} + 2\frac{G\sqrt{\varepsilon_i}c_i}{a_i} A + A^2 \\ &\quad + \frac{G^2\varepsilon_i}{a_i^2} \left(b_i^2 + c_i^2 - \frac{1}{4} \right). \end{aligned} \quad (8.103)$$

Multiplying both sides by a_i and summing over i , we solve for A to find

$$A = \frac{1}{2}R^{3/2} + 2m\sqrt{R} + \frac{\psi}{\sqrt{R}} + \gamma, \quad (8.104)$$

where

$$\begin{aligned} 2m &\equiv \sum_i \frac{a_i \varepsilon_i^2}{2} + GH, \\ \psi &\equiv \sum_i \frac{G^2 \varepsilon_i}{2a_i} \left(b_i^2 + c_i^2 - \frac{1}{4} \right), \\ \gamma &\equiv - \sum_i G \varepsilon_i^{3/2} c_i. \end{aligned} \quad (8.105)$$

Eq. (8.103) becomes

$$\begin{aligned} -\frac{1}{4R}\dot{R}^2 &= \frac{R^3}{4} + 2mR^2 + \left(\frac{c_i\sqrt{\varepsilon_i}G}{a_i} + \gamma \right) R^{3/2} \\ &\quad + \mathcal{O}(R). \end{aligned} \quad (8.106)$$

We therefore require that $\frac{c_i\sqrt{\varepsilon_i}G}{a_i} \equiv \kappa$, independent of i . Eq. (8.105) implies that

$$\gamma = -\kappa. \quad (8.107)$$

Eq. (8.103) reduces to

$$-\dot{R}^2 = R^4 + 8mR^3 + 8\rho R^2 + 4\sigma R + 4\psi^2, \quad (8.108)$$

where

$$\begin{aligned} \rho &\equiv \frac{G\varepsilon_i b_i}{a_i} + \frac{1}{2}(\psi + \tilde{\varepsilon}_i^2) \\ \sigma &\equiv \frac{G^2 \varepsilon_i}{a_i^2} \left(b_i^2 - \frac{1}{4} \right) - 2\psi\tilde{\varepsilon}_i, \\ \tilde{\varepsilon}_i &\equiv \varepsilon_i - 2m. \end{aligned} \quad (8.109)$$

Solving for a_i and b_i , we obtain

$$a_i = \frac{G\varepsilon_i\zeta_i}{2\sqrt{\left[\frac{1}{2}(\tilde{\varepsilon}_i^2 + \psi) - \rho\right]^2 - \varepsilon_i(2\psi\tilde{\varepsilon}_i + \sigma)}}, \quad (8.110a)$$

$$b_i = \frac{a_i}{G\varepsilon_i} \left[\rho - \frac{1}{2}(\tilde{\varepsilon}_i^2 + \psi) \right], \quad (8.110b)$$

where $\zeta_i = \pm 1$. Eq. (8.108) is an elliptic equation of motion for the pairing energy R .

Comparing Eqs. (8.96) and (8.110a), we determine that the two-spin spectral polynomial can be expressed as

$$\mathcal{Q}_4(u) = 4 \left\{ \begin{aligned} &\left[\left(\frac{u}{2} - \mathbf{m} \right)^2 + \frac{\psi}{4} - \frac{\rho}{2} \right]^2 \\ &- u \left[\psi \left(\frac{u}{2} - \mathbf{m} \right) + \frac{\sigma}{4} \right] \end{aligned} \right\}. \quad (8.111)$$

Eqs. (8.108) and (8.111) express the pairing energy dynamics and the reduced spectral polynomial in terms of a common set of parameters $\{\mathbf{m}, \rho, \sigma, \psi\}$.

For most quenches in phase **III** (to the left of the β_{MR} line in Fig. 8.15), the isolated roots $u_{1,\pm}$ and $u_{2,\pm}$ take the form of two complex conjugate pairs [Eqs. (8.90) and (8.91)]. Expanding Eq. (8.111) and matching powers of u to the anticipated form, we find that

$$\begin{aligned} \mathbf{m} &= \frac{1}{4}(u_{1,\text{r}} + u_{2,\text{r}}), \\ \rho &= -\frac{1}{4}(u_{1,\text{i}}^2 + u_{2,\text{i}}^2) + \frac{1}{2}U_{\text{r}}^2 - \frac{3\psi}{2}, \\ \sigma &= \frac{U_{\text{r}}}{2}(u_{2,\text{i}}^2 - u_{1,\text{i}}^2) - 4\psi\mathbf{m}, \end{aligned} \quad (8.112)$$

where

$$U_{\text{r}} = \frac{1}{2}(u_{1,\text{r}} - u_{2,\text{r}}). \quad (8.113)$$

The parameter ψ has two solutions in terms of the roots,

$$\psi_{\pm} = \frac{1}{8} \left[-(u_{1,\text{i}}^2 + u_{2,\text{i}}^2) - 2u_{1,\text{r}}u_{2,\text{r}} \pm 2|u_1||u_2| \right], \quad (8.114)$$

where $|u| = \sqrt{u_{\text{r}}^2 + u_{\text{i}}^2}$ is the modulus the complex root u . The physical solution is $\psi = \psi_+$, since this gives positive turning points for the positive-definite pairing energy R , as shown below.

Pairing energy dynamics

We first consider a quench confined to the weak-pairing BCS region with $\{\Delta_0^{(i)}, \Delta_0^{(f)}\} \ll \Delta_{\text{QCP}}$. In phase **III**, the corresponding roots take the form $u_{\{1,2\},\pm} \simeq 2\mu_0^{(i)} \pm 2i\delta_{1,2}$, where $\delta_{1,2}$ is of order $\sqrt{\mu_0^{(i)}\Delta_0^{(i)}} \ll \mu_0^{(i)}$. Here $\mu_0^{(i)} \simeq 2\pi n$ denotes the chemical potential in the initial state; see Sec. 8.3.5 for details. To leading order, Eqs. (8.112)–(8.114) simplify as follows:

$$\begin{aligned} \mathbf{m} &\simeq \mu_0^{(i)}, \quad \rho \simeq -\frac{1}{4}(u_{1,\text{i}}^2 + u_{2,\text{i}}^2), \quad \sigma \simeq \frac{(u_{1,\text{i}}^2 - u_{2,\text{i}}^2)^2}{32\mu_0^{(i)}}, \\ U_{\text{r}} &\sim \mathcal{O}(\Delta_0^{(i)})^2, \quad \psi_+ \sim \mathcal{O}(\Delta_0^{(i)})^4. \end{aligned} \quad (8.115)$$

Given that $R \sim \mathcal{O}(\Delta_0^{(i)})^2$ and retaining only the leading terms, Eq. (8.108) reduces to

$$\begin{aligned}\dot{R}^2 &\simeq 8\mu_0^{(i)} R (R_+ - R) (R - R_-), \\ R_{\pm} &= \frac{1}{8\mu_0^{(i)}} (u_{1,i} \pm u_{2,i})^2.\end{aligned}\tag{8.116}$$

This has the same structure as the previously-studied s-wave case.[6, 10, 12, 13] The turning points of the modulus $|\Delta_{\pm}| \equiv \sqrt{R_{\pm}}$ are proportional to the sum and difference of the isolated root pairs' imaginary parts. At the boundary of phase **III** marked $\beta_c^{(-)}$ in Fig. 8.15, the imaginary part of pair one vanishes $|u_{1,i}| \rightarrow 0$, leading to the collapse of the oscillatory amplitude.

Eq. (8.116) has the solution

$$\begin{aligned}|\Delta|(t) &= \frac{u_i}{\sqrt{2\mu_0^{(i)}}} dn\left(u_i t \left| \frac{u_{1,i}u_{2,i}}{u_i^2} \right| \right), \\ u_i &\equiv \frac{1}{2} (u_{1,i} + u_{2,i}),\end{aligned}\tag{8.117}$$

where $|\Delta| = \sqrt{R}$ and $dn(z|M)$ denotes the Jacobi elliptic function ($M = k^2$ is the modulo parameter). Just inside of phase **III** near the boundary with **II**, the period of $|\Delta|(t)$ is $T \simeq 2\pi/u_{2,i} \sim \mathcal{O}(\sqrt{\mu_0^{(i)}}\Delta_0^{(i)})^{-1}$, valid in the weak pairing limit $\{\Delta_0^{(i)}, \Delta_0^{(f)}\} \ll \Delta_{\text{QCP}}$.

Next we consider general phase **III** quenches. Using Eqs. (8.112)–(8.114) and taking $\psi = \psi_+$, the fourth-order polynomial in Eq. (8.108) can be factored. The result is

$$\dot{R}^2 = (R_+ - R)(R - R_-)(R + \tilde{R}_+)(R + \tilde{R}_-),\tag{8.118a}$$

where

$$\begin{aligned}R_{\pm} &\equiv \frac{1}{2} \left[\sqrt{(|u_1| - u_{1,r})} \pm \sqrt{(|u_2| - u_{2,r})} \right]^2, \\ \tilde{R}_{\pm} &\equiv \frac{1}{2} \left[\sqrt{(|u_1| + u_{1,r})} \pm \sqrt{(|u_2| + u_{2,r})} \right]^2.\end{aligned}\tag{8.118b}$$

The above is an elliptic EOM for the pairing energy R , which executes undamped periodic motion between the turning points $R_- \leq R \leq R_+$.

In Figs. 8.7 and 8.16, representative order parameter oscillations for phase **III** quenches are shown. The blue solid curves are the results of numerical simulations of the BCS Hamiltonian in Eq. (8.13) for 5024 classical Anderson pseudospins. The red dashed curves in these figures are solutions to Eq. (8.118a), with parameters in Eq. (8.118b) extracted from the roots.

We define

$$R_0 \equiv \frac{R_+ + R_-}{2}, \quad R_d \equiv \frac{R_+ - R_-}{2},\tag{8.119}$$

and introduce dimensionless amplitude y via

$$\begin{aligned}R(t) &\equiv R_0 + R_d y(t), \\ y_1 &\equiv \frac{R_0 + \tilde{R}_-}{R_d}, \quad y_2 \equiv \frac{R_0 + \tilde{R}_+}{R_d}.\end{aligned}\tag{8.120}$$

The relative amplitude y is constrained to $-1 \leq y \leq 1$, while $1 \leq y_1 \leq y_2$. Eq. (8.118a) becomes

$$\dot{y}^2 = R_d^2 (1 - y^2)(y + y_1)(y + y_2).\tag{8.121}$$

The solution may be written as

$$y(t) = \frac{2y_2 cn^2(\alpha t|M) - (y_2 + 1)}{(y_2 + 1) - 2cn^2(\alpha t|M)}, \quad (8.122)$$

where $cn(z|M)$ denotes the Jacobi elliptic function ($M = k^2$ is the modulo parameter). In terms of the roots,

$$\begin{aligned} R_0 &= \frac{1}{2} (|u_1| + |u_2| - u_{1,r} - u_{2,r}), \\ R_d &= \sqrt{|u_1| - u_{1,r}} \sqrt{|u_2| - u_{2,r}}, \\ y_2 &= \frac{|u_1| + |u_2| + \sqrt{|u_1| + u_{1,r}} \sqrt{|u_2| + u_{2,r}}}{R_d}, \\ M &= \frac{u_{1,i} u_{2,i}}{\alpha^2}, \\ \alpha &= \frac{1}{2} \sqrt{(u_{1,r} - u_{2,r})^2 + (u_{1,i} + u_{2,i})^2}. \end{aligned} \quad (8.123)$$

The physical period T of $R(t)$ is

$$T = \frac{4K(M)}{2\alpha}, \quad (8.124)$$

where $K(M)$ is the complete elliptic integral of the first kind (and $M = k^2$).

Consider quenches near the phase boundary within **III** such that $u_{1,i} \rightarrow 0$, i.e. quench coordinates $\{\Delta_0^{(i)}, \Delta_0^{(f)}\}$ lying just below the curve $\beta_c^{(-)}$ in Fig. 8.15. Here Eqs. (8.120) and (8.122) simplify to

$$\begin{aligned} R(t) &\simeq R_0 + R_d \cos(\Omega_c t), \\ R_0 &\simeq \frac{|u_2| - u_{2,r}}{2} + \mathcal{O}(u_{1,i})^2, \\ R_d &\simeq \sqrt{\frac{|u_2| - u_{2,r}}{2u_{1,r}}} u_{1,i} + \mathcal{O}(u_{1,i})^3, \\ \Omega_c &\simeq \sqrt{(u_{1,r} - u_{2,r})^2 + u_{2,i}^2} + \mathcal{O}(u_{1,i}). \end{aligned} \quad (8.125)$$

The orbit collapses for $u_{1,i} \rightarrow 0$, wherein isolated pair one merges with the continuum along the real axis. This is the phase boundary marked $\beta_c^{(-)}$ in Fig. 8.15.

Finally, we consider quenches along the small segment of the β_{MR} line in Fig. 8.15 which intrudes into phase **III** near $\{\Delta_0^{(i)}, \Delta_0^{(f)}\} = \{0, \Delta_{MR}\}$. For a discussion of the β_{MR} line and its termination at $\Delta_0^{(i)} = 0$, see Sec. 8.3.6, Eqs. (8.81) and (8.86). Along this line, the second root pair becomes negative real and degenerate: $u_{2,\pm} = -v_2 < 0$. The solution in Eq. (8.122) reduces to

$$\begin{aligned} y(t) &= \frac{y_2 \cos(\Omega_{MR} t) - 1}{y_2 - \cos(\Omega_{MR} t)}, \\ \Omega_{MR} &= \sqrt{(u_{1,r} + v_2)^2 + u_{1,i}^2}. \end{aligned} \quad (8.126)$$

Order parameter phase dynamics

Eqs. (8.100) and (8.104) imply that the pairing amplitude phase ϕ in Eq. (8.92) satisfies

$$\dot{\phi} = \frac{3}{2}R + 2m - \frac{\psi_+}{R}. \quad (8.127)$$

From Eqs. (8.112), (8.114), and (8.125), one can show that at the boundary separating phases **II** and **III** where $u_{1,i} = 0$,

$$\dot{\phi} = |u_2| = 2\mu_\infty,$$

which is the expected result. Here we have used Eq. (8.89) to relate the remaining pair of isolated roots $u_{2,\pm}$ to μ_∞ .

8.5 Non-equilibrium winding numbers and observables

In this section we calculate the long-time asymptotic spin distribution function from the conservation of the Lax norm. Using this result, we compute the winding numbers Q and W and the *approach* of $\Delta(t)$ to its asymptotic constant value in phases **I** and **II** of the phase diagram (Fig. 8.4). We relate the parity of zeroes in the Cooper pair distribution (introduced in Sec. 8.2.4) to Q and W , and to the RF spectroscopy amplitude in Eq. (8.29). Additional results, including the Bogoliubov amplitudes $u_{\mathbf{k}}(t)$ and $v_{\mathbf{k}}(t)$ as well as single-particle Green's functions are relegated to Appendix 8.10.

As discussed in Sec. 8.4.1, in phase **II** the asymptotic behavior of the order parameter is $\Delta(t) = \Delta_\infty \exp(-2i\mu_\infty t)$. It is the modulus of Δ that goes to a constant, but the phase winds at the frequency $2\mu_\infty$. This includes the ground state (zero quench) with $\mu_\infty = \mu_0^{(i)}$, due to the chemical potential shift of the field relative to $\vec{B}_i$ in Eq. (8.14). Unless otherwise noted, in this section we will work in the rotating frame $s_i^-(t) \rightarrow s_i^-(t) \exp(2i\mu_\infty t)$ when discussing phase **II**, c.f. Ref. [60]. In this frame, $\Delta(t) \rightarrow \Delta_\infty$ (constant).

8.5.1 Pseudospin distribution function

In phases **I** and **II** of the quench phase diagram in Fig. 8.4, $\Delta(t)$ asymptotes to a constant Δ_∞ (equal to zero in **I**). In the long-time limit, the effective magnetic field $\vec{B}_i$ seen by Anderson pseudospin $\vec{s}_i$ is given by Eq. (8.24). This is identical to the field in an “effective” ground state with pairing amplitude Δ_∞ and chemical potential μ_∞ . In the actual BCS or BEC ground state, each spin is parallel to its associated field.[60] For a quench, the situation is different. As $t \rightarrow \infty$, each spin *precesses* about its field with an energy-dependent frequency, as in Eq. (8.25). The pseudospin distribution function γ_i determines the projection of the spin onto the field in this equation. [In phase **I**, we should take $\hat{B}_i = -\hat{z}$ and $\mu_\infty = 0$; this reconciles Eqs. (8.18) and (8.25). In phase **II**, γ is referred to as the Cooper pair distribution in Sec. 8.2.4.]

The conservation of the Lax norm (spectral polynomial) allows the determination of γ_i in Eq. (8.25). As $t \rightarrow \infty$, the Lax components in Eq. (8.34) become

$$\begin{aligned} L^\pm(u; t) &= \nu_0 \int_0^{\varepsilon_\Lambda} d\varepsilon \frac{\sqrt{\varepsilon} [s^x(\varepsilon; t) \pm i s^y(\varepsilon; t)]}{\varepsilon - u}, \\ L^z(u; t) &= \nu_0 \int_0^{\varepsilon_\Lambda} d\varepsilon \left[\frac{\varepsilon s^z(\varepsilon; t)}{\varepsilon - u} \right] + \frac{\pi \nu_0}{g_f}. \end{aligned} \tag{8.128}$$

In these equations, we have converted to the continuum via Eq. (8.32); g_f is the post-quench coupling strength [Eq. (8.17)], and $s^a(\varepsilon_i; t) \equiv \hat{a} \cdot \vec{s}_i(t)$ [$a \in \{x, y, z\}$] is the continuum version of the precessing spin in Eq. (8.25). The energy cutoff in Eq. (8.128) is

$$\varepsilon_\Lambda \equiv 2 [\Lambda + \mu_0^{(i)}] \tag{8.129}$$

[see Eqs. (8.32) and (8.67)].

For $\Delta_\infty \neq 0$, all three Lax components in Eq. (8.128) contain both oscillating and constant terms. Our procedure to determine $\gamma(\varepsilon)$ is as follows: We sit at some fixed complex value of u

away from the positive real axis. Next, we take $t \rightarrow \infty$. The oscillatory terms in $L^{\{\pm, z\}}(u; t)$ vanish in this limit, as can be seen through repeated integration-by-parts.[65] The result is

$$\begin{aligned} L^x(u; \infty) &= \frac{\nu_0}{2} \int_0^{\varepsilon_\Lambda} d\varepsilon \gamma(\varepsilon) \frac{\varepsilon \Delta_\infty}{(\varepsilon - u) E_\infty(\varepsilon)}, \\ L^y(u; \infty) &= 0, \\ L^z(u; \infty) &= \frac{\nu_0}{2} \int_0^{\varepsilon_\Lambda} d\varepsilon \gamma(\varepsilon) \frac{\varepsilon \left(\frac{\varepsilon}{2} - \mu_\infty\right)}{(\varepsilon - u) E_\infty(\varepsilon)} + \frac{\pi \nu_0}{g_f}. \end{aligned} \quad (8.130)$$

Finally, we let u approach the positive real axis and evaluate the Lax components at $u \rightarrow u \pm i\eta$, with u and η positive and real on the right-hand side. We obtain

$$\begin{aligned} L^x(u \pm i\eta; \infty) &= \frac{\nu_0}{2} \left[\mathcal{J}(u) \pm i\pi \frac{u \Delta_\infty}{E_\infty(u)} \gamma(u) \right], \\ L^z(u \pm i\eta; \infty) &= \frac{\nu_0}{2} \left\{ \mathcal{K}(u) \pm i\pi \left[\frac{u \left(\frac{u}{2} - \mu_\infty\right)}{E_\infty(u)} \right] \gamma(u) \right\}, \end{aligned} \quad (8.131a)$$

where

$$\begin{aligned} \mathcal{J}(u) &\equiv P \int_0^{\varepsilon_\Lambda} d\varepsilon \left(\frac{1}{\varepsilon - u} \right) \frac{\varepsilon \gamma(\varepsilon) \Delta_\infty}{E_\infty(\varepsilon)}, \\ \mathcal{K}(u) &\equiv P \int_0^{\varepsilon_\Lambda} d\varepsilon \left(\frac{1}{\varepsilon - u} \right) \frac{\varepsilon \gamma(\varepsilon) \left(\frac{\varepsilon}{2} - \mu_\infty\right)}{E_\infty(\varepsilon)} + \frac{2\pi}{g_f}. \end{aligned} \quad (8.131b)$$

In these equations, P denotes the principal value. Combining Eqs. (8.131) and (8.35) determines the Lax norm at infinite time, $L_2(u \pm i\eta; t = \infty)$. We equate this to Eq. (8.64b), which gives $L_2(u)$ in terms of the pre-quench state, leading to

$$\begin{aligned} &u \left\{ \mathcal{J}(u) \pm i\pi \frac{u \Delta_\infty}{E_\infty(u)} \gamma(u) \right\}^2 \\ &+ \left\{ \mathcal{K}(u) \pm i\pi \left[\frac{u \left(\frac{u}{2} - \mu_\infty\right)}{E_\infty(u)} \right] \gamma(u) \right\}^2 = I_\mp(u), \end{aligned} \quad (8.132)$$

where the initial state is encoded in

$$\begin{aligned} I_\mp(u) &\equiv \beta^2 + [E^{(i)}(u) f^{(i)}(u; 1) \mp i\pi u]^2 \\ &+ \beta \frac{(u - 2\mu_0^{(i)})}{E^{(i)}(u)} [E^{(i)}(u) f^{(i)}(u; 1) \mp i\pi u]. \end{aligned} \quad (8.133)$$

The form of $f^{(i)}(u; 1)$ is given by Eq. (8.68). Eq. (8.132) implies that

$$\begin{aligned} s_\pm \sqrt{I_\mp(u) - \frac{u [\mathcal{J}(u) \left(\frac{u}{2} - \mu_\infty\right) - \mathcal{K}(u) \Delta_\infty]^2}{E_\infty^2(u)}} \\ = \frac{\left(\frac{u}{2} - \mu_\infty\right) \mathcal{K}(u) + u \Delta_\infty \mathcal{J}(u)}{E_\infty(u)} \pm i\pi u \gamma(u), \end{aligned} \quad (8.134)$$

with $\{s_+, s_-\} \in \pm 1$.

We take the difference of the $\pm i\eta$ prescriptions in Eq. (8.134) to obtain

$$\gamma(u) = \frac{1}{2i\pi u} \left\{ \begin{aligned} &s_+ \sqrt{I_-(u) - u \left[\frac{\Xi(u)}{E_\infty(u)} \right]^2} \\ &- s_- \sqrt{I_+(u) - u \left[\frac{\Xi(u)}{E_\infty(u)} \right]^2} \end{aligned} \right\}, \quad (8.135)$$

where

$$\begin{aligned}\Xi(u) &\equiv \mathcal{J}(u) \left(\frac{u}{2} - \mu_\infty \right) - \mathcal{K}(u) \Delta_\infty \\ &= -\Delta_\infty \left\{ \frac{2\pi}{g_f} - \int_0^{\varepsilon_\Lambda} d\varepsilon \left[-\gamma(\varepsilon) \right] \frac{\varepsilon}{2E_\infty(\varepsilon)} \right\} \\ &= 0.\end{aligned}\tag{8.136}$$

That $\Xi = 0$ is explained as follows. Clearly this holds in phase **I**, wherein $\Delta_\infty = 0$. To see why Ξ vanishes for $\Delta_\infty > 0$ (phase **II**), we note that the term in brackets on the second line of Eq. (8.136) is the continuum version of the BCS equation, Eqs. (8.50) and (8.161a), for effective spins of “length” $-\gamma(\varepsilon)/2$. Indeed, the Lax components at infinite time in Eq. (8.130) appear as though evaluated for a ground state with $\{\Delta_0, \mu\} = \{\Delta_\infty, \mu_\infty\}$, for effective spins aligned *along* the field as in Eq. (8.49), but with a renormalized spin length set by $\gamma(\varepsilon)$ (which is the projection onto the field of the physical, precessing pseudospins).

We therefore conclude that

$$\gamma(\varepsilon) = \frac{s}{2i\pi\varepsilon} \left[\sqrt{I_-(\varepsilon)} - \sqrt{I_+(\varepsilon)} \right],\tag{8.137}$$

which is *independent* of $\{\Delta_\infty, \mu_\infty\}$. Relative to Eq. (8.135), we set $s_+ = s_- \equiv s \in \pm 1$ to obtain a real amplitude. Eq. (8.137) holds throughout phases **I** and **II**. Note that $\gamma(\varepsilon) \rightarrow -1$ as $\varepsilon \rightarrow \infty$ for any quench, since the particle density is finite and all spins are aligned along $-\hat{z}$ for sufficiently large energies. Subject to this boundary condition, the physical branch (sign s) of Eq. (8.137) changes at an energy ε whenever $\gamma(\varepsilon) \rightarrow 0$ with a non-zero slope, so as to produce a continuous distribution function.

A more useful but equivalent expression is

$$\gamma(\varepsilon) = s \sqrt{1 - \frac{1}{2(\pi\varepsilon)^2} \left[\mathcal{N}(\varepsilon) - \sqrt{\left\{ \frac{\mathcal{N}^2(\varepsilon)}{-\varepsilon \left[\frac{2\pi\varepsilon\Delta_0^{(i)}\beta}{E_\infty(\varepsilon)} \right]^2} \right\}} \right]},\tag{8.138a}$$

where

$$\begin{aligned}\mathcal{N}(\varepsilon) &\equiv \left[\left(\frac{\varepsilon}{2} - \mu_0^{(i)} \right) f^{(i)}(\varepsilon; 1) + \beta \right]^2 \\ &\quad + \varepsilon \left[\Delta_0^{(i)} f^{(i)}(\varepsilon; 1) \right]^2 + (\pi\varepsilon)^2.\end{aligned}\tag{8.138b}$$

Eq. (8.138a) gives a manifestly real formula for $\gamma(\varepsilon)$; one must still choose the branch $s \in \pm 1$ as a function of energy so as to produce a continuous distribution function.

Finally, we note that the expression for $\gamma(\varepsilon)$ in Eqs. (8.137) or (8.138a) also applies in phase **III**, if suitably interpreted. In this case, $-\gamma(\varepsilon)/2$ denotes the projection of spin $\vec{s}(\varepsilon)$ in the post-quench asymptotic state onto the *reduced* spin solution $\vec{s}_{\text{red}}(\varepsilon)$. The reduced spin solution is defined such that $\vec{s}_{\text{red}}(\varepsilon)$ satisfies Eq. (8.61b) in terms of the collective variables $\vec{\sigma}_{1,2}$, as discussed in Sec. 8.4.2.

8.5.2 Winding numbers

Green’s function winding W

The winding number W in Eq. (8.10) depends upon the asymptotic form of the retarded Green’s function $\mathcal{G}_{\mathbf{k}}(t, t')$. As discussed in Sec. 8.2.3, this function satisfies the Bogoliubov-de

Gennes equation (8.21), subject to the initial condition in (8.22). It is therefore independent of the distribution function $\gamma(\varepsilon)$, which does not appear in these equations. This is confirmed by a calculation in Appendix 8.10, which yields the explicit form for $\mathcal{G}_{\mathbf{k}}(t, t')$ in Eq. (8.190). This result is identical to that for a system in its ground state, except that here the order parameter Δ_∞ and the chemical potential μ_∞ are determined by the quench through the single isolated pair of roots [Eq. (8.89)].

The winding number W therefore depends only upon $\text{sgn}(\mu_\infty)$ in phase **II**, and takes the values shown in Fig. 8.9, as discussed in Sec. 8.3.6. By the argument in Sec. 8.2.3, $W = 1$ ($W = 0$) signals the presence (absence) of edge states in the Bogoliubov-de Gennes quasiparticle spectrum following a quench in phase **II**. By contrast, W is ill-defined in the gapless phase **I**.

Pseudospin winding Q

As explained in Sec. 8.2.3, starting from an initial $p + ip$ state in either the BCS or BEC phases, the evolving spin distribution can be parameterized at any time t as in Eq. (8.7), where $\varrho(k)$ and $\Theta(k)$ are time-dependent. Eq. (10.17) then implies that the pseudospin winding number Q is given by

$$Q = \frac{1}{2} \{ \text{sgn}[\varrho(k=0)] + 1 \}. \quad (8.139)$$

As $t \rightarrow \infty$, the momentum space pseudospin texture is reconstructed from Eqs. (8.18) and (8.25) in phases **I** and **II**, respectively. The latter is transcribed in Eq. (8.178) of Appendix 8.10.

We consider first the gapless phase **I**. The winding $Q = \frac{1}{2} \{ \gamma(0) + 1 \}$. From Eq. (8.138), one can check that

$$\lim_{\varepsilon \rightarrow 0} |\gamma(\varepsilon)| = \begin{cases} 1, & \mu_0^{(i)} \neq 0 \\ 0, & \mu_0^{(i)} = 0 \end{cases}, \quad (8.140)$$

where $\mu_0^{(i)}$ is the pre-quench chemical potential, with $\mu_0^{(i)} > 0$ ($\mu_0^{(i)} < 0$) indicating a BCS (BEC) initial state. We conclude that Q is well-defined throughout the gapless phase **I**, except for a quench starting from the quantum critical point $\Delta_0^{(i)} = \Delta_{\text{QCP}}$.

To compute Q , we must determine the branch of Eq. (8.138a) (i.e., $s = \pm 1$) relevant for $\varepsilon \rightarrow 0$. We know that $\lim_{\varepsilon \rightarrow \infty} \gamma(\varepsilon) = -1$, so that the branch is $s = -1$ at large ε . The branch switches every time γ goes to zero with a non-zero slope, so as to preserve the continuity. We find that in the gapless phase **I**,

$$\lim_{\varepsilon \rightarrow 0} \gamma(\varepsilon) = \begin{cases} -1, & \mu_0^{(i)} < 0 \Rightarrow Q = 0 \\ +1, & \mu_0^{(i)} > 0 \Rightarrow Q = 1 \end{cases}. \quad (8.141)$$

In other words, Q is conserved in the gapless phase. This can be understood in various ways. The pseudospin winding number cannot change unless **(a)** the spin distribution develops a discontinuity or a diabolical point,[24] or **(b)** a skyrmion-number changing process (hedgehog) occurs in *momentum-time*. Scenario **(a)** cannot occur within a finite time interval, because the time evolution is a smooth deformation. Scenario **(b)** cannot happen for the reduced p-wave BCS Hamiltonian dynamics [which are identical for Eqs. (8.1) and (8.11), as shown in Appendix 8.8].

The conservation of Q in phase **I** wherein $\Delta(t) \rightarrow 0$ leads to the notion of a “gapless topological phase.” This occurs for quenches in the region marked **B**, Fig. 8.8. Those in **A** are topologically trivial. Corresponding topological and trivial pseudospin textures appear similar to those in Figs. 8.2(a) and 8.2(c), but now these textures undulate in time: The spins at radius k precess about $\hat{z}$ with frequency k^2 [Eq. (8.18)]. In Figs. 8.11 and 8.12, $\gamma(\varepsilon)$ is plotted against $k = \sqrt{\varepsilon}$ for representative quenches in **A** and **B**, respectively.

Next we consider phase **II**. As discussed above and in Sec. 8.2.3, in the limit $t \rightarrow \infty$, the retarded Green's function winding number W is completely determined by μ_∞ . This appears in $\vec{B}(\varepsilon)$, the continuum version of Eq. (8.24). The latter can be viewed as an effective ground state field, which “winds” whenever $W \neq 0$ (recall that spins are aligned along the field in the actual ground state). For $W = 1$, we have $\mu_\infty > 0$ and $\hat{B}(0) = \hat{z}$ (“winding”), while $W = 0$ implies that $\mu_\infty < 0$ and $\hat{B}(0) = -\hat{z}$ (“non-winding”). As shown in Fig. 8.9, W undergoes a dynamical topological transition for quenches across the quantum critical point. In particular, W evolves from trivial to non-trivial or vice-versa for quenches in the regions marked **C** and **H** in Fig. 8.10.

The pseudospin winding Q is determined by $s^z(0)$. Since $|\gamma(0)| = 1$ for $\mu_0^{(i)} \neq 0$ [Eq. (8.140)], Eq. (8.25) implies that $Q = 0$ for $\gamma(0)B^z(0) = 1$ and $Q = 1$ for $\gamma(0)B^z(0) = -1$. For a given phase **II** quench $\{\Delta_0^{(i)}, \Delta_0^{(f)}\}$, let us denote the initial value of the Green's function winding as W_0 , while W_∞ is the asymptotic value as $t \rightarrow \infty$. Imposing continuity on the function $\gamma(\varepsilon)$, we find that

$$\gamma(0) = \begin{cases} +1, & W_\infty \neq W_0 \\ -1, & W_\infty = W_0 \end{cases} . \quad (8.142)$$

In other words, the Cooper pair distribution $\gamma(\varepsilon)$ “winds” from -1 at $\varepsilon \rightarrow \infty$ to $+1$ at $\varepsilon = 0$ whenever W in the asymptotic post-quench state differs from its value in the initial state. As a result, we determine that Q is conserved for all quenches in phase **II**, so that Eq. (8.142) can be rewritten as

$$\gamma(0) = \begin{cases} +1, & W \neq Q \\ -1, & W = Q \end{cases} , \quad (8.143)$$

where both winding numbers are computed in the asymptotic steady-state.

Plots of $\gamma(\varepsilon)$ for phase **II** quenches in regions **{C, D, E, H}** marked in Fig. 8.10 appear in Figs. 8.13 and 8.17. Because quenches in regions **C** and **H** have $W \neq Q$ and therefore $\gamma(0) = +1$, we deduce that the number of zeroes in $\gamma(\varepsilon)$ is odd for quenches in these regions. By contrast, $\gamma(\varepsilon)$ must exhibit an even number of zeroes for quenches wherein $W = Q$, including those in **D** and **E**. A quench therefore imprints a new $\mathbb{Z}_2$ index upon the Cooper pair distribution function, in the form of its parity of zeroes. By contrast, $\gamma(\varepsilon) = -1$ for all mode energies ε in both the BCS and BEC ground states.

For an ultracold atomic realization of the $p + ip$ superfluid, the Cooper pair distribution can in principle be measured in RF spectroscopy, as discussed in Sec. 8.2.4.

8.5.3 Approach to the steady-state

Phase I: Decay to zero

In the continuum limit, the post-quench order parameter in Eq. (8.15) is given by

$$\Delta(t) = -\frac{g_f}{2\pi} \int_0^{\varepsilon_\Lambda} d\varepsilon \sqrt{\varepsilon} s^-(\varepsilon; t). \quad (8.144)$$

The order parameter can be self-consistently determined by linearizing the spin equations of motion in Eq. (8.14). In phase **I**, $\Delta(t)$ decays to zero as $t \rightarrow \infty$, and $s^-(\varepsilon; t)$ can be written as a sum of the pure precession in Eq. (8.18), plus a fluctuation:

$$\begin{aligned} s^-(\varepsilon; t) &\equiv s_\infty^-(\varepsilon; t) + \delta s^-(\varepsilon; t), \\ s_\infty^-(\varepsilon; t) &= \frac{1}{2} \sqrt{1 - \gamma^2(\varepsilon)} \exp \{ -i [\varepsilon t + \Theta(\varepsilon)] \} . \end{aligned} \quad (8.145)$$

The continuum limit of Eq. (8.14) can be written as

$$\begin{aligned}\dot{s}^z(\varepsilon; t) &= i\sqrt{\varepsilon} [\Delta^* s^-(\varepsilon; t) - \Delta s^+(\varepsilon; t)], \\ \dot{s}^-(\varepsilon; t) &= i [2\sqrt{\varepsilon} \Delta s^z(\varepsilon; t) - \varepsilon s^-(\varepsilon; t)],\end{aligned}\tag{8.146}$$

where $\dot{x} = dx/dt$. Eq. (8.146) is invariant under the *effective* time-reversal transformation

$$\begin{aligned}s^z(\varepsilon; t) &\rightarrow s^z(\varepsilon; -t), \quad s^\pm(\varepsilon; t) \rightarrow s^\mp(\varepsilon; -t), \\ \Delta(t) &\rightarrow \Delta^*(-t).\end{aligned}$$

These relations are also satisfied by the initial condition, Eq. (8.49) for the pre-quench $p + ip$ state with real Δ_0 . Therefore we can set $\Theta(\varepsilon) = 0$ in Eqs. (8.18), (8.25), and (8.145).

To linear order in the smallness of $\Delta(t)$,

$$\delta \dot{s}^-(\varepsilon; t) = i [\sqrt{\varepsilon} \gamma(\varepsilon) \Delta(t) - \varepsilon \delta s^-(\varepsilon; t)],\tag{8.147a}$$

$$\Delta(t) = \Delta_\infty(t) + \delta \Delta(t),\tag{8.147b}$$

$$\Delta_\infty(t) \equiv -\frac{g_f}{4\pi} \int_0^{\varepsilon_\Lambda} d\varepsilon \mathcal{A}(\varepsilon) e^{-i\varepsilon t},\tag{8.147c}$$

$$\delta \Delta(t) = -\frac{g_f}{2\pi} \int_0^{\varepsilon_\Lambda} d\varepsilon \sqrt{\varepsilon} \delta s^-(\varepsilon; t),\tag{8.147d}$$

$$\mathcal{A}(\varepsilon) \equiv \sqrt{\varepsilon} \sqrt{1 - \gamma^2(\varepsilon)}.\tag{8.147e}$$

In these equations, $\Delta_\infty(t)$, $\delta \Delta(t)$, and $\delta s^-(\varepsilon; t)$ all vanish by assumption as $t \rightarrow 0$.

We first calculate $\Delta_\infty(t)$, which exhibits the same power-law decay as the full $\Delta(t)$. The latter is also computed explicitly below. For an initial state not at the quantum critical point ($\mu_0^{(i)} \neq 0$), Eqs. (8.138) and (8.147e) imply that

$$\mathcal{A}(\varepsilon) = c\varepsilon + \mathcal{O}(\varepsilon^2),\tag{8.148}$$

$$c \equiv \frac{\beta \Delta_0^{(i)}}{|\mu_0^{(i)}| \left(\beta + 2\mu_0^{(i)} \log \left[\frac{2\Lambda}{(\Delta_0^{(i)})^2 + 2|\mu_0^{(i)}| \theta(-\mu_0^{(i)})} \right] \right)},$$

where $\{\Delta_0^{(i)}, \mu_0^{(i)}\}$ characterize the initial state and $\beta > 0$ is the quench parameter [Eq. (8.16)]. Because $\mathcal{A}(\varepsilon)$ is a regular (if complicated) function over the integration interval, we can evaluate Eq. (8.147c) via repeated integration-by-parts. The leading result is

$$\Delta_\infty(t) = \frac{g_f}{4\pi} \left[\frac{c}{t^2} + \frac{1}{it} \mathcal{A}(\varepsilon_\Lambda) \exp(-i\varepsilon_\Lambda t) \right].$$

The energy cutoff ε_Λ was defined in Eq. (8.129). Via Eqs. (8.138) and (8.147e), one can show that $\mathcal{A}(\varepsilon_\Lambda) = d/\Lambda + \mathcal{O}(\Lambda)^{-2}$ with d a cutoff-independent constant, leading to

$$\Delta_\infty(t) = \frac{g_f}{4\pi} \left[\frac{c}{t^2} - i \frac{d}{\Lambda t} \exp(-i\varepsilon_\Lambda t) \right].\tag{8.149}$$

This has the form given by Eq. (8.19), which consists of a cutoff-independent $1/t^2$ decay plus a “non-universal” oscillating term proportional to $1/\Lambda t$. The latter is technically beyond the logarithmic accuracy to which we have been working throughout, but can be important when comparing against numerics.

By contrast, for a quench starting from the quantum critical point (with $\Delta_0^{(i)} = \Delta_{\text{QCP}}$ and $\mu_0^{(i)} = 0$), one finds that

$$\mathcal{A}(\varepsilon) = \sqrt{\varepsilon} - \tilde{c}\varepsilon^{3/2} + \mathcal{O}(\varepsilon^{5/2}), \quad (8.150)$$

with $\tilde{c}$ a constant. The square root leads to the slower $t^{-3/2}$ decay law in Eq. (8.20); the ultraviolet gives the same non-universal contribution.

We can also determine the precise form of $\Delta(t)$. Ignoring the cutoff-dependent term, the $1/t^2$ decay of $\Delta_\infty(t)$ in Eq. (8.149) enters as a source in the right-hand side of Eq. (8.147a). As a result, $\delta s^-(\varepsilon; t) \propto 1/t^2$ and we can drop the time-derivative on the left-hand side, which decays faster. We thereby obtain

$$\delta s^-(\varepsilon; t) = \frac{1}{\sqrt{\varepsilon}} \gamma(\varepsilon) \Delta(t),$$

and using Eq. (8.147d)

$$\delta \Delta(t) = \frac{g_f \Delta(t)}{2\pi} (\varepsilon_\Lambda - 8\pi n). \quad (8.151)$$

Here we have expressed the conserved particle density n through the distribution function,

$$n = \frac{1}{8\pi} \int_0^{\varepsilon_\Lambda} d\varepsilon [1 + 2s^z(\varepsilon)] = \frac{1}{8\pi} \int_0^{\varepsilon_\Lambda} d\varepsilon [1 + \gamma(\varepsilon)].$$

Using Eq. (8.147b), Eq. (8.151) simplifies to

$$\begin{aligned} \Delta(t) &= \frac{\Delta_\infty(t)}{\left[1 + \frac{g_f}{2\pi} (8\pi n - \varepsilon_\Lambda)\right]} \\ &= \frac{1}{\left\{ \mu_0^{(f)} \log \left[\frac{2\Lambda e}{(\Delta_0^{(f)})^2} \right] - \mu_0^{(i)} \right\}} \frac{c}{4t^2}. \end{aligned} \quad (8.152)$$

On the second line, we have employed the BCS equation (8.162). Here $\mu_0^{(i)}$ ($\mu_0^{(f)}$) denotes the chemical potential associated to $\Delta_0^{(i)}$ ($\Delta_0^{(f)}$) in the BCS ground state. For a quench in phase I, we have $\mu_0^{(f)} \simeq 2\pi n$, while $\mu_0^{(i)} \leq 2\pi n$. We note that only logarithmic dependence upon the cutoff appears in the final expression.

Phase II: Decay to $\Delta_\infty > 0$

We will evaluate the counterpart of Eq. (8.147c) for quenches wherein the order parameter asymptotes to a non-zero constant. The precessing spin in Eq. (8.25) has the minus component

$$s_\infty^-(\varepsilon; t) = \frac{1}{\sqrt{\varepsilon}} \left\{ \alpha_1(\varepsilon) \cos [2E_\infty(\varepsilon)t] - i\alpha_2(\varepsilon) \sin [2E_\infty(\varepsilon)t] + \alpha_3(\varepsilon) \right\}, \quad (8.153)$$

where

$$\begin{aligned} \alpha_1(\varepsilon) &\equiv \mathcal{A}(\varepsilon) \frac{\left(\frac{\varepsilon}{2} - \mu_\infty\right)}{2E_\infty(\varepsilon)}, \\ \alpha_2(\varepsilon) &\equiv \frac{1}{2} \mathcal{A}(\varepsilon), \\ \alpha_3(\varepsilon) &\equiv \gamma(\varepsilon) \frac{\varepsilon \Delta_\infty}{2E_\infty(\varepsilon)}, \end{aligned} \quad (8.154)$$

and the amplitude $\mathcal{A}(\varepsilon)$ was defined in Eq. (8.147e).

Inserting Eq. (8.153) into Eq. (8.144), the static term involving $\alpha_3(\varepsilon)$ evaluates to

$$-\frac{g_f}{2\pi} \int_0^{\varepsilon_\Lambda} d\varepsilon \alpha_3(\varepsilon) = \Delta_\infty, \quad (8.155)$$

where we have used Eq. (8.136). Thus the spin distribution reconstructed from the conservation of the Lax norm is consistent with the pairing amplitude Δ_∞ computed from the isolated root pair. The time-dependent part of the order parameter is given by

$$\delta\Delta_\infty(t) = -\frac{g_f}{2\pi} \int_0^{\varepsilon_\Lambda} d\varepsilon \left\{ \alpha_1(\varepsilon) \cos[2E_\infty(\varepsilon)t] - i\alpha_2(\varepsilon) \sin[2E_\infty(\varepsilon)t] \right\}. \quad (8.156)$$

To compute Eq. (8.156), we must distinguish two regimes. For quenches with $\Delta_\infty^2 < \mu_\infty$, the dominant contribution comes from a saddle-point at non-zero ε . This is the region of phase **II** to the left of the line marked β_{coh} in Fig. 8.15, as discussed above Eq. (8.82) in Sec. 8.3.6. For quenches in this regime, $E_\infty(\varepsilon)$ reaches its minimum value (the non-equilibrium spectral gap) $E_{\min} = \Delta_\infty \sqrt{2\mu_\infty - \Delta_\infty^2}$ at $\varepsilon_0 = 2(\mu_\infty - \Delta_\infty^2) > 0$:

$$E_\infty(\varepsilon) = E_{\min} + \frac{(\varepsilon - \varepsilon_0)^2}{8E_{\min}} + \dots$$

The saddle-point gives

$$\delta\Delta_\infty(t) \simeq -g_f \sqrt{\frac{E_{\min}}{2\pi t}} \begin{bmatrix} \alpha_- \cos(2E_{\min}t) \\ -\alpha_+ \sin(2E_{\min}t) \end{bmatrix}, \quad (8.157)$$

$$\alpha_\mp \equiv \alpha_1(\varepsilon_0) \mp i\alpha_2(\varepsilon_0).$$

This is qualitatively the same behavior as obtained for weak BCS-to-BCS quenches in the s-wave case.[66, 12]

The saddle-point contribution in Eq. (8.157) vanishes when $\varepsilon_0 \rightarrow 0$ ($\mu_\infty = \Delta_\infty^2$). For $\Delta_\infty^2 > \mu_\infty$, the minimum of $E_\infty(\varepsilon)$ occurs at $\varepsilon = 0$ (see Sec. 8.3.2 for further discussion of the spectrum). Eq. (8.156) can then be evaluated by repeated integration-by-parts. Note that

$$\alpha_1(\varepsilon) = \alpha_2(\varepsilon) [-\text{sgn}(\mu_\infty) + \mathcal{O}(\varepsilon)],$$

leading to

$$\delta\Delta_\infty(t) \simeq \frac{g_f \text{sgn}(\mu_\infty)}{4\pi} \int_0^{\varepsilon_\Lambda} d\varepsilon \mathcal{A}(\varepsilon) e^{2iE_\infty(\varepsilon) \text{sgn}(\mu_\infty)t}. \quad (8.158)$$

This is the phase **II** generalization of Eq. (8.147c), valid for $\Delta_\infty^2 > \mu_\infty$. Eqs. (8.148) and (8.150) imply that the cutoff-independent part of the decay is $1/t^2$ for $\Delta_0^{(i)} \neq \Delta_{\text{QCP}}$ and $1/t^{3/2}$ for $\Delta_0^{(i)} = \Delta_{\text{QCP}}$. At $\varepsilon = 0$, the phase factor $e^{2iE_\infty(\varepsilon) \text{sgn}(\mu_\infty)t} \rightarrow e^{2i\mu_\infty t}$; this is eliminated by moving back to the “lab” frame.

We conclude that Eqs. (8.19) and (8.20) also describe the decay of $\delta\Delta(t) \equiv [\Delta(t) - \Delta_\infty]$ for phase **II** quenches to the right of the β_{coh} line in Fig. 8.15. By contrast, quenches to the left of this line within phase **II** exhibit slower, oscillatory decay according to Eq. (8.157), due to the saddle-point contribution.

8.6 Conclusion

8.6.1 Pair-breaking processes

In this paper, we have computed the quench dynamics of a $p+ip$ superfluid in the collisionless regime. This is a non-adiabatic evolution of the initial state in which pair-breaking processes are neglected. The preconditions necessary to observe our results in an experiment are that

$$t_{\text{quench}} \ll \frac{1}{E_{\text{min}}} \ll t_{\text{pb}}, \quad (8.159)$$

where t_{quench} is the duration of the quench (zero for the instantaneous quench studied here), E_{min} is the minimum quasiparticle energy (quasiparticle gap), and t_{pb} is the time scale associated to inelastic pair breaking processes. The various predictions presented in this paper describe the post-quench asymptotic steady-state. Provided the bounds in Eq. (8.159) are met, we expect our results to hold for times t such that $\frac{1}{E_{\text{min}}} \ll t \lesssim t_{\text{pb}}$.

For BCS-to-BCS quenches entirely confined to the weak pairing regime,

$$E_{\text{min}} = \Delta_0^{(i)} \sqrt{2\mu_0^{(i)} - (\Delta_0^{(i)})^2} \simeq \Delta_0^{(i)} \sqrt{4\pi n}.$$

Fermi liquid theory then implies the order-of-magnitude estimate

$$t_{\text{pb}} \sim \frac{1}{E_{\text{min}}} \left[\frac{\mu_0^{(i)}}{E_{\text{min}}} \right] \gg \frac{1}{E_{\text{min}}}, \quad (8.160)$$

implying the existence of a large window over which the collisionless dynamics computed in this paper can be observed. The investigation of pair-breaking processes upon quenches originating or terminating beyond the weak-coupling regime remains an important subject for future work.

8.6.2 Summary and open questions

In summary, we have investigated quantum quenches in 2D topological p-wave superfluids. The post-quench dynamics have been computed via classical integrability. Within the classical approximation, our treatment is exact. Because of the infinite-ranged nature of the interactions in the reduced BCS Hamiltonian, we expect that our results apply to the quantum model in the thermodynamic limit.

We constructed the quench phase diagram, and extracted the exact asymptotic order parameter $\Delta(t)$ dynamics, finding that either (1) $\Delta(t)$ goes to zero, (2) $\Delta(t)$ goes to non-zero constant, or (3) $\Delta(t)$ exhibits persistent oscillations. These results are qualitatively the same as the s-wave case.[6, 19, 18, 9, 11, 10, 12, 13, 14]

The key difference from previous work is that here we have characterized the quench-induced dynamics of the system topology. We found that the pseudospin winding number Q is unchanged by the quench, leading to the prediction of a “gapless topological state.” By contrast, the retarded Green’s function winding number W can undergo a dynamical transition. This happens e.g. for quenches across the quantum critical point separating the topologically non-trivial BCS and trivial BEC phases. In the asymptotic steady state wherein the order parameter goes to a constant, the corresponding Bogoliubov-de Gennes Hamiltonian is expected to possess edge states in a finite geometry whenever $W \neq 0$.

While W determines the existence of edge modes following a quench, we have not determined the *occupancy* of these states. The difficulty is that introducing an edge breaks the integrability of our momentum-space BCS model. A fundamental question is whether these non-equilibrium topological steady states support the kind of quantized thermal conductance expected in an equilibrium p-wave superconductor.[54, 55] A related question is the formation, preservation,

or destruction of Majorana zero modes following a quench in 1D topological superconductor; this was studied numerically for a non-interacting model in Ref. [61].

Another interesting open problem relates to the role of topological defects in thermalization. Once pair-breaking processes are included, the theory is no longer integrable. One therefore expects thermalization at the longest times. How does this occur? One possibility is that topological defects, which can appear either as phase vortices in real space, or hedgehog instantons in momentum-time, proliferate and scramble the topological order.

Finally, we have determined that the parity of zeroes in the Cooper pair distribution is odd whenever $Q \neq W$, i.e. whenever W undergoes a dynamical transition. We have argued that the Cooper pair distribution should be observable in RF spectroscopy in an ultracold atomic or molecular realization of the 2D $p + ip$ superfluid. By contrast, the same response does not distinguish the BCS from BEC phases in the ground state.

Probing the Cooper pair distribution can therefore provide a bulk signature of the topological properties of the system when it is driven far from equilibrium by a quench. In this way, a quantum quench can be used to transfer topological entanglement normally hidden from experiment into a physical observable, i.e. a non-equilibrium distribution function.

8.7 Ground state

8.7.1 BCS equations

In the thermodynamic limit, the BCS equations for the pairing amplitude Δ_0 and particle density n are

$$\frac{1}{g} = \frac{1}{2\pi} \int_0^{2(\Lambda+\mu)} d\varepsilon \frac{\varepsilon/2}{\sqrt{(\frac{\varepsilon}{2} - \mu)^2 + (\Delta_0)^2 \varepsilon}}, \quad (8.161a)$$

$$n = \frac{1}{8\pi} \int_0^{2(\Lambda+\mu)} d\varepsilon \left[1 - \frac{(\frac{\varepsilon}{2} - \mu)}{\sqrt{(\frac{\varepsilon}{2} - \mu)^2 + (\Delta_0)^2 \varepsilon}} \right], \quad (8.161b)$$

where $\varepsilon = k^2$, and we have cut these integrals off at a single particle energy $k^2/2 = \Lambda + \mu$. (The inclusion of μ simplifies the analysis; results are obtained to logarithmic accuracy in $\Lambda \gg |\mu|$.) The BCS coupling G in Eq. (8.11) is related to g via Eq. (8.17). The dimensionful interaction strength g is non-zero in the thermodynamic limit, and carries units of inverse density. Eq. (8.161a) is the continuum version of Eq. (8.50), using Eq. (8.32).

Integrating Eq. (8.161) and discarding terms proportional to inverse powers of Λ , one obtains Eq. (8.5) for the chemical potential and

$$\frac{1}{g} - \frac{1}{g_{\text{QCP}}} = \frac{\mu}{\pi} \log \left[\frac{2\Lambda e}{\Delta_0^2 + 2|\mu|\theta(-\mu)} \right]. \quad (8.162)$$

In this equation, g_{QCP} is the coupling strength at the BCS-BEC transition $\mu = 0$,

$$\frac{1}{g_{\text{QCP}}} = \frac{\Lambda}{\pi} - 4n.$$

The linear divergence in Eq. (8.161a) has been absorbed into $1/g_{\text{QCP}}$. Because a quench is completely specified by the initial order parameter $\Delta_0^{(i)}$ and the difference of the initial and final coupling strengths [Eq. (8.16)], g_{QCP} plays no role in the dynamics.

8.7.2 Spectral transitions and tunneling density of states

We first note three special values of Δ_0 , defined implicitly through the chemical potential equation (8.5):

$$\begin{aligned}\mu(n, \Delta_{\text{Coh}}) &= (\Delta_{\text{Coh}})^2, \\ \mu(n, \Delta_{\text{MR}}) &= \frac{1}{2}(\Delta_{\text{MR}})^2, \\ \mu(n, \Delta_{\text{QCP}}) &= 0.\end{aligned}\tag{8.163}$$

Since μ is a monotonically-decreasing function of Δ_0 [Eq. (8.5) and Fig. 8.3], we have $\Delta_{\text{Coh}} < \Delta_{\text{MR}} < \Delta_{\text{QCP}}$. Each of these values corresponds to a particular transition or anomalous point in the shape of the quasiparticle energy spectrum. To see this, we rewrite the quasiparticle energy E_k in Eq. (8.4) in terms of $\varepsilon = k^2$ [Eq. (8.26)]:

$$E(\varepsilon; \Delta_0) = \sqrt{\left(\frac{\varepsilon}{2} - \mu\right)^2 + (\Delta_0)^2 \varepsilon}.$$

For sufficiently weak pairing, the minimum of E with respect to ε occurs slightly below $2\mathcal{E}_F$, where $\mathcal{E}_F$ denotes the Fermi energy. As Δ_0 (or equivalently, the coupling strength) is increased, this minimum moves to smaller energies. At $\Delta_0 = \Delta_{\text{Coh}}$, it reaches zero. For $\Delta_0 < \Delta_{\text{Coh}}$ [$\mu > (\Delta_{\text{Coh}})^2$], the tunneling density of states exhibits a coherence peak (van Hove singularity) above its threshold value, see Fig. 8.14(b) and Eq. (8.166), below. The coherence peak disappears for $\Delta_0 \geq \Delta_{\text{Coh}}$. At the special point $\Delta_0 = \Delta_{\text{MR}}$, the curvature of $E(\varepsilon)$ vanishes everywhere:

$$E(\varepsilon; \Delta_{\text{MR}}) = \frac{1}{2} [\varepsilon + (\Delta_{\text{MR}})^2].$$

The condition $\Delta_0 = \Delta_{\text{MR}}$ for variable density n was termed the ‘‘Moore-Read’’ line in Ref. [41]. Finally, the BCS-BEC quantum phase transition occurs at $\mu = 0$, $\Delta_0 = \Delta_{\text{QCP}}$. Here the spectrum exhibits a gapless Dirac node at $\varepsilon = 0$,

$$E(\varepsilon; \Delta_{\text{QCP}}) = \Delta_{\text{QCP}} \sqrt{\varepsilon + \varepsilon^2/4(\Delta_{\text{QCP}})^2}.$$

Eqs. (8.5) and (8.163) can be solved to obtain

$$\begin{aligned}\Delta_{\text{Coh}} &= \sqrt{\Upsilon(n; 1)}, \quad \Delta_{\text{MR}} = \sqrt{\Upsilon(n; 0)}, \\ \Delta_{\text{QCP}} &= \sqrt{\Upsilon(n; -1)},\end{aligned}\tag{8.164}$$

where

$$\begin{aligned}\Upsilon(n; x) &\equiv -\frac{4\pi n}{\mathcal{W}_{-1}\left[-\frac{2\pi n}{\Lambda} \exp(-x)\right]} \\ &\simeq \frac{4\pi n}{\ln\left(\frac{\Lambda}{2\pi n}\right) + x + \ln\left[\ln\left(\frac{\Lambda}{2\pi n}\right) + x\right]}.\end{aligned}$$

Here, $\mathcal{W}_{-1}(z)$ is the $k = -1$ branch of Lambert’s W function.

The tunneling density of states measured at a tip potential V is given by [Schrieffer(1989)]

$$\begin{aligned}\nu(V) &\equiv \int \frac{d^2\mathbf{k}}{(2\pi)^2} |u_{\mathbf{k}}|^2 \delta(E_k - V) \\ &= \frac{1}{8\pi V} \int d\varepsilon \left(\frac{\varepsilon}{2} - \mu + V\right) \delta[E(\varepsilon; \Delta_0) - V],\end{aligned}\tag{8.165}$$

where $u_{\mathbf{k}} = \frac{1}{\sqrt{2}}\sqrt{1 + (k^2/2 - \mu)/E_k}$ is a coherence factor. Performing the integration, one obtains

$$\begin{aligned} \nu(V) = & \frac{1}{2\pi} \left[\frac{V - (\Delta_0)^2}{\sqrt{V^2 - V_{\min}^2}} \right] \theta(V - V_{\min}) \theta(\mu - V) \\ & \times \theta(\Delta_{\text{Coh}} - \Delta_0) \\ & + \frac{1}{4\pi} \theta(V - |\mu|) \left[1 + \frac{V - (\Delta_0)^2}{\sqrt{V^2 - V_{\min}^2}} \right]. \end{aligned} \quad (8.166)$$

In this equation, $\theta(\varepsilon)$ denotes the unit step function. The first term in Eq. (8.166) is non-zero only for weak pairing strengths such that $\Delta_0 \leq \Delta_{\text{Coh}}$. In this range, the single particle excitation gap is [Eq. (8.53)]

$$V_{\min} = \Delta_0 \sqrt{2\mu - (\Delta_0)^2}, \quad (8.167)$$

and $\nu(V)$ exhibits a coherence peak above this energy, as shown in Fig. 8.14(b). For $\Delta_0 > \Delta_{\text{Coh}}$, only the second term in Eq. (8.166) contributes. The minimum of $E(\varepsilon)$ occurs at $\varepsilon = 0$, where the single particle excitation gap is $|\mu|$ [Eq. (8.54)]. This is non-zero on both sides of the BCS-BEC transition. On the BCS side ($\Delta_{\text{Coh}} < \Delta_0 < \Delta_{\text{QCP}}$), $\nu(V)$ in Eq. (8.166) vanishes continuously at $V = \mu$; on the BEC side ($\Delta_0 > \Delta_{\text{QCP}}$), there is a discontinuous jump, see Fig. 8.14(b). The difference is a coherence factor effect due to $|u_{\mathbf{k}}|^2$ in Eq. (8.165).

8.8 Classical dynamics in the chiral p-wave model

In this Appendix, we establish the equivalence of dynamics generated from a $p + ip$ initial state using the “real” p-wave Hamiltonian in Eq. (8.1) and the chiral one in Eq. (8.11). For the Hamiltonian in Eq. (8.1), the equations of motion are

$$\begin{aligned} \dot{s}_{\mathbf{k}}^z &= \frac{i}{2} [B_{\mathbf{k}} s_{\mathbf{k}}^+ - B_{\mathbf{k}}^* s_{\mathbf{k}}^-], \\ \dot{s}_{\mathbf{k}}^- &= -i [k^2 s_{\mathbf{k}}^- + B_{\mathbf{k}} s_{\mathbf{k}}^z], \end{aligned} \quad (8.168)$$

where

$$B_{\mathbf{k}} \equiv 4G \sum_{\mathbf{q}}' \mathbf{k} \cdot \mathbf{q} s_{\mathbf{q}}^-. \quad (8.169)$$

For a time-dependent $p + ip$ state, we can write

$$s_{\mathbf{k}}^- \equiv e^{-i\phi_k} s_k^-, \quad s_{\mathbf{k}}^z \equiv s_k^z. \quad (8.170)$$

Eq. (8.168) becomes

$$\begin{aligned} \dot{s}_k^z &= iG \sum_{\mathbf{q}}' kq [s_q^- s_k^+ - s_k^- s_q^+], \\ \dot{s}_k^- &= -i \left(k^2 s_k^- + 2G \sum_{\mathbf{q}}' kq s_q^- s_k^z \right), \end{aligned} \quad (8.171)$$

where we have used the fact that

$$\sum_{\mathbf{q}}' \exp(-2i\phi_q) \rightarrow \nu_0 \int d\varepsilon \int_0^\pi \frac{d\phi_q}{\pi} \exp(-2i\phi_q) = 0.$$

In this last equation, we convert to the continuum via Eq. (8.32). Defining $\vec{s}_i \equiv \vec{s}_{k_i}$, Eq. (8.171) takes the form

$$\begin{aligned}\dot{s}_i^z &= i\sqrt{\varepsilon_i} [\Delta^* s_i^- - \Delta s_i^+], \\ \dot{s}_i^- &= i \{ 2\sqrt{\varepsilon_i} s_i^z \Delta - \varepsilon_i s_i^- \}, \\ \Delta &\equiv -G \sum_j \sqrt{\varepsilon_j} s_j^-. \end{aligned} \quad (8.172)$$

These are identical to Eq. (8.14).

8.9 Phase III dynamics for negative real roots

In Sec. 8.4.2, we computed the asymptotic dynamics for $\Delta(t)$ through the bulk of phase **III**. Eqs. (8.120)–(8.124) give the evolution of the squared modulus $R(t) \equiv |\Delta(t)|^2$ everywhere in **III** to the left of the line marked β_{MR} in Fig. 8.15. All coefficients are determined by the two pairs of isolated roots $u_{1,\pm}$ and $u_{2,\pm}$, which come in complex conjugate pairs [Eqs. (8.90) and (8.91)]. For a given quench $\{\Delta_0^{(i)}, \beta\}$, these solve Eq. (8.66).

These results do not apply to a very narrow phase **III** sliver of width $\sim \sqrt{n}[\log(\Lambda/2\pi n)]^{-5/2} \ll \{\Delta_{\text{Coh}}, \Delta_{\text{MR}}, \Delta_{\text{QCP}}\}$ in Fig. 8.15. This is the region bounded on the left (right) by the β_{MR} ($\beta_c^{(-)}$) curve. Within this sliver, the roots $u_{2,\pm}$ are non-degenerate, negative, and real. Quenches in phases **II** and **III** between the lines marked β_{MR} and β_{QCP} in Fig. 8.15 are non-equilibrium versions of the BCS ground state with $\Delta_{\text{MR}} < \Delta_0 < \Delta_{\text{QCP}}$. The corresponding root configurations lie between those marked (3) and (4) in Fig. 8.14(a).

In this Appendix, we transcribe the order parameter dynamics for quenches in this sliver. Instead of Eq. (8.91), the second isolated pair is

$$u_{2,\{a,b\}} \equiv -v_{2,\{a,b\}} < 0.$$

Eqs. (8.118a) and (8.118b) are replaced by

$$\dot{R}^2 = (R_+ - R)(R - R_-) [(R - R_{\text{r}})^2 + R_{\text{i}}^2], \quad (8.173a)$$

$$\begin{aligned} R_{\pm} &\equiv \frac{1}{4} \left[\sqrt{v_{2,b}} + \sqrt{v_{2,a}} \pm \sqrt{2(|u_1| - u_{1,\text{r}})} \right]^2, \\ R_{\text{r}} + iR_{\text{i}} &\equiv \frac{1}{4} \left[\sqrt{v_{2,b}} - \sqrt{v_{2,a}} + i\sqrt{2(|u_1| + u_{1,\text{r}})} \right]^2. \end{aligned} \quad (8.173b)$$

The solution is

$$R(t) \equiv R_0 + R_d z(t), \quad (8.174)$$

$$z(t) = \frac{(1 + z_0^2) \left\{ z_{c,\text{r}} [1 + cn^2(\theta t|N)] + (1 + |z_c|^2) cn(\theta t|N) \right\}}{\left\{ (1 + z_0 z_{c,\text{i}})^2 + 2z_{c,\text{r}} (1 + z_0^2) cn(\theta t|N) \right\} + z_0^2 z_{c,\text{r}}^2 + [(z_0 - z_{c,\text{i}})^2 + z_{c,\text{r}}^2] cn^2(\theta t|N)}, \quad (8.175)$$

where $z_c \equiv z_{c,r} + iz_{c,i}$. In terms of the roots,

$$\begin{aligned}
R_0 &= \frac{1}{4} \left[2(|u_1| - u_{1,r}) + (\sqrt{v_{2,a}} + \sqrt{v_{2,b}})^2 \right], \\
R_d &= \frac{1}{\sqrt{2}} \sqrt{|u_1| - u_{1,r}} (\sqrt{v_{2,a}} + \sqrt{v_{2,b}}), \\
z_{c,r} &= -\frac{\sqrt{2}(|u_1| + \sqrt{v_{2,a}}\sqrt{v_{2,b}})}{\sqrt{|u_1| - u_{1,r}}(\sqrt{v_{2,a}} + \sqrt{v_{2,b}})}, \\
z_{c,i} &= \frac{\sqrt{|u_1| + u_{1,r}}(\sqrt{v_{2,b}} - \sqrt{v_{2,a}})}{\sqrt{|u_1| - u_{1,r}}(\sqrt{v_{2,b}} + \sqrt{v_{2,a}})}, \\
z_0 &= \frac{\left\{ \begin{aligned} &u_{1,i}^2 + (u_{1,r} + v_{2,a})(u_{1,r} + v_{2,b}) \\ &+ \sqrt{[u_{1,i}^2 + (u_{1,r} + v_{2,a})^2]} \\ &\times [u_{1,i}^2 + (u_{1,r} + v_{2,b})^2] \end{aligned} \right\}}{u_{1,i}(v_{2,b} - v_{2,a})}, \\
N &= \frac{1}{1 + z_0^2}, \\
\theta &= \left\{ \begin{aligned} &[u_{1,i}^2 + (u_{1,r} + v_{2,a})^2] \\ &\times [u_{1,i}^2 + (u_{1,r} + v_{2,b})^2] \end{aligned} \right\}^{1/4}.
\end{aligned} \tag{8.176}$$

The physical period T of $R(t)$ is

$$T = \frac{4K(N)}{\theta}, \tag{8.177}$$

where $K(N)$ is the complete elliptic integral of the first kind (and $N = k^2$).

8.10 Green's functions

In this Appendix, we compute single particle Green's functions in the long time limit for quenches in phase **II** of the diagram in Fig. 8.4. Throughout this Appendix, we work in the rotating frame employed in Sec. 8.5 such that the order parameter itself (and not only its modulus) asymptotes to a constant.

8.10.1 Post-quench coherence factors

In phase **II**, the asymptotic spin configuration is given by Eq. (8.25), where $\gamma_i = \gamma(\varepsilon_i)$ in Eq. (8.138a). Following the discussion surrounding Eq. (8.31), the spins in the 2D $\mathbf{k}$ -plane evolving from an initial $p + ip$ state are reconstructed as follows:

$$\begin{aligned}
2s_{\mathbf{k}}^-(t) &= \left\{ \begin{aligned} &\sqrt{1 - \gamma_k^2} \left[\left(\frac{\xi_k}{E_k} \right) \cos(2E_k t) - i \sin(2E_k t) \right] \\ &+ \gamma_k \left(\frac{k\Delta_\infty}{E_k} \right) \end{aligned} \right\} \\
&\times \exp(-i\phi_k), \\
2s_{\mathbf{k}}^z(t) &= -\sqrt{1 - \gamma_k^2} \left(\frac{k\Delta_\infty}{E_k} \right) \cos(2E_k t) + \gamma_k \left(\frac{\xi_k}{E_k} \right), \\
\xi_k &\equiv \left(\frac{k^2}{2} - \mu_\infty \right), \quad E_k \equiv \sqrt{\xi_k^2 + k^2\Delta_\infty^2},
\end{aligned} \tag{8.178}$$

where ϕ_k is the polar angle in momentum space, and Δ_∞ and μ_∞ refer to the post-quench steady-state (*not* ground state) values. Comparing to Eq. (8.25), we have set the phase shifts $\Theta_i = 0$, see Sec. 8.5.3. The pre-quench parameters $\Delta_0^{(i)}$ and $\mu_0^{(i)}$ enter through the distribution function

$$\gamma_k \equiv \gamma(\varepsilon = k^2), \quad (8.179)$$

the latter evaluated in Eq. (8.138a).

In the thermodynamic limit, the many-body wavefunction assumes a BCS product form with time-dependent coherence factors,

$$|\Psi(t)\rangle = \prod_{\mathbf{k}} [u_{\mathbf{k}}(t) + v_{\mathbf{k}}(t) s_{\mathbf{k}}^+] |0\rangle, \quad (8.180)$$

where $|0\rangle$ is the vacuum (all pseudospins down). In this state, the expectations of Anderson pseudospin Schrödinger picture operators are given by

$$\begin{aligned} \langle \Psi(t) | s_{\mathbf{k}}^+ | \Psi(t) \rangle &= v_{\mathbf{k}}^* u_{\mathbf{k}}(t), \\ \langle \Psi(t) | s_{\mathbf{k}}^- | \Psi(t) \rangle &= u_{\mathbf{k}}^* v_{\mathbf{k}}(t), \\ \langle \Psi(t) | s_{\mathbf{k}}^z | \Psi(t) \rangle &= \frac{1}{2} (|v_{\mathbf{k}}|^2 - |u_{\mathbf{k}}|^2)(t). \end{aligned} \quad (8.181)$$

The coherence factors solve the same Bogoliubov-de Gennes equation as the retarded Green's function $\mathcal{G}_{\mathbf{k}}(t, t')$ [Eq. (8.21)]. In the large time limit,

$$i \frac{d}{dt} \begin{bmatrix} u_{\mathbf{k}}(t) \\ v_{\mathbf{k}}(t) \end{bmatrix} = \begin{bmatrix} -\xi_k & k^* \Delta_\infty \\ k \Delta_\infty & \xi_k \end{bmatrix} \begin{bmatrix} u_{\mathbf{k}}(t) \\ v_{\mathbf{k}}(t) \end{bmatrix}, \quad (8.182)$$

where $\mathbf{k} \equiv k^x - ik^y$. Different from $\mathcal{G}_{\mathbf{k}}(t, t')$, the coherence factors “remember” details of the pre-quench state through the initial condition at $t = 0$. The general solution to Eq. (8.182) is

$$\begin{aligned} u_{\mathbf{k}}(t) &= (E_k - \xi_k) A_k e^{-iE_k t} + (E_k + \xi_k) B_k e^{iE_k t}, \\ v_{\mathbf{k}}(t) &= [A_k e^{-iE_k t} - B_k e^{iE_k t}] \Delta_\infty k e^{-i\phi_k}, \end{aligned} \quad (8.183)$$

where the undetermined complex constants A_k and B_k satisfy

$$\begin{aligned} 1 &= [(E_k - \xi_k)^2 + k^2 \Delta_\infty^2] |A_k|^2 \\ &\quad + [(E_k + \xi_k)^2 + k^2 \Delta_\infty^2] |B_k|^2. \end{aligned} \quad (8.184)$$

For the p-wave problem, one coherence factor must have odd parity; we have chosen $v_{\mathbf{k}}$ in Eq. (8.183).

Using Eq. (8.183), one can compute the expectation values in Eq. (8.181). Comparing the results to that of the Lax calculation in Eq. (8.178), we determine that

$$\begin{aligned} u_{\mathbf{k}}(t) &= \frac{1}{2} \sqrt{\frac{(1 + \gamma_k)(E_k - \xi_k)}{E_k}} e^{-iE_k t + i\Gamma_k} \\ &\quad + \frac{1}{2} \sqrt{\frac{(1 - \gamma_k)(E_k + \xi_k)}{E_k}} e^{iE_k t + i\Gamma_k}, \\ v_{\mathbf{k}}(t) &= \frac{1}{2} \sqrt{\frac{(1 + \gamma_k)(E_k + \xi_k)}{E_k}} e^{-iE_k t - i\phi_k + i\Gamma_k} \\ &\quad - \frac{1}{2} \sqrt{\frac{(1 - \gamma_k)(E_k - \xi_k)}{E_k}} e^{iE_k t - i\phi_k + i\Gamma_k}, \end{aligned} \quad (8.185)$$

where Γ_k is an undetermined time-independent phase.

The ground state (zero quench) has $\gamma_k = -1$ for all k , leading to $u_{\mathbf{k}}(t) = \sqrt{\frac{1}{2}(1 + \xi_k/E_k)} \exp(iE_k t + i\Gamma_k)$ and $v_{\mathbf{k}}(t) = -\sqrt{\frac{1}{2}(1 - \xi_k/E_k)} \exp(iE_k t - i\phi_k + i\Gamma_k)$, as expected.

8.10.2 One particle Green's functions and structure factors

The dynamic single particle Green's functions can be computed from the coherence factors. For example,

$$\begin{aligned} \mathcal{G}_{\mathbf{k},>}(t, t') &\equiv -i \langle \Psi(t) | c_{\mathbf{k}} e^{-iH(t-t')} c_{\mathbf{k}}^\dagger | \Psi(t') \rangle \\ &= -i u_{\mathbf{k}}^*(t) u_{\mathbf{k}}(t') \\ &\quad \times \langle 0 | c_{\mathbf{k}} \prod_{\mathbf{q} \neq \mathbf{k}}' [u_{\mathbf{q}}^*(t) + v_{\mathbf{q}}^*(t) c_{-\mathbf{q}} c_{\mathbf{q}}] e^{-iH_{\mathbf{k}} t} \\ &\quad \times e^{iH_{\mathbf{k}} t'} \prod_{\mathbf{q}' \neq \mathbf{k}}' [u_{\mathbf{q}'}(t') + v_{\mathbf{q}'}(t') c_{\mathbf{q}'}^\dagger c_{-\mathbf{q}'}^\dagger] c_{\mathbf{k}}^\dagger | 0 \rangle, \end{aligned} \quad (8.186)$$

where $H_{\mathbf{k}}$ is the interacting (and time-independent) BCS Hamiltonian in Eq. (8.11) *excluding the mode* $\{\mathbf{k}, -\mathbf{k}\}$:

$$H_{\mathbf{k}} = \sum_{\mathbf{q} \neq \mathbf{k}}' q^2 s_{\mathbf{q}}^z - G \sum_{\mathbf{q}_1, \mathbf{q}_2 \neq \mathbf{k}}' \mathbf{q}_1 \mathbf{q}_2^* s_{\mathbf{q}_1}^+ s_{\mathbf{q}_2}^-. \quad (8.187)$$

Above we have used the fact that $e^{ik^2 s_{\mathbf{k}}^z} c_{\mathbf{k}}^\dagger | 0 \rangle = c_{\mathbf{k}}^\dagger | 0 \rangle$. Eq. (8.186) becomes

$$\begin{aligned} \mathcal{G}_{\mathbf{k},>}(t, t') &= -i u_{\mathbf{k}}^*(t) u_{\mathbf{k}}(t') \\ &\quad \times \langle 0 | c_{\mathbf{k}} \prod_{\mathbf{q} \neq \mathbf{k}}' [u_{\mathbf{q}}^*(0) + v_{\mathbf{q}}^*(0) c_{-\mathbf{q}} c_{\mathbf{q}}] \\ &\quad \times \prod_{\mathbf{q}' \neq \mathbf{k}}' [u_{\mathbf{q}'}(0) + v_{\mathbf{q}'}(0) c_{\mathbf{q}'}^\dagger c_{-\mathbf{q}'}^\dagger] c_{\mathbf{k}}^\dagger | 0 \rangle \\ &= -i u_{\mathbf{k}}^*(t) u_{\mathbf{k}}(t'). \end{aligned} \quad (8.188)$$

In these manipulations, we have used the fact that in mean field theory, the many-body BCS state can always be expressed as product over modes of either a coherent admixture of empty and doubly-occupied levels, or alternatively a singly-occupied (“blocked”) level.

We thereby obtain the following Green's functions

$$\begin{aligned} i\mathcal{G}_{\mathbf{k},>}(t, t') &= \langle \Psi_i | c_{\mathbf{k}}(t) c_{\mathbf{k}}^\dagger(t') | \Psi_i \rangle \\ &= \left(\frac{E_k - \xi_k \gamma_k}{2E_k} \right) \cos[E_k(t - t')] \\ &\quad + i \left(\frac{E_k \gamma_k - \xi_k}{2E_k} \right) \sin[E_k(t - t')] \\ &\quad + \frac{k\Delta_\infty}{2E_k} \sqrt{1 - \gamma_k^2} \cos[E_k(t + t')], \end{aligned} \quad (8.189a)$$

$$\begin{aligned}
-i\mathcal{G}_{\mathbf{k},<}(t,t') &= \langle \Psi_i | c_{\mathbf{k}}^\dagger(t') c_{\mathbf{k}}(t) | \Psi_i \rangle \\
&= v_{\mathbf{k}}^*(t') v_{\mathbf{k}}(t) \\
&= \left(\frac{E_k + \xi_k \gamma_k}{2E_k} \right) \cos [E_k(t - t')] \\
&\quad - i \left(\frac{E_k \gamma_k + \xi_k}{2E_k} \right) \sin [E_k(t - t')] \\
&\quad - \frac{k\Delta_\infty}{2E_k} \sqrt{1 - \gamma_k^2} \cos [E_k(t + t')],
\end{aligned} \tag{8.189b}$$

$$\begin{aligned}
\mathcal{G}_{\mathbf{k}}^+(t,t') &= \langle \Psi_i | c_{\mathbf{k}}^\dagger(t) c_{-\mathbf{k}}^\dagger(t') | \Psi_i \rangle \\
&= v_{\mathbf{k}}^*(t) u_{\mathbf{k}}(t') \\
&= \left\{ \begin{aligned} &\frac{\xi_k}{2E_k} \sqrt{1 - \gamma_k^2} \cos [E_k(t + t')] \\ &+ \gamma_k \frac{k\Delta_\infty}{2E_k} \cos [E_k(t - t')] \\ &+ \frac{i}{2} \sqrt{1 - \gamma_k^2} \sin [E_k(t + t')] \\ &+ i \frac{k\Delta_\infty}{2E_k} \sin [E_k(t - t')] \end{aligned} \right\} e^{i\phi_k},
\end{aligned} \tag{8.189c}$$

$$\begin{aligned}
\mathcal{G}_{\mathbf{k}}^-(t,t') &= \langle \Psi_i | c_{-\mathbf{k}}(t) c_{\mathbf{k}}(t') | \Psi_i \rangle \\
&= u_{\mathbf{k}}^*(t) v_{\mathbf{k}}(t') = [\mathcal{G}_{\mathbf{k}}^+(t',t)]^*.
\end{aligned} \tag{8.189d}$$

In these equations, $|\Psi_i\rangle$ denotes the initial pre-quench BCS, BEC, or quantum critical state.

Using these results, the retarded Green's function in Eq. (8.9) evaluates to

$$\mathcal{G}_{\mathbf{k}}(t,t') = -i \begin{bmatrix} \mathcal{G}_{\mathbf{k}}^{(1)}(t-t') & \mathcal{G}_{\mathbf{k}}^{(2)}(t-t') \\ -[\mathcal{G}_{\mathbf{k}}^{(2)}(t-t')]^* & [\mathcal{G}_{\mathbf{k}}^{(1)}(t-t')]^* \end{bmatrix} \theta(t-t'), \tag{8.190a}$$

where

$$\begin{aligned}
\mathcal{G}_{\mathbf{k}}^{(1)}(t) &\equiv \cos(E_k t) + i \left(\frac{\xi_k}{E_k} \right) \sin(E_k t), \\
\mathcal{G}_{\mathbf{k}}^{(2)}(t) &\equiv -i \left[\frac{(k^x + ik^y)\Delta_\infty}{E_k} \right] \sin(E_k t).
\end{aligned} \tag{8.190b}$$

The retarded function in Eq. (8.190) is a function only of the time difference $(t - t')$, and is independent of γ_k . It satisfies Eq. (8.21) with the initial condition in Eq. (8.22). By contrast, the other Green's functions in Eq. (8.189) depend upon both the relative and average $(t + t')$ times, and upon the non-thermal Cooper pair distribution function γ_k .

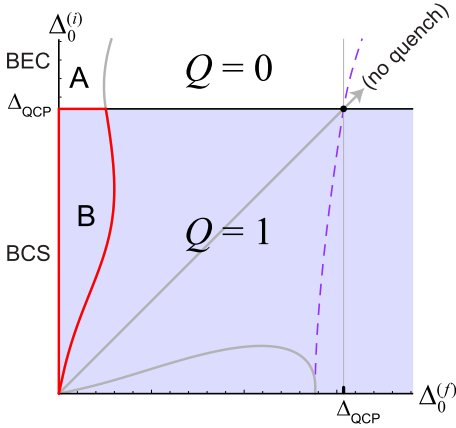


Figure 8.8: Quench phase diagram: Pseudospin winding number Q . We find that Q is unchanged from its initial value, such that $Q = 1$ ($Q = 0$) for an initial BCS $\Delta_0^{(i)} < \Delta_{\text{QCP}}$ (BEC $\Delta_0^{(i)} > \Delta_{\text{QCP}}$) state. The highlighted region **B** is a “topological gapless” phase. Throughout phase **I** of Fig. 8.4, $\Delta(t)$ vanishes as $t \rightarrow \infty$. In subregion **B**, the pseudospin texture nevertheless retains a non-zero winding $Q = 1$, Fig. 8.12 and Eq. (8.18). The state can be visualized as a skyrmion texture as in Fig. 8.2(a), but now the tilted pseudospins *precess* at different frequencies about $\hat{z}$ in spin space—see Eq. (8.18). The notion of smooth topology for the evolving texture remains well-defined for times up to the inverse level spacing.

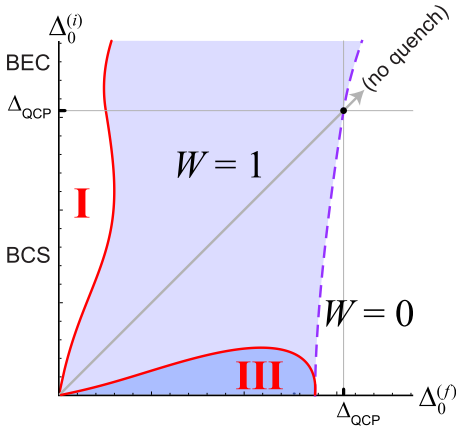


Figure 8.9: Quench phase diagram: Retarded Green's function winding number W and Majorana edge modes. In phase **II** of Fig. 8.4, $\{\Delta(t), \mu(t)\} \rightarrow \{\Delta_\infty, \mu_\infty\}$ as $t \rightarrow \infty$, with Δ_∞ a non-zero constant. Along the dashed purple line, $\mu_\infty = 0$; this is a non-equilibrium extension of the ground state quantum phase transition; see also Fig. 8.6. To the left (right) of this line, $\mu_\infty > 0$ ($\mu_\infty < 0$), leading to $W = 1$ ($W = 0$). W is ill-defined in the gapless phase **I**. Phase **III**, wherein $\Delta(t)$ exhibits persistent amplitude and phase oscillations [Figs. 8.5(b) and 8.7] is topologically non-trivial.[46] Both phase **III** and the $W = 1$ region of phase **II** support gapless Majorana edge modes. The former is confirmed by the Floquet analysis in Ref. [46], while the latter is established here via Eq. (8.21) and the surrounding discussion.

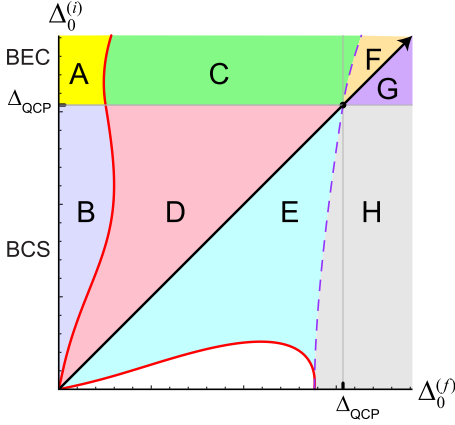


Figure 8.10: Sectioned phase diagram. Regions **C** and **H** include strong-to-weak and weak-to-strong quenches across the quantum critical point at $\Delta_0 = \Delta_{\text{QCP}}$.

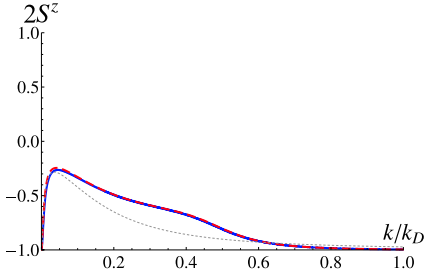


Figure 8.11: Asymptotic (infinite time) pseudospin distribution function for a representative quench in the gapless region **A** of Fig. 8.8. The solid blue curve is the result of numerical simulation for 5024 classical pseudospins; the red dashed curve is the analytical solution obtained from the Lax construction. The dotted gray line gives the initial ground state distribution. Although the initial and asymptotic distributions share the same winding $Q = 0$, in the latter case $\Delta(t)$ has decayed to zero, due to the dephasing of precessing pseudospins [Eqs. (8.18) and (8.19)]. By contrast, in the ground state each pseudospin is aligned along its magnetic field. The quench coordinates are $\{\Delta_0^{(i)}, \Delta_0^{(f)}\} = \{1.56, 0.00211\}$, and $k_D \equiv 4\sqrt{4\pi n} = 12.9$.

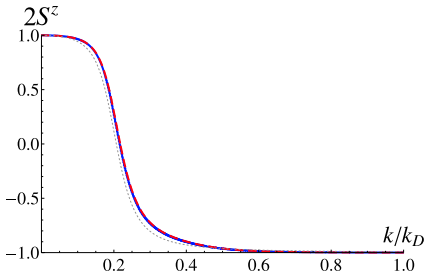


Figure 8.12: The same as Fig. 8.11, but for a point in the gapless region **B** of Fig 8.8. In this case $Q = 1$, and the decay of $\Delta(t)$ to zero implies that the fluctuating region **B** is a “topological gapless” phase. The quench coordinates are $\{\Delta_0^{(i)}, \Delta_0^{(f)}\} = \{0.750, 0.00224\}$.

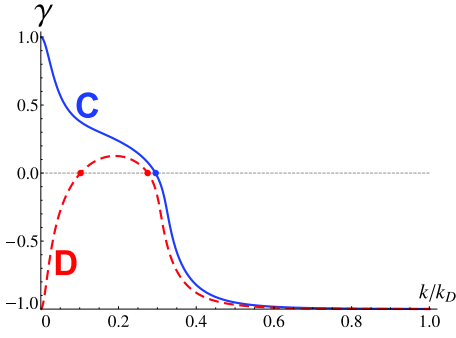


Figure 8.13: Non-equilibrium winding number: Parity of zeroes in the Cooper pair distribution function of phase **II**, for representative quenches in regions **C** and **D** [Fig. 8.10]. The Cooper pair distribution function is the pseudospin projection $\gamma(k)$ at each momentum k onto the effective field $-\vec{B}(\varepsilon = k^2)$. The latter encodes the asymptotic global parameters $\{\Delta_\infty, \mu_\infty\}$, see Eqs. (8.24) and (8.25). Distributions **C** (blue solid line) and **D** (red dashed line) belong to quenches in the corresponding regions highlighted in Fig. 8.10. Whenever $Q \neq W$ ($Q = W$), $\gamma(k)$ exhibits an odd (even) number of zeroes, illustrated here by curve **C** (**D**); c.f. Figs. 8.8 and 8.9. The parity of these zeroes is a $\mathbb{Z}_2$ -valued quantum number that can in principle be extracted from an RF spectroscopic measurement in a cold atomic realization of the quench. The quench coordinates for curves C and D are $\{\Delta_0^{(i)}, \Delta_0^{(f)}\} = \{1.65, 0.359\}$ and $\{1.47, 0.365\}$, respectively. Both curves were obtained from the analytical solution.

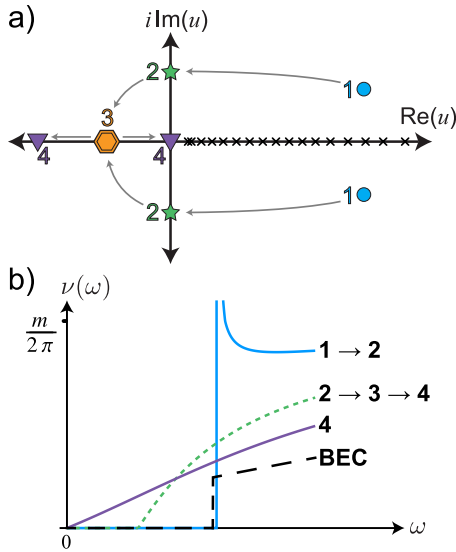


Figure 8.14: Isolated roots and ground state spectral transitions. Subfigure (a) shows the positions of the isolated root pair $u_0^{(\pm)}$ [Eq. (8.52)] in the ground state spectral polynomial, for increasing values of Δ_0 . Particular root configurations are labeled (1)–(4). For $\Delta_0 < \Delta_{\text{Coh}}$ (1), the isolated roots are a conjugate pair with positive real part. At $\Delta_0 = \Delta_{\text{Coh}}$ (2), the roots become purely imaginary. At $\Delta_0 = \Delta_{\text{MR}}$ (“Moore-Read,” see text) (3), the roots become negative real and degenerate. For $\Delta_{\text{MR}} < \Delta_0 < \Delta_{\text{QCP}}$, the roots split and travel along the negative real axis. At $\Delta_0 = \Delta_{\text{QCP}}$ (4), the retreating root hits zero. The thresholds $\{\Delta_{\text{Coh}}, \Delta_{\text{MR}}\}$ lie within the BCS phase ($\mu > 0$), while Δ_{QCP} marks the topological transition. All are defined explicitly in Appendix 8.7.2, Eqs. (8.163) and (8.164). For $\Delta_0 > \Delta_{\text{QCP}}$ (BEC), both roots are again non-degenerate and negative real. The zero temperature tunneling density of states $\nu(\omega)$ is shown for the corresponding root positions in subfigure (b). The weak pairing coherence peak visible in the trace (1 $\rightarrow$ 2) disappears for $\Delta_0 \geq \Delta_{\text{Coh}}$. The difference between the “soft” and “hard” gaps in the BCS (2 $\rightarrow$ 3 $\rightarrow$ 4) and BEC regimes is a coherence factor effect. The spectrum is gapless at $\Delta_0 = \Delta_{\text{QCP}}$ (4).

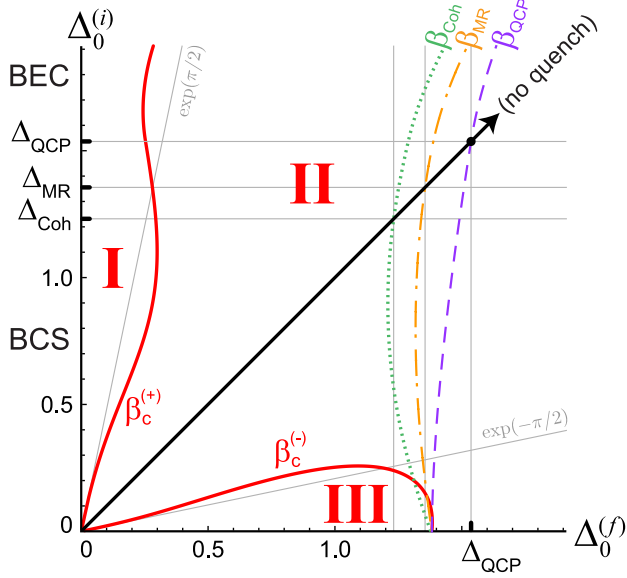


Figure 8.15: Detailed quench phase diagram. The dynamical phase boundaries $\beta_c^{(\pm)}$ were obtained from a numerical solution to Eq. (8.70), using Eq. (8.69). The pairing amplitude values Δ_{Coh} , Δ_{MR} , and Δ_{QCP} mark spectral transitions in the equilibrium ground state, as described in Sec. 8.3.2 and illustrated in Fig. 8.14. Explicit values appear in Eqs. (8.163) and (8.164). The lines marked β_{Coh} , β_{MR} , and β_{QCP} are the non-equilibrium extensions of these ground state spectral transitions, as discussed in Sec. 8.3.6.

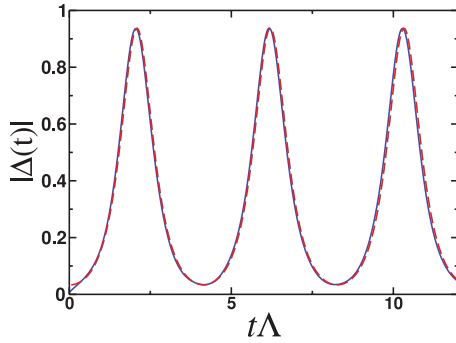


Figure 8.16: Persistent order parameter oscillations following a quench. The same as Fig. 8.7, but for quench coordinates $\{\Delta_0^{(i)}, \Delta_0^{(f)}\} = \{0.00503, 0.961\}$.

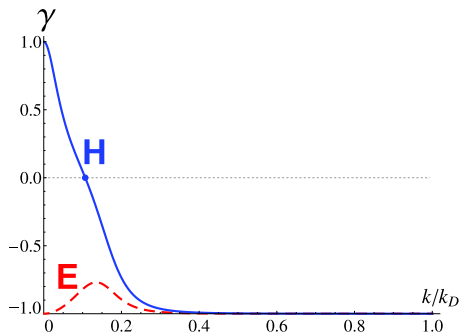


Figure 8.17: The Cooper pair distribution function $\gamma(k)$ as in Fig. 8.13, but plotted for representative quenches in regions **E** (red dashed line) and **H** (blue solid line) of phase **II**. The quench coordinates for **E** are $\{\Delta_0^{(i)}, \Delta_0^{(f)}\} = \{0.972, 1.25\}$, while the quench coordinates for **H** are $\{\Delta_0^{(i)}, \Delta_0^{(f)}\} = \{0.972, 1.55\}$. Regions **E** and **H** are specified in Fig. 8.10.

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- $$\Delta(t) = \sqrt{R(t)} \exp[-i\phi(t) + 2i\bar{\mu}_\infty t], \quad 0 \leq t \leq T.$$
- In this equation, $R(t)$ has the explicit form in Eqs. (8.120)–(8.123), the period T is given by Eq. (8.124), and the phase $\phi(t)$ is obtained by numerically integrating Eq. (8.127). Finally, the parameter $2\bar{\mu}_\infty \equiv [\phi(T) - \phi(0)]/T$; the orbits in Fig. 8.5(b) are thus plotted in the frame rotating with frequency $2\bar{\mu}_\infty$.
- [60] In a BCS ground state (which includes the Fermi liquid as a special case), the Anderson pseudospins are aligned to the magnetic field that incorporates the chemical potential. The ground state configuration has
- $$2s_i^a = \frac{b_i^a}{|\vec{b}_i|}, \quad \vec{b}_i = \vec{B}_i + 2\mu\hat{z},$$
- with μ determined by Eq. (8.5). The equation of motion in Eq. (8.14) instead involves $\vec{B}_i$. Due to the mismatch, all spins precess in the ground state around $\hat{z}$ with frequency 2μ ; similarly, $\Delta(t) = \Delta_0 \exp(-2i\mu t)$. One can eliminate this spurious evolution (and align the spins to the field) by boosting to the rotating frame $s_i^-(t) \rightarrow \exp(2i\mu t) s_i^-(t)$.
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9 Dynamical vanishing of the order parameter in a confined Bardeen-Cooper-Schrieffer Fermi gas after an interaction quench by Hannibal et al.

Abstract

We present a numerical study of the Higgs mode in an ultracold confined Fermi gas after an interaction quench and find a dynamical vanishing of the superfluid order parameter. Our calculations are done within a microscopic density-matrix approach in the Bogoliubov-de Gennes framework which takes the three-dimensional cigar-shaped confinement explicitly into account. In this framework, we study the amplitude mode of the order parameter after interaction quenches starting on the BCS side of the BEC-BCS crossover close to the transition and ending in the BCS regime. We demonstrate the emergence of a dynamically vanishing superfluid order parameter in the spatiotemporal dynamics in a three-dimensional trap. Further, we show that the signal averaged over the whole trap mirrors the spatiotemporal behavior and allows us to systematically study the effects of the system size and aspect ratio on the observed dynamics. Our analysis enables us to connect the confinement-induced modifications of the dynamics to the pairing properties of the system. Finally, we demonstrate that the signature of the Higgs mode is contained in the dynamical signal of the condensate fraction, which, therefore, might provide a new experimental access to the nonadiabatic regime of the Higgs mode.

9.1 Conclusion

In this paper, we have presented a numerical study of the dynamical vanishing of the order parameter in a cigar-shaped ultracold Fermi gas in the framework of the fully microscopic BdG equations. We have calculated the spatiotemporal dynamics of the order parameter after an interaction quench. We have observed a rapid initial decay as well as a revival which both affect the whole trap. Thus, both can be characterized by the spatially averaged gap. Whereas, after the initial decay we find an oscillation of the order parameter in the longitudinal direction of the harmonic trap which is not seen in the average value due to the continuity equation.

Exploiting the spatially averaged order parameter, we have demonstrated that the dynamical vanishing in inhomogeneous systems is robust with respect to the size and aspect ratio of the system while modifications to this vanishing arise from the finite size due to the trap potential. The occurrence of a vanishing order parameter predicted in the homogeneous case is maintained but the precise onset of the vanishing is altered by the pairing properties in the trap. We find that the pairing strength, characterized by the gap at the chemical potential or equivalently by the smallest BdG eigenenergies, increases with the aspect ratio as $r^{2/3}/\tau_{\text{trap}}$ and with the particle number as $N^{1/3}$.

We have found that the transition to a vanishing order parameter in a cigar-shaped ultracold Fermi gas is determined by two criteria which both need to be fulfilled and which depend contrarily on the aspect ratio of the trap. For small aspect ratios the position of the transition is determined by the initial decay rate which needs to be sufficiently fast compared to the characteristic time scale of the trap. In this case we observe that the quench strength necessary for the transition decreases with increasing aspect ratio. However, for smaller quench strengths and larger aspect ratios the signature of the linear Higgs mode from "phase II" becomes more prominent which prevents a vanishing to be visible. Then, the transition is determined by the linear Higgs mode which leads to an increasing transition quench strength with increasing aspect ratio.

Furthermore, we have shown that the condensate fraction on the BCS side of the BCS-BEC crossover can be a suitable measure to access the Higgs mode in such a cigar-shaped ultracold

Fermi gas. Nevertheless, we also find that the condensate fraction will reduce the amplitude of the signal compared to the order parameter and does not mirror the Higgs mode exactly. We expect that this does not introduce new effects but will rather only lead to a shift of the observed transition.

9.2 Introduction

Due to a remarkable control over many relevant system parameters which, in most cases, can be tuned at will, ultracold quantum gases are an ideal testbed for many-body theories and for concepts known from solid-state theory. These include, e.g., lattice symmetries [Bloch *et al.*(2008)], topological properties [Goldman *et al.*(2016)], spin-orbit coupling [Galitski and Spielman(2013), Wu *et al.*(2016)], and non-homogeneous superconductivity [Liao *et al.*(2010)]. For these reasons ultracold Fermi gases have received great attention both from an experimental and from a theoretical point of view [Bloch *et al.*(2008), Giorgini *et al.*(2008)].

Using a Feshbach resonance [Chin *et al.*(2010)] to adjust the interaction strength a Bardeen-Cooper-Schrieffer (BCS) superfluid state of Cooper pairs has been achieved for small attractive interactions, whereas in the limit of large interactions between the fermions repulsive bosonic dimers form and a Bose Einstein condensate (BEC) of the latter has been observed [Regal *et al.*(2004)]. Those two limiting superfluid states are connected by the BCS-BEC crossover featuring strong interactions and unitary properties [Zwerger(2011)].

The implementation of ultracold Fermi gases always requires an external confinement, e.g., optical traps. The control of this external confinement on the one hand provides means for the outstanding control over the system [Bloch *et al.*(2008), Grimm *et al.*(2000)]. On the other hand, this implies a finite system size which can have a major impact on the physics of a superfluid system. The pairing properties strongly depend on the structure of the energy spectrum of the system and, hence, on the dimensionality of the system. Theoretical studies predict an atypical BCS-BEC crossover due to confinement effects [Shanenko *et al.*(2012)].

Recently, efforts have been devoted to the non-equilibrium properties of the superfluid state [Polkovnikov *et al.*(2011), Yin and Radzihovsky(2013), Yin and Radzihovsky(2016)]. Due to the spontaneously broken U(1) symmetry two fundamental modes of the complex order parameter evolve: the (massive Higgs) amplitude mode and the (massless Goldstone) phase mode. The observation of the collective Higgs excitation in a system with spontaneous symmetry breaking (SSB) is of fundamental importance to gain a better understanding of the physical system at hand, as was recently demonstrated in the case of the standard model of particle physics [Collaboration(2012)].

The first evidence of the Higgs mode was reported by Raman scattering in a superconducting charge density wave compound [Sooryakumar and Klein(1980), Littlewood and Varma(1981)]. Furthermore, in ultracold gases the Higgs mode has been observed in the spectral response of a 2D optical lattice excited by an amplitude modulation of the lattice [Endres *et al.*(2012)]. A time-resolved observation of the nonadiabatic regime of the Higgs mode [Papenkort *et al.*(2007), Papenkort *et al.*(2008)], as has been achieved in a superconducting NbN film [Matsunaga *et al.*(2014), Matsunaga *et al.*(2013), Matsunaga and Shimano(2012)], is still pending for ultracold gases. In order to reach the nonadiabatic regime the challenging implementation of an interaction quench has been proposed to be done either by an RF flip of one species of the condensate (cf. [Fröhlich *et al.*(2011)]), which leads, in the vicinity of a Feshbach resonance, to the desired almost instantaneous change of the scattering length. Alternatively the quench could be implemented by an optical control of the interaction in quantum gases [Clark *et al.*(2015)]. However, there are several theoretical studies addressing an ultracold Fermi gas after an interaction quench [Yuzbashyan *et al.*(2015), Yuzbashyan *et al.*(2006), Yuzbashyan and Dzero(2006), Yuzbashyan *et al.*(2005), Barankov and Levitov(2006), Bruun *et al.*(1999), Bruun(2014), Scott

et al.(2012), Hannibal *et al.*(2015)]. In the case of a homogeneous system three dynamical phases depending on the excitation condition have been predicted analytically [Yuzbashyan *et al.*(2015)]. They include persistent oscillations of the BCS gap (“phase III”), damped oscillations with inverse square root decay (“phase II”) and a dynamical vanishing of the order parameter (“phase I”). Previous studies in confined systems focused on the damped oscillations and showed that for superconducting BCS nanowires the decay exponent is changed from $-1/2$ to $-3/4$ due to quantum size resonances [Zachmann *et al.*(2013)], while for even tighter confined superconducting nanorods the decaying oscillation becomes irregular [Kettmann *et al.*(2017)]. In a BCS Fermi gas confined in a box with periodic boundary conditions in two dimensions and a harmonic confinement in the third dimension the inverse square-root decay has been confirmed [Scott *et al.*(2012)] and for a three-dimensional harmonic confinement an additional fragmentation of the damped oscillations has been predicted [Hannibal *et al.*(2015)].

In this paper we show that the dynamical vanishing of the superfluid order parameter, i.e., phase I, also emerges in a Fermi gas confined in a three dimensional harmonic trapping potential after an interaction quench. The emergence of phase I has not only been shown in ultracold Fermi gases but also in BCS superconductors [Papenkort *et al.*(2009), Chou *et al.*(2017)]. Here, we analyze the transition to a dynamically vanishing order parameter in dependence of the confinement properties. To this end, we present a numerical study based on the previously published microscopic density matrix approach within the Bogoliubov-de Gennes framework [Hannibal *et al.*(2015)]. This allows for a full microscopic and coherent quantum mechanical treatment of the system and provides access to the spatiotemporal dynamics of the order parameter as well as to the condensate fraction which is a possible experimental candidate to carry the signature of the Higgs mode.

This paper is organized as follows. In Sec. 19.3 we will give a short summary of the used theoretical model, while Sec. 9.4 is devoted to the results of our numerical calculations. We start by discussing the spatiotemporal dynamics of the order parameter in Sec. 9.4.1. In the next step, we will give a detailed analysis of the dynamical vanishing in a cigar-shaped trap in Secs. 9.4.2 and 9.4.3 based on the spatially averaged order parameter. In Secs. 9.4.4 and 9.4.5 we will provide extrapolations to typical experimental particle numbers. Finally, in Sec. 9.4.6 we present the dynamics of the condensate fraction after an interaction quench. In Sec. 9.1 we will briefly summarize our findings.

9.3 Theoretical Model

We employ the Bogoliubov-de Gennes (BdG) formalism [Hannibal *et al.*(2015), Datta and Bagwell(1999), De Gennes(1989)] to calculate the dynamics of the superfluid order parameter $\Delta(\mathbf{r}, t)$ of an ultracold Fermi gas confined in an axially symmetric harmonic trapping potential after an interaction quench. In this section, we provide a short summary of the most essential aspects of the used formalism and the relevant system parameters. For a comprehensive discussion of the formalism we refer the reader to our previous work [Hannibal *et al.*(2015)].

We consider the gas to be composed of fermionic ^6Li atoms in two different internal spin states (labeled by $\uparrow$ and $\downarrow$) with equal particle numbers $N_\uparrow = N_\downarrow = N/2$. The atoms in the two spin states interact via a contact interaction with $V_{\text{contact}} = (4\pi\hbar^2 a/m) \delta(\mathbf{r}_1 - \mathbf{r}_2) = g \delta(\mathbf{r}_1 - \mathbf{r}_2)$, where m is the mass of a ^6Li atom and $a < 0$ is the scattering length. Hence we only consider systems on the BCS side of the BEC-BCS crossover.

We use a cigar-shaped harmonic confinement potential given by $V_{\text{conf}} = \frac{1}{2}m\omega_\perp^2(x^2 + y^2) + \frac{1}{2}m\omega_\parallel^2 z^2$, where $\omega_\perp$ ($\omega_\parallel$) is the radial (longitudinal) confinement frequency, respectively. We denote the eigenfunctions of the confinement potential by $\langle \hat{r} | k \rangle = \phi_k(\mathbf{r})$ and the equidistant single-particle energies of the confinement potential are labeled by ξ_k , where we use the tuple $k = (k_x, k_y, k_z)$ with $k_i \in \{0, 1, 2, \dots\}$. In this paper, we consider elongated traps characterized

by an aspect ratio of the cloud $r = \omega_{\perp}/\omega_{\parallel} \gg 1$. In such a geometry one-dimensional subbands form resulting in quantum size oscillations in the case of small particle numbers [Shanenko *et al.*(2012), Hannibal *et al.*(2015)]. In order to characterize the number and positions of these subbands it is instructive to introduce the subband parameter $s = E_F/(\hbar\omega_{\perp})$ where the Fermi energy E_F is given by the chemical potential in the non-interacting case. That is, for a system with $s = j$, where $j \in \mathbb{N}$, the minimum of subband j is located at the Fermi energy. The choice $s = j + 0.5$ implies that the physics of the system are not dominated by a single band and the effects of the quantum size resonances are negligible also in small systems (cf. [Hannibal *et al.*(2015)]).

In order to describe the system, we use the introduced contact interaction and confinement potential and we obtain the BdG Hamiltonian with the BCS-like mean-field approximation

$$H_{\text{BdG}} = \sum_{\sigma} \int d^3r \Psi_{\sigma}^{\dagger}(\mathbf{r}) H_0 \Psi_{\sigma}(\mathbf{r}) + \int d^3r \Delta^*(\mathbf{r}) \Psi_{\downarrow}(\mathbf{r}) \Psi_{\uparrow}(\mathbf{r}) + h.c., \quad (9.1)$$

where H_0 is the single-particle Hamiltonian. By doing so we introduce the superfluid spatially dependent order parameter

$$\Delta(\mathbf{r}) = g \langle \Psi_{\downarrow}(\mathbf{r}) \Psi_{\uparrow}(\mathbf{r}) \rangle. \quad (9.2)$$

Here, $\Psi_{\sigma}(\mathbf{r})$ is the field operator annihilating an atom with spin σ at position $\mathbf{r}$. The corresponding eigenvalue problem to the Hamiltonian in Eq. (9.1) is the BdG equation which reads

$$\begin{pmatrix} H_0 & \Delta(\mathbf{r}) \\ \Delta^*(\mathbf{r}) & -H_0^* \end{pmatrix} \begin{pmatrix} u_K(\mathbf{r}) \\ v_K(\mathbf{r}) \end{pmatrix} = E_K \begin{pmatrix} u_K(\mathbf{r}) \\ v_K(\mathbf{r}) \end{pmatrix}. \quad (9.3)$$

The solution of this equation yields two branches (labeled by $K \rightarrow ka$ and $K \rightarrow kb$) for the eigenenergies and eigenfunctions which result from the two spin species. However, the eigenenergies and eigenfunctions can be expressed by one another [Datta and Bagwell(1999)]. Therefore, we drop the index a/b wherever possible and imply thereby the use of the states and energies corresponding to branch a .

Since the BdG equation has the form of a one-particle Schrödinger equation we use the BdG wave functions to introduce Bogoliubov's quasiparticles which diagonalize H_{BdG} . The transformation reads

$$\begin{aligned} \gamma_{ka}^{\dagger} &= \int u_k(\mathbf{r}) \Psi_{\uparrow}^{\dagger}(\mathbf{r}) + v_k(\mathbf{r}) \Psi_{\downarrow}(\mathbf{r}) d^3r \\ \gamma_{kb}^{\dagger} &= \int u_k(\mathbf{r}) \Psi_{\downarrow}^{\dagger}(\mathbf{r}) - v_k(\mathbf{r}) \Psi_{\uparrow}(\mathbf{r}) d^3r. \end{aligned} \quad (9.4)$$

In order to obtain the solution of the BdG equation we make use of Anderson's approximation [Anderson(1959)], i.e., we assume the BdG wave functions to be proportional to the bare atomic wave functions with $u_k(\mathbf{r}) = u_k \phi_k(\mathbf{r})$ and $v_k(\mathbf{r}) = v_k \phi_k(\mathbf{r})$. This yields the BdG eigenenergies

$$E_k = \sqrt{(\xi_k - \mu)^2 + \Delta_k^2} \quad (9.5)$$

and BdG coefficients

$$u_k = \sqrt{\frac{1}{2} \left(1 + \frac{\xi_k - \mu}{E_k} \right)} \quad v_k = \sqrt{\frac{1}{2} \left(1 - \frac{\xi_k - \mu}{E_k} \right)} \quad (9.6)$$

where $\Delta_k = \langle k | \Delta(\mathbf{r}) | k \rangle$. Exploiting this solution and substituting Bogoliubov's quasiparticles in the definition of the order parameter in Eq. (9.2) we obtain the well-known self-consistency

equations for the order parameter and the chemical potential μ in the ground state of the system at $T = 0$ K. They read

$$\Delta_k = -\frac{1}{2} \sum_{k'} V_{kk'} \frac{\Delta_{k'}}{E_{k'}} \chi_{k'} \quad (9.7)$$

$$N = 2 \sum_k \left(1 - \frac{\xi_k - \mu}{E_k} \right), \quad (9.8)$$

where we have used $V_{kk'} = g \int d^3r |\phi_k(\mathbf{r})|^2 |\phi_{k'}(\mathbf{r})|^2$. The self-consistent solution Δ_k of these equations characterizes the pairing properties in the ground state of the system, where the pairing takes place predominantly in an interval around the chemical potential as known from bulk BCS theory. The factor $\chi_{k'} = (1 - E_{k'}/\xi_{k'})$ in Eq. (9.7) regularizes the well-known ultraviolet divergence of the contact interaction [Bruun et al.(1999), Bruun and Heiselberg(2002), Shanenko et al.(2012)]. This regularization is introduced into all terms which result from summations over states k . Additionally, the numerically necessary cutoff is chosen such that all physical effects are qualitatively correctly accounted for.

In order to excite the nonadiabatic dynamics of the system we implement an interaction quench, i.e., an instantaneous change of the scattering length a , which leaves the particle density $n(\mathbf{r}) = n_\uparrow(\mathbf{r}) + n_\downarrow(\mathbf{r}) = 2 \langle \Psi_\uparrow^\dagger \Psi_\uparrow \rangle$ unchanged. In doing so, only occupations $x_{kl} = \langle \gamma_{ka}^\dagger \gamma_{la} \rangle = \langle \gamma_{kb}^\dagger \gamma_{lb} \rangle$ and coherences $y_{kl} = \langle \gamma_{ka}^\dagger \gamma_{lb} \rangle = \langle \gamma_{lb} \gamma_{ka} \rangle^*$ which are diagonal in terms of the quasiparticles, i.e., $x_{kl} = x_k \delta_{kl}$ and $y_{kl} = y_k \delta_{kl}$, are excited.

In the next step, we utilize Heisenberg's equations of motion in order to obtain equations of motion for the occupations x_k and coherences y_k of the density matrix. For our dynamical calculations we choose a time-independent basis, i.e., the basis given by the final system (u_m, v_m) . This leads to a time-dependent Hamiltonian and the appearance of terms specifying the deviation between the dynamical gap Δ and the ground-state value Δ_{GS} . The obtained equations of motion read

$$i\hbar \frac{d}{dt} x_k = a_k y_k^* - a_k^* y_k \quad (9.9)$$

$$i\hbar \frac{d}{dt} y_k = -2 E_k^{(\text{ren})} y_k + a_k (1 - 2x_k), \quad (9.10)$$

where

$$E_k^{(\text{ren})} = E_k + 2 u_k v_k \text{Re} [(\Delta - \Delta_{GS})_k] \quad (9.11)$$

$$a_k = v_k^2 (\Delta - \Delta_{GS})_k - u_k^2 (\Delta - \Delta_{GS})_k^* \quad (9.12)$$

and the deviation of the current gap is given by

$$(\Delta - \Delta_{GS})_k = - \sum_l [2v_l u_l x_l + u_l^2 y_l^* - v_l^2 y_l] V_{kl} \chi_l. \quad (9.13)$$

Here, χ_l regularizes this term in analogy to Eq. (9.7). We solve these equations of motion numerically with the Runge-Kutta-Fehlberg method which provides good numerical stability for all investigated sets of parameters.

Finally, we invert Bogoliubov's transformation and we arrive at the space- and time-dependent order parameter in terms of the quasiparticles which reads

$$\begin{aligned} \Delta(\mathbf{r}, t) = g \sum_k 2 v_k(\mathbf{r}) u_k(\mathbf{r}) & \left(x_k(t) - \frac{1}{2} \right) \\ & + u_k^2(\mathbf{r}) y_k^*(t) - v_k^2(\mathbf{r}) y_k(t). \end{aligned} \quad (9.14)$$

9.4 Results

In this section, we will first discuss the full spatiotemporal dynamics of the modulus of the order parameter $|\Delta(\mathbf{r}, t)|$ and in the next step we turn to the temporal evolution of the modulus of the spatially averaged order parameter given by

$$\bar{\Delta}(t) = \left| \frac{1}{V} \int d^3r \Delta(r, t) \right|, \quad (9.15)$$

where the volume $V = (\hbar/m)^{3/2} \omega_{\perp}^{-1} \omega_{\parallel}^{-1/2} = l_{\perp}^2 l_{\parallel}$ is determined by the harmonic trapping frequencies. This quantity will enable us to carry out a systematic study of the obtained dynamics in dependence on the trap parameters.

In this paper, we investigate interaction quenches for which the initial ground state, characterized by $1/(k_F a_i)$ and the ground-state gap $\bar{\Delta}_i$, is situated in the crossover regime of the BEC-BCS crossover and the final ground state $[1/(k_F a_f), \bar{\Delta}_f]$, is situated deep in the BCS regime. Here, we assume the well-known dispersion relation of the homogeneous Fermi gas in order to obtain the Fermi wave vector k_F . Throughout the paper we choose $1/(k_F a_f) = -1.45$ and characterize the strength of the quench by the initial coupling parameter $1/(k_F a_i)$. We note that it would also be possible to characterize the quench strength by $\bar{\Delta}_f/\bar{\Delta}_i$ in analogy to [Yuzbashyan *et al.*(2015)]. However, with the introduced regularization $\bar{\Delta}_f/\bar{\Delta}_i$ converges much slower than Δ_k when increasing the numerical cutoff. Since Δ_k determines the main properties of the system in our formalism it is sufficient to choose the numerical cutoff according to Δ_k , which substantially reduces the numerical effort. Bearing this in mind, we characterize the quench strength by $1/(k_F a_i)$, which is independent of the numerical cutoff. In the case of a homogeneous system a study of similar quenches has been done before and our quenches here are connected to the emergence of phase I [Yuzbashyan *et al.*(2015), Yuzbashyan and Dzero(2006)].

9.4.1 Spatiotemporal dynamics

In this section, we will investigate the spatiotemporal dynamics of the order parameter. To this end, we keep here, and in Secs. 9.4.2 and 9.4.3, the subband parameter $s = 3.5$ fixed. Furthermore, we choose $\omega_{\parallel} = 2\pi \cdot 120$ Hz, which then determines together with a given aspect ratio r and subband parameter s the transverse trapping frequency $\omega_{\perp}$ and the particle number N .

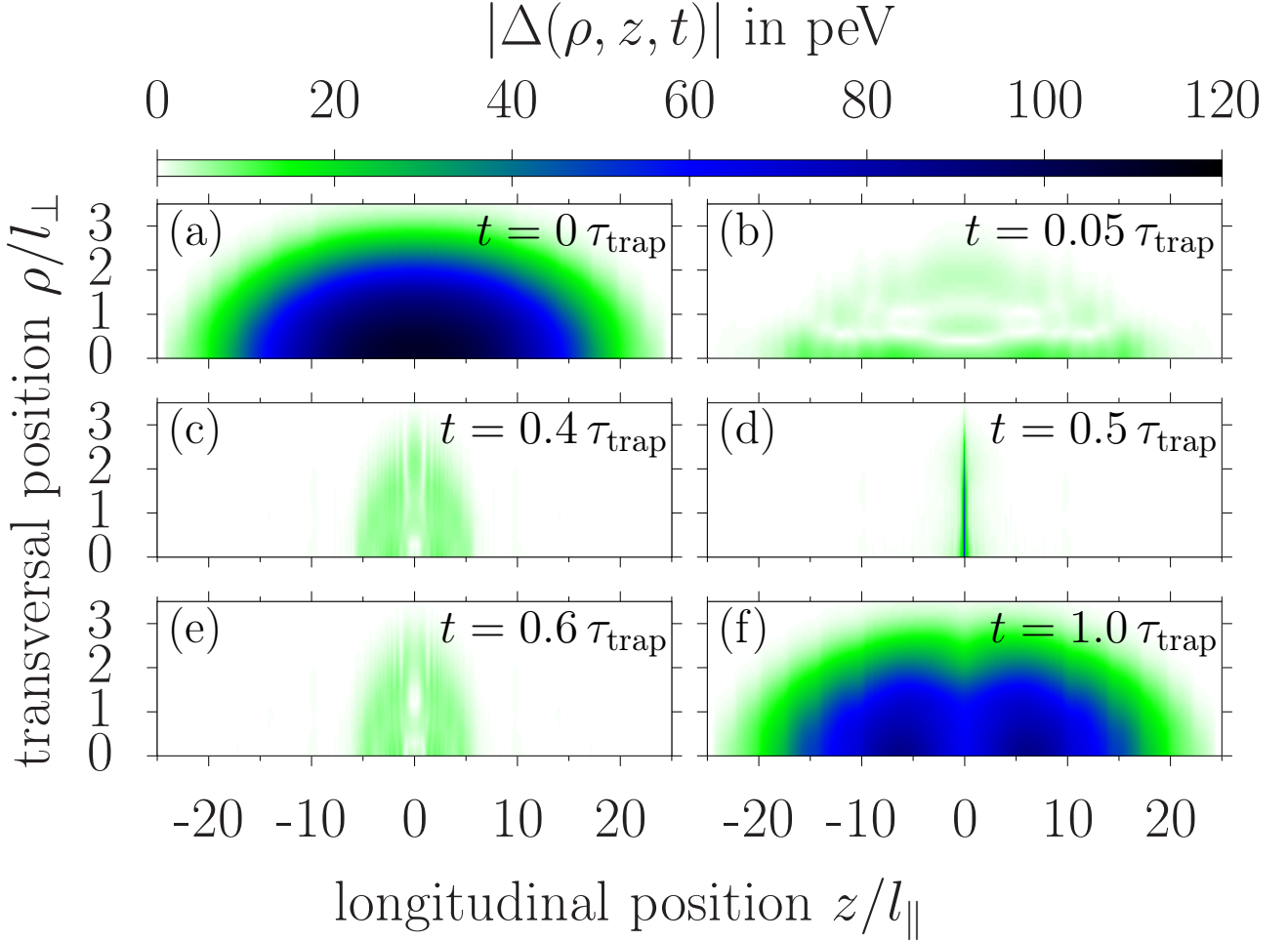


Figure 9.1: Spatiotemporal dynamics of the modulus of the order parameter after a sudden change of the scattering length from $1/(k_F a_i) = -0.4$ to $1/(k_F a_f) = -1.45$ and an aspect ratio of $r = 50$, where $\tau_{\text{trap}} = h/\delta E = 1/(2f_{\parallel})$ is the characteristic time scale of the trap [Hannibal et al.(2015)].

In Fig. 9.1 we show the spatially dependent modulus of the order parameter $|\Delta(\mathbf{r}, t)|$ at six different times after an excitation by an interaction quench. Figure 9.1(a) shows the initial spatial distribution of the gap at $t = 0$ after the instantaneous quench, i.e., the excitations introduced by the quench are already included. We find a symmetrical distribution which features a maximum in the center of the trap. After $t = 0.05\tau_{\text{trap}}$, we obtain a rapid decay of the order parameter of approximately one order of magnitude which affects the whole trap (cf. Fig. 9.1(b)). In addition, we find that the relative suppression of the order parameter with respect to the ground-state value of the final system is strongest in the center of the trap. Subsequently, we find that the remaining condensate moves towards the center of the trap where the distribution of the order parameter becomes very narrow with a large amplitude at $t = 0.5\tau_{\text{trap}}$ (cf. Fig. 9.1(d)). Comparing Figs. 9.1(c) and (e), which are almost identical, we see that this behavior occurs oscillatory and we find that the frequency is given by $f_{\text{collapse}} = 2f_{\parallel}$. Finally, in Fig. 9.1(f) we see a revival of the order parameter at $t = \tau_{\text{trap}}$, which shows two maxima located symmetric to the center of the trap.

Overall, from Fig. 9.1 we extract two effects. The first one is characterized by the rapid initial decay of the order parameter and the revival after τ_{trap} . The initial distribution of $|\Delta(\mathbf{r}, t = 0)|$ is set by the ground-state order parameter of the initial system while at the time of the revival we see that the order parameter shows two maxima which result from –as our data shows– the spatial profile of the order parameter in the final system. The second effect is an

oscillation taking place in the remaining superfluid after the initial decay. We see from our data that the suppression of the order parameter with respect to the ground-state value is strongest in the center of the trap. This causes the superfluid to start oscillating in the harmonic trap by moving towards the center. An analysis of the complex phase of the order parameter $\Phi(\mathbf{r}, t)$ shows that this oscillation fulfills the general relation between the superfluid velocity and the gradient of the phase, i.e., $v_s = \hbar/m \nabla \Phi(\mathbf{r}, t)$. This implies that the order parameter obeys a continuity equation and, hence, this oscillation will not be visible in the spatially averaged order parameter, as we will see in the next section.

Before we proceed, we point out that during the dynamics the average occupation number of the fermionic atoms stays smeared out around the chemical potential and the coherences y_k of the density matrix stay finite, indicating the existence of pair correlations at all times. In equilibrium these properties are directly linked to a superfluid behavior of the system. In a dynamical situation these properties are typically related to a superfluid behavior of the system even in the case of a vanishing order parameter as was also pointed out in Ref. [Yuzbashyan and Dzero(2006)].

In the following, we will systematically analyze the dynamical vanishing of the order parameter in a harmonic trap. To this end, we will discuss the modulus of the order parameter averaged over the trap. First, we will show that this is a suitable quantity to characterize the dynamical vanishing. Further, we will investigate its dependence on the aspect ratio r of the trap for small systems with fixed parameter s . In the next step we will show how the results scale for a larger system and, finally, we will present the dynamics of the condensate fraction for larger systems after the same type of quench.

9.4.2 Dynamical vanishing in the spatially averaged order parameter

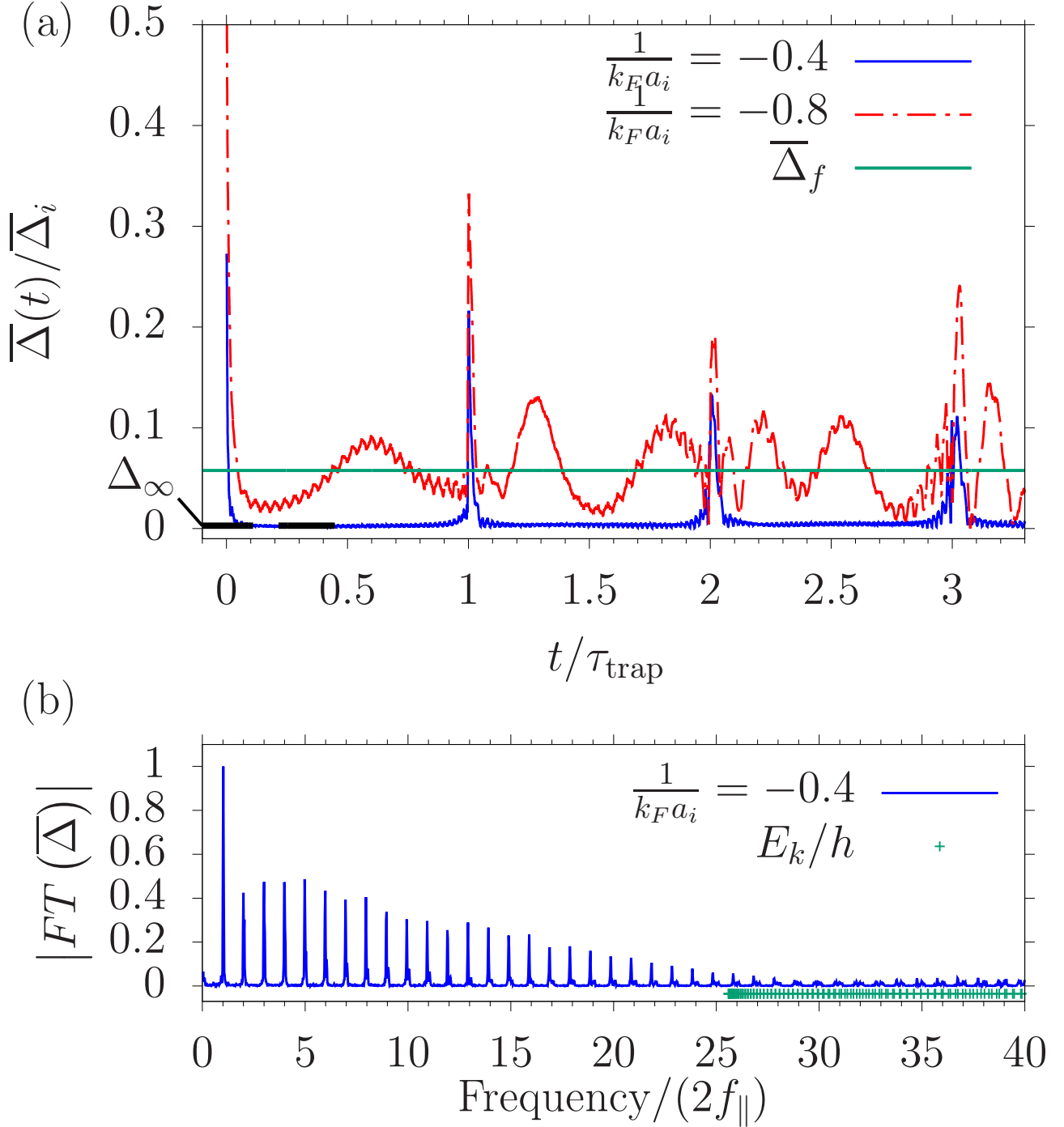


Figure 9.2: (a) Dynamics of the spatially averaged gap after a sudden change of the scattering length from $1/(k_F a_i) = -0.4$ (blue solid line) and $1/(k_F a_i) = -0.8$ (red dash-dotted line) to $1/(k_F a_f) = -1.45$ and an aspect ratio of $r = 50$. The gap is normalized to the spatially averaged ground-state gap of the system before the quench $\overline{\Delta}_i$. $\overline{\Delta}_f$ is the spatially averaged ground-state gap of the final system and Δ_∞ marks the height of the plateau for $1/(k_F a_i) = -0.4$; (b) Fourier transform of the gap dynamics with $1/(k_F a_i) = -0.4$ and the BdG eigenenergies E_k .

Figure 9.2(a) shows the dynamics of the modulus of the spatially averaged order parameter in a system with $r = 50$ after quenches with $1/(k_F a_i) = -0.4$ (blue solid curve) and $1/(k_F a_i) =$

-0.8 (red dash-dotted curve). After the stronger quench (i.e., $1/(k_F a_i) = -0.4$, being identical to Fig. 9.1) a decay of the modulus of the order parameter to a value which is much smaller than the ground-state value of the final system $\bar{\Delta}_f$ (solid green line) is seen. Following this decay a plateau evolves where the averaged value of the height of the plateau is labeled by Δ_∞ . Although the height of the plateau is very small compared to $\bar{\Delta}_f$ there are still some oscillations with the transverse confinement frequency $\omega_\perp$ visible. Since we cannot expect an exact vanishing of our numerical solution of a nondissipative model, we will, nevertheless, refer to such a behavior as dynamical vanishing of the gap. A technical discussion of the classifications of the numerical solutions is provided in the Appendix. Overall, our definition of the dynamical vanishing of the gap is in close analogy to what has been labeled phase I in a homogeneous system [Yuzbashyan *et al.*(2015)].

Additionally, in contrast to the homogeneous system, there are pronounced spikes visible at τ_{trap} . These result from the rephasing of the oscillators, i.e., the quasiparticle occupations and coherences, with equidistant energies spaced by $\delta E = 2\hbar f_\parallel$ that contribute to the dynamics. Therefore, the time is given by $\tau_{\text{trap}} = \hbar/\delta E = 1/(2f_\parallel)$ as has been discussed for the dynamics dominated by a dephasing of these oscillators, also labeled by phase II [Hannibal *et al.*(2015), Yuzbashyan *et al.*(2015)].

In comparison to the spatiotemporal dynamics in Fig. 9.1 we find that $\bar{\Delta}(t)$ precisely maps the rapid decay and the revival of the order parameter. As expected the oscillations towards the center of the trap are not visible; instead we observe an oscillation with the transverse confinement frequency. In general, this oscillation is also visible in $\Delta(\mathbf{r})$. However, it is not visible in Fig. 9.1 due to the temporal spacing of frames since $f_\perp \gg f_\parallel$. Thus, we establish that $\bar{\Delta}(t)$ is suitable to analyze both the initial decay and the evolution of a plateau in the dynamics of the order parameter in a cigar-shaped trap.

The weaker quench, i.e., $1/(k_F a_i) = -0.8$, belongs to phase II and shows an oscillation around a non-vanishing average, slightly smaller than $\bar{\Delta}_f$. The higher-frequency parts are given by the transverse trapping energy $\hbar\omega_\perp$. For an exhaustive discussion of phase II in an inhomogeneous system we refer the reader to our previous publication [Hannibal *et al.*(2015)]. In addition, comparing the two quenches we see that a larger quench leads to a faster initial decay of the order parameter which we will come back to later.

In order to gain further insight into the dynamics we show the Fourier transform of the dynamically vanishing amplitude of the order parameter in Fig. 9.2(b). We observe a spectrum with equidistant Fourier components spaced by $2f_\parallel$ which decay with increasing energy. The equidistant peaks evolve as a result of a nonlinear behavior of the equations of motion. While for a small quench, i.e., in the linear phase II, the Fourier components are given by the BdG quasiparticle energies (green dots) the frequencies for a large quench are given by the difference between the energies of the quasiparticle states contributing to the dynamics. To be precise, each occupation and coherence oscillates dominantly with a frequency given by twice the difference between the corresponding single-particle energy and the chemical potential μ . Since the final system is located in the BCS regime, i.e., $\mu \approx E_F$, this difference is always a multiple of $\hbar\omega_\parallel$. Overall, we obtain in phase I a spectrum with equidistant peaks dominated by non-linear effects. We refer to such a spectrum as “quasi-ungapped” spectrum, since the gap of the spectrum is given by the energy spacing of the equidistant peaks which is much smaller than the quasiparticle gap of the ground state.

In such a case of an equidistant quasi-ungapped spectrum a Fourier series directly connects the initial decay in the time domain with the width of the distribution of the frequency components. We label this width by β and determine it by an exponential fit in the time domain, which we discuss in the Appendix. In frequency space a physical understanding of the width of the decay can be obtained when considering the BCS pairing properties: Cooper pairing of atoms takes place in an interval around the chemical potential determined by the BdG wave

functions. From the solution of the BdG equation we see that the width of this interval is determined by the gap Δ_k of state k at the chemical potential μ . Further, the pairing strength continuously decreases in the given interval with increasing distance to the chemical potential due to the form of the BdG wave functions. This implies that the amplitude of the oscillation of the occupations and coherences decays with increasing distance to the chemical potential. Therefore, considering Eq. (9.14) also the Fourier coefficients of the dynamical vanishing gap decay with increasing energy over a width determined by Δ_k .

Overall, we find that the width β is connected to the pairing properties at the chemical potential which can be characterized by the smallest BdG eigenenergy $E_{\min}$, since $\xi_k \approx \mu$ at the chemical potential. Hence, we obtain that $\beta \propto E_{\min}$ for the dynamical vanishing of phase I. We have confirmed numerically that there is no additional dependence on the aspect ratio, i.e., for a given $E_{\min}$ β is identical for any aspect ratio r .

Summarizing, from Fig. 9.2 we have extracted the relevant physical properties to understand the dynamical vanishing in an inhomogeneous system. We have shown that the dynamics has a nonlinear character and we have related the dynamical behavior to the ground-state properties of the final system for a given quench strength and aspect ratio. We have shown that for a given quench the initial decay characterized by β is proportional to the pairing properties in the final state, i.e., $E_{\min}$ which, thus, will be an important variable for the subsequent analysis. With this fundamental understanding we now turn to the transition between the two different “phases” and investigate the impact of different aspect ratios.

9.4.3 Impact of the aspect ratio

In the following we will systematically investigate the impact of the aspect ratio of the cloud on the emergence of a dynamical vanishing of the spatially averaged order parameter. We will first establish numerical criteria necessary for the analysis and then evaluate the dynamics for various aspect ratios and again connect the observed behavior to the ground-state pairing properties characterized by $E_{\min}$.

For the subsequent analysis of the dynamical vanishing of the order parameter Δ_∞ will be the central quantity. We obtain Δ_∞ from our numerical data by fitting an exponential decay $f(t) = \exp(-\beta t) + \alpha$ in the beginning and by checking whether a flat plateau evolves afterwards. For this we require β to be large enough (criterion A) and an oscillatory behavior to be nonvisible in the subsequent dynamics (criterion B). If a plateau is identified we take Δ_∞ as the height of this plateau otherwise Δ_∞ is the arithmetic mean over the whole calculation time. A discussion of details and implications of our numerical procedure is provided in the Appendix.

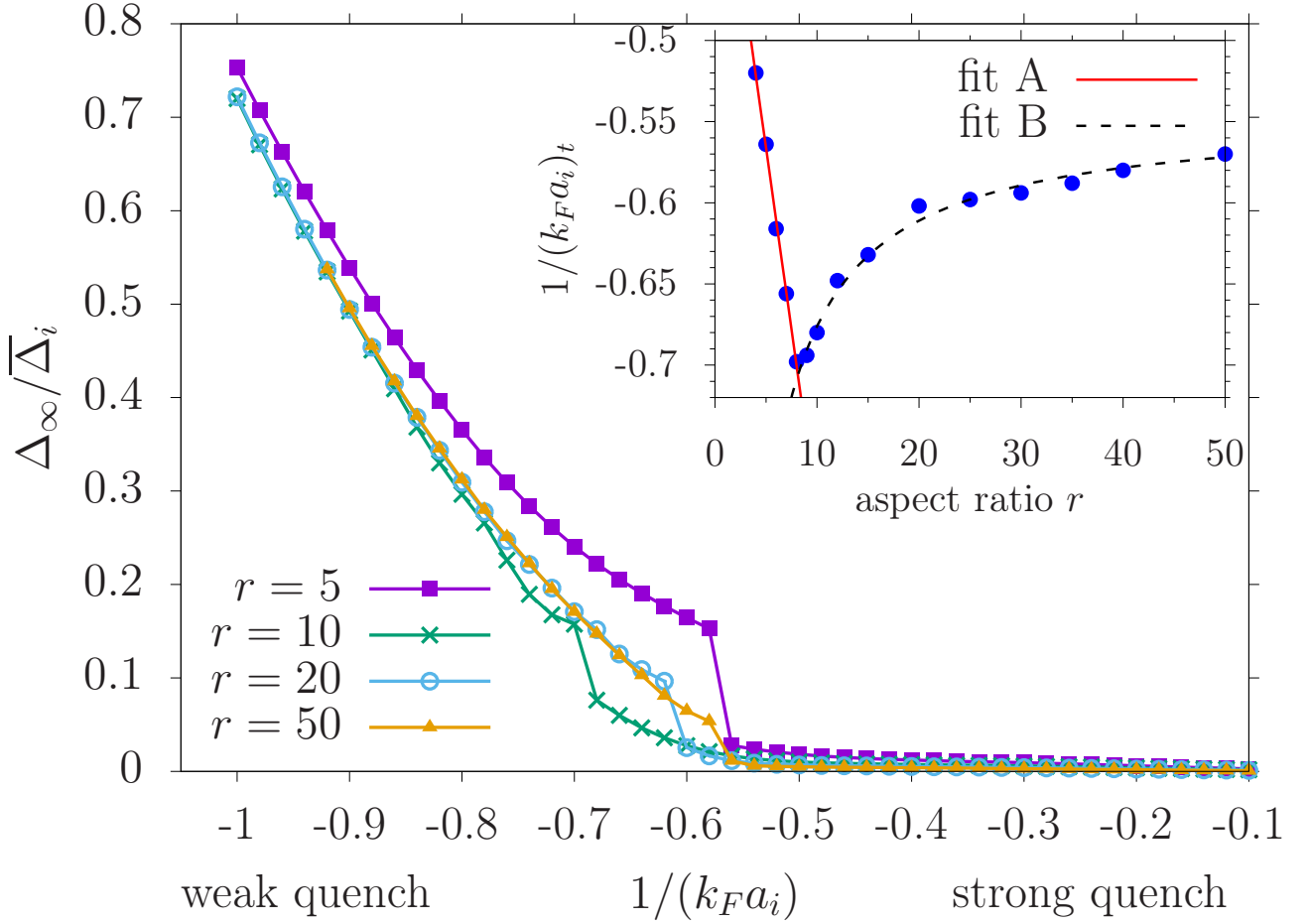


Figure 9.3: Δ_∞ normalized to the ground-state gap of the initial system $\bar{\Delta}_i$ for different quenches with $1/(k_F a_f) = -1.45$ and aspect ratios r . Inset: quench parameter $1/(k_F a_i)_t$ at the transition to the dynamical vanishing vs. the aspect ratio r . The red solid line (fit A) shows a linear fit to data for $r \leq 8$ and the black dashed line (fit B) is an inverse proportional fit to data for $r \geq 9$.

In the following, we carry out an analysis of Δ_∞ for a range of quenches which all end at the same $1/(k_F a_f) = -1.45$ where we vary the aspect ratio of the systems and keep the parameters $s = 3.5$ and $\omega_\parallel = 2\pi \cdot 120$ Hz fixed. In Fig. 9.3 we show results for four different aspect ratios which represent the effects obtained from our numerical data. For small quenches, i.e., small $1/(k_F a_i)$, we find a decreasing of Δ_∞ with increasing quench strength, i.e., larger $1/(k_F a_i)$, as has been predicted before [Yuzbashyan *et al.*(2015), Scott *et al.*(2012)]. Increasing the quench strength further, we observe a drop in Δ_∞ where the position of this visible drop is altered depending on the aspect ratio. The drop implies that the specified conditions for a vanishing order parameter are met and Δ_∞ is taken as the height of the identified plateau. Furthermore, we see that the height of the obtained plateau continuously tends to zero where it almost coincides for all aspect ratios. In the following, we will investigate the dependence of the drop on the aspect ratio which marks the transition to a vanishing order parameter.

For $r = 5$ (purple boxes) we find a distinct drop at $1/(k_F a_i)_t \approx -0.56$ to a plateau with a small height. When increasing the aspect ratio to $r = 10$ (green crosses) we observe that the transition to a plateau behavior is shifted to smaller quench strength, i.e., $1/(k_F a_i)_t \approx -0.69$. At the same time the height of the plateau after the transition is increased. In contrast, increasing the aspect ratio even further to $r = 20$ (blue circles) and $r = 50$ (orange triangles) leads to a reversal in the shift of the necessary quench strength for the transition. For $r = 50$ it is given by $1/(k_F a_i)_t \approx -0.56$, while the height of the plateau is reduced, since it is determined by the same curve for all aspect ratios which is set by the nonlinear behavior of the equations

of motion.

In order to investigate the shifts of the transition to a dynamical vanishing in more detail, we plot the initial coupling parameter at the transition point $1/(k_F a_i)_t$ in dependence of the aspect ratio in the inset of Fig. 9.3. For small aspect ratios $r \leq 8$ we find that $1/(k_F a_i)_t$ decreases linearly which is illustrated by the linear fit A. In contrast, for $r \geq 9$ the transition quench strength increases again and shows an $a/r + b$ (fit B) dependence, where $b \approx -0.55$ is in agreement with the maximum value of $1/(k_F a_i)_t$ in the main part of Fig. 9.3. It provides a limit of the quench strength necessary for aspect ratios $r \geq 50$. In the following, we will investigate these two confinement-induced shifts of the critical value of the quench strength characterized by $1/(k_F a_i)_t$, responsible for the transition to the dynamical vanishing. To this end, we will analyze the dominant energy and, hence, time scales of the system. This includes the dependence of the initial decay constant β , which turns out to control the behavior according to fit A, as well as the period of the Higgs mode in phase II, which will control the emergence of a dynamical vanishing and the visibility of a plateau in the range of aspect ratios where fit B applies.

In order to observe a vanishing of the order parameter the initial decay needs to be sufficiently fast compared to the trap time τ_{trap} (cf. criterion A). Therefore, we start by investigating the initial decay characterized by β with respect to the trap time τ_{trap} in dependence of the initial coupling $1/(k_F a_i)$ and the aspect ratio r . Figure 9.4 shows $\beta\tau_{\text{trap}}$ for a range of different quenches which are identical to those in Fig. 9.3. We find a nonlinear increase of the initial decay rate with an increased quench strength for all aspect ratios. That is, a stronger quench leads to a faster decay as expected for our initial value problem: the initially excited occupations and coherences depend on the order parameter in the initial system. Since we keep the final system identical the initial system for a larger quench is located closer to the BCS-BEC crossover and, hence, features a larger gap. This leads to a wider pairing interval determined by the BdG coefficients u_m, v_m and, therefore, to larger initial values which are distributed over a wider range of energies. This carries over to the spectral properties of the dynamics and thus leads to a larger β and, hence, a faster initial decay. Therefore, we establish the criterion $\beta\tau_{\text{trap}} > 10$ as a necessary condition for the vanishing to occur, as discussed in the Appendix.

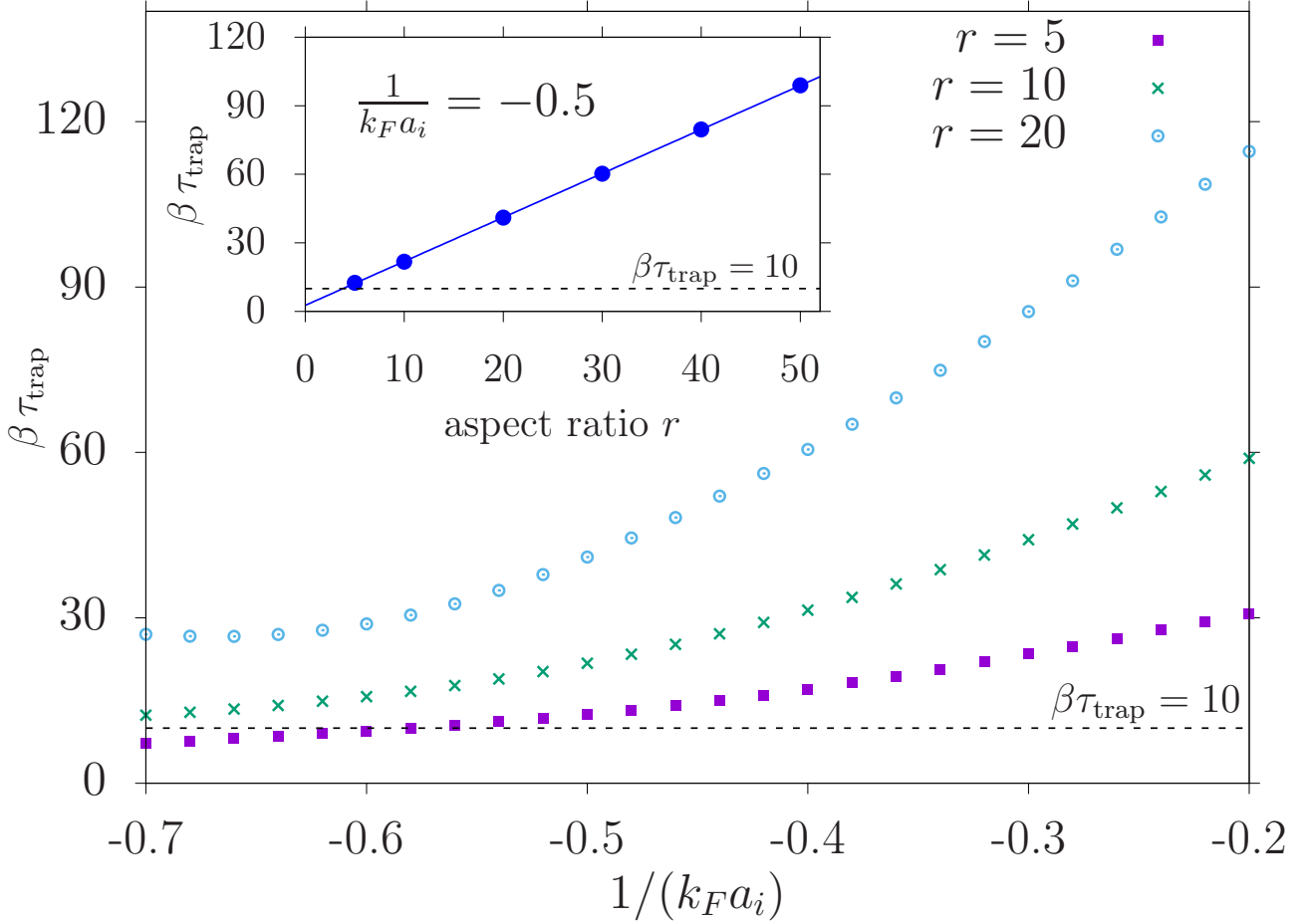


Figure 9.4: Time scale of the initial decay with respect to the trap time $\beta\tau_{\text{trap}}$ for various quenches to $1/(k_F a_f) = -1.45$ and $\omega_{\parallel} = \text{const.}$ for $r = 5, 10$ and 20 . Inset: $\beta\tau_{\text{trap}}$ vs. the aspect ratio r at fixed quench conditions; the dashed black line marks the condition $\beta\tau_{\text{trap}} = 10$ in either figure.

In the case of a system with an aspect ratio of $r = 5$ this criterion is met for $1/(k_F a_i) > -0.56$. This matches with the onset of the dynamical vanishing as seen in Fig. 9.3 and applies to all systems which show the behavior of fit A. The linear behavior of fit A is inherited from the linear dependence of $\beta\tau_{\text{trap}}$ on the aspect ratio which we show in the inset of Fig. 9.4 at a fixed quench strength. Here, the solid blue line is a linear fit to the data points. Since the dependence of $\beta\tau_{\text{trap}}$ on the quench strength is rather small in the range of the shift of $1/(k_F a_i)_t$ in Fig. 9.3 the dependence of $1/(k_F a_i)_t$ on r is dominated by the dependence of $\beta\tau_{\text{trap}}$ on the aspect ratio which results in the observed linear dependence of fit A due to criterion A.

In contrast, for the other shown aspect ratios in Fig. 9.4 ($r = 10, 20$) the condition $\beta\tau_{\text{trap}} > 10$ is fulfilled for any quench strength with $1/(k_F a_i) > -0.7$. Nevertheless, in comparison with Fig. 9.3 one finds that this does not necessarily lead to a dynamical vanishing of the gap. Furthermore, in the inset of Fig. 9.3 we found a different dependence of $1/(k_F a_i)_t$ on the aspect ratio r for these systems. This suggests that a different effect, which turns out to be connected to criterion B, is responsible for the transition point to the dynamical vanishing. Since these systems are located at the transition between phase I and phase II we go back to a system from phase II [cf. red dash-dotted line in Fig. 9.2(a)] and discuss the dependence of the relevant energy scale on the aspect ratio in this phase.

If the period of the Higgs mode T_{Higgs} is large compared to the trap time, i.e., $T_{\text{Higgs}}/\tau_{\text{trap}} \gg 1$, a plateau after the initial decay can become visible (cf. Fig. 9.7 in the Appendix). Additionally to the criterion A of a fast initial decay, characterized by $\beta\tau_{\text{trap}}$, we require the features of the linear Higgs mode of phase II to be non-visible in the dynamics. This is expressed through

the condition $T_{\text{Higgs}}/\tau_{\text{trap}} \gg 1$ (criterion B). If this second condition is not met, we observe an oscillation of the spatially averaged order parameter around the mean value Δ_∞ even if the initial decay is sufficiently fast.

In order to determine the period of the linear Higgs mode we consider the dominant energy scale in phase II which is given by the main frequency of the amplitude (Higgs) mode f_{Higgs} . Previous studies [Hannibal et al.(2015), Yuzbashyan et al.(2015)] showed that f_{Higgs} depends on the quench strength: in the case of a very small quench f_{Higgs} is given by $2E_{\text{min}}$. Further, for all quenches from "phase II" f_{Higgs} is closely connected to the mean value Δ_∞ of the oscillations¹. Therefore, f_{Higgs} decreases with increasing excitation strength as can be seen from Fig. 9.3. However, the decrease of Δ_∞ occurs in the same manner for all aspect ratios and, hence, does not introduce a significant dependence of f_{Higgs} on the aspect ratio. As a result, the dependence of f_{Higgs} on the aspect ratio is in good approximation solely inherited from the dependence of E_{min} on the aspect ratio.

We investigate this dependence in the following sections and find that the minimal quasiparticle energy E_{min} increases with the aspect ratio. Therefore, the period of the Higgs mode T_{Higgs} decreases with increasing aspect ratio and it turns out that this decrease is in good agreement with the $1/r$ behavior of fit B in the inset of Fig. 9.3, which we will come back to in Sec. 9.4.5. Hence now the transition is determined by the criterion B.

Summarizing, we have analyzed the confinement-induced effects on the continuous transition to the dynamical vanishing of the order parameter in a cigar-shaped ultracold Fermi gas. Exploiting our introduced classification, we have formulated two criteria which need to be fulfilled in order to observe a dynamical vanishing characterized by the emergence of a plateau. These two criteria depend contrarily on the aspect ratio and, hence, we distinguish the following two cases. For small aspect ratios $r \lesssim 8$ the transition is determined by the initial decay constant $\beta \propto E_{\text{min}}$. This leads to a linear shift of the transition in dependence of the order parameter according to fit A. For larger aspect ratios $r \gtrsim 9$ (fit B) the transition is determined by the period of the Higgs mode where again the relevant dependence is solely given by E_{min} . Since we have shown that both criteria depend on E_{min} we will provide a more general study of this key variable in the next sections.

9.4.4 Scaling properties

In this section we will consider the dependence of E_{min} on the aspect ratio r in a more general situation where the aspect ratio $r = \omega_\perp/\omega_\parallel$ is changed as $r \rightarrow \lambda r$, where we now keep a fixed particle number of $N = 1000$ as opposed to the parameter s . In order to achieve this scaling of the aspect ratio r we consider three possible options which are shown in Table 1. For each option we numerically calculate E_{min} , E_F , and β and extract the scaling behavior by fits analog to those in the inset of Fig. 9.4.

Option one is to keep $\omega_\parallel$ fixed, which is similar to the previous method but now we keep the particle number N fixed, option two ensures a constant $\omega_\perp$, while option three leaves the volume of the trap unchanged. Strictly speaking, all scaling possibilities could already be deduced from options one and two but we add option three for illustration. Table 1 lists the exponents p of the corresponding dependence of $E_{\text{min}}\tau_{\text{trap}} \rightarrow \lambda^p E_{\text{min}}\tau_{\text{trap}}$ for each option.

Our numerical study shows that the scaling of both $\beta\tau_{\text{trap}}$ and $E_{\text{min}}\tau_{\text{trap}}$ is –independent of the way in which the scaling of the aspect ratio r is performed– proportional to $\lambda^{2/3}$. The scaling of $E_{\text{min}}\tau_{\text{trap}}$ follows directly from the gap equation (9.7): we consider Δ_k to be located at the bottom of a subband, i.e., $\xi_k \approx \mu$, and omit the term introduced for regularization. Then, according to Eq. (9.5) $\Delta_k/E_k \approx 1$ holds true and the effect of the scaling is determined

¹In the homogeneous dynamical BCS theory both the frequency of the oscillations and the average value is given by the gap.

way of scaling	$\omega_{\parallel}$	$\omega_{\perp}$	τ_{trap}	E_F	$E_{\text{min}}\tau_{\text{trap}}$
1: $\omega_{\parallel} = \text{const}$	0	1	0	$2/3$	$2/3$
2: $\omega_{\perp} = \text{const}$	-1	0	1	$-1/3$	$2/3$
3: $\text{Vol} = \text{const}$	$-2/3$	$1/3$	$2/3$	0	$2/3$

Table 1: Scaling exponents p of λ for the relevant quantities of a cigar-shaped cloud when scaling the aspect ratio as $r \rightarrow \lambda r$ and fixed particle number N .

by $V_{kk'} \propto a \omega_{\perp} \omega_{\parallel}^{1/2}$. Bearing in mind, that we keep $1/(k_F a)$ fixed which implies a scaling of the scattering length as $a \propto k_F^{-1} \propto E_F^{-1/2} \propto V^{-1/3}$, this yields in all three options $E_{\text{min}}\tau_{\text{trap}} \propto \lambda^{2/3}$, as one can deduce from Table 1. Additionally, this also implies $\beta\tau_{\text{trap}} \propto \lambda^{2/3}$, since $\beta \propto E_{\text{min}}$ as argued in Sec. 9.4.2.

However, we point out that from the inset of Fig. 9.4 we find a different scaling of $\beta\tau_{\text{trap}}$ which can be explained by the fact that the particle number is changed when keeping the parameter $s = \text{const}$. If we keep $s = \text{const}$. and scale the aspect ratio as $r \rightarrow \lambda r$ we find from our analysis that $N \rightarrow \lambda N$ and, hence, $E_F \rightarrow \lambda E_F$. However, in order to generalize our previous findings to any scaling of the aspect ratio thus to experimentally more relevant systems in a cigar-shaped trap, we will discuss the behavior for larger particle numbers in the next subsection.

9.4.5 Large system

We will now consider a cloud of ^6Li atoms in a cigar-shaped trap with larger particle numbers and calculate the smallest BdG quasiparticle energy E_{min} of the system. As argued above the relevant time scales of the dynamics are set by this pairing property for a fixed quench. Hence, in this section we extend our systematic analysis to much larger particle numbers up to $N \sim 10^5$ by only calculating the ground-state properties of the given system. A full dynamical calculation of these systems is not possible due to numerical constraints.

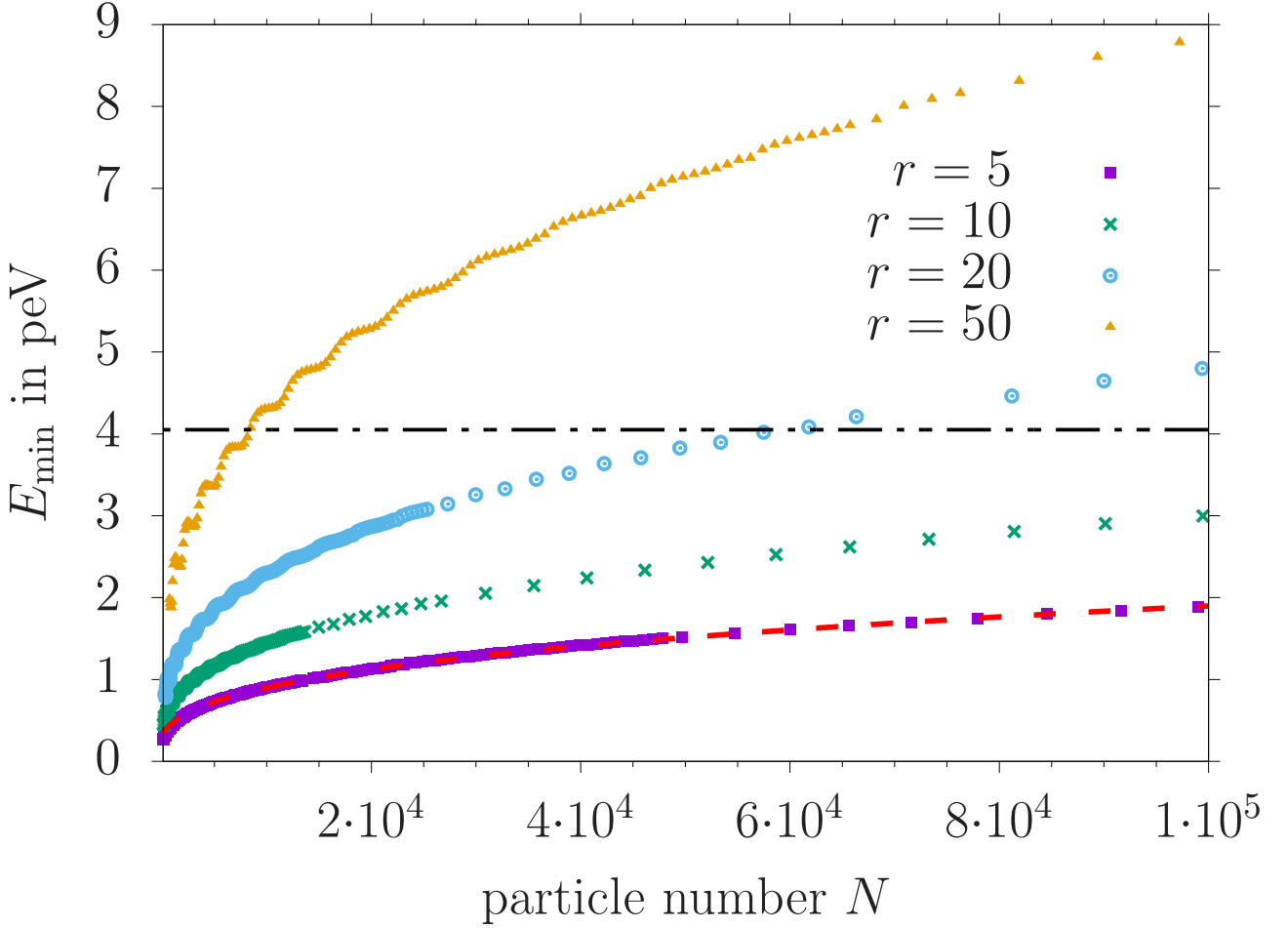


Figure 9.5: Minimal BdG quasiparticle energy $E_{\min}$ for various aspect ratios in dependence of the particle number N . The red dashed line is a fit $E_{\min} = a N^{1/3} + b$ to the data for $r = 5$ and the dash-dotted black line is the value of $E_{\min}$ for $N = 1 \cdot 10^6$ obtained from the fit.

Figure 9.5 shows the increase of the smallest BdG eigenenergy $E_{\min}$ in dependence of the particle number N , where the aspect ratio is scaled such that $\omega_{\parallel} = \text{const.}$. The dashed red line shows a fit $E_{\min} = a N^{1/3} + b$ for an aspect ratio $r = 5$ and the oscillations visible for small particle numbers are the aforementioned quantum size oscillations [Shanenko *et al.*(2012)], which can be neglected for the general analysis of the scaling in this paper. Overall, we extract from our numerical data that $E_{\min} \propto N^{1/3}$ for all aspect ratios r which also implies that $\beta \propto N^{1/3}$.

With this result, we now go back to the case discussed in Sec. 9.4.3 where the parameters s and $\omega_{\parallel}$ are fixed which implies that the particle number scales with $N \propto r$. Combining the general scaling properties of $E_{\min} \tau_{\text{trap}} \propto r^{2/3}$ in Table 1 with the dependence on the particle number yields $T_{\text{Higgs}} \propto 1/f_{\text{Higgs}} \propto 1/E_{\min} \propto 1/r$ and $\beta \propto E_{\min} \propto r$. This is in good agreement with the inset of Fig. 9.3 and the inset of Fig. 9.4, respectively.

Furthermore, the black dash-dotted line shows the value of $E_{\min}$ for $N = 1 \cdot 10^6$ as obtained from the shown fit. From the intersections with the black dash-dotted line obtained in Fig. 9.5 we can choose system parameters with rather small particle numbers which resemble closely the situation in a system with a far larger particle number but smaller aspect ratio. In the next section we present the calculation of the condensate fraction for a system with $r = 50$ and $N = 8428$ for this reason.

9.4.6 Condensate fraction

We obtain the condensate fraction of the system following the usual definition of the condensate fraction c of the order parameter [Leggett(2006)], which yields

$$c = \frac{V}{Ng^2} \int d^3r |\Delta(\mathbf{r})|^2, \quad (9.16)$$

where V is the volume, N the particle number, and g the interaction strength. In Fig. 9.6 the dynamics of the condensate fraction and the modulus of the averaged gap is shown for a system with an aspect ratio of $r = 50$ and a particle number $N = 8428$. This is the system that energetically resembles a system with an aspect ratio of $r = 5$ and a particle number of $N = 1 \cdot 10^6$.

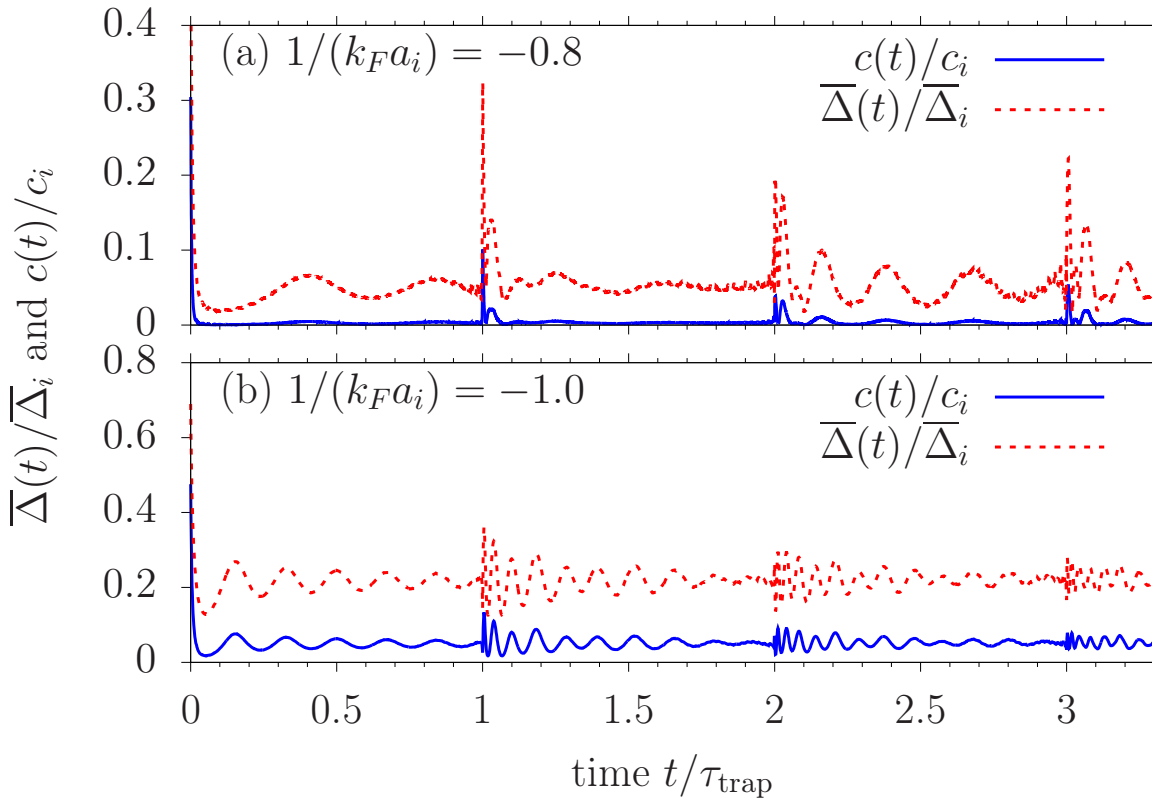


Figure 9.6: Condensate fraction (blue solid line) and modulus of averaged order parameter (red dashed line) after a quench to $1/(k_F a_f) = -1.45$ from (a) $1/(k_F a_i) = -0.8$ and from (b) $1/(k_F a_i) = -1.0$, with an aspect ratio $r = 50$, and a particle number $N = 8428$. The data is normalized to the ground-state value of the initial system c_i and $\bar{\Delta}_i$, respectively.

From Fig. 9.6(a) it becomes apparent that the visibility of the signature of the Higgs mode is reduced in the signal of the condensate fraction due to the quadratic dependence on the order parameter. While the signal still contains all frequencies inherited from the Higgs mode, the amplitude of these oscillations is barely visible and the resulting dynamics of the condensate fractions fulfills all requirements to be identified as vanishing. In the case of a smaller quench, i.e., further away from the transition, the signal of the condensate fraction is large enough to maintain the visibility of the Higgs mode, i.e., its shape and, hence, main frequency, as is shown in Fig. 9.6(b).

Overall, the condensate fraction could provide a measure to detect amplitude oscillations of the superfluid gap in an ultracold Fermi gas. However, due to the quadratic dependence on the gap the condensate fraction is already for rather small quenches reduced to $\lesssim 10\%$ of the

initial condensate fraction. This can make it challenging to detect oscillations in a time-resolved manner. Further, when analyzing the transition to a vanishing order parameter this will lead to observing a shift of the transition to smaller quench strength. However, we expect the finite-size effects to be qualitatively identical for the condensate fraction as previously discussed for the modulus of the spatially averaged order parameter since all main frequencies are inherited.

9.5 Fitting procedure

In this Appendix we discuss our fitting routine in order to obtain Δ_∞ in "phase I" and "phase II" from our numerical data $\bar{\Delta}(t)$.

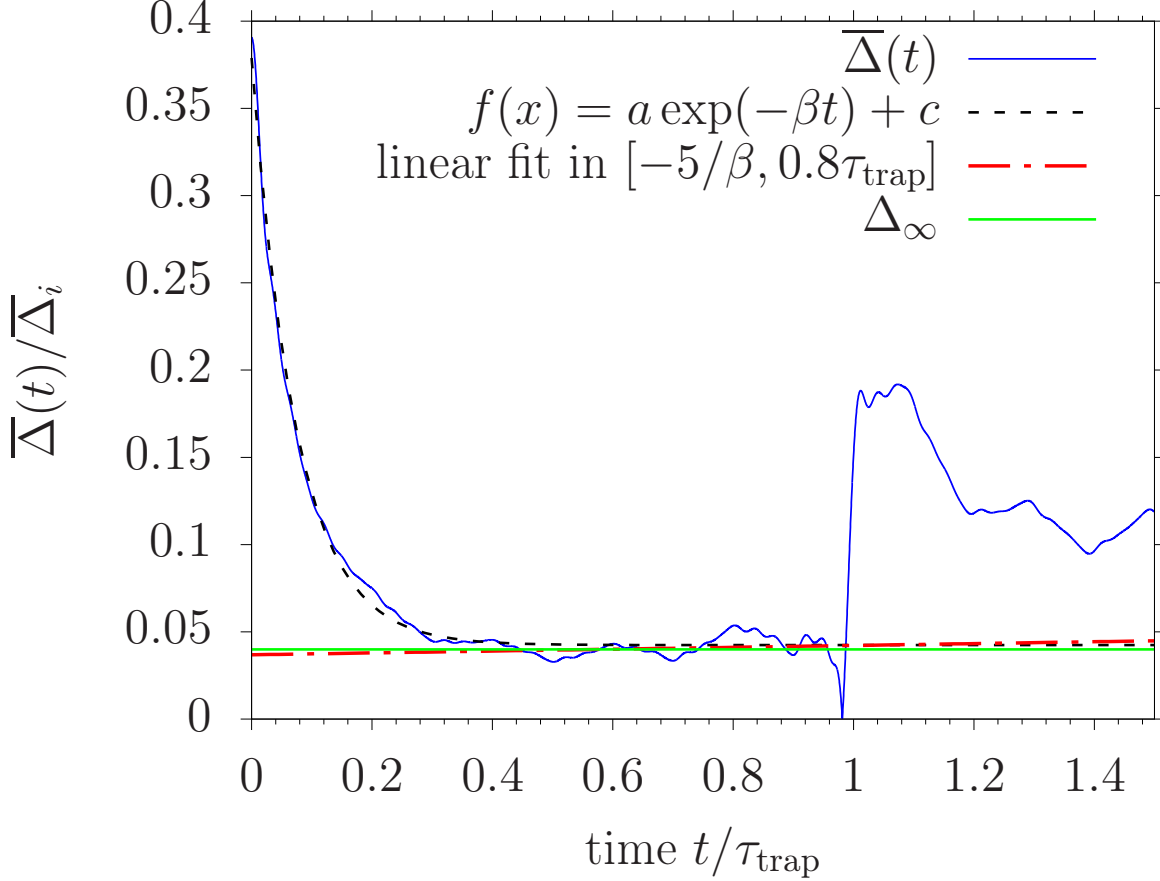


Figure 9.7: Exemplary illustration of our fitting routines in order to obtain Δ_∞ for a system with $r = 10$ and a quench with $1/(k_F a_i) = -0.66$. In this system we find a plateau behavior before τ_{trap} , which is characterized by the non-vanishing mean value of the plateau.

In Fig. 9.7 we show the results of the fitting routines carried out for an exemplary case of $r = 10$ and $1/(k_F a_i) = -0.66$, i.e., a system with a plateau with a non-vanishing height (cf. Fig. 9.3). First, we fit $f(t) = a \exp(-\beta t) + c$ to our data (black dotted line) and check whether $\beta \tau_{\text{trap}}/2 > 5$ holds true. If that is the case the initial decay is fast enough that at $t = 0.5\tau_{\text{trap}}$ the amplitude of the oscillations described by an exponential decay contributes with less than 1% to the total signal and a plateau can become visible before τ_{trap} as shown in Fig. 9.7.

Since the exponential fit is only carried out for the initial decay the offset does not identify the visibility of a plateau correctly in the transition between the two phases. Therefore, if $\beta \tau_{\text{trap}}/2 > 5$ is fulfilled, we carry out a linear fit from $5/\beta$ to $0.8\tau_{\text{trap}}$ and require the relative incline to be less than 10%. If these criteria are met we find a plateau and choose Δ_∞ to be the mean value for $5/\beta \leq t \leq 0.8\tau_{\text{trap}}$; otherwise, we take Δ_∞ as the arithmetic mean over the whole calculation time. The lower bound of the interval is motivated by the initial exponential

decay while the upper bound ensures that the emergence of the peak at τ_{trap} does not hinder a detection of a visible plateau. A linear fit is necessary to rule out oscillatory behavior as, for example, seen in the red dash-dotted line in Fig. 9.2. This routine enables us to systematically distinguish between the dynamical vanishing of the gap including the height of the plateau and linear dephasing dynamics of "phase II".

In mean-field theory a phase transition is obtained in the thermodynamic limit, which has previously been done on the case of the homogeneous system [Yuzbashyan *et al.*(2015)]. In order to investigate the confinement-induced effects to the dynamics in such an ultracold Fermi gas the thermodynamic limit is not suitable. This directly implies that the transition between the "phase I" and "phase II" takes place continuously in the inhomogeneous system. The above introduced classification quantifies the effects of the two "phases" and, hence, enables us to investigate the effects of the aspect ratio to this continuous transition. Although the criteria are well motivated from our observations of the dynamical behavior the exact numerical values for each criteria are within a reasonable range arbitrarily. However, changing the criteria within a reasonable range does not have a qualitative effect on the results presented in this paper. Solely, small quantitative changes such as slightly shifted transition points to the vanishing regime or heights of the plateau occur.

10 Solution for the dynamics of the BCS and central spin problems by Yuzbashyan, Altshuler, Kuznetsov, Enolskii

Abstract

We develop an explicit description of a time-dependent response of fermionic condensates to perturbations. The dynamics of Cooper pairs at times shorter than the energy relaxation time can be described by the BCS model. We obtain a general explicit solution for the dynamics of the BCS model. We also solve a closely related dynamical problem – the central spin model, which describes a localized spin coupled to a "spin bath". Here, we focus on presenting the solution and describing its general properties, but also mention some applications, e.g. to nonstationary pairing in cold Fermi gases and to the issue of electron spin decoherence in quantum dots. A typical dynamics of the BCS and central spin models is quasi-periodic with a large number of frequencies and stable under small perturbations. We show that for certain special initial conditions the number of frequencies decreases and the solution simplifies. In particular, periodic solutions correspond to the ground state and excitations of the BCS model.

10.1 Introduction

Recent experiments [7] on cold fermion pairing in the vicinity of a Feshbach resonance offer a unique opportunity to control the strength of pairing interactions between fermions by a magnetic field. This opens up an exciting possibility to explore fundamentally new aspects of the nonequilibrium pairing following an abrupt change of the coupling strength.

Interestingly, the dynamical pairing in this regime can be linked to a seemingly unrelated spin model. The latter describes the interaction of a localized (central) spin with a "spin bath" of environmental spins. The central spin model has reemerged recently in the context of experiments [2, 3] on electron spin dynamics due to the hyperfine interaction with nuclei in GaAs quantum dots. Since single electron quantum dots are now experimentally accessible[3], and are considered one of the most promising candidates for solid state qubits, it is also important to understand the effect of this interaction on the coherence of the electron spin.

In this paper, we present a theory that provides an accurate description for a range of phenomena in the dynamical pairing and central spin problems. As we will see below, there is an intimate connection between these two problems that will enable us to treat them on an equal footing.

The study of the dynamics of the superconducting state in metals founded on microscopic principles has begun during the decade following the advent[20] of the BCS theory (see Ref. [1] for a review). The simplest theory for nonstationary processes in superconductors is based on the time-dependent Ginzburg-Landau (TDGL) equation [2, 3, 4] for the order parameter $\Delta(t)$. The TDGL approach is valid when quasiparticles are able to reach a local equilibrium quickly on the characteristic time scale of the order parameter variation τ_Δ ($\simeq \Delta^{-1}$ at zero temperature). This requirement usually limits the applicability of the TDGL theory to situations where mechanisms destroying Cooper pairs are effective, such as a narrow vicinity of T_c or a large concentration of magnetic impurities.

An alternative theory [5, 6] is the Boltzmann kinetic approach, which describes the dynamics in terms of a kinetic equation for the quasiparticle distribution function coupled to a self-consistent equation for $\Delta(t)$. However, this scheme is justified only when external parameters change slowly on the τ_Δ time scale.

As pointed out in Ref. [15], cold fermionic gases can be in a regime where the notion of the excitation spectrum is irrelevant and neither TDGL nor the Boltzmann kinetic equations are valid. Indeed, in these systems external parameters such as the position of a Feshbach resonance can change on a time scale $\tau_0 \ll \tau_\Delta, \tau_\epsilon$, where τ_ϵ is the quasiparticle energy relaxation time. On the other hand, the energy relaxation is slow, while the lifetime of the samples is limited. It is therefore desirable to develop a theory that describes nonstationary pairing effects in this regime.

The nonequilibrium Cooper pairing at times $t \ll \tau_\epsilon$ is non-dissipative and in a translationally invariant system can be described by the reduced BCS model

$$\hat{H}_{BCS} = \sum_{\mathbf{p}, \sigma} \epsilon_{\mathbf{p}} \hat{c}_{\mathbf{p}\sigma}^\dagger \hat{c}_{\mathbf{p}\sigma} - g \sum_{\mathbf{p}, \mathbf{k}} \hat{c}_{\mathbf{p}\uparrow}^\dagger \hat{c}_{-\mathbf{p}\downarrow}^\dagger \hat{c}_{-\mathbf{k}\downarrow} \hat{c}_{\mathbf{k}\uparrow} \quad (10.1)$$

It is interesting to note that the use of this model for describing the dynamics of a homogeneous superconducting state in metals in the non-dissipative regime, $t \ll \tau_\epsilon$, has been justified [12] quite generally starting from nonstationary Eliashberg equations.

Here, we are interested in a situation when a system at zero temperature is out of an equilibrium at $t = 0$. The goal is to determine the subsequent evolution of the initial state. In particular, this includes the case when at $t = 0$ the coupling constant has been abruptly changed from g' to g .

In the absence of translational invariance, e.g. in a dirty superconductor or in a finite system, Hamiltonian (10.1) has to be appropriately generalized[21]. First, let us discuss the familiar case of electron-electron interactions. If there are no spatial symmetries, but the time-reversal invariance is preserved, single-particle orbitals ϵ_j are degenerate only with respect to spin. For each orbital ϵ_j there is a pair of states, $|j \uparrow\rangle$ and $|j \downarrow\rangle$, related by time reversal symmetry. The pairing occurs between time reversed states, i.e.

$$\hat{H}_{BCS} = \sum_{j, \sigma} \epsilon_j \hat{c}_{j\sigma}^\dagger \hat{c}_{j\sigma} - g \sum_{j, q} \hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow}^\dagger \hat{c}_{q\downarrow} \hat{c}_{q\uparrow} \quad (10.2)$$

Deviations of the coupling strength g from a constant in q and j are small and sample-dependent[21, 22].

For a two species Fermi system, such as a two-component mixture of fermionic atoms in a trap, spin up and down states in (11.18) have to be identified with the two species of fermions.

With this replacement, arguments[21, 22, 15] leading to Hamiltonian (11.18) are valid in the weak coupling regime.

Hamiltonian (10.1) is obtained from (11.18) by specializing to a translationally invariant situation. In this case, time reversed states are $|\mathbf{p} \uparrow\rangle$ and $|\mathbf{p} \downarrow\rangle$. One can also take a thermodynamic limit by taking the number of orbital levels to infinity. However, in view of applications where the effective number of levels is finite and even small (see below), we deal with the general Hamiltonian (11.18) here.

Remarkably, the BCS model turns out to be closely related to a model describing the Heisenberg exchange interaction of a single localized spin (the central spin) with a number of environmental spins. The Hamiltonian reads

$$\hat{H}_0 = \sum_{j=1}^{n-1} \gamma_j \hat{\mathbf{K}}_0 \cdot \hat{\mathbf{K}}_j - B \hat{K}_0^z \quad (10.3)$$

where $\hat{\mathbf{K}}_0$ is the central spin, $\hat{\mathbf{K}}_j$ are environmental spins, γ_j are (nonuniform) coupling constants, and B represents the magnetic field [16].

The central spin model has an interesting history of its own. For example, it emerged in the studies of electron spin dynamics in disordered insulators [17] and of coherent spin tunnelling in ferromagnetic grains[14]. More recently, it attracted considerable attention as a model for decoherence of qubits [19, 17, 19, 20]. In particular, it can be used to model[19, 17] the hyperfine interaction of a localized electron spin with nuclear spins and the spin-dependent transport[23] in GaAs quantum dots. In these cases, it provides an adequate description of spin dynamics at times shorter than the nuclear spin relaxation time in the regime where the orbital level spacing in the dot is much larger than the temperature and typical interaction energies.

The main result of this paper is an explicit general solution for the dynamics of the BCS and central spin models. In particular, we determine as functions of time the order parameter $\Delta(t)$ and the expectation value of the central spin (equations (10.35) and (10.37)) as well as the dynamics of the remaining degrees of freedom for arbitrary initial conditions. Here, we concentrate on presenting the solution and describing its general properties. We also mention several applications to specific issues such as nonequilibrium pairing and that of decoherence due to the hyperfine interaction, but leave a detailed discussion for the future.

The solution is based on the integrability [24, 25, 26] of the BCS and central spin models. This important property has been largely underestimated in part because it was discovered outside the main physical context of these models. Besides, due to the infinite range of interactions in Hamiltonians (11.18) and (10.3), the mean field approximation is exact [11, 10] for these models in the limit of large number of particles. In this approximation, the BCS and central spin models can be mapped onto a classical nonlinear system (see below). However, while the mean field simplifies the description of equilibrium properties to an extent where no advanced techniques are required, it is the integrability of the resulting classical system that enables us to solve dynamical problems. The BCS solution[20] for the ground state and excitations of the BCS model is recovered as a periodic case of the general solution (see the discussion following (11.20) and Section 10.4). It is also interesting to note that, as we will see, models (11.18) and (10.3), being in many respects classical, have distinct robust features that are preserved when the integrability is destroyed.

The Dicke model [9] that describes an ensemble of two-level systems coupled to a bosonic mode is also within the scope of our construction. In fact, it belongs to the same class [25] of integrable models as the BCS and central spin models. The Dicke model has been recently adopted [16] to the study of a dynamical coexistence of atoms and molecules in cold Fermi gases. In the strong coupling regime, it seems to be an appropriate alternative to the BCS model for these systems.

The general nonstationary solution of the BCS and central spin models is in terms of hyperelliptic functions – multiple variable generalizations of ordinary elliptic functions. These functions frequently arise as solutions of integrable equations and have well known analytical properties [31].

We show that for most initial conditions the dynamics is *quasi*-periodic with a number of independent frequencies equal to the number of degrees of freedom, n , and evaluate the frequencies in terms of the integrals of motion. The typical motion uniformly explores an invariant torus – an n -dimensional subspace of the $2n$ -dimensional phase space allowed by the conservation laws. These features of the typical motion are stable against small perturbations that destroy integrability.

Further, we identify certain special values of integrals of motion for which the number of independent frequencies, m , becomes less than the number of degrees of freedom. For these degenerate cases, we were able to explicitly reduce the motion of the BCS and central spin models with n degrees of freedom to that of same systems with only m degrees of freedom. The solution progressively simplifies as the number of independent frequencies decreases. For example, solutions characterized by a single frequency ($m = 1$), i.e. periodic trajectories, are given by trigonometric functions; degenerate solutions with two independent frequencies ($m = 2$) are given by a combination of trigonometric and elliptic functions (see also Refs. [15, 16] and the end of Section 10.4) etc.

Periodic solutions ($m = 1$) occupy a special place among degenerate solutions. As we show in Section 10.4, they reproduce the BCS solution for the ground state and excitations of the BCS model.

However, it should be emphasized that, unlike the general solution, degenerate solutions are nonrepresentative of the *dynamics* of the BCS and central spin models and, with the exception of the ground state, are expected to be unstable.

The paper is organized as follows. In Section 10.2, we use a mean field approximation to map the BCS and central spin models onto equivalent classical models. Section 11.6 contains an explicit general solution for the dynamics of both problems. In Section 10.4, we consider degenerate solutions. Section 10.5 discusses some open problems and possible applications of our results.

10.2 Mean field approximation

Our starting point is the mean field approximation, which enables the mapping of the BCS and central spin models onto equivalent classical nonlinear systems. We also derive the equations of motion and describe the relationship between the BCS and central spin models.

The discussion of the mean field is facilitated by representing BCS Hamiltonian (11.18) in terms of Anderson pseudospin-1/2 operators [11].

$$\hat{H}_{BCS} = \sum_{j=0}^{n-1} 2\epsilon_j \hat{K}_j^z - g \hat{L}_+ \hat{L}_- \quad \hat{\mathbf{L}} = \sum_{q=0}^{n-1} \hat{\mathbf{K}}_q \quad (10.4)$$

where n is the number of single-particle orbitals. Pseudospin operators are related to fermion creation and annihilation operators via

$$\begin{aligned} \hat{K}_j^z &= \frac{c_{j\uparrow}^\dagger \hat{c}_{j\uparrow} + \hat{c}_{j\downarrow}^\dagger \hat{c}_{j\downarrow} - 1}{2} \\ \hat{K}_j^- &= \hat{c}_{j\downarrow} \hat{c}_{j\uparrow} \quad \hat{K}_j^+ = \hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow}^\dagger, \end{aligned} \quad (10.5)$$

Pseudospins are defined on unoccupied and doubly occupied pairs of states $|j \uparrow\rangle$ and $|j \downarrow\rangle$, where they have all properties of spin-1/2. Singly occupied pairs of states, on the other hand, do not participate in pair scattering and are decoupled from the dynamics.

Our goal is to determine the time evolution according to Hamiltonian (11.19) of an arbitrary initial distribution of pseudospins. The mean field approximation consists in the replacement of the effective field seen by each pseudospin in BCS Hamiltonian (11.19) with its quantum mechanical average, $\mathbf{b}_j(t) = (-2\Delta_x(t), -2\Delta_y(t), 2\epsilon_j)$, where $\Delta(t)$ is the BCS gap function

$$\Delta(t) \equiv \Delta_x(t) - i\Delta_y(t) \equiv g \sum_p \langle \hat{K}_j^-(t) \rangle \quad (10.6)$$

In this approximation, each spin evolves in the self-consistent field created by other spins

$$\dot{\hat{\mathbf{K}}}_j = \mathbf{i} [\hat{\mathbf{H}}_{\text{BCS}}, \hat{\mathbf{K}}_j] = \mathbf{b}_j \times \hat{\mathbf{K}}_j \quad (10.7)$$

Since equations (11.5) are linear in $\hat{\mathbf{K}}_j$, we can take their quantum mechanical average with respect to the time-dependent state of the system to obtain

$$\dot{\mathbf{s}}_j = \mathbf{b}_j \times \mathbf{s}_j \quad \mathbf{b}_j = (-gJ_x, -gJ_y, 2\epsilon_j) \quad \mathbf{J} = \sum_{q=0}^{n-1} \mathbf{s}_q \quad (10.8)$$

where $\mathbf{s}_j(t) = 2\langle \hat{\mathbf{K}}_j(t) \rangle$. Evolution equations (10.8) conserve the square of the average for each spin, i.e. $\mathbf{s}_j^2 = \text{const.}$ If spins initially were in a product state [32], $\mathbf{s}_j^2 = 1$.

Note from equation (10.5) the following correspondence between components of $\mathbf{s}_j(t)$ and the normal and anomalous (Keldysh) Green functions at coinciding times:

$$G_j(t) = -i\langle [c_{j\uparrow}(t), \hat{c}_{j\uparrow}^\dagger(t)] \rangle = i s_j^z(t)$$

$$F_j(t) = -i\langle [c_{j\uparrow}(t), \hat{c}_{j\downarrow}(t)] \rangle = i s_j^-(t)$$

Equations (10.8) were derived [12] for phonon superconductors within the general framework of nonstationary Eliashberg theory in the collisionless regime $t \ll \tau_\epsilon$. A linearized version of these equations was considered in Refs [11, 12] and [13].

Due to the infinite range of interactions between spins in (11.19), the mean field approximation is exact in the thermodynamic limit. For a system with a finite number, N , of particles (spins) we expect, based on an analysis of leading finite size corrections [35] to the mean field, equations (10.8) to be accurate for large N at times $t < t^* N^\eta$, where t^* and $\eta > 0$ do not depend on N .

We see that $\mathbf{s}_j = 2\langle \hat{\mathbf{K}}_j \rangle$ have all properties of classical spins governed by a Hamiltonian

$$H_{\text{BCS}} = \sum_{j=0}^{n-1} 2\epsilon_j s_j^z - \frac{g}{2} J_+ J_- \quad (10.9)$$

and usual angular momentum Poisson brackets. Thus, the problem reduces to determining the time evolution of n classical spins according to Hamiltonian (11.20).

The BCS solution for the ground state corresponds to the minimum of (11.20) and is obtained by aligning each spin in (11.20) antiparallel to the field acting upon it. A pair excitation[34] (excitation of the condensate) of energy $2\sqrt{(\epsilon_j - \mu)^2 + \Delta_0^2}$, where μ and Δ_0 are the chemical potential and the equilibrium gap respectively, is obtained by flipping the spin $\mathbf{s}_j$. The $U(1)$ symmetry of the BCS order parameter Δ is equivalent to the symmetry of (11.20) with respect to uniform rotations of all spins around the z -axis. Due to this symmetry, spin

configurations that determine the ground state and excitations can rotate around the z -axis with a frequency 2μ at no energy cost. In the presence of a particle-hole symmetry $\mu = 0$ and these configurations are stationary with respect to Hamiltonian (11.20) (see Section 10.4 where we recover the BCS solution as a periodic case of the general time-dependent solution).

Next, we turn to the central spin model and its relationship to the BCS model. The central spin Hamiltonian (10.3) turns out to be a member of an integrable family of Gaudin magnets [25] – n Hamiltonians of the form:

$$\hat{H}_q = 2 \sum_{j=0}^{n-1} \frac{\hat{\mathbf{K}}_q \cdot \hat{\mathbf{K}}_j}{\epsilon_q - \epsilon_j} - \alpha \hat{K}_q^z \quad q = 0, \dots, n-1 \quad (10.10)$$

where ϵ_j and α are arbitrary parameters. All Hamiltonians $\hat{H}_q$ commute with each other and with the z -component of the total spin, $\hat{L}_z \propto \sum_q \hat{H}_q$, for arbitrary ϵ_j and α . The central spin model (10.3) coincides with $\hat{H}_0$ in equation (10.10) if we choose $\epsilon_0 - \epsilon_j = 2/\gamma_j$ and $\alpha = B$.

Note that if a number of orbitals ϵ_j in (11.19) are degenerate, the magnitude of their total spin $\sum_{\epsilon_j=\text{const}} \hat{\mathbf{K}}_j$ is conserved by Hamiltonian (11.19). In this case one can replace $\hat{\mathbf{K}}_j \rightarrow \sum_{\epsilon_j=\text{const}} \hat{\mathbf{K}}_j$ in Hamiltonians (11.19) and (10.10) and sum over nondegenerate orbitals only.

Now consider the following linear combination of $\hat{H}_q$:

$$\sum_{q=0}^{n-1} \epsilon_q \hat{H}_q = -\frac{\alpha}{2} \left[\sum_{q=0}^{n-1} 2\epsilon_q \hat{K}_q^z - \frac{2}{\alpha} \hat{L}_+ \hat{L}_- \right] + \text{const}$$

We see that for $\alpha = 2/g$ the expression in the square brackets coincides with BCS Hamiltonian (11.19). Therefore, for $\alpha = 2/g$, the BCS Hamiltonian commutes with all $\hat{H}_q$ and thus belongs to the same class of integrable models [25, 26].

In the same way as we did for the BCS model, we employ the mean field approximation to derive classical Hamiltonians that govern the evolution of $\mathbf{s}_j(t) = 2\langle \hat{\mathbf{K}}_j(t) \rangle$ for Hamiltonians (10.10):

$$\begin{cases} H_q = \sum_{j=0}^{n-1} \frac{\mathbf{s}_q \cdot \mathbf{s}_j}{\epsilon_q - \epsilon_j} - \alpha s_q^z \\ H_{BCS} = \sum_{j=0}^{n-1} 2\epsilon_j s_j^z - \frac{J_+ J_-}{\alpha} \end{cases} \quad (10.11)$$

In particular, the classical counterpart of central spin Hamiltonian (10.3) is

$$H_0 = \sum_{j=0}^{n-1} \frac{\gamma_j}{2} \mathbf{s}_0 \cdot \mathbf{s}_j - B s_0^z \quad (10.12)$$

The validity of the mean field approximation for central spin model (10.3) is subject to similar considerations as for the BCS model (see the second paragraph following equation (10.8) and also Ref. [17]).

Thus, the problem of determining the dynamics of the BCS and central spin models reduces to solving equations of motion for Hamiltonians H_0 and H_{BCS} in system (10.11). To obtain solutions for BCS (11.18) and central spin (10.3) models, one has to choose

$$\begin{aligned} \alpha &= \frac{2}{g} \quad \text{BCS} \\ \frac{2}{\epsilon_0 - \epsilon_j} &= \gamma_j \quad \alpha = B \quad \text{central spin} \end{aligned} \quad (10.13)$$

Hamiltonians (10.11) Poisson-commute with each other and with the z -component of the total spin $J_z \propto \sum_q H_q$. Each Hamiltonian in system (10.11) describes an evolution of n spins in an $2n$ -dimensional phase space (two angles for each spin) and has n integrals of motion (the energy and the remaining Hamiltonians from system (10.11)). Therefore, all Hamiltonians (10.11) are classical integrable models.

10.3 The solution

In the previous section we mapped the BCS and central spin models onto a classical integrable system (10.11). Nevertheless, even though Liouville's theorem [28] guarantees a formal integrability in quadratures of classical integrable models, an explicit solution for the evolution of the original dynamical variables is not always possible. Fortunately, this turns out to be not the case for the BCS and central spin problems. For these models, as detailed below, we were able to obtain an explicit general solution of equations of motion.

Before we proceed, let us discuss generic features [28] expected of the dynamics of a classical integrable model with n degrees of freedom. The motion is confined by conservation laws to an n -dimensional subspace of the $2n$ -dimensional phase space. This subspace (invariant torus) is determined by initial values of integrals of motion and is topologically equivalent to an n -dimensional torus. The motion on the invariant torus is characterized by n angle variables, ϕ_k and the corresponding angular frequencies, $\phi_k(t) = \Omega_k t + \phi_k(0)$. The frequencies depend only on integrals of motion, while constants $\phi_k(0)$ are determined by initial values of remaining degrees of freedom. Since for most initial conditions all frequencies Ω_k are independent [37], typical trajectories uniformly explore the entire torus. All these properties are not affected by small perturbations destroying integrability (Kolmogorov-Arnold-Moser theorem).

Now we turn to the solution for the dynamics of the BCS and central spin models. The solution consists of two main steps. The first one is a change of variables that casts equations of motion into the form of a known mathematical problem. In the second step, we use the solution of this problem to obtain dynamical variables $\mathbf{s}_j(t) = 2\langle \hat{\mathbf{K}}_j(t) \rangle$ as explicit functions of time.

The change of variables is facilitated by defining a 2×2 matrix [38] that depends on dynamical variables $\mathbf{s}_j$ and also on an auxiliary parameter u

$$\mathcal{L} = \begin{pmatrix} A(u) & B(u) \\ B^*(u) & -A(u) \end{pmatrix} \quad (10.14)$$

where matrix elements of $\mathcal{L}$ are

$$A(u) = \alpha - \sum_{j=0}^{n-1} \frac{s_j^z}{u - \epsilon_j} \quad B(u) = \sum_{j=0}^{n-1} \frac{s_j^-}{u - \epsilon_j} \quad (10.15)$$

The eigenvalues of this matrix, $\pm v(u)$, depend on dynamical variables only through integrals of motion – H_j and $\mathbf{s}_j^2$

$$v^2(u) = \alpha^2 + \sum_{j=0}^{n-1} \left(\frac{2H_j}{u - \epsilon_j} + \frac{\mathbf{s}_j^2}{(u - \epsilon_j)^2} \right) \equiv \frac{\alpha^2 Q_{2n}(u)}{P_n^2(u)} \quad (10.16)$$

where we defined two polynomials, $Q_{2n}(u)$ and $P_n(u)$ that will be frequently used in subsequent calculations

$$Q_{2n}(u) \equiv \prod_{j=0}^{2n-1} (u - E_j) \equiv \sum_{k=0}^{2n} \alpha_k u^k \quad \alpha_{2n} = 1 \quad (10.17)$$

$$P_n(u) \equiv \prod_{q=0}^{n-1} (u - \epsilon_q) \quad (10.18)$$

Note that because $\mathcal{L}$ is Hermitian for real u , its eigenvalues are real and therefore the polynomial $Q_{2n}(u)$ is positive definite on the real axis. We will often refer to the polynomial $Q_{2n}(u)$ as the spectral polynomial.

Following an algebraic version of variable separation method,[29, 23] we introduce $n - 1$ variables, u_j , as zeroes of $B(u)$ – one of the off-diagonal matrix elements of $\mathcal{L}$. Variables v_j , canonically conjugate to u_j , are given by one of the eigenvalues of matrix $\mathcal{L}$ at $u = u_j$.

$$B(u_j) = 0 \quad v_j = -A(u_j) \quad j = 1, \dots, n - 1 \quad (10.19)$$

Although we will not use the Hamilton-Jacobi method, we remark that variables (u_j, v_j) separate[23] Hamilton-Jacobi equations for Hamiltonians (10.11). Since u_j are zeroes of $B(u)$, we can rewrite the matrix element $B(u)$ as

$$B(u) = J_- \frac{\prod_{j=1}^{n-1} (u - u_j)}{\prod_{j=0}^{n-1} (u - \epsilon_j)} \equiv J_- \frac{R_{n-1}(u)}{P_n(u)} \quad (10.20)$$

This equation is useful for expressing original dynamical variables – components of spins $\mathbf{s}_j$ – through separation variables u_j and J_- . For example, using equations (10.15) and (10.20), we have

$$s_j^- = \text{res}_{u=\epsilon_j} B(u) = J_- \frac{R_{n-1}(\epsilon_j)}{P'_n(\epsilon_j)} \quad (10.21)$$

where the prime denotes a derivative with respect to u .

Equations of motion for central spin Hamiltonian (10.12) in terms of new variables u_j and J_- can be derived by evaluating Poisson brackets between the Hamiltonian and matrix elements of $\mathcal{L}$. We have

$$\dot{u}_j = \frac{iB\sqrt{Q_{2n}(u_j)}}{u_j \prod_{m \neq j} (u_j - u_m)} \frac{s_0^-}{J_-} \quad (10.22)$$

$$\dot{J}_- = iBs_0^- = i\lambda J_- R_{n-1}(0) \quad \lambda = \frac{B}{2^n} \prod_{j=1}^n \gamma_j$$

Similarly, one obtains equations of motion for BCS Hamiltonian (11.20)

$$\begin{aligned} \dot{u}_j &= \frac{2i\sqrt{Q_{2n}(u_j)}}{\prod_{m \neq j} (u_j - u_m)} \\ \dot{J}_- &= -2iJ_- \left(\sum_{j=0}^{n-1} \epsilon_j + \frac{gJ_z}{2} - \sum_{j=1}^{n-1} u_j \right) \end{aligned} \quad (10.23)$$

In equations (10.22) and (11.44) as well as in the rest of the paper, with no loss of generality, we shifted parameters ϵ_j in the BCS and central spin Hamiltonians by a constant so that $\epsilon_0 = 0$.

In the reminder of this section we obtain an explicit general solution for the expectation value of each (pseudo)spin, $\mathbf{s}_j(t) = 2\langle \hat{\mathbf{K}}_j(t) \rangle$, as a function of time. In particular, this includes the expectation value of the central spin, $\mathbf{s}_0(t)$, and the BCS gap function, $\Delta(t) = gJ_-(t)$.

The solution is based on the observation that, with a help of elementary algebra, equations of motion (10.22) and (11.44) can be cast into a form of a known mathematical problem. Specifically, one can rewrite equations (10.22) and (11.44) for variables u_j as

$$\sum_{j=1}^{n-1} \frac{u_j^{l-1} du_j}{\sqrt{Q_{2n}(u_j)}} = dx_l \quad l = 1, \dots, n - 1 \quad (10.24)$$

where the polynomial $Q_{2n}(u)$ is defined in equation (10.16). To obtain equations of motion (11.44) for the BCS Hamiltonian one has to choose

$$\mathbf{x}^T = i(c_1, \dots, c_{n-2}, 2t + c_{n-1}) \quad (10.25)$$

while for the central spin model one has

$$\mathbf{x}^T = -i\lambda(t + c_1, \dots, c_{n-2}, c_{n-1}) \quad (10.26)$$

where $c_1, \dots, c_{n-1}$ are arbitrary constants.

Differential equations (10.24) constitute a well-known mathematical problem called Jacobi's inversion problem (see, for example, Ref. [31] and references therein). The solution of equations (10.24) can be expressed [32] through hyperelliptic Abelian functions – multiple variable generalizations of ordinary elliptic functions. These functions are often encountered as solutions of integrable equations and have well known analytical properties. They are also implemented in standard mathematical software packages [42].

Riemann theta function of *genus* g (in our case $g = n - 1$) is defined as the following sum over all g -dimensional integer vectors

$$\theta(\mathbf{x}|\tau) = \sum_{\mathbf{m} \in \mathbb{Z}^g} \exp [i\pi(\mathbf{m}^T \tau \mathbf{m} + 2\mathbf{x}^T \mathbf{m})] \quad (10.27)$$

where $\tau = \omega' \omega^{-1}$, and ω and ω' are $g \times g$ matrices to be specified below. Klenian σ - and ζ -functions of genus g are defined through the Riemann theta function as [43]

$$\begin{aligned} \sigma(\mathbf{x}) &= C \exp[\mathbf{x}^T \eta (2\omega)^{-1} \mathbf{x}] \theta((2\omega)^{-1} \mathbf{x} | \tau) \\ \zeta_l(\mathbf{x}) &= \frac{\partial \ln \sigma(\mathbf{x})}{\partial x_l} \end{aligned} \quad (10.28)$$

In our case, the $g \times g$ matrices ω and η (matrices of periods) that appear in the definition of hyperelliptic functions (10.27) and (10.28) are

$$\begin{aligned} 2\omega_{kl} &= \oint_{b_k} \frac{u^{l-1} du}{\sqrt{Q_{2n}(u)}} \\ 2\eta_{kl} &= - \oint_{b_k} \frac{du}{4\sqrt{Q_{2n}(u)}} \sum_{k=l+1}^{2n-l} (k-l) \alpha_{k+l} u^{k-1} \end{aligned} \quad (10.29)$$

where contours of integration b_k go around branch cuts of a hyperelliptic curve $z(u) = \sqrt{Q_{2n}(u)}$ as shown on Fig. 1. Matrices ω' and η' are also defined by equations (10.29) with the replacement of the contours of integration $b_k \rightarrow h_k$ (see Fig. 1).

The solution [32] of differential equations (10.24) for arbitrary initial conditions states that functions $u_p(\mathbf{x})$ are zeroes of the following polynomial of a degree $g = n - 1$:

$$R_{n-1}(u, t) = u^{n-1} - \sum_{k=1}^{n-1} f_k(\mathbf{x}) u^{k-1} \quad (10.30)$$

where coefficients of $R_{n-1}(u, t)$ are

$$f_k(\mathbf{x}) = \zeta_k(\mathbf{x} + \mathbf{d}) - \zeta_k(\mathbf{x} - \mathbf{d}) + a_k \quad (10.31)$$

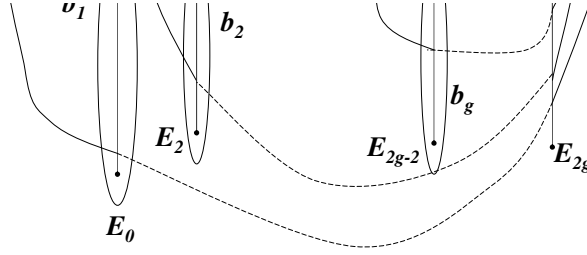


Figure 10.1: Riemann surface of a hyperelliptic curve $z(u) = \sqrt{Q_{2n}(u)}$ of genus [44] $g = n - 1$ showing contours of integration b_k and h_k with $k = 1, \dots, g$. These contours appear in the definition of matrices of periods (10.29) of hyperelliptic functions. Contours b_k go around $g = n - 1$ branch cuts connecting first $2g = 2n - 2$ roots E_k of the spectral polynomial $Q_{2n}(u)$. Integration along the contours is clockwise. Note that since the spectral polynomial is positively defined the branch cuts are parallel to the imaginary axis.

Here vector $\mathbf{x}$ is defined by equations (10.25) and (10.26) for the BCS and central spin problems respectively. The argument of ζ -functions in equations (10.31) is shifted by a vector of constants $\mathbf{d}$ that has the following components:

$$d_l = \int_{E_0}^{\infty} \frac{u^{l-1} du}{\sqrt{Q_{2n}(u)}} \quad l = 1, \dots, n - 1 \quad (10.32)$$

where E_0 is one of the roots of the spectral polynomial $Q_{2n}(u)$ (see Fig. 1 and equation (10.17)). Note that, according to its definition (10.16), the spectral polynomial $Q_{2n}(u)$ depends only on integrals of motion and on parameters ϵ_j and α .

Finally, constants a_k in equations (10.31) are obtained from the expansion

$$\sqrt{\sum_{k=0}^{2n} \frac{\alpha_k}{z^k}} = \frac{1}{z^n} + \frac{a_1}{z^{n-1}} + \dots + \frac{a_{n-1}}{z} + O(1) \quad (10.33)$$

where α_k are coefficients of the spectral polynomial $Q_{2n}(u)$ defined by equation (10.17). For example, $a_{n-1} = \alpha_{2n-1}/2$, $a_{n-2} = \alpha_{2n-2}/2 - \alpha_{2n-1}^2/8$ etc.

To obtain quantum mechanical expectation values of original spins $\mathbf{s}_j(t) = 2\langle \hat{\mathbf{K}}_j(t) \rangle$ for central spin model (10.3) and pseudospins (10.5) for BCS model (11.18), we observe that, according to equation (10.21), they can be expressed through the polynomial $R_{n-1}(u, t)$ given by expression (10.30). We have

$$s_j^-(t) = J_-(t) \frac{R_{n-1}(\epsilon_j, t)}{P'_n(\epsilon_j)} \quad (10.34)$$

Recall that coefficients of the polynomial $P_n(u)$, defined by equation (10.18), depend only on parameters ϵ_j .

Similarly, one can use an expression for the matrix element $A(u)$ defined in equation (10.15) in terms of variables u_j to obtain z -components of (pseudo)spins as functions of time. However, these components are not independent and can be also obtained from the equation $(s_j^z)^2 + |s_j^-|^2 = s_j^2 = 1$, where the sign of s_j^z is determined by initial conditions.

Integrating equations (10.22) and (11.44) for the evolution of x and y components of the total (pseudo)spin with the help of equation (10.31), we obtain

$$J_-(t) = c_n e^{-i\beta t} \frac{\sigma(\mathbf{x} + \mathbf{d})}{\sigma(\mathbf{x} - \mathbf{d})} \quad \Delta(t) = gJ_-(t) \quad (10.35)$$

where vector $\mathbf{x}$ is defined by equations (10.25) and (10.26), the frequency β is different for the BCS (β_{BCS}) and central spin (β_0) models

$$\beta_{BCS} = gJ_z + 2 \sum_{j=0}^{n-1} \epsilon_j - \alpha_{2n-1} \quad \beta_0 = a_1 \lambda \quad (10.36)$$

and λ is defined in equation (10.22).

Average components of the electron (central) spin, $\mathbf{s}_0(t) = 2\langle \hat{\mathbf{K}}_0(t) \rangle$ are obtained from equation (10.34) (recall that $\epsilon_0 = 0$)

$$s_0^-(t) = c'_n [\zeta_1(\mathbf{x} - \mathbf{d}) - \zeta_1(\mathbf{x} + \mathbf{d}) - a_1] \frac{\sigma(\mathbf{x} + \mathbf{d})}{\sigma(\mathbf{x} - \mathbf{d})} \quad (10.37)$$

where $c'_n = (\lambda c_n)/B$.

To summarize, quantum mechanical averages of (pseudo)spins for the BCS and central spin models, including the electron (central) spin and the BCS gap function $\Delta(t)$ are hyperelliptic functions (equations (10.27) and (10.28)) of the time variable and $n-1$ constants c_k (equations (10.34), (10.35), and (10.37)). These functions are specified by their matrices of periods ω , ω' , and η . The latter depend only on integrals of motion according to equations (10.16) and (10.29). The remaining n out of $2n$ initial conditions fix constants c_k in equations (10.25) and (10.26).

An important characteristic of a quasi-periodic motion are its (Fourier) frequencies. To determine the frequencies for the BCS and central spin models, we observe from equation (10.28) that the time variable enters the solution only through combinations of $e^{i\Omega_k t}$, where

$$\begin{aligned} \Omega_k &= i\pi\omega_{kg}^{-1} & \Omega_0 &= \beta_{BCS} & \text{BCS} \\ \Omega_k &= -i\pi\omega_{k1}^{-1} & \Omega_0 &= \beta_0 & \text{central spin} \end{aligned} \quad (10.38)$$

$k = 1, \dots, g$, and $g = n-1$. Note that, since the spectral polynomial $Q_{2n}(u)$ in equation (10.29) is positively defined, the matrix ω is pure imaginary and, therefore, all frequencies Ω_k are real (see Fig. 1).

10.4 Degenerate solutions

Here we identify a class of “resonant” nonstationary solutions for the BCS and central spin models with n degrees of freedom (n spins or pseudospins) but only $m < n$ independent frequencies.

While a typical dynamics of the BCS and central spin models with n (pseudo)spins is quasi-periodic with n independent frequencies, it is clear already on physical grounds that there must be solutions with fewer frequencies. For example, the only “spatial” symmetry of the BCS ground state is the $U(1)$ symmetry of the order parameter that translates into rotations of all pseudospins in Hamiltonians (11.19) and (11.20) around the z -axis. The corresponding trajectory in the phase space is a circle and, therefore, the motion is periodic with a single frequency (see also the discussion following equation (11.20) and below in this section).

Generally, one can show[28] that there are (resonant) tori with an arbitrary number $m < n$ of independent frequencies in a vicinity of any point in the phase space. It should be noted though that the set of points for which $m < n$ has a zero measure in the phase space just like the set of rational numbers on the real axis. Moreover, such points are typically unstable[28] and provide seeds of chaotic behavior in integrable systems. This occurs because small perturbations are able to destroy resonant tori and let the corresponding trajectories escape to other regions of the energy shell. In contrast, the majority of invariant tori with n independent frequencies are only slightly deformed by perturbations.

Interestingly, for the resonant (degenerate) cases considered here we were able to reduce the BCS and central spin models with n spins to same models with only m spins. For example, solutions with $m = 2$ frequencies can be obtained from the BCS and central spin models for only two spins etc. Further, we will see that one spin solutions (periodic trajectories) are special among degenerate solutions in that they correspond to the BCS energy spectrum. The solution that corresponds to the ground state is further exceptional in that it is stable against conservative perturbations because it minimizes the total energy. In this case, even though there is only one frequency, the trajectory cannot leave the resonant torus because the latter coincides with the energy shell. The same applies to the solution that minimizes the central spin Hamiltonian.

We start by determining conditions under which the number of independent frequencies is reduced. Since the frequencies are fixed by initial values of integrals of motion, degeneracies occur only for special values of the integrals. On the other hand, a complete information about integrals of motion is encoded in the spectral polynomial $Q_{2n}(u)$ defined in equations (10.16) and (10.17). Specifically, we have seen in the previous section that the number of frequencies Ω_k is equal to the number of branch cuts of the hyperelliptic curve $z(u) = \sqrt{Q_{2n}(u)}$. Evidently, the latter decreases by one when two roots of the polynomial $Q_{2n}(u)$ coincide. Thus, as detailed below, merging $2(n - m)$ roots of $Q_{2n}(u)$ we obtain solutions with $m < n$ frequencies.

Indeed, consider a situation when $2(n - m)$ roots coincide, i.e. the spectral polynomial $Q_{2n}(u)$ in equation (10.16) has the following special form:

$$Q_{2n}(u) = \tilde{Q}_{2m}(u) \prod_{k=m}^{n-1} (u - E_k)^2 = \tilde{Q}_{2m}(u) W_{n-m}^2(u) \quad (10.39)$$

We assume that double roots in equation (10.39) are complex conjugate to each other so that coefficients of polynomials $\tilde{Q}_{2m}(u)$ and $W_{n-m}(u)$ are real. Note that, since the polynomial $Q_{2n}(u)$ is positively defined (see the remark following equation (10.18)), so is the polynomial $\tilde{Q}_{2m}(u)$. We will see below that degenerate solutions are completely determined by the “residual” spectral polynomial $\tilde{Q}_{2m}(u)$.

Now let us choose $m - n$ separation variables u_j to coincide with double roots of the polynomial $Q_{2n}(u)$, i.e.

$$u_j = E_j \quad j = m, \dots, n-1 \quad (10.40)$$

$$W_{n-m}(u) = \prod_{k=m}^{n-1} (u - E_k) = \prod_{j=m}^{n-1} (u - u_j)$$

We observe that equations of motion for these variables are automatically satisfied because both sides of equations (10.22) and (11.44) vanish.

Next, we use equations (10.39) and (10.16) to express eigenvalues of matrix $\mathcal{L}$ in terms of the residual spectral polynomial $\tilde{Q}_{2m}(u)$ as follows

$$v^2(u) = \alpha^2 \left(\sum_{q=0}^{n-1} \frac{C_q}{u - \epsilon_q} \right)^2 \tilde{Q}_{2m}(u) \quad (10.41)$$

where we expanded the ratio of polynomials $W_{n-m}(u)$ and $P_n(u)$ into elementary fractions

$$\frac{W_{n-m}(u)}{P_n(u)} = \sum_{q=0}^{n-1} \frac{C_q}{u - \epsilon_q} \quad (10.42)$$

Since the degrees of polynomials $W_{n-m}(u)$ and $P_n(u)$ differ by m , there are m constraints on the values of C_q , which can be derived by expanding equation (10.42) in $1/u$. We have

$$\sum_{q=0}^{n-1} C_q \epsilon_q^{k-1} = \delta_{km} \quad k = 1, \dots, m \quad (10.43)$$

Values of C_q can be expressed through coefficients of $\tilde{Q}_{2m}(u)$ by comparing residues at double poles at $u = \epsilon_q$ in equations (10.16) and (10.41).

$$C_q = \frac{e_q}{\alpha \sqrt{\tilde{Q}_{2m}(\epsilon_q)}} \quad \text{where } e_q = \pm 1 \quad (10.44)$$

Note that equations (10.43) provide the following m constraints

$$\sum_{q=0}^{n-1} \frac{e_q \epsilon_q^{k-1}}{\sqrt{\tilde{Q}_{2m}(\epsilon_q)}} = \alpha \delta_{km} \quad k = 1, \dots, m \quad (10.45)$$

on $2m$ coefficients of a positively defined polynomial $\tilde{Q}_{2m}(u)$.

Values of integrals of motion H_q for which degeneracies occur can be determined in terms of coefficients of the residual spectral polynomial $\tilde{Q}_{2m}(u)$ by comparing residues at simple poles at $u = \epsilon_q$ in equations (10.16) and (10.41).

$$H_q = \alpha^2 \sum_{j=0}^{n-1} \frac{C_q C_j \tilde{Q}_{2m}(\epsilon_q)}{\epsilon_q - \epsilon_j} + \frac{\alpha^2}{2} C_q^2 \tilde{Q}'_{2m}(\epsilon_q) \quad (10.46)$$

In particular, the value of H_0 – twice the energy[45] of the central spin model on degenerate solutions is

$$\mathcal{E}_0 = h \sqrt{\tilde{\alpha}_0} e_0 \sum_{q=1}^{n-1} \frac{C_q \gamma_j}{4} + \frac{\tilde{\alpha}_1}{2 \tilde{\alpha}_0} \quad (10.47)$$

where $\tilde{\alpha}_k$ are coefficients of $\tilde{Q}_{2m}(u)$ defined as

$$\tilde{Q}_{2m}(u) \equiv \sum_{k=0}^{2m} \tilde{\alpha}_k u^k \quad \tilde{\alpha}_{2m} = 1 \quad (10.48)$$

The BCS energy $\mathcal{E}_{BCS} \propto \sum_q \epsilon_q H_q + \text{const}$ and the value of $J_z \propto \sum_q H_q$ can be obtained from the knowledge of H_q . However, a more convenient derivation is to compare expansions of equations (10.16) and (10.41) in powers of $1/u$. We have

$$J_z + \alpha \sum_{q=0}^{n-1} C_q \epsilon_q^m = -\frac{\alpha \tilde{\alpha}_{2m-1}}{2} \quad (10.49)$$

$$g \mathcal{E}_{BCS} = - \sum_{q=0}^{m-1} C_q (2 \epsilon_q^{m+1} + \tilde{\alpha}_{2m-1} \epsilon_q^m) - \quad (10.50)$$

$$\frac{\tilde{\alpha}_{2m-1}^2}{4} + \tilde{\alpha}_{2m-2}$$

So far, we determined initial conditions for which the number of independent frequencies for a system with n (pseudo)spins drops from n to m . We saw that these conditions are fixed by the residual spectral polynomial $\tilde{Q}_{2m}(u)$. Note that separation variables u_j with $j \geq m$ are also fixed by this polynomial because they are zeroes of equation (10.42).

Thus, we are left with m dynamical variables – u_j with $j = 1, \dots, m-1$ and J_- . Equations of motion (10.22) and (11.44) for these u_j can be cast into a standard form (10.24) where n is now replaced with m and the original spectral polynomial $Q_{2n}(u)$ is replaced with the residual polynomial $\tilde{Q}_{2m}(u)$. On the other hand, it is clear that we would obtain similar equations of motion for m spins $\tilde{\mathbf{s}}_j$ governed by the BCS or central spin Hamiltonians. This analogy can be made precise if we identify the residual polynomial $\tilde{Q}_{2m}(u)$ with the spectral polynomial of a system of m spins (cf. equation (10.16)), i.e.

$$\tilde{v}^2 = \alpha^2 + \sum_{j=0}^{m-1} \left(\frac{2\tilde{H}_j}{u - \tilde{\epsilon}_j} + \frac{\tilde{\mathbf{s}}_j^2}{(u - \tilde{\epsilon}_j)^2} \right) = \frac{\alpha^2 \tilde{Q}_{2m}(u)}{\tilde{P}_m^2(u)} \quad (10.51)$$

where we replaced parameters ϵ_j with new ones

$$\tilde{P}_m(u) = \prod_{q=0}^{m-1} (u - \tilde{\epsilon}_q) \quad (10.52)$$

and chose $\tilde{\epsilon}_0 = 0$, consistent with the choice $\epsilon_0 = 0$ (see below equation (11.44)). The rest of $\tilde{\epsilon}_j$ are arbitrary real numbers.

BCS (11.20) and central spin (10.12) Hamiltonians for new classical spins $\tilde{\mathbf{s}}_j$ are

$$\begin{aligned} \tilde{H}_{BCS} &= \sum_{j=0}^{m-1} 2\tilde{\epsilon}_j \tilde{s}_j^z - \frac{g}{2} \tilde{J}_+ \tilde{J}_- \quad \tilde{\mathbf{J}} = \sum_{q=0}^{m-1} \tilde{\mathbf{s}}_q \\ \tilde{H}_0 &= \sum_{j=0}^{m-1} \frac{\tilde{\gamma}_j}{2} \tilde{\mathbf{s}}_0 \cdot \tilde{\mathbf{s}}_j - B \tilde{s}_0^z \quad \tilde{\gamma}_j = \frac{2}{\tilde{\epsilon}_0 - \tilde{\epsilon}_j} \end{aligned} \quad (10.53)$$

Now, comparing equations of motion (10.22) and (11.44) for m separation variables $\tilde{u}_j$ for new spins with those for the remaining variables u_j for $j = 1, \dots, m-1$ in the original problem with n spins, we see that they coincide if we re-scale the time variable for the central spin problem as follows

$$\tilde{t} = \frac{(-1)^n e_0 \prod_{k=1}^{m-1} \tilde{\epsilon}_m}{B \sqrt{\tilde{\alpha}_0}} t \quad \text{central spin} \quad (10.54)$$

$$\tilde{t} = t \quad \text{BCS}$$

No re-scaling is necessary for the BCS model. Further, analyzing equations of motion for x and y components of the total spin (10.22) and (11.44), we conclude that they coincide for original and new spins, i.e.

$$\tilde{J}_- = J_-$$

To express original n (pseudo)spins $\mathbf{s}_j(t) = 2\langle \hat{\mathbf{K}}_j(t) \rangle$ in terms of m new spins $\tilde{\mathbf{s}}_j(\tilde{t})$, we use equations (10.42) and (11.51) to rewrite the matrix element $B(u)$ given by equation (10.20) as

$$\begin{aligned} B(u) &= J_- \prod_{j=1}^{m-1} (u - u_j) \sum_{q=0}^{n-1} \frac{C_q}{u - \epsilon_q} = \\ &\tilde{B}(u) \tilde{P}_m(u) \sum_{q=0}^{n-1} \frac{C_q}{u - \epsilon_q} \end{aligned} \quad (10.55)$$

Evaluating residues at $u = \epsilon_j$ in equation (10.55), we obtain according to equation (10.21)

$$s_j^-(t) = C_j \tilde{P}_m(\epsilon_j) \sum_{q=0}^{m-1} \frac{\tilde{s}_j^-(\tilde{t})}{\epsilon_j - \tilde{\epsilon}_q} \quad (10.56)$$

where constants C_j are given by equation (10.44). Similarly, one can relate z -components of original spins to new ones

$$s_j^z(t) = C_j \tilde{P}_m(\epsilon_j) \left(-\alpha + \sum_{q=0}^{m-1} \frac{\tilde{s}_q^z(\tilde{t})}{\epsilon_j - \tilde{\epsilon}_q} \right) \quad (10.57)$$

The solution for new spins $\tilde{s}_j(\tilde{t})$ as functions of time can be read off directly from the general solution – equations (10.34), (10.35), and (10.37) of the previous section. We only need to replace n with m , re-scale the time variable for the central spin model according to equation (10.54), and replace the spectral polynomial $Q_{2n}(u)$ with its reduced counterpart $Q_{2m}(u)$ in the definition of matrices of periods (10.29) of hyperelliptic functions.

For future references, let us also write down equations for components of original spins $\mathbf{s}_j(t) = 2\langle \hat{\mathbf{K}}_j(t) \rangle$ and evolution of $J_-(t)$ for degenerate solutions in terms of m variables $\tilde{u}_j = u_j$. Evaluating residues at $u = \epsilon_j$ in equation (10.55) and using equation (10.21), we find

$$s_j^- = C_j J_- \prod_{j=1}^{m-1} (\epsilon_j - u_j) \quad (10.58)$$

With the help of equation (10.39) and (10.58), evolution equations (10.22) and (11.44) for J_- become

$$\begin{aligned} \frac{d \ln J_-}{dt} &= \frac{(-1)^{m_i e_0}}{\sqrt{\tilde{\alpha}_0}} \prod_{j=1}^{m-1} u_j \quad \text{central spin} \\ \frac{d \ln J_-}{dt} &= i \left(\tilde{\alpha}_{2m-1} + 2 \sum_{j=1}^{m-1} u_j \right) \quad \text{BCS} \end{aligned} \quad (10.59)$$

We see that degenerate solutions with $m < n$ frequencies can be parameterized by m auxiliary spins both for the BCS and central spin model. Therefore, corresponding trajectories live on an m -dimensional torus. By symmetry, for the same initial conditions, trajectories of all Hamiltonians H_q in system (10.11) live on the same torus. This implies, for example, that a slight change of initial values of integrals of motion (10.46) will induce small oscillations along the remaining $n - m$ directions.

Let us consider several examples of degenerate solutions.

One spin solutions, $m = 1$. These solutions reproduce the BCS solution for the ground state and excitations of the BCS model (see also the discussion following equation (11.20)).

Since the residual spectral polynomial is positively defined, we can parameterize it as $\tilde{Q}_2(u) = (u - \mu)^2 + \Delta_0^2$. For $m = 1$, there is only one condition in equation (10.45)

$$\sum_{q=0}^{n-1} \frac{e_q}{\sqrt{(\epsilon_q - \mu)^2 + \Delta_0^2}} = \alpha \quad e_q = \pm 1 \quad (10.60)$$

For the BCS model $\alpha = 2/g$ and we recognize equation (10.60) as the BCS gap equation. Similarly, equation (10.49) becomes the BCS equation for the chemical potential.

The energy of BCS model (10.50) for $m = 1$ is

$$\mathcal{E}_{BCS} = \frac{\Delta_0^2}{g} - \mu J_z - \sum_{j=0}^{n-1} e_j \sqrt{(\epsilon_j - \mu)^2 + \Delta_0^2} \quad (10.61)$$

We see that a choice of signs $e_j = 1$ for all j yields the BCS ground state energy, while if we choose one of e_j to be negative, $e_q = -1$ and $e_j = 1$ for $j \neq q$, the solution corresponds to a pair excitation of energy $2\sqrt{(\epsilon_q - \mu)^2 + \Delta_0^2}$. Similarly, one obtains solutions with two and more excitations by choosing several signs e_j to be negative.

According to equations (10.53), the BCS and central spin models for $m = 1$ reduce to the following single spin models:

$$\tilde{H}_0 = -B\tilde{s}_0^z \quad \tilde{H}_{BCS} = -\frac{g}{2}\tilde{s}_0^+\tilde{s}_0^- \quad (10.62)$$

Then, equation (10.51) provides an alternative parametrization of the residual spectral polynomial $\tilde{Q}_2(u)$

$$\tilde{Q}_2(u) = u^2 - \frac{2\tilde{s}_0^z}{\alpha}u + \frac{\tilde{s}_0^2}{\alpha^2} \quad (10.63)$$

The solution of equations of motion is

$$J_-(t) = \tilde{s}_0^-(\tilde{t}) = \alpha\Delta_0 e^{-i\beta t + i\phi} \quad (10.64)$$

where the frequency β is different for BCS (β_{BCS}) and central spin (β_0) models

$$\beta_{BCS} = 2\mu \quad \beta_0 = \frac{-e_0}{\sqrt{\mu^2 + \Delta_0^2}} \quad e_0 = \pm 1$$

Components of original spins $\mathbf{s}_j(t) = 2\langle \hat{\mathbf{K}}_j(t) \rangle$ are given by equations (10.56) and (10.57). We have

$$s_j^-(t) = \frac{e_j J_-(t)}{\alpha \sqrt{(\epsilon_j - \mu)^2 + \Delta_0^2}} \quad s_j^z = -\frac{e_j(\epsilon_j - \mu)}{\sqrt{(\epsilon_j - \mu)^2 + \Delta_0^2}} \quad (10.65)$$

Note that one spin solutions constructed above do not minimize central spin Hamiltonian (10.12). Instead, we observe that in this case the configuration of spins with minimum energy is $s_j^z = -\text{sgn}\gamma_j$. Similarly, we see from equations (10.11) that configurations that minimize Hamiltonians H_q are $s_j^z = -\text{sgn}(\epsilon_q - \epsilon_j)$. Interestingly, according to the definition of pseudospins (10.5), one of these configurations is the unperturbed Fermi ground state. Since all spins in these configurations are oriented along the z -axis, they are also stationary with respect to BCS Hamiltonian (11.20).

Two spin solutions, $\mathbf{m} = 2$. In this case there are two constraints (10.45) on four coefficients of a positively defined polynomial $\tilde{Q}_4(u)$.

$$\sum_{q=0}^{n-1} \frac{e_q}{\sqrt{\tilde{Q}_4(\epsilon_q)}} = 0 \quad \sum_{q=0}^{n-1} \frac{e_q \epsilon_q}{\sqrt{\tilde{Q}_4(\epsilon_q)}} = \alpha \quad e_q = \pm 1$$

$$\tilde{Q}_4(u) = \sum_{k=0}^4 \tilde{\alpha}_k u^k \quad \tilde{\alpha}_4 = 1$$

Equations (10.47) and (10.50) show that the energy of these solutions can take arbitrary values in the range of energies allowed for the BCS and central spin models.

The equivalent two spin problems (10.53) are

$$\tilde{H}_{BCS} = 2\tilde{\epsilon}_1\tilde{s}_1^z - \frac{g}{2}\tilde{J}_+\tilde{J}_- \quad \tilde{H}_0 = \frac{\tilde{\gamma}_0}{2}\tilde{\mathbf{s}}_0 \cdot \tilde{\mathbf{s}}_1 - B\tilde{s}_0^z \quad (10.66)$$

We are left with only one separation variable $u_1 = \tilde{u}_1$ and equations (10.24) now read

$$\begin{aligned} \dot{u}_1^2 + 4\tilde{Q}_4(u_1) &= 0 \quad \text{BCS} \\ \tilde{\alpha}_0\dot{u}_1^2 + \tilde{Q}_4(u_1) &= 0 \quad \text{central spin} \end{aligned} \quad (10.67)$$

while for the total (pseudo)spin, using equation (10.59), we obtain

$$\begin{aligned} \frac{d \ln J_-}{dt} &= \pm \frac{i}{\sqrt{\tilde{\alpha}_0}} u_1 \quad \text{central spin} \\ \frac{d \ln J_-}{dt} &= i\tilde{\alpha}_3 + 2iu_1 \quad \text{BCS} \end{aligned} \quad (10.68)$$

Components of individual spins can be expressed either through u_1 and J_- or in terms of new spins using equations (10.58) and (10.56)

$$s_j^- = C_j J_- (\epsilon_j - u_1) = C_j (\epsilon_j - \tilde{\epsilon}_1) \tilde{s}_0^-(\tilde{t}) + C_j \epsilon_j \tilde{s}_1^-(\tilde{t}) \quad (10.69)$$

A special case of two spin solutions when the four coefficients of a positively defined polynomial $\tilde{Q}_4(u)$ can be parameterized by three numbers, $\tilde{Q}_4(u) = ((u - \omega)^2 + \Delta_-^2 + \Delta_+^2)^2 - 4\Delta_-^2\Delta_+^2$, has been previously discovered in Ref. [15]. The general two spin solution is obtained from equations (10.67-10.69), which are solved by setting $n = 2$ ($g = 1$) in equations (10.30), (10.34) and (10.35) where hyperelliptic functions now become ordinary elliptic functions.

Similarly, one can obtain three, four etc. spin solutions (degenerate solutions with $m = 3, 4$ etc. frequencies) in terms of hyperelliptic functions of genus $g = 2, 3$ etc. However, as discussed in the beginning of this section, solutions with $m < n$ frequencies are nonrepresentative of the dynamics anywhere in the phase space and are expected to be unstable in the sense of KAM theorem.

Moreover, we emphasize that degenerate solutions derived here are not all solutions with $m < n$ independent frequencies. Indeed, it is clear that for a generic point in the phase space all roots of the spectral polynomial $Q_{2n}(u)$ are distinct and, by continuity, the distance to the closest point where two roots coincide is finite. On the other hand, there are points with any number of frequencies arbitrarily close to any point in the phase space. Other solutions with $m < n$ frequencies can be constructed using the reduction theory for hyperelliptic functions[46].

Note also that degenerate solutions with m frequencies contain ones with fewer frequencies and can be further reduced to $m - 1$, $m - 2$ etc. Geometrically, they live on a $2m$ -dimensional surface in the $2n$ -dimensional phase space (see the remark below equation (10.59)). Surfaces with smaller m are embedded into ones with larger m . Interestingly, periodic trajectories that correspond to the BCS energy spectrum ($m = 1$) belong to all these surfaces.

10.5 Applications and Open Problems

Our results can be directly applied to the problem of decoherence within the central spin model. In this case, we believe, a complete exact answer can be obtained using asymptotics of hyperelliptic functions.

Another possible application is to experiments[2] on electron transport in spin blockaded semiconductor double dots. In this connection, it might be useful to analyze first a simpler

setup of a single dot connected to two polarized leads. This setup leads[47] to an integrable model very similar to the central spin model.

Further, it might be possible to use the BCS model (11.20) with few classical spins to describe a number of grains connected to each other by Josephson junctions.

Other interesting applications include pairing phenomena in cold fermion gases[7] modeled either by BCS or Dicke model and in superconducting metals (see e.g. the discussion in Ref. [13]). In these cases, a careful identification of observable robust features of the solution is needed.

An interesting open problem is the evaluation of leading finite size (quantum) corrections to the general solution obtained in this paper.

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11 Integrable dynamics of coupled Fermi–Bose condensates by Yuzbashyan, Kuznetsov, Altshuler

Abstract

We study the mean-field dynamics of a fermionic condensate interacting with a single bosonic mode (a generalized Dicke model). This problem is integrable and can be mapped onto a corresponding BCS problem. We derive the general solution and a full set of integrals of motion for the time evolution of coupled Fermi-Bose condensates. The present paper complements our earlier study of the dynamics of the BCS model (cond-mat/0407501 and cond-mat/0505493). Here we provide a self-contained introduction to the variable separation method, which enables a complete analytical description of the evolution of the generalized Dicke, BCS, and other similar models.

11.1 Introduction

A number of closely related non-stationary problems have come up recently in different contexts. In these problems the goal is to describe the dynamics of a many-body system following a sudden perturbation that drove the system out of an equilibrium. The system in question can be a BCS superconductor[11, 2, 12, 13, 14, 6, 18, 19, 20, 10, 11], coupled Fermi-Bose condensates[16, 15], or a single electronic spin interacting with many nuclear spins (the central spin model)[14, 15, 16, 17, 18, 19, 20]. A common feature of all these problems is that they can be formulated in terms of spin Hamiltonians, which belong to a class of integrable systems known as Gaudin magnets[25, 29, 23]. It turns out that this fact enables one to solve for the evolution of all these systems using the same approach.

In the present paper we focus on a particular example of such a problem. Namely, we study the dynamics of a fermionic condensate interacting with a single bosonic mode. Recent studies of this problem[16, 15] were motivated by experiments[24, 25, 26, 27] on cold fermion pairing. The idea was that near the Feshbach resonance [28, 29] these systems can be modeled by a fermionic condensate of atoms strongly coupled to the lowest energy bosonic mode.

We assume that the system has been prepared in a nonequilibrium state at $t \leq 0$ and study the subsequent evolution for $t > 0$. Our main result is a complete solution for the dynamics of

the system. We also derive a full set of its integrals of motion. It turns out that the mean-field time evolution of the coupled Fermi-Bose condensates can be mapped onto the corresponding BCS problem, which was solved previously in Refs. [20, 10]. For this reason, the emphasis in the present paper is on this mapping and detailed or alternative derivations that have been largely omitted in Refs. [20, 10].

The fermion-boson condensate is described by the following Hamiltonian:

$$\hat{H} = \sum_{j,\sigma} \epsilon_j \hat{c}_{j\sigma}^\dagger \hat{c}_{j\sigma} + \omega \hat{b}^\dagger \hat{b} + g \sum_j \left(\hat{b}^\dagger \hat{c}_{j\downarrow} \hat{c}_{j\uparrow} + \hat{b} \hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow}^\dagger \right), \quad (11.1)$$

where ϵ_j are the single-particle energy levels and the operators $\hat{c}_{j\sigma}^\dagger$ ($\hat{c}_{j\sigma}$) create (annihilate) a fermion of one of the two species $\sigma = \uparrow$ or $\downarrow$ in an orbital eigenstate of energy ϵ_j . Eigenstates $|j \uparrow\rangle$ and $|j \downarrow\rangle$ are related by the time-reversal symmetry[30]. For example, if the single-particle potential is translationally invariant, $|j \uparrow\rangle = |\mathbf{p} \uparrow\rangle$ and $|j \downarrow\rangle = |-\mathbf{p} \downarrow\rangle$. Operators $\hat{b}^\dagger$ ($\hat{b}$) create (annihilate) quanta of the bosonic field.

We study the dynamics of the fermion-oscillator model (11.1) in the mean-field approximation. This amounts to treating the bosonic field classically, i.e. replacing operators $\hat{b}^\dagger$ and $\hat{b}$ with c -numbers in the Heisenberg equations of motion for Hamiltonian (11.1). This procedure is expected to be exact as long as the bosonic mode is macroscopically populated. It turns out that in this approximation the dynamics coincides with that of a classical Hamiltonian system, which can be mapped onto the corresponding BCS problem (see below). The classical dynamical variables are the time-dependent quantum-mechanical expectation values $\langle \hat{c}_{j\downarrow} \hat{c}_{j\uparrow} \rangle$, $\langle \hat{b} \rangle$, and $\langle \sum_\sigma \hat{c}_{j\sigma}^\dagger \hat{c}_{j\sigma} \rangle$. We will see below that the mean-field approximation is also equivalent to the usual way of obtaining the classical limit – by replacing operators with classical variables and their commutators with Poisson brackets.

Here we assume that the number of energy levels, n , interacting with the bosonic field in the system (11.1) is arbitrary large, but *finite*. In this case the typical evolution is *quasi*-periodic with $n+1$ incommensurate frequencies. The system exhibits irregular multi-frequency oscillations ergodically exploring the part of the phase-space allowed by the conservation laws, returning arbitrarily close to its initial state at irregular time intervals. For finite n , the solution, albeit explicit, is rather complicated. However, it considerably simplifies in the thermodynamic limit $n \rightarrow \infty$. In this limit, the return time diverges for most physical initial conditions, while the solution *decays* to a simple limiting dynamics at large times[31]. For example, following an initial decay, $|\langle \hat{b}(t) \rangle|$ asymptotes either to a constant value (cf. Refs. [2, 12, 18, 11]) or to an oscillatory behavior characterized by only a few frequencies (cf. Fig. 2 in Refs. [13, 6]).

The fermion-oscillator model (11.1) can be viewed as a *generalization* of the Dicke (Tavis-Cummings) model of Quantum Optics. The latter can be obtained from (11.1) in the zero fermionic bandwidth limit, i.e. when all single-particle levels ϵ_j are degenerate, $\epsilon_j = \mu$. To see this, it is useful to reformulate the model (11.1) as a spin-oscillator model using Anderson's pseudospin representation[11]. Pseudospins are defined as

$$2\hat{K}_j^z = \hat{n}_j - 1, \quad \hat{K}_j^- = \hat{c}_{j\downarrow} \hat{c}_{j\uparrow}, \quad \hat{K}_j^+ = \hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow}^\dagger, \quad (11.2)$$

where $\hat{n}_j = \sum_\sigma \hat{c}_{j\sigma}^\dagger \hat{c}_{j\sigma}$. Operators $\hat{\mathbf{K}}_j$ have all properties of spin-1/2 on the subspace of unoccupied and doubly occupied (unblocked) levels ϵ_j . Singly occupied (blocked) levels do not interact with the bosonic field and are decoupled from the dynamics. In terms of pseudospins the Hamiltonian (11.1) for n unblocked levels takes the form

$$\hat{H} = \sum_{j=0}^{n-1} 2\epsilon_j \hat{K}_j^z + \omega \hat{b}^\dagger \hat{b} + g \sum_{j=0}^{n-1} \left(\hat{b}^\dagger \hat{K}_j^- + \hat{b} \hat{K}_j^+ \right). \quad (11.3)$$

In the zero bandwidth limit the Hamiltonian (11.3) describes an interaction of a single collective spin $\hat{\mathbf{T}} = \sum_j \hat{\mathbf{K}}_j$ with a harmonic oscillator

$$\hat{H}_{Dicke} = 2\mu\hat{T}_z + \omega\hat{b}^\dagger\hat{b} + g(\hat{b}^\dagger\hat{T}_- + \hat{b}\hat{T}_+). \quad (11.4)$$

This model was suggested by R. H. Dicke in 1953 [9]. Its spectrum was obtained exactly by M. Tavis and F. W. Cummings in 1967 [33]. In the mean-field approximation it becomes a classical model for the dynamical variables $\langle\hat{\mathbf{T}}(t)\rangle$ and $\langle\hat{b}(t)\rangle$ with only two degrees of freedom (see also below). The mean-field solution for the time evolution was outlined by Dicke[9] and derived in detail by R. Bonifacio and G. Preparata in 1969 [33]. In fact, the resulting classical problem is that of a spherical pendulum. The solution for $\langle\hat{\mathbf{T}}(t)\rangle$ and $\langle\hat{b}(t)\rangle$ is in terms of elliptic functions and (apart from the azimuthal motion of the pendulum) is simply periodic [33].

Remarkably, a more general many-body model (11.1,11.3) also turns out to be integrable. This was established by M. Gaudin in 1972[25] and later by B. Jurco[35] who used a different approach. First explicit nonlinear solutions for the mean-field dynamics of the model (11.1,11.3) were constructed in Refs. [16, 15]. Interestingly, the initial conditions for which these solutions occur are such that the dynamics of the model (11.1,11.3) reduces to that of the Dicke model (11.4), i.e. it can be described in terms of a single *collective* spin coupled to the bosonic field. Similar solutions for the mean-field dynamics of the BCS model were discovered by R. A. Barankov et. al. in Ref. [6] and in an unpublished work by V. S. Shumeiko [14].

Here we solve for the dynamics of the model (11.1,11.3) in the mean-field approximation for arbitrary initial conditions. We also derive a full set of conservation laws for the mean-field dynamics. The typical evolution of the system is quasi-periodic with $n + 1$ incommensurate frequencies. However, for certain special choices of initial conditions the dynamics is characterized by $m + 1 < n + 1$ incommensurate frequencies and can be described in terms of m *collective* spins coupled to the bosonic field.

Our approach to the mean-field dynamics of the fermion-oscillator model (11.1) employs the method of *separation of variables*. This method was suggested by I. V. Komarov[37] and later developed by E. Sklyanin and V. Kuznetsov[29, 23]. It allows us to derive a full set of integrals of motion for both the quantum model (11.1,11.3) and its classical counterpart. We also use it to derive and analyze equations of motion in terms of separation variables.

Upon the replacement of the quantum bosonic field $\hat{b}$ with its time-dependent expectation value, the Hamiltonian (11.1) becomes similar to the mean-field BCS Hamiltonian. In this analogy, $g\langle\hat{b}(t)\rangle$ plays the role of the BCS gap $\Delta(t)$. The difference is that in the case of the BCS model $\Delta(t)$ is not an independent dynamical variable, but is related to $\langle\hat{c}_{j\downarrow}(t)\hat{c}_{j\uparrow}(t)\rangle = \langle\hat{K}_j^-(t)\rangle$ by the self-consistency condition $\Delta(t) = g\sum_j\langle\hat{K}_j^-(t)\rangle$. Nevertheless, it turns out to be possible to map the mean-field dynamics of the fermion-oscillator model (11.1) with n energy levels onto the corresponding BCS problem with $n + 1$ levels. This mapping is facilitated by the variable separation and enables us to obtain the general solution for the time dependence of the expectation values $\langle\hat{\mathbf{K}}_j(t)\rangle$ and $\langle\hat{b}(t)\rangle$ in terms of *hyperelliptic* functions[31, 39] using the known solution for the mean-field dynamics of the BCS model[20, 10].

The rest of the paper is organized as follows. In section 11.2, we derive the classical Hamiltonian that governs the mean-field dynamics of the quantum model (11.1,11.3). We show that the dynamics of the resulting classical model is integrable and derive a full set of its conservation laws in section 11.3. In section 11.4, we perform a transformation to a new set of variables, which facilitates the solution of the equations of motion. The mapping to the corresponding BCS problem and the general solution for the mean-field dynamics of the model (11.1,11.3) are derived in section 11.6. Finally, section 11.7 is devoted to an interesting class of particular solutions that include mean-field equilibrium states and special solutions of Refs. [16, 15, 9, 33].

11.2 Classical Model

Here we derive the classical Hamiltonian model that describes the mean-field dynamics of the quantum model (11.1,11.3).

We start with the Heisenberg equations of motion for the spin-oscillator model (11.3).

$$\begin{aligned}\dot{\hat{\mathbf{K}}}_j &= i [\hat{\mathbf{H}}, \hat{\mathbf{K}}_j] = \hat{\mathbf{B}}_j \times \hat{\mathbf{K}}_j & \hat{\mathbf{B}}_j &= (2g\hat{\mathbf{b}}_x, 2g\hat{\mathbf{b}}_y, 2\epsilon_j), \\ \dot{\hat{b}} &= i[\hat{H}, \hat{b}] = -i\omega\hat{b} - ig\hat{T}_- & \hat{\mathbf{T}} &= \sum_{q=0}^{n-1} \hat{\mathbf{K}}_q,\end{aligned}\tag{11.5}$$

where the operators $\hat{b}_x$ and $\hat{b}_y$ are defined by

$$\hat{b} = \hat{b}_x - i\hat{b}_y \quad \hat{b}^\dagger = \hat{b}_x + i\hat{b}_y.$$

In the regime when the bosonic mode is macroscopically populated, we can replace operators $\hat{b}(t)$ and $\hat{b}^\dagger(t)$ in Eqs. (11.5) with c -numbers: $b(t) = \langle \hat{b}(t) \rangle$ and $\bar{b}(t) = \langle \hat{b}^\dagger(t) \rangle$, where $\langle \dots \rangle$ stands for the time-dependent quantum-mechanical expectation value. After this replacement, Eqs. (11.5) become linear in operators. Taking their quantum-mechanical expectation value, we obtain for $\mathbf{s}_j(t) = \langle \hat{\mathbf{K}}_j(t) \rangle$

$$\begin{aligned}\dot{\mathbf{s}}_j &= \mathbf{B}_j \times \mathbf{s}_j & \mathbf{B}_j &= (2gb_x, 2gb_y, 2\epsilon_j), \\ \dot{b} &= -i\omega b - igJ_- & \mathbf{J} &= \sum_{j=0}^{n-1} \mathbf{s}_j.\end{aligned}\tag{11.6}$$

An important observation is that Eqs. (11.6) are Hamiltonian equations for the following classical model:

$$H = \sum_{j=0}^{n-1} 2\epsilon_j s_j^z + \omega \bar{b}b + g \sum_{j=0}^{n-1} (\bar{b}s_j^- + bs_j^+),\tag{11.7}$$

where $s_j^\pm = s_j^x \pm is_j^y$. This Hamiltonian governs n *classical spins* interacting with a classical harmonic oscillator[40].

Classical spins are similar to the angular momentum and have the same Poisson brackets

$$\begin{aligned}\{s_j^a, s_j^b\} &= -\varepsilon_{abc} s_j^c, \\ \{b, \bar{b}\} &= i \quad \{b_x, b_y\} = \frac{1}{2},\end{aligned}\tag{11.8}$$

where a , b , and c stand for the spatial indexes x , y , and z . All other Poisson brackets between components of $\mathbf{s}_j$ and b vanish. Equations (11.8) determine the fundamental Poisson brackets in our problem in a sense that Poisson brackets between any other pair of dynamical variables (functions of $b, \bar{b}$ and components of $\mathbf{s}_j$) can be obtained from Eqs. (11.8) using the standard properties of Poisson brackets[41]. For example, Eqs. (11.6) can be obtained from the Hamiltonian equations of motion $\dot{\mathbf{s}}_j = \{\mathbf{H}, \mathbf{s}_j\}$ and $\dot{b} = \{H, b\}$ using Eqs. (11.8) and (11.7).

Equations of motion (11.6) conserve the length of each spin, $\mathbf{s}_j^2 = \text{const}$. This fact can also be viewed as a property of the brackets (11.8) – since $\mathbf{s}_j^2$ Poisson-commutes with all other dynamical variables, its bracket with the Hamiltonian (11.7) also vanishes. In the mean-field approximation eigenstates of the Hamiltonian (11.3) are product states. In a pure state of a spin-1/2, there is always an axis, $\mathbf{n}$, such that the projection of the spin onto it is 1/2, i.e.

$\hat{\mathbf{K}}_j \cdot \mathbf{n} = 1/2$. In this case, $\mathbf{s}_j^2 = \langle \hat{\mathbf{K}}_j \rangle^2 = 1/4$. Therefore, if the system was in a product state at $t = 0$

$$\mathbf{s}_j^2 = (s_j^z)^2 + s_j^+ s_j^- = \frac{1}{4}. \quad (11.9)$$

If a number of orbitals ϵ_j in (11.7) are degenerate, the magnitude of their total classical spin $\sum_{\epsilon_j=\epsilon} \mathbf{s}_j$ is conserved by the equations of motion (11.6). In this case, one can replace the corresponding spins $\mathbf{s}_j$ with a single classical spin of a larger magnitude,

$$\sum_{\epsilon_j=\epsilon} \epsilon_j \mathbf{s}_j = \epsilon \mathbf{S}_\epsilon, \quad \text{where} \quad \mathbf{S}_\epsilon = \sum_{\epsilon_j=\epsilon} \mathbf{s}_j, \quad (11.10)$$

in the Hamiltonian (11.7) and sum over nondegenerate orbitals only. Below we will assume whenever necessary that such a replacement has been made, i.e. that orbitals ϵ_j are nondegenerate.

Finally, comparing Hamiltonians (11.3) and (11.7) and brackets (11.8) to the corresponding quantum commutators, we note that the mean-field approximation is equivalent to the standard procedure of going from quantum to classical mechanics by replacing commutators with Poisson brackets, $i[A, B] \rightarrow \{A, B\}$ (here $\hbar = 1$).

11.3 Integrability

Here we demonstrate that the classical model (11.7), as well as its quantum counterparts (11.3) and (11.1), are integrable and introduce a useful tool to analyze and solve for their dynamics.

The Hamiltonian (11.7) depends on $2(n+1)$ dynamical variables: 2 angles for each of n spins plus the coordinate and the momentum of the oscillator[40]. Therefore, its phase space is $2(n+1)$ -dimensional and the number of the degrees of freedom (the number of generalized coordinates) is $n+1$. To show that the classical model (11.7) is integrable, we have to show that it has $n+1$ independent integrals of motion (see e.g. Refs. [28, 41]).

Consider the following vector-functions of an auxiliary parameter u :

$$\mathbf{L}_j(u) = \frac{\mathbf{s}_j}{u - \epsilon_j} \quad j = 0, \dots, n-1$$

$$\mathbf{L}_n(u) = \frac{1}{g^2} \begin{pmatrix} 2gb_x \\ 2gb_y \\ 2u - \omega \end{pmatrix}.$$

It follows from the above definitions and Eqs. (11.8) that components of $\mathbf{L}_k(u)$ have the following Poisson brackets:

$$\{L_k^a(v), L_k^b(w)\} = \frac{\varepsilon_{abc}}{v-w} (L_k^c(v) - L_k^c(w)) \quad k = 0, \dots, n \quad (11.11)$$

Because components of different $\mathbf{L}_j(u)$ Poisson-commute with each other, their sum

$$\mathbf{L}(u) = \frac{1}{g^2} \begin{pmatrix} 2gb_x \\ 2gb_y \\ 2u - \omega \end{pmatrix} + \sum_{j=0}^{n-1} \frac{\mathbf{s}_j}{u - \epsilon_j} \quad (11.12)$$

also satisfies Eqs. (11.11). One can check using only Eq. (11.11) that the square of this vector commutes with itself at any two values, v and w , of the auxiliary parameter, i.e.

$$\{\mathbf{L}^2(v), \mathbf{L}^2(w)\} = 0. \quad (11.13)$$

The function $\mathbf{L}^2(u)$ acts as a generating function for the model (11.7) and its integrals of motion. Indeed, evaluating $\mathbf{L}^2(u)$ from Eq. (11.12), we obtain

$$\mathbf{L}^2(u) = \frac{(2u - \omega)^2}{g^4} + \frac{4H_n}{\omega g^2} + \sum_{j=0}^{n-1} \left[\frac{2H_j}{g^2(u - \epsilon_j)} + \frac{\mathbf{s}_j^2}{(u - \epsilon_j)^2} \right]. \quad (11.14)$$

where

$$H_j = (2\epsilon_j - \omega)s_j^z + g(\bar{b}s_j^- + bs_j^+) + g^2 \sum_{k \neq j} \frac{\mathbf{s}_j \cdot \mathbf{s}_k}{\epsilon_j - \epsilon_k}, \quad (11.15)$$

$$H_n = \omega(\bar{b}b + J_z) \quad \mathbf{J} = \sum_{j=0}^{n-1} \mathbf{s}_j. \quad (11.16)$$

Hamiltonians H_j defined by Eqs. (11.15,11.16) Poisson-commute with each other

$$\{H_j, H_k\} = 0 \quad \text{for all } j \text{ and } k.$$

This follows directly from the fact that Eq. (11.13) holds for any v and w . On the other hand, it is straightforward to verify that the spin-oscillator Hamiltonian (11.7) is a linear combination of H_j

$$H = \sum_{j=0}^n H_j. \quad (11.17)$$

Since H_j Poisson-commute with each other, the Hamiltonian (11.7) commutes with all $n + 1$ Hamiltonians H_j . Therefore, H_j given by Eqs. (11.15,11.16) are integrals of motion for the classical spin-oscillator model, i.e. this model is integrable.

The above construction based on Eqs. (11.11) and (11.12) constitutes the so-called Lax vector representation for the spin-oscillator model (11.7). Its main advantages are that it provides a powerful tool to study the dynamics and can also be extended to quantum models. In particular, note that integrals (11.15) and (11.16) can be trivially generalized to the quantum case

$$\hat{H}_j = (2\epsilon_j - \omega)\hat{K}_j^z + g(\hat{b}^\dagger \hat{K}_j^- + \hat{b} \hat{K}_j^+) + g^2 \sum_{j \neq k} \frac{\hat{\mathbf{K}}_j \hat{\mathbf{K}}_k}{\epsilon_j - \epsilon_k},$$

$$\hat{H}_n = \omega(\hat{b}^\dagger \hat{b} + \hat{T}_z) \quad \hat{\mathbf{T}} = \sum_{q=0}^{n-1} \hat{\mathbf{K}}_q.$$

Operators $\hat{H}_j$ pairwise commute with each other and the spin-oscillator Hamiltonian (11.3), which is their linear combination as in Eq. (11.17).

The same construction with classical dynamics in the mean-field approximation and Lax vector for resulting classical models applies to a number of other models. In particular, a closely related model is the BCS model

$$\hat{H}_{BCS} = \sum_{j,\sigma} \epsilon_j \hat{c}_{j\sigma}^\dagger \hat{c}_{j\sigma} - g \sum_{j,q} \hat{c}_{j\uparrow}^\dagger \hat{c}_{j\downarrow}^\dagger \hat{c}_{q\downarrow} \hat{c}_{q\uparrow}. \quad (11.18)$$

Notations here are the same as in Eq. (11.1). In terms of Anderson's spins (see above Eq. (11.3)), the Hamiltonian reads

$$\hat{H}_{BCS} = \sum_{j=0}^n 2\epsilon_j \hat{K}_j^z - g \sum_{j,q} \hat{K}_j^+ \hat{K}_q^-. \quad (11.19)$$

The usual BCS mean-field is equivalent to the procedure (Eqs. (11.5) and (11.6)) we used to derive the classical Hamiltonian (11.7), only now the role of $\hat{b}$ is played by $\sum_j \hat{K}_j^-$ (see e.g. Refs. [6, 20]). The mean-field dynamics is described by the following classical Hamiltonian:

$$H_{BCS} = \sum_{j=0}^n 2\epsilon_j s_j^z - g \sum_{j,q} s_j^+ s_q^- \quad (11.20)$$

with the Lax vector

$$\mathbf{L}_{BCS}(u) = -\frac{\hat{\mathbf{z}}}{g} + \sum_j \frac{\mathbf{s}_j}{u - \epsilon_j}, \quad (11.21)$$

where $\hat{\mathbf{z}}$ is a unit vector along the z -axis. The Lax vector $\mathbf{L}_{BCS}(u)$ has the same properties (11.11) and (11.13) as before and generates integrals of motion for the BCS model.

11.4 Separation of variables

In this section we perform a transformation to a new set of variables, which facilitates the solution of the equations of motion (11.6). The choice of new variables naturally follows from the Lax vector construction of the previous section. The new variables are canonical and also separate the Hamilton-Jacobi equations for the classical Hamiltonian (11.7) in the usual sense [28, 41].

We define n variables u_j as zeros of

$$L_-(u) = L_x(u) - iL_y(u) = \frac{2b}{g} + \sum_{j=0}^{n-1} \frac{s_j^-}{u - \epsilon_j}. \quad (11.22)$$

Note that the nominator of this expression is a polynomial of degree n and therefore there are n roots u_j with $j = 0, \dots, n-1$. The coefficients of this polynomial are functions of the dynamical variables s_j^- and b . Accordingly, its roots are also functions of s_j^- and b and therefore define a new set of dynamical variables.

Variables u_j play a role of canonical coordinates for the classical Hamiltonian (11.7). The corresponding momenta are defined as $v_j = L_z(u_j)$. Thus, we have

$$L_-(u_j) = 0, \quad v_j = L_z(u_j), \quad j = 0, \dots, n-1. \quad (11.23)$$

Because our system has $2(n+1)$ degrees of freedom, we need two additional variables u_n and v_n , which can be introduced as

$$u_n = b, \quad v_n = \frac{H_n}{\omega b} = \bar{b} + \frac{J_z}{b}, \quad (11.24)$$

where H_n is given by Eq. (11.16).

Separation variables (u_j, v_j) are canonical, i.e.

$$\{u_j, u_k\} = 0 \quad \{v_j, v_k\} = 0 \quad \{v_j, u_k\} = -i\delta_{jk} \quad j, k = 0, \dots, n. \quad (11.25)$$

The first relation in Eq. (11.25) follows from the fact that by Eqs. (11.23) and (11.24) variables u_j depend only on mutually Poisson-commuting variables s_k^- and b . The second relation follows from $\{L_z(v), L_z(w)\} = \{L_z(v), b\} = 0$. To derive Poisson brackets between v_j and u_k for $j, k = 0, \dots, n-1$; we use the following equation obtained from Eq. (11.11)

$$\{L_z(v), L_-(w)\} = \frac{i}{v-w} (L_-(v) - L_-(w)). \quad (11.26)$$

Evaluating this expression at $v = u_j$ and $w = u_k$, we obtain

$$\{L_z(u_j), L_-(w)\}_{w=u_k} = \lim_{w \rightarrow u_k} \frac{-i}{u_j - w} L_-(w) = iL'_-(u_k) \delta_{jk}. \quad (11.27)$$

where $L'(u) = \partial L / \partial u$. On the other hand,

$$\{L_z(u_j), L_-(w)\}_{w=u_k} = \{L_z(u_j), L_-(u_k)\} - L'_-(u_k) \{L_z(u_j), u_k\} = -L'_-(u_k) \{L_z(u_j), u_k\}, \quad (11.28)$$

where we used $L_-(u_k) = 0$. Comparing Eqs. (11.27) and (11.28), we obtain the last relation in Eq. (11.25) for $j, k = 0, \dots, n-1$.

The original dynamical variables $\mathbf{s}_j = \langle \hat{\mathbf{K}}_j(t) \rangle$ and $b = \langle \hat{b}(t) \rangle$ that we are interested in, can be explicitly expressed in terms of u_j (the inverse map). Indeed, note that $L_-(u)$ has its zeros at $u = u_j$ and poles at $u = \epsilon_j$. Therefore, using Eqs. (11.23) and (11.22), we can write it as

$$L_-(u) = \frac{2b}{g} + \sum_{j=0}^{n-1} \frac{s_j^-}{u - \epsilon_j} = \frac{2b}{g} \frac{\prod_k (u - u_k)}{\prod_j (u - \epsilon_j)}, \quad (11.29)$$

where we have also used $L_-(u) = 2b/g + O(1/u)$ for large u . Equating residues at $u = \epsilon_j$ and $u = \infty$, we obtain

$$s_j^- = \frac{2b}{g} \frac{\prod_k (\epsilon_j - u_k)}{\prod_{l \neq j} (\epsilon_j - \epsilon_l)}, \quad (11.30)$$

$$J_- = \sum_j s_j^- = \frac{2b}{g} \sum_{j=0}^{n-1} (\epsilon_j - u_j). \quad (11.31)$$

Similarly, using

$$L_z(u_j) = v_j \quad L_z(u) = \frac{2u - \omega}{g^2} + O(1/u),$$

one can express $L_z(u)$ in terms of u_j . The z -components of classical spins, s_j^z , are residues of $L_z(u)$ at $u = \epsilon_j$ (see Eq. (11.12))

$$s_j^z = s_j^- \left[\frac{\epsilon_j - \omega/2 + \sum_k (u_k - \epsilon_k)}{bg} + \sum_k \frac{v_k}{(\epsilon_j - u_k) L'_-(u_k)} \right]. \quad (11.32)$$

It also follows from Eq. (11.23) that $v_j^2 = \mathbf{L}^2(u_j)$. This allows us to express variables v_j through u_j and the integrals of motion H_j

$$v_j^2 = \frac{(2u_j - \omega)^2}{g^4} + \frac{4H_n}{\omega g^2} + \sum_{k=0}^{n-1} \left[\frac{2H_k}{g^2(u_j - \epsilon_k)} \frac{\mathbf{s}_k^2}{(u_j - \epsilon_k)^2} \right] \quad j = 0, \dots, n-1 \quad (11.33)$$

$$v_n = \frac{H_n}{\omega u_n} = \frac{H_n}{\omega b}. \quad (11.34)$$

Thus, to determine the evolution of $\mathbf{s}_j(t) = \langle \hat{\mathbf{K}}_j(t) \rangle$ and $b(t) = \langle \hat{b}(t) \rangle$ we only need to derive and solve the equations of motion for $u_j(t)$.

11.5 Equations of motion for separation variables

In order to derive the equations of motion for separation variables u_j , we first determine the brackets $u_{l,k} = \{H_k, u_l\}$ for $l, k = 0, \dots, n-1$ using Eqs. (11.33) and (11.25). This is done by taking Poisson brackets of both sides of Eq. (11.33) with u_l and solving the resulting system of linear equations for $u_{l,k}$.

As soon as $u_{l,k}$ are found in terms of u_j and v_j , we can use the expression (11.17) for the Hamiltonian in terms of H_j to determine $\dot{u}_j$

$$\dot{u}_j = \{H, u_j\} = \sum_k u_{j,k}. \quad (11.35)$$

Taking Poisson brackets of both sides of Eqs. (11.33) and (11.34) with u_l , we obtain

$$-ig^2 v_j \delta_{jl} = \sum_{k=0}^{n-1} \frac{u_{l,k}}{u_j - \epsilon_k} \quad j, l = 0, \dots, n-1. \quad (11.36)$$

In order to determine $u_{n,k}$ and $u_{l,k}$ for $l, k = 0, \dots, n-1$, we need to invert the matrix

$$A_{jk} = \frac{1}{u_j - \epsilon_k}, \quad (11.37)$$

which is called the Cauchy matrix.

This can be done with the help of the following identity coming from partial fraction decomposition of the LHS:

$$\frac{\prod_{l=0}^{n-1} (u - u_l)}{(u - u_p) \prod_{l=0}^{n-1} (u - \epsilon_l)} = \sum_{j=0}^{n-1} \frac{1}{u - \epsilon_j} \frac{\prod_{l \neq p} (\epsilon_j - u_l)}{\prod_{l \neq j} (\epsilon_j - \epsilon_l)}. \quad (11.38)$$

To verify this identity note that both sides have the same poles and that the residues at these poles coincide. Substituting $u = u_q$ in Eq. (11.38), one derives

$$\delta_{qp} = \sum_{j=0}^{n-1} \frac{1}{u_q - \epsilon_j} \frac{\prod_{l=1}^n (u_p - \epsilon_l)}{\prod_{l \neq j} (\epsilon_j - \epsilon_l)} \prod_{l \neq p} \frac{\epsilon_j - u_l}{u_p - u_l}. \quad (11.39)$$

Hence

$$(A^{-1})_{jk} = \frac{\prod_{l=1}^n (u_k - \epsilon_l)}{\prod_{l \neq j} (\epsilon_j - \epsilon_l)} \prod_{l \neq k} \frac{\epsilon_j - u_l}{u_k - u_l}. \quad (11.40)$$

It follows from Eqs. (11.36) and (11.30) that

$$u_{j,k} = -ig^2 v_j \frac{\prod_{l=0}^{n-1} (u_j - \epsilon_l)}{\prod_{l \neq j} (u_j - u_l)} \frac{gs_k^-}{2b(\epsilon_k - u_j)}. \quad (11.41)$$

Finally, noticing that $0 = L_-(u_j) = 2b/g + \sum_k s_k^- / (u_j - \epsilon_k)$ and using Eqs. (11.35) and (11.33), we derive

$$\begin{aligned} \dot{u}_j &= -\frac{2i\sqrt{Q_{2n+2}(u_j)}}{\prod_{m \neq j} (u_j - u_m)} \quad j = 0, \dots, n-1 \\ b &= -2ib \left(\frac{\omega}{2} + \sum_{j=0}^{n-1} \epsilon_j - \sum_{k=0}^{n-1} u_k \right), \end{aligned} \quad (11.42)$$

where the *spectral polynomial* $Q_{2n+2}(u)$ is defined as

$$Q_{2n+2}(u) = g^4 \mathbf{L}^2(u) \prod_{j=0}^{n-1} (u - \epsilon_j)^2. \quad (11.43)$$

By Eq. (11.14), the coefficients of $Q_{2n+2}(u)$ depend only on the integrals of motion H_j and parameters ϵ_j and g . The equation of motion for $u_n = b$ is obtained by substituting Eq. (11.31) into Eq. (11.6).

An almost identical derivation of equations of motion for separation variables can be performed for the mean-field BCS model (11.20) using its vector Lax representation (11.21) leading to the following equations of motion:

$$\begin{aligned} \dot{u}_j &= -\frac{2i\sqrt{Q_{2n+2}^{BCS}(u_j)}}{\prod_{m \neq j} (u_j - u_m)} \quad j = 0, \dots, n-1 \\ \dot{J}_- &= -2iJ_- \left(gJ_z + \sum_{j=0}^n \epsilon_j - \sum_{k=0}^{n-1} u_k \right), \end{aligned} \quad (11.44)$$

where the spectral polynomial for the BCS model is defined as

$$Q_{2n+2}^{BCS}(u) = \frac{1}{g^2} \mathbf{L}_{BCS}^2(u) \prod_{j=0}^n (u - \epsilon_j)^2. \quad (11.45)$$

11.6 The solution

Here we obtain the general solution for the mean-field dynamics of the fermion-oscillator (spin-oscillator) model (11.1, 11.3) by mapping it onto the corresponding BCS model. The solution therefore can be read off from the known general solution of the mean-field BCS problem [20].

By comparing equations of motion (11.42) for the spin-oscillator model with those for BCS model (11.44), we observe that they coincide upon a replacement

$$\sum_{j=0}^n \epsilon_j + gJ_z \rightarrow \sum_{j=0}^{n-1} \epsilon_j + \frac{\omega}{2}, \quad J_- \rightarrow b, \quad Q_{2n+2}^{BCS}(u) \rightarrow Q_{2n+2}(u). \quad (11.46)$$

Thus, the mean-field dynamics of the spin-oscillator model (11.3) with n spins and a bosonic field coincides with that of the BCS model (11.19) with $n+1$ spins. This allows us to immediately write down the solution for the time-dependent averages $\mathbf{s}_j(t) = \langle \hat{\mathbf{K}}_j(t) \rangle$ and $b(t) = \langle \hat{b}(t) \rangle$

$$b(t) = [J_-]_{BCS}, \quad [\mathbf{s}_j(t)]_{\text{Dicke}} = [\mathbf{s}_j(t)]_{BCS}. \quad (11.47)$$

The explicit form of $[J_-]_{BCS}$ and $[\mathbf{s}_j(t)]_{BCS}$ in terms of hyperelliptic functions can be found in Ref. [20] (see Eqs. (3.22 – 3.24) of Ref. [20]).

The dynamics of $\mathbf{s}_j(t) = \langle \hat{\mathbf{K}}_j(t) \rangle$ and $b(t) = \langle \hat{b}(t) \rangle$ is typical of a classical integrable system with $n+1$ degrees of freedom [28] (recall that n is the number of non-degenerate single particle levels ϵ_j in the fermion-oscillator model (11.1)). The typical motion is quasi-periodic with $n+1$ incommensurate periods. For example, a Fourier transform of $b(t) = \langle \hat{b}(t) \rangle$ shows $n+1$ basic frequencies. The system uniformly (ergodically) explores the invariant torus – the $n+1$ -dimensional portion of the $2n+2$ -dimensional phase space allowed by the integrals of motion (11.15) and (11.16), returning arbitrarily close to the initial point at irregular intervals.

We found that in the thermodynamic limit $n \rightarrow \infty$ the solution simplifies for most physical initial conditions. The return time diverges in this limit. The dynamics of $b(t)$ is particularly simple – it decays, typically as a power-law, to a steady state where $|b(t)|$ is either constant or is characterized by a few independent frequencies[31]. However, the motion of the spin system still contains a continuum of frequencies. The final steady state of $|b(t)|$ depends on the initial conditions.

11.7 Few spin solutions

The evolution described in the previous section occurs for most initial conditions and is stable against small perturbations of the Hamiltonian by the KAM theorem even if these perturbations destroy the integrability. However, there always exists a set of points of measure zero in the phase space, where the motion is characterized by only a few incommensurate frequencies, while the stability is not guaranteed [28]. Below we consider a family of such solutions, which we call *m-spin solutions* with $m < n$. The reason is that in these cases the dynamics of n spins and the bosonic field in Hamiltonian (11.3) degenerates to that of $m < n$ spins and the bosonic field.

Few spin solutions are constructed by choosing integrals of motion H_j (i.e. the initial conditions) so that $2(n - m)$ roots of the spectral polynomial $Q_{2n+2}(u)$ defined in Eq. (11.14) become double degenerate. Suppose $u = E_0$ is a double root[43] of $Q_{2n+2}(u)$ and note that, since $Q_{2n+2}(E_0) = \mathbf{L}^2(E_0) = 0$, setting e.g. $u_{n-1}(t) = E_0$ solves the equation of motion (11.42) for the variable u_{n-1} . Therefore, we can "freeze" one of the separation variables in this root. Then $L_-(E_0) = 0$, and it follows from $L_z(u) + L_-(u)L_+(u) = \mathbf{L}^2(u)$ that $L_z(E_0) = 0$. Thus, one can factor out $(u - E_0)$ from all components of the Lax vector and show that it is proportional to the Lax vector of the spin-oscillator model with $n - 1$ spins. Similarly, if there are 2 pairs of double degenerate roots, the number of spins reduces to $n - 2$ etc. This procedure is followed in detail in Ref. [20] using a different method.

First, let us consider the general case when the number of spins effectively reduces from n to $m < n$. We have

$$\mathbf{L}(u) = \left(1 + \sum_j \frac{d_j}{u - \epsilon_j} \right) \mathbf{L}_t(u), \quad (11.48)$$

where d_j are time-independent constants and $\mathbf{L}_t(u)$ is the Lax vector of the m -spin problem

$$\mathbf{L}_t(u) = \frac{1}{g^2} \begin{pmatrix} 2gb_x \\ 2gb_y \\ 2u - \omega' \end{pmatrix} + \sum_{k=0}^{m-1} \frac{\mathbf{t}_k}{u - \mu_k}. \quad (11.49)$$

The m -spin system has its own m arbitrary "energy levels" μ_k . Its dynamics is governed by the spin-oscillator Hamiltonian (11.7) for m spins with new parameters replacing ϵ_j and ω .

$$H_m = \sum_{k=0}^{m-1} 2\mu_k t_k^z + \omega' \bar{b}b + g \sum_{k=0}^{m-1} (\bar{b}t_k^- + bt_k^+). \quad (11.50)$$

Matching the residues at poles at $u = \epsilon_j$ on both sides of Eq. (11.48), we express the original spins $\mathbf{s}_j$ in terms of *collective* spins $\mathbf{t}_k$

$$\mathbf{s}_j = d_j \mathbf{L}_t(\epsilon_j). \quad (11.51)$$

Constants d_j are determined from the above equation using $\mathbf{s}_j^2 = 1/4$. We have

$$d_j = \frac{|\mathbf{s}_j|e_j}{\sqrt{\mathbf{L}_t^2(\epsilon_j)}} = \frac{e_j}{2\sqrt{\mathbf{L}_t^2(\epsilon_j)}} \quad e_j = \pm 1. \quad (11.52)$$

Finally we have to match the residues at $u = \mu_k$ and the $u \rightarrow \infty$ asymptotic. This leads to the following $m + 1$ equations

$$1 + \sum_{j=0}^{n-1} \frac{d_j}{\mu_k - \epsilon_j} = 0 \quad k = 0, \dots, m-1 \quad (11.53)$$

$$\omega = \omega' - 2 \sum_{j=0}^{n-1} d_j.$$

Equations (11.53) constrain the lengths of new spins $\mathbf{t}_k$, i.e. the coefficients of the spectral polynomial $Q_{2m+2}(u) = g^4 \mathbf{L}_{\mathbf{t}}^2(u) \prod_{k=0}^{m-1} (u - \mu_k)^2$ of the m -spin system (see Eq. (11.14)). Indeed, using Eq. (11.52), one can cast the constraints Eq (11.53) into the form

$$\sum_{j=0}^{n-1} \frac{e_j \epsilon_j^{l-1}}{\sqrt{Q_{2m+2}(\epsilon_j)}} = \frac{2}{g^2} \delta_{lm} \quad l = 1, \dots, m \quad (11.54)$$

$$\omega' = \omega + 2 \sum_{k=0}^{m-1} \mu_k - \sum_{j=0}^{n-1} \frac{g^2 e_j \epsilon_j^m}{\sqrt{Q_{2m+2}(\epsilon_j)}}. \quad (11.55)$$

To obtain m -spin solutions explicitly, one has to choose parameters μ_k , resolve the $m + 1$ constraints (11.54) and (11.55) for the lengths of spins $\mathbf{t}_k$ and frequency ω' , and solve for the dynamics of the m -spin Hamiltonian (11.50). The dynamics can be obtained from the general solution (11.47) by replacing $n \rightarrow m$ and the set of $\{\epsilon_j\}$ with $\{\mu_k\}$.

Let us illustrate the construction of m -spin solutions by considering the cases $m = 0$ and $m = 1$ in more detail.

m=0. The 0-spin solutions correspond to the mean-field eigenstates of quantum Hamiltonians (11.1) and (11.3). They are equilibrium states for the classical Hamiltonian (11.7).

For $m = 0$ the dynamics is governed by the Hamiltonian $H_0 = \omega' \bar{b} b$ with an obvious solution $b = b_0 e^{-i(\omega' t + \phi)}$, where $b_0 = |b(t)| = \text{const}$. Further, using Eqs. (11.49), (11.51), and (11.52), we obtain

$$\mathbf{L}_{\mathbf{t}}^2(u) = \frac{1}{g^4} (4g^2 b_0^2 + (2u - \omega')^2), \quad (11.56)$$

$$\langle \hat{c}_{j\downarrow} \hat{c}_{j\uparrow} \rangle = s_j^- = \frac{e_j g b}{\sqrt{4g^2 b_0^2 + (2\epsilon_j - \omega')^2}}, \quad \langle \hat{n}_j \rangle - 1 = 2s_j^z = \frac{2e_j (2\epsilon_j - \omega')}{\sqrt{4g^2 b_0^2 + (2\epsilon_j - \omega')^2}}. \quad (11.57)$$

The frequency ω' and the gap $\Delta_0 = g b_0$ have to satisfy a single constraint (11.55) that now reads

$$\omega' = \omega - \sum_{j=0}^{n-1} \frac{g^2 e_j}{\sqrt{4g^2 b_0^2 + (2\epsilon_j - \omega')^2}}. \quad (11.58)$$

We see from Eq. (11.57), that all spins rotate uniformly around the z -axis with a frequency ω' . This rotation can be eliminated, i.e. ω' can be set to zero, by an appropriate choice of the chemical potential. Then, Eq. (11.58) is the analog of the BCS gap equation. The configuration of spins (11.57) corresponds to the product BCS wavefunction. The latter can be straightforwardly reconstructed from the knowledge of $\mathbf{s}_j = \langle \hat{\mathbf{K}}_j \rangle$ as fixing the average of spin-1/2 uniquely fixes its quantum state. The choice of signs $e_j = +1$ for all j corresponds to the ground state, while choosing one of the signs to be -1 corresponds to an excited state – pair excitation of the fermionic condensate.

There is also an important class of 0-spin solutions that is obtained by setting $b_0 = 0$. We see from Eq. (11.57) that in this case all spins are along the z -axis, $s_j^z = e_j/2 = \pm 1/2$. The

constraint (11.58) is now irrelevant, because all xy -components vanish. The choice of signs $e_j = \text{sign } \epsilon_j$, where ϵ_j are counted from the Fermi level, yields the Fermi ground state. Other choices correspond to the excitations of the Fermi gas. All these states are stationary with respect to the mean-field dynamics. For a finite number of pairs, they are non-stationary with respect to the quantum Hamiltonian (11.1). Thus, their initial evolution is entirely governed by quantum corrections (cf. Refs. [33, 11, 10]).

m=1. In this case the dynamics reduces to that of a single collective spin $\mathbf{t}$ coupled to an oscillator, i.e. it is governed by the classical counterpart of the Dicke model (11.4)

$$H_1 = 2\mu t_z + \omega' \bar{b}b + g(\bar{b}t_- + bt_+). \quad (11.59)$$

Using Eq. (11.57), one can express original spins in terms of the collective spin and the bosonic field

$$s_j^- = \frac{e_j(\epsilon_j - \mu)b + g e_j t_-}{2g\sqrt{Q_4(\epsilon_j)}} \quad s_j^z = \frac{e_j(2\epsilon_j - \omega')(\epsilon_j - \mu) + g^2 t_z}{2g^2\sqrt{Q_4(\epsilon_j)}}. \quad (11.60)$$

To complete the construction of 1-spin solutions, we need to choose a positively defined polynomial $Q_4(u)$ so that it satisfies two constraints (11.54) (11.55) and solve for the dynamics of the classical Dicke model (11.59). In this case there is only one non-stationary separation variable u_0 and equations of motion (11.42) take the following form:

$$\begin{aligned} \dot{u}_0^2 + 4Q_4(u_0) &= 0, \\ \dot{b} &= -ib(\omega + 2\mu - 2u_0). \end{aligned} \quad (11.61)$$

These equations can be solved in terms of elliptic functions. This is not surprising, since the model (11.59) is, in fact, equivalent to a spherical pendulum and its solution has already been obtained in Refs. [9, 33].

The 1-spin solutions were originally obtained in Refs. [16, 15] where they were used to describe the evolution beginning from a state infinitesimally close to the normal ground state.

11.8 Conclusion

In this paper we solved for the mean-field dynamics of the fermion-oscillator model (11.1). In the mean-field approximation, the problem reduces to a classical Hamiltonian model. We derived integrals of motion for both the classical (11.7) and the quantum (11.1) models and showed that the dynamics of the fermion-oscillator model (11.1) maps onto that of the BCS model. This was used to derive an explicit general solution for the mean-field dynamics of the fermion-oscillator model with an arbitrary finite number n of degrees of freedom.

The typical dynamics is quasi-periodic with n incommensurate frequencies. The system ergodically explores the part of the phase-space allowed by integrals of motion, returning arbitrarily close to its initial state at irregular time intervals. We have also constructed a class of particular, few-spin, solutions, for which the dynamics reduces to that of m collective spins governed by the same classical model. The case $m = 0$ corresponds to the mean-field eigenstates of the fermion-oscillator Hamiltonian (11.1).

An interesting problem is to obtain and describe the thermodynamic limit, $n \rightarrow \infty$, of the solution. For most physical initial conditions the dynamics considerably simplifies in this limit [31]. In particular, we found that $\langle \hat{b}(t) \rangle$ decays, typically as a power-law, to a steady state where $|\langle \hat{b}(t) \rangle|$ is either a constant (cf. Refs. [2, 12, 18, 11]) or oscillates with few independent frequencies (cf. Fig. 2 in Refs. [13, 6]). The motion of the spin system still contains a continuum of frequencies. The steady state of $|\langle \hat{b}(t) \rangle|$ depends on the initial conditions.

It is also interesting to study quantum and finite temperature effects. Quantum corrections to the mean-field dynamics can become important when the bosonic mode is weakly populated [33]. For example, normal states where $\langle \hat{b}^\dagger \hat{b} \rangle = \langle \hat{c}_{j\downarrow} \hat{c}_{j\uparrow} \rangle = 0$ are stationary in the mean-field approximation (11.6), but are not stationary with respect to quantum dynamics (11.5).

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- [40] The coordinate, x , and the momentum, p of the oscillator are related to the fields b and $\bar{b}$ as $x = \sqrt{2}b_x = (b + \bar{b})/\sqrt{2}$ and $p = \sqrt{2}b_y = (\bar{b} - b)/(\sqrt{2}i)$. Then, we have the usual coordinate-momentum (canonical) brackets $\{x, p\} = 1$, while the second term in the Hamiltonian (11.7) reads $\bar{b}b = p^2/2 + x^2/2$.
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- [43] Note that if the root E_0 is real, it has to be degenerate, since according to Eq. (11.43) $Q_{2n+2}(u) \geq 0$ for real u . In this case $\mathbf{L}^2(E_0) = 0$ implies $L_z(E_0) = L_-(E_0)$ and, since zeros of $L_-(u)$ are separation variables, one of them *has to be* frozen at E_0 .

Part X

TDGL: Other Theory

12 Onset of superconductivity in a voltage-biased NSN microbridge by Serbyn, Skvortsov

Abstract

We study the stability of the normal state in a mesoscopic NSN junction biased by a constant voltage V with respect to the formation of the superconducting order.

Using the linearized time-dependent Ginzburg-Landau equation, we obtain the temperature dependence of the instability line, $V_{\text{inst}}(T)$, where nucleation of superconductivity takes place.

For sufficiently low biases, a stationary symmetric superconducting state emerges below the instability line.

For higher biases, the normal phase is destroyed by the formation of a non-stationary bimodal state with two superconducting nuclei localized near the opposite terminals.

(bimodal state ???)

The low-temperature and large-voltage behavior of the instability line is highly sensitive to the details of the inelastic relaxation mechanism in the wire.

Therefore, experimental studies of $V_{\text{inst}}(T)$ in NSN junctions may be used as an effective tool to access parameters of the inelastic relaxation in the normal state.

Nonequilibrium superconductivity has been attracting significant experimental and theoretical attention over decades [Kopnin(2001), Gray(1981), Gulian and Zharkov(1999)], ranging from vortex dynamics [Langenberg and Larkin(1986)] to the physics of the resistive state in current-carrying superconductors [Vodolazov *et al.*(2003), Xiong *et al.*(1997), Rogachev *et al.*(2006), Li *et al.*(2011), Astafiev *et al.*(2012)].

It was recognized long ago [Ivlev and Kopnin(1984)] that a superconducting wire typically has a hysteretic current voltage characteristic specified by several “critical” currents.

(seems that I need to read their paper!)

In an up-sweep, a current exceeding the thermodynamic depairing current, $I_c(T)$, does not completely destroy superconductivity but drives the wire into a nonstationary resistive state [Meyer and Minnigerode(1972)], with the excess phase winding relaxing through the formation of phase slips [Langer and Ambegaokar(1967), McCumber and Halperin(1970)].

(todo: study ones phase slips..)

The resistive state continues until $I_2(T) > I_c(T)$, when the wire eventually becomes normal. In the down-sweep of the current voltage characteristic, the wire remains normal until $I_1(T) < I_2(T)$ when an emerging order parameter leads to the reduction of the wire resistance.

The theoretical description of a nonequilibrium superconducting state is a sophisticated problem, requiring a simultaneous account of the nonlinear order parameter dynamics and quasiparticle relaxation under nonstationary conditions. The resulting set of equations is extremely complicated [Langenberg and Larkin(1986), Kopnin(2001)] and can be treated only numerically [Keizer *et al.*(2006), Vercruyssen *et al.*(2012), Snyman and Nazarov(2009)] (even then the stationarity of the superconducting state is often assumed [Keizer *et al.*(2006), Vercruyssen *et al.*(2012)]).

(???????)

A more intuitive but somewhat oversimplified approach is based on the a time-dependent Ginzburg-Landau (TDGL) equation for the order parameter field $\Delta(\mathbf{r}, t)$.

The TDGL approach which is generally inapplicable in the gapped phase [Gorkov and Eliashberg(1968)], can be justified only in a very narrow vicinity of the critical temperature, T_c , provided that the electron-phonon (e -ph) interaction is sufficiently strong to thermalize quasiparticles [Kramer and Watts-Tobin(1978)].

(I'll read it later)

These generalized TDGL equations are analyzed numerically in Refs. [Vodolazov *et al.*(2003), Michotte *et al.*(2004), Vodolazov(2007), Elmurodov *et al.*(2008)].

(I'll read them later)

While the applicability of the TDGL equation in the superconducting region is a controversial issue, its linearized form can be safely employed to find the line $I_{\text{inst}}(T)$ of the absolute instability of the normal state with respect to the appearance of an infinitesimally small order parameter $\Delta(\mathbf{r}, t)$ [Ivlev and Kopnin(1984), Rubinstein *et al.*(2007), Chtchelkatchev and Vinokur(2009)].

(I'll read them later)

If the transition to the superconducting state is second order, then $I_1(T)$ coincides with $I_{\text{inst}}(T)$. Otherwise the actual instability takes place at a larger $I_1(T) > I_{\text{inst}}(T)$. In both cases, $I_{\text{inst}}(T)$ gives the lower bound for $I_1(T)$.

Previous results [Ivlev and Kopnin(1984), Rubinstein *et al.*(2007)] for the instability line of a superconducting wire connected to normal reservoirs (NSN microbridge) have been obtained in the limit of quasi-equilibrium, when strong e -ph relaxation renders the distribution function locally thermal.

This approximation breaks down for low- T_c superconducting wires shorter than the e -ph relaxation length, $l_{e\text{-ph}}(T_c)$ (e.g., for aluminium, $l_{e\text{-ph}}(T_c) \approx 40 \mu\text{m}$ [Giazotto *et al.*(2006)]).

(????!!! why??? so length of a wire plays a role???)

Such systems have recently been experimentally studied in Refs. [Vercruyssen *et al.*(2012)] (Al) and [Chen *et al.*(2009), Tian *et al.*(2005)] (Zn; reservoirs may be driven normal by a magnetic field).

It was found that for sufficiently large biases superconductivity arises near the terminals through a second-order phase transition, with $I_1(T) = I_{\text{inst}}(T)$ [Vercruyssen *et al.*(2012)].

(todo: check it)

In this Letter we study the normal state instability line in an NSN microbridge biased by a DC voltage V , relaxing the assumption of strong thermalization. For small biases, $eV \ll T_c$, the instability line is universal and we reproduce the results of Refs. [Ivlev and Kopnin(1984), Rubinstein *et al.*(2007)].

(check, how exactly??)

The universality breaks down for larger biases, where we obtain $V_{\text{inst}}(T)$ as a functional of the normal state distribution function and analyze it for various types of inelastic interactions.

We model the NSN microbridge as a diffusive wire of length L coupled at $x = \pm L/2$ to large normal reservoirs via transparent interfaces. The terminals are biased by a constant voltage V . The wire length, L , is assumed to be larger than the zero-temperature coherence length, $\xi_0 = \sqrt{\pi D/8T_{c0}}$, where D is diffusion coefficient, and T_{c0} is the critical temperature of the infinite wire.

(why???)

The equilibrium critical temperature, $T_c = T_{c0}(1 - \pi^2 \xi_0^2/L^2)$, is smaller than T_{c0} due to the finite size effect [Boogaard *et al.*(2004)].

(this is interesting!!! I'll read about it, never have taken it into account)

In what follows we neglect superconducting fluctuations [Larkin and Varlamov(2002)].

General stability criterion.

An arbitrary nonequilibrium normal state becomes absolutely unstable with respect to superconducting fluctuations if an infinitesimally small order parameter, $\Delta(\mathbf{r}, t)$, does not decay with time but evolves to finite values.

(trivial statement?)

Evolution of $\Delta(\mathbf{r}, t)$ is described by the TDGL equation, and it suffices to keep only the linear term to judge its stability.

In dirty superconductors, the linearized TDGL equation can be readily derived from the Keldysh σ -model formalism [Levchenko and Kamenev(2007), Feigel'man *et al.*(2000), Kamenev(2011)] or dynamic Usadel equations [Langenberg and Larkin(1986)] by expanding in Δ .

(I have no idea, how to do it?)

It takes the form $(L_R)^{-1} * \Delta = 0$, where $(L_R)^{-1}$ is the inverse fluctuation propagator, and convolution in time and space indices is implied. In the frequency representation, $(L_\omega^R)^{-1}$ is an integral operator in real space specified by the kernel

$$(L_\omega^R)^{-1}_{\mathbf{r}, \mathbf{r}'} = -\frac{\delta_{\mathbf{r}, \mathbf{r}'}}{\lambda} + i \int_{-\omega_D}^{\omega_D} dE F(E, \mathbf{r}) C_{\omega-2E}(\mathbf{r}, \mathbf{r}'), \quad (12.1)$$

where λ is the dimensionless BCS interaction constant, ω_D is the Debye frequency, and C stands for the retarded Cooperon, $C_\epsilon(\mathbf{r}, \mathbf{r}') = \langle \mathbf{r} | (-D\nabla^2 - i\epsilon)^{-1} | \mathbf{r}' \rangle$, vanishing at the boundary with the terminals.

(a complicated formula...)

The operator (12.1) is a functional of the normal-state nonequilibrium electron distribution function, $F(E, \mathbf{r})$, which contains information about the mechanism for inelastic relaxation in the wire.

(so, in which limit are we working, such that we have kinetics??? and we don't consider Cooper pairs separately at all, only allelectrons and phonons)

The distribution function should be determined from the kinetic equation

$$D\nabla^2 F(E, \mathbf{r}) + \mathcal{I}^{e-e}[F] + \mathcal{I}^{e-ph}[F] = 0, \quad (12.2)$$

with $\mathcal{I}^{e-e}[F]$ and $\mathcal{I}^{e-ph}[F]$ being the electron-electron ($e-e$) and e -ph collision integrals, respectively.

(and form of them is unknown...)

The corresponding energy relaxation lengths, $l_{e-e}(T) \propto T^{-1/4}$ and $l_{e-ph}(T) \propto T^{-3/2}$, behave as a negative power of the temperature T in quasi-equilibrium [Giazotto *et al.*(2006)].

(to understand it, read this article....)

In the absence of inelastic collisions, the kinetic equation (12.2) is solved by the “two-step” function [Nagaev(1995), Pothier *et al.*(1997)]:

$$F(E, x) = (1/2 - x/L)F_L(E) + (1/2 + x/L)F_R(E). \quad (12.3)$$

The distribution functions in the terminals, $F_{L,R}(E) = F_0(E \pm eV/2)$, are given by the equilibrium distribution function, $F_0(E) = \tanh(E/2T)$, shifted by $\pm eV/2$ ($e > 0$). In the opposite case of strong inelastic relaxation, the distribution function takes the form

$$F_{\text{in}}(E, x) = \tanh[(E - e\phi(x))/2T(x)], \quad (12.4)$$

where $\phi(x) = Vx/L$ is the potential in the normal state, and $T(x)$ is the effective temperature. For strong lattice thermalization ($l_{e\text{-ph}} \ll L \ll l_{e\text{-e}}$), $T(x) = T$. For the dominating $e\text{-e}$ scattering ($l_{e\text{-e}} \ll L \ll l_{e\text{-ph}}$), $T^2(x) = T^2 + (3/4\pi^2)[1 - (2x/L)^2](eV)^2$ [Nagaev(1995)].

The order parameter evolution governed by the linearized TDGL operator (12.1) can be naturally described in terms of the eigenmodes $\Delta_k(\mathbf{r})e^{-i\omega_k t}$ annihilated by $(L_\omega^R)^{-1}$.

The normal state is stable provided $\text{Im}\omega_k < 0$ for all eigenmodes. Below we analyze the spectrum of the operator (12.1) and determine the instability line in the coordinates $T\text{--}V$. In the general case, the spectrum can be obtained only numerically. Analytical considerations become possible when the operator (12.1) may be linearized in ω : $(L_\omega^R)^{-1} = i\tau\omega - \mathcal{H}$. The instability occurs when the real part of the lowest eigenvalue of the operator $\mathcal{H}$ turns to zero.

Weak-nonequilibrium regime.

In the limit of low biases, $eV \ll T_c$, the deviation from equilibrium is small everywhere in the wire and the distribution function acquires a universal form, $F(E, x) \approx F_0(E) - F'_0(E) e\phi(x)$, regardless of the relaxation mechanism. Then Eq. (12.1) takes the form

$$(L_\omega^R)^{-1} = i\pi\omega/8T - \ln(T/T_{c0}) - (\xi_0/L)^2 \mathcal{H}_v,$$

with

$$\mathcal{H}_v = -\partial_{\tilde{x}}^2 + 2iv\tilde{x}, \quad \tilde{x} \in [-1/2, 1/2]. \quad (12.5)$$

The Hamiltonian $\mathcal{H}_v$ describes quantum-mechanical motion *in an imaginary electric field*, $v = eV/E_{\text{Th}}$, on the interval $\tilde{x} \equiv x/L \in [-1/2, 1/2]$, where $E_{\text{Th}} = D/L^2$ is the Thouless energy. Hard-wall boundary conditions, $\psi(\pm 1/2) = 0$, imposed on wave functions correspond to the complete suppression of superconductivity at the contacts with highly-conducting terminals due to the inverse proximity effect.

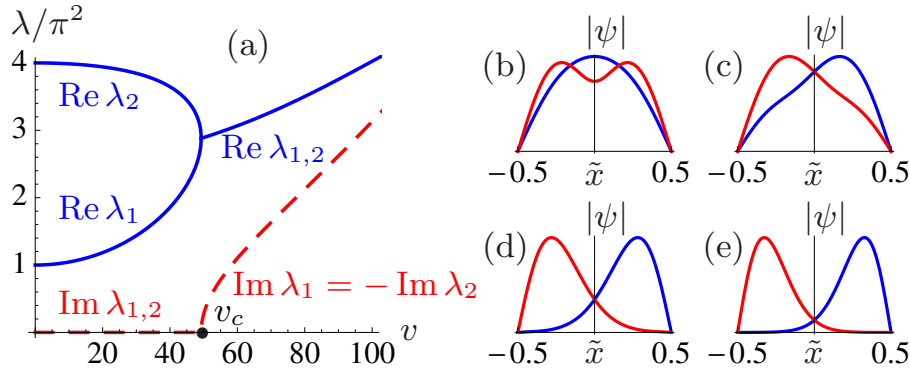


Figure 12.1: (Color online) (a) Real (solid blue line) and imaginary (dashed red line) parts of the lowest eigenvalues $\lambda_{1,2}(v)$ of the Hamiltonian (12.5). The spectrum is entirely real until $v = v_c \approx 49.25$. (b)–(e) Spatial dependence of the absolute values of the eigenfunctions, $|\psi_1(\tilde{x})|$ (blue) and $|\psi_2(\tilde{x})|$ (red), for $v = 0.8v_c$, $1.2v_c$, $5v_c$, and $10v_c$ respectively.

The Hamiltonian (12.5) has been recently analyzed in Ref. [Rubinstein *et al.*(2007)]. It belongs to a class of non-Hermitian Hamiltonians invariant under the combined action of the time-reversal, $\mathcal{T}$: $f(x) \mapsto f^*(x)$, and parity, $\mathcal{P}$: $f(x) \mapsto f(-x)$, transformations. The $\mathcal{PT}$ -symmetry of the Hamiltonian (12.5) ensures that its eigenvalues $\lambda_n(v)$ are either real or form complex-conjugated pairs, providing a complex extension of the notion of Hermiticity [Bender and Boettcher(1998), Bender(2007)]. At $v = 0$, the spectrum is non-degenerate: $\lambda_n(0) = \pi^2 n^2$ ($n = 1, 2, \dots$). It evolves continuously with v and a nonzero $\text{Im}\lambda(v)$ arises only when the two

lowest eigenvalues, $\lambda_1(v)$ and $\lambda_2(v)$, coalesce [see Fig. 12.1 (a)]. This happens at $v = v_c \approx 49.25$ [Rubinstein *et al.*(2007)], indicating the transition to a complex-valued spectrum. For $v < v_c$, the ground state of (12.5) is $\mathcal{PT}$ -symmetric, and hence $|\psi_1(\tilde{x})| = |\psi_1(-\tilde{x})|$. For $v > v_c$, the $\mathcal{PT}$ -symmetry is spontaneously broken and there is a pair of states with the lowest $\text{Re } \lambda(v)$: $\psi_L(\tilde{x}) = \psi_1(\tilde{x})$ and $\psi_R(\tilde{x}) = \psi_2(\tilde{x}) = \psi_1^*(-\tilde{x})$, shifted to the left (right) from the midpoint [see Fig. 12.1 (b)–(e)].

Spontaneous breaking of the $\mathcal{PT}$ -symmetry associated with the spectral bifurcation at $v = v_c$ explains the appearance of asymmetric superconducting states observed in numerical simulations [Vodolazov and Peeters(2007)] and recent experiments [Vercruyssen *et al.*(2012)]. The normal-state instability line, $V_{\text{inst}}(T)$, is specified implicitly by the relation

$$1 - T/T_{c0} = (\xi_0/L)^2 \text{Re } \lambda_1(eV_{\text{inst}}(T)/E_{\text{Th}}), \quad (12.6)$$

and exhibits a singular behavior at the critical bias V_* given by $eV_* = v_c E_{\text{Th}} \approx 50 E_{\text{Th}}$ (see inset in Fig. 12.2). The bifurcation of the instability line occurs at the temperature $T_* \approx T_{c0}(1 - 28.44 \xi_0^2/L^2)$. For long wires ($L \gg \xi_0$), T_* is very close to T_c .

The time dependence of the emergent superconducting state is determined by $\text{Im } \lambda_1(v)$. Below the bifurcation threshold, for $V_{\text{inst}}(T) < V_*$, the system undergoes at $V = V_{\text{inst}}(T)$ the transition to a *stationary* superconducting state, with the superconducting chemical potential being the half-sum of the chemical potentials in the terminals. This state is supercurrent-carrying, and can withstand a maximum phase winding of π achieved at the critical bias V_* . For larger voltages, $V_{\text{inst}}(T) > V_*$, two modes, $\psi_L(x)$ and $\psi_R(x)$, nucleate simultaneously at $V_{\text{inst}}(T)$. The resulting bimodal superconducting state is *non-stationary* since the left and right modes feel different electrochemical potentials and their phases rotate with opposite frequencies, $\Omega_{L,R}(V) = \mp E_{\text{Th}} \text{Im } \lambda_1(eV/T)$. This will result in the Josephson generation with the differential frequency $\Omega(V) = \Omega_R(V) - \Omega_L(V) \propto \sqrt{V - V_*}$ as $V_{\text{inst}}(T) \rightarrow V_*$. Though these supercurrent oscillations are locked to the superconducting part they may excite oscillations of the normal current in the whole circuit. Thus the dc biased NSN microbridge may act as a voltage-tunable generator of an ac current, with the maximal amplitude of oscillations expected in the *coherent* regime, $V - V_* \sim V_*$. The possibility of experimental observation of such a generation remains an open problem.

Incoherent regime.

As the voltage is increased far above the bifurcation threshold, $V_{\text{inst}}(T) \gg V_*$, the eigenmodes $\psi_{L,R}(x)$ gradually localize near the corresponding terminals, with their size, $a(V)$, becoming much smaller than the wire length [see Fig. 12.1 (b)–(e)]. This is the *incoherent* regime, where the overlap between $\psi_L(x)$ and $\psi_R(x)$ is exponentially small, supercurrent oscillations are suppressed, and nucleation of superconductivity near each terminal can be described independently [Vercruyssen *et al.*(2012)].

Using $a(V)/L$ as a small parameter and still working in the vicinity of T_c , we linearize $F(E, x)$ near the left terminal and reduce Eq. (12.1) to the form: $(L_\omega^R)^{-1} = i\pi(\omega + eV)/8T - \ln(T/T_{c0}) - \mathcal{H}_\alpha$, where the operator

$$\mathcal{H}_\alpha = -\xi_0^2 \partial_{x_L}^2 + \alpha x_L, \quad x_L \geq 0, \quad (12.7)$$

acts on the semiaxis $x_L \equiv x + L/2 \geq 0$ with the boundary condition $\psi(0) = 0$. The complex parameter α is a functional of the distribution function:

$$\alpha\left(\frac{eV}{T}\right) = - \int \frac{dE \partial_x F(E - eV/2, x)|_{x=-L/2+a(V)}}{2(E - i0)}. \quad (12.8)$$

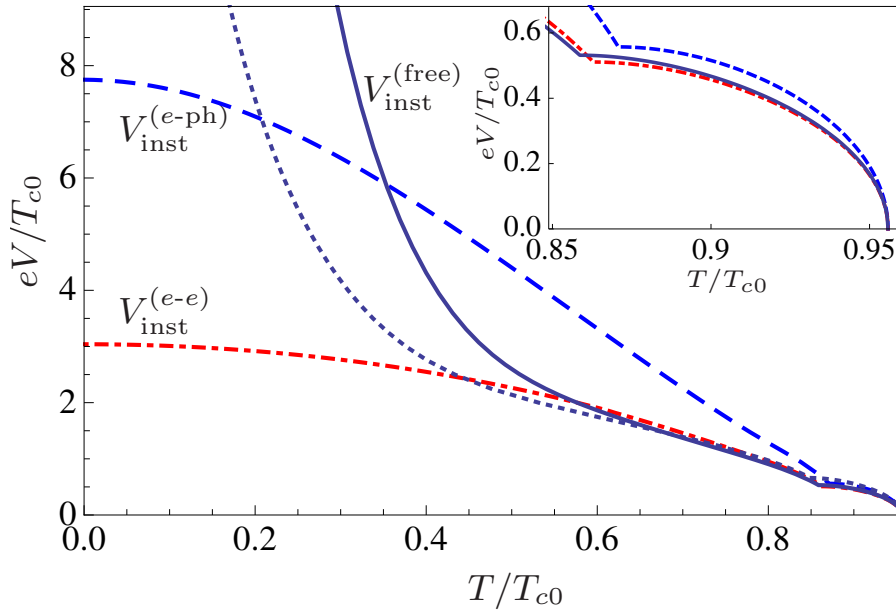


Figure 12.2: (Color online) Instability voltage as a function of temperature, $V_{\text{inst}}(T)$, obtained numerically for a wire of length $L = 15\xi_0$ for three limiting types of the distribution function: without inelastic relaxation (solid blue line), and with dominant $e-e$ (dot-dashed line) or $e-ph$ (dashed line) relaxation. Dotted curve illustrates the suppression of $V_{\text{inst}}^{(\text{free})}(T)$ by a finite terminal resistance, $\beta = 0.1$ (see text). Inset shows the behavior in the vicinity of the bifurcation point $eV_* \approx 50E_{\text{Th}} \approx 0.57 T_{c0}$.

The ground state of the Hamiltonian (12.7) has the energy $\gamma_0(\alpha\xi_0)^{2/3}$ and the wave function $\psi_0(x_L) = \text{Ai}(x_L/a - \gamma_0)/\sqrt{a} \text{Ai}'(-\gamma_0)$, where $-\gamma_0 \approx -2.34$ is the first zero of the Airy function, and $a = (\xi_0^2/\alpha)^{1/3}$ is the nucleus size [Ivlev and Kopnin(1984)]. For the instability line we get:

$$1 - T/T_{c0} = \gamma_0 \xi_0^{2/3} \text{Re} \alpha^{2/3} (eV_{\text{inst}}(T)/T). \quad (12.9)$$

The left and right unstable states rotate with the frequencies $\Omega_{L,R}(V) = \mp[eV - \Omega_1(V)]$, where $\Omega_1(V) = (8T_{c0}/\pi)\gamma_0\xi_0^{2/3} \text{Im} \alpha^{2/3}(eV/T)$ is a small correction to the Josephson frequency determined by the electrochemical potential of the corresponding terminal. At the instability line, $V = V_{\text{inst}}(T)$, the size a of the unstable mode is of the order of the temperature-dependent superconducting coherence length $\xi(T) \sim (1 - T/T_{c0})^{-1/2}\xi_0$.

For long wires ($L \gg \xi_0$), the incoherent regime partly overlaps with the weak-nonequilibrium regime. Then for $eV_* \ll eV_{\text{inst}}(T) \ll T_c$, Eq. (12.9) gives a universal answer

$$\frac{eV_{\text{inst}}(T)}{T_{c0}} = \frac{2^{7/2}}{\pi} \frac{L}{\xi_0} \left(\frac{T_{c0} - T}{\gamma_0 T_{c0}} \right)^{3/2}, \quad (12.10)$$

which could have also been deduced from Eq. (12.6) using the quasiclassical approximation for $\lambda_1(v)$ at $v \gg 1$. Equation (12.9) exactly coincides with the result of Ref. [Ivlev and Kopnin(1984)], predicting that superconductivity nucleates near the terminals at a finite current $I_{\text{inst}}(T) \approx 0.356 I_c(T)$.

The position of the instability line in the incoherent regime at large biases, $eV_{\text{inst}}(T) \gg T_c$, depends on the relation between the inelastic lengths l_{e-e} and l_{e-ph} , the wire length L , and the nucleus size $a(V)$. The presence of the latter scale, which probes the distribution function near the boundaries of the wire, leads to a rich variety of regimes realized at different temperatures.

For the three limiting distributions [Eqs. (12.3) and (12.4)], the function $\alpha(u)$ can be found analytically: (i) $\alpha_{\text{free}}(u) = [\psi(1/2 + iu/2\pi) - \psi(1/2)]/L$ for the non-interacting case, $L \ll l_{e-e}, l_{e-\text{ph}}$, where $\psi(x)$ is the digamma function; (ii) $\alpha_{e-\text{ph}}(u) = i\pi u/4L$ for strong lattice thermalization, $l_{e-\text{ph}} \ll a(V) \ll L \ll l_{e-e}$; and (iii) $\alpha_{e-e}(u) = [i\pi u/4 + 3u^2/2\pi^2]/L$ for the dominant $e-e$ interaction, $l_{e-e} \ll a(V) \ll L \ll l_{e-\text{ph}}$. In case (ii), the instability line $V_{\text{inst}}^{e-\text{ph}}(T)$ is given by Eq. (12.10). In the vicinity of T_c , the instability lines in cases (i) and (iii) are given by:

$$\frac{eV_{\text{inst}}^{(\text{free})}(T)}{T_{c0}} = 1.13 \exp \left\{ \frac{L}{\xi_0} \left(\frac{T_{c0} - T}{\gamma_0 T_{c0}} \right)^{3/2} \right\}, \quad (12.11)$$

$$\frac{eV_{\text{inst}}^{(e-e)}(T)}{T_{c0}} = \left(\frac{2\pi^2}{3} \frac{L}{\xi_0} \right)^{1/2} \left(\frac{T_{c0} - T}{\gamma_0 T_{c0}} \right)^{3/4}. \quad (12.12)$$

Counterintuitively, in cases (i) and (iii) the instability current $I_{\text{inst}}(T) \propto V_{\text{inst}}(T)/L$ has a nontrivial dependence on the system size, as opposed to Eq. (12.10). Such a behavior is a consequence of strong nonequilibrium in the wire. The limiting curves $V_{\text{inst}}^{(\text{free})}(T)$, $V_{\text{inst}}^{(e-\text{ph})}(T)$, and $V_{\text{inst}}^{(e-e)}(T)$ for all temperatures obtained numerically from Eq. (12.1) for the wire with $L/\xi_0 = 15$ are shown in Fig. 12.2. The universal behavior at small biases can be easily seen (inset). Since the ratio L/ξ_0 is not very large, the instability line becomes strongly dependent on the distribution function already for $V \gtrsim V_*$.

The most exciting feature of our results is the exponential growth of $V_{\text{inst}}(T)$ with decreasing temperature in the non-interacting case, Eq. (12.11). Hence, even a small deviation of the distribution function from the two-step form (12.3) will drastically modify $V_{\text{inst}}(T)$. As an example, consider the effect of a finite resistance of the normal terminals. Then the function $F_L(E)$ in Eq. (12.3) will be replaced by $F_L(E) = \beta F_0(E + eV/2) + (1 - \beta)F_0(E - eV/2)$, where V is the voltage applied to the NSN microbridge, and $\beta = R_T/(R_N + 2R_T)$ [R_T and R_N are the resistances of the N and S part of the junction, respectively]. The resulting $V_{\text{inst}}(T)$ for $\beta = 0.1$ is shown by the dotted blue line in Fig. 12.2. While $V_{\text{inst}}(T)$ is unchanged for small biases, it is strongly suppressed compared to $V_{\text{inst}}^{(\text{free})}(T)$ for large biases.

Low-temperature behavior.

—The exponential growth of $V_{\text{inst}}^{(\text{free})}(T)$ in the non-interacting case formally implies that superconductivity at $T = 0$ might persist up to exponentially large voltages, $\ln(eV_{\text{inst}}(0)/T_{c0}) \sim L/\xi_0 \gg 1$. This conclusion is wrong, since inelastic relaxation and heating become important with increasing V , even if they were negligible at $V = 0$. To study the low- T part of the instability line, we consider here a model of the e -ph interaction (e - e relaxation neglected) when the phonon temperature is assumed to coincide with the base temperature of the terminals and e -ph relaxation is weak at T_c : $l_{e-\text{ph}}(T_c) \gg L$ (as in the experiment [Vercruyssen *et al.* (2012)]).

With decreasing T below T_c , the instability line first follows Eq. (12.11). At the same time, $l_{e-\text{ph}}$ decreases and eventually the distribution function in the middle of the wire becomes nearly thermal with the effective temperature T_{eff} . This happens when T_{eff} obtained from the heat balance equation [Giazotto *et al.* (2006)], $(eV/L)^2 \sim T_{\text{eff}}^5/T_c^3 l_{e-\text{ph}}^2(T_c)$, becomes so large that $l_{e-\text{ph}}(T_{\text{eff}}) \sim L$. The corresponding voltage, V_{ph} , can be estimated as $eV_{\text{ph}}/T_c \sim [l_{e-\text{ph}}(T_c)/L]^{2/3}$. Consequently, the exponential growth (12.11) persists for voltages $V_* \lesssim V \lesssim V_{\text{ph}}$, corresponding to the temperature range $T_{\text{ph}} \lesssim T \lesssim T_*$, where with logarithmic accuracy $1 - T_{\text{ph}}/T_{c0} \sim (\xi_0/L)^{2/3}$.

For higher biases, $V > V_{\text{ph}}$, electrons in the central part of the wire have the temperature T_{eff} . However, the parameter α , Eq. (12.8), is determined by the distribution function in the vicinity of the terminals which is not thermal. Matching solution of the collisionless kinetic equation for $0 < x_L < l_{e\text{-ph}}(T_{\text{eff}})$ at the effective right “boundary”, $x_L = l_{e\text{-ph}}(T_{\text{eff}})$, with the function (12.4) with $T(x) = T_{\text{eff}}$, we obtain $\alpha \sim 1/l_{e\text{-ph}}(T_{\text{eff}})$. Therefore, for $V \gtrsim V_{\text{ph}}$ we get with logarithmic accuracy:

$$\frac{eV_{\text{inst}}(T)}{T_{c0}} \sim \frac{L}{\xi_0} \left(\frac{l_{e\text{-ph}}(T_c)}{\xi_0} \right)^{2/3} \left(\frac{T_{c0} - T}{T_{c0}} \right)^{5/2}. \quad (12.13)$$

Equation (12.13) corresponding to the case $a(V) \ll l_{e\text{-ph}} \ll L$ is different from the expression (12.10) when phonons are important already at T_c , and $l_{e\text{-ph}} \ll a(V) \ll L$. The scaling dependence of Eq. (12.13) on L indicates that the stability of the normal state is controlled by the applied current, similar to Eq. (12.10). At zero temperature the instability current exceeds the thermodynamic depairing current by the factor of $(l_{e\text{-ph}}(T_c)/\xi_0)^{2/3} \gg 1$.

Discussion.

—Our general procedure locates the absolute instability line, $V_{\text{inst}}(T)$, of the normal state for a voltage-biased NSN microbridge. Following experimental data [Vercruyssen *et al.*(2012)] we assumed that the onset of superconductivity is of the second order. While non-linear terms in the TDGL equation are required to determine the order of the phase transition [Serbyn and Skvortsov()], we note that were it of the first order, its position would be shifted to voltages higher than $V_{\text{inst}}(T)$.

In the vicinity of T_c , the problem of finding $V_{\text{inst}}(T)$ can be mapped onto a one-dimensional quantum mechanics in some potential $U(x)$. For small biases, $eV \ll T_{c0}$, the potential $U(x)$ does not depend on the distribution function details, explaining universality of the instability line, including the bifurcation from the single-mode to the bimodal superconducting state at $eV \sim 50E_{\text{Th}}$ [Rubinstein *et al.*(2007)] and nucleation of superconductivity in the vicinity of the terminals for larger biases [Ivlev and Kopnin(1984)].

For $eV \gtrsim T_{c0}$, the potential $U(x)$ becomes a functional of the normal-state distribution function, producing $V_{\text{inst}}(T)$ that is strongly sensitive to inelastic relaxation mechanisms in the wire. For the dominant $e\text{-ph}$ interaction, the instability is controlled by the electric field $\mathcal{E} = V/L$ [Eqs. (12.10) and (12.13)], while in the opposite case [Eqs. (12.11) and (12.12)], the instability cannot be solely interpreted as current or voltage-driven. At zero temperature, the (nonuniform) superconducting state can withstand a current which is parametrically larger than the thermodynamic depairing current.

High sensitivity of $V_{\text{inst}}(T)$ to the details of the distribution function opens avenues for its use as a probe of inelastic relaxation in the normal state. The shape of $V_{\text{inst}}(T)$ can be further used to determine the dominating relaxation mechanism and extract the corresponding inelastic scattering rate.

Part XI

Nonequilibrium Vortex Dynamics

13 Vortex Motion in Superconductors by Larkin, Ovchinnikov

13.1 1. Introduction

According to their behaviour in a magnetic field all the superconductors can be classified under two large groups: type-I and type-II superconductors. In a sufficiently strong magnetic field type-II superconductors transfer into a special state, in which superconducting properties, for instance, magnetic moment, weaken with increasing magnetic field. This state is called the mixed state or the Shubnikov phase (Shubnikov et al. 1937). As has been shown by Abrikosov (1957), a magnetic field penetrates into a superconductor in the mixed state in the form of separate threads. In this case the threads form a regular triangular lattice. The magnetic field breaks the space homogeneity of the order parameter and generates vortices.

In an electric field flux lines move with a velocity proportional to the electric field strength. With such a motion the distribution function of normal excitations deviates from its equilibrium value, which results in energy dissipation.

In a weak electric field the deviation of the distribution function from the equilibrium value is proportional to the electric field $\mathbf{E}$.

In an ideal homogeneous superconductor the transport current is also proportional to $\mathbf{E}$. However, nonlinearity of the $I - V$ characteristic of a superconductor sets in rather early. This can be accounted for by the fact that in metals energy relaxation is slow and the distribution function is thus strongly dependent on the electric field. In a normal metal the conductivity is determined by the total number of electrons and is independent of details of their distribution with energy. Therefore, the IV' characteristic of a normal metal remains linear up to large fields. In a superconductor the conductivity depends on the order parameter, which is very sensitive to the shape of the distribution function. As a result, the $I - V$ characteristic of a superconductor becomes nonlinear at a current density much below the depairing critical current.

The initial part of the IV characteristic is determined by inhomogeneities always present in real samples. Inhomogeneities distort the vortex lattice and lead to arising of metastable current states. In such a state a current below some critical value does not lead to the motion of flux lines and energy dissipation. The value of the critical current, its dependence on magnetic field and temperature, and also the $I - V$ characteristic for current densities exceeding the critical one, depend essentially on the type of inhomogeneities.

In section 2, the equation of motion of a vortex structure is derived. Each term in this equation has a simple physical meaning and describes a force acting on the vortex. Here the viscosity force is expressed through variation of the Green function and the distribution function of excitations in an electric field.

In section 3 we consider ideal homogeneous superconductors in a weak electric field. The dependence of the conductivity of a superconductor on the values of magnetic field, temperature and the electron free path has been determined. The conductivity of superconductors has been investigated theoretically for all the cases when the solution of a static equation for the order parameter is known.

Section 4 is devoted to investigation of nonlinear effects arising upon motion of flux lines.

In the vicinity of the transition temperature excitations are put out of the region of the vortex core. The vortex compresses and viscosity shows a drastic drop. At low temperatures the process of generation of new excitations is more essential, which may result in an intensive emission of almost monochromatic phonons.

Section 5 presents a description of elastic properties of a vortex lattice, which are important for understanding the phenomenon of pinning. Unlike the common elastic media, the elastic moduli of a vortex lattice have a strong space dispersion (Brandt 1977). Another difference is that even weak inhomogeneities cause the vanishing of the long-range order in the vortex lattice (Larkin 1970).

In section 6, pinning processes are investigated. If the interaction of a defect with the lattice is sufficiently strong, then metastable states may form on a single defect (strong pinning). In this case the critical current is proportional to the defect concentration (Labusch 1969b). For defects weakly interacting with the vortex lattice, the critical current arises due to collective effects and is proportional to the square of the concentration (Larkin and Ovchinnikov 1973c).

The friction force depends on the velocity of the vortex motion, which results in deviation of the $I - V$ characteristic from the linear law. When the pinning is strong, the friction in motion is less than the friction at rest. When the pinning is weak, the friction force grows with increasing vortex lattice velocity.

The Appendix presents a derivation of kinetic equations for the excitation distribution function. These equations are used for studying the dynamic characteristics of superconductors.

13.2 2. Equation of flux line motion in inhomogeneous superconductors

13.2.1 2.1. Phenomenological derivation

Many phenomena arising on the flux line motion in superconductors have a macroscopic character.

In this case a flux line lattice can be regarded as a continuous medium. The equations of motion of such a medium can be obtained by averaging microscopic equations over the volume containing a large number of flux lines. After such averaging the state of a flux line lattice is described by a two-dimensional displacement vector u . The forces acting on an isolated unit volume are different in physical nature and have different dependences on the displacement vector u . The viscosity force f_v at small velocities v of the flux line lattice is proportional to the velocity

$$f_v = -n_v \eta v, \quad v = \partial u / \partial t \quad (1)$$

where n_v is the flux line density related to the induction B by the formula

$$B = n_i \phi, \quad \phi = \pi \hbar / e \quad (2)$$

ϕ is the flux quantum. A mean electric field is associated with the motion of flux lines:

$$E = -\frac{1}{c} [v \times B]. \quad (3)$$

In the coordinate system moving together with flux lines, the mean electric field is zero.

The viscosity coefficient η in eq. (1) is, generally speaking, a tensor. The nondiagonal elements of the tensor describe the Hall effect. With increasing velocity, the viscosity coefficient decreases. The value η changes essentially at a comparatively small velocity, when the transport current j is essentially less than the critical current of depairing. This phenomenon is connected

with the fact that the electron distribution function becomes nonequilibrium upon flux line motion.

If a transport current with the density $\langle j \rangle$ passes through a superconductor, then the Lorentz force acts on the flux lines:

$$f_L = \frac{1}{c}[j \times B]. \quad (4)$$

In homogeneous superconductors the Lorentz force is equal to the viscosity force. This allows us to express the current density through the electric field intensity

$$j = \sigma E, \quad \sigma = \phi\eta/B. \quad (5)$$

In real superconductors there always exist inhomogeneities of different kinds: pores, dislocation loops, grain boundaries and other phase regions. Such inhomogeneities are called pinning centres and lead to a dependence of the free energy F on the position of the flux line lattice and, namely, to a pinning force arising,

$$f_{\text{pin}} = -\delta F_{\text{pin}}/\delta u \quad (6)$$

Usually the distances between pinning centers are large. This makes it possible to distinguish the volume containing a separate defect and the pinning force (6) connected with it.

In the space between the pinning centers the flux line lattice is deformed. Its deformation is weak and may be described by means of the theory of elasticity. The elastic force is equal to (Labusch 1967)

$$f_{\text{el}} = -\frac{\delta F_{\text{el}}}{\delta u} = (C_{11} - C_{66}) \frac{\partial}{\partial \rho} \left(\frac{\partial u}{\partial \rho} \right) + \left(C_{66} \frac{\partial^2}{\partial \rho^2} + C_{44} \frac{\partial^2}{\partial z^2} \right) u; \quad (7)$$

the z axis is chosen in the direction of the magnetic field, ρ is a twodimensional vector in the xy plane, and C_{1K} are elastic moduli. In many cases elasticity theory is applicable in all the volume involving the vicinities of the pinning center. When the centers are very strong, plastic deformation arises in their vicinity. In this case eq. (7) is applicable at sufficiently large distances from the pinning center. A triangular lattice has only three independent moduli. They may be considered to be constant only in the case when the displacement vector varies slowly at distances of the order of the magnetic field penetration depth. Otherwise, the bending of the lattice is stronger than that of magnetic lines of force. This effect results in a strong space dispersion of the elastic moduli C_{11} and C_{44} (Brandt 1977).

The sum of all the forces acting on the flux line lattice should be zero. Substituting for these forces eqs. (1), (4), (6) and (7) we get an equation of motion of the flux line lattice:

$$f_{\text{el}} + f_v + f_L + f_{\text{pin}} = 0 \quad (8)$$

$$(C_{11} - C_{66}) \frac{\partial}{\partial \rho} \left(\frac{\partial u}{\partial \rho} \right) + \left(C_{66} \frac{\partial^2}{\partial \rho^2} + C_{44} \frac{\partial^2}{\partial z^2} \right) u - \sigma B^2 \frac{\partial u}{\partial t} + [j \times B] + f_{\text{pin}} = 0 \quad (9)$$

(Schmid and Hauger 1973, Larkin and Ovchinnikov 1973c). The parameter which is most difficult to determine, both theoretically and experimentally, is the pinning force f_{pin} . Usually, microscopic parameters of the pinning center are known poorly. In the most interesting case, strong pinning, the lattice deformation is plastic and quantitative calculation is difficult. But even when

the force of interaction with one pinning center is known, the problem of summing arises as well. The problem is that f_{pin} depends on the relative positions of the lattice and the pinning center.

When the interaction force with one pinning center achieves a maximum, the interaction force with other centers may be less or even may have an opposite direction. Therefore, the mean pinning force is less than the sum of the maximum values of individual forces. To calculate the mean force it is necessary to solve eq. (9). Therefore, the mean force may depend on the vortex velocity, i.e., on the electric field. The friction of motion is not equal to that of the state of rest. Averaging eq. (9) over the volume containing a large number of pinning centers, one may get the I V characteristic of the sample

$$\langle j(\mathbf{E}) \rangle = \sigma(\mathbf{E})\mathbf{E} + [\langle f_{\text{pin}} \rangle \times \mathbf{B}] / B^2 \quad (10)$$

where σ is the conductivity without pinning.

At zero electric field the superconducting current may be nonzero. The state with a current is metastable. Thermal fluctuations lead to the damping of a superconducting current with time. At temperatures not very close to the critical one, the energy of the vortex interaction with the pinning centers far exceeds the temperature. Such damping ("creep") is thus exponentially small and will not be taken into account below.

13.2.2 2.2. Microscopic derivation of the equation of motion

Equation (9) can be derived from microscopic equations of the theory of superconductivity. Therewith, the microscopic nature of the parameters involved in eq. (9) is elucidated and in many cases it is possible to find their dependence on temperature, magnetic field and on microscopic characteristics of the superconductor.

The microscopic equations for the order parameter Δ and for the vector potential A have the form of eqs. (A.20) and (A.21):

$$\begin{aligned} \frac{1}{\lambda_{\text{eff}}} \Lambda_{1,2} &= \pm \frac{1}{8} \int d\Omega_p (g_p^K(t, t))_{12,21} \\ \text{rot rot } A &= \frac{4\pi}{c} j = -\frac{e\pi v}{2} \int d\Omega_p \text{Tr } \tau_2 \frac{p}{m} g_p^K(t, t) \end{aligned} \quad (11)$$

where $\nu = mp/2\pi^2$ is the density of states on the Fermi surface, λ_{eff} is the effective constant of the electron electron interaction, and y_p^K is the matrix composed of single-particle Green functions. The equation expressing y_p^K through Δ and $\mathbf{A}$ is given in the Appendix (A.12).

Suppose the solutions of the system of equations (11) in the static case $A_0(r)$, $A_0(r)$ are known. The change in the free energy δF at small static deviations from equilibrium is

$$\delta F = v \int d^3r \int \frac{d\Omega_p}{4\pi} \text{Tr} \left(\frac{\hat{\Delta}}{\lambda_{\text{eff}}} - \frac{\pi}{2} g_p^K(t, t) \right) \delta \hat{\Delta} + \frac{1}{4\pi} \int d^3r \delta A (\text{rot rot } A - 4\pi j), \quad (12)$$

where $\hat{A}$ is determined by eq. (A.20).

At small velocities $\mathbf{V}$ of the vortex lattice motion, the solution of eqs. (11) may be written in the form

$$\begin{aligned} \Delta^{(1,2)}(r, t) &= \Delta_0^{(1,2)}(r - u) \exp(\pm 2ie\chi) + \Delta_1^{(1,2)} \\ A(r, t) &= A_0(r - u) + \frac{\partial \chi}{\partial r} + \tilde{A} + A_1 \\ \frac{\partial \chi}{\partial t} &= \frac{\partial u}{\partial t} A_0(r - u) \end{aligned} \quad (13)$$

The vector potential $\tilde{A}$ describes the magnetic field created by the transport current $\langle j \rangle$. This field varies only slightly at distances of the order of the lattice period. Therefore, one may regard it as included into the induction $\mathbf{B}$, so that Δ_0 and A_0 are determined by the local value of the induction $\mathbf{B}$ at each point of the sample.

The phase χ of the gauge transformation is chosen so that at an infinitesimal displacement u the increment of the order parameter A would not contain terms increasing with distance. The small corrections Δ_1 and A_1 do not contain increasing terms either and satisfy the complicated equations obtained by linearization of system (11). It is shown that for calculating the mean forces involved in eq. (9), it is not required to find A_1 and A_1 . This method is similar to the perturbation theory in quantum mechanics, where it is sufficient to know the wave functions of a zero-order approximation to calculate the energy in first order. The function y may be presented in the form $g_0 + g_1$ with an accuracy up to the first-order term in velocity. The function g_0 is found by solving eqs. (A.12) in a static case. In these equations instead of Δ and A one should insert their expansion (13) together with the zero- and first-order terms. The function g_1 is found by solving eqs. (A.12) to first order with respect to velocity of motion, using a zero-order approximation for the potentials Δ and A .

Then this expansion and (13) are substituted into system (11), the first equations of system (11) are multiplied by $\partial_+ A^*$ and $\partial_- A$, the latter is multiplied by $\text{rot } A_0$ vectorially and the summation of all equations is performed. As a result, taking into account formula (12), we arrive at

$$\begin{aligned} & \frac{\delta}{\delta U} \left[F + \left(\frac{\delta F}{\delta \Delta^{(1)}} \Delta_1^{(1)} + \frac{\delta F}{\delta \Delta^{(2)}} \Delta_1^{(2)} + \frac{\delta F}{\delta \mathbf{A}} \mathbf{A}_1 \right) \right] + [j \times \mathbf{B}] \\ & - \frac{mp}{4\pi} \int \frac{d\varepsilon}{2\pi} \int \frac{d\Omega_p}{4\pi} \left\langle \text{Tr} \left(e[\mathbf{H} \times \mathbf{v}] \tau_z - \left(\frac{\partial}{\partial \mathbf{r}} - 2ie\mathbf{A} \tau_z \right) \hat{\Delta} \right) g_1^K(\boldsymbol{\rho}, \varepsilon, t) \right\rangle = 0. \end{aligned} \quad (14)$$

The free energy F of a homogeneous superconductor does not change when the flux line lattice is subjected to a homogeneous displacement. The first term in eq. (14) is thus nonzero only due to inhomogeneities and elastic deformations and is equal to the sum of the pinning forces f_{pin} and the elastic force f_{el} . The second and third terms in square brackets give small corrections, proportional to the velocity, to these forces and are not taken into account in this chapter. Therefore, to calculate f_{el} and f_{pin} it is necessary to obtain the deformation dependence of the free energy. The next term in eq. (14) is equal to the Lorentz force and arises due to the term $\tilde{A}$ in eq. (13).

The last term in eq. (14) gives a microscopic expression for the viscosity force. In this expression the Green function g_1 appears due to the motion of flux lines. The viscosity force at small velocities is proportional to velocity.

The relationship of the mean electric field with the flux line velocity is obtained from eq. (13) for the vector potential. Differentiating A with respect to time and averaging over the volume, we have

$$\mathbf{E} = - \left\langle \frac{\partial \mathbf{A}}{\partial t} \right\rangle = - \left[\frac{\partial \mathbf{u}}{\partial t} \text{rot } \mathbf{A} \right] = [\mathbf{B} \times \mathbf{v}]. \quad (15)$$

13.3 3. Conductivity of superconductors in a mixed state

In this section we consider the vortex motion in homogeneous superconductors free from pinning forces. Actually this means that the transport current passing through a superconductor much exceeds the pinning critical current.

When the transport current flows, a Lorentz force arises, causing flux line motion. The flux line motion is accompanied by heating of normal excitations and energy dissipation. Two

mechanisms of energy dissipation can be considered. One of the dissipation mechanisms is related with the Joule heating of normal excitations by an electric field (Strnad et al. 1964, Bardeen and Stephen 1965). The second mechanism is connected with inhomogeneity of the absolute value of the order parameter in the flux line (Tinkham 1964). Such inhomogeneity leads to a force affecting normal excitations. Therefore, the moving flux line drags the normal component.

The conductivity of a superconductor in a mixed state can be found from the condition of equality of the Lorentz and viscosity forces. The Lorentz force is proportional to the transport current density, the viscosity force to the velocity of the flux line lattice. As follows from eq. (15), the flux line velocity is proportional to the electric field $\mathbf{E}$. So, in the region of weak currents we have

$$\mathbf{j} = \sigma \mathbf{E} \quad (16)$$

In homogeneous superconductors the pinning force and elastic deformation of a vortex lattice are absent. Therefore, the first term in eq. (14) is zero. Making a cross-product of eq. (14) by $\mathbf{B}$ we get:

$$\mathbf{j} = -\frac{mp}{4\pi B^2} \int \frac{d\varepsilon}{2\pi} \int \frac{d\Omega_p}{4\pi} \left\langle \text{Tr} \left(ev(\mathbf{B} \times \mathbf{H})\tau_z + \left[\mathbf{B}, \left(\frac{\partial}{\partial r} - 2ie\mathbf{A}\tau_z \right) \hat{\Delta} \right] \right) g_1^K(p, r, t) \right\rangle. \quad (17)$$

Equation (17) expresses the transport current through the Green function $g_1^K(p, t)$.

Equations for the Green function $y_1^K(p, r, t)$ are derived in the Appendix. It is possible to obtain an analytical expression for the conductivity for some limiting cases only.

13.3.1 3.1. Temperatures close to T_c

In the vicinity of the transition temperature the order parameter Δ can be found from the Ginzburg Landau equation. For superconductors with a small electron free path the range within which the order parameter varies is of the order of $\xi(T) = (\pi D/8(T_c - T))^{1/2}$. Therefore, the Green functions of the local approximation (A.39) satisfy eq. (A.37).

The omitted gradient terms are small with respect to the parameter $(1 - T/T_c)^{1/2}$. In a local approximation the contribution to the current arising from the distribution function f is anomalously large. Retaining only this contribution in the Green function g_p^K (eq. A.47), eq. (17) is reduced to

$$\mathbf{j} = \frac{mp}{4\pi B^2} \left\langle \int_{|\varepsilon| > |\Delta|} \frac{d\varepsilon}{2\pi} (\varepsilon^2 - |\Delta|^2)^{1/2} f \cdot \left[\mathbf{B} \frac{\partial |\Delta|^2}{\partial r} \right] \right\rangle. \quad (18)$$

For fields close to H_{c2} the term $\sigma_N \mathbf{E}$ arising from the current part of the Green function should be added to the right-hand side of eq. (18): $\sigma_N = e^2 p^2 l_{tr}/3\pi^2$ is the conductivity of a metal in the normal state, with l_{tr} , the electron mean free path of transport. The distribution function f is determined from eqs. (A.43) and (A.44), in which $\partial/\partial t$ should be substituted by $-(v\partial/\partial r)$. Applying formula (15) the lattice velocity v is expressed through the electric field E . As a result, from eq. (18) we find the conductivity of a superconductor in a mixed state:

$$\frac{\sigma}{\sigma_N} - 1 = \frac{1}{4e^2 D^2 T B^2} \int_0^x d\varepsilon \phi(\varepsilon) \quad (19)$$

where

$$\phi(\varepsilon) = \frac{eB}{\pi} \int_{V_j} d^2 \mathbf{r} \frac{\partial (\varepsilon^2 - |\Delta|^2)^{1/2}}{\partial \mathbf{r}} \left(-\frac{\partial^2}{\partial \mathbf{r}^2} \right)^{-1} \frac{\partial (\varepsilon^2 - |\Delta|^2)^{1/2}}{\partial \mathbf{r}}, \quad (20)$$

π/eB is the volume of an elementary cell, V_ε is the region where $|\varepsilon| > |\Delta|$. The boundary condition in expression (20) for $(-\hat{c}^2/\hat{c}r^2)^{-1}$ is given by eq. (A.42). For $|\varepsilon| > |\Delta|_{\max}$ the volume V_ε coincides with the whole cell and for $\phi(\varepsilon)$ we get

$$\phi(\varepsilon) = \left\langle \left((\varepsilon^2 - |\Delta|^2)^{1/2} - \left\langle (\varepsilon^2 - |\Delta|^2)^{1/2} \right\rangle \right)^2 \right\rangle, \quad |\varepsilon| > |\Delta|_{\max}. \quad (21)$$

In eqs. (19) and (20), only ε of the order of Δ are important so $\phi(\varepsilon) \sim A^2 B/H_{c2}$. Expression (19) for the conductivity has thus the form

$$\frac{\sigma}{\sigma_N} - 1 = \frac{1}{(1 - T/T_c)^{1/2}} \frac{H_{c2}}{B} \tilde{f}(B/H_{c2}). \quad (22)$$

We now substitute into formulas (19), (21) the value of the order parameter Δ , found from the Ginzburg Landau equation in a weak magnetic field $B \ll H_{c2}$ and for fields close to H_{c2} . As a result, for the function $\tilde{f}$ we have

$$\tilde{f}(x) = \begin{cases} 4.04, & x \ll 1, \quad \kappa \gg 1, \\ 0.31 \left(\frac{1-H/H_{c2}}{1-\frac{1}{2}\kappa^2} \right)^{3/2}, & 1-x \ll 1, \end{cases} \quad (23)$$

where κ is the Ginzburg-Landau parameter. The numerical coefficients in formula (23) differ from those obtained in earlier papers (Gorkov and Kopnin 1973a, Larkin and Ovchinnikov 1973b), since in these papers an erroneous boundary condition for the operator $(-\partial^2/\partial r^2)^{-1}$ was used. To calculate the function $\tilde{f}$ in an arbitrary magnetic field, the hexagonal lattice is replaced by a round cell. Such an approximation is exact for weak fields. Even for fields close to H_{c2} the error in determining the derivative of $|\Delta|$ at $|\Delta| = 0$ is less than 1% in the round-cell method. One may thus hope that the error is small in calculation of the conductivity as well.

In the approximation of a round cell the expression for conductivity has been derived by Danilov et al. (1974). However, in their paper erroneous boundary conditions were used. In the present chapter, we applied the boundary condition (A.42) for $|\varepsilon| < |\Delta|_{\max}$, and for $|\varepsilon| > |\Delta|_{\max}$ we used expression (22) for the function $\phi(\varepsilon)$. The results of numerical calculation are presented in table 1. These values can be given by an interpolation formula

$$\tilde{f}(x) = \begin{cases} 4.04 - \left(\frac{B}{H_{c2}} \right)^{1/4} \left(3.96 + 2.38 \frac{B}{H_{c2}} \right), & \frac{B}{H_{c2}} < 0.315, \\ \left(1 - \frac{B}{H_{c2}} \right)^{3/2} \left(0.43 + 0.69 \left(1 - \frac{B}{H_{c2}} \right) \right), & \frac{B}{H_{c2}} > 0.315. \end{cases} \quad (24)$$

Table 1
Numerical results for the function $f(x)$, calculated with eq. (24).

B/H_{c2}	$\tilde{f}(x)$	B/H_{c2}	$\tilde{f}(x)$
0	4.04	0.1633	1.2
0.02	2.53	0.2222	0.92
0.0247	2.449	0.32	0.604
0.03125	2.347	0.5	0.2776
0.04082	2.214	0.08889	0.0182
0.05556	2.036	0.9512	$4.987 \cdot 10^{-3}$
0.08	1.814	0.978	$1.473 \cdot 10^{-3}$
0.125	1.435	0.99187	$3.179 \cdot 10^{-4}$

It should be noted that the function $\tilde{f}$ essentially decreases already in weak fields.

Growth of conductivity on approaching the transition temperature is caused by a strong change in the distribution function of excitations. Such growth continues while the size of a flux line is not too large, and the time of diffusion of excitation through the core of the flux line is smaller than the time of the energy relaxation. These time scales become of the same order when $T_c - T \sim \tau_\varepsilon^{-1}$. With further approaching T_c the deviation of the distribution function from equilibrium decreases. In this case the first term in eq. (A.40) is small as compared to I^{ph} . The collision integral I^{ph} in this case can be taken in the form of a τ_ε -approximation (formula eq. A.34), because the essential ε in eq. (19) for the conductivity are of the order of Δ and are thus much smaller than T_c . In a linear expansion in the velocity from eq. (A.40) we find

$$f = \tanh\left(\frac{\varepsilon}{2T}\right) + \frac{\tau_\varepsilon}{2\varepsilon} \frac{\partial \tanh(\varepsilon/2T)}{\partial \varepsilon} \left(v \frac{\partial |\Delta|^2}{\partial \mathbf{r}} \right). \quad (25)$$

Substituting expression (25) for the distribution function into eq. (18), we get the following expression for the conductivity:

$$\sigma - \sigma_N = \frac{mp\tau_\varepsilon}{8\pi TB^2} \left\langle |\Delta| \left| \left(\frac{\partial |\Delta|}{\partial r} \right)^2 \right| \right\rangle. \quad (26)$$

For small fields the solution for A for a single flux line can be inserted into eq. (26). As a result, we have (Larkin and Ovchinnikov 1977)

$$\begin{aligned} \frac{\sigma}{\sigma_N} &= \frac{H_{c2}}{B} \frac{\pi^2 \tau_\varepsilon}{16T(T_c - T)} \int_0^\infty d\rho \rho |\Delta| \left(\frac{\partial |\Delta|}{\partial \rho} \right)^2 \\ &= 2.73 \frac{H_{c2}}{B} \tau_\varepsilon T_c (1 - T/T_c)^{1/2} \end{aligned} \quad (27)$$

While approaching T_c the conductivity determined by eq. (26) decreases. Therefore, there is a maximum at $T_c - T \sim \tau_\varepsilon^{-1}$ in the temperature dependence of the conductivity.

13.3.2 3.2. Fields close to H_{c2}

Let us consider now fields close to H_{c2} . As follows from eq. (24), the contribution to the conductivity connected with a change of the distribution function decreases as $(1 - H/H_{c2})^{3/2}$. The change of the Green function $g^{\text{R,A}}$ becomes more essential. In the vicinity of H_{c2} the order parameter Δ is small and the Green functions can be found by expanding eq. (A.37) in powers of Δ . In main approximation $\alpha^{\text{R,A}} = \pm 1$, and eq. (A.37) becomes a linear equation for the function $\beta^{\text{R,A}}$. The coordinate dependence of the function $\beta^{\text{R,A}}$ is the same as that of the order parameter Δ . They are eigenfunctions of the operator $-\partial_-^2$, corresponding to the minimal eigenvalue equal to $2eH_{c2}$:

$$\beta_1^{\text{R,A}} = -\frac{i\Lambda}{\tilde{\lambda} \mp i\varepsilon}, \quad \beta_2^{\text{R,A}} = -\frac{i\Delta^*}{\tilde{\lambda} \mp i\varepsilon}, \quad (28)$$

where

$$\tau = eH_{c2}D + \tau_s^{-1} + (2\tau_\varepsilon)^{-1}. \quad (29)$$

Approximation (A.38) for the collision integral holding in the vicinity of T_c has been applied for deriving formulas (28) and (29). At low temperatures, in eq. (29) the last term should be omitted completely, since it is small compared to the first term. Now insert expressions (28) into eq. (A.47) for the function g^K . As a result, for the conductivity σ from eq. (17) we get

$$\frac{\sigma}{\sigma_N} - 1 = \frac{\langle |\Delta|^2 \rangle}{4\pi T} \left\{ \frac{1}{eH_{c2}D} \psi' \left(\frac{1}{2} + \frac{\tilde{\lambda}}{2\pi T} \right) + \frac{1}{\tilde{\lambda}} \psi' \left(\frac{1}{2} + \frac{\tilde{\lambda}}{2\pi T} \right) + \frac{1}{2\pi T} \psi'' \left(\frac{1}{2} + \frac{\tilde{\lambda}}{2\pi T} \right) \right\} = \tilde{\alpha} (1 - H/H_{c2}). \quad (30)$$

The first term in eq. (30) (Maki 1966) arises from the first term in expression (A.47) for y^K . This term is related with drag of electrons by the moving lines of constant absolute value of the order parameter. Magnetic impurities and energy relaxation only weakly affect this term. The last two terms are connected with changes in the spectrum of excitations at small energies (Thompson 1970). Magnetic impurities and energy relaxation decrease this contribution.

The expression for $\langle |\Delta|^2 \rangle$ near H_{c2} can be obtained by expanding eqs. (A.20) and (A.37) into series in Δ (Caroli et al. 1966). As a result, for the value $\tilde{\alpha}$ determined by formula (30) we have (Ovchinnikov and Brandt 1975)

$$\tilde{\alpha} = \frac{2}{\beta_A} \psi'(x) \frac{\psi'(x) + \frac{eH_{c2}D}{\tilde{\lambda}} \psi'(x) + \frac{eH_{c2}D}{2\pi T} \psi''(x)}{-\psi''(x) - \frac{1}{6\pi T} \left(\frac{1}{\tau_s} + \frac{1}{2\tau_\varepsilon} \right) \psi'''(x) - \frac{8e^2 p^2 l_{tr}^2 v}{9\pi} (\psi'(x))^2}, \quad (31)$$

where $\beta_A = 1.1596$, $x = 0.5 + \tilde{i}/2\pi T$.

In the vicinity of the transition temperature eq. (31) simplifies to

$$\tilde{\alpha} = \frac{2.496}{1 - \frac{1}{2}\kappa^2} \left[1 + \left(1 + \frac{\pi}{4(T_c - T)} (\tau_s^{-1} + (2\tau_\varepsilon)^{-1}) \right)^{-1} \right] \quad (32)$$

where

$$\kappa^2 = 63\zeta(3)/2\pi^3 e^2 p^2 l_{tr}^2 v$$

Equation (30) is valid for superconductors with a small electron mean free path. Near the transition temperature the limitation for the free path is weaker: $l \ll \xi(T)$. In this case $\langle |\Delta|^2 \rangle$ should be found from the Ginzburg-Landau static equation, its coefficients depending on the ratio $l/\xi(0)$ (Gor'kov 1959).

The expression for the conductivity near H_{c2} , valid at an arbitrary temperature and the ratio $l/\xi(0)$, has been obtained by Ovchinnikov (1974).

The first terms in eqs. (30) and (32) depend only slightly on the energy relaxation time. The second term in eq. (32) decreases rapidly on approaching T_c .

Figure 1 shows the dependence of the coefficient $\tilde{\alpha}$ on the reduced temperature T/T_c at $\kappa^2 \gg 1$ and different values of the parameter

$$\Gamma = (\tau_s^{-1} + (2\tau_\varepsilon)^{-1}) / 2\pi T_c$$

The same figure shows experimental data (Meier-Hirmer et al. 1976) on the conductivity in the vicinity of H_{c2} for $\text{Pb}_{0.8}\text{In}_{0.2}$ and $\text{Pb}_{0.9}\text{In}_{0.1}$. It follows from fig. 1 that for the energy relaxation time

$$(T_c \tau_\varepsilon)^{-1} \sim 0.2$$

13.3.3 3.3. Low temperatures

Consider now the case of low temperatures $T \ll T_c$, weak fields, and a small electron free path (Gor'kov and Kopnin 1973b). At low temperatures, in eqs. (17) and (A.48) low energies $\varepsilon \sim T$ are essential. Therefore, it is sufficient to solve eqs. (A.37) for the Green functions $\beta^{R,A}, \alpha^{R,A}$ at $\varepsilon = 0$. In this case

$$\alpha^R = -\alpha^A = \sin \psi, \quad \beta^R = \beta^A = -i \cos \psi \exp(i\varphi), \quad (33)$$

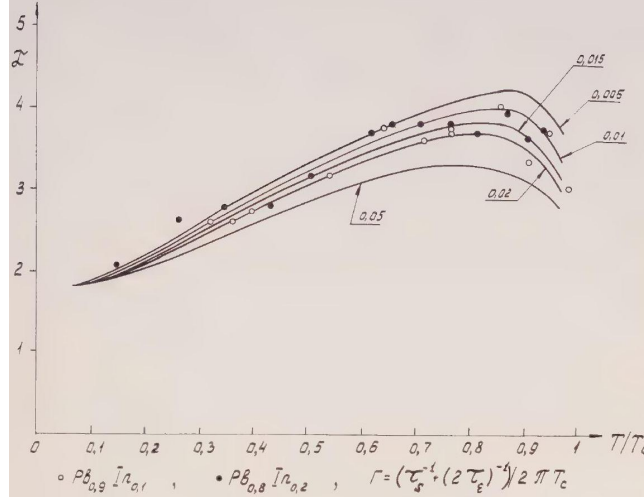


Fig. 1. Solid lines: dependence of $\tilde{x}$ on reduced temperature according to eqs (30)-(32), for different Γ . Symbols: experimental results of Meier-Hirmer et al. (1976).

where φ is the phase of the order parameter, and ψ is a real function of the coordinates, satisfying the equation

$$\frac{r^2 \psi}{\partial r^2} + \frac{\sin 2\psi}{2r^2} - \frac{2|A|}{D} \sin \psi = 0 \quad (34)$$

with the boundary condition

$$\psi(\infty) = 0. \quad (35)$$

From eqs. (17) and (A.48), taking account of eq. (33), we find

$$j = mp_{2\pi B^2} \int_{2\pi}^{d\varepsilon} \left\langle D \left[\mathbf{B}^{i\varphi i\varphi} \right]_{\partial r^2}^{\hat{r}^2 f_1} + \frac{if}{\partial \varepsilon} \left[\mathbf{B}^{i|1|} \right] \left(v^i \frac{\cos \psi}{\partial \mathbf{r}} \right) \right\rangle. \quad (36)$$

where v is the velocity of the flux line lattice.

It should be noted that in eq. (36) the distribution function f should be taken in zero approximation. The function f_1 is found from eq. (A.32), which, taking account of eq. (33), is reduced to

$$-D \left[\frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial}{\partial \rho} \right) - \frac{1}{\rho^2} \right] \tilde{f}_1 + 2|\Delta| \cos \psi \tilde{f}_1 = \frac{2|\Delta| \cos \psi}{\rho}, \quad (37)$$

where

$$f_1 = -\frac{\partial f}{\partial \varepsilon} \frac{\rho}{2} \left(\frac{\partial \varphi}{\partial \mathbf{r}} v \right) \tilde{f}_1 \quad (38)$$

From eq. (36) we find (Gor'kov and Kopnin 1973).

$$\frac{\sigma}{\sigma_N} = \frac{H_{c2}(0)}{B} \tilde{\beta}, \quad (39)$$

$$\tilde{\beta} = \frac{D}{\Delta_\infty} \frac{\partial \tilde{f}_1}{\partial \rho} \Big|_0 + \frac{1}{\Delta_\infty} \int_0^\infty d\rho \rho \frac{\partial |\Delta|}{\partial \rho} \frac{\partial \cos \psi}{\partial \rho} = 0.9. \quad (40)$$

The conductivity of superconductors with a large electron mean free path has been less investigated. By order of magnitude the vortex viscosity at low temperatures for pure and dirty superconductors has the same form (39) (Bardeen and Stephen 1965). In pure superconductors the derivative $\partial \Delta / \partial \rho$ at $\rho \rightarrow 0$ increases with decreasing temperature (Kramer and Pesch 1974). One may assume that a decrease in temperature causes a compression of the vortex up to the size of vT/T_c^2 . The conductivity is determined by the behaviour of the absolute value of the order parameter Δ at distances $r \sim v/T_c$. From the assumption on the compression of the flux line core it follows that conductivity grows logarithmically with decreasing temperature (Bardeen and Sherman 1975, Larkin and Ovchinnikov 1976). The Green function of a superconductor with a large electron free path at low energies is

$$g^{R, A} = \chi^{R, A} \begin{pmatrix} 1 & -i \exp(i\varphi + i\eta(t)) \\ i \exp(-i\varphi - i\eta(t)) & -1 \end{pmatrix} \quad (41)$$

(Kramer and Pesch 1974), where the phase $\eta(t)$ is

$$\eta(t) = \frac{2e}{v \sin \theta} \int_{-\infty}^t dt_1 \nu \left(A - \frac{1}{2e} \frac{\partial \varphi}{\partial r} \right), \quad (42)$$

$t/\sin \theta$ is the distance on the trajectory counted from the point which is nearest to the axis of the flux line, and θ is the angle between the velocity and the magnetic field.

$$\begin{aligned} \chi^{R, A} = & \frac{iv \sin \theta}{2} \exp \left(-\frac{2}{v \sin \theta} \left| \int_0^t \Delta \cos \eta \, dt_1 \right| \right) \\ & \times (\varepsilon \varphi_1(\theta, x) - \varphi_2(\theta, x) \pm i\delta)^{-1} \end{aligned} \quad (43)$$

where x is the impact parameter,

$$\begin{aligned} \varphi_1(\theta, x) &= \int_0^\infty dt \exp \left(-\frac{2}{v \sin \theta} \left| \int_0^t dt_1 \Delta \cos \eta \right| \right) \\ \varphi_2(\theta, x) &= \int_0^\infty dt \Delta \sin \eta \exp \left(-\frac{2}{v \sin \theta} \left| \int_0^t dt_1 \Delta \cos \eta \right| \right) \end{aligned} \quad (44)$$

It follows from the system of equations (A.52) that the distribution function f_1 is small, and the function f is constant on the trajectory. Integrating the first equation (A.52) along the trajectory we get

$$\begin{aligned} & i \frac{\partial f}{\partial \varepsilon} \int_{-\infty}^\infty dl \, \text{Tr} \frac{\partial \hat{H}}{\partial t} (g_p^R - g_p^A) \\ &= \frac{m}{2} \int_{-\infty}^\infty dl \int d\Omega_{p_1} \sigma_{pp_1} (f_p - f_{p_1}) \text{Tr} (g_p^R - g_p^A) (g_{p_1}^R - g_{p_1}^A) \end{aligned} \quad (45)$$

where

$$\hat{H}(t) = -\frac{ie}{m}(p \cdot A)\tau_z - i\hat{\Delta}.$$

The current density is determined by eq. (17) in which

$$g_1^K(p) = (g_p^R - g_p^A) f(p).$$

It follows from eq. (43) that $g_p^R - g_p^A$ is proportional to $\delta(\varepsilon\varphi_1 - \varphi_2)$. Therefore, for $|\varepsilon| \ll |\Delta|$, φ_2 is small, which corresponds to the trajectories passing at small impact parameters from the flux line axis.

Provided the Kramer and Pesch (1974) hypothesis is valid and compression of the flux line core takes place, then essentially at distances of the order of v/Δ from the flux line axis the order parameter may be regarded to be constant equal to Δ_∞ . To logarithmic accuracy eq. (45) for isotropic scattering is reduced to an algebraic equation, and for the distribution function we get the expression

$$f_p = \tanh\left(\frac{\varepsilon}{2T}\right) + \frac{2\tau\Delta^2}{v} \partial \frac{\tanh(\varepsilon/2T)}{\partial \varepsilon} \frac{e(v \cdot \mathbf{E})}{|e[\mathbf{B} \times v]|} \ln\left(\frac{v \sin \theta}{\Delta|\bar{X}|}\right). \quad (46)$$

Substituting this distribution function into expression (17) for the current, we have (Larkin and Ovchinnikov 1976).

$$\sigma/\sigma_N = 0.23 (H_{c2}/B) \ln(\Delta_0/T) \quad (47)$$

In case the hypothesis on the compression of the flux line core is invalid, then in eq. (47) the logarithm is replaced by a number of the order of unity. For determining this number it is required to know the coordinate dependence of the order parameter.

Thus, the problem of the flux line lattice conductivity is solved for all the cases when the solution of a static problem for the order parameter Δ is known.

13.4 4. Nonlinear effects in the motion of vortices in superconductors

In metals at low temperatures the energy relaxation time $\tau_\varepsilon \sim \theta_D^2/T^3$ is large. Therefore, in comparatively weak electric fields a strong change of the energy distribution function of electrons takes place. In a normal metal such a change does not affect the conductivity. In superconductors the energy gap strongly depends on the shape of the distribution function, hence, deviation from Ohm's law occurs even for slowly moving flux lines.

13.4.1 4.1. Nonlinear conductivity

-For a rough estimation of the dependence of the conductivity on the electric field we consider the case when the time of inelastic electron-electron collisions is small, as compared to the time of electron-phonon or phonon-thermostat interactions. In this case the temperature of electrons may noticeably differ from the thermostat temperature T . Thus, the conductivity, which is determined by the electron temperature, is

$$\sigma(E, T) = \sigma(0, T + \sigma E^2 \tau^*/c) \quad (48)$$

where c is the electron heat capacity. It may be seen from this expression that at large τ^* nonlinear effects may be strong in a rather weak electric field.

Below we consider another limiting case, when heat is removed effectively and the lattice is in equilibrium with the thermostat, and the time of energy relaxation due to interelectron collisions is larger than that of the electronphonon interaction or of the same order. In this case, due to the slow energy relaxation, the change in conductivity is large and is determined by the variation of the distribution function which now is not reduced to variation in the electron temperature and is more complicated.

The effect is strongest at temperatures close to the critical one, when a slight variation in the distribution function causes a large variation in the order parameter or in its coordinate dependence. This in turn causes a decrease in conductivity, and the voltage dependence of the current has an N-type shape. The parameters of the IV characteristic depend essentially on the magnetic field value and on the magnetic impurity concentration. The strongest nonlinear effects arise in weak magnetic fields at low concentrations of magnetic impurities, when there is a gap in the excitation spectrum and the excitation is captured in the core of the vortex.

In the vicinity of the transition temperature strong variations of conductivity and of the order parameter arise in a weak electric field, when the distribution function varies only slightly. This is connected with the fact that a nonequilibrium correction to the distribution function in the equation for Δ must be compared with the terms proportional to $T_c - T$. In this case the distribution function $\langle f(\varepsilon) \rangle$, in the left-hand side of eq. (A.45) can be replaced by $\tanh(\varepsilon/2T)$. The time derivative in eq. (A.45) should be replaced by $-(v\partial/\partial\mathbf{r})$. Equations (A.45), (A.46) and (A.50) form a closed system of equations for the functions $\Delta(r)$ and $f(\varepsilon)$. However, to determine the function $f(\varepsilon)$ at $\varepsilon \ll \Delta$ it is sufficient to know the shape of the function $A(r)$ near the flux line axis, where $|\Delta(\rho)| = A\rho$. The left- and right-hand sides of formula (A.45) are proportional to A^{-2} , and so the correction to the distribution function at $\varepsilon \ll |\Delta|$ is independent of the slope A and is equal to

$$f(\varepsilon) - \tanh(\varepsilon/2T) = \frac{17}{36}\tau_\varepsilon v^2 \varepsilon / DT \quad (49)$$

In a weak magnetic field $B \ll H_{c2}$ it is possible to find the analytical dependence of $f(\varepsilon)$ at $\varepsilon \rightarrow \Delta_\infty$. In this case in eq. (A.45) the essential distances much exceed the correlation length where $\Delta(r) = \Delta_\gamma(1 - A/\rho^2)$. Substituting such value into eq. (A.45) yields

$$f(\varepsilon) - \tanh(\varepsilon/2T) = \tau_\varepsilon v^2 (\Delta_\infty^2 - \varepsilon^2)^{1/2} / \pi DT \quad (50)$$

At $\varepsilon > A_\gamma$ in a weak magnetic field $B \ll H_{c2}$, the correction to the distribution function is small, since the heating occurs in the region $\rho \sim \xi$, and relaxation proceeds in the whole volume.

In the region $\varepsilon < \Delta_\infty$ we use the interpolation formula

$$f(\varepsilon) - \tanh(\varepsilon/2T) = 0.4\tau_\varepsilon v^2 \frac{\varepsilon (\Delta_\infty^2 - \varepsilon^2)^{1/2}}{\Delta_\infty DT}. \quad (51)$$

The chosen form of the function $f(\varepsilon)$ has a correct dependence at $\varepsilon \ll \Delta$ and near the threshold $\varepsilon = \Delta_\infty$.

The numerical coefficient 0.4 is equal to the average of the limiting values 0.32 and 0.48. For the distribution function determined by eq. (51) the nonequilibrium correction to the Ginzburg Landau equation (A.50) is calculated analytically and equals

$$\int_{\Delta}^{A_\infty} \frac{d\varepsilon \Delta}{(\varepsilon^2 - \Delta^2)^{1/2}} (f(\varepsilon) - \tanh(\varepsilon/2T)) = 0.314\tau_\varepsilon v^2 \frac{\Delta (\Delta_\infty^2 - \Delta^2)}{\Delta_\infty DT}. \quad (52)$$

The correction determined by eq. (52) has the same form as the non-gradient terms in the Ginzburg-Landau equation. The coordinate dependence of the order parameter is thus retained, $\Delta(r) = \Delta(\rho/\xi)$ with the substitution

$$\xi^2 \rightarrow \xi^2(v) = \xi^2(0) [1 + (E/E^*)^2]^{-1} \quad (53)$$

where

$$(E^*)^2 = D \left(1 - \frac{T}{T_c}\right)^{1/2} \left(\frac{B}{C}\right)^2 \tau_\varepsilon^{-1}, \quad \xi_{(0)}^2 = \frac{\pi D}{8(T_c - T)} \quad (54)$$

So, the motion of flux lines leads to a decrease of the vortex core. The physical reason for this is in the following: normal excitations inside the vortex are heated and diffuse into the energy range $\varepsilon > \Delta$. Their effective pressure on the walls of the vortex drops and the vortex compresses. At temperatures close to T_c generation of excitations in the center of the vortex appears inessential. It should be noted that strong compression of the vortex takes place in a comparatively weak electric field, when the correction to the distribution function is still small. Therefore, for the conductivity one may apply eqs. (19) and (20) of the linear approximation. The order parameter $\Delta(r)$ in these formulas should be found with allowance made for an electric field. In an approximation, when the compression of vortex is described by a decrease in the correlation length ξ (eq. 53), in eq. (20) it is possible to use dimensionless variables. Since the function $\phi(\varepsilon)$ is proportional to ξ^2 , the expression for conductivity in weak magnetic fields acquires the form

$$\frac{\sigma}{\sigma_N} = \frac{4.04}{(1 - T/T_c)^{1/2}} \frac{H_{c2}}{B} \frac{1}{1 + (E/E^*)^2}. \quad (55)$$

It follows from eq. (55) that with increasing electric field the conductivity decreases. This decrease continues up to fields of the order

$$E = E^* / (1 - T/T_c)^{1/4} \quad (56)$$

With further increase of the electric field the diffusion of excitations in the energy inside the vortex becomes strong and the distribution function for $0 < \varepsilon < A_x$ acquires a value independent of the electric field. The compression of the vortex stops at the size

$$\xi = \check{\zeta}(T) (1 - T/T_c)^{1/4}.$$

In this case the conductivity becomes independent of the electric field:

$$\sigma/\sigma_N = CH_{c2}/B, \quad (57)$$

where the constant C is of the order of unity. For such small sizes of the vortex one cannot use the local approximation equations (A.38) for the Green functions. The conductivity determined by eq. (57) exceeds that of a normal metal. A complete destruction of superconductivity occurs at a larger electric field, when the distribution function changes not only for the electrons captured in vortices, but also for the electrons in the whole volume. From eq. (48) for this field we get

$$eE = T(1 - T/T_c)^{1/2} / (D\tau_\varepsilon)^{1/2}. \quad (58)$$

To determine the numerical coefficient in eq. (58) it is necessary to solve the integral equation for the electron distribution function in a normal metal. In this case one cannot use the τ_ε approximation for the collision term.

From eqs. (55) and (57) it follows that the I-V characteristic has an N-type shape,

$$j = \sigma_N \frac{H_{c2}}{B} \left\{ \frac{4.04}{(1 - T/T_c)^{1/2}} \frac{1}{1 + (E/E^*)^2} + C \right\} \mathbf{E} \quad (59)$$

and on the dependence $j(E)$ there is a maximum lying in the vicinity of the point $E = E^*$. In the regime of a given current a hysteresis should be observed. A jump in the $I - V$ characteristic has been observed experimentally (Musienko et al. 1980). By the location of the jump the value of E^* was determined, its dependence on the magnetic field and temperature coinciding with the theoretical one (eq. 54). For the energy relaxation time the following values are obtained:

$$\tau_\varepsilon(\text{Sn}) = 6.6 \times 10^{-10} \text{ s}, \quad \tau_\varepsilon(\text{Al}) = 4 - 7 \times 10^{-8} \text{ s}$$

Equations (55)-(59) are valid for the case of a weak magnetic field. As follows from eqs. (22) and (24), the linear conductivity decreases rapidly with increasing magnetic field. Such a decrease is described by the function $\hat{f}(B/H_{c2})$. In weak magnetic fields the action of a magnetic field is reduced to compression of the vortex. Therefore, besides the effect of a decrease in conductivity described above, the change in the argument of the function $\hat{f}$ should also be taken into account,

$$\frac{\sigma}{\sigma_N} = \frac{1}{(1 - T/T_c)^{1/2}} \frac{H_{c2}}{B} \frac{1}{1 + (E/E^*)^2} \tilde{f}\left(\frac{B}{H_{c2}} \frac{1}{1 + (E/E^*)^2}\right) \quad (60)$$

Equation (60) is valid in magnetic fields which are small compared to H_{c2} . As the magnetic field grows, not only the cooling of electrons captured by the vortex, but also the heating of electrons with energy $\varepsilon > \Delta$ becomes essential. In magnetic fields close to H_{c2} , the heating of electrons and the decrease in T_c and in H_{c2} caused by this heating, becomes most essential. This region of magnetic fields has been considered in detail by Larkin and Ovchinnikov (1975). In this case the electric field suppresses essentially the superconducting addition to conductivity. However, the main contribution, coinciding with the conductivity of a normal metal, is not sensitive to the electric field. Therefore, the $I - V$ characteristic in magnetic fields close to H_{c2} displays only slight deviation from the linear dependence. At low temperatures the conductivity is weakly sensitive to a change in the distribution function. Therefore, nonlinear effects in the $I - V$ characteristic are weak (Larkin and Ovchinnikov 1977).

13.4.2 4.2. Emission of monochromatic phonons in the motion of vortices

As a result of the vortex motion there arises a nonequilibrium distribution of electrons with energies causing emission of nonequilibrium phonons. The most interesting phenomena arise at low temperatures. In rather weak electric fields the dependence of the distribution function on ε is determined by the electric field and not by the temperature. As long as the energy of heated electrons is low as compared with the gap Δ , they do not leave the core of the vortex, and a wide spectrum of phonons with a frequency ω of the order of the nonequilibrium excitation energy is emitted. In a sufficiently strong electric field, electrons heated inside the vortex reach the energy $\varepsilon = \Delta$. Then they leave the region of the vortex core and, if the vortex density is low, they are, essentially, not heated further. Accumulation of excitations with energy ε near Δ takes place, and these excitations recombine with emission of phonons with a frequency close to 2Δ . The width and intensity of emission can be found from eq. (A.45) for the distribution function. In the narrow region essential for us, in the vicinity of the threshold $\varepsilon = \Delta$, the energy diffusion coefficient D_ε depends logarithmically on energy and we replace it by the constant

$$D_\varepsilon = \begin{cases} 2\tau\Delta^2 eB(E/B)^2, & \tau\Delta > 1, \\ \Delta eB(E/B)^2, & \tau\Delta < 1. \end{cases}$$

For widely spaced flux lines, the recombination of excitations accompanied by emission of a phonon with energy close to 2Δ is the main relaxation process in eq. (A.45) at $\varepsilon > \Delta$. Equation

(A.45) for $\varepsilon > \Delta$ is reduced to

$$\frac{\partial}{\partial \varepsilon} \left(D_\varepsilon \frac{\partial f}{\partial \varepsilon} \right) = - \frac{2\Lambda^3 m g_{\text{eff}}^2}{\pi (sp)^2 (\varepsilon - \Delta)^{12}} (1 - f(\varepsilon)) \int_1^\infty \frac{d\varepsilon_1}{(\varepsilon_1 - \Delta)^{1/2}} (1 - f(\varepsilon_1)). \quad (61)$$

The solution of eq. (61) is the Bessel function

$$1 - f = Ay^{2/3} K_{2/3}(y), \quad (62)$$

where

$$y^{4/3} = \frac{\varepsilon - \Delta}{\tilde{\mu}}, \quad \tilde{\mu}^{-2} = \frac{\Delta^3 m p g_{\text{eff}}^2}{(sp)^2 D_\varepsilon} A \cdot 2.55$$

$$A = (1 - f(\Delta)) \cdot 2.33 \quad (63)$$

The boundary value of the distribution function $f(\Delta)$ is determined by the diffusion of excitations inside the flux line.

The distribution of phonons with frequencies in the vicinity of $\omega = 2\Delta$ has the form

$$\frac{\partial N_\omega}{\partial t} = \frac{2\Delta^3 m p}{\pi (sp)^2} \lambda_{\text{eff}} \iint_\Delta \frac{d\varepsilon d\varepsilon_1}{(\varepsilon - \Delta)^{1/2} (\varepsilon_1 - \Delta)^{1/2}} \delta(\varepsilon + \varepsilon_1 - \omega) (1 - f(\varepsilon)) (1 - f(\varepsilon_1)) \quad (64)$$

The characteristic width of this distribution is of the order of $\tilde{\mu}$.

If the energy of heated electrons is small, as compared to Δ , the number of electrons leaving the vortex core is exponentially small (Larkin and Ovchinnikov 1977):

$$1 - f = A \exp \left[- (\varepsilon/T^*)^{5/2} \right] \quad (65)$$

where

$$T^* \sim \Delta [T_c \tau_\varepsilon (T_c) (j/j_c)^2]^{1/5} \quad (66)$$

and j_c is the critical depairing current. In this case the number of monochromatic phonons with energy 2Δ and the phonon distribution width are exponentially small. For T^* of the order of Δ the distribution width is

$$\tilde{\mu}^3 = \left(\frac{0.5T^*}{\Delta} \right)^5 \frac{T_c T^*}{\tau} \frac{B}{H_{c2}}, \quad \tau \Delta > 1 \quad (67)$$

Equations (65)-(67) are valid for superconductors with a large electron mean free path $\tau \Delta \gg 1$.

Therefore, at low temperatures and widely spaced flux lines the total energy of the electric field is converted into the energy of almost monochromatic phonons with frequency $\omega = 2\Delta$.

13.5 5. Elasticity of a vortex lattice

13.5.1 5.1. Elastic moduli

Smooth deformations of a vortex lattice can be described by means of elasticity theory. For a triangular lattice only three elastic moduli are nonzero: the compression modulus $C_1 = C_{11} - C_{66}$, shear modulus C_{66} , and tilting modulus C_{44} . The correction to the free energy resulting from elastic deformations of a vortex lattice is equal to

$$\delta F_{\text{el}} = \frac{1}{2} \int d^3r \left\{ (C_{11} - C_{66}) \left(\frac{\partial u}{\partial r} \right)^2 + C_{66} \left(\frac{\partial u_\beta}{\partial r_x} \right)^2 + C_{44} \left(\frac{\partial u}{\partial z} \right)^2 \right\}. \quad (68)$$

Upon compression and tilting deformation, the shape of a vortex lattice does not change and the free energy of a deformed lattice is equal to that of an undeformed lattice in the renormalized magnetic field. On compression we have $\mathbf{B} = \mathbf{B}_0(1 + \text{div } \mathbf{u})$, on tilting we have

$$H_\perp^2 = H_\parallel^2 \left(\frac{\partial u}{\partial z} \right)^2 \quad (69)$$

Therefore, for C_{11} and C_{44} we get (Labusch 1967)

$$C_{44} = \frac{\mathbf{H} \cdot \mathbf{B}}{4\pi}, \quad C_{11} - C_{66} = \frac{B^2}{4\pi} \frac{\partial H}{\partial \mathbf{B}}. \quad (70)$$

The shape of a triangular lattice changes with shear deformations. And the elastic modulus C_{66} is expressed through the free energy of the lattice, in which the angle between the vectors of an elementary cell is not 60° (Labusch 1967, 1969a, Brandt 1976).

13.5.2 5.2. Space dispersion of elastic moduli

As a rule, elasticity theory is applicable when the size of a deformed region is large as compared to the lattice period. For the vortex lattice there exist physical reasons for the moduli C_{11} and C_{44} having a strong space dispersion at rather small wave vectors (Brandt 1977). This comes about as a result of trapping the magnetic field by a long-wave deformation of the lattice and the fact that the elastic energy is determined by variation of the magnetic field energy. But the magnetic field cannot vary at distances less than the effective penetration depth δ_{eff} . So, under deformation of a lattice with the wave vectors $\mathbf{K}$ exceeding δ_{eff}^{-1} , the magnetic field does not vary and the magnetic force lines break away from the vortex lattice. In this case the elastic energy does not increase with K . For description of such deformations the vector potential and the displacement vector u should be regarded as independent variables. In fields close to H_{c2} there exists another reason for there to arise space dispersion in the elastic modulus C_{11} . The thing is that the shape of a vortex lattice is determined by terms of the order of Δ^2 in the free energy, and the value Δ itself by terms of the order of Δ^4 . Thus, the absolute value of Δ should also be regarded as an independent variable. We seek the order parameter Δ and the vector potential in the form

$$\Delta(r) = \Delta_0(r - u) \exp [2ie(\mathbf{u} \cdot \mathbf{A}_0) + i\chi + s], \quad \mathbf{A} = \mathbf{A}_0 + \mathbf{A}_1. \quad (71)$$

Expanding expression (12) for the free energy in a power series of u, s, χ and A_1 up to quadratic terms we get

$$\begin{aligned} \delta F_{\text{el}} = \frac{k_h^2}{8\pi} \int d^3r \left\{ \frac{1}{k_h^2} (\text{rot } A_1)^2 + \left(A_1 + [\mathbf{B} \cdot \mathbf{u}] - \frac{1}{2e} \frac{\partial \chi}{\partial r} \right)^2 \right. \\ \left. + a_1 \left(A_z^{(1)} - \frac{1}{2e} \frac{\partial \chi}{\partial z} \right)^2 + a_2 \left(\left(\frac{\partial s}{\partial r} \right)^2 + k_\psi^2 s^2 \right) + a_3 \left(\frac{\partial s}{\partial z} \right)^2 \right\} \end{aligned}$$

$$\begin{aligned}
& -a_4 \left(s \frac{\partial \mathbf{u}}{\partial \mathbf{r}} \right) - a_5 s \left(\frac{\partial}{\partial \mathbf{r}} [\mathbf{B} \cdot \mathbf{A}_1] \right) \Big\} \\
& + \int d^3 r \left\{ \langle j_1 \rangle \mathbf{A}_1 + \frac{C_{66}}{2} \frac{\partial u_\alpha}{\partial r_\beta} \frac{\partial u_\alpha}{\partial r_\beta} + \frac{a_6}{2} \left(\frac{\partial u}{\partial z} \right)^2 \right\}, \quad \alpha, \beta = 1, 2.
\end{aligned} \tag{72}$$

The coefficients in this formula have been found for T close to T_c :

$$\begin{aligned}
a_1 = a_3 = a_5 = 0, \quad a_6 = \mathbf{B}(\mathbf{H} - \mathbf{B})/4\pi, \quad a_2 = 1/4e^2 \\
a_4 = B/e, \quad k_h^2 = \frac{2e^2 p^2 t_{tr}}{3T} \eta \langle |\Delta|^2 \rangle \\
k_\psi^2 = k_h^2 \left[1 - \frac{\partial H}{\partial B} \right]^{-1}, \quad \eta = 1 - \frac{8T\tau_{tr}}{\pi} \left(\psi \left(\frac{1}{2} + \frac{1}{4\pi T\tau_{tr}} \right) - \psi \left(\frac{1}{2} \right) \right)
\end{aligned} \tag{73}$$

and for magnetic fields close to H_{c2} (Larkin and Ovchinnikov 1979)

$$\begin{aligned}
k_h^2 &= \frac{8\pi e v \langle |\Delta|^2 \rangle}{B} \gamma_1, \quad k_\psi^2 = \frac{e B \langle |\Delta|^2 \rangle}{2\pi^2 T^2} \gamma_{\gamma_1} \\
a_2 &= 1/4e^2, \quad a_1 = 4e^2 a_3 = \gamma_3/\gamma_1, \quad a_4 = B/e \\
a_5 &= \gamma_3/eB\gamma_1, \quad a_6 = B(H - B)/4\pi \\
C_{66} &= 0.48 \frac{v \langle |\Delta|^2 \rangle^2}{32\pi^2 \beta_A T^2} \left\{ 1 - \frac{8e^2 p^2 v^3 \tau_{tr}^2}{9\pi \gamma_2} \left(\psi' \left(\frac{1}{2} + \frac{eBD}{2\pi T} \right) \right)^2 \right\} \\
\gamma_1 &= \psi \left(\frac{1}{2} + \frac{3eBD}{2\pi T} \right) - \psi \left(\frac{1}{2} + \frac{eBD}{2\pi T} \right) \\
\gamma_2 &= -\beta_A \psi'' \left(\frac{1}{2} + \frac{eBD}{2\pi T} \right) - \frac{8e^2 p^2 v^3 \tau_{tr}^2}{9\pi} (\beta_A - 1) \left(\psi' \left(\frac{1}{2} + \frac{eBD}{2\pi T} \right) \right)^2, \\
\gamma_3 &= \frac{eBD}{\pi T} \psi' \left(\frac{1}{2} + \frac{eBD}{2\pi T} \right) - \gamma_1, \\
\langle |\Delta|^2 \rangle &= \frac{2\pi T}{\gamma_2} e D (H_{c2} - B) \psi' \left(\frac{1}{2} + \frac{eBD}{2\pi T} \right)
\end{aligned} \tag{74}$$

Minimizing expression (72) for the free energy with respect to variables s and $A_1 - (1/2e)(\partial\chi/\partial r)$ yields

$$\begin{aligned}
\delta F_{el} &= \frac{1}{2} \int \frac{d^3 \mathbf{K}}{(2\pi)^3} \{ (C_{11}(\mathbf{K}) - C_{66}) K_\alpha K_\beta \\
&+ (C_{66} \mathbf{K}_\perp^2 + C_{44}(\mathbf{K}) K_z^2) \delta_{\alpha\beta} \} U_\alpha(\mathbf{K}) U_\beta(-\mathbf{K})
\end{aligned} \tag{75}$$

$$\begin{aligned}
C_{44} &= \frac{B^2 k_h^2}{4\pi} (1 + a_1) [\mathbf{K}^2 + k_h^2 + a_1 (\mathbf{K}_z^2 + k_h^2)]^{-1} + a_6 \\
C_{11} - C_{66} &= \frac{B^2 k_h^2}{4\pi} \{ (\mathbf{K}^2 + k_h^2 (1 + a_1)) [\mathbf{K}^2 + k_h^2] \times (\mathbf{K}^2 + k_h^2 + a_1 (\mathbf{K}_z^2 + k_h^2)) \}^{-1} \\
&- \left(1 + \frac{a_5}{a_4} \frac{B^2 k_h^2}{\mathbf{K}^2 + k_h^2} \right) \left(\frac{a_4^2}{4B^2 a_2} + \frac{a_4 a_5}{4a_2} \frac{k_h^2}{\mathbf{K}^2 + k_h^2} \right) \\
&\times \left(\mathbf{K}^2 + k_\psi^2 + \frac{a_3}{a_2} K_z^2 - \frac{a_5}{4a_2} \mathbf{K}_\perp^2 \frac{B^2 k_h^2}{\mathbf{K}^2 + k_h^2} \right)^{-1} \Big\}
\end{aligned} \tag{76}$$

The space dispersion of the elastic modulus C_{66} is small and the elastic moduli C_{44} and C_{11} decrease rapidly with increasing K , provided $K > \delta_{\text{eff}}^{-1}$. Plastic deformation of a vortex lattice is often observed experimentally. Particularly, separate dislocations in such a lattice have been observed (Truble and Essmann 1968). The relative easiness of formation of plastic deformations is likely to be a consequence of the fact that the minimum of the free energy F for a triangular lattice is shallow and its energy is only 2° less than that of a square lattice. Plastic deformation has hardly been studied theoretically.

13.5.3 5.3. Absence of long-range order in a vortex lattice

Vanishing of the long-range order due to weak inhomogeneities (Larkin 1970) is an important property of a vortex lattice. And it is this property that distinguishes the vortex lattice from an ordinary solid body, in which the point defects do not break the long-range order. This results from the fact that defects shift together with the lattice in a solid body. The energy of interaction of defects with the lattice is independent of homogeneous shift and is determined by deformation. The vortex lattice does not drag the defects of an atomic structure. And the shift of vortices caused by such a defect decreases slowly with the distance from the defect. At large distances the shifts caused by different defects accumulate and increase with distance so that the long-range order vanishes.

Proceeding from simple energy considerations it is possible to estimate the size of the region where there exists short-range order. Suppose the force of interaction of a lattice with one pinning center is $\tilde{f}_{\text{pin}}$. Then the interaction energy changes by the value $\tilde{f}_{\text{pin}} a$, if the lattice is shifted by the period a . Pinning centers are located in random points and forces are randomly orientated. Therefore, upon shifting a region containing N pinning centers, the change in energy is determined by fluctuations and is equal to

$$\delta F_{\text{pin}} = -\tilde{f}_{\text{pin}} a N^{1/2} = -\tilde{f}_{\text{pin}} a (nV_c)^{1/2} \quad (77)$$

where n is the concentration of pinning centers, V_c is the volume of the region where a short-range order exists and the relative shift is less than a period. The shift of the region results in an increase of the elastic energy. Thus, the change of the unit volume energy δF due to inhomogeneities is

$$\delta F = C_{66} (a/R_c)^2 + C_{44} (a/L_c)^2 - \tilde{f}_{\text{pin}} a (n/V_c)^{1/2}, \quad (78)$$

where R_c and L_c are the transverse and longitudinal dimensions of the region in which there is a short-range order, $V_c = R_c^2 L_c$. Let us find R_c and L_c from the condition that expression (78) has a minimum

$$\begin{aligned} R_c &= 32^{1/2} \frac{C_{66}^{3/2} C_{44}^{1/2} a^2}{n \tilde{f}_{\text{pin}}^2}, & L_c &= \frac{8a^2 C_{66} C_{44}}{n \tilde{f}_{\text{pin}}^2} \\ V_c &= \frac{256a^6 C_{44}^2 C_{66}^4}{n^3 \tilde{f}_{\text{pin}}^6} \end{aligned} \quad (79)$$

For weak inhomogeneities the pinning force can be found with perturbation theory. For superconductors with randomly varying electron-electron interaction we have

$$\lambda_{\text{eff}}^{-1} = \langle \lambda_{\text{eff}}^{-1} \rangle + \lambda_1, \quad (80)$$

where λ_1 is a random value, its correlator being

$$\varphi(\mathbf{r} - \mathbf{r}_1) = \langle \hat{\lambda}_1(\mathbf{r}) \hat{\lambda}_1(\mathbf{r}_1) \rangle. \quad (81)$$

The variation of the free energy δF in first order with respect to λ_1 is equal to

$$\delta F = v \int d^3 r \lambda_1(r) |\Delta(r)|^2. \quad (82)$$

The free energy (82) implies the following local force of interaction of the pinning center with the vortex lattice:

$$f_{\text{pin}} = v \hat{\nu}_1(r) \frac{i|\Delta|^2}{ir}. \quad (83)$$

Equation (9) makes it possible to express the correlation function of shifts $U(r)$ through the correlator $\varphi(r)$

$$\begin{aligned} \langle (u(r) - u(0))^2 \rangle &= \int \frac{d^3 \mathbf{K}}{(2\pi)^3} W(\mathbf{K}) (1 - \cos \mathbf{K} \cdot \mathbf{r}) \\ &\times \left\{ (C_{66} \mathbf{K}_\perp^2 + C_{44}(\mathbf{K}) K_z^2)^{-2} + (C_{11}(\mathbf{K}) K_\perp^2 + C_{44}(\mathbf{K}) K_z^2)^{-2} \right\}, \end{aligned} \quad (84)$$

where

$$W(\mathbf{K}) = v^2 \int d^3 \mathbf{r} \varphi(\mathbf{r}) \left\langle \frac{\partial |\Delta(r_1)|^2}{\partial \mathbf{r}_1} \frac{\partial |\Delta(r + \mathbf{r}_1)|^2}{\partial \mathbf{r}} \right\rangle \exp(-i\mathbf{K} \cdot \mathbf{r}). \quad (85)$$

The elastic modulus C_{11} always exceeds the modulus C_{66} , and the second term in formula (84) is small as compared to the first term. Using expression (76) for the modulus C_{44} and calculating the integral in formula (84) to logarithmic accuracy, yields

$$\begin{aligned} \langle (u(r) - u(0))^2 \rangle &= \frac{w(0)}{4\pi^{1/2} B C_{66}^{3/2}} \\ &\times \left\{ \left(\rho^2 + \frac{4\pi C_{66}}{B^2} z^2 \right)^{1/2} + \frac{1}{2k_h} \ln \left(\frac{\rho^2}{\xi^2} + \frac{4\pi C_{66} z^2}{B^2 k_h^2 \xi^4} \right) \right\} \end{aligned} \quad (86)$$

The region of the vortex lattice in which the relative shift is less than the interaction radius of pinning forces r_f is called a correlation region. The linear dimensions R_c of the correlation region are determined by eq. (86):

$$\langle (u(R_c) - u(0))^2 \rangle = r_f^2. \quad (87)$$

If the linear dimensions R_c of the correlation region satisfy the condition

$$R_c k_h \gg 1 \quad (88)$$

then the dispersion of elastic moduli is inessential. In this case the values L_c and R_c determined by eqs. (86) and (87) go into expression (79) with the substitution

$$a \rightarrow r_f, \quad n \tilde{f}_{\text{pin}}^2 = W(0). \quad (89)$$

In another limiting case,

$$\xi < R_c < k_h^{-1}, \quad (90)$$

the second term in eq. (86) is essential. In this case the volume of the correlation region V_c depends exponentially on the magnetic field

$$V_c \sim \exp \left\{ b \frac{12\pi^{1/2} B C_{66}^{3/2} k_h r_f^2}{W(0)} \right\} \quad (91)$$

where b is a number of the order of unity.

13.6 6. Pinning

13.6.1 6.1. Critical current

In superconductors with atomic lattice inhomogeneities (hard superconductors) the transport current may flow without flux line motion and energy dissipation. The Lorentz force in this case is balanced against the interaction with defects (pinning force f_{pin}). The current state arising from this interaction is metastable.

The problem of calculating the mean pinning force can be divided into two parts: calculation of the force of interaction of the vortex lattice with a separate defect and averaging the force over different randomly arranged pinning centers. To linear approximation in the defect concentration it is sufficient to calculate the mean force with which one defect acts on the vortex lattice. Averaging eq. (9) over coordinates we get

$$[j_c \cdot \mathbf{B}] = -n \left\langle \int f_{\text{pin}} d^3r \right\rangle \quad (92)$$

where n is the concentration of defects, and $\langle \int f_{\text{pin}} d^3r \rangle$ is the mean force of interaction of one defect with the vortex lattice. On the other hand, the force arising when the defect shifts relative to the lattice is

$$-\partial F / \partial \mathbf{u}_\infty \quad (93)$$

where F is the free energy of a superconductor with a defect. From eq. (93) it follows that the mean force is nonzero only when there exist metastable states and, consequently, F is a many-valued function of the displacement (Labusch 1969b). Averaging eq. (93) over the value of displacement yields

$$j_c B = n \delta F / \tilde{a}_1 \quad (94)$$

where δF is the free energy jump on passing from one metastable state to another, and $\tilde{a}_1$ is the mean distance where such a transition takes place.

For sufficiently weak interaction of pinning centers with the vortex lattice the interaction energy can be found with perturbation theory (eq. 82) and is a single-valued function of the vortex lattice location. In this case the critical current is determined by collective effects, and the term proportional to the concentration of defects is zero.

A qualitative picture of the formation of metastable states can be obtained. if eq. (83) of the perturbation theory is utilized for the pinning force, but account is taken of a change of the argument of the order parameter as a result of a shift caused by the defect. Then

$$u = C^{-1} f_{\text{pin}} (r - u), \quad C^{-1} = \int \frac{d^3 \mathbf{K}}{(2\pi)^3} [C_{66} \mathbf{K}_\perp^2 + C_{44}(\mathbf{K}) K_2^2]^{-1} \quad (95)$$

is the displacement of the lattice caused by the force.

If the force f_{pin} is small, eq. (95) has only one solution and metastable states are absent. Another case is realized for a sufficiently large force f_{pin} . If the maximum shift exceeds some

critical value u_c , the dependence $u(r)$ and, consequently, the functions of $f(\mathbf{r} - \mathbf{u})$ become many-valued. In this case the pinning force depends on history, i.e., hysteresis appears.

If the range where the function f_{pin} changes is equal to r_r , the solution of eq. (95) becomes many-valued under the condition that

$$u > r_f, \quad f_{\text{pin}} > Cr_f \quad (96)$$

Condition (96) can be fulfilled in the whole range of magnetic fields. In this case the dependence of the pinning force on the magnetic field has a smooth dome-shaped form.

There exist superconductors for which condition (96) is satisfied only in weak magnetic fields or in fields close to H_{c2} , where elastic moduli are small. In this case in the dependence of the critical current on the magnetic field a sharp maximum is observed near H_{c2} (peak-effect). Estimates for the pinning force f_{pin} and the position of the critical-current maximum for different types of defects have been obtained by Larkin and Ovchinnikov (1979). With increasing temperature the position of the current maximum may be invariable or moves in the direction of large or small fields. All these types of temperature dependence have been observed in experiment (Osborne and Kramer 1974, Borka et al. 1977, Vinnikov et al. 1976).

A quantitative expression for the value of the free energy jump and, hence, for the critical current density, has been obtained for some particular cases only.

For large-size defects, whose properties differ from those of the matrix, the metastable state of a vortex lattice can be studied by elasticity theory (Larkin and Ovchinnikov 1981). The critical current is high for cases where there is a barrier for the vortex to enter into the defect region. In this case the layer of vortices is adjacent to the surface of a defect in a large area. The vortex-defect interaction force is

$$f_{\text{pin}} = \frac{2}{3}\pi ab p_c, \quad (97)$$

where the adjoining area is

$$\pi ab = \frac{\pi p_c^2 R_2^{3/2} R_1^{1/2}}{4C_{44}C_{11}^{3/4}C_{66}^{1/4}}, \quad R_1/R_2 > (C_{66}C_{11})^{1/2}/C_{44} \quad (98)$$

where a, b are radii of the adjacent ellipse in the x, z directions, respectively, p_c is the maximum pressure exerted by the defect boundary onto the vortex layer,

$$p_c = \int f_{\text{pin}} dy \quad (99)$$

and $R_{1,2}$ are the radii of the defect curvature.

If the maximum displacement of vortices, u , is large compared to the lattice period, the force of interaction of each defect with the lattice is close to the maximum one, and averaging over defect positions is reduced to simple summation. In the general case the averaging of the force leads to eq. (94). Therefore, for calculating the mean force it is necessary to find the free energy of the vortex lattice with a defect as a function of a u , -displacement of the lattice at large distances relative to the defect and calculate the jump of this function.

If the size of a defect is comparable with or less than the lattice period, then metastable states on a single defect may arise, provided the properties of the defect are considerably different from those of the matrix. In this case distortion of the vortex lattice near the defect is strong and cannot be found by means of elasticity theory. To calculate the functions $F(u,)$ it is necessary to solve numerically the Ginzburg-Landau equation or the more general eqs. (11) and (12). Such calculations have been performed for cylindrical pores of small radius in a field close to H_{c2} (Ovchinnikov 1980). The dependence of the critical current on the magnetic field in this case turns out to be nonmonotonous and by order of magnitude is

$$j_c B \sim (1 - T/T_c)^{1.5} (1 - H/H_{c2})^2. \quad (100)$$

In case no metastable states arise on a separate pinning center, the pinning force is much weaker and appears due to collective effects, when a large volume of the order of V_c turns into a metastable state (eqs. 79, 80).

If the flowing current is less than the critical one, each of the volumes V_c shifts independently by a distance less than the size of the vortex core under the Lorentz force, so that the arising pinning force would compensate the Lorentz force. Since inside the volume V_c the lattice is almost regular, and the pinning centers are randomly arranged, the pinning forces acting on the lattice from each center almost compensate each other. The maximum force acting on the volume V_c is equal to $\tilde{f}_{\text{pin}} N^{1/2}$, where $N = nV_c$ is the number of pinning centers in the volume V_c . The critical current is

$$Bj = \tilde{f}_{\text{pin}} \frac{n}{N^{1/2}} = \tilde{f}_{\text{pin}} (n/V_c)^{1/2} \quad (101)$$

When the space dispersion of elastic moduli is not important, the volume V_c is determined by eq. (79), and the critical current (Larkin and Ovchinnikov 1973) is

$$Bj = \frac{n^2 \tilde{J}_{\text{pin}}^4}{16a^3 C_{44} C_{66}^2}. \quad (102)$$

In a magnetic field close to H_{c2} the elastic modulus C_{44} reaches a constant limit, $C_{66} \sim (H_{c2} - B)^2$, and the force $\tilde{f}_{\text{pin}}$ is in many cases proportional to Δ^2 , i.e. $H_{c2} - B$. As a result, the critical current only slightly depends on the magnetic field.

While the magnetic field approaches H_{c2} , its effective penetration depth $\delta = k_h^{-1}$ increases and exceeds the correlation radius R_c . In this case one should take into account the space dispersion of the modulus C_{++} , and for the volume V_c expression (91) must be used. The critical current increases sharply (peak-effect) with increasing magnetic field (Larkin and Ovchinnikov 1978). The current stops increasing as R_c becomes less than the lattice period a . There is no further decrease in the volume V_c , and due to the decrease in the force $\tilde{f}_{\text{pin}}$ the current decreases proportionally with $H_{c2} - B$. Another, more reasonable cause of the peak-effect has been described in the foregoing and is connected with the fact that in the vicinity of H_{c2} the elastic modulus C_{66} drops, the lattice softens and strong pinning with formation of metastable states on a single defect becomes possible.

Above, the case is considered when defects only slightly distort the vortex lattice in the main part of the volume. In case the field or temperature is close to the critical one and there are many defects, the phase transition is smeared and in the area of smearing there arises a structure strongly distinguished from the periodic one (Ioffe and Larkin 1981). In this case superconductivity arises in separate, widely spaced regions connected thanks to the proximity effect. The real transition temperature and the critical current in such a structure are found by means of percolation theory.

The transport current or thermal fluctuations break the weakest bonds. The critical current is determined from the condition that unbroken bonds form a continuous cluster. In the region of a smeared phase transition the critical current decreases exponentially with increasing magnetic field. For instance, for defects which are inclusions of another phase with a critical magnetic field H_{c2} , exceeding the critical field of the matrix H_m , for magnetic fields $H > H_m$ the critical current is

$$j = j_0 \zeta_c^{-1.8} \xi \xi_{11} n^{2/3} \exp(-\zeta_c) \quad (103)$$

$$\zeta_c^{-2} = 2, 7\pi n \xi^2 \check{\zeta}_{11}, \quad \check{\zeta}_{11}^2 = \zeta^2 \frac{H_m}{H - H_m}$$

where n is the concentration of inclusions, ξ is the size of a pair (correlation radius) of the matrix and j_0 is the depairing current in the inserted substance.

Two types of critical current dependence on the magnetic field value have been observed experimentally. In the case of strong pinning this dependence has a smooth maximum at a magnetic field $(0.3-0.5) H_{c2}$. Usually in the case of weak pinning, generally in a wide range of magnetic fields the pinning force only slightly depends on the magnetic field and only in the vicinity of H_{c2} there is a narrow high maximum (peak-effect). In magnetic fields not close to H_{c2} , a collective pinning is likely to be realized. This fact is indicated both by the small value of the critical current and by its nonlinear dependence on the concentration of defects (Kpfer and Manuel 1979).

It would be natural to suppose that in the region of magnetic fields beyond the peak strong pinning is realized in the current density, as in the case of the dome-shaped dependence of the current density on the magnetic field. A most simple picture of strong pinning has been described above and consists in formation of metastable states on a single pinning center. In this case the mean force is obtained by simple summation. However, such a simple picture does not always agree with experiment. A quantitative comparison of theory and experiment can be carried out for the pinning centers produced as a result of irradiation. Under irradiation it is possible to check the number and the magnitude of defects. In some cases (Kramer 1978) a linear dependence of the mean pinning force on concentration of small-size defects is observed, for which the Labusch criterion is not fulfilled even when allowance is made for the space dispersion of elastic moduli. In other cases (Kpfer and Manuel 1979, Reichert et al. 1981) the pinning force does not generally depend on the concentration of defects. May be in this case plastic flow of the vortex lattice takes place.

13.6.2 6.2. I-V characteristic

At a transport current density exceeding the critical value flux lines start moving, and in a superconductor there arises a voltage proportional to the velocity of the flux line motion. The transport current in this case is determined by the sum of the pinning and viscous forces. The viscous force has been considered above and for a weak electric field $\mathbf{E}$ is proportional to $\mathbf{E}$. The pinning force is also dependent on the velocity of the flux line motion which affects the shape of the $I - V$ characteristic. If metastable states are formed on a single defect (strong pinning), the pinning force decreases with increasing velocity, since the moving lattice badly fits the defect shape. The mutual influence of defects diminishes with increasing velocity. This usually decreases the mean pinning force. Therefore, the collective part of the pinning force increases with velocity.

Let us first consider weak pinning. Quantitative results in this case can be obtained at a sufficiently high velocity of the flux line motion, when the pinning force in eq. (9) can be taken into account by perturbation theory (Larkin and Ovchinnikov 1973c, Schmid and Hauger 1973). It is convenient to write eq. (9) in the integral form

$$\begin{aligned}
 u &= u_0 + u_1 \\
 u_1(r, t) &= \int_{-\infty}^t dt_1 \int d^3r M(t - t_1, r - r_1) \\
 &\quad \times \left\{ f_{\text{pin}}(r_1, r_1 - u_0(t_1) - u_1(r_1, t_1)) + [\langle j \rangle \mathbf{B}] - \sigma B^2 \frac{\partial u_0}{\partial t} \right\}.
 \end{aligned} \tag{104}$$

The retarded Green function M satisfies the equation

$$\begin{aligned}
& - \left[(C_{11} - C_{66}) \frac{\partial}{\partial \rho} \left(\frac{\partial M}{\partial \rho} \right) + \left(C_{66} \frac{\partial^2}{\partial \rho^2} + C_{44} \frac{\partial^2}{\partial z^2} \right) M \right] + \sigma B^2 \frac{\partial M}{\partial t} \\
& = \delta(t - t_1) \delta(r - r_1).
\end{aligned} \tag{105}$$

In a time-independent electric field $\mathbf{E}$ the lattice moves with a constant velocity v and $u_0 = vt$. The electric field E is connected with the velocity v by eq. (15) and the current density $\langle j \rangle$ is expressed in $\mathbf{E}$ and the pinning force by eq. (10).

In zero order with respect to u_1 the mean force f_{pin} is zero and the current density $\mathbf{j} = \sigma \mathbf{E}$. In first order with respect to $\mathbf{u}_1$ we have

$$[j_1 \mathbf{B}] = \left\langle \left(u_1 \frac{\partial}{\partial u_0} \right) f_{\text{pin}} \right\rangle. \tag{106}$$

For the pinning force determined by inhomogeneities of the interaction constant, eq. (83) can be easily rewritten as

$$f_{\text{pin}}(r, r - u_0) = i\nu \lambda_1(r) \sum_{\mathbf{n}} \mathbf{K}_n |\Delta|_n^2 \exp(i\mathbf{K}_n(r - u_0)) \tag{107}$$

where $\mathbf{K}_n$ are reciprocal lattice vectors and $|\Delta|_n^2$ are the Fourier components of the function $|\Delta(r)|^2$. Substituting expression (107) into eqs. (104) and (106) yields

$$\begin{aligned}
[j_1 \mathbf{B}] = & \frac{iv^2}{2} \sum_{\mathbf{n}} \int \frac{d^3 g}{(2\pi)^3} \varphi_n |(|\Delta|_n^2)|^2 \left\{ (C_{66} g_{\perp}^2 + C_{44} g_z^2 + i\sigma B^2 (\mathbf{K}_n v))^{-1} \right. \\
& \left. + (C_{11} g_{\perp}^2 + C_{44} g_z^2 + i\sigma B^2 (\mathbf{K}_n v))^{-1} \right\},
\end{aligned} \tag{108}$$

where

$$\varphi_n = \int d^3 r \exp(-i\mathbf{K}_n \cdot r) \langle \lambda_1(r) \lambda_1(0) \rangle \tag{109}$$

At small velocities v in the integral (108) small g are essential:

$$C_{66} g_{\perp}^2 \sim \sigma B^2 v/a, \tag{110}$$

where a is the linear size of the cell. If the value $g_{\perp}^2$ determined by formula (110) satisfies the condition

$$g_{\perp}^2 < k_h^2 \tag{111}$$

then the space dispersion of the elastic modulus C_{44} is unimportant and the correction j_1 to the current density is proportional to $v^{1/2}$

$$\begin{aligned}
[j_1 \mathbf{B}] = & \frac{v^2}{8\pi (2C_{44})^{1/2}} \left(\frac{1}{C_{66}} + \frac{1}{C_{11}} \right) \sum_{\mathbf{n}} \varphi_n |(|\Delta|_n^2)|^2 \mathbf{K}_n \mathbf{K}_n^2 \\
& \times (\sigma B^2 |(\mathbf{K}_n v)|)^{1/2} \text{sign}(\mathbf{K}_n v).
\end{aligned} \tag{112}$$

It should be noted that the pinning force increases with velocity. With increasing velocity, the value $g_{\perp}$ increases, condition (111) is not

satisfied anymore and space dispersion of the elastic modulus C_{44} becomes important. In this case the correction to the current density depends logarithmically on velocity

$$[j_1 \mathbf{B}] = \frac{v^2 \sigma B}{16\pi^{1/2} k_h C_{66}^{3/2}} \sum_n \varphi_n |(\Delta_n^2)|^2 \mathbf{K}_n \mathbf{K}_n^2 (\mathbf{K}_n v) \ln \left(\frac{C_{66}}{a^2 \sigma B^2 |(\mathbf{K}_n v)|} \right). \quad (113)$$

Equations (112) and (113) are obtained by expanding in $\mathbf{u}_1$. Such an expansion is valid for sufficiently high velocities. Investigation of the subsequent terms of the expansion shows that the IV characteristic of a superconductor with a weak pinning can be presented in the form

$$j(\mathbf{E}) = j_c \mathcal{T}(\sigma E / j_c), \quad (114)$$

where $\mathcal{T}(x)$ is some universal function. The critical current density j_c is determined by eq. (101).

Now we would like to find the $I - V$ characteristic for strong pinning. As has been shown above, the critical current in this case is determined by the free energy of a superconductor with one defect. The free energy is a function of lattice shift relative to defect.

The shape of the $I - V$ characteristic in some region of the electric field is also determined by the dependence of the free energy on the value of the displacement.

When the lattice moves with a small velocity, its deformation far from the defect is described by the elasticity equation (9), even if in the vicinity of the defect the deformation is plastic and this equation is inapplicable. It is important that in the vicinity of the defect deformation is the same as that in the static case and the force of interaction of the defect with the lattice is determined by expression (93). However, the argument in this formula is not the displacement at infinity $\mathbf{u}$, but the displacement that would have existed in a static problem. The difference in these displacements is determined by eq. (104) and is equal to

$$u_1(t) = \int dt_1 (M_v(t - t_1) - M_{t=0}(t - t_1)) \tilde{f}_{\text{pin}}(t_1) \quad (115)$$

where the Green function M is determined by eq. (105). We are interested in the displacement u_1 at the location of the defect. At this point expression (115) is reduced to the form

$$u_1(t) = -\frac{1}{8\pi C_{44}^{1/2}} \left(\frac{1}{C_{66}} + \frac{1}{C_{11}} \right) \left(\frac{\sigma B^2}{\pi} \right)^{1/2} \frac{\partial}{\partial t} \int_{-\infty}^t \frac{dt_1}{(t - t_1)^{1/2}} \hat{f}_{\text{pin}}(t_1). \quad (116)$$

The mean force is

$$[j \mathbf{B}] = -\frac{n}{T} \int_0^T dt f_{\text{pin}}(vt - u_1(t)) \quad (117)$$

where $T = \hat{a}_1/v$ is the time between jumps. In zero-order approximation at $u_1 = 0$ we obtain eq. (94) for the current density from eqs. (93) and (117). In this approximation the time-averaged value coincides with the result of averaging over a random distribution of defects. To calculate the next approximation we perform a substitution of the argument in eq. (117)

$$vt - u_1(t) \rightarrow v\tilde{t}$$

Using expression (116) for $\mathbf{u}_1$, we find the correction to the current density which depends on the velocity of the vortex lattice motion

$$[j_1 \mathbf{B}] = - \frac{1}{8\pi C_{44}^{1/2}} \left(\frac{1}{C_{66}} + \frac{1}{C_{11}} \right) \left(\frac{\sigma B^2}{\pi} \right)^{1/2} \frac{n}{v^2 T} \times \int_0^T dt \left(\tilde{f}_{\text{pin}}(vt) - \tilde{f}_{\text{pin}}(0) \right) \frac{\partial^2}{\partial t^2} \int_{-\infty}^t \frac{dt_1}{(t-t_1)^{1/2}} \left(v \tilde{f}_{\text{pin}}(vt_1) \right) \quad (118)$$

where $\tilde{f}_{\text{pin}}(0)$ is the value of the force at the moment immediately after the jump. The subtraction in eq. (118) has been performed to eliminate spurious divergence. It follows from eq. (118) that the correction to the current density is proportional to v^{12} . The proportionality coefficient depends on the explicit form of the function $\tilde{f}_{\text{pin}}$. For large-size defects the force $\tilde{f}_{\text{pin}}$ is always close to its maximum value. Deviation from the maximum value for the whole period of motion is a linear function of time.

$$\tilde{f}_{\text{pin}}(t) = \mathcal{J}_m \left(1 - \frac{T-t}{T} C_1 \right), \quad 0 < t < T \quad (119)$$

The coefficient C_1 is of the order of the ratio of the lattice period to the displacement caused by the defect. The IV characteristic is obtained from the formulas (118) and (119):

$$j = \sigma E + j_c \left\{ 1 - \frac{\tilde{f}_m C_1^2}{8 \cdot 2^{1/2} \frac{\pi^2 \tilde{a}_1^{3/2}}{C_{44}^{1/2}}} \left(\frac{1}{C_{66}} + \frac{1}{C_{11}} \right) (-\zeta_{112}) (\sigma B^2 2\pi i)^{12} \right\} \quad (120)$$

Equation (120) has also been obtained by another method (Larkin and Ovchinnikov 1981). There the dependence of the coefficient C_1 on the defect characteristics has been found as well.

Thus, with increasing velocity of the lattice motion the mean pinning force decreases. This decrease in the pinning force at small velocities exceeds the viscosity force. So, voltage jumps should exist on the $I - V$ characteristic. This has been observed in experiment (Chang and McKinnon 1968, Chang et al. 1969).

The square root dependence of the current density on the velocity of the flux line lattice motion is connected with deformation of the lattice at large distances of order $v^{-1/2}$. While deriving eq. (120) the concentration of defects has been assumed to be small and their mutual effect is not taken into consideration. The initial part of the $I - V$ characteristic is determined by collective effects. Therefore, at sufficiently high defect concentration no jumps on the $I - V$ characteristic can be observed.

An interesting phenomenon may occur when a high-frequency current is passing through a superconducting film. In this case the $I - V$ characteristic can be obtained, provided in eqs. (104) and (107) the mean velocity v is regarded as a function of time

$$\frac{\partial u_0}{\partial t} = v_0 + v_1 \cos \omega t \quad (121)$$

Analogously as eq. (108) has been obtained, we derive the following equation for the current density in an alternating field (Schmid and Hauger 1973, Larkin and Ovchinnikov 1973c)

$$[j_i \mathbf{B}] = \frac{v^2}{16dC_{66}} \sum_{n,m} \varphi_m \mathbf{K}_m \mathbf{K}_m^2 I_n^2 \left(\frac{(\mathbf{K}_m \mathbf{v}_1)}{\omega} \right) |(|\Lambda_m^2|)^2 \text{sign}((\mathbf{K}_m v_0) - n\omega). \quad (122)$$

Therefore on the $I - V$ characteristic steps arise at voltages satisfying the Josephson relation $2\pi E m = B \omega n a$ (a is the lattice period, m, n are integers). The value of these steps is proportional to the critical current. Such steps have been observed experimentally (Fiory 1971).

13.7 7. Conclusion

In this chapter the theory of flux line motion in superconductors has been reviewed. Experimental results have been mentioned only when they could provide information confirming or contradicting the theoretical results.

The theory of flux line motion in homogeneous superconductors without pinning forces has been developed fairly well. The dependence of conductivity on temperature and magnetic field and non-linear effects on an electric field are investigated for all the cases where the solution of a static equilibrium problem is known. Usually, these results are in agreement with experiment, provided the pinning force effect is separated. The experimental data on critical pinning currents are very numerous, but they are poorly systematized because no fundamental theory of this phenomenon is yet available. The pinning theory is developed sufficiently well for those cases where the lattice deformation may be assumed to be elastic. In this case it is possible to explain

the peak on the dependence of a critical current on a magnetic field near H_{c2} ("peak-effect") and some singularities on the I V characteristic. In many cases the vortex lattice deformation is likely to be plastic. The theory of plastic deformation exists only for some simple cases. A quantitative comparison of theory and experiment is also difficult, since in most cases the character of defects and the force of the single-defect interaction with a vortex lattice is unknown.

13.8 Appendix. Kinetic equation for dirty superconductors

In normal metals the classical kinetic equation is applicable when the electron mean free path is large in comparison with the wavelength of electrons. In superconductors, in addition to the excitation wavelength there is another parameter: the size of a pair or the correlation length. Therefore, the standard kinetic equation is applicable only in the cases when the mean free path and the range in which the order parameter changes, greatly exceeds the correlation length.

Here we obtain a more general but more complicated quasi-classical equation. While deriving this equation we use only the smallness of the electron wavelength compared to all other characteristic lengths (Eliashberg 1971, Larkin and Ovchinnikov 1973a, 1975, 1977, Shelankov 1980, Schmid 1981, Watts-Tobin et al. 1981). This equation is a generalization of static quasi-classical equations for the Green functions (Eilenberger 1968, Larkin and Ovchinnikov 1968).

It is convenient to represent the Green functions of a superconductor in matrix form, the elements of the matrix being ordinary and anomalous Green functions:

$$\begin{aligned} G^>(1, 2) &= -i \left\langle \begin{pmatrix} \psi_{\uparrow}(1)\psi_{\uparrow}^+(2) & \psi_{\uparrow}(1)\psi_{\downarrow}(2) \\ -\psi_{\downarrow}^+(1)\psi_{\uparrow}^+(2) & -\psi_{\downarrow}^+(1)\psi_{\downarrow}(2) \end{pmatrix} \right\rangle, \\ G^<(1, 2) &= i \left\langle \begin{pmatrix} \psi_{\uparrow}^+(2)\psi_{\uparrow}(1) & \psi_{\downarrow}(2)\psi_{\uparrow}(1) \\ -\psi_{\uparrow}^+(2)\psi_{\downarrow}^+(1) & -\psi_{\downarrow}(2)\psi_{\downarrow}^+(1) \end{pmatrix} \right\rangle. \end{aligned} \quad (\text{A.1})$$

The averaging is performed over the dynamic state. It is supposed that at $t \rightarrow -x$ the system is in equilibrium, and then the external field transforms it into some dynamic state. In a static case, using the functions $G^>$ and $G^<$ it is possible to compose chronologically ordered averages; the equations for these have been obtained by Gorkov (1958). In the dynamic case we apply the methods proposed by Keldysh (1964) for deriving the kinetic equation in a normal metal. Keldysh showed that for the dynamic case the Dyson equation is replaced by a system of equations for three values:

$$\begin{aligned}
G^R(1, 2) &= \theta(t_1 - t_2) \{G^>(1, 2) - G^<(1, 2)\}, \\
G^A(1, 2) &= -\theta(t_2 - t_1) \{G^>(1, 2) - G^<(1, 2)\}, \\
G^K(1, 2) &= G^>(1, 2) + G^<(1, 2).
\end{aligned} \tag{A.2}$$

This system can be written in matrix form, provided the following matrices are introduced:

$$\check{G} = \begin{pmatrix} G^R & G^K \\ 0 & G^A \end{pmatrix}, \quad \check{\Sigma} = \begin{pmatrix} \Sigma^R & \Sigma^K \\ 0 & \Sigma^A \end{pmatrix}. \tag{A.3}$$

Each element of the matrix (A.3) in the Keldysh space is a matrix in the Gor'kov space as well. These matrices satisfy the equation

$$(\check{G}_0^{-1} - \check{\Sigma}) \check{G} = \check{1}. \tag{A.4}$$

Here the product means multiplication of matrices and convolution with respect to the coordinates and time,

$$\int d^4x_3 (\check{G}_0^{-1} - \check{\Sigma})(x_1, x_3) \check{G}(x_3, x_2) = \check{\delta}(x_1 - x_2) \tag{A.5}$$

where

$$\check{G}_0^{-1}(x_1, x_2) = \left(i\tilde{\tau}_z \frac{\partial}{\partial t_1} - e\varphi + \mu + \frac{1}{2m} \check{\partial}^2 \right) \check{\delta}(x_1 - x_2), \tag{A.6}$$

$$\check{\partial} = \frac{\partial}{\partial r} - ieA\tilde{\tau}_z, \tag{A.7}$$

where φ is the scalar potential, $\mathbf{A}$ is the vector potential, and μ is the chemical potential.

The two-row Pauli matrices τ act in the space of the Gor'kov matrices and are unit matrices in the space of the Keldysh matrices. In a normal metal the quasiclassical approximation makes it possible to obtain an equation for the density matrix which is equal to the Green function in coinciding times $t_1 = t_2$. In superconductors the equation for the Green function at close coordinate points is more general. As is seen from eq. (A.5), the Green function is a rapid function of the relative variable $r_1 - r_2$, and it is convenient to perform the Fourier transformation with respect to this variable:

$$\begin{aligned}
\check{G}(\mathbf{p}, \mathbf{r}) &= \int d^3\mathbf{r}' \check{G}(\mathbf{r}_1, \mathbf{r}_2) \exp(-i\mathbf{p} \cdot \mathbf{r}') \\
\mathbf{r}_{1,2} &= \mathbf{r} \pm \frac{1}{2}\mathbf{r}'
\end{aligned} \tag{A.8}$$

After such a transformation the main term in G_0^{-1} is $p^2, 2m - \mu$. Therefore, $\check{G}(p)$ has a sharp maximum near the Fermi surface. To cancel large terms we subtract from eq. (A.5) the equation conjugated to it,

$$(\check{G}_0^{-1} - \check{\Sigma}) \check{G} - \check{G} (\check{G}_0^{-1} - \check{\Sigma}) = 0. \tag{A.9}$$

In this equation only $\check{G}$ has a sharp dependence on p^2 . Therefore, it can be integrated with respect to $\xi_p = p^2/2m - \mu$ and one may obtain an equation for the quasiclassical Green function,

$$\check{g}(\mathbf{p}, \mathbf{r}) = \frac{i}{\pi} \int d\xi_p \check{G}(\mathbf{p}, r) \quad (\text{A.10})$$

The dependence on p in the left-hand side of formula (A.10) means only the dependence on the directions of the vector p , its value being equal to $p = p_F$;

$$\left[\tau_z \frac{\partial}{\partial t} \delta(t - t_1) + ie\varphi(t) + \frac{p}{m} \check{\partial} + i\check{\Sigma}, \circ \check{g} \right]_- = 0 \quad (\text{A.11})$$

[,] denotes a commutator, and the sign $\circ$ is convolution with respect to time, so that in detailed notation eq. (A.11) has the form

$$\begin{aligned} \check{\tau}_z \frac{\partial \check{g}}{\partial t} + \frac{\partial \check{g}}{\partial t'} \check{\tau}_z + \frac{p}{m} \left(\frac{\partial \check{g}}{\partial r} - ie\mathbf{A}(t) \check{\tau}_z \check{g} + ie\mathbf{A}(t') \check{g} \check{\tau}_z \right) + ie\varphi(t) \check{g} - i\check{y}\varphi(t') \\ + i \int_{-\infty}^{\infty} dt_1 \{ \check{\Sigma}(t, t_1) \check{g}(t_1, t') - \check{g}(t, t_1) \check{\Sigma}(t_1, t') \} = 0 \end{aligned} \quad (\text{A.12})$$

Equation (A.12) is homogeneous relative to the function $\dot{g}$. An additional condition fixing normalization of the function and essentially simplifying the solution of the equation is

$$\check{g} \circ \check{g} = \check{1}. \quad (\text{A.13})$$

Direct calculation shows that this condition is fulfilled in the equilibrium case and that it is compatible with eq. (A.12). It is possible to express the self-energy operator $\check{\Sigma}$ in $\check{g}$, taking into account the interaction with phonons and impurities by perturbation theory. To first order we have

$$\begin{aligned} \Sigma^>(1, 2) = G^>(1, 2) D^<(1, 2), \quad \Sigma^<(1, 2) = G^<(1, 2) D^>(1, 2), \\ D^>(1, 2) = D^<(2, 1) = -i \langle \phi(x_1) \phi(x_2) \rangle, \end{aligned} \quad (\text{A.14})$$

where ϕ is the potential of the interaction of electrons with phonons or impurities. The impurity field is time-independent and classical. Therefore, $D_{\text{imp}}^> = D_{\text{imp}}^<$,

$$\check{\Sigma}_p^{\text{imp}}(t, t') = -\frac{1}{2} inv \int d\Omega_{p_1} \sigma_{pp_1} \check{g}_{p_1}(t, t') \quad (\text{A.15})$$

where $\sigma_{pp'}$ is the impurity scattering cross-section, and n is the impurity concentration.

If there exist spin-flip impurities, then these would contribute to the selfenergy

$$\check{\Sigma}_s(t, t') = -\frac{i}{2\tau_s} \int \frac{d\Omega_{p_1}}{4\pi} \check{\tau}_- \check{g}_{p_1}(t, t') \check{\tau}_-$$

Representing in formula (A.14) $G^{\geq}$ and $D^{\geq}$ through $G^{\text{R.A.K}}$, $D^{\text{R.A.K}}$ we obtain

$$\begin{aligned} \sum_p^{\text{ph(R, A)}}(t, t') &= \frac{1}{8} \lambda_{\text{eff}} \int d\Omega_{p_1} \left\{ D_{p-p_1}^{\text{K}}(t' - t) g_{p_1}^{\text{R, A}}(t, t') + D_{p-p_1}^{\text{A, R}}(t' - t) g_{p_1}^{\text{K}}(t, t') \right\} \\ \sum_p^{\text{ph(K)}}(t, t') &= \frac{1}{8} \lambda_{\text{eff}} \int d\Omega_{p_1} \left\{ D_{p-p_1}^{\text{K}}(t' - t) g_{p_1}^{\text{K}}(t, t') \right. \\ &\quad \left. - (D_{p-p_1}^{\text{R}}(t' - t) - D_{p-p_1}^{\text{A}}(t' - t)) (g_{p_1}^{\text{R}}(t, t') - g_{p_1}^{\text{A}}(t, t')) \right\} \end{aligned} \quad (\text{A.16})$$

where $i_{\text{eff}} = v g_{\text{eff}}^2$, and g_{eff} is the constant of the electron-phonon interaction. The phonon Green functions $D^{\text{R,A}}$ are equal to

$$D_k^{\text{R}}(\omega) = D_k^{\text{A}}(\omega)^* = -\frac{\omega^2(\mathbf{k})}{\omega^2(\mathbf{k}) - (\omega + i\delta)^2} \quad (\text{A.17})$$

In thermodynamic equilibrium the phonon function $D^K(k, \omega)$ is given by the expression

$$D^K(k, \omega) = (D_k^{\text{R}}(\omega) - D_k^{\text{A}}(\omega)) \coth(\omega/2T) \quad (\text{A.18})$$

If the phonons are not in a thermodynamic equilibrium state, then in eq. (A.18) one should substitute $\coth(\omega/2T)$ by $1 + 2n(\omega)$, where $n(\omega)$ is the phonon distribution function. The terms in Σ^{ph} , proportional to $D^{\text{R}} - D^{\text{A}}$ or to D^K , describe the processes accompanied by emission or absorption of phonons. The term proportional to $D^{\text{R}} + D^{\text{A}}$ corresponds to virtual phonon exchange. The frequencies of virtual phonons are of the order of ω_{D} and are large compared to the external field frequency and temperature. Therefore,

$$D_k^{\text{R}}(t - t') + D_k^{\text{A}}(t - t') = -2\delta(t - t') \quad (\text{A.19})$$

Such a term is inserted with the same coefficient into $2^{\text{ph(R,A)}} \text{ }^{\text{(R)}}$ and is absent from $\Sigma^{\text{ph(K)}}$. The contribution of virtual phonons to Σ^{ph} is thus proportional to a unit matrix in the Keldysh space and equals $-\delta(t - t') \bar{\Delta}$, where

$$\bar{\Delta} = \begin{pmatrix} \hat{\Delta} & 0 \\ 0 & \hat{\Delta} \end{pmatrix}, \quad \hat{\Delta}(t) = \frac{1}{8} \lambda_{\text{eff}} \int d\Omega_p g_p^K(t, t). \quad (\text{A.20})$$

The current density is also expressed through the Green function, integrated over the energy variable

$$j = -\frac{1}{8} e v \int d\Omega_p \text{Tr} \tau_z \frac{p}{m} g_p^K(t, t). \quad (\text{A.21})$$

The diagonal elements $y^K(t, t)$ in formula (A.21) give renormalization of the chemical potential and velocity on the Fermi surface. Below, these effects are neglected.

For superconductors with a small electron free path the system of equations (A.12) can be simplified and reduced to an equation in the Green function $\check{g}$, averaged over the directions of the vector p .

In this case the Green function $\check{g}_p$ can be presented in the form

$$\check{g}_p = \check{g} + p \check{g} \check{g}_1, \quad p \check{g}_1 \ll \check{g}, \quad (\text{A.22})$$

where the functions $\check{g}$ and $\check{g}_1$ are independent of the direction of the vector p . From the normalization conditions (A.13) it follows that

$$\check{g} \circ \check{g} = \check{1}, \quad \check{g} \circ \check{g}_1 + \check{g}_1 \circ \check{g} = 0. \quad (\text{A.23})$$

Substituting expression (A.22) for the function $\check{g}_p$ into eq. (A.12) and taking into account relation (A.23) we get

$$\check{g}_1 = -\frac{\tau_{\text{tr}}}{m} \{ i e \mathbf{A} \tau_z \delta(t - t') + \check{g} \circ \check{\partial} g \}. \quad (\text{A.24})$$

Substituting expressions (A.22) and (A.24) for the function $\check{g}_p$ into eq. (A.12) and averaging over the directions of the vector $\mathbf{p}$ yields

$$\begin{aligned}
& -D \frac{\partial}{\partial r} \{ie \mathbf{A} \tilde{\tau}_z \delta(t-t') + \check{g} \circ \check{\partial} \check{g}\} + \tilde{\tau}_z \frac{\partial \check{g}}{\partial t} + \frac{\partial \check{g}}{\partial t'} \tilde{\tau}_z + (-i\check{\Delta}(t) + ie\varphi(t))\check{g} \\
& - \check{g} (-i\check{\Delta}(t') + ie\varphi(t')) + ieD \{ \mathbf{A}(t) \tilde{\tau}_z \check{g} \circ \check{\partial} \check{g} - (\check{g} \circ \check{\partial} \check{g}) \mathbf{A}(t') \tilde{\tau}_z \} \\
& = -i (\check{\Sigma}^{\text{ph}} \circ \check{g} - \check{g} \circ \check{\Sigma}^{\text{ph}})
\end{aligned} \tag{A.25}$$

where $D = vl_{\text{tr}}$, 3 is the diffusion coefficient. Equation (A.25) contains equations for $y^{R,A}$ and y^K . The diagonal elements of the matrix equation (A.25) give equations for $g^{K,A}$. They have the same form as eq. (A.25), with $\check{g}$ substituted by $g^{R,A}$. The normalization equation (A.25) denotes that

$$g^{R,A} \circ g^{R,A} = 1, \quad g^R \circ g^K + g^K \circ g^A = 0. \tag{A.26}$$

Any matrix satisfying the second condition of eq. (A.26) can be presented in the form

$$g^K = g^R \circ \hat{f} - \hat{f} \circ g^A. \tag{A.27}$$

The non-diagonal element of the matrix equation (A.25) determines the matrix y^K . We insert it into this equation in the form of (A.27). From the equation obtained we subtract the equation for y^R , multiplied by $\hat{f}$, from the right-hand side, and add the equation for g^A , multiplied by $\hat{f}$, from the left-hand side. As a result we have

$$\begin{aligned}
& -D \frac{\partial}{\partial r} \left(\frac{\partial \hat{f}}{\partial r} - g^R \circ \frac{\partial \hat{f}}{\partial r} \circ g^A \right) + g^R \circ \left(\tau \frac{\partial \hat{f}}{\partial t_i} + \frac{\partial \hat{f}}{\partial t'} \tau_z \right) - \left(\tau_z \frac{\partial \hat{f}}{\partial t} + \frac{\partial \hat{f}}{\partial t_i} \tau_z \right) \circ g^A \\
& + D \frac{\partial \hat{f}}{\partial r} \circ g^A \circ \partial g^A - D (g^R \circ \partial g^R) \circ \frac{\partial \hat{f}}{\partial r} + [\hat{f} - i\hat{\Delta} + ie\varphi) - (-i\hat{\Delta} + ie\varphi)\hat{f}] \circ g^A \\
& + g^R \circ [(-i\hat{\Delta} + ie\varphi)\hat{f} - \hat{f}(-i\hat{\Delta} + ie\varphi)] + ieD \left(\mathbf{A} \tau_z \frac{\partial \hat{f}}{\partial \mathbf{r}} - \frac{\partial \hat{f}}{\partial \mathbf{r}} \mathbf{A} \tau_z \right) \\
& + ieD \left(g^R \circ \frac{\partial \hat{f}}{\partial \mathbf{r}} \circ g^A \mathbf{A} \tau_z - \mathbf{A} \tau_z g^R \circ \frac{\partial \hat{f}}{\partial \mathbf{r}} \circ g^A \right) \\
& + ieD \left(\hat{f} \mathbf{A} \tau_z - \mathbf{A} \tau_z \hat{f} \right) \circ g^A \partial g^A - ieD (g^R \circ \partial g^R) \circ \left(\hat{f} \mathbf{A} \tau_z - \mathbf{A} \tau_z \hat{f} \right) \\
& + ieD \left(\frac{\partial \mathbf{A}}{\partial \mathbf{r}} \tau_z \hat{f} - \hat{f} \frac{\partial \mathbf{A}}{\partial \mathbf{r}} \tau_z \right) \\
& = -i \left\{ - \sum^{\text{ph(R)}} \circ \hat{f} \circ g^A + \sum^{\text{ph(K)}} \circ g^A - g^R \circ \sum^{\text{ph(K)}} \right. \\
& \quad \left. - g^R \circ \hat{f} \circ \sum^{\text{ph(A)}} + g^R \circ \sum^{\text{ph(R)}} \circ \hat{f} + \hat{f} \circ \sum^{\text{ph(A)}} \circ g^A \right\}.
\end{aligned} \tag{A.28}$$

Of the four equations for the matrix $\hat{f}$ only two are linearly independent. Following the papers by Schmid and Schön (1975) and Larkin and Ovchinnikov (1977), we choose the matrix $\hat{f}$ to be diagonal,

$$\hat{f} = f + f_1 \tau_z. \tag{A.29}$$

In a paper by Shelankov (1980) another choice for the matrix $\hat{f}$ has been proposed:

$$\hat{f} = f + g^R f_1. \quad (\text{A.30})$$

For fields slowly varying with time (frequency $(1) \ll 1$), the Green functions $y^{R,\uparrow}$ and the distribution function $\hat{f}$ are sharp functions of time difference and slowly varying functions of the half-sum of times. Therefore, it is convenient to perform the Fourier transformation over the time difference. The Fourier component of convolution over the time variable is determined by the formula

$$A \circ B = \exp \left[\frac{1}{2} i (\partial_\varepsilon^A \partial_t^B - \partial_t^A \partial_\varepsilon^B) \right] A(\varepsilon, t) B(\varepsilon, t), \quad (\text{A.31})$$

where the differential operator λ^4 affects A only. Substituting expression (A.29) for $\hat{f}$ into eq. (A.28) we get

$$\begin{aligned} & -D \frac{\hat{c}}{\partial r} \text{Tr} \left(\frac{\partial f}{\partial r} (1 - g^R g^A) \right) + \frac{\partial f}{\partial t} \text{Tr} (g^R \tau_z - \tau_z g^A) \\ & - D \frac{\partial f_1}{\partial \mathbf{r}} j_\varepsilon - \frac{\partial f}{\partial \varepsilon} \text{Tr} \frac{\partial \hat{\Delta}}{\partial t} (g^R - g^A) - 2i f_1 \text{Tr} \hat{\Delta} \tau_z (g^R - g^A) \\ & + \frac{\partial f_1}{\partial t} \text{Tr} \hat{\Delta} \tau_z \frac{\partial (g^R + g^A)}{\partial \varepsilon} + e D \frac{\partial A}{\partial t} \frac{\partial f}{\partial \varepsilon} j_\varepsilon = -4I_1^{\text{ph}}(f) \\ & - D \frac{\partial}{\partial \mathbf{r}} \text{Tr} \left(\frac{\partial f_1}{\partial z} (1 - g^R \tau_z g^A \tau_z) \right) + \frac{\partial f_1}{\partial t} \text{Tr} (g^R \tau_z - \tau_z g^A) \\ & - D \frac{\partial f}{\partial \mathbf{r}} j_\varepsilon + e \frac{\partial \varphi}{\partial t} \frac{\partial f}{\partial \varepsilon} \text{Tr} (g^R \tau_z - \tau_z g^A) \\ & + \frac{\partial f}{\partial \varepsilon} \text{Tr} \tau_z \frac{\partial \hat{\Delta}}{\partial t} (g^R + g^A) + \frac{i}{2} \frac{\partial f}{\partial \varepsilon} \text{Tr} \tau_z \frac{\partial^2 \Delta}{\partial t^2} \frac{\partial (g^R - g^A)}{\partial \varepsilon} \\ & - 2i f_1 \text{Tr} (g^R \hat{\Delta} + \hat{\Delta} g^A) + \frac{\partial f_1}{\partial t} \text{Tr} \left(\hat{\Delta} \frac{\partial (g^R - g^A)}{\partial \varepsilon} \right) \\ & + 2e D \frac{\partial^2 A}{\partial \mathbf{r} \partial t} \frac{\partial f}{\partial \varepsilon} = -4I_2^{\text{ph}}(f_1) \end{aligned} \quad (\text{A.32})$$

where

$$j_\varepsilon = \text{Tr} \tau_z (g^R \partial g^R - g^A \partial g^A)$$

The collision integrals $I_1^{\text{ph}}, I_2^{\text{ph}}$ in formula (A.32) are rather cumbersome and we shall not write them out. They are presented in the paper by Larkin and Ovchinnikov (1977). In the static case the solution of eq. (A.32) has the form

$$f = \tanh(\varepsilon/2T), \quad f_1 = 0 \quad (\text{A.33})$$

The dependence of the distribution function on energy is determined from the condition that the electron -phonon collision integral should be zero. Near the transition temperature, for energies $\varepsilon \ll T$ the collision integral I_1^{ph} is equal to

$$I_1^{\text{ph}}(f) = \tau_\varepsilon^{-1} \frac{|\varepsilon|}{(\varepsilon^2 - |\Delta|^2)^{1/2}} \left(f(\varepsilon) - \tanh \left(\frac{\varepsilon}{2T} \right) \right), \quad |\varepsilon| > |\Delta|, \quad (\text{A.34})$$

where

$$\tau_\varepsilon^{-1} = \frac{\pi\lambda_{\text{eff}}}{2(sp)^2} \int_0^\infty \frac{d\varepsilon \varepsilon^2}{\cosh\left(\frac{\varepsilon}{2T}\right) \sinh\left(\frac{\varepsilon}{2T}\right)} = \frac{7\zeta(3)\pi\lambda_{\text{eff}}T^3}{2(sp)^2} \sim T^3 \quad (\text{A.35})$$

where s is the velocity of sound, and θ_D is the Debye temperature. τ_ε is large; consequently, in relatively weak electric fields a strong change arises in the electron distribution function.

In many problems the electric field acts stronger on the distribution function than on the Green function $g^{\text{R,A}}$. In these cases it is possible to restrict oneself to a static approximation for the functions $g^{\text{R,A}}$. Writing down the Green function $g^{\text{R,A}}$ in the form

$$g^{\text{R,A}} = \begin{pmatrix} \alpha & \beta_1 \\ -\beta_2 & -\alpha \end{pmatrix}^{\text{R,A}} \quad (\text{A.36})$$

then from formula (A.25) we get the following system of equations for the functions $\alpha, \beta_{1,2}$:

$$-D\partial \mp \left(\alpha^{\text{R,A}} \partial \mp \beta_{1,2}^{\text{R,A}} - \beta_{1,2}^{\text{R,A}} \frac{\partial \alpha^{\text{R,A}}}{\partial r} \right) + 2i\Delta_{1,2} \alpha^{\text{R,A}} - 2i\varepsilon \beta_{1,2}^{\text{R,A}} + \frac{2}{\tau_\varepsilon} \alpha^{\text{R,A}} \beta_{1,2}^{\text{R,A}} = -I_{1,2}^{\text{ph(R,A)}}. \quad (\text{A.37})$$

The electron-phonon collision integral $I_{1,2}^{\text{ph(R,A)}}$ is determined by formula (A.25). In the vicinity of T_c and for $\varepsilon \ll T_c$ it is equal to

$$I_{1,2}^{\text{ph(R)}} = \frac{\beta_{1,2}^{\text{R}}}{\tau_\varepsilon}, \quad I_{1,2}^{\text{ph(A)}} = -\frac{\beta_{1,2}^{\text{A}}}{\tau_\varepsilon}. \quad (\text{A.38})$$

If the gradient terms in eq. (A.37) are small, one may use a local approximation for the functions $\alpha, \beta_{1,2}$:

$$\frac{\alpha^{\text{R,A}}}{\varepsilon \pm i/2\tau_\varepsilon} = \frac{\beta_{1,2}^{\text{R,A}}}{\Delta_{1,2}} = \pm \left(\left(\varepsilon \pm \frac{i}{2\tau_\varepsilon} \right)^2 - |\Delta|^2 \right)^{-1/2}. \quad (\text{A.39})$$

In such cases the kinetic equation for the function f acquires the form (Larkin and Ovchinnikov 1977)

$$-D \frac{\partial^2 f}{\partial r^2} + \frac{\partial f}{\partial t} \frac{\partial (\varepsilon^2 - |\Delta|^2)^{1/2}}{\partial \varepsilon} - \frac{\partial f}{\partial \varepsilon} \frac{\partial (\varepsilon^2 - |\Delta|^2)^{1/2}}{\partial t} = -I_1^{\text{ph}}(f). \quad (\text{A.40})$$

For the distribution function f in eq. (A.40) it is necessary to obtain a boundary condition at $|\varepsilon| = |\Delta|$. Integrating eq. (A.32) we obtain that in the narrow region near the boundary the value

$$\frac{\partial f}{\partial r} \text{Tr} (1 - g^{\text{R}} g^{\text{A}}) \quad (\text{A.41})$$

is continuous. Expression (A.41) under the barrier is zero. Consequently,

$$\left. \frac{\partial f}{\partial n} \right|_{\text{on bound.}} = 0. \quad (\text{A.42})$$

In many problems the range of variation of the order parameter Δ is much less than the energy relaxation length $R_\varepsilon = (D\tau_\varepsilon)^{1/2}$. In this case it is possible to obtain an equation for the coordinate-averaged distribution function $\langle f \rangle$. The distribution function f in this case has the form

$$f = \langle f \rangle + \tilde{f} \quad (\text{A.43})$$

The coordinate-dependent $\tilde{f}$ is small and can be found in a linear approximation in the electric field without the phonon collision integral.

When eq. (A.40) is valid, $\tilde{f}$ satisfies the equation

$$-D \frac{\partial^2 \tilde{f}}{\partial r^2} = \frac{\partial \langle f \rangle}{\partial \varepsilon} \frac{\partial (\varepsilon^2 - |\Delta|^2)^{1/2}}{\partial t}. \quad (\text{A.44})$$

Averaging eq. (A.40) over coordinates and using the expression for $\tilde{f}$ determined by eq. (A.44), we obtain

$$\frac{\partial}{\partial \varepsilon} \left(D_\varepsilon \frac{\partial \langle f \rangle}{\partial \varepsilon} \right) = \langle I_1^{\text{ph}}(f) \rangle, \quad (\text{A.45})$$

where

$$D_\varepsilon = \frac{eB}{\pi} \int_{V_\varepsilon} d^2r \frac{\partial (\varepsilon^2 - |\Delta|^2)^{1/2}}{\partial t} \left(-D \frac{\partial^2}{\partial r^2} \right)^{-1} \frac{\partial (\varepsilon^2 - |\Delta|^2)^{1/2}}{\partial t}, \quad (\text{A.46})$$

V_ε is the region in which $|\varepsilon| > |\Delta|$.

In the general case, eq. (A.45) for the function $\langle f \rangle$ retains its form. However, for the function D_ε a more complicated expression is obtained (Larkin and Ovchinnikov 1975, 1977).

All the physical values are determined by the Green function g^K . By means of eqs. (A.24), (A.27), (A.29) and (A.31) g^K is expressed through the distribution function. In slowly varying fields in the convolution operation (A.31) it is possible to retain only the first time-derivatives. As a result we get

$$\begin{aligned} g_p^K &= g^K + p g_1^K \\ g^K &= g^R \hat{f} - \hat{f} g^A - \frac{i}{2} \frac{\partial f}{\partial \varepsilon} \frac{\partial}{\partial t} (g^R + g^A) \\ g_1^K &= -\frac{\tau_{1r}}{m} \left\{ \frac{\partial \hat{f}}{\partial r} - g^R \frac{\partial \hat{f}}{\partial r} g^A + (g^R \partial g^R) \hat{f} - \hat{f} (g^A \partial g^A) \right. \\ &\quad + \frac{e}{2} \frac{\partial A}{\partial t} \left[\frac{\partial g^R}{\partial \varepsilon} \tau = g^K - g^R \tau = \frac{\partial g^K}{\partial \varepsilon} + \frac{\partial g^K}{\partial \varepsilon} \tau = g^A - g^K \tau = \frac{\partial g^A}{\partial \varepsilon} \right] \\ &\quad \left. + \frac{i}{2} \left[-\frac{\partial g^R}{\partial t} \frac{\partial g^K}{\partial \varepsilon} + \frac{\partial g^R}{\partial \varepsilon} \frac{\partial g^K}{\partial t} - \frac{\partial g^K}{\partial t} \frac{\partial g^A}{\partial \varepsilon} + \frac{\partial g^K}{\partial \varepsilon} \frac{\partial g^A}{\partial t} \right] \right\}. \end{aligned} \quad (\text{A.47})$$

Near the transition temperature, from eqs. (A.21), (A.37) for the order parameter Δ one may obtain a Ginzburg Landau type equation with an additional term arising due to deviation of the distribution function $\langle f \rangle$ from the equilibrium one:

$$\begin{aligned} &\left\{ (1 - T/T_c) + \frac{\pi D}{8T} \partial_-^2 - \frac{7\zeta(3)|\Delta|^2}{8\pi^2 T^2} \right\} \Delta + \\ &+ \frac{1}{4} \int_{-\infty}^{\infty} d\varepsilon (\langle f(\varepsilon) \rangle - \tanh(\varepsilon/2T)) (\beta_1^R - \beta_1^A) = 0 \end{aligned} \quad (\text{A.49})$$

In the particular case of a weak current and a coordinate-independent absolute value of the order parameter we have

$$\begin{aligned}
& \left\{ (1 - T/T_c) - \frac{7\zeta(3)|\Delta|^2}{8\pi^2 T^2} - \frac{\pi D e^2 A^2}{2T} \right. \\
& \quad \left. + \int_{|\Delta|}^{\infty} \frac{d\varepsilon}{(\varepsilon^2 - |\Delta|^2)^{1/2}} \left(\langle f(\varepsilon) \rangle - \tanh\left(\frac{\varepsilon}{2T}\right) \right) \right\} \Delta = 0 \\
& j = -\frac{e^2 p^2 I_{tr}}{3\pi} \Delta f(\Delta) A
\end{aligned} \tag{A.50}$$

Let us consider now in short the case of an arbitrary electron mean free path. As in the case of superconductors with a small electron mean free path, we search for the Green function g^K in the form of eq. (A.27). However, now the Green function $g^{R,A}$ and the distribution function f, f_1 depend also on the direction of momentum p .

Analogously as eq. (A.28) has been obtained, from formula (A.12) we get an equation for the distribution function $\hat{f}_p$:

$$\begin{aligned}
& g^R \circ \left(\mathbf{v} \frac{\partial \hat{f}}{\partial \mathbf{r}} \right) - \left(\mathbf{v} \frac{\partial f}{\partial \mathbf{r}} \right) \circ g^A + g^R \circ \left(\frac{\partial}{\partial t} + \frac{\partial}{\partial t'} \right) \hat{f} \tau_z - \tau_z \left(\left(\frac{\partial}{\partial t} + \frac{\partial}{\partial t_1} \right) \hat{f} \right) \circ g^A \\
& + g^R \cdot \left(\hat{H}(t_1) \hat{f} - \hat{f} \hat{H}(t') \right) + \left(\hat{f} \hat{H}(t_1) - \hat{H}(t) \hat{f} \right) g^A \\
& + i \left\{ \sum^{\text{Ph(K)}} \circ g^A - g^R \circ \sum^{\text{Ph(K)}} - \sum^{\text{Ph(R)}} \circ \hat{f} \circ g^A - g^R \circ \hat{f} \circ \sum^{\text{Ph(A)}} \right. \\
& \quad \left. + g^R \circ \sum^{\text{Ph(R)}} \circ \hat{f} + \hat{f} \circ \sum^{\text{Ph(A)}} \circ g^A \right\} = 0
\end{aligned} \tag{A.51}$$

$$\hat{H}(t) = -\frac{ie}{m} (pA(t)) \tau_z - i\hat{\Delta}(t) + ie\varphi(t)$$

Using eq. (A.31) for the convolution operation, from eq. (A.51) in a slowly-varying electromagnetic field we get

$$\begin{aligned}
& \text{Tr} \left\{ \left[\left(\mathbf{v} \frac{\partial f_1}{\partial \mathbf{r}} \right) + \frac{\partial f}{\partial t} - \frac{\partial f}{\partial \varepsilon} e \left(\mathbf{v} \frac{\partial \mathbf{A}}{\partial t} \right) \right] \tau_z (g^R - g^A) - \frac{\partial f}{\partial \varepsilon} \frac{\partial \hat{\Delta}}{\partial t} (g^R - g^A) \right. \\
& \quad + 2if_1 \hat{\Delta} (\tau_z g^A + g^R \tau_z) \\
& \quad \left. + \frac{nv}{2} \int d\Omega_{p_1} \sigma_{pp_1} (\hat{f}_p - \hat{f}_{p_1}) [g_p^R g_{p_1}^R + g_{p_1}^A g_p^A - g_p^A g_{p_1}^R - g_{p_1}^A g_p^R] \right\} = \text{St}_1^{\text{Ph}} \\
& \text{Tr} \left\{ \left[\left(\mathbf{v} \frac{\partial f}{\partial \mathbf{r}} \right) + \frac{\partial f_1}{\partial t} + e \frac{\partial \varphi}{\partial t} \frac{\partial f}{\partial \varepsilon} \right] \tau_z (g^R - g^A) + \frac{\partial f_1}{\partial t} \hat{\Delta} \frac{\partial (g^R - g^A)}{\partial \varepsilon} \right. \\
& \quad + \frac{\partial f}{\partial \varepsilon} \left[\frac{\partial \hat{\Delta}}{\partial t} (g^A \tau_z - \tau_z g^R) - \frac{i}{2} \frac{\partial^2 \hat{\Delta}}{\partial t^2} \left(\frac{\partial g^A}{\partial \varepsilon} \tau_z + \tau_z \frac{\partial g^R}{\partial \varepsilon} \right) \right] - 2if_1 \hat{\Delta} (g^R + g^A) \\
& \quad \left. + \frac{nv}{2} \int d\Omega_{p_1} \sigma_{pp_1} (\hat{f}_p - \hat{f}_{p_1}) (\tau_z g_p^R g_{p_1}^R + \tau_z g_{p_1}^A g_p^A - g_p^A \tau_z g_{p_1}^R - g_{p_1}^A \tau_z g_p^R) \right\} = \text{St}_2^{\text{Ph}}. \tag{A.52}
\end{aligned}$$

The kinetic equations (A.51) are written as applied to the problem on the flux line motion. More complete equations with electron-phonon collision integrals have been presented by Larkin and Ovchinnikov (1977).

Part XII

Other Experiments with Simulations of Superconductors

Part XIII

Other Experiments with Superconductors

Part XIV

Hubbard Model and KBE for Nonequilibrium Superconductivity

14 Chirped amplitude mode in photo-excited superconductors by Blommel, Kaye, Murakami, Gull, Golež

Abstract

Using a state-of-art numerical scheme, we show that the Higgs mode under excitation exhibits chirped oscillations and exponential decay when fluctuations are included. This is in stark contrast to conventional BCS collisionless dynamics which predict power-law decay and the absence of the chirping. The chirped amplitude mode enables us to determine the local modification of the effective potential even when the system is in a long-lived pre-thermal state. We then show that this chirped amplitude mode is an experimentally observable quantity since the photo-induced (super)-current in pump-probe experiments serves as an efficient proxy for the order parameter dynamics, including the chirped dynamics. Our result is based on the attractive Hubbard model using dynamical mean-field theory within the symmetry-broken state after a excitation across the superconducting gap. Since the collective response involves long timescales, we extend the hierarchical low-rank compression method for nonequilibrium Green's functions to symmetry-broken states and show that it serves as an efficient representation despite long-lived memory kernels.

14.1 Introduction

Soon after the seminal paper on BCS superconductivity [Bardeen *et al.*(1957)], Anderson pointed out that the electromagnetic response of a superconductor leads to a collective response [Anderson(1958a), Anderson(1958b), Anderson(1963)]. This realization had a profound impact on condensed matter physics, as well as particle physics, with the prediction of the Higgs mode [1, Higgs(1964b)]. It took almost two decades before such an excitation was observed in 2H-NbSe₂ with the coexistence of superconductor and charge-density-wave [Sooryakumar and Klein(1980), 2, 12, Pekker and Varma(2015)] phases, and another three decades for direct observation using terahertz spectroscopy either in a pump-probe setup [Matsunaga *et al.*(2013)] or with third-harmonics generation [Matsunaga *et al.*(2014), Matsunaga *et al.*(2017)].

This progress opened the field of Higgs spectroscopy, which has now been applied to numerous superconducting materials, including superconductors with running supercurrents [Moor *et al.*(2017), Nakamura *et al.*(2019), Crowley and Fu(2022)] and unconventional superconductors [Katsumi *et al.*(2018), Chu *et al.*(2020), Poniatowski *et al.*(2022)]. Ever more precise measurements of collective mode response enable the study of their long-time behavior after excitation both in solids [Matsunaga *et al.*(2013)] as well as quantum simulators [Young *et al.*(2024), Lewis-Swan *et al.*(2021), Behrle *et al.*(2018)].

The theoretical description of the time evolution of the Higgs mode is difficult. Time-dependent Landau-Ginzburg theory assumes an ad-hoc rapid modification of the effective potential and damping is considered on the phenomenological level [Aranson and Kramer(2002), Zong *et al.*(2019a)]. Microscopic time-dependent BCS theory predicts richer dynamics, including a power-law decay [Volkov and Kogan(1973), Tsuji and Aoki(2015)] or solitonic solution [Barankov *et al.*(2004) Barankov, Levitov, and Spivak, Yuzbashyan *et al.*(2006), Barankov and Levitov(2006), Yuzbashyan *et al.*(2015)]. However, all these theoretical statements assume

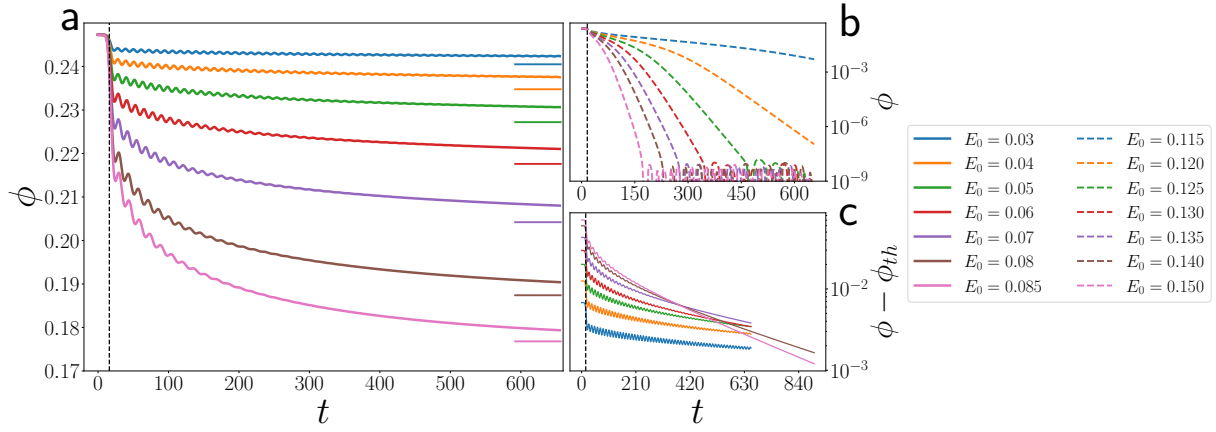


Figure 14.1: Time evolution of the order parameter ϕ after photoexcitation with pump amplitude E_0 in the (a) weak excitation regime, in which amplitude mode oscillations are present, and (b) strong excitation regime, in which we observe superexponential decay. The noise at small values in (b) is consistent with the truncation tolerance 10^{-8} used in the HODLR compression scheme. (c) The order parameter of the system exponentially approaches the expected thermal value, ϕ_{th} , which is the value of the order parameter in the equilibrium system that has the same energy (horizontal bars on right-hand side of (a)). $E_0 = 0.08, 0.085$ are integrated further in time to emphasize the exponential decay.

dissipationless electronic dynamics, which is questionable in real materials and opens up a question on the decay of collective modes upon the inclusion of electronic scattering.

The main difficulty of the theoretical description of electronic scattering on collective mode dynamics stems from the fact that collective response involves long timescales. This has limited previous studies to dissipationless dynamics [Volkov and Kogan(1973), Tsuji and Aoki(2015), Barankov et al.(2004) Barankov, Levitov, and Spivak, Yuzbashyan *et al.*(2006), Barankov and Levitov(2006)], or when fluctuations were included, the description was limited to relatively short times with few coherent oscillations and limited information about amplitude mode lifetime [Kemper *et al.*(2017), Nosarzewski *et al.*(2017), Kumar and Kemper(2019), Murakami *et al.*(2016a)].

Only recently, progress in numerical solvers for nonequilibrium Green's functions [Schüler *et al.*(2020)], including the generalized Kadanoff-Baym ansatz [Lipavský *et al.*(1986), Tuovinen *et al.*(2020), Schlünzen *et al.*(2017), Pavlyukh *et al.*(2022)] and memory kernel truncation methods [Schüler *et al.*(2018), Stahl *et al.*(2022)], has allowed access to long enough times to examine these questions. However, the application of these techniques to symmetry-broken phases is much less understood due to the long-lived power-law correlators acting as memory kernels.

In this work, we study the long-time evolution of collective modes by solving the full Kadanoff-Baym equations (KBE), and show that the order parameter displays an extremely slow approach to a thermal state. Within such a nonthermal state, the amplitude mode exhibits chirping (dynamical decrease in the frequency). The velocity of the chirping increases as we excite the system with larger field strengths. We show that the effect is experimentally accessible in terahertz pump-probe experiments by measuring the probe-induced currents, which can be used to extract the dynamics of the order parameter with a small imprint of the quasiparticle response.

We also highlight our use of the hierarchical off-diagonal low-rank (HODLR) compression method for the numerical solution of the KBE [Kaye and Golež(2021)]. The standard $\mathcal{O}(N^3)$ scaling of the direct solution of the KBE with the number of time steps N , resulting from memory terms, typically prevents access to the time scales necessary to analyze the lifetime of col-

lective excitations [Schüler *et al.*(2020), Kemper *et al.*(2017), Nosarzewski *et al.*(2017), Kumar and Kemper(2019)]. A new class of data compression-based methods, including the HODLR approach, have recently been proposed to overcome this bottleneck [Yin *et al.*(2022), Reeves *et al.*(2023), Shinaoka *et al.*(2023), Murray *et al.*(2023), Meirinhos *et al.*(2022)]. Viewing two-time Green's functions as matrices, the HODLR method decomposes these matrices into blocks, refined towards the diagonal, which in many cases have been observed to be numerically low-rank. A truncated singular value decomposition of these blocks is systematically updated on-the-fly during time-stepping and used to compute history integrals with controllable accuracy, yielding an $\mathcal{O}(k^2 N^2 \log N)$ computational complexity and $\mathcal{O}(k N \log N)$ memory complexity, for k the maximum block rank, without modifying the underlying KBE. Using this method, we demonstrate a 100-fold decrease in the computational cost required to propagate to the time scales studied compared with direct time stepping, with 50-200 times less memory.

14.2 Model and method

We study superconductivity within the attractive Hubbard model

$$H = -t_0 \sum_{\langle j,k \rangle \sigma} e^{iqA(t)} d_{j\sigma}^\dagger d_{k\sigma} - U \sum_j (n_{j\uparrow} - 1/2)(n_{j\downarrow} - 1/2), \quad (14.1)$$

where $d_{i\sigma}$ is the annihilation operator at site i and spin σ , $n_{i\sigma}$ is the spin-dependent density operator and q is the charge. The energy scale is set via the hopping amplitude $t_0 = 1$, the Coulomb attraction is set to $U = 2$, and we fix the occupation at half-filling. Units are measured in $\hbar = q = 1$.

We solve the problem within time-dependent dynamical mean-field theory (DMFT) on the Bethe lattice with infinite coordination number, using the Nambu formalism to describe the s-wave superconducting phase. This leads to matrix-valued Green's functions $G_{\alpha\beta}(t, t') = -i\langle \mathcal{T} \psi_\alpha(t) \psi_\beta^\dagger(t') \rangle$, with the Nambu spinor $\psi = \{d_\uparrow, d_\downarrow\}$. For the impurity solver, we use the self-consistent second-order perturbation theory known as second Born, or GF2 [van Leeuwen and Dahlen(2006), Dong *et al.*(2022), Rusakov and Zgid(2016)]. Collective order parameter dynamics take place on long time scales, and such long-time integration of the KBE is made possible by using a compressed representation of the Green's function to reduce both the computational and memory complexity [Kaye and Golež(2021)]. We refer to the SM for further implementation details.

We induce a dynamical perturbation by a short pump pulse introduced by a Peierls substitution in Eq. 14.1. The electric field $E_p = -\partial_t A$ is parameterized as

$$E_P(t) = E_0 \exp\left(-\frac{(t - t_c)^2}{\sigma^2}\right) \sin(\omega(t - t_c)), \quad (14.2)$$

with pulse center t_c , pulse width σ , driving frequency ω , and amplitude E_0 .

14.3 Results

For an interaction strength of $U = 2$ the attractive Hubbard model in equilibrium has a phase transition at inverse temperature $1/T_c = \beta_c \approx 10.6$ within the GF2 approximation. Across this transition, the system develops a nonzero superconducting pair amplitude $\phi(t) = \sum_k \langle c_{k\downarrow} c_{-k\uparrow} \rangle$, which from now on we consider as the order parameter. We choose a temperature $\beta = 18$ deep in the ordered state, and find that the system has a band gap of $2\Delta \approx 0.65$.

To excite the amplitude mode, we pump the system with a pulse which has a frequency twice the band gap ($t_c^{\text{Pump}} = 16$, $\sigma = 6.5$, $\omega = 4\Delta$). The time evolution of the order parameter

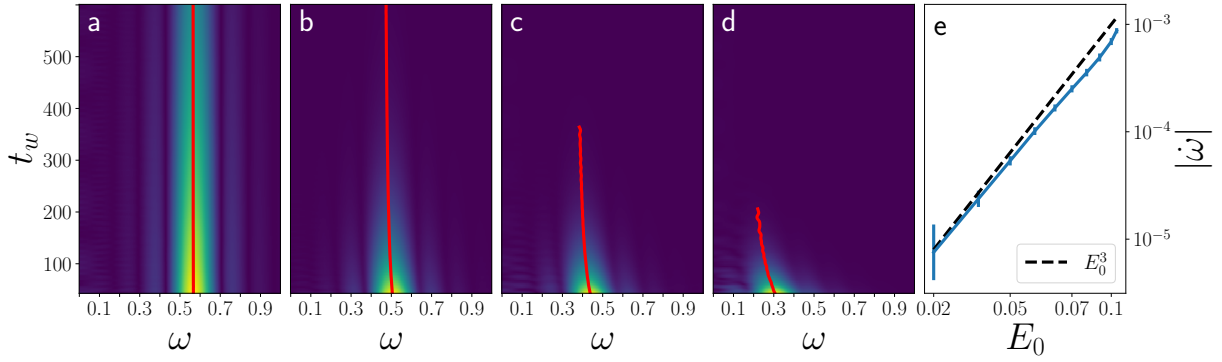


Figure 14.2: (a-d): Windowed Fourier transform $F[\tilde{\phi}](t_w)$ of the order parameter centered around t_w with exponential background subtracted, for pump amplitudes $E_0 = 0.02, 0.06, 0.08$, and 0.105 . The red line tracks the maximum at each value of t_w . (e) Initial chirping velocity, defined as the slope of the red lines in (a-d) for a small window of time immediately after the pump pulse. Error bars are obtained by varying width of FT window and number of time-points used for the linear fit.

is presented in Fig. 14.1. At weak excitation strength, the order parameter decays at an extremely slow exponential rate [Tsuji *et al.*(2013), Werner *et al.*(2012)]. The expected thermal values ϕ_{th} , marked by the horizontal lines, are not reached on the time scale of our simulation; see also Fig. 14.1(c), in which the thermal values are subtracted. For intermediate pump strengths, we observe the decay of the order parameter through several orders of magnitude. On top of the exponential relaxation, well-defined oscillations correspond to the amplitude mode excitation [Shimano and Tsuji(2020), Pekker and Varma(2015)]. As the pump strength is increased, the initial amplitude of these oscillations increases; however, they also have shorter lifetimes. Increasing the pump amplitude further, we reach a point at which the lifetimes of these oscillations hinder our ability to study their frequency [Tsuji *et al.*(2013), Golež *et al.*(2016), Picano and Eckstein(2021)], see Fig. 14.6(a,b) in the SM.

We observe that the frequency of the collective mode oscillations gradually chirps to smaller values. To extract the chirping velocity, we first subtract the background exponential decay from the data in Fig. 14.1, denoted as $\tilde{\phi}$, and then compute a windowed Fourier transform $FT[\tilde{\phi}](t_w) = \int_0^{T_{max}} dt \tilde{\phi}(t) e^{-i\omega t - (t-t_w)^6/\sigma_w^6}$. The width of the window, $\sigma_w = 24$, is set such that at least 4 oscillations of the order parameter are captured at any time, t_w . We define the chirping velocity $\dot{\omega}$ as the slope of the maximum of $FT[\tilde{\phi}](t_w)$, which is tracked in Fig. 14.2(a-d) for increasing pump amplitudes. Fig. 14.2(e) shows that the chirping velocity follows a power-law scaling $\dot{\omega} \propto E_0^3$ with the pump amplitude, which is the same as the scaling $j \propto E_0^3$ of the current due to photo-induced amplitude mode oscillations [Tsuji and Aoki(2015), Shimano and Tsuji(2020)].

Whereas the dissipationless BCS dynamics lead to either power-law decay [Volkov and Kogan(1973), Gurarie(2009)] or persistent oscillations [Barankov *et al.*(2004)Barankov, Levitov, and Spivak, Barankov and Levitov(2006), Yuzbashyan *et al.*(2015), Yuzbashyan *et al.*(2006)] of the superconducting amplitude depending on the initial condition, the inclusion of fluctuations modifies the Higgs mode dynamics into an exponential decay, see SM. This is consistent with the effect of thermal noise on the dissipationless BCS dynamics [Barankov *et al.*(2004)Barankov, Levitov, and Spivak, Murakami *et al.*(2016b)]. The lifetime of the amplitude mode decays with increasing pump strength (see Fig. 14.2(a-d)), and changes by more than an order of magnitude from the weak to the strong excitation regime.

The chirping of the amplitude mode is the main result of this work. Observing such phenomena requires microscopic treatment of fluctuations beyond the mean-field theory, see SM for

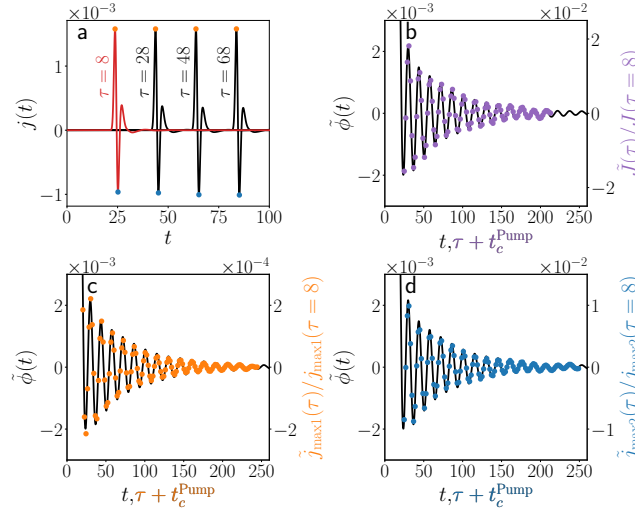


Figure 14.3: (a) Current, j , induced by a monocycle probe pulse with parameters $E_0 = 0.1E_0^{\text{Pump}}$, $t_c = t_c^{\text{Pump}} + \tau$, $\sigma = 1$, $\omega = 1$, $E_0^{\text{Pump}} = 0.08$. The markers denote the location of the maximal current response. (b) Integrated current response with background subtracted off, $\tilde{j}$, compared to the order parameter oscillations, $\tilde{\phi}$. (c) Oscillations of the first maximum of the current response to the probe field, j_{max1} , corresponding to the orange points in (a). (d) Oscillations of the second maximum of the current response to the probe field, j_{max2} , corresponding to the blue points in (a).

comparison, as the resulting state is highly nonthermal. Moreover, the chirping phenomenon is not limited to superconductors but should be observed in many systems with broken symmetry. For example, the superconducting solution analyzed in this work can be exactly mapped to the evolution of the antiferromagnetic state in the repulsive Hubbard model [Shiba(1972)], and it is degenerated with the charge density waves. Both of these phases have been addressed experimentally [Hong *et al.*(2017), Demsar *et al.*(1999), Yusupov *et al.*(2010), Yoshikawa *et al.*(2021)]. In actual experimental setups, there is a trade-off in pump strength between the chirping velocity and the lifetime of the amplitude mode, which depends on the details of the pulse and the material studied.

Now, we interpret the microscopic dynamics within the time-dependent Landau-Ginzburg theory. The chirping of the amplitude mode corresponds to a dynamical reduction of the local curvature due to the feedback of the electronic scattering on the collective order. While it would be tempting to interpret the chirping as a heating effect within the Landau-Ginzburg theory, we emphasize that the system is closed and in a prethermal regime, which the temperature change cannot describe. In principle, one could construct a time-dependent Landau-Ginzburg theory of chirping. However, it is difficult in practice to extract the evolution of global structure of the Landau-Ginzburg potential, even when microscopic dynamics are available. An important consequence of our analysis is that the chirped frequency can act as a microscopic measure of the dynamically modified local curvature.

We next discuss how this physics could be observed experimentally. The amplitude mode does not couple linearly to light; however, it can be observed in nonlinear processes, such as pump-probe setups [Matsunaga *et al.*(2013)] or the third-harmonics response [Matsunaga *et al.*(2014), Shimano and Tsuji(2020), Tsuji and Aoki(2015)]. Our numerical tests agree with previous observations that it is difficult to extract the amplitude mode signal from the optical response due to the overlap with the quasiparticle continuum [Kumar and Kemper(2019)]. Following previous studies [Matsunaga *et al.*(2013)], we rather focus on the photo-induced current and show that it serves as an excellent proxy for the dynamics of the order parameter. While, in principle, the optical conductivity and photo-induced current carry similar information, we

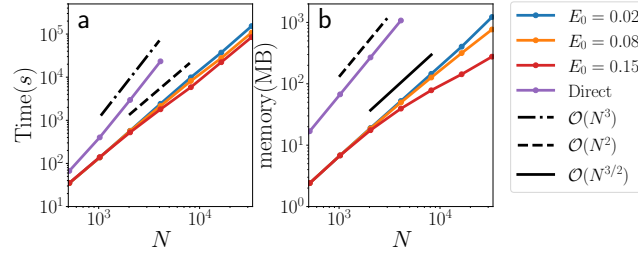


Figure 14.4: (a) Wall clock time and (b) required memory with increasing number of timesteps for HODLR propagation scheme versus direct time stepping, for various pump amplitudes, E_0 , starting from the equilibrium state at $\beta = 18$.

found that in practice it is much easier to extract coherent oscillations from the time-dependent current, compared with spectral analysis of the optical conductivity.

We evaluate the probe-induced current by directly simulating an additional probe pulse at a time τ after the pump pulse, and measuring the induced current [Lenarčič *et al.*(2014), Shao *et al.*(2016), Eckstein and Kollar(2008)]. This approach is theoretically convenient as it includes vertex corrections to the response function, which represents the contribution from the collective modes [Murakami *et al.*(2016c), Golež *et al.*(2020)]. We propose two measurement protocols and show that the current response in both cases could be connected with the collective dynamics of the amplitude mode. The probe pulse is modelled as a small monocycle pulse, given by Eq. 14.2 with parameters $\sigma = 1$, $t_c = t_c^{\text{Pump}} + \tau$, $\omega = 1$, and $E_0 = 0.1E_0^{\text{Pump}}$. Fig. 14.3(a) shows the current response to such probe field for several different delay times τ . We track the dependence of the maximal current response $j(t_{\text{max}})$ versus the delay time and observe decaying oscillations that align to high accuracy with the order parameter dynamics, see Figs. 14.3(c-d). The maximum current $j(t_{\text{max}})$ thus gives information about both the chirped frequency and the lifetime of the amplitude mode. We also confirmed that the time-averaged current response $J(\tau) = \hat{j}(\omega = 0; \tau)$ shows excellent agreement with the dynamics of the order parameter, shown in Fig. 14.3(b).

Based on these results, we propose that the direct measurement of the photo-induced current is a convenient way to measure both the chirping of the amplitude mode as well as to decay envelope of the Higgs mode which would be power-law (exponential) in the dissipationless (fluctuating) description. The photo-induced current and chirping can be measured in the pump-probe setup using either free-space electro-optic sampling, like in Ref. [Matsunaga *et al.*(2013)], or transmission line experiments, which were recently used to detect superconducting nonlinear transport [Wang *et al.*(2023)], the light-induced anomalous Hall effect due to circularly polarized light [McIver *et al.*(2020)], and ultrafast resistive switching in 1T-TaS₂ [Venturini *et al.*(2022)]. In these experiments, one measures the variation of the transmitted field induced by the probe field in the material. The modification of the transmitted field originates from the variation of the microscopic polarization in the material which is connected with the microscopic currents [Katayama *et al.*(2012), Udina *et al.*(2019), Felderhof and Marowsky(1987)].

Since our analysis was enabled by the HODLR compression technique for the numerical solution of the KBE [Kaye and Golež(2021)], we consider the compressibility of the dynamics of symmetry-broken states, and the performance of the scheme compared with the direct solution of KBE. In Fig. 14.4(a), we observe approximately quadratic scaling of the computational cost with the propagation time over a range of pump intensities and equilibrium state temperatures. We reach 32768 time steps in roughly one day on a single-core of a workstation using 0.5 GB of memory. By contrast, we extrapolate that cubic-scaling direct time stepping using the NESSi code [Schüler *et al.*(2020)] would have taken 129 days with 137 GB of memory. Fig. 14.4(b) shows the different scaling between the full Green's function and the HODLR representation.

At $N = 32768$ we have a factor of 50-200 memory compression of the Green's function and self-energy compared with direct storage on a two-time grid. The SM considers the compressibility in more detail.

We have shown that the collective amplitude mode displays chirped dynamics after the above gap excitation and have analyzed a direct experimental signature of the response based on the photo-induced current within the pump-probe setup. Future applications include the nonlinear response of unconventional superconductors [Katsumi *et al.*(2018), Chu *et al.*(2020)], and extensions beyond dynamical mean-field theory to study spatial fluctuations [Stahl and Eckstein(2021)] and the build-up of the fluctuating order in systems with competing orders [Sun and Millis(2020), Zong *et al.*(2019b)]. A particularly appealing question is the instability of the system to form inhomogeneous patterns after excitation [Dzero *et al.*(2009), Chern and Barros(2019), Barresi *et al.*(2023)] that should be extremely sensitive to the decay mechanism of the Higgs mode presented in this work. Methodologically, this work represents the first application of the HODLR compression technique for the KBE to symmetry-broken states, and demonstrates that it can access long enough time scales for the practical study of a broad field of photo-induced phase transitions [Nasu(2004), Murakami *et al.*(2023), de la Torre *et al.*(2021), Giannetti *et al.*(2016)].

14.4 Implementation details

The attractive Hubbard Hamiltonian in Eq. (14.1) can be rewritten as

$$H = -t_0 \sum_{\langle j,k \rangle \sigma} \sigma e^{iqA(t)\sigma} \psi_{j\sigma}^\dagger \psi_{k\sigma} + U \sum_i \psi_{i\downarrow}^\dagger \psi_{i\downarrow} \psi_{i\uparrow}^\dagger \psi_{i\uparrow} - \frac{U}{2} \sum_{i\sigma} \psi_{i\sigma}^\dagger \psi_{i\sigma}, \quad (14.3)$$

where we have used the Nambu spinors

$$\psi_i = \begin{pmatrix} d_{i\uparrow} \\ d_{i\downarrow}^\dagger \end{pmatrix}. \quad (14.4)$$

We again use the Nambu spinors to define the anomalous Green's function

$$G(t, t') = \langle \psi(t) \psi^\dagger(t') \rangle = \begin{pmatrix} \langle d_{\uparrow} d_{\uparrow}^\dagger \rangle(t, t') & \langle d_{\uparrow} d_{\downarrow} \rangle(t, t') \\ \langle d_{\downarrow}^\dagger d_{\uparrow} \rangle(t, t') & \langle d_{\downarrow}^\dagger d_{\downarrow} \rangle(t, t') \end{pmatrix}, \quad (14.5)$$

which is the quantity we compute by solving the KBE.

We solve this system within the DMFT approximation on a Bethe lattice, giving the hybridization function

$$\Delta(t, t') = \Delta_R(t, t') + \Delta_L(t, t') \quad (14.6)$$

$$\Delta_R(t, t') = \frac{1}{2} \bar{t}_0(t) \sigma_z G(t, t') \sigma_z \bar{t}_0^*(t') \quad (14.7)$$

$$\Delta_L(t, t') = \frac{1}{2} \bar{t}_0^*(t) \sigma_z G(t, t') \sigma_z \bar{t}_0(t'), \quad (14.8)$$

with hopping matrix elements given by

$$\bar{t}_0 = \begin{pmatrix} e^{iA(t)} & 0 \\ 0 & e^{-iA(t)} \end{pmatrix}, \quad (14.9)$$

$$A(t) = - \int_0^t E(s) ds \quad (14.10)$$

due to the Peierls substitution. We set the charge and the lattice distance to unity, $q = a = 1$.

The impurity problem is solved using the fully self-consistent second Born approximation. In this approximation, the Fock term is given by

$$\Sigma_{ij}^F(t) = iU(t)G_{ij}^<(t, t^-)\delta_{i\bar{j}}, \quad (14.11)$$

and the Hartree term cancels the chemical potential since we are at half-filling. There are two second-order diagrams:

$$\Sigma_{ij}^B(t, t') = U(t)U(t')G_{ij}(t, t')G_{i\bar{j}}(t, t')G_{ij}(t', t), \quad (14.12)$$

$$\Sigma_{ij}^E(t, t') = -U(t)U(t')G_{i\bar{j}}(t, t')G_{i\bar{j}}(t, t')G_{i\bar{j}}(t', t). \quad (14.13)$$

The hybridization $\Delta[G](t, t')$ and the self-energy $\Sigma[G](t, t')$ depend on the Green's function, and we maintain self-consistency using fixed-point iteration terminating when the maximum difference of the Green's function at subsequent iterates is less than 10^{-10} .

We use the fifth-order backward differentiation formula multistep method to discretize the KBE. Gregory quadrature is used to discretize memory integrals to fifth-order accuracy. The discrete Lehmann representation [Kaye *et al.*(2022a)], implemented in `libdlr` [Kaye *et al.*(2022b)], is used to discretize the imaginary time variables appearing in the vertical leg of the Keldysh contour, as in Ref. [Kaye and Strand(2023)]. We have confirmed that the compression of the Green's function using the HODLR scheme preserves conservation laws, namely the density and energy, to within the SVD truncation tolerance used in the scheme, which we set to 10^{-8} .

In the pump-probe experiments, we measure the current, given by

$$j(t) = \text{Im}(\text{Tr}[\sigma_z(\Gamma_L(t) - \Gamma_R(t))]) \quad (14.14)$$

$$\Gamma_{R/L}(t) = -i[G * \Delta_{L/R}]^<(t, t), \quad (14.15)$$

with $*$ the convolution operator on the three-legged Keldysh contour.

14.5 Hartree-Fock

Solving the KBE allows for inclusion of dynamical fluctuations absent in a Hartree-Fock treatment of superconductors. We show that including these correlations leads to qualitatively different dynamics, demonstrating the necessity of calculations beyond the mean field level. In the main text, we observed chirping of the amplitude mode after the photo-excitation and we contrast these dynamics with the Hartree-Fock approximation, see Fig. 14.5. Already in equilibrium, the critical inverse temperature for the transition is $\beta_c = 2$ which is much higher than in the 2nd Born approximation.

We can use this order parameter data to confirm one of the main predictions of a Hartree-Fock treatment, which is a power-law decay of the Higgs mode [Volkov and Kogan(1973)]. The amplitude of the Higgs oscillation in the Hartree-Fock and second Born approximation is presented in Fig. 14.6. In the Hartree-Fock case, the expected power-law decay of $\phi \propto t^{-1/2}$ is shown to hold true over two orders of magnitude. However, when fluctuations are introduced through the 2nd Born self-energy, the Higgs mode instead decays exponentially [Tsuji and Aoki(2015)], with the lifetime strongly depending on the excitation strength.

Fig. 14.5 shows that when the pump field is turned off, the order parameter oscillates around a fixed value, in stark contrast to the continual decay present in the second order theory. This suggests an immediate alteration of the potential energy surface within the system, suggesting an absence of chirping. This absence of chirping is confirmed in Fig. 14.7, where the time dependence of the Higgs oscillation frequency is presented. The frequency is constant for all pump strengths, with only noise less than 1% of the original frequency.

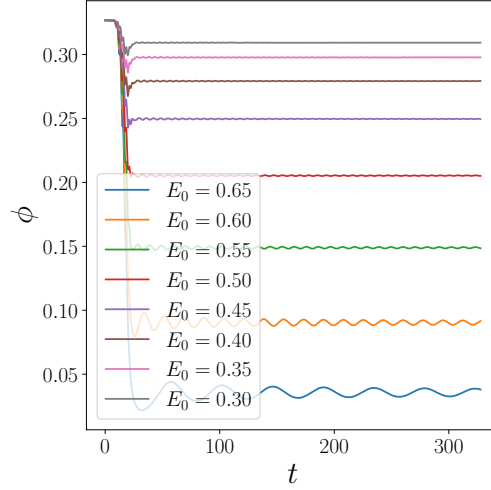


Figure 14.5: Time evolution of the order parameter ϕ within the Hartree-Fock approximation for increasing strength of the external pulse E_0 .

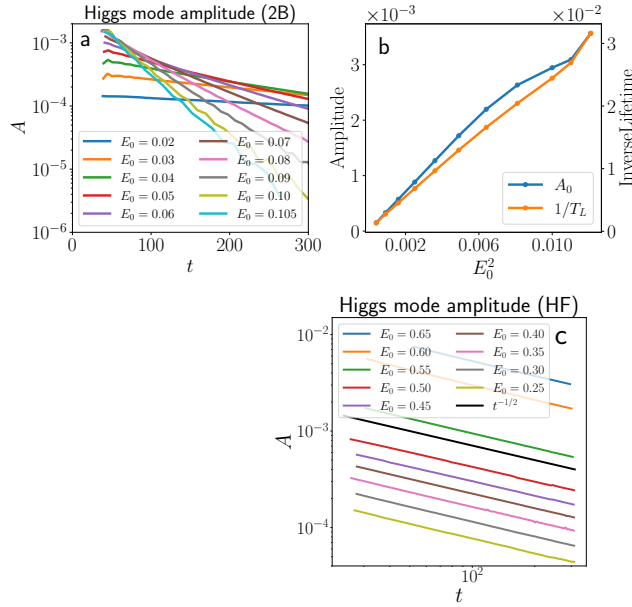


Figure 14.6: (a) Decay of the oscillating amplitude of the order parameter, A , for different excitation strengths E_0 within the 2nd Born (a) and the Hartree-Fock approximation (c). The amplitude of the Hartree-Fock results is compared with the power-law decay $t^{-1/2}$. (b) The initial amplitude, A_0 , and inverse lifetime, $1/T_L$, of the Higgs mode with increasing pump intensity for the 2B results.

14.6 Numerical ranks in HODLR scheme

Our time propagation scheme compresses the Green's function using a HODLR decomposition [Kaye and Golež(2021)]. A truncated SVD is performed on each block of the decomposition, and updated on-the-fly, with singular values below 10^{-10} discarded. The numerical ranks of these blocks (within this precision) determine the compressibility of the Green's functions, and the performance of the scheme.

Fig. 14.8(a) shows the growth of the maximum block rank k with the number of time steps. At inverse temperature $\beta = 6$, the system is in a disordered state, and the Green's functions decay exponentially on their off-diagonal. This leads to a rapid saturation of the ranks, giving an $\mathcal{O}(N^2 \log N)$ computational complexity of the algorithm, and an $\mathcal{O}(N \log N)$ memory

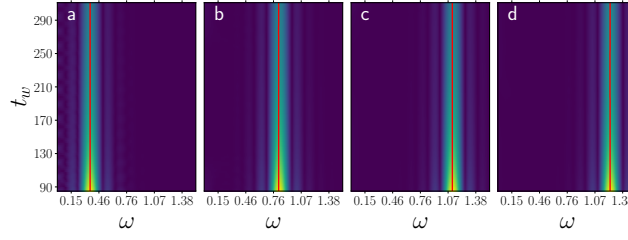


Figure 14.7: Windowed Fourier transform $F[\phi](t_w)$ of the order parameter centered around t_w with background subtracted from the Hartree-Fock dynamics. Results for pump amplitudes (a) $E_0 = 0.6$, (b) $E_0 = 0.5$, (c) $E_0 = 0.4$, (d) $E_0 = 0.3$. The red line tracks the maximum of the spectrum, and varies by at most one unit of the frequency resolution.

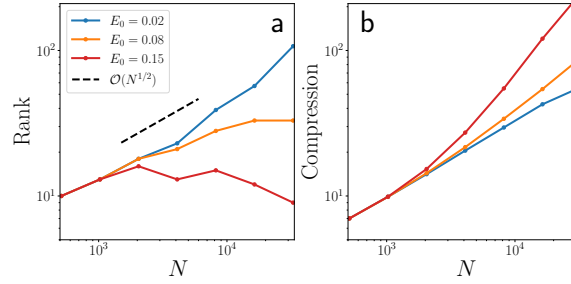


Figure 14.8: (a) Rank of the truncated SVD of the largest block in a HODLR decomposition of Green's function at timestep T . Results for several different pump strengths are presented. (b) Compression factor for Green's function and self-energy for the presented pump strengths.

complexity, with N the number of time steps. The behavior is similar for the simulations in which a low-temperature state is excited with a strong pump (i.e. $E_0 = 0.15$), destroying the superconducting state. At low temperature, in the ordered state, although the blocks are still numerically low rank with $k \ll N$, they grow as $k \propto \sqrt{N}$, leading to an $\mathcal{O}(N^3 \log N)$ computational complexity and $\mathcal{O}(N^{3/2} \log N)$ memory complexity. Nevertheless, while the memory usage, shown in Fig. 14.8(b), reflects the expected scaling, the wall clock time shown in Fig. 14.4 reflects approximately $\mathcal{O}(N^2)$ scaling, suggesting that lower-scaling steps of the algorithm dominate throughout the time scale of this simulation.

Part XV

Exotic Approaches and Unusual Phenomena

15 Resistive States of Type-I Superconductors by Dolgoplov

15.1 1. Introduction

In this chapter we shall consider the resistive states which arise in bulk samples of type-I superconductors. The main factor which governs the structure of resistive states in bulk samples is the magnetic field distribution in the bulk of a sample, and this distinguishes, in principle, the present case from resistive states in thin superconducting channels or films. The intermediate state, the two-dimensional mixed state and the resistive state of the surface superconductivity layer, which may be observed in superconductors with Ginzburg-Landau parameter $\kappa > 0.42$, all refer to the resistive states in type-I superconductors.

The first part of this chapter considers the problem related to resistive phenomena in the intermediate state, and the second one deals completely with the study of the two-dimensional mixed state of type-I superconductors. The problems regarding the resistive state of the surface superconductivity layer will not be considered here because they have been discussed in detail in the review by Park (1969).

The intermediate state has now been under investigation for several decades. After the first experiments it became clear that a sample in the intermediate state represents a more or less dispersed mixture of regions of superconducting and normal phases; several years later, after the Meissner effect had been discovered (Meissner and Ochsenfeld 1933), a macroscopic theory of the static intermediate state was developed (Peierls 1936, London 1936). A year later L. Landau (1937) solved the problem of the shape of interphase boundaries and obtained an expression for the period of the layered structure of the intermediate state in a plate in a transverse magnetic field. Then, World War II interrupted rapid progress in the study of the intermediate state, and the structure of this state was discovered only after WWII in experiments performed by Meschkovskii and Shal'nikov (1947).

The period of the intermediate state structure is determined by the surface tension at the boundary between superconducting and normal phases, so study of the structures of the intermediate state can yield the coefficient of surface tension, which is an important parameter of a superconductor. Yet, the structures which arise in an external magnetic field appeared to be much more complicated than the idealized pattern of plane-parallel layers used in theoretical calculations. For instance, layers which are plane-parallel in the bulk of a sample can twist and branch out quaintly near the surface* (Balashova and Sharvin 1956, Faber 1958), and if the concentration of one of the phases is small, the normal or superconducting layers may be in the form of filaments extended along the magnetic field. The crystallographic anisotropy of the surface tension also significantly affects the formation of structures of the intermediate state.

While studying the intermediate state in a spherical tin sample, Balashova and Sharvin (1956) noticed that near the equatorial line where the magnetic field is strongly tilted to the sample surface the structure is in the form of parallel layers. Using then a magnetic field tilted to the surface of plane samples at an angle of 10° – 15° , Sharvin (1957, 1960) obtained sufficiently regular layered structures of the intermediate state to determine the interphase surface tension for tin (1957) and indium (1960). As was shown theoretically by Dzyaloshinskii (1955), such

structures in a tilted magnetic field correspond to a minimal energy.

Thus, in the middle of the fifties the phenomenon of the destruction of superconductivity by an external magnetic field seemed to be understood almost completely. The resistance of samples in the intermediate state was associated with a current flowing perpendicular to the boundaries between the phases, and the electric current was assumed to align the domains appropriately. However, a paper by Gorter appeared in 1957, in which he noticed that even in the presence of a direct current a continuous motion of the domains is possible - a dynamic intermediate state. The dynamic intermediate state was discovered experimentally by Sharvin (1965b).

The discovery of the dynamic intermediate state stimulated a large number of theoretical and experimental works dealing with resistive states in type-I superconductors, because rather specific features of the intermediate state are fully manifested precisely under such nonequilibrium conditions. Below we shall try to elucidate the present state of the art in this field of physics of superconductors.

Of course, in a single chapter it is impossible to give equal attention to all the aspects of the problem, so we shall concentrate, above all, on the least studied processes occurring in the destruction of superconductivity by a current; more briefly we shall touch upon such phenomena as the dynamic intermediate state and propagation of electromagnetic waves^{**}; problems concerning the resistance of the phase boundaries will not be considered at all, i.e. we shall confine ourselves to sufficiently low temperatures where the Andreev reflection of electrons at the boundary with the superconducting phase (Andreev 1966) leads to the fact that the conductivity tensor of the normal domains coincides with the conductivity tensor for a bulk normal metal, even if the free path of the charge carriers exceeds the thickness of the normal regions.

The two-dimensional mixed state of type-1 superconductors has been observed experimentally quite recently (I. Landau and Sharvin 1969). This specific resistive state plays a rather essential role in many cases where superconductivity is destroyed by a current. Unfortunately, so far even the qualitative picture of processes occurring in the mixed state of type-1 superconductors is unclear, not to say about a consistent theory of this state. Therefore, we shall concentrate on the discussion of experimental results, and at the end of the paper will present a brief analysis of theoretical models now available.

15.2 2. Resistive phenomena in the intermediate state

15.2.1 2.1. The dynamic intermediate state

As was already noted, Gorter (1957) was the first to point out the possibility of a continuous motion of the domains in the intermediate state in the presence of a direct current. He considered the case where the superconductivity of a cylindrical sample is destroyed by a current. Problems concerning the destruction of superconductivity by a current will be discussed below; here we shall dwell upon the dynamic intermediate state, which arises when a relatively weak current flows through a sample whose superconductivity is partly destroyed by an external magnetic field.

Let us consider how the domains are to move in the simplest case where the structure of the intermediate state is in the form of alternating layers of superconducting and normal phases extended along the current (fig. 1). It may seem at first sight that all the current must flow

^{1*}Thus, a large dispersity of the structure is attained near the sample surface, which leads to a decrease in the energy related to an inhomogeneity of the magnetic field above the surface.

^{**} Problems concerning the propagation of electromagnetic waves in the intermediate state have been considered in detail in the review by Krylov (1980).

in the superconducting layers and the resistance of the sample must be zero. However, had such a current distribution taken place, the magnetic field due to the current flowing in the superconducting layer would have been added to the external magnetic field on the one side and would have been cancelled out of it on the other side (fig. 1b). Under such conditions the superconducting layer cannot, obviously, remain at rest, but will start moving to the side where the magnetic field is weaker, which, in turn, leads to an electric field and a current in the normal regions.

The velocity of the interphase boundaries, v , can easily be determined from the continuity equation for the tangential component of the electric field at the phase boundary, which must be satisfied in the boundary-fixed coordinate system. In the superconducting region the electric field vanishes and the boundary condition can be written in the form

$$\left(e + \frac{1}{c} v \times h \right) \times n = 0 \quad (1)$$

where e and h are the values of the electric and magnetic fields in the normal region, and n is a vector normal to the boundary. Since in the intermediate state

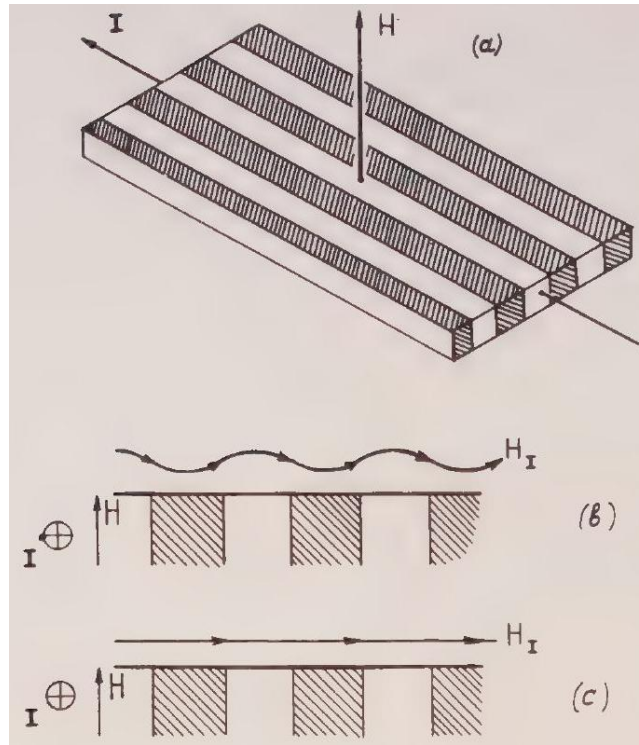


Fig. 1. Motion of the layers oriented parallel to the transport current: (a) Diagram of the experiment. (b) Distribution of the magnetic field due to the current (curve H_I) near the sample surface; (c) in the case where the current flows only through superconducting layers; (d) in the case where the current density in the normal domains is I/S .

$|h| = H_c$, we obtain

$$v = c \frac{\mathbf{e} \times \mathbf{h}}{H_c^2}. \quad (2)$$

The electric field in the normal regions is $\mathbf{e} = \rho j_n$ where ρ is the resistivity of the normal metal, and j_n is the current density in the normal phase. The equality between the values of

the magnetic field on the opposite sides of a superconducting layer implies that $|j_n| = I/S$ where I is the total current in the sample, and S is its cross-section (see fig. 1c)*.

Observation of the domain motion requires a regular structure of the intermediate state of a known configuration and a technique which makes it possible to detect the motion of superconducting and normal layers; moreover, the samples must be sufficiently perfect (otherwise, the domain boundaries will be hold up by the defects, which may cause the domains to turn and stop). All these conditions have been satisfied in the experiments performed by Sharvin (1965b).

In those experiments a tilted magnetic field was used to create a layered structure of the intermediate state extended along the current, and a microcontact technique designed by Sharvin (1965a) was employed to detect the domain motion.

The microcontacts were obtained by touching the sample surface with a thin ($20 - 50 \mu\text{m}$ in diameter) wire. When the sample goes over (under the contact) from the superconducting to the normal state the contact increases its resistance by $10^{-1} - 10^{-3}$ Ohm, which corresponds to contact diameters $\sim 0.1 - 1 \mu\text{m}$ (Sharvin 1965a). Such microcontacts appeared to be a simple and convenient tool for recording domain motion in the intermediate state. Mounting several microcontacts on the sample surface, one could observe the motion of the domains from one contact to another and, thus, determine the domain velocity and the structure period. Figure 2 shows simultaneous recording of temporal oscillations of the resistance of two contacts when a moving structure of the intermediate state is present in the sample.

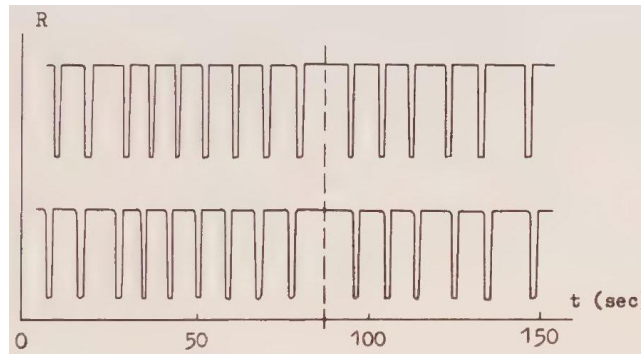


Fig. 2. Simultaneous recording of voltage oscillations across two contacts 0.1 mm apart in the direction of the layer motion. A tin sample at $T=2.93 \text{ K}$, $H=105 \text{ Oe}$, $I=2 \text{ A}$. At the moment shown by the dashed line the current in the sample is reversed. (Sharvin and I. Landau 1970.)

A theory of the dynamic intermediate state has been constructed by Andreev and Sharvin (1967). They showed, in particular, that the motion of the domain should necessarily take place if an electric field component tangential to the interface exists in the normal regions. For instance, the layers aligned normally to the current can move due to the Hall effect in the normal domains*; substituting the Hall electric field in the form $\mathbf{e}_x = \mathbf{j}_n \times \mathbf{h}/Nq$ (N is the difference between the number of electrons and holes per unit volume, q is the electron charge) into eq.

(2), we obtain for the velocity of motion in the current direction

$$v = j_n/Nq. \quad (3)$$

*This statement refers only to the case where the current in the sample is perpendicular to the magnetic field (for a more rigorous derivation see the paper by Andreev 1966).

*Such a motion of the layers is also possible in anisotropic metals, provided the current direction does not coincide with the direction of one of the principal axes of the conductivity tensor.

Thus, in this case the layers move at a velocity equal to the charge carrier drift velocity.

An arbitrary configuration of the superconducting and normal domains will be displaced as a whole in the direction normal to the resultant electric field at the velocity defined by eq. (2), and the only possible static structure is a layered structure of the intermediate state with the layers aligned perpendicular to the electric field. The main conclusions of the theory of the dynamic intermediated state have been confirmed experimentally in the work of Sharvin and I. Landau (1970).

Thus, the motion of the domains in the intermediate state, when a direct current flows through the sample, is the rule rather than the exception. Let us emphasize once more that sufficiently perfect samples are required to observe the dynamic intermediate state*. By perfectness of the samples we mean in this case the homogeneity at distances of the order of the coherence length $\xi(T)$, rather than chemical purity of sample material. So, uniformly distributed impurities should not hinder the motion of domains; this was actually observed in the experiments of Sharvin and Gal'perina (1971), in which they found that even a significant amount of tin introduced into indium samples did not lead to any obstacles for the motion of the domains.

It should also be noted that in imperfect samples the pinning of layers extended along the current may lead to a decrease in the sample resistance or even to its vanishing. The opposite effect would be observed if in a metal with nonzero Hall effect the layers perpendicular to the transport current are at rest. Actually, the continuity of the tangential component of the electric field at the interface will lead to transverse currents in the normal layers that are shortened through the superconducting regions, and this should result in an increase of dissipation in the sample.

In the first work on the observation of the dynamic intermediate state Sharvin pointed out that the motion of domains in the intermediate state could, in principle, be used for constructing a continuously operating refrigerating machine. In fact, at low temperatures almost all the entropy is connected with the conduction electrons and, therefore, is confined to the normal regions, while the superconducting layers play the part of sufficiently effective heat insulators. Hence, the motion of the layers should lead to entropy transfer from one edge of a sample to the other. If one makes use of the motion of layers caused by the Hall effect, the dissipation due to the release of Joule heat can be reduced by employing

samples with lower residual resistance; in this case, owing to the specific mechanism of electron reflection at the interface (Andreev 1966) the scattering of charge carriers at the sample surface is practically eliminated in the intermediate state. Such a "freezing" effect has been observed in the work of Sharvin and Bronevoi (1977); it should, however, be noted that practical use of this effect requires samples whose residual resistance must be several orders of magnitude less than that obtained by present-day experimental techniques.

At present, the dynamic intermediate state has been observed in most of the type-I superconductors*. Microcontacts, magnetic field micro-probes and, probably the most obvious, the magneto-optic method were used to observe the motion of domains. The work of Lebedev and Khotkevich (1980) is worth mentioning here, in which they observed softening of lead in the intermediate state caused by the motion of domains under the flow of a direct current.

15.2.2 2.2. Macroscopic electrodynamics of the intermediate state

Since the surface tension at the boundaries between the normal and superconducting phases is small, the period of the structure of the intermediate state is also small in comparison with the sample size**. This makes it possible to use a macroscopic description of the intermediate

*Apparently, Chandrasekhar et al. (1967) failed to observe the dynamic intermediate state in experiments similar to those of Sharvin because of the imperfection of the samples used.

state. The complete system of macroscopic equations describing the electromagnetic properties of the intermediate state has been derived by Andreev (1966).

The macroscopic approach employs quantities averaged over regions whose dimensions are large in comparison with the thickness of the layers, but small in comparison with distances over which the structure changes significantly; these quantities are the magnetic induction $\mathbf{B}$ and the electric field intensity $\mathbf{E}$. They are defined as mean values of h and e , the "microscopic" intensities of the magnetic and electric fields. The fields h and e themselves are quantities which have already been averaged over volumes whose dimensions are much larger than the interatomic distances, but smaller than the thickness of the layers. In the superconducting phase h and e vanish; in the normal phase one may not distinguish between the microscopic magnetic induction and h . because the magnetic susceptibility of the normal metal is small.

The fields e and h in the normal domains obey ordinary Maxwell's equations. and $|h| = H_c$. Following the paper by Andreev, let us introduce the macroscopic magnetic field intensity H in such a way that in normal layers the value of H at a given "point" coincides with the value of h . The vectors of magnetic induction B

and magnetic moment per unit volume $\mathbf{M}$ will be in this case:

$$B = x_n H, \quad M = \frac{1 - x_n}{4\pi} H, \quad |H| = H_c, \quad (4)$$

where x_n is the concentration of the normal phase. The averaged microscopic equations yield the ordinary (for any conducting medium) formulae, which relate B, H and the mean value of the electric field $E = x_n e$ by

$$\operatorname{div} B = 0, \quad \operatorname{rot} E = -\frac{1}{c} \frac{\partial B}{\partial t}, \quad \operatorname{rot} B = \frac{4\pi}{c} \bar{i}. \quad (5)$$

Here $\bar{i}$ denotes the macroscopically averaged total current density in the intermediate state. Separating, as usual, out of the total current density the term $c \operatorname{rot} M$, which is related to magnetization currents but does not lead to actual charge transfer through the sample, we can rewrite the last of eqs. (5) as

$$\operatorname{rot} H = \frac{4\pi}{c} j \quad (6)$$

where $j = \bar{i} - c \operatorname{rot} M$ is the transport current density in the sample*. Let us emphasize that besides currents in the normal regions j also includes superconducting currents flowing along the phase boundaries.

The electrodynamics of the intermediate state differs, in principle, from the electrodynamics of a normal metal in that there is not a universal relationship between the transport current density j and the intensities of the fields E and H . However, there exists a relationship between the field intensities and the component of the transport current perpendicular to the magnetic field, $j_\perp$

$$j_{\perp\alpha} = \frac{1}{x_n} \sigma_{\alpha\beta} E_\beta \quad (7)$$

where $\sigma_{\alpha\beta}$ is the two-dimensional conductivity tensor of a bulk normal metal placed in the magnetic field $\mathbf{H}$. The subscripts α and β take two values in the plane perpendicular to the magnetic field.

*See, for example, Salomon and Harris (1971), Farrel (1972), Farrel and Isett (1972), Gupta and Farrel (1973), Chien and Farrel (1974), Wolf et al. (1974), Farrel and Chien (1975), Laeng et al. (1975), and Lerski (1975).

**The surface tension is usually written in the form $\Delta \cdot H_c^2 / 8\pi$ where the parameter Δ has the dimension of length and lies in the range $10^{-6} - 10^{-4}$ cm.

In order to obtain a closed system of equations, Andreev used the transversality of the electric and magnetic fields in the intermediate state, $\mathbf{E} \cdot \mathbf{H} = 0$, which directly follows from the "microscopic" boundary conditions for the electric (eq. 1) and magnetic ($\mathbf{h} \times \mathbf{n} = 0$) fields.

Since the component of the transport current $j_{\parallel}$ along the magnetic field is not directly related to the field intensities, j should be eliminated from the equations. To this end, eq. (6) has to be vector-multiplied by $\mathbf{H}$. Thus, the

complete system of electrodynamic equations of the intermediate state is

$$\begin{aligned} \operatorname{div} \mathbf{B} &= 0, \quad \operatorname{rot} \mathbf{E} = -\frac{1}{c} \frac{\partial \mathbf{B}}{\partial t}, \quad (\mathbf{H} \cdot \nabla) \mathbf{H} = \frac{4\pi}{c} j_{\perp} \mathbf{H}, \\ \mathbf{E} \cdot \mathbf{H} &= 0, \quad |\mathbf{H}| = H_c, \quad \mathbf{B} = x_n \mathbf{H}, \quad j_{\perp \alpha} = \frac{1}{x_n} \sigma_{\alpha\beta} E_{\beta}. \end{aligned} \quad (8)$$

Let us note that this is non-linear even if σ does not depend on $\mathbf{H}$. Once these equations are solved, the total transport current density can be found by calculating $(c/4\pi) \operatorname{rot} \mathbf{H}$.

Solution of system (8) makes it possible to determine macroscopic characteristics of the intermediate state in samples of various geometry for different configurations of external magnetic fields and current in the sample; for instance, Andreev (1966) has solved the problem for a plate with a current in a transverse magnetic field for an arbitrary relationship between the current and the intensity of the external magnetic field. Yet, of the most interest is the possibility to solve in a general way the problem of propagation of electromagnetic waves in the intermediate state. The results of such calculations are presented in the next section.

15.2.3 2.3. Propagation of electromagnetic waves and hydrodynamic effects in the intermediate state

The general equations of macroscopic electrodynamics (8) have been used by Andreev (1966) to study the propagation of electromagnetic waves. He showed that weakly decaying electromagnetic waves can propagate in the intermediate state of a sample with a unequal number of electrons and holes. These waves are circularly polarized and are similar, in their properties, to helicons, which exist in normal metals (see, for instance, the review by Kaner and Skobov 1966).

Following Andreev (1966), we put

$$\mathbf{H} = \mathbf{H}_0 + \mathbf{H}', \quad x_n = x_n^{(0)} + x'_n, \quad |\mathbf{H}_0| = H_c \quad (9)$$

to explain the dispersion law of the waves. Small increments to the equilibrium values, $\mathbf{H}'$, x'_n and also $\mathbf{E}$, are due to the presence of the wave and proportional to $\exp\{i(k'z - \omega t)\}$. We shall assume that the equilibrium values H_0 and $x_n^{(0)}$ themselves change at distances much larger than $|k|^{-1}$ and linearize the equations (8) with respect to x'_n , $\mathbf{H}'$ and $\mathbf{E}$:

$$\begin{aligned} k \cdot \mathbf{H}_0 x'_n + x_n^{(0)} k \cdot \mathbf{H}' &= 0, \quad k \times \mathbf{E} = -\frac{1}{c} (\omega (\mathbf{H}_0 x'_n + x_n^{(0)} \mathbf{H}')), \\ \mathbf{E} \cdot \mathbf{H}_0 &= 0, \quad \mathbf{H}' \cdot \mathbf{H}_0 = 0, \quad (\mathbf{k} \cdot \mathbf{H}_0) \mathbf{H}' = -\frac{4\pi}{c} i (j_{\perp} \times \mathbf{H}_0). \end{aligned} \quad (10)$$

Taking vector products of the second and last of eqs. (10) with H_0 and eliminating $\mathbf{H}'$ from the relations thus obtained, we find

¹*Here we employ the now common term "transport current", instead of the term "conduction current" used in the original paper by Andreev.

$$E_\alpha = \frac{4\pi i\omega}{c^2 (k \cdot H_0)^2} H_c^2 \sigma_{\alpha\beta} E_\beta. \quad (11)$$

Equating the determinant of this system to zero, we obtain the dispersion relation

$$\omega^2 (\sigma_{\alpha\beta} \sigma_{\beta\alpha} - \sigma_{\alpha\alpha} \sigma_{\beta\beta}) - i\omega \frac{c^2 (k \cdot H_0)^2}{4\pi H_c^2} (\sigma_{\alpha\alpha} + \sigma_{\beta\beta}) + \left(\frac{c^2 (k \cdot H_0)^2}{4\pi H_c^2} \right)^2 = 0. \quad (12)$$

Weakly decaying waves arise in the case where the non-diagonal elements of the conductivity tensor are large as compared to the diagonal elements, i.e. if

$$\sigma_{\alpha\beta} = -\sigma_{\beta\alpha} = \frac{Nqc}{H_c} \gg \sigma_{\alpha\alpha}, \sigma_{\beta\beta}. \quad (13)$$

The spectrum of helicons in the intermediate state can be written in the form

$$\omega = \frac{cH_c}{4\pi Nq} k_z^2 \quad (14)$$

where $k_z = (k \cdot H_0) / H_c$ is the projection of the wave vector of the wave onto the magnetic field direction. The attenuation of electromagnetic waves,

$$\gamma = 2\pi \frac{\text{Im}\{\omega\}}{\omega} = \frac{\sigma_{\alpha\alpha} + \sigma_{\beta\beta}}{\sigma_{\alpha\beta}}, \quad (15)$$

is determined by the two-dimensional conductivity tensor of a bulk normal metal in the plane perpendicular to the magnetic field.

These calculations correspond to the case where the macroscopic description is applicable, and this means that the helicon wavelength must be large in comparison with the period of the intermediate state structure. Thus, the propagation of electromagnetic waves is associated with the motion of interphase boundaries, and their experimental investigation as well as the study of other dynamic effects in the intermediate state require sufficiently perfect samples.

Helicons in the intermediate state were first observed by Maxfield and Johnson (1965): a more detailed investigation was then performed by Kushnir et al. (1971). In those works, however, samples of poor quality were used, which led to difficulties in the interpretation of the experimental results and apparent discrepancy with the theory. These experimental difficulties were overcome by Krylov (1976) who studied helicons in high-purity single-crystal indium samples. All the experimental results obtained by Krylov are in excellent agreement with Andreev's theory. Let us note here the work of

Krylov, Bronevoi and Sharvin (1974), in which the measurement of helicon attenuation was used to examine indium purity by determining the free path of charge carriers; this method appeared to be especially convenient for highpurity samples where the mean free path of the electrons is comparable with the sample size, because the attenuation of helicons in the intermediate state (see eq. 15) is determined by the conductivity tensor of the bulk metal.

Andreev (1967) has also considered the effect of a direct current on the spectrum of helicons in the intermediate state. In fact, the electrodynamic equations of the intermediate state are nonlinear and the presence of a direct current alters the spectrum of helicons in such a way that their erenkov radiation from sample inhomogeneities becomes possible. As a result, a fraction of the energy of the current is transferred to the energy of radiated waves and dissipated, which leads to an increase in the sample resistance. Andreev has called this additional resistance "wave resistance"*. Let us emphasize that the wave resistance does not vanish when the free path of charge carriers increases infinitely, so it must dominate for sufficiently pure samples.

If inhomogeneities in a sample are distributed regularly, the current-voltage characteristic of the sample should exhibit a large number of resonance peaks**. In fact there exists only a

discrete set of wavelengths λ_n of helicons, which may be radiated from static inhomogeneities, these wavelengths being dependent on the current value. If one of the values λ_n coincides with the period of the inhomogeneity distribution, the radiation is especially intensive. It should be noted, however, that the observation of resonance wave resistance is rather difficult, because this requires that the mode of oscillations excited from the inhomogeneities should be attenuated little in the time the structure is displaced by one inhomogeneity period. This period must be large in comparison to that of the intermediate state structure and small in comparison with the sample thickness.

So far, only nonresonant increase in the sample resistance in the intermediate state has been observed in the presence of regularly distributed inhomogeneities (Krylov 1974, 1975a,b); also, the "hydrodynamic" character of the flow of a moving intermediate state structure around a stationary obstacle has been observed directly by magneto-optic techniques (Laeng et al. 1978).

15.2.4 2.4. On the structure of the intermediate state under the destruction of superconductivity by a current

Let us now turn to the case where the superconductivity of a bulk cylindrical sample is destroyed by a current flowing through it. In this case the destruction of superconductivity starts when the magnetic field (due to the current) at the surface exceeds a critical value—Silsbee's rule (1916)*. For currents I exceeding the critical value $I_c = cr_0 H_c/2$ (r_0 is the sample radius) one or another structure of the intermediate state must exist in the sample.

The first model of the structure of the intermediate state for this case was suggested by London (1937). In this model the intermediate state represents a periodic (along the sample axis) system of alternating superconducting and normal layers (fig. 3). However, the conclusions of the London model agreed

poorly with experimental results, and twenty years later Gorter (1957) and Gorter and Potters (1958) suggested an entirely new model of the intermediate state. The Gorter structure is dynamic, and represents a system of coaxial cylindrical layers of superconducting and normal phases, which arise at the boundary of the intermediate-state core and collapse at the sample axis (fig. 4).

It will be shown below that it is the Gorter structure that must be observed in experiment. So, we shall concentrate on processes taking place in a sample in the presence of the Gorter structure of the intermediate state. The magnetic field in the normal metal outside the intermediate-state core, $H(r)$, is the sum of the magnetic field due to the current flowing through the intermediate-state core and the magnetic field due to the current in the normal part of the sample. If the current in the normal part is distributed uniformly, then

$$H(r) = Ar + B/r. \quad (16)$$

1

- The nonlinearity $(\mathbf{H} \cdot \nabla)\mathbf{H}$, which appears in the electrodynamic equation of the intermediate state, is similar to the term $(\mathbf{v} \cdot \nabla)\mathbf{v}$ in Euler's hydrodynamic equations, and the radiation of helicons from sample inhomogeneities resembles the radiation of sound when a supersonic flow goes round an obstacle. This analogy will be even more complete if we introduce the velocity of the structure in the current direction $\mathbf{v} = \mathbf{j}/Nq$ (see eq. 3); then the radiation of helicons from static inhomogeneities can be considered as the formation of waves under the motion of the intermediate state structure. In this case, radiation of helicons of wavelength λ necessarily takes place when the velocity v becomes equal to that of propagation of electromagnetic waves of this wavelength.

**Andreev (1967) has solved this problem for a plate in a transverse magnetic field.

¹*The accuracy to which this rule is to be fulfilled will be considered at the end of this section.

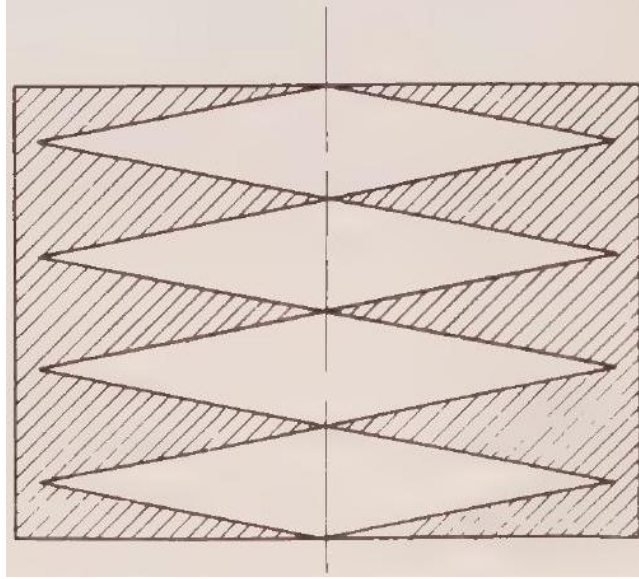


Fig. 3. Schematic diagram of the London structure (normal domains are hatched).

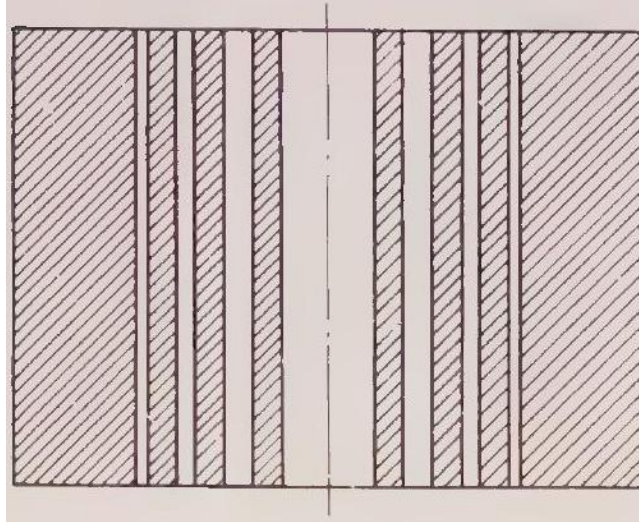


Fig. 4. Schematic diagram of the Gorter structure (normal domains are hatched).

The constants A and B are determined from the conditions $H = H_c$ at the boundary of the intermediate-state core (at $r = r_i$), and $H = (I/I_c) H_c$ at the sample surface. It can easily be seen that the radius of the intermediate state region cannot exceed the value

$$r_{i0} = r_0 \left[I/I_c - (I^3/I_c^2 - 1)^{1/2} \right]. \quad (17)$$

In this case the magnetic field, which is equal to H_c at the boundary of the region of the intermediate state, increases towards the sample surface, and $(\partial H / \partial r) |_{r = r_{i0}} = 0$. For the Gorter structure the radius of the intermediate-state core cannot remain constant, but must decrease because of the motion of the outer boundary of the external superconducting layer towards the sample axis. If the radius of the intermediate-state core $r_1 < r_{i0}$, a region is formed in the normal part of the sample (for $r > r_1$) where the magnetic field is less than the critical one (fig. 5), and the difference $H_c - H_m$ and the size of the region in which $H < H_c$ increase with decreasing r_1 . This process goes on until a new superconducting layer is formed in the normal part of the sample at the minimum of the magnetic field.

Out of the two above-mentioned structures only the London structure can be calculated completely. The condition that the magnetic field intensity everywhere in the normal metal must not be less than the critical value makes it possible to obtain (London 1937) the radius of the intermediate-state core in a sample carrying a current I (eq. 17), the dependence of the concentration of the normal phase on the distance from the sample axis $x_n = r/r_{i0}$, and the resistance of the sample in the intermediate state

$$R = \frac{R_n}{2} \left[1 + (1 - I_c^2/I^2)^{1/2} \right] \quad (18)$$

(R_n is the sample resistance in the normal state). In particular, for $I = I_c$ we have $R = 0.5R_n^*$.

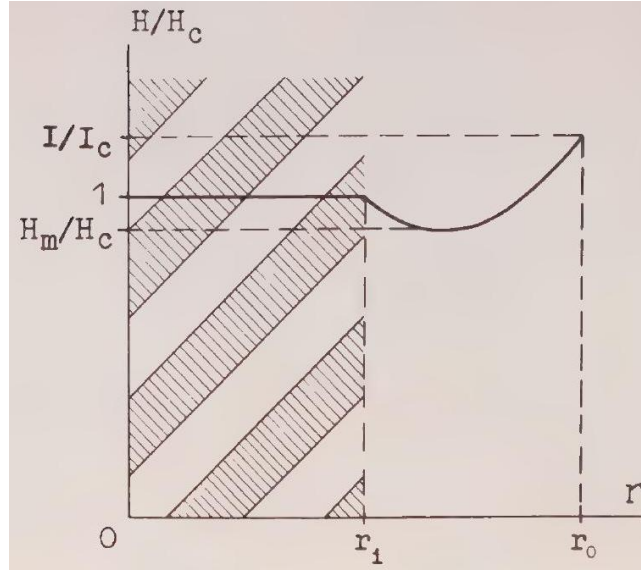


Fig. 5. Diagram of the distribution of the magnetic field in the sample; region $r < r_i$ is occupied by the intermediate state.

The Gorter structure implies continuous formation of new superconducting layers in the bulk of a normal metal, and this occurs with some delay, relative to the equilibrium situation, due to supercooling effects. Therefore, the average (in time) radius of the intermediate-state core, for the Gorter structure, is always less than r_{i0} , and the resistance is larger than the value determined by eq. (18). The exact values of these quantities cannot, however, be calculated.

The experimental value of the resistance jump at $I = I_c$ (see fig. 6) is always much higher than the value $0.5R_n$, which follows from the London model; since the Gorter model can, in principle, give any value of the resistance jump (higher than $0.5R_n$) one might conclude that the Gorter model is more suitable here. However, some modifications of the London model of the periodic structure have appeared (Andreev 1968, Baird and Mukherjee 1968) which agree much better with experimental resistance values.

It is worth noting here that no conclusion should be made about the validity of one or another model only on the basis of data on the resistance in the intermediate state. For instance, Andreev (1968) and Baird and Mukherjee (1968) have suggested models for the structures of

- The same results can be obtained by solving the system of equations (8).

the intermediate state based on quite different ideas and assumptions, but these models give practically the same dependences $R(I)$ for actual experiments*. Another important fact is the

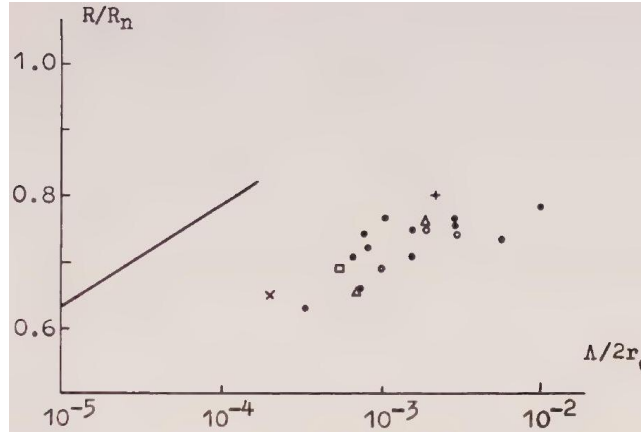


Fig. 6. Resistance jump at $I=I_{\mathrm{c}}$ as a function of surface tension. Solid line is calculation for a rounded-off periodic structure (I. Landau 1979). Symbols correspond to various experimental data: (+) Alekseevskii (1938), () H. Meissner (1958), H. Meissner and Zdanis (1958), (Δ) E. Troinar (1960); ($\square$) Freud et al. 1968), ($\times$) I. Landau (1973), () Watson and Huebener (1974.)}

large scatter of experimental data on the resistance jump obtained by various authors, which considerably exceeds possible experimental errors (see fig. 6)*. This fact is usually ignored, though it seems obvious that a correct model of the structure of the intermediate state must not only agree with individual experimental results but also explain the scatter of experimental data**.

The London structure of the intermediate state has been developed for zero surface tension at the phase boundary. However, in type-I superconductors the phase boundaries have a positive surface tension, and this circumstance is so important here that it makes impossible the existence of any periodic structures under ordinary experimental conditions. In the London structure the interface has a break at point A (see fig. 7). Surface tension prevents the existence of breaks, so the interface near point A must be rounded. At a curved interface the magnetic field must satisfy the following condition:

$$H = H_c (1 - \Lambda/2r_c), \quad (19)$$

where r_c is the radius of curvature of the interface at a given point. Thus, at point A' the magnetic field must be slightly less than the critical one; and the closer the rounded interface is to point A, the larger is the difference $H_c - H(A')$.*.

Here it is pertinent to recall that under the destruction of superconductivity by a current the magnetic field at a distance r' from the sample axis is determined by the total current flowing in the region $r < r'$. Hence, the relationship between the shape of the interface and

*A considerable discrepancy between the curves corresponding to these models presented in the paper by Mukherjee (1973), is connected with the fact that the wrong value of one of the parameters was used in constructing the curve corresponding to Andreev's theory (see the paper by I. Landau 1978).

**In this figure the values of R/R_n are plotted as a function of the parameter $\Lambda/2r_0$; it is worth emphasizing that the scatter depends weakly on the way of representation of experimental data.

** Let us note that for the Gorter structure a considerable scatter of experimental data is quite natural because the degree of magnetic supercooling of the normal phase, necessary for the formation of a new superconducting layer, determines the resistance of a sample, and this degree may change significantly from sample to sample, depending on the preparation method, the presence and composition of impurities, etc.

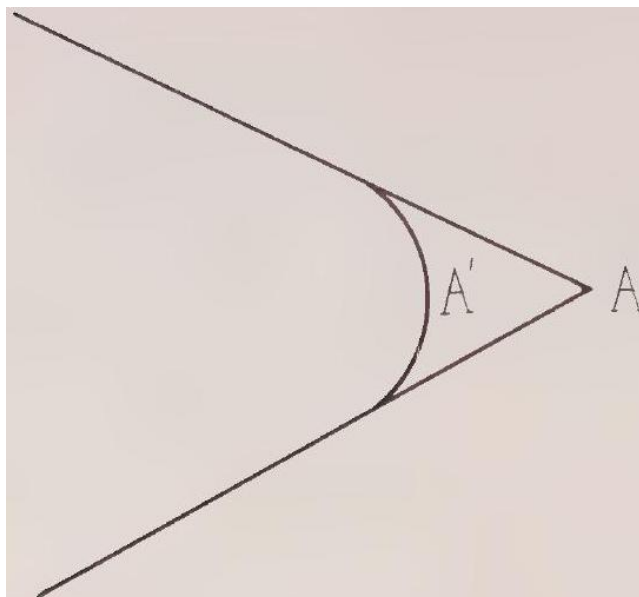


Fig. 7. Schematic diagram of a superconducting layer near the edge of the intermediate state core.

the magnetic field on it is essentially nonlocal, so in order to alter the magnetic field near the boundary of the intermediate-state core, one has to rearrange, to a greater or lesser extent, the structure in the central part of the sample**.

The shape of the interface, at which the magnetic field obeys condition (19) and the currents in the normal phase are distributed in accordance with the conditions at the boundary with the superconducting phase, can be calculated only numerically. Such calculations were performed by I. Landau (1978) for several values of surface tension and it was found that in all actual situations the radius of the intermediate-state core for the rounded structure, r_i , is appreciably less than r_{i0} . This means that in the normal metal outside the intermediate-state core, like the case of the Gorter structure, there is a region in which $H < H_c$ (see fig. 5). Therefore, the periodic structure of the intermediate state is unstable relative to the appearance of new superconducting

domains in the normal part of the sample, so it might exist only as a metastable state. Comparison of calculated values of the sample resistance with experiment (see fig. 6) has shown, however, that the Gorter-type structure occurs in practice*.

Let us briefly consider other models of the intermediate state now available. In the models suggested by Andreev (1968) and by Baird and Mukherjee (1968), as well as in the London model, interfaces have breaks, which makes the existence of such structures impossible. In a paper by Mel'nikov (1981) dealing with the same problem, the rounded periodic structure of the intermediate state was calculated numerically. But in these calculations surface tension was not taken into account; the radius obtained for the intermediate state core is not only much smaller than the sample radius, but also much smaller than the radius of the rounded periodic structure considered above**. So, that paper seems to be of mathematical rather than physical interest.

¹*The important role of surface tension in the formation of the structure of the intermediate state under the destruction of superconductivity by a current was noted by Sharvin (1977).

**This case differs, in principle, from the situation where superconductivity is destroyed by an external magnetic field. In the latter case the relationship between the shape of the interface and the magnetic field on it is local and difficulties similar to those mentioned above (e.g., the presence of breaks at the interface when the domains exit onto the surface of the sample) are not significant.

Let us now discuss why it was determined experimentally, which structure of the intermediate state arises under the destruction of superconductivity by a current. The main difficulty is that the intermediate-state core is covered with a layer of the normal metal, and the structure can only be judged from various indirect data. As was already noted, data on the sample resistance carry little information. Observation of the periodic structure of the intermediate state on the surface of a sample in the presence of a transverse magnetic field is, in fact, a different experimental situation, since the transverse magnetic field will, naturally, lead to a considerable rearrangement of the structure. From this point of view, the work of Kan and Frolov (1974) was of interest, in which the periodic structure of the intermediate state in the bulk of a sample seemed to have been observed directly. However, as was established later by the same authors (Frolov and Kan 1978), the work appeared to be erroneous, and the oscillations observed were caused by inhomogeneity of the sample parameters along its length. Thus, so far there have been no experimental observations yielding direct, or even indirect, evidence of the existence of a periodic structure of the intermediate state; moreover, some experimental results agree poorly with this assumption.***.

On the other hand, direct indications of the existence of the Gorter structure are not available either (if experiments with a longitudinal magnetic field are not considered****). Indeed, if the Gorter structure were present the periodic

formation of new superconducting layers would lead to oscillations of the sample resistance. This circumstance, noted in Gorter's first work (1957) seemed to lead to a prompt discovery of the Gorter structure. However, even experiments specially designed to observe resistance oscillations (see, for example, the paper by Watson and Huebener 1974) yielded, in fact, negative results: though some samples did reveal an alternating component of the voltage, its amplitude was much lower than theoretical predictions. The absence of clearly pronounced resistance oscillations in the intermediate state is, apparently, one of the main arguments against the existence of the Gorter structure.

In practice, however, a small amplitude of the alternating component of the voltage across the sample is by no means an indication that there is no Gortertype structure in the sample. The fact is that in actual samples such a structure may differ significantly from the idealized system of coaxial cylindrical layers of superconducting and normal phases. As was already noted at the beginning of this section, a new superconducting region arises when the magnetic field in the normal metal decreases to a certain value $H_1 < H_c$. If the sample is sufficiently homogeneous, then for $H < H_1$ a large number of nuclei of the superconducting phase arise simultaneously in the bulk of the normal metal; these merge and form a circular superconducting layer, so we have the case of a typical Gorter structure. If the sample is far from perfect, the nuclei of the superconducting phase will be formed in various regions of the sample at different degrees of supercooling, i.e. at different moments. Small superconducting regions, which arise in this case, simultaneously grow and move to the axis of the sample, so the structure formed in the sample will not even be axially symmetric, and the amplitude of resistance oscillations may be many order of magnitude less than what follows from theoretical calculations.

Most of the experiments dealing with the destruction of superconductivity by a current have been performed with thin wires made by extruding through draw plates. Such samples

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- A rounded periodic structure may, probably, exist only in samples of a very large diameter (with the ratio $\Lambda/2r_0 \lesssim 10^{-5}$). Experiments on such samples have not been performed because of considerable experimental difficulties.

** In the model suggested by Mel'nikov, the region in which $H < H_c$ for $I = I_c$ occupies more than 95% of the sample volume, and the minimal value of the magnetic field in the normal metal is less than $0.2H_c$.

*** A detailed analysis of various experimental results has been presented by I. Landau (1978).

****These experiments will be discussed below, in the section dealing with the paramagnetic effect in type-I superconductors.

are polycrystalline, and since ordinary objects of investigation have a considerable anisotropy of the thermal expansion coefficient, this leads to the concentration of stresses at grain boundaries and makes such samples absolutely unfit for studying the dynamic intermediate state.

Let us touch upon another circumstance to be taken into account in the experimental study of the intermediate state appearing under the destruction of superconductivity of a cylindrical sample by a current. The fact is that an ideal Gorter structure is unstable with respect to a longitudinal magnetic field. Obviously, even a very weak longitudinal field will be pumped into a sample through successive motions of superconducting layers; as time goes on, this process will lead to the destruction of the intermediate state and the formation of a more complicated structure, in which the layers of the two-dimensional mixed state of type-I superconductors play an important role. This situation will be considered in greater detail below; here we shall only estimate the value of an external longitudinal magnetic field, which is sufficient to destroy the intermediate state in the bulk of the sample.

The longitudinal field will not be pumped into the sample only in the case where the external field is so weak that the normal regions cannot contain even one magnetic flux quantum, $\phi_0 = 2 \times 10^{-7} \text{Oecm}^2$. For a sample several millimetres in diameter (one can hardly imagine a high-quality single crystal of much smaller diameter) this estimate yields a value of the external longitudinal field $H < 10^{-6} \text{Oe}$. Thus, the ideal Gorter structure can occur only under thorough screening and compensation of external magnetic fields. However, the time during which the structure of the intermediate state in the bulk of the sample is destroyed by pumping of a longitudinal magnetic field, can be very long. For instance, in pure samples $\sim 1 \text{cm}$ in diameter this time can reach several days for reasonable values of the external magnetic field, $H_{||} = 10^{-4} H_c = 10^{-2} \text{Oe}$; in other words, the structure which arises under such conditions practically does not differ from the purely Gorter structure during the time of an ordinary experiment.

Let us now determine the corrections to the critical current connected with the surface tension at the interfaces having a finite value. Surface tension causes the destruction of superconductivity not to start at exactly the critical value of the magnetic field at the surface of the sample, but at somewhat stronger currents. In a cylindrical sample the destruction of superconductivity should start with the formation of a normal shell along the cylinder circumference. The correction to the critical current can easily be found from the condition that the gain due to the energy of magnetic field in the normal metal must be equal to the surface energy of the cylindrical interface formed:

$$\Delta I_c = 0.5 (\Lambda/r_0)^{2/3} I_c \quad (20)$$

After the transition of the surface to the normal state the intermediate state structure is gradually formed in the bulk of the sample, and the free energy of the samples increases due to the work done by the current supply. However, although the free energy of the sample in the intermediate state is appreciably greater than in the superconducting state, the inverse transition to the superconducting state is impossible because the radius of the intermediate state region is smaller than the sample radius.

The calculated value of ΔI_c is very small, which is in excellent agreement with experiments of Posada and Rinderer (1975) and also of I. Landau (1980). There it was shown that the destruction of superconductivity with increasing current occurs practically at $I = I_c$, while the restoration of superconductivity with decreasing current may be "delayed" down to currents considerably lower than I_c . The existence of the intermediate state for $I < I_c$ is further evidence of the fact that the radius of the intermediate state region at $I = I_c$ is less than the sample radius.

Andreev (1968) calculated a similar correction to the critical current, by comparing the total

energy of the sample in the intermediate state with the energy of the superconducting sample, and obtained, naturally, a much larger value $\Lambda I_c = 0.8 (\Lambda/r_0)^{4/9} I_c$. These calculations would be applicable if an intermediate state structure existed in the sample whose radius tended to the sample radius as $I \rightarrow I_c$. Let us also note that these calculations are inconsistent with experimental data because such corrections to I_c exceed possible experimental errors (see I. Landau 1980).

15.3 3. Two-dimensional mixed state of type-I superconductors

15.3.1 3.1. Introductory remarks

In the preceding sections we have considered resistive phenomena associated with the existence of the intermediate state, but it was found that under the destruction of superconductivity by a current a resistive state may exist in type-I superconductors, which is not the intermediate one, i.e. which does not represent a mixture of macroscopic regions of superconducting and normal phases*.

Let us consider, for example, the destruction of superconductivity of a hollow cylindrical sample by a current. If the axial symmetry conditions are satisfied, the magnetic field due to the current is zero in the sample hole and, hence, remains small inside the metal near the internal surface. Therefore, the normal state for $T < T_c$ must be unstable even if the current in the sample considerably exceeds the critical value $I_c = cr_2 H_c/2$ (r_2 is the radius of the external surface of the sample). On the other hand, the sample has a resistance for $I > I_c$, so an uninterrupted superconducting layer cannot be formed along the internal surface. It may be assumed that in this case a thin macroscopically homogeneous layer of a certain special state arises at the internal surface of the sample. The thickness of this layer must be of the order of $\xi(T)$. The existence of a special resistive state under such conditions was first proposed by L. Landau (1938) who called this a mixed state.

In this case the term "mixed state" means a certain new state in which superconducting and normal properties are "mixed up" so intimately that it does not make sense to speak about individual regions of superconducting or normal phases. It was assumed initially that the mixed state should also arise under the destruction of superconductivity by an external magnetic field as a result of infinite branching of normal domains near the surface of the sample (L. Landau 1943).

The possibility of the existence of the mixed state under the destruction of superconductivity by a current has not been considered until recently, and after it has been found that a non-branched intermediate state structure arises in the destruction of superconductivity by an external magnetic field (Lifshits and Sharvin 1977), the term "mixed state" was used to denote the Shubnikov phase in type-II superconductors. Therefore, to avoid confusion, and taking into account that the mixed state of type-I superconductors is in the form of thin layers, we shall call it the two-dimensional mixed state or, in brief, the tm state.

A number of experiments have been performed so far to study the tm state. The macroscopic properties of tm layers have been investigated and some effects have been observed caused by the existence of such layers. It has been shown experimentally that the thickness of a tm layer is, in fact, of the order of the coherence length. Nevertheless, it is impossible to reconstruct, even qualitatively, the picture of processes occurring in the layer only on the basis of experimental results. It seems to us that progress in the study of the tm state is held back by the absence of a theory capable to describe the actual experimental situation.

Let us also note that at currents only slightly exceeding the critical one a region occupied

*In the present section the term "macroscopic" will be used to describe regions whose size considerably exceeds the coherence length $\xi(T)$.

by the intermediate state may, in principle, exist in the bulk of the sample. However, if the current increases up to the values

$$I > I_i = I_c \frac{r_1^2 + r_2^2}{2r_1 r_2}$$

(r_1 is the radius of the hole) the intermediate state cannot exist in the bulk of the sample. Here we shall consider only this relatively simple case, where the difference between the properties of the sample and a normal metal is only due to the presence of the tm layer.

15.3.2 3.2. The tm state in hollow cylinders

The tm state was first observed in the work of I. Landau and Sharvin (1969). The existence of this state is revealed most clearly in the current-voltage characteristics of hollow cylindrical samples (fig. 8). As can be seen from the figure, at $T < T_c$ and $I > I_c$ the sample carries an excess current in comparison with that of the normal sample. It is natural to assume that this excess current flows through the tm layer (we shall denote it by I_{1m}). The value of I_{tm} can easily be determined from experimental current voltage characteristics as the difference between the abscissae of curves 2 and 3 (or curves 1 and 3 for a lower temperature). Within the accuracy of those experiments the current in the tm layer was $I_c r_1 / r_2$, i.e. the surface density of the current in the layer is

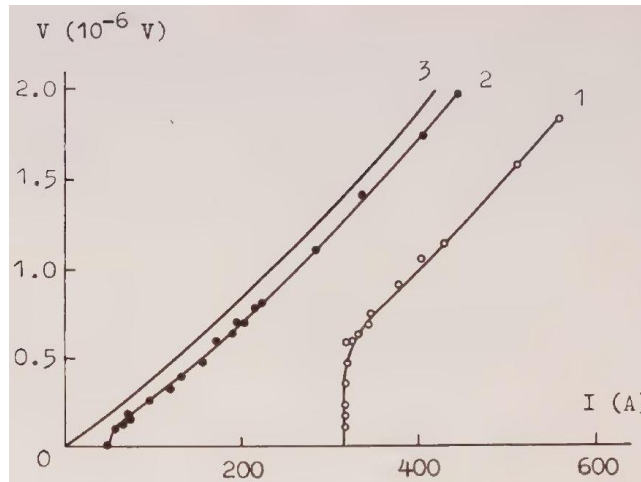


Fig. 8. Current-voltage characteristics of a hollow cylindrical sample: (1) $T = 1.39 \text{ K}$; (2) $T = 3.16 \text{ K}$; (3) extrapolation of the current-voltage characteristic in the normal state to $T = 3.16 \text{ K}$. The deviation of the characteristic from a straight line is due to the dependence of the resistance of the normal metal on the magnetic field of the current. (I. Landau and Sharvin 1969.)

close to $H_c/4\pi$, which allows the metal adjacent to the tm layer to be in the normal state*.

In the experiments of I. Landau and Sharvin (1972) conditions were realized under which the tm layer could be created not only on the inner surface of a hollow cylindrical sample, but also on the outer surface and in the bulk of the sample at an arbitrary distance from the axis. To this end, a superconducting wire extended along the cylinder axis was placed in the cavity of the sample. The current in this central wire (it will be denoted by I_b) could flow independently of the current in the sample. Small inductance coils were placed near the inner and outer surfaces of the cylinder to monitor the state of the sample surface. These coils made

it possible to observe changes in the surface impedance, thereby detecting the appearance or disappearance of the tm layer**.

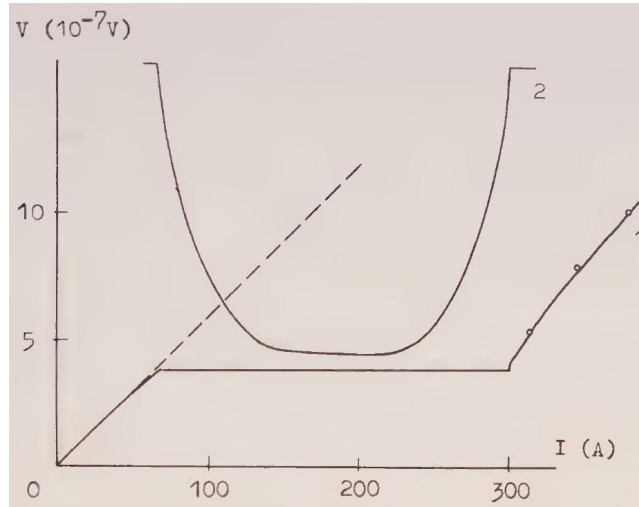


Fig. 9. Dependence on the current of (curve 1) the voltage across the sample and (curve 2) the impedance of the outer surface of the sample. $T=2.95 \text{ }^{\circ}\text{K}$, $I_{\text{b}}=-181 \text{ A}$. Points are calculated values of the voltage across the sample. (I. Landau and Sharvin 1972.)

The most interesting is the case where the currents in the central wire and the sample flow in opposite directions. Figure 9 shows the current voltage characteristic of the sample at constant $I_{\text{b}} < 0$ (curve 1) and the dependence on the current of the impedance of the outer surface of the sample (curve 2). The experimental conditions were such that $|I_{\text{b}}| > I_{\text{c}}$, so in the current range $0 \leq I < -(I_{\text{b}} + I_{\text{c}})$ the sample was completely in the normal state and its resistance and impedance were practically constant. As the current flowing through the sample increased further, a tm layer occurred on the outer surface of the sample, and the impedance of this surface then dropped rapidly, reaching practically the value for a purely superconducting surface near the point $I = -I_{\text{b}}$ where the magnetic field at the outer surface vanishes. As the current varied in the range

$$-(I_{\text{b}} + I_{\text{c}}) < I < -(I_{\text{b}} - I_{\text{c}}).$$

the voltage across the sample remained constant, hence the current flowing through the normal part of the sample was also constant, and the magnetic field at the boundary between the normal metal and the tm layer remained equal to H_{c} . At the same time, the current in the tm layer increased from zero to $2I_{\text{c}}$ and the field at the outer surface of the sample varied from $-H_{\text{c}}$ to H_{c} . At point $I + I_{\text{f}} = I_{\text{c}}$ the surface of the sample passed to the normal state, yet the tm layer did not disappear but started moving into the bulk of the sample, continuing to carry a considerable portion of the current. The points in fig. 9 represent the calculation of the

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- More accurate measurements performed later (I. Landau 1976, 1979) have shown that at currents slightly exceeding I_{c} the difference between I_{tm} and $I_{\text{c}}r_1/r_2$ is negligible, but an increase in the current density in the normal part of the sample leads to a decrease in I_{tm} , this decrease being more considerable in samples of metal with a larger resistivity.

**These coils were connected in the tank circuit of a RF oscillator, so the change in frequency of the oscillator, Δf (the measured value), was proportional to change in the imaginary part of the surface impedance.

current-voltage characteristic under the assumption that the tm layer is a cylindrical surface of radius r' and carries a current $cr'H_c$, so the magnetic field jumps from $-H_c$ to H_c at the layer.

The foregoing assumptions that the tm state should exist on the inner surface of hollow cylindrical samples remain, in principle, valid for any value of the current. On the other hand, it is obvious that sufficiently strong current will destroy the tm state and the entire sample will become normal. The current I_{c2} corresponding to the destruction of the tm state was measured by I. Landau and Svistov (1980) for two single-crystal indium samples. The samples had the same geometrical dimensions and crystallographic orientation, but different residual resistance. The results of the experiments are shown in fig. 10. As can be seen from the figure, the tm state is destroyed at sufficiently strong currents and different factors had to be taken into account, which might lead to considerable errors in determining I_{c2} . So, the temperature of the samples was monitored directly during the measurements, and the transverse magnetic field, which appeared in the hole of the sample because the geometry of the sample and current leads was not ideally axisymmetric, was compensated. The high value of I_{c2} restricted the possibility of measuring its temperature dependence, and although sufficiently strong currents (up to 1200 A) were used

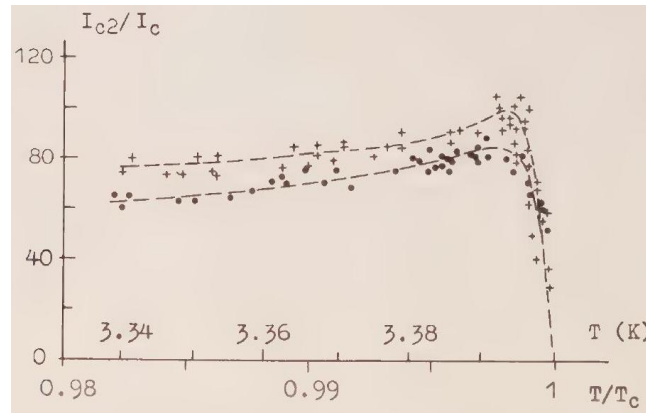


Fig. 10. Dependence of I_{c2}/I_c on temperature; () sample 1, (+) sample 2. (I. Landau and Svistov 1980.) The results presented in figs. 10 and 13-17 are obtained with four single-crystal indium samples with a slight admixture of tin; the dimensions of the samples were $r_1 = 2$ mm, $r_2 = 4$ mm; the resistance ratio $R(300\text{ K})/R(0\text{ K})$ was 1700 for sample 1, 3300 for sample 2, 1800 for sample 3 and 2200 for sample 4.

in the experiments the current at which the tm state was destroyed could be measured only near T_c .

The sharp decrease in the ratio I_{c2}/I_c on approaching T_c can be explained by the following simple considerations. In fact, a tm layer must, at any rate, be destroyed if the current density in the sample is so high that the magnetic field due to the current flowing in the completely normal sample reaches the critical value for a thickness equal to that of the tm layer. This condition can be written in the form

$$I_{c2} \sim I_c \frac{r_2^2 - r_1^2}{2r_1 d_m}, \quad (21)$$

where d_m is the minimum possible thickness of a tm layer. The value of d_m cannot be less than $\zeta(T)$, so it grows when T_c is approached, thereby causing I_{c2}/I_c to decrease. Such an estimate gives, obviously, too high values. More interesting is a slight decrease in I_{c2}/I_c as the temperature falls below $0.998T_c$ and the electric field rises (for sample with higher resistance).

15.3.3 3.3. Paramagnetic effect in type-I superconductors

The paramagnetic effect observed by Steiner and Schoeneck (1943) originates from the fact that the magnetic permeability of a bulk cylindrical sample whose superconductivity is destroyed by a current exceeds unity for a magnetic field parallel to the sample axis.

Here we shall not touch upon various attempts to explain the paramagnetic effect by the formation of special intermediate state structures, since it can easily be shown from the general equations of electrodynamics of the intermediate state (8) that no intermediate state structure can arise in a cylindrical sample with a current placed in a longitudinal magnetic field*.

In a cylindrical coordinate system the fourth of equations (8) can be written in the form

$$E_\varphi H_\varphi + E_r H_r + E_z H_z = 0 \quad (22)$$

The mean value, with respect to time, of E_φ is zero ($\bar{E}_\varphi \neq 0$ would mean a continuous increase or decrease of the longitudinal magnetic field in the sample); the components E_r and H_r are also zero. Thus, we obtain from eq. (22) $E_z H_z = 0$, but since $E_z \neq 0$ we have $H_z = 0$, i.e. the intermediate state cannot describe the situation which arises in a sample in the presence of a longitudinal magnetic field.

On the other hand, the Gorter structure of the intermediate state** that

must take place in the destruction of superconductivity by a current seems to be quite suitable for explaining the paramagnetic effect. Actually, a successive motion of cylindrical superconducting regions must lead to a pumping of longitudinal magnetic flux into the sample, which, in fact, is observed in the paramagnetic effect. It can easily be seen, however, that using only superconducting and normal domains one cannot describe a steady-state situation where the mean longitudinal magnetic flux does not change with time.

Quite a natural way out is to suggest that a cylindrical superconducting layer, in its motion towards the sample axis, is transformed into a tm layer before a new superconducting layer is formed around it* (here we mean a steady-state situation). As long as the superconducting layer exists the longitudinal magnetic flux is pumped into the sample; after the superconducting layer is transformed into a tm layer the magnetic flux starts decreasing (because the tm state is resistive) and the tm layer moves further on towards the sample axis. The decrease in the longitudinal magnetic flux in the sample proceeds until a new superconducting layer is formed. Thus, in the model considered the magnetic flux in the sample and its resistance must oscillate with time**.

Regular oscillations of the magnitude of the longitudinal magnetic flux and resistance were observed by I. Landau (1973) in the study of the paramagnetic effect in indium samples (fig. 11). The above-considered model of the paramagnetic effect does not allow the magnitude of the longitudinal magnetic flux to be calculated for any sample because the moment of formation of a new superconducting layer cannot be determined exactly from a priori considerations. However, quite simple ideas (see I. Landau 1973) make it possible to relate the observed resistance of the sample and its effective permeability for a longitudinal magnetic field $\mu^* = \phi / \pi r_0^2 H_e$ (ϕ is the longitudinal magnetic flux in the sample, H_e is the external magnetic field); as is frequently the case, calculations can be performed only for two limiting cases: weak external field ($H_e/H_c \ll 1$) and strong external field ($1 - H_e/H_c \ll 1$). The calculated curves are compared with experimental values of μ^* in fig. 12.

The observation of correlated oscillations of the resistance and magnetic moment of a sample, and also the good agreement between the calculated and experimental values of μ^* , are

*The simple proof of this statement given below comes from Andreev (1972).

**In this paragraph we shall discuss only an ideal Gorter structure, assuming that new superconducting regions arise already in the form of cylindrical surfaces.

convincing evidence of the validity of the dynamic model of the paramagnetic effect.

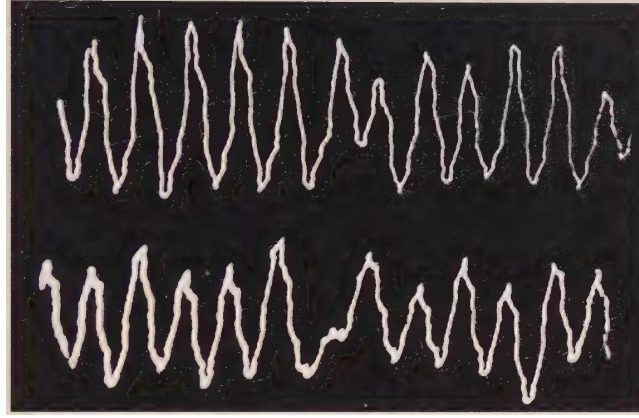


Fig. 11. Oscillations of magnetic flux (upper curve) and resistance (lower curve) of indium sample 7.8 mm in diameter, $R(300 \text{ K}) / R(0 \text{ K}) = 2500$, $T = 3.24 \text{ K}$, $H_e = 0.18 H_c$, $H_c = 4.5 \text{ Oe}$, $I = 1.05 I_c = 44 \text{ A}$. Oscillation frequency is 10 Hz ; amplitude of magnetic flux oscillations is $\sim 4\%$ of mean magnetic flux in the sample; amplitude of resistance oscillations is $\sim 2\%$ of the mean value. (I. Landau 1973.)

15.3.4 3.4. Study of the tm state in a longitudinal magnetic field

There are two types of external magnetic fields which do not distort the axial symmetry in the study of the tm state in hollow cylindrical samples, namely, circular and longitudinal magnetic fields. In section 3.2 we have already discussed the results obtained in the presence of an external circular magnetic field. In this paragraph we shall consider the results of experimental investigations of the tm state in hollow cylindrical samples in the presence of a longitudinal magnetic field and also under simultaneous action of a longitudinal and a circular magnetic field (I. Landau 1974, 1979; I. Landau and Svistov 1980).

Figure 13 shows how the longitudinal magnetic field H_i in the hole of the sample depends on the external longitudinal magnetic field H_e . As can be seen from the figure, H_i considerably exceeds H_e in the entire range of external

magnetic fields up to $H_e \approx H_c$ when the whole sample becomes normal. At $H_e \approx 0.65 H_c$ the $H_i(H_e)$ curve revealed a sharp drop of H_i with the increasing H_e , and hysteresis was observed for this part of the curve. Measurements of the impedance of the inner surface showed that the singularity in the $H_i(H_e)$ curve corresponds to a sharp increase in the surface impedance (fig. 14). As the magnetic field increased further, the impedance started oscillating with time at a frequency of 1 – 2 Hz (traces presented in the figure were obtained at a constant sweep rate of the magnetic field).

The dependences of the ratio H_i/H_e on the current in the sample and the temperature are shown in figs. 15 and 16.

¹*Let us emphasize that this assumption follows directly from the analysis of the motion of a cylindrical superconducting layer in the presence of a longitudinal magnetic field. The fact is that as this superconducting layer moves the longitudinal magnetic field $H_{\parallel}$ increases at the inner surface of the layer; an increase in $H_{\parallel}$ must, in turn, lead to a decrease in the velocity of the interface, and since the velocity of the outer boundary of the layer remains unchanged the thickness of the layer will decrease until it is transformed into a tm layer.

**The tm state is resistive, so the coexistence in a sample of longitudinal electric and magnetic fields in the presence of a structure of tm layers is not in contradiction with the laws of electrodynamics.

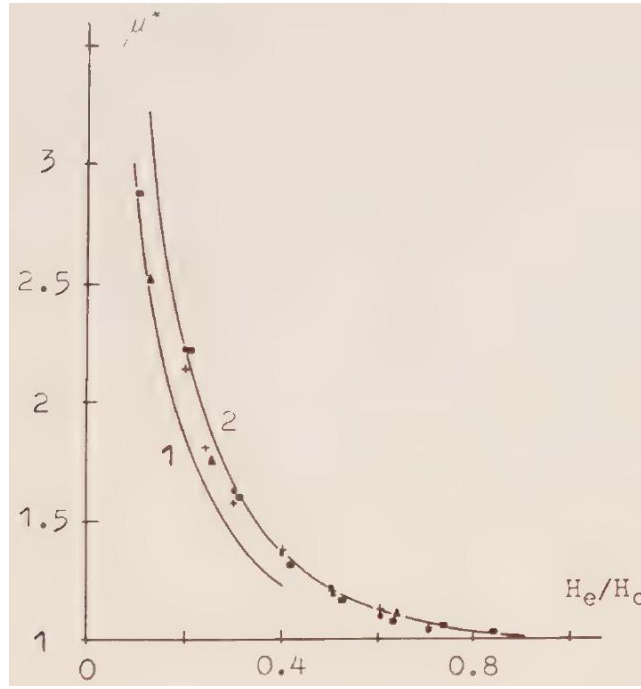


Fig. 12. The dependence μ^* on H_e/H_c for two indium samples. First sample: (○) $T = 3.335$ K, (□) $T = 3.27$ K, (△) $T = 3.24$ K. Second sample: (+) $T = 3.24$ K. Solid curves: calculation based on the observed values of the sample resistance: (1) for $H_e/H_c \ll 1$; (2) for $1 - H_e/H_c \ll 1$. (I. Landau 1973.)

Figure 17 presents the longitudinal magnetic field in the hole of the sample as a function of current in the central wire. In that interval of I_b values for which the tm layer is on the inner surface of the sample, $H_i > H_e$; the tm layer in the bulk of the sample did not affect the magnetic field in the hole, so $H_i = H_e$; when the layer reached the outer surface the effect changed its sign and $H_i < H_e$. The dependence $H_i(H_e)$ for the latter case is also shown in fig. 13.

The results presented in this section show that in the presence of a longitudinal magnetic field the current in the layer is not parallel to the electric field. Lines of current are in the form of helices whose sign is determined by the mutual orientation of the electric and magnetic fields at the boundary between the layer and the normal metal, the helical pitch decreasing with increasing longitudinal magnetic field.

15.3.5 3.5. The tm state under non-stationary external conditions

What was considered above referred to the tm state which arise when a direct current flows through a sample. It was found, however, that tm layers may arise under fast changes in an external magnetic field.

The first experiments of such kind were those of Dolgoplov and Murzin (1976, 1977, 1979) in which they studied the behaviour of cylindrical tin samples placed in a longitudinal magnetic field that varied as $H_e = H_+ + H_- \sin(\omega t)$. At all frequencies ω used in the experiments the skin-layer depth in the normal state δ was small in comparison with the sample radius r_0 . It was shown in those experiments that superconductivity in the bulk of the sample is destroyed even in the case where $H_+ < H_c$ (provided $H_+ + H_- > H_c$), and the magnetic flux through the sample may even exceed $\pi r_0^2 H_0$ (fig. 18).

This pumping of the longitudinal magnetic flux into the sample is related to the fact that

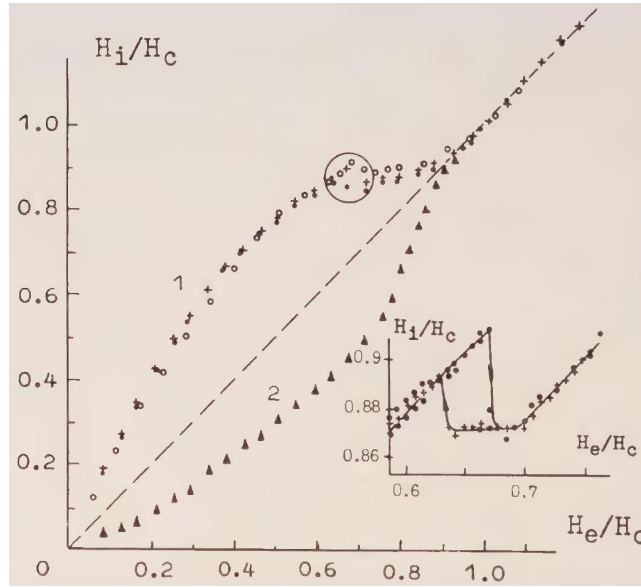


Fig. 13. Dependence on the external magnetic field of the longitudinal magnetic field in the hole of sample. Curve 1: the tm layer is on the inner surface of the sample ($I_{\text{b}} = 0$): ($\bigcirc$) sample 3, $T = 3.194$ K, $I = 1.25I_c = 81$ A; (+, -) sample 4, $T = 3.27$ K; (+) corresponds to $I = 1.15I_c = 51$ A and (-) to $I = 2.08I_c = 85$ A. Curve 2: the tm layer is on the outer surface of sample 4, $I = 85$ A, $I_b = -93$ A, $T = 3.27$ K. The insert is a large-scale view of the part in the circle; sample 4, $T = 3.194$ K, $I = 81$ A. (I. Landau 1974.)}

during the time when $H_e > H_c$ the surface of the sample becomes normal and the external magnetic field can freely penetrate into the sample,

while for the time when the external magnetic field decreases to values less than H_c a tm layer is formed on the surface, which makes it difficult for the magnetic field to get out of the central part of the sample. At the boundary between the tm layer and the normal metal the magnetic field is equal to the critical one, therefore, the magnetic field for the normal region inside the sample never drops below H_c (fig. 19).

Measurements of the average (over the sample surface) electric field have shown that at least for sufficiently large values of H the layer which arises on the sample surface, is really a resistive one. Arrow A in fig. 20 indicates the formation of the layer and arrow B to its destruction. As can be seen from the figure, the electric field inside the layer (region AB) is not zero.

The peak at B (fig. 20) is due to the penetration of the magnetic field into the region that was previously occupied by the tm layer. The thickness of the layer, d , can be estimated from the area of the peak. Figure 21 shows the experimental dependence of d on ω . In the frequency range investigated the thickness of the layer decreased with the increasing ω and amounted to about $3\xi(T)$ at maximum frequencies. This character of the dependence $d(\omega)$ was caused, most likely, by the nonequilibrium conditions under which the tm layer was formed. In fact, the formation of a tm layer on the surface of a normal sample, like any phase transition of the first order, proceeds with a certain delay relative to the equilibrium situation (at high frequencies the layer is, probably, to arise only after the magnetic field decreases to a value corresponding to absolute instability of the normal phase). At that moment the magnetic field in the normal metal is less than H_c inside a region of the order

of the skin-layer thickness and the time the tm layer exists on the sample surface is too short for the layer to reach equilibrium.

More equilibrium conditions can be obtained if a strong external magnetic field ($H_e > H_c$

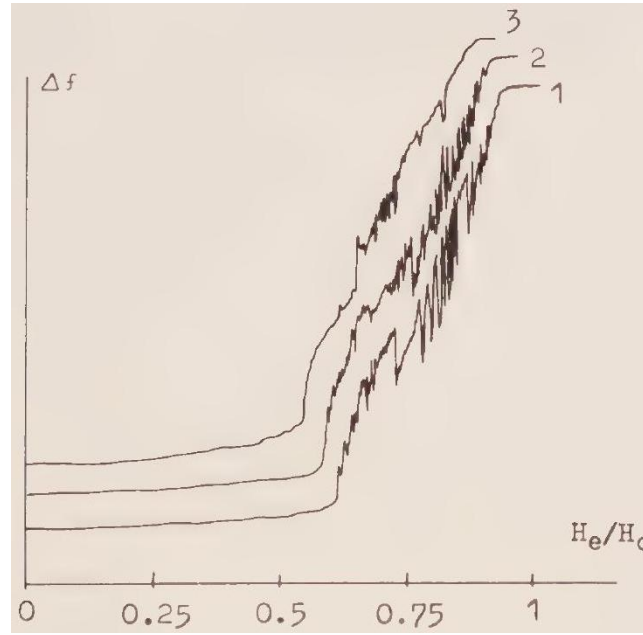


Fig. 14. Dependence of the inner-surface impedance of sample 4 on the longitudinal magnetic field. $T = 3.375 \text{ K}$

\$: (1) $I = 8.6 \text{ A}$, (2) $I = 15 \text{ A}$, (3) $I = 29 \text{ A}$; $I_c = 6.6 \text{ A}$. (I. Landau 1974.)}

) parallel to the sample surface is switched off. After the external field is switched off eddy currents are induced in the sample and initially the magnetic field in the bulk of the sample remains larger than H_c ; the magnetic field outside the sample is zero and conditions on the sample surface are completely analogous to those on the inner surface of hollow cylinders when superconductivity is destroyed by a current. Under such experimental conditions suggested by Dolgoplov and Dorozhkin (1978) there is no need for strong currents, which makes the experiment much simpler from a technical viewpoint. On the other hand, the attenuation of eddy currents leads to the fact that the tm layer appears to be under nonstationary external conditions, and this restricts, to some extent, experimental possibilities.

The tm state arising on switching off the external magnetic field was studied by Dorozhkin and Dolgoplov (1979a,b) and by I. Landau and Svistov (1980). The experimental conditions of these investigations were somewhat different: Dorozhkin and Dolgoplov's measurements were performed with a rectangular aluminium sample with dimensions $1 \times 1 \times 2 \text{ cm}$ (the larger dimension being directed along the magnetic field), while I. Landau and Svistov investigated a long cylindrical indium sample (the diameter of the sample was 8 mm). The main goal of those experiments was to study the structure of the layer with the aid of microcontacts.

If a magnetic field component perpendicular to the surface under investigation was present, the tm layer was inhomogeneous, that is, it contained a structure of moving normal domains. When these normal domains passed under the microcontacts the resistance of the contacts oscillated. By placing several microcontacts on the surface of the sample and comparing the signals from different contacts one could observe the motion of the normal regions between them and determine the direction and velocity of motion, and the characteristic dimensions of the structure.

The results obtained for aluminium and indium were slightly different. Two types of structures, namely layered and island, were observed in experiments with aluminium. The structure of the first type appeared on switching off a relatively weak external field [$(H_e - H_c)/H_c \lesssim 3$]. The normal regions in this case were in the form of strips extended along the projection of the external magnetic field on the sample surface; the direction of motion of these regions coincided

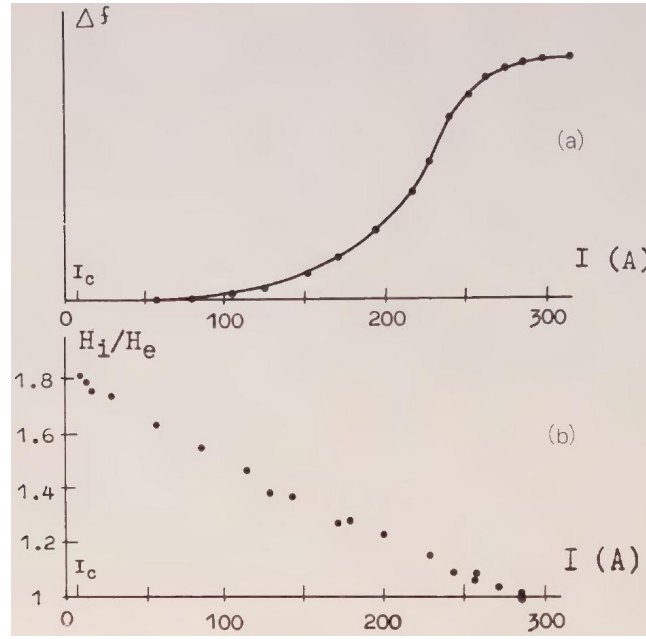


Fig. 15. Dependences of (a) the inner-surface impedance and (b) the magnitude of the paramagnetic effect on the current in the sample. Sample 4, $T=3.375 \text{ K}$, $H_{\perp}/H_e=0.22$, $H_c=0.73 \text{ Oe}$. (I. Landau 1974.)

with that of the currents flowing in the sample.

After switching off stronger magnetic fields H_e island structures were observed in aluminium, as well as in indium, which moved perpendicular to the current*. The velocity of the normal regions increased with decreasing transverse component of the magnetic field approximately as $1/H_{\perp}$. At small values of $H_{\perp}$ the signals from different contacts were not correlated any more, so the parameters of the structure could not be measured. Figure 22 shows several oscillograms of the voltage across the contacts.

In the absence of a transverse magnetic field no structure in the tm layer was observed in experiments with indium. In aluminium, under similar conditions, voltage pulses across the contacts were observed that corresponded to the motion of normal regions. The pulses from different contacts did not correlate in this case, and asymmetric pulses were frequently observed. This behaviour was interpreted as the formation of non-stationary structure with spontaneously appearing and annihilating normal regions carrying transverse magnetic fields of opposite signs. The measurements on aluminium, however, were performed with relatively short samples, so it is quite probable that the normal regions observed for the field H_e parallel to the surface were due to edge effects.

In experiments with aluminium the thickness of the tm layer was also measured. The measurements were performed as follows. After a strong magnetic field had been switched off, a tm layer appeared on the surface of the sample; the magnetic field was then switched on and the voltage across a coil wound around the sample was measured. The signal from this coil revealed a characteristic voltage pulse associated with the destruction of the layer and penetration of the magnetic field into this region. Determining the magnetic flux penetrating into the sample, one could estimate the thickness of the layer.

The experiments performed yielded the value $\sim \xi(T)$ for the thickness of the layer.

* The absence of the layered structure in indium samples is most likely to be related to the fact that measurements for small values of H_e were not performed in these experiments.

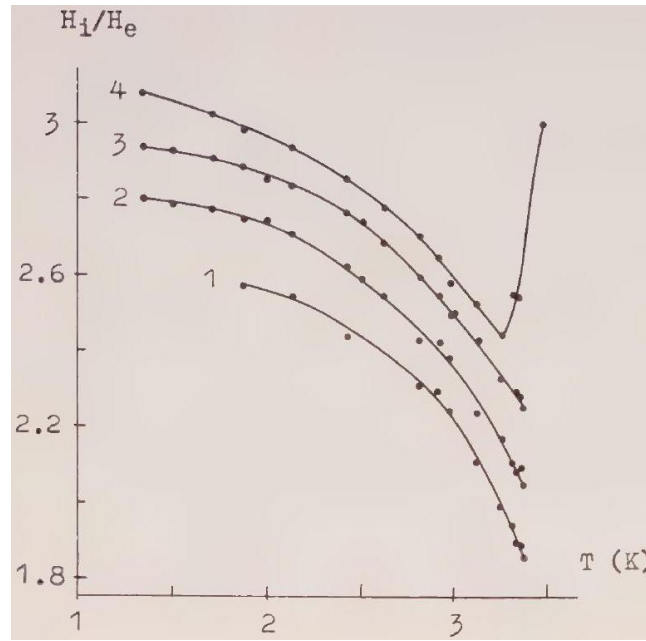


Fig. 16. The magnitude of the paramagnetic effect as a function of temperature. Sample 1, H_1/H_e : (curve 1) $I = 4I_c$; (2) $I = 2.8I_c$; (3) $I = 1.9I_c$; (4) $I = 1.2I_c$. (I. Landau and Svistov 1980.)

Among the results concerning the structure arising in the tm layer in the presence of a transverse magnetic field one should note the relatively large period of the structure (fig. 23), which considerably exceeds $\xi(T)^*$. This means that an electric field should also exist between the normal regions because the arguments that a resistive layer exists on the surface of a sample as a whole are completely applicable to individual "superconducting" regions in the tm layer whose dimensions are much larger than the layer thickness. Thus, the case where a transverse magnetic field is present can be interpreted as a peculiar intermediate state, in which regions occupied by the tm state play the role of superconducting regions.

15.3.6 3.6. Discussion of theoretical models

At present there are a number of theoretical papers dealing with the tm state, the theoretical concepts being developed via the construction of various models. In some papers periodic structures are proposed for the tm layer, which consist, in analogy with the intermediate state, of purely superconducting and purely normal domains; models of a homogeneous tm layer also exist. Nevertheless, all these models either are inconsistent with experimental results or have a region of applicability of the theoretical calculations which is far from actual experimental situation.

Let us first consider the models with periodic structures (Gor'kov and Dorokhov 1974, Andreev and Dzhikaev 1977). These structures are shown in fig. 24. As can be seen from the figure, both the structures have breaks on the

interface, which makes the existence of such structures impossible (see section 2.4). For the structure in a thin tm layer this circumstance appears to be much more essential than for the structure of the intermediate state in a bulk cylindrical sample. Let us consider an

^{1*}This fact by no means contradicts the estimates of the thickness of the tm layer because in superconducting films with a thickness of the order of the coherence length an island-type structure is observed whose characteristic dimensions significantly exceed the thickness of the films (Goren and Tinkham 1971).

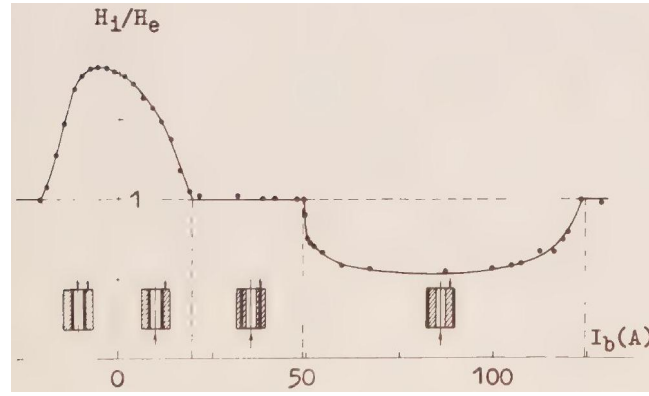


Fig. 17. Dependence on the current in the central wire of the longitudinal magnetic field in the hole of sample 4 ; $T=3.27 \text{ K}$, $I=-85 \text{ A}$, $H_{\text{e}}=0.4 H_{\text{c}}=8.2 \text{ Oe}$. The bottom part of the figure is a schematic diagram of the sample cross-section, arrows show directions of the currents, bold-face line indicates the position of the im layer. (I. Landau 1974).}

arbitrary superconducting region at the inner surface of a hollow cylindrical sample (fig. 25). The magnetic field in a

hole of the sample is zero, while the magnetic field at the boundary between the superconducting region and the normal part of the sample is determined by currents flowing into this region; therefore this field increases when moving from point 1 to point 2. In the case of a periodic structure of the intermediate state under the destruction of superconductivity of a bulk cylindrical sample such inflow of the currents is compensated by the increasing distance from the sample axis (for the cylindrical geometry the magnetic field decreases as $1/r$). In the case of the tm state the change in the distance from the sample axis can be neglected and equilibrium at the interface can be attained only if the equilibrium magnetic field at the interface also increases when moving from point 1 to point 2. There are, probably, only two ways to alter equilibrium conditions at the interface: first, one can use the dependence of the equilibrium magnetic field on the thickness of the superconducting layer (this possibility was employed by Gor'kov and Dorokhov 1974); second, one can use the fact that the equilibrium magnetic field at the interface depends on the curvature of the interface at a given point (this way of constructing the structure of the tm state was employed by Andreev and Dzhikaev 1977). Both these ways lead to acute angles at the interface where the concentration of the superconducting phase vanishes; and this is impossible, as was mentioned above.

Let us also note that the absence of a regular periodic structure in the tm layer directly follows from experimental study of the layer by microcontacts (see section 3.5).

Ivlev and Kopnin (1980) have suggested a model of the tm layer constructed in analogy with the resistive state of thin superconducting channels. In this model the layer consists of "superconducting" regions separated by narrow regions in which the order parameter vanishes (phase-slip centres). Quotation marks have been put around the word superconducting here, because in these regions an electric field exists along with the superconducting condensate current and, hence, in these regions energy is consumed from a current supply. This model is not truly periodic, since the phase-slip centres do not exist permanently but arise from time to time in various parts of the layer. Unfortunately, the applicability of this model requires electric fields and current densities considerably exceeding those used in real experiments. The restriction on the current density is most essential here, because the structure of the above type cannot exist for the relatively small current densities attained in experiments.

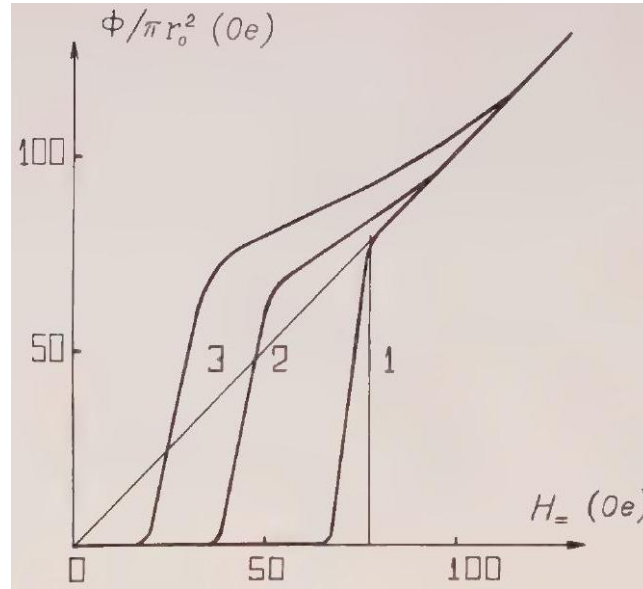


Fig. 18. Dependence on external magnetic field of the longitudinal magnetic flux in a sample: (1) $H_{\sim} = 0$; (2) $H_{=} = 45.5 \text{ Oe}$; (3) $H_{\sim} = 67.5 \text{ Oe}$. (Dolgoplov and Murzin 1974.)

As far as the models of homogeneous tm layers are concerned, they cannot be compared with experiment because the region of their applicability is very far from the real experimental situation. For instance, the model developed by Andreev and Tekel (1972) corresponds to the case where the electric field in the sample $E \equiv 0$ and describes, in fact, a thin superconducting layer. The model developed by Andreev and Bestgen (1973) refers to the opposite limiting case where the superconductivity in the tm layer is destroyed to such a degree that the small amount of superconducting pairs in the layer can be considered as fluctuations; in this case the superconducting current in the layer I_s^* must be significantly less than the normal current I_n^* , while in experiments, on the contrary, we have $I_s^* \gg I_n^*$.

Nevertheless, it is interesting to note that Bestgen (1973, 1979) considered the effect of a longitudinal magnetic field on the tm layer within the framework of the model of Andreev and Bestgen (1973), and showed that many of the theoretical conclusions are in qualitative agreement with experiment. Among these conclusions one may note the correct sign of the effect when a tm layer is either on the inner or on the outer surface of the sample, the absence of the effect when the tm layer is in the bulk of the sample, instabilities observed for large values of the longitudinal magnetic field (see fig. 14), and, finally, a decrease in the magnitude of the paramagnetic effect with increasing temperature (see fig. 16). Although the theoretical calculations are formally not applicable to the experimental situation, it is difficult to believe that the aforementioned qualitative agreement between theory and experiment for various aspects of the behaviour of the tm layer in a longitudinal magnetic field is simply occasional. It is, probably, an indication of the fact that such a behaviour of the superconducting condensate in the presence of parallel electric and magnetic fields is, in essence, a more general circumstance, and is retained for any relationship between I_s^* and I_n^* .

The structure of the tm state is of great interest because it represents a certain new type of resistive state of superconductors. Probably, kinetic and non-stationary process, related to the coexistence in the tm layer of superconductivity and electric field, should play an important role.

Let us also note that experimental investigations alone, without a profound development of

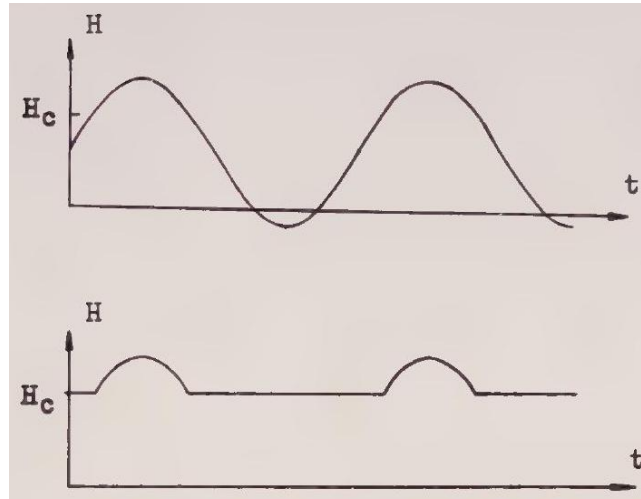


Fig. 19. Temporal dependence of (top) the external magnetic field and (bottom) the magnetic field at the boundary of the normal region.

the theoretical concepts, are unlikely to lead to considerable progress in the understanding of processes occurring in the tm layer. The fact is that of most interest here are rather sophisticated and laborious experiments, for example, creation of a tunnel or Josephson junction on the surface of a sample and investigation of the characteristics of such junctions in the presence of the tm layer. Such experiments, however, usually yield rather complicated results which, apparently, are difficult to interpret unambiguously without an appropriate theoretical model. On the other hand, experimental data now available are, of course, quite sufficient to verify the validity of one theoretical model or another.

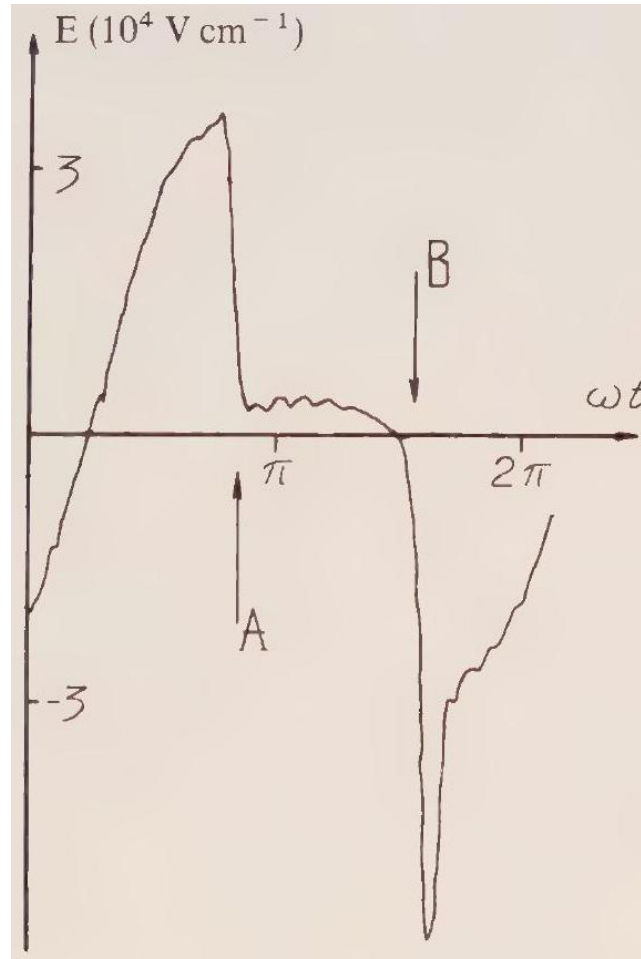


Fig. 20: Electric field strength, averaged over the sample surface, as a function of the phase of the alternating field. $\omega / 2\pi = 146 \text{ kHz}$, $H_{\perp} = 40 \text{ Oe}$, $H_{\parallel} = 54 \text{ Oe}$, $H_{\sim} = 40 \text{ Oe}$. (Dolgoplov and Murzin 1979.)

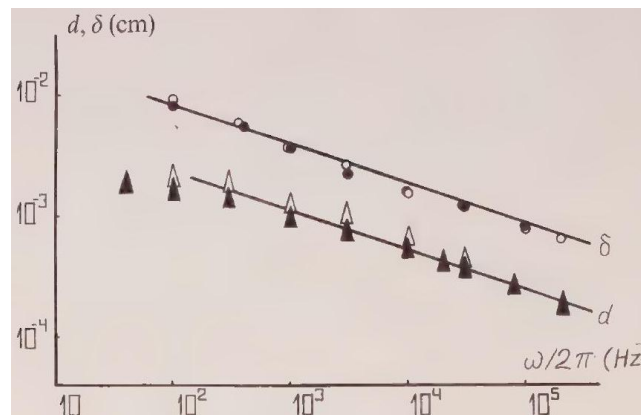


Fig. 21. Skin-layer depth and the thickness of the tm layer as functions of the alternating field frequency. Dots for d are obtained at $H_{\sim} = 225 \text{ Oe}$, $H_{\parallel} = 230 \text{ Oe}$, $H_{\perp} = 37 \text{ Oe}$. Straight lines correspond to the dependence $\omega^{1/3}$. (Dolgoplov and Murzin 1979.)

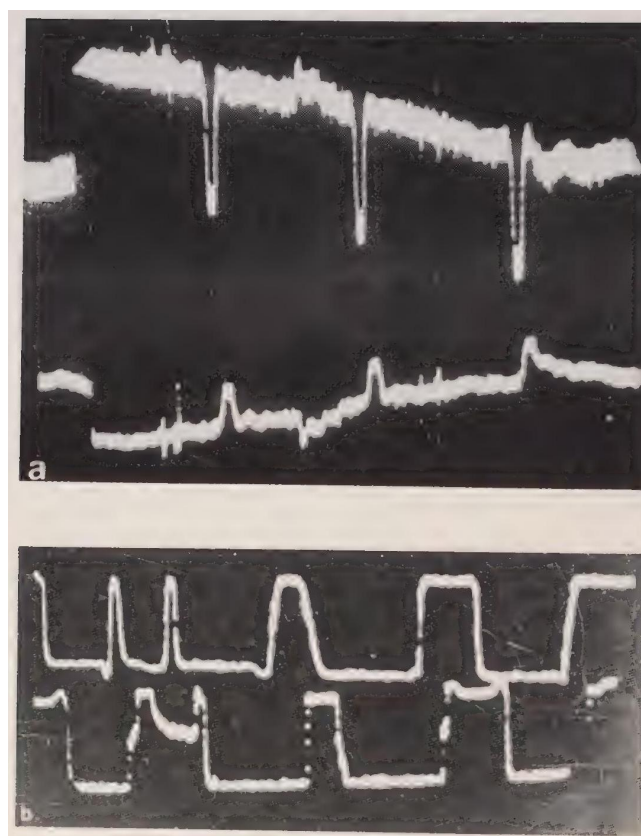


Fig. 22. Oscillograms of the voltage across the microcontacts. (a) From Dorozhkin and Dolsopolon i 1970 i The magnetic field H , that somithed off ated to the sample surface at an angle of 111.11° $2.7 \cdot 10^{-4} \text{ K}$

-1 K the light an incleste in the sample resistance corresponds to deflection of the tas upward for the lawe wallestam and to deflection downward for the upper oscillogram. (b) From I Landau and Svistov (1980) H_e is parallel to the sample surface; $H_e/H_c = 8.4$; the external static magnetic field is $H_e = 19 \text{ Oe}$; $T = 2.8 \text{ K}$ (sweep width is 15 ms).

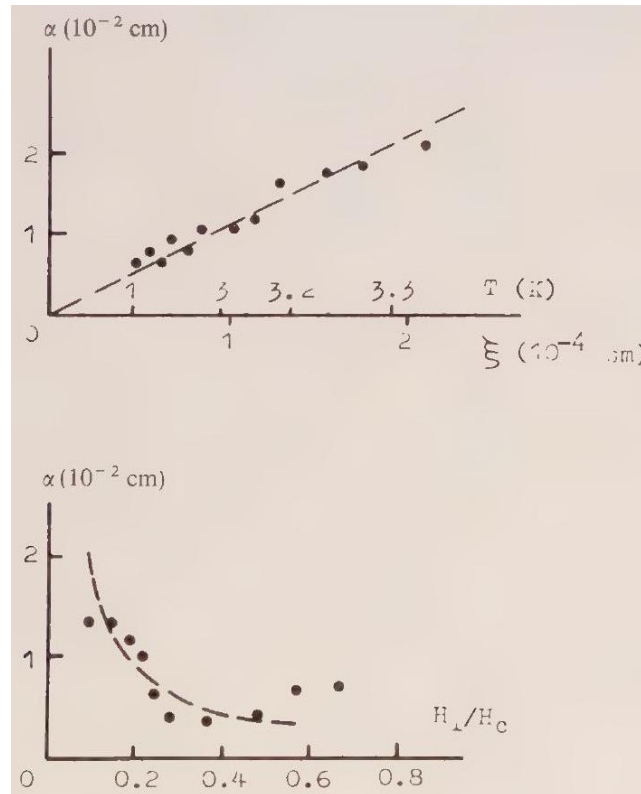


Fig. 23. Dependence of the mean period of the structure on temperature and the transverse magnetic field; (a) $\alpha(T)$ at $H_{\perp} = 0.18 H_c$; (b) $\alpha(H_{\perp})$ at $T = 3.06 \text{ K}$; $H_c = 700 \text{ Oe}$; $H_c = 52 \text{ Oe}$; the density of the circular current near the sample surface is $6.9cH_c/4\pi r_0$. The dashed curve corresponds to the dependence $\alpha \propto 1/H_{\perp}$. (1. Landau and Svistov 198(.).)

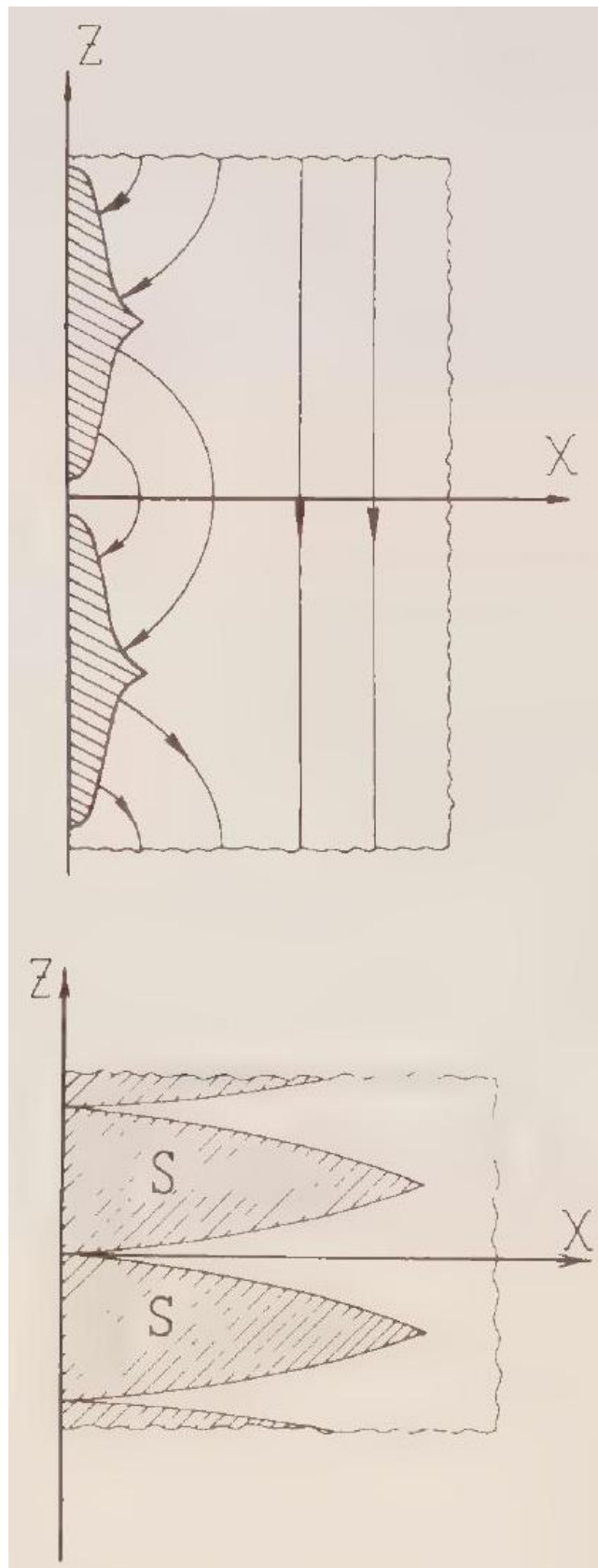


Fig. 24. Patterns of periodic structures suggested for the tm layer: (a) Gor'kov and Dorokhov (1974); (b) Andreev and Dzhikaev (1977). Superconducting domains are hatched. Arrows schematically show the direction of lines of the current.

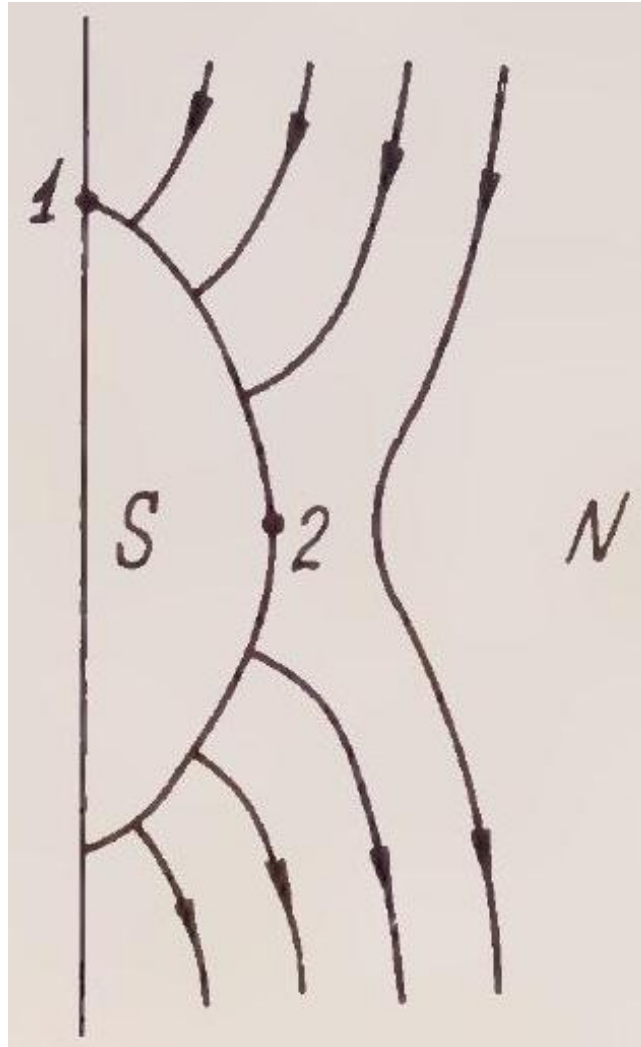


Fig. 25. Schematic diagram of the superconducting region on the inner surface of a hollow cylindrical sample; arrows show the direction of lines of current.

16 Theory of Anderson pseudospin resonance with Higgs mode in superconductors by Tsuji, Aoki

(finally, what is this effect???)

Abstract

A superconductor illuminated by an ac electric field with frequency Ω is theoretically found to generate a collective precession of Anderson's pseudospins, and hence a coherent amplitude oscillation of the order parameter, with a doubled frequency 2Ω through a nonlinear light-matter coupling. (read, learn, which?)

We provide a fundamental theory, based on the mean-field formalism, to show that the induced pseudospin precession resonates with the Higgs amplitude mode of the superconductor at $2\Omega = 2\Delta$ with 2Δ being the superconducting gap. (write key formula for it)

The resonant precession is accompanied by a divergent enhancement of the third-harmonic generation (THG). By decomposing the THG susceptibility into the bare one and vertex correction, we find that the enhancement of the THG cannot be explained by individual quasiparticle excitations (pair breaking), so that the THG serves as a smoking gun for an identification of the collective Higgs mode. (???)

We further explore the effect of electron-electron scattering on the pseudospin resonance by applying the nonequilibrium dynamical mean-field theory to the attractive Hubbard model driven by ac electric fields. (???)

The result indicates that the pseudospin resonance is robust against electron correlations, although the resonance width is broadened due to electron scattering, which determines the lifetime of the Higgs mode. (???)

16.1 Introduction

Dynamical control of quantum many-body states of matter without destroying quantum coherence is becoming a central challenge in condensed matter physics (this shouldn't be taken too seriously, there are no arguments for this statement) .

While recent developments in ultrafast laser experiments have enabled one to study relaxation dynamics of quantum systems after pulse excitation, an alternative direction we can pursue is to look at far-from-equilibrium quantum states that are realized *during photoirradiation*. (what is the photoirradiation and why it is not connected to ultrafast pulses?)

From this viewpoint, superconductivity is an intriguing ground to look for a novel optical control. (give an example of optical control)

A superconducting state can be described in terms of pseudospins introduced by Anderson in 1958[Anderson(1958)].

Indeed, a collective precession of the pseudospins represents a Higgs amplitude mode [Anderson(1958), Volkov,Kogan (1973), Littlewood (1981), 27, Pekker and Varma(2015)] (read these articles, learn, are they relevant? VolkovKogan1973 is maybe definitely not) , i.e., a coherent amplitude oscillation of the superconducting order parameter with a frequency 2Δ (the superconducting gap), which is a condensed matter analog of the Higgs boson in elementary particle physics,[Eng(), 1, 13] and the σ meson in nuclear physics [Nam(a)].

(how exactly precession of the pseudospins represents a Higgs amplitude???)

This naturally emerges as a massive mode along the radial direction in the Mexican-hat potential profile when a spontaneous symmetry breaking occurs in systems coupled to gauge fields [9, Eng(), 1, 13] (it is known statement, but what is in that articles?) .

The Higgs mode in superconductors has been experimentally observed by Raman [Sooryakumar and Klein(1980), 28] and THz pump-probe[4] spectroscopies (how?? read these papers at least ones) .

A natural question then is whether one can manipulate the dynamics of the pseudospins like one does for real spins by applying a magnetic field. (maybe no, because it is just a mathematical description, no?)

Usually, however, it has been supposed to be difficult to photo-control the pseudospins (what is at all photo-control the pseudospins? we have an Anderson model, so what are we proposed to do?) , since the pseudospins do not directly couple to electromagnetic fields (in the linear-response regime).

In this paper, we theoretically show that, if we go over to a *nonlinear* regime, an ac electric field with frequency Ω does indeed generate a collective precession of Anderson's pseudospins with frequency 2Ω through the nonlinear light-matter coupling, which results in a 2Ω amplitude oscillation of the superconducting order parameter.

We further find that a *resonance* between the induced pseudospin precession and the Higgs mode emerges when $2\Omega = 2\Delta$. (in other words, if we shine light with frequency Δ , what will be??)

This is remarkable, since this occurs *not* at $\Omega = 2\Delta$ but at Ω well below 2Δ (subgap regime), where quasiparticle excitations are suppressed.

We may call the phenomenon "Anderson pseudospin resonance" (APR). APR may seem analogous to the nuclear magnetic resonance (NMR) or electron spin resonance (ESR), but APR is distinct in that the effect is essentially a *collective phenomenon as a resonance with the Higgs amplitude mode*. We show that APR should appear as a divergent enhancement of the third-order nonlinear optical response [third harmonic generation (THG)]. We further find that the enhancement of THG cannot be explained by quasiparticle excitations, which hence distinguishes the collective Higgs mode from individual pair breaking processes, both of which lie at the same energy scale. APR has been experimentally observed very recently by a THz laser experiment [46].

16.2 Phenomenological time-dependent Ginzburg-Landau theory

To understand how the order parameter and the Higgs amplitude mode dynamically respond to electromagnetic fields, it is instructive to first overview the time-dependent Ginzburg-Landau (GL) theory. This gives a simple macroscopic (and phenomenological) description of the superconducting order parameter as a low-energy effective field theory, although we have to mention that the time-dependent GL theory has a serious problem in describing the Higgs mode and its resonance in superconductors as we shall stress toward the end of this section, which makes us opt for a microscopic theory in later sections.

Let us consider the GL "Lagrangian density" as a functional of the complex order parameter $\Psi(\mathbf{r}, t)$ in a general form of

$$\mathcal{L} = - \left[a|\Psi|^2 + \frac{b}{2}|\Psi|^4 + \frac{1}{2m^*} |(-i\nabla - e^* \mathbf{A})\Psi|^2 \right] + c|(i\partial_t - e^* \phi)\Psi|^2 + d\Psi^\dagger(i\partial_t - e^* \phi)\Psi \quad (16.1)$$

where a, b, c and d are coefficients, ϕ and $\mathbf{A}$ are the scalar and vector potentials, and e^* and m^* are the effective electric charge and effective mass, respectively. The Lagrangian density (16.1) is invariant under the gauge transformation $\Psi(\mathbf{r}, t) \rightarrow e^{ie^*\chi(\mathbf{r}, t)}\Psi(\mathbf{r}, t)$, $\phi(\mathbf{r}, t) \rightarrow \phi(\mathbf{r}, t) - \partial_t\chi(\mathbf{r}, t)$,

$\mathbf{A}(\mathbf{r}, t) \rightarrow \mathbf{A}(\mathbf{r}, t) + \nabla\chi(\mathbf{r}, t)$. At temperatures $T < T_c$, $a = a_0(T - T_c)$ becomes negative, and the global $U(1)$ symmetry $[\Psi(\mathbf{r}, t) \rightarrow e^{ie^*\chi}\Psi(\mathbf{r}, t)$ with a constant $\chi]$ is spontaneously broken. The other coefficients are taken to be positive. To describe the dynamics of the order parameter, we have included the kinetic terms, one with a coefficient c that represents the kinetic term of Klein-Gordon-type equations, and another with d that represents the kinetic term of Gross-Pitaevskii-type equations.

Now, we expand $\mathcal{L}$ (16.1) around the ground state $\Psi_0 = \sqrt{-a/b}$ (the phase is chosen as such without loss of generality). There are two kinds of elementary excitations from the ground state: the variation along the radial direction and another along the circumferential direction on the complex plane of the order parameter. We write them as $\Psi(\mathbf{r}, t) = [\Psi_0 + H(\mathbf{r}, t)]e^{i\theta(\mathbf{r}, t)}$, where H and θ denote the Higgs and Nambu-Goldstone (NG) fields, respectively. The expansion gives us

$$\begin{aligned} \mathcal{L} = & c(\partial_t H)^2 + ce^{*2} \left(\phi + \frac{1}{e^*} \partial_t \theta \right)^2 (\Psi_0 + H)^2 \\ & - de^* \left(\phi + \frac{1}{e^*} \partial_t \theta \right) (\Psi_0 + H)^2 + 2aH^2 - \frac{1}{2m^*} (\nabla H)^2 \\ & - \frac{e^{*2}}{2m^*} \left(\mathbf{A} - \frac{1}{e^*} \nabla \theta \right)^2 (\Psi_0 + H)^2 + \dots \end{aligned} \quad (16.2)$$

in which we have dropped total-derivative terms as well as higher-order interactions.

The terms proportional to $\phi \partial_t \theta$ and $\mathbf{A} \cdot \nabla \theta$ in Eq. (16.2) indicate that the NG phase mode turns into a longitudinal component of the gauge field. As a result of the Anderson-Higgs mechanism, [9, Eng(), 1, 13] the NG mode is absorbed to the gauge field, and is pushed to very high energy scale of the plasma frequency ω_p . We can thus regard θ in Eq. (16.2) to be an unphysical degree of freedom, which one can eliminate by taking the unitary gauge,

$$\begin{aligned} \mathcal{L} = & c(\partial_t H)^2 + (ce^{*2}\phi^2 - de^*\phi)(\Psi_0^2 + 2\Psi_0 H) + 2aH^2 \\ & - \frac{1}{2m^*} (\nabla H)^2 - \frac{e^{*2}\Psi_0^2}{2m^*} \mathbf{A}^2 + \frac{e^{*2}\Psi_0}{m^*} \mathbf{A}^2 H + \dots \end{aligned} \quad (16.3)$$

One can see that the terms $ce^{*2}\phi^2$ and $\frac{e^{*2}\Psi_0^2}{2m^*} \mathbf{A}^2$ represent the mass of the gauge field generated via the Anderson-Higgs mechanism.

In the case of electrically neutral superfluids ($e^* = 0$), the Anderson-Higgs mechanism does not occur, so that the Higgs field mixes with the NG field via the term proportional to $d(\partial_t \theta)H$ in Eq. (16.2), and the Higgs mode is no longer considered to be an isolated excitation. Furthermore, there are interactions between H and θ via the terms proportional to $c(\partial_t \theta)^2 H$ and $(\nabla \theta)^2 H$ in Eq. (16.2), which causes the relaxation of the Higgs into lower energy NG bosons, which makes the Higgs mode unstable. At this point, it has been often emphasized that the particle-hole symmetry is important in forcing $d \sim 0$ to suppress such a mixing between the Higgs and NG modes. [27, Pekker and Varma(2015)] In other words, the Higgs mode is to be protected by the particle-hole symmetry. However, this argument for the stability of the Higgs mode is not needed in the case of charged superconductors (although the particle-hole symmetry is a good symmetry near the Fermi surface in superconductors), since *the NG field decouples from the Higgs field [in Eq. (16.3)] due to the Anderson-Higgs mechanism* as stated above.

Equation (16.3) suggests that the interaction between the Higgs and gauge fields is given by ϕH , $\phi^2 H$ and $\mathbf{A}^2 H$. The linear coupling ϕH is suppressed in superconductors due to the inherent particle-hole symmetry ($d \sim 0$). The leading interaction is the second-order process $\phi^2 H$ and $\mathbf{A}^2 H$, the latter of which, e.g., is represented by a Feynman diagram shown in Fig. 16.1. The nonlinear Higgs-gauge coupling implies that H describes a scalar boson having no electric

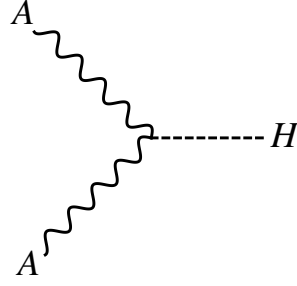


Figure 16.1: The Feynman diagram for the interaction vertex that connects the Higgs field H (dashed line) and electromagnetic field $\mathbf{A}$ (wavy lines).

charge. These nonlinear couplings ($\phi^2 H$ and $\mathbf{A}^2 H$) have indeed been used in the discovery of the Higgs particle at the LHC experiment [23, 24] (where $\mathbf{A}$ corresponds to the vector bosons W or Z).

From Eq. (16.3) (with $d = 0$), we can derive the equation of motion for the Higgs field,

$$\left(c\partial_t^2 - \frac{1}{2m^*}\nabla^2\right)H = 2aH + e^{*2}\Psi_0\left(c\phi^2 - \frac{1}{2m^*}\mathbf{A}^2\right), \quad (16.4)$$

which is “relativistic” (with an emergent Lorentz symmetry), [Pekker and Varma(2015)] meaning that the first time-derivative is absent even though we started from the non-relativistic GL Lagrangian (16.1). Let us first look at the case of $\phi = \mathbf{A} = 0$. By putting $H(\mathbf{r}, t) \sim e^{i\mathbf{q}\cdot\mathbf{r} - i\omega t}$, we obtain the dispersion relation for the Higgs mode,

$$\omega(\mathbf{q})^2 = -\frac{2a}{c} + \frac{q^2}{2m^*c} = \omega_H^2 + \frac{q^2}{2m^*c}, \quad (16.5)$$

where the mode is a gapped (massive) excitation with a characteristic frequency (mass)

$$\omega_H = \sqrt{-\frac{2a}{c}}. \quad (16.6)$$

From this, one can see that $\omega_H \propto (T_c - T)^{1/2} \propto \Delta$. In fact, the microscopic calculation [Anderson(1958), Littlewood (1981)] shows that $\omega_H = 2\Delta$, which exactly coincides with the lowest energy necessary to create a pair of Bogoliubov quasiparticles. Using the microscopic result [Gor()] of $b(\Psi_0/\Delta)^2 = 3/(4\epsilon_F)$ (with ϵ_F the Fermi energy), we have $c = -2a/(2\Delta)^2 = 2b\Psi_0^2/(2\Delta)^2 = 3/(8\epsilon_F)$. With this and $m^* = 2m$, we reproduce the well-known relation, [Littlewood (1981)]

$$\omega(\mathbf{q})^2 = (2\Delta)^2 + \frac{1}{3}v_F^2q^2, \quad (16.7)$$

where $v_F = \sqrt{2\epsilon_F/m}$ is the Fermi velocity. From Eq. (16.2), it is obvious that the dispersion for the NG mode (or Bogoliubov mode) shares the same form $\omega(\mathbf{q})^2 = q^2/(2m^*c) = v_F^2q^2/3$ with the Higgs mode besides the mass term. This agrees with the previously known result. [Bogoliubov et al.(1958)]

Next, we turn to a situation where the system is driven by a continuous and homogeneous ac electric field $\mathbf{A}(t) = \mathbf{A}e^{-i\Omega t}$. The problem becomes equivalent to a forced oscillation of a harmonic oscillator, and the solution for Eq. (16.4) is given by

$$H(t) = \frac{1}{(2\Omega)^2 - \omega_H^2} \frac{e^{*2}\Psi_0 A^2}{2m^*c} e^{-2i\Omega t}. \quad (16.8)$$

This captures the fundamental aspect of the resonance phenomenon discussed in the paper. Due to the nonlinear coupling to the electric field, the elementary frequency of the oscillation of the Higgs field is 2Ω (rather than Ω). When 2Ω matches with the eigenfrequency of the Higgs field ω_H , the resonance occurs and the oscillation amplitude diverges as $(2\Omega - \omega_H)^{-1}$. From a microscopic point of view, this phenomenon can be understood as a resonant precession of Anderson pseudospins as we shall discuss in Sec. 16.3.

The current $\mathbf{j} = \partial\mathcal{L}/\partial\mathbf{A}$ is expressed as

$$\mathbf{j} = -\frac{ie^*}{2m^*}[\Psi^\dagger\nabla\Psi - (\nabla\Psi^\dagger)\Psi] - \frac{e^{*2}}{m^*}\mathbf{A}\Psi^\dagger\Psi.$$

Expanding Ψ around Ψ_0 , we obtain the leading nonlinear current response against $\mathbf{A}$,

$$\mathbf{j}_{\text{NL}}(t) = -\frac{2e^{*2}\Psi_0}{m^*}\mathbf{A}(t)H(t).$$

This takes the form of a London equation, where the current is proportional to $\mathbf{A}(t)$. Remarkably, the nonlinear current is also *proportional to the Higgs field* $H(t)$, so that the current can, and does indeed, sensitively reflect the temporal change of the Higgs field. Since $\mathbf{A}(t)$ oscillates with frequency Ω , while $H(t)$ oscillates with 2Ω , the current $[\propto \mathbf{A}(t)H(t)]$ ends up with oscillating with frequency 3Ω . This implies that a *giant third harmonic generation* (THG) is induced near the resonance ($2\Omega \sim \omega_H$) with the Higgs mode.

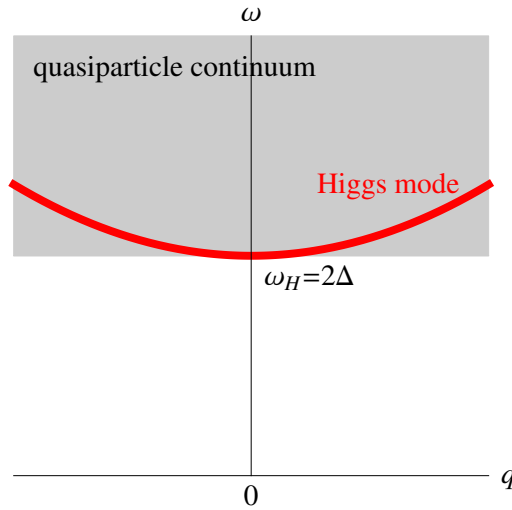


Figure 16.2: (Color online) Schematic excitation spectrum of *s*-wave superconductors. The red curve is the collective Higgs mode, while the shaded region represents the quasiparticle excitation continuum.

So far, we have discussed the Higgs mode and its resonance with electromagnetic waves based on the time-dependent GL theory (16.1). Apart from the fact that the characteristic frequency ω_H of the Higgs mode cannot be determined within GL theory, the problem of GL theory is that it does not take account of relaxations of the Higgs mode into quasiparticles. As shown in Fig. 16.2, the Higgs mode is degenerate with the lower bound of the quasiparticle excitation continuum ($\omega_H = 2\Delta$). The coincidence of the two energies is known as the Nambu relation.[21, Vol(b)] Since the Higgs mode lies at the same energy scale as the quasiparticle excitations, it can easily decay into individual quasiparticles. Furthermore, at low temperatures the relaxation time of quasiparticles becomes much longer than the time scale of the order-parameter variation in clean superconductors. As a result, one cannot neglect quasiparticle

excitations, and the dynamics of the order parameter is necessarily entangled with those of quasiparticles. The low-energy effective theory of the Higgs mode may not be expressed only in terms of Ψ , but may involve fermionic degrees of freedom. The crucial questions that arise are (i) whether the Higgs resonance discussed here would survive or not after we take account of the relaxation to quasiparticles (pair-breaking process), and (ii) if it would survive, then how one can distinguish the collective Higgs mode from individual quasiparticle excitations, both of which are energetically degenerate. These motivate us to move on to the underlying microscopic theory in the subsequent sections.

16.3 Microscopic theory for Anderson pseudospin resonance

Having identified the necessity of going beyond GL theory, we start from the pairing Hamiltonian for an s -wave superconductor coupled to a dynamical electric field,

$$H_{\text{pair}} = \sum_{\mathbf{k}, \sigma} \epsilon_{\mathbf{k} - e\mathbf{A}(t)} c_{\mathbf{k}\sigma}^\dagger c_{\mathbf{k}\sigma} - U \sum_{\mathbf{k}, \mathbf{p}} c_{\mathbf{k}\uparrow}^\dagger c_{-\mathbf{k}\downarrow}^\dagger c_{-\mathbf{p}\downarrow} c_{\mathbf{p}\uparrow}, \quad (16.9)$$

where $\epsilon_{\mathbf{k}}$ is the band dispersion measured from Fermi energy ϵ_F , e the elementary charge, $\mathbf{A}(t) = \mathbf{A} \sin \Omega t$ the vector potential for the ac electric field $\mathbf{E}(t)$ introduced by Peierls substitution (in the temporal gauge), $c_{\mathbf{k}\sigma}^\dagger$ the creation operator for electrons, and $-U (< 0)$ is the attractive pairing interaction. We consider a superconducting thin film, into which the electric field can penetrate. For H_{pair} (16.9), the BCS mean-field description becomes exact. We define the superconducting gap function,

$$\Delta = \Delta' + i\Delta'' = U \sum_{\mathbf{k}} \langle c_{\mathbf{k}\uparrow}^\dagger c_{-\mathbf{k}\downarrow}^\dagger \rangle, \quad (16.10)$$

that serves as the order parameter in the BCS theory. We can replace the momentum sum with an integral $D(\epsilon_F) \int_{-\omega_D}^{\omega_D} d\epsilon$ with $D(\epsilon_F)$ the density of states at the Fermi energy and ω_D the energy cut off (e.g., the Debye frequency of the bosonic pairing glue such as phonons). The interaction strength is characterized by a dimensionless $\lambda = UD(\epsilon_F)$. In the following we set $\hbar = 1$, and use ω_D as the unit of energy.

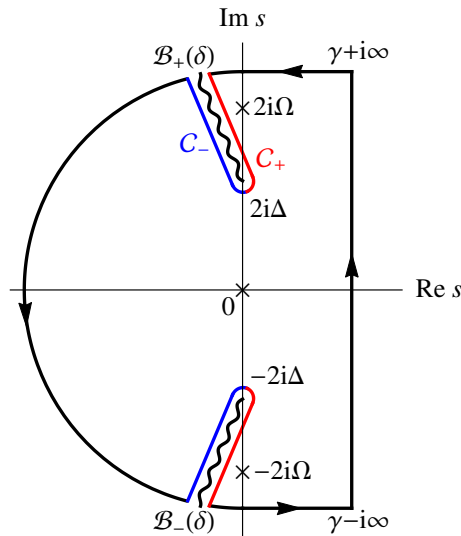


Figure 16.3: (Color online) The integral contour (solid closed curve) that we take on the complex plane to evaluate the integral (16.22). Crosses and wavy lines represent poles and branch cuts, respectively.

Anderson's pseudospin [Anderson(1958)] is defined by

$$\boldsymbol{\sigma}_{\mathbf{k}} = \frac{1}{2} \Psi_{\mathbf{k}}^{\dagger} \cdot \boldsymbol{\tau} \cdot \Psi_{\mathbf{k}}, \quad (16.11)$$

where $\Psi_{\mathbf{k}} = (c_{\mathbf{k}\uparrow}, c_{-\mathbf{k}\downarrow}^{\dagger})^t$ is the Nambu spinor, and $\boldsymbol{\tau} = (\tau^x, \tau^y, \tau^z)$ are the Pauli matrices. The pseudospin satisfies the usual commutation relations for angular momentum, $[\sigma_{\mathbf{k}}^j, \sigma_{\mathbf{k}}^k] = i\varepsilon_{jkl}\sigma_{\mathbf{k}}^l$. With this, the pairing Hamiltonian (16.9) is recast in a form,

$$H_{\text{pair}} = 2 \sum_{\mathbf{k}} \mathbf{b}_{\mathbf{k}} \cdot \boldsymbol{\sigma}_{\mathbf{k}}, \quad (16.12)$$

which can be regarded as a spin system in an effective magnetic field,

$$\mathbf{b}_{\mathbf{k}} = \left(-\Delta', -\Delta'', \frac{\epsilon_{\mathbf{k}-e\mathbf{A}(t)} + \epsilon_{\mathbf{k}+e\mathbf{A}(t)}}{2} \right). \quad (16.13)$$

The z component of $\mathbf{b}_{\mathbf{k}}$ represents the light-matter coupling involving contributions from both the particle and hole sectors. Since $\mathbf{b}_{\mathbf{k}}$ is a function even in $\mathbf{A}(t)$ if the system is parity symmetric ($\epsilon_{-\mathbf{k}} = \epsilon_{\mathbf{k}}$), we can readily recognize that the linear coupling vanishes, so that the leading effect of the electric field starts from $O(\mathbf{A}(t)^2)$. The self-consistency condition (16.10) reads

$$\Delta = U \sum_{\mathbf{k}} (\sigma_{\mathbf{k}}^x + i\sigma_{\mathbf{k}}^y) \quad (16.14)$$

in the pseudospin notation. While the dynamics cannot be described by the conventional GL equation, which would be valid only when the time scale of the order-parameter motion is much longer than that of quasiparticle relaxations, in the present formalism the time evolution is determined by a Bloch equation for the pseudospins, [Volkov,Kogan (1973), Barankov et al.(2004), 32]

$$\partial_t \boldsymbol{\sigma}_{\mathbf{k}} = i[H_{\text{pair}}, \boldsymbol{\sigma}_{\mathbf{k}}] = 2\mathbf{b}_{\mathbf{k}} \times \boldsymbol{\sigma}_{\mathbf{k}}. \quad (16.15)$$

Anderson pseudospins have been recently used to analyze the dynamics of charge fluctuations in a time-resolved Raman experiment for high- T_c cuprates. [102, Lor()]

We can analytically solve Eq. (16.15) up to the leading (second) order in $\mathbf{A}(t)$. This is achieved by linearizing Eq. (16.15) with the time-independent and time-dependent parts separated as $\boldsymbol{\sigma}_{\mathbf{k}}(t) = \boldsymbol{\sigma}_{\mathbf{k}}(0) + \delta\boldsymbol{\sigma}_{\mathbf{k}}(t)$ and $\Delta(t) = \Delta + \delta\Delta(t)$. We assume that the initial state is superconducting at zero temperature. The initial Δ may be taken to be real positive without loss of generality. Thus the initial condition reads $\sigma_{\mathbf{k}}^x(0) = \Delta/\omega_{\mathbf{k}}$ and $\sigma_{\mathbf{k}}^z(0) = -\epsilon_{\mathbf{k}}/\omega_{\mathbf{k}}$ with $\omega_{\mathbf{k}} = 2(\epsilon_{\mathbf{k}}^2 + \Delta^2)^{1/2}$. The linearized equations of motion are

$$\partial_t \delta\sigma_{\mathbf{k}}^x(t) = -2\epsilon_{\mathbf{k}} \delta\sigma_{\mathbf{k}}^y(t), \quad (16.16)$$

$$\begin{aligned} \partial_t \delta\sigma_{\mathbf{k}}^y(t) &= 2\epsilon_{\mathbf{k}} \delta\sigma_{\mathbf{k}}^x(t) + 2\Delta \delta\sigma_{\mathbf{k}}^z(t) \\ &+ \frac{1}{\omega_{\mathbf{k}}} \left[e^2 \Delta \sum_{ij} \partial_{k_i} \partial_{k_j} \epsilon_{\mathbf{k}} A_i(t) A_j(t) - 2\epsilon_{\mathbf{k}} \delta\Delta(t) \right], \end{aligned} \quad (16.17)$$

$$\partial_t \delta\sigma_{\mathbf{k}}^z(t) = -2\Delta \delta\sigma_{\mathbf{k}}^y(t). \quad (16.18)$$

Note that $\partial_t(\Delta\delta\sigma_{\mathbf{k}}^x - \epsilon_{\mathbf{k}}\delta\sigma_{\mathbf{k}}^z) = 0$. From this, along with the initial condition $\delta\boldsymbol{\sigma}_{\mathbf{k}}(0) = 0$, it turns out that the relation $\Delta\delta\sigma_{\mathbf{k}}^x = \epsilon_{\mathbf{k}}\delta\sigma_{\mathbf{k}}^z$ holds all the time, which helps us to reduce the number of the equations.

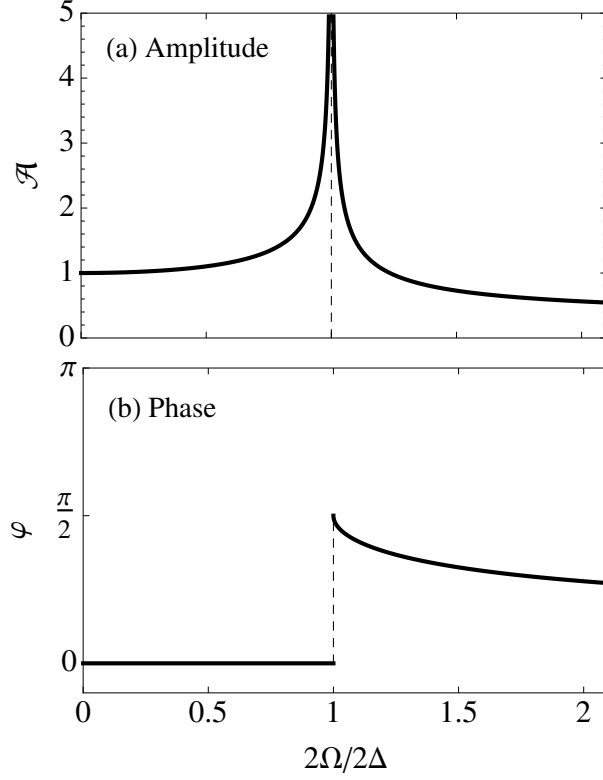


Figure 16.4: (a) The amplitude $\mathcal{A}$ and (b) phase shift φ of the 2Ω oscillation of the superconducting order parameter $\delta\Delta(t)$ [Eq. (16.24)] against $2\Omega/2\Delta$.

We solve the equations by a Laplace transformation, $\mathcal{L}[\delta\Delta(t)](s) =: \delta\Delta(s)$, etc. Let us call the direction of the electric field x . Then we have $\sum_{ij} \partial_{k_i} \partial_{k_j} \epsilon_{\mathbf{k}} A_i(t) A_j(t) = \partial_{k_x}^2 \epsilon_{\mathbf{k}} |\mathbf{A}(t)|^2$. When the crystallographic directions are equivalent, we have $\partial_{k_x}^2 \epsilon_{\mathbf{k}} \rightarrow d^{-1} \nabla_{\mathbf{k}}^2 \epsilon_{\mathbf{k}}$ with d the spatial dimension. If the band dispersion is isotropic with $\epsilon_{\mathbf{k}} = \epsilon(|\mathbf{k}|)$, we expand $\epsilon_{\mathbf{k}}$ around the Fermi wave number k_F as $\epsilon_{\mathbf{k}} = \sum_{n=1}^{\infty} c_n (|\mathbf{k}| - k_F)^n$. With this, we can define a series expansion,

$$d^{-1} \nabla_{\mathbf{k}}^2 \epsilon_{\mathbf{k}} = \alpha_0 + \alpha_1 \epsilon_{\mathbf{k}} + \alpha_2 \epsilon_{\mathbf{k}}^2 + \cdots, \quad (16.19)$$

where $\alpha_0 = 2c_2 d^{-1} + c_1(1 - d^{-1})k_F^{-1}$, $\alpha_1 = c_1^{-1}[6c_3 d^{-1} + (1 - d^{-1})(2c_2 k_F^{-1} - c_1 k_F^{-2})]$, etc., with each coefficient $\alpha_n \sim O(\epsilon_F^{1-n})$. Since the α_0 term just gives a trivial phase $\exp(i\alpha_0 e^2 \int_0^t dt' A(t')^2)$ to $\Delta(t)$, which can be gauged out, the α_1 term provides the leading contribution around the Fermi surface (with $\omega_D \ll \epsilon_F$). For anisotropic band structures, the same expansion is still sometimes possible. For instance, the d -dimensional cubic lattice (with cosine bands $\epsilon_{\mathbf{k}} = -2 \sum_i \cos k_i - \epsilon_F$) has $d^{-1} \nabla_{\mathbf{k}}^2 \epsilon_{\mathbf{k}} = \alpha_0 + \alpha_1 \epsilon_{\mathbf{k}}$ with $\alpha_0 = -\epsilon_F d^{-1}$ and $\alpha_1 = -d^{-1}$.

Thus, in most cases of our interest, we arrive at

$$\frac{\delta\Delta(s)}{\alpha_1 e^2 A^2 \Delta} = \frac{\Omega^2}{s(s^2 + 4\Omega^2)} \left[1 - \frac{1}{\lambda(s^2 + 4\Delta^2) F(s)} \right], \quad (16.20)$$

where $A = |\mathbf{A}|$ and

$$\begin{aligned} F(s) &= \int_{-\infty}^{\infty} d\epsilon \frac{1}{2\sqrt{\epsilon^2 + \Delta^2}(s^2 + 4\epsilon^2 + 4\Delta^2)} \\ &= \frac{1}{s\sqrt{s^2 + 4\Delta^2}} \sinh^{-1} \left(\frac{s}{2\Delta} \right). \end{aligned} \quad (16.21)$$

In the above, we have replaced the range of integration from $\int_{-\omega_D}^{\omega_D}$ to $\int_{-\infty}^{\infty}$, which is allowed in the BCS regime ($\omega_D \gg \Delta$). $F(s)$ can be analytically continued on the complex plane, where

branch cuts $\mathcal{B}_{\pm}(\delta) = \{\pm 2i\Delta \pm ire^{\pm i\delta} | r \in [0, \infty)\}$ with δ small but nonzero are introduced (Fig. 16.3).

To obtain $\delta\Delta(t)$ with an inverse Laplace transformation, we need to evaluate a Bromwich integral,

$$I(t) = \frac{1}{2\pi i} \int_{\gamma-i\infty}^{\gamma+i\infty} ds e^{st} \frac{\Omega^2}{s(s^2 + 4\Omega^2)(s^2 + 4\Delta^2)F(s)}, \quad (16.22)$$

where $\gamma \in \mathbb{R}$ is taken to be larger than any of the real parts of the poles in the integrand. There are three first-order poles at $s = 0, \pm 2i\Omega$ and two branching points at $s = \pm 2i\Delta$ (Fig. 16.3) in the integrand, where $s = \pm 2i\Delta$ corresponds to the Higgs amplitude mode, while $s = \pm 2i\Omega$ to the forced precession of the Anderson pseudospins driven by the electric field. As one changes Ω , the poles merge with the branching points at $2\Omega = 2\Delta$, which causes a *resonance* between the forced pseudospin precession and the Higgs mode.

To make it more explicit, we evaluate the integral (16.22) by taking the contour as depicted in Fig. 16.3, which surrounds the three poles but avoids the branch cuts. This kind of contour is often used to calculate similar integrals (see, e.g., Ref. [Volkov,Kogan (1973)]). We take $\delta > 0$ so that the contours along the branch cuts $\mathcal{B}_{\pm}(\delta)$ do not touch the poles $s = \pm 2i\Omega$ when $\Omega > \Delta$. Since the contributions from infinity vanish, we are left with the residues of the poles and the line integrals ($\mathcal{C}_{\pm}$ in Fig. 16.3 and their Hermitian conjugates) along the branch cuts. The asymptotic behavior of the integrals $\mathcal{C}_{\pm}$ for $t \rightarrow \infty$ is evaluated by the saddle-point method. Finally we end up with long-time asymptotic forms of the order parameter,

$$\frac{\delta\Delta(t)}{\alpha_1 e^2 A^2 \Delta} \sim \frac{1}{4\lambda} \left[\frac{2}{\pi^{3/2}} \frac{\Omega^2}{\Omega^2 - \Delta^2} \frac{1}{\sqrt{\Delta t}} \cos\left(2\Delta t + \frac{\pi}{4}\right) - 1 \right] + \frac{1 - \cos 2\Omega t}{4} \quad (16.23)$$

$$+ \frac{1}{4\lambda} \times \begin{cases} \frac{\Omega}{\sqrt{\Delta^2 - \Omega^2}} \frac{\cos 2\Omega t}{\sin^{-1}\left(\frac{\Omega}{\Delta}\right)} & \Omega < \Delta \\ \frac{\Omega}{\sqrt{\Omega^2 - \Delta^2}} \frac{\cos(2\Omega t - \varphi)}{\sqrt{[\cosh^{-1}\left(\frac{\Omega}{\Delta}\right)]^2 + \left(\frac{\pi}{2}\right)^2}} & \Omega > \Delta \end{cases}, \quad (16.24)$$

where φ is the phase shift given by

$$\varphi = \tan^{-1} \left(\frac{\pi/2}{\cosh^{-1}\left(\frac{\Omega}{\Delta}\right)} \right). \quad (16.25)$$

The first term in Eq. (16.24) can be interpreted as the Higgs amplitude mode induced by an effective change of the interaction parameter due to the ac field, $U \rightarrow U_{\text{eff}} = (1 - \frac{1}{2}\alpha_1 e^2 A^2)U$. [Tsuji et al.(2009)] Indeed, it approaches the result for the interaction-quench problem [Barankov and Levitov(2006), Yuzbashyan and Dzero(2006)] in the limit of $\Omega \rightarrow \infty$. The Higgs mode is amplified by the ac electric field around $2\Omega = 2\Delta$. The term decays algebraically as $t^{-1/2}$, [Volkov,Kogan (1973)] which suggests that the Higgs mode effectively has an infinite lifetime within the BCS approximation.

In the long-time limit, the constant term and the term oscillating with frequency 2Ω survive. The constant term in $\delta\Delta(t)$ is proportional to $\alpha_1(1 - \lambda^{-1})$, which implies, intriguingly, that we can attain an *amplification* of superconductivity on time average when this term is positive. The 2Ω oscillation term represents the APR. If we write the last term in Eq. (16.24) as $\frac{1}{4\lambda} \mathcal{A} \cos(2\Omega t - \varphi)$, the amplitude $\mathcal{A}$ and the phase shift φ are universal functions that depend only on the ratio $2\Omega/2\Delta$ (Fig. 16.4). The amplitude $\mathcal{A}$ diverges as $|2\Omega - 2\Delta|^{-1/2}$ at $2\Omega = 2\Delta$ (resonance condition). It clearly differs from the result of the time-dependent GL (16.8), $|2\Omega - 2\Delta|^{-1}$. The reduction of the power from 1 to 1/2 signifies that the Higgs mode is a bit less stable, where

each pseudospin precession gradually dephases. Physically we can interpret this as coming from Landau damping; that is, the collective mode decays into individual quasiparticle excitations even in the collision-less equation (16.15). An anomaly is also found in the phase shift φ : for $2\Omega < 2\Delta$, φ is locked to zero, i.e., the 2Ω oscillation of the order parameter is in-phase with $\mathbf{E}(t)^2$. As soon as 2Ω exceeds 2Δ , the φ discontinuously jumps to $\pi/2$ and starts to drift (Fig. 16.4). Along with the order-parameter oscillation, the pseudospin itself continues to precess around the axis parallel to $\boldsymbol{\sigma}_{\mathbf{k}}(0)$, with two modes of frequencies $\omega_{\mathbf{k}}$ and 2Ω surviving in $t \rightarrow \infty$ (the former of which dephases). By numerically simulating Eq. (16.15), we also confirmed that APR generally occurs for finite-temperature initial states and for pulsed electric fields that contain large enough number of oscillation cycles.

APR appears in various physical quantities. What is readily accessible experimentally is the electric current,

$$\mathbf{j} = e \sum_{\mathbf{k}, \sigma} \mathbf{v}_{\mathbf{k}-e\mathbf{A}(t)} c_{\mathbf{k}\sigma}^\dagger c_{\mathbf{k}\sigma} \quad (16.26)$$

($\mathbf{v}_{\mathbf{k}} = \nabla_{\mathbf{k}} \epsilon_{\mathbf{k}}$ is the group velocity). The current is expressed in the pseudospin notation as $\mathbf{j} = e \sum_{\mathbf{k}} [\mathbf{v}_{\mathbf{k}-e\mathbf{A}(t)} - \mathbf{v}_{\mathbf{k}+e\mathbf{A}(t)}] \sigma_{\mathbf{k}}^z + \frac{e}{2} \sum_{\mathbf{k}\sigma} [\mathbf{v}_{\mathbf{k}-e\mathbf{A}(t)} + \mathbf{v}_{\mathbf{k}+e\mathbf{A}(t)}] c_{\mathbf{k}\sigma}^\dagger c_{\mathbf{k}\sigma}$. If we expand it in $\mathbf{A}(t)$, the linear response is given by $\mathbf{j}^{(1)} = -2\alpha_1 e^2 \mathbf{A}(t) \sum_{\mathbf{k}} \epsilon_{\mathbf{k}} \sigma_{\mathbf{k}}^z(0) + e \sum_{\mathbf{k}\sigma} \mathbf{v}_{\mathbf{k}} c_{\mathbf{k}\sigma}^\dagger c_{\mathbf{k}\sigma}$, which is irrelevant to APR. In fact, the linear-response optical conductivity does not show any divergence for $\Omega \neq 0$. [108] The leading term that reflects the change of the order parameter is the third-harmonic generation (THG),

$$\mathbf{j}^{(3)}(t) = -2\alpha_1 e^2 \Delta U^{-1} \delta\Delta(t) \mathbf{A}(t), \quad (16.27)$$

where we have used $\epsilon_{\mathbf{k}} \delta\sigma_{\mathbf{k}}^z = \Delta \delta\sigma_{\mathbf{k}}^x$. The consequence is remarkable: although the frequency Ω is below the energy gap, we do obtain the colossal nonlinear response due to divergence of $\delta\Delta(t)$. It may be used as an efficient THz harmonic emitter.

16.4 Response function for Anderson pseudospin resonance

To reinforce our picture for APR phenomenon, we can approach it from an alternative, diagrammatic point of view. This allows one to decompose the THG susceptibility into the bare and vertex-correction diagrams, each of which contains individual and collective excitations, respectively. Thus we can unambiguously distinguish the Higgs mode from quasiparticle excitations that are degenerate at the superconducting gap energy. To this end, we take the Nambu-Gor'kov Green's function defined by

$$\hat{G}_{\mathbf{k}}(t, t') = \begin{pmatrix} -i\langle \mathcal{T} c_{\mathbf{k}\uparrow}(t) c_{\mathbf{k}\uparrow}^\dagger(t') \rangle & -i\langle \mathcal{T} c_{\mathbf{k}\uparrow}(t) c_{-\mathbf{k}\downarrow}(t') \rangle \\ -i\langle \mathcal{T} c_{-\mathbf{k}\downarrow}^\dagger(t) c_{\mathbf{k}\uparrow}^\dagger(t') \rangle & -i\langle \mathcal{T} c_{-\mathbf{k}\downarrow}^\dagger(t) c_{-\mathbf{k}\downarrow}(t') \rangle \end{pmatrix},$$

where $\mathcal{T}$ represents the time ordering. With this, the gap function is expressed as

$$\Delta(t) = -\frac{i}{2} U \sum_{\mathbf{k}} \text{Tr} \left[\tau_1 \hat{G}_{\mathbf{k}}^<(t, t) \right], \quad (16.28)$$

where $\hat{G}_{\mathbf{k}}^<$ is the lesser Green's function, and $\Delta(t)$ is assumed to be real.

Now we take variations of both sides of Eq. (16.28) with respect to the external field $\mathbf{A}(t)$. In this section, we consider the monochromatic wave $\mathbf{A}(t) = \mathbf{A} e^{-i\Omega t}$. Since the leading change of the order parameter $\delta\Delta(t) = \delta\Delta e^{-2i\Omega t}$ is the second order in $\mathbf{A}(t)$, we have

$$\delta\Delta = \delta_A^2 \Delta = -\frac{i}{2} U \sum_{\mathbf{k}} \text{Tr} \left[\tau_1 \delta_A^2 \hat{G}_{\mathbf{k}}^< \right], \quad (16.29)$$

where δ_A represents the functional derivative with respect to $\mathbf{A}$. In the BCS theory, the Nambu-Gor'kov Green's function is given by the Dyson equation $\hat{G}_{\mathbf{k}} = (i\partial_t - \hat{\xi}_{\mathbf{k}} + \Delta\tau_1)^{-1}$ with $\hat{\xi}_{\mathbf{k}} \equiv \xi_{\mathbf{k}} - e\mathbf{A}\tau_3$. Hence the variation of the order parameter reads

$$\begin{aligned} \delta_A^2 \Delta = & -iU \sum_{\mathbf{k}} \text{Tr} \left[\tau_1 \hat{G}_{\mathbf{k}} (\delta_A \hat{\xi}_{\mathbf{k}}) \hat{G}_{\mathbf{k}} (\delta_A \hat{\xi}_{\mathbf{k}}) \hat{G}_{\mathbf{k}} \right]^< \\ & - \frac{i}{2} U \sum_{\mathbf{k}} \text{Tr} \left[\tau_1 \hat{G}_{\mathbf{k}} (\delta_A^2 \hat{\xi}_{\mathbf{k}} - \delta_A^2 \Delta \tau_1) \hat{G}_{\mathbf{k}} \right]^<. \end{aligned} \quad (16.30)$$

Here the lesser component of the products of the Green's functions should be understood by Langreth's rule [e.g., $(GG)^< = G^R G^< + G^< G^A$]. Equation (16.30) determines $\delta\Delta = \delta_A^2 \Delta$ self-consistently. This is diagrammatically represented in the first line of Fig. 16.5. One can show that the first term on the right hand side of Eq. (16.30) vanishes in the BCS theory. By solving Eq. (16.30) in the frequency domain, we end up with

$$\delta\Delta = \frac{\frac{i}{2} \alpha_1 U \sum_{\mathbf{k}, \omega} \xi_{\mathbf{k}} \text{Tr} \left[\tau_1 \hat{G}_{\mathbf{k}}(\omega + 2\Omega) \tau_3 \hat{G}_{\mathbf{k}}(\omega) \right]^<}{1 - \frac{i}{2} U \sum_{\mathbf{k}, \omega} \text{Tr} \left[\tau_1 \hat{G}_{\mathbf{k}}(\omega + 2\Omega) \tau_1 \hat{G}_{\mathbf{k}}(\omega) \right]^<}. \quad (16.31)$$

Note that $\langle \tau_1 \hat{G} \tau_1 \hat{G} \rangle$ appearing in the denominator is the dynamical pair-pair correlation function.

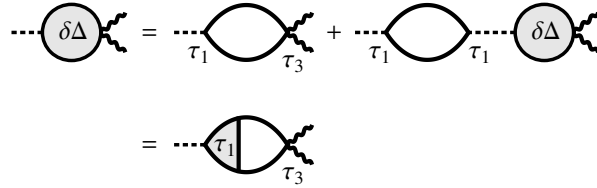


Figure 16.5: The diagrammatic representation of the self-consistent Eq. (16.30) for $\delta\Delta$ and its relation to the τ_1 vertex. The wavy and dashed lines represent the gauge field $\mathbf{A}$ and the interaction vertices, respectively.

The amplitude of the 2Ω oscillation of $\delta\Delta$ diverges (i.e., APR occurs) when the denominator of Eq. (16.31) vanishes due to fluctuations in the τ_1 channel. Thus the resonance sensitively reflects the structure of the pair-pair correlation function. We should emphasize that the τ_1 fluctuation appears without considering self-energy corrections beyond the mean-field BCS formalism. Namely, the τ_1 fluctuation is already present in the response to electromagnetic fields in the BCS regime before we further include fluctuations by, e.g., the random phase approximation.

As indicated in the second line of Fig. 16.5, $\delta\Delta$ can also be expressed in terms of the τ_1 vertex. Formally, the Higgs mode is defined as a pole of the τ_1 vertex. Therefore, the divergence of $\delta\Delta$ can indeed be rephrased as a resonance with the Higgs mode.

The explicit calculation for the correlation functions within the BCS theory enables one to write down $\delta\Delta$ (16.31) analytically as

$$\delta\Delta = \frac{1}{2} \alpha_1 e^2 A^2 \Delta \left[\frac{1}{\lambda R(\Omega, T)} - 1 \right], \quad (16.32)$$

where the resonance function $R(\Omega, T)$ is given by

$$\begin{aligned} R(\Omega, T) = & \mathcal{P} \int_{\Delta}^{\infty} d\omega \frac{\Omega^2 - \Delta^2}{(\Omega^2 - \omega^2) \sqrt{\omega^2 - \Delta^2}} \tanh \left(\frac{\omega}{2T} \right) \\ & - \frac{i\pi}{2} \theta(\Omega - \Delta) \frac{\sqrt{\Omega^2 - \Delta^2}}{\Omega} \tanh \left(\frac{\Omega}{2T} \right) \end{aligned} \quad (16.33)$$

with $\mathcal{P}$ denoting the principal value of the integral.

In the limit of $2\Omega \rightarrow 2\Delta$ (i.e., $\Omega \rightarrow \Delta$), $R(\Omega, T)$ has an asymptotic form of

$$R(\Omega, T) \sim \begin{cases} \frac{\pi}{2} \frac{\sqrt{\Delta^2 - \Omega^2}}{\Delta} \tanh\left(\frac{\Delta}{2T}\right) & \Omega \rightarrow \Delta - 0 \\ -\frac{i\pi}{2} \frac{\sqrt{\Omega^2 - \Delta^2}}{\Delta} \tanh\left(\frac{\Delta}{2T}\right) & \Omega \rightarrow \Delta + 0. \end{cases} \quad (16.34)$$

Hence $|\delta\Delta|$ diverges in this limit as

$$|\delta\Delta| \sim \frac{\alpha_1 e^2 A^2 \Delta^2}{\pi \lambda \tanh\left(\frac{\Delta}{2T}\right)} \frac{1}{|\Omega^2 - \Delta^2|^{1/2}}. \quad (16.35)$$

Remarkably, the divergence persists at arbitrary temperatures $T < T_c$ with the fixed critical exponent $\frac{1}{2}$, which indicates that the APR is robust against thermal fluctuations.

In the zero-temperature limit, $R(\Omega, T)$ is reduced to

$$R(\Omega, T) \xrightarrow{T \rightarrow +0} \begin{cases} \frac{\sqrt{\Delta^2 - \Omega^2}}{\Omega} \sin^{-1}\left(\frac{\Omega}{\Delta}\right) & \Omega < \Delta \\ \frac{\sqrt{\Omega^2 - \Delta^2}}{\Omega} \left[\cosh^{-1}\left(\frac{\Omega}{\Delta}\right) - i\frac{\pi}{2} \right] & \Omega > \Delta. \end{cases} \quad (16.36)$$

Plugging this into Eq. (16.32), one can see that it precisely reproduces the result (16.24) derived in the previous section [note that the seeming difference of the factor $\frac{1}{2}$ is due to the assumption of $\mathbf{A}(t) = \mathbf{A}e^{-i\Omega t}$ in the present section and $\mathbf{A}(t) = \mathbf{A} \sin \Omega t$ in the previous section].

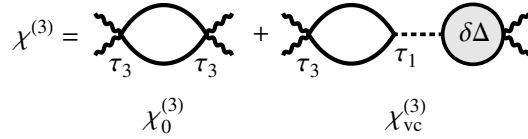


Figure 16.6: Feynman diagram for the THG susceptibility of superconductors. The first and second terms on the right-hand side correspond to $\chi_0^{(3)}$ and $\chi_{vc}^{(3)}$, respectively.

In the language of the Green's function, the current is given by

$$j(t) = ie \sum_{\mathbf{k}} \text{Tr} \left[\hat{v}_{\mathbf{k}} \hat{G}_{\mathbf{k}}^<(t, t) \right], \quad (16.37)$$

where $\hat{v}_{\mathbf{k}} \equiv v_{\mathbf{k} - e\mathbf{A}(t)\tau_3}$. If we focus on the THG response that is relevant to APR, we can take the third derivative with respect to $\mathbf{A}(t)$ to obtain

$$\begin{aligned} j^{(3)} &= ie \sum_{\mathbf{k}} \text{Tr} \left[(\delta_A^3 \hat{v}_{\mathbf{k}}) \hat{G}_{\mathbf{k}}^< \right] \\ &+ ie \sum_{\mathbf{k}} \text{Tr} \left[(\delta_A \hat{v}_{\mathbf{k}}) \hat{G}_{\mathbf{k}} (\delta_A^2 \hat{\xi}_{\mathbf{k}} - \delta_A^2 \Delta \tau_1) \hat{G}_{\mathbf{k}} \right]. \end{aligned} \quad (16.38)$$

Here we have used the fact that odd-order derivatives of $\hat{G}_{\mathbf{k}}^<(t, t)$ vanish as indicated by the pseudospin analysis [see Eqs. (16.16)-(16.18)].

The first term in Eq. (16.38) is of higher order in (ω_D/ϵ_F) than the other terms, so that it is negligible. Then the leading contributions to $j^{(3)}$ are the second and third terms, which we write as

$$j^{(3)} = \chi_{\text{tot}}^{(3)} A^3 = (\chi_0^{(3)} + \chi_{vc}^{(3)}) A^3. \quad (16.39)$$

$\chi_{\text{tot}}^{(3)}$ is the total THG susceptibility, which comprises the bare susceptibility $\chi_0^{(3)}$ and vertex correction $\chi_{\text{vc}}^{(3)}$. The Feynman diagram for these is depicted in Fig. 16.6. Here one can single out the contribution of the 2Ω collective oscillation of the order parameter to the THG signal, which is $\chi_{\text{vc}}^{(3)}$. In other words, we can distinguish the effect of quasiparticle excitations from the contribution of the Higgs mode.

We can evaluate the components of the susceptibility explicitly in the BCS theory with the function $R(\Omega, T)$ (16.33) as

$$\chi_0^{(3)} = \alpha_1^2 e^4 \frac{\Delta^2}{U} [\lambda R(\Omega, T) - 1], \quad (16.40)$$

$$\chi_{\text{vc}}^{(3)} = -\alpha_1^2 e^4 \frac{\Delta^2}{U} \frac{[\lambda R(\Omega, T) - 1]^2}{\lambda R(\Omega, T)}. \quad (16.41)$$

Taking the sum of the two susceptibilities, we obtain the total contribution,

$$\chi_{\text{tot}}^{(3)} = \chi_0^{(3)} + \chi_{\text{vc}}^{(3)} = -\alpha_1^2 e^4 \frac{\Delta^2}{U} \left[\frac{1}{\lambda R(\Omega, T)} - 1 \right]. \quad (16.42)$$

With Eqs. (16.32) and (16.42), we reproduce the previous relation (16.27). Note that the term $\lambda R(\Omega, T)$ appears in the denominator for $\chi_{\text{tot}}^{(3)}$ and in the numerator for $\chi_0^{(3)}$ in the opposite ways.

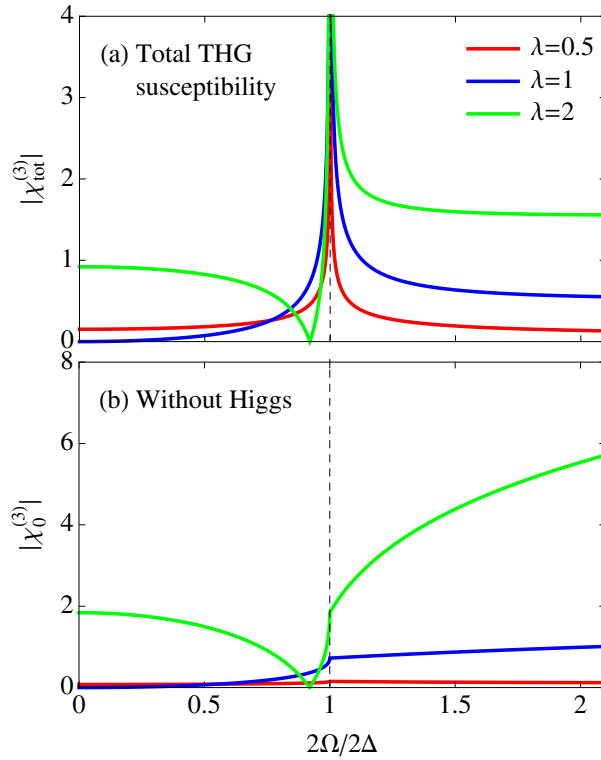


Figure 16.7: (Color online) The amplitude of (a) the full and (b) bare THG susceptibilities for superconductors at $T = 0$ with $\lambda = 0.5, 1, 2$ in units of $\alpha_1^2 e^4 \omega_D^2 D(\epsilon_F)$.

In Fig. 16.7, we plot $\chi_{\text{tot}}^{(3)}$ along with $\chi_0^{(3)}$ for several values of λ at $T = 0$. When we change λ , we evaluate the gap by $\Delta = \omega_D / \sinh(1/\lambda)$. We can see that $\chi_{\text{tot}}^{(3)}$ diverges in a similar manner as $\delta\Delta$ (Fig. 16.4) at $2\Omega = 2\Delta$, while $\chi_0^{(3)}$ only shows kink structures. This endorses that the resonance peak is indeed a manifestation of the effect of the Higgs mode, and cannot be explained by quasiparticle excitations or pair breaking contained in $\chi_0^{(3)}$. As one increases λ ,

the amplitude of the divergence becomes larger. For $\lambda > 1$, $\chi_{\text{tot}}^{(3)}$ and $\chi_0^{(3)}$ vanish at the value of $2\Omega/2\Delta$ at which $\lambda R(\Omega, T) = 1$ is satisfied. Consequently, the susceptibility spectrum shows a sharp dip structure. This can be exploited to experimentally discern whether $\lambda > 1$ or not. We also notice that the shape of the resonance peak is significantly asymmetric about $2\Omega = 2\Delta$, which becomes more prominent for larger λ . The asymmetry originates from the mixing of the collective mode (discrete level) with quasiparticle excitations (continuous levels). While this is reminiscent of the Fano resonance, the form of the resonance function is not identical with the Fano form.

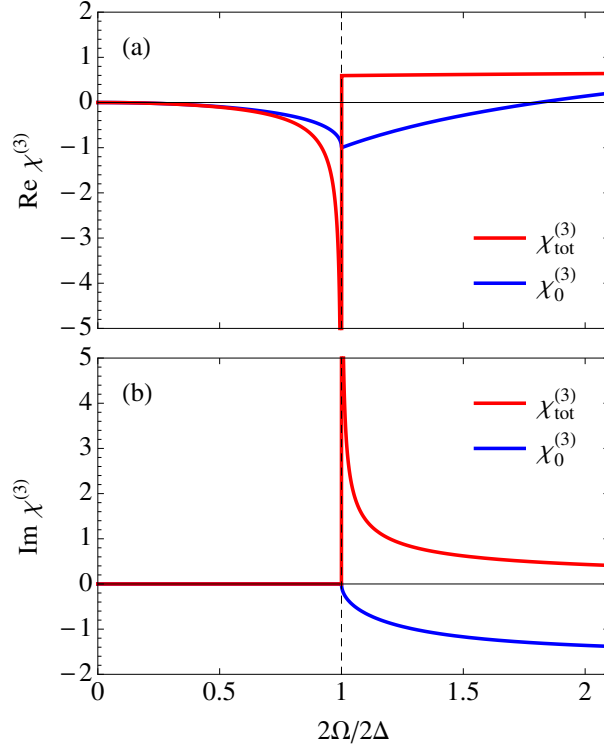


Figure 16.8: (Color online) (a) The real and (b) imaginary parts of the THG susceptibility for superconductors at $T = 0$ with $\lambda = 1$ in units of $\alpha_1^2 e^4 \omega_D^2 D(\epsilon_F)$.

If we look at the real and imaginary parts of the THG susceptibility in Fig. 16.8, the spectral features are again very different between $\chi_{\text{tot}}^{(3)}$ and $\chi_0^{(3)}$. $\text{Re } \chi_{\text{tot}}^{(3)}$ and $\text{Im } \chi_{\text{tot}}^{(3)}$ diverge as $\Omega \rightarrow \Delta - 0$ and $\Omega \rightarrow \Delta + 0$, respectively, whereas $\text{Re } \chi_0^{(3)}$ remains finite and $\text{Im } \chi_0^{(3)}$ vanishes in these limits. Both $\chi_{\text{tot}}^{(3)}$ and $\chi_0^{(3)}$ have zero imaginary parts at $\Omega < \Delta$. This simply reflects that photo-absorption is not allowed with frequencies below the energy gap even in the nonlinear-response regime.

If we turn to the temperature dependence of the THG susceptibility in Fig. 16.9, $\chi_{\text{tot}}^{(3)}$ diverges for temperatures $T < T_c$, similarly to the behavior of $\delta\Delta$ (16.35). The shape of the resonance peak does not change significantly against temperature. The insensitivity of the THG signal against temperature should facilitate experiments where a scan of the pump frequency Ω is difficult: one can instead scan temperature to change Δ for a fixed Ω . In Ref. [46], the THG resonance peak was in fact mapped out in this way.

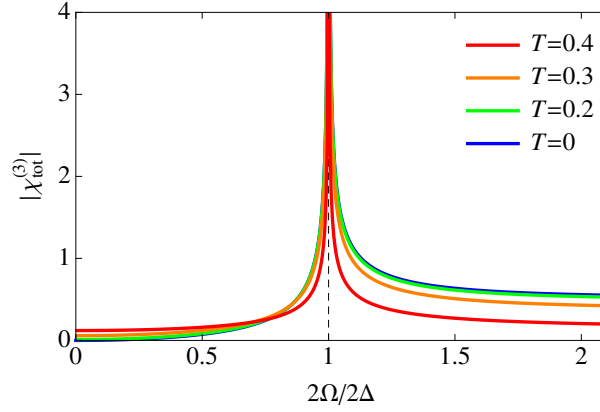


Figure 16.9: (Color online) The temperature dependence of the THG susceptibility for superconductors with $\lambda = 1$ in units of $\alpha_1^2 e^4 \omega_D^2 D(\epsilon_F)$. Here, $T_c = 0.446$.

16.5 Effect of electron-electron scattering

So far, the argument has been based on the pairing Hamiltonian (16.9), which has the long-range interaction in real space. For more realistic models of superconductivity with short-range interactions, the analysis above is considered to be a static mean-field approximation, whose validity is restricted to the weak-coupling regime. Furthermore, the equation of motion (16.15) does not involve thermalization processes, which correspond to changes in the pseudospin length $|\sigma_{\mathbf{k}}|$ due to correlation effects. Thus let us go beyond the static mean field by considering the attractive Hubbard model with a driving ac field,

$$H_{\text{Hubbard}} = \sum_{\mathbf{k}\sigma} \epsilon_{\mathbf{k}-e\mathbf{A}(t)} c_{\mathbf{k}\sigma}^\dagger c_{\mathbf{k}\sigma} - U \sum_i c_{i\uparrow}^\dagger c_{i\uparrow} c_{i\downarrow}^\dagger c_{i\downarrow}, \quad (16.43)$$

where i labels the lattice sites and U is an attractive Hubbard interaction. We take, as an example, a one-dimensional dispersion $\epsilon_k = -2 \cos k$ with the bandwidth $W = 4$ and $\alpha_1 = -1$ (later in this section we also consider an infinite-dimensional lattice). We calculate the time evolution by means of the nonequilibrium dynamical mean-field theory (DMFT) [Freericks et al.(2006), Aoki et al.(2014), not()], which is extended here to the Nambu formalism for treating superconductors. For an impurity solver for DMFT, we employ the third-order perturbation theory [Tsuji and Werner(2013)], which is supposed to be reliable in the region $U < W$. The system is set at half filling with $U = 3.5$, which belongs to a strong-coupling regime ($2\Delta_{T=0}/T_c \approx 5.0$ well above the BCS value).

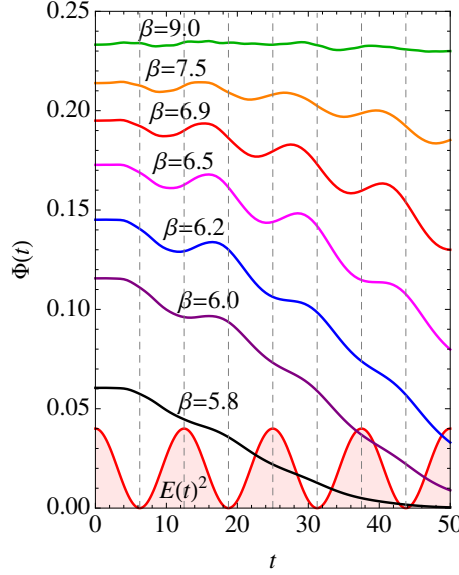


Figure 16.10: (Color online) Temporal evolution of the superconducting order parameter $\Phi(t)$ calculated with the nonequilibrium DMFT for the attractive Hubbard model with the 1D density of states at half filling driven by an ac field with $U = 3.5$, $A = 0.15$, and $\Omega = 2\pi/25$ for several temperatures (β^{-1}) for the initial states. The sinusoidal curve represents $\mathbf{E}(t)^2 \propto \cos^2 \Omega t$. Dashed lines are a guide to the eye.

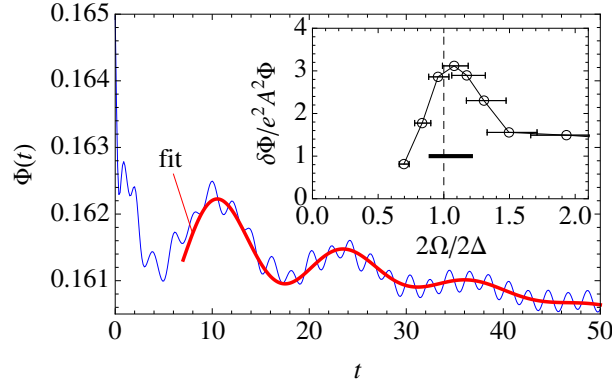


Figure 16.11: (Color online) Temporal evolution of the order parameter $\Phi(t)$ after a quench $U \rightarrow U - \delta U$ at $t = 0$ with $U = 3.5$ and $\delta U = 0.01$ in the attractive Hubbard model at $\beta = 6.4$. The rapid oscillation comes from a band-edge effect, while the slower one corresponds to the Higgs mode. Thick (red) curve is a fit (see text). Inset: The amplitude of the 2Ω oscillating component of the order parameter $\delta\Phi$ for the attractive Hubbard model driven by an ac field with $U = 3.5$, $\Omega = 2\pi/25$, and various Δ . The bar shows the width estimated from the lifetime of the Higgs mode.

The time evolution of the local superconducting order parameter, $\Phi(t) = \langle c_{\uparrow}^{\dagger} c_{\downarrow}^{\dagger} \rangle$, for various initial temperatures (β^{-1}) is shown in Fig. 16.10. With increased total energy due to the continuous excitation, the overall value of the order parameter gradually decreases. On top of that, the coherent oscillation of the order parameter with frequency 2Ω emerges [with the same oscillation period of $\mathbf{E}(t)^2$ shown in Fig. 16.10]. The oscillation is particularly enhanced around $\beta = 6.5$, and becomes invisible for $\beta = 9.0$. The phase-shift anomaly is not clearly observed in this interaction regime. We evaluate the energy gap 2Δ in equilibrium from the single-particle spectral function $A(\omega)$, which is calculated by Fourier transformation of the real-time simulation. If we measure the amplitude of the 2Ω oscillation of the order parameter, $\delta\Phi$, at the

third cycle, we can clearly see in the inset of Fig. 16.11 that a resonance peak indeed emerges at $2\Omega = 2\Delta$ (the error bars represent inaccuracy in measuring Δ). The peak position corresponds to $\beta \approx 6.4$. The result indicates that APR indeed exists beyond the static mean-field level.

However, we do notice a deviation from the BCS result; i.e., the resonance has a finite width (the inset of Fig. 16.11). There are several factors that determine the resonance width. Besides extrinsic experimental factors such as the limited measurement time scale or energy dissipation to external environment (which is absent in our calculations), one intrinsic factor is the finite lifetime τ of the Higgs amplitude mode, which can decay into individual excitations [note that Higgs does not decay into the Nambu-Goldstone (NG) mode in charged superconductors, since the energy of the NG mode is lifted to the plasma frequency, at least away from the critical regime near $T = T_c$]. If the Higgs mode decays exponentially, the poles $s = \pm 2i\Delta$ acquire a real part on the complex plane (Fig. 16.3), and are thus prevented from meeting the branching points $\pm 2i\Omega$, resulting in broadening of the resonance peak. We can numerically evaluate the decay rate by generating the Higgs mode at $\beta = 6.4$ with a small perturbation (here we use an interaction quench [Tsuji et al.(2013)]), where $\Phi(t)$ is fitted with $\Phi_0 e^{-t/\tau} \cos(2\Delta t + \theta)$ on top of a linear drift (Fig. 16.11). A rapid oscillation in Fig. 16.11 comes from the divergence of the 1D density of states at band edges, and is irrelevant to the Higgs mode. From the derived τ , we estimate the resonance width as indicated by the bar in the inset of Fig. 16.11, which roughly coincides with the peak width of APR with the background subtracted (Fig. 16.11).

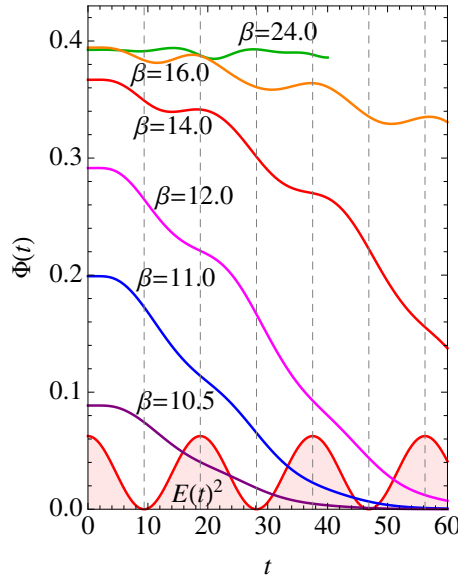


Figure 16.12: (Color online) Temporal evolution of the superconducting order parameter $\Phi(t)$ calculated with the nonequilibrium DMFT for the attractive Hubbard model with the infinite-dimensional (Gaussian) density of states at half filling driven by the ac field with $U = 2.25$, $A = 0.2$, and $\Omega = 2\pi/37.5$, for several temperatures (β^{-1}) for the initial states. The sinusoidal curve represents $\mathbf{E}(t)^2 \propto \cos^2 \Omega t$. Dashed lines are a guide to the eye.

While we have applied the nonequilibrium DMFT to a system with 1D density of states for simplicity, we can actually confirm that the result does not change qualitatively for the infinite-dimensional hypercubic lattice with the Gaussian density of states $D(\epsilon) = e^{-\epsilon^2}/\sqrt{\pi}$, where the DMFT formalism is no longer an approximation but becomes exact. Let us consider the hypercubic lattice with the electric field applied along the diagonal direction, $\mathbf{A}(t) = A(t)(1, 1, 1, \dots)$. The energy dispersion reads[Turkowski and Freericks(2005), Aoki et al.(2014)]

$$\epsilon_{\mathbf{k}-\mathbf{A}(t)} = \epsilon_{\mathbf{k}} \cos A(t) + \bar{\epsilon}_{\mathbf{k}} \sin A(t), \quad (16.44)$$

where $\epsilon_{\mathbf{k}} = -1/\sqrt{d} \sum_{i=1}^d \cos k_i$ and $\bar{\epsilon}_{\mathbf{k}} = -1/\sqrt{d} \sum_{i=1}^d \sin k_i$. This makes the momentum summation of the lattice Green's function in the nonequilibrium DMFT a double integral with respect to ϵ and $\bar{\epsilon}$. The double integral becomes computationally very heavy, especially in the present case where we have to keep track of the system evolving over a long enough interval to capture the slow order-parameter dynamics. To overcome the difficulty here we make use of the following formula,

$$\begin{aligned}
& \sum_{\mathbf{k}} G_{\mathbf{k}}(t, t') \\
&= \int d\epsilon d\bar{\epsilon} D(\epsilon) D(\bar{\epsilon}) (i\partial_t + \mu - \epsilon \cos A - \bar{\epsilon} \sin A - \Sigma)^{-1} \\
&= \frac{1}{2} \left[\int d\epsilon D(\epsilon) \left(i\partial_t + \mu - \epsilon \cos A - \frac{1}{\sqrt{2}} \sin A - \Sigma \right)^{-1} \right. \\
&\quad \left. + \int d\epsilon D(\epsilon) \left(i\partial_t + \mu - \epsilon \cos A + \frac{1}{\sqrt{2}} \sin A - \Sigma \right)^{-1} \right] \\
&\quad + O(A^3),
\end{aligned} \tag{16.45}$$

to reduce the double integral to a single one, where μ is the chemical potential, Σ is the self-energy, and we have used $\int d\bar{\epsilon} D(\bar{\epsilon}) \bar{\epsilon}^2 = 1/2$. The formula is valid up to the second order in A , which is sufficient for the present purpose, since our interest is in the order-parameter oscillation arising from the second-order nonlinear effect. An advantage of the above formula is that the first and second terms are in the form of the Green's function with $\bar{\epsilon}$ replaced by $\pm 1/\sqrt{2}$, so that the implementation is straightforward. We also remark that keeping the form of the Green's function is vital for maintaining the numerical stability. As an impurity solver for the nonequilibrium DMFT for the hypercubic lattice, here we employ the second-order iterative perturbation theory [Tsuji and Werner(2013)] to further reduce the computational cost. If we look at the time evolution of the order parameter in the infinite-dimensional hypercubic lattice in Fig. 16.12, the 2Ω oscillation of the order parameter is prominent around $\beta = 14.0$ and 16.0 , which is close to the resonance condition $2\Omega (= 4\pi/37.5 = 0.335) = 2\Delta (\approx 0.32, \text{evaluated from the nonequilibrium DMFT calculation for the spectral function})$. Away from this, the oscillation tends to be suppressed and the oscillation becomes incoherent. This shows that APR also occurs in the nonequilibrium DMFT calculation for the infinite-dimensional lattice where DMFT becomes exact. Thus the essential features of APR do not depend on a particular form of the density of states.

16.6 Summary

To summarize, we theoretically propose a phenomenon that may be called Anderson pseudospin resonance (APR) for a superconductor driven by an ac electric field, which is confirmed by solving the equation of motion analytically within the BCS approximation, and by solving the attractive Hubbard model via the nonequilibrium DMFT. APR can be distinguished from quasiparticle excitations or pair breaking processes near the superconducting gap energy by looking at the divergent enhancement of third harmonic generation. APR provides not only a new pathway of controlling superconductors, but also provides an avenue offering information about dynamical aspects of the order parameter and the Higgs mode in superconductors. Important future problems include whether APR occurs for other pairing symmetries such as the anisotropic d -wave pairing.

17 Amplitude-mode dynamics of polariton condensates by Brierley, Littlewood

Abstract

We study the stability of collective amplitude excitations in non-equilibrium polariton condensates. These excitations correspond to renormalized upper polaritons and to the collective amplitude modes of atomic gases and superconductors. They would be present following a quantum quench or could be created directly by resonant excitation. We show that uniform amplitude excitations are unstable to the production of excitations at finite wavevectors, leading to the formation of density-modulated phases. The physical processes causing the instabilities can be understood by analogy to optical parametric oscillators and the atomic Bose supernova.

In addition to the phase mode responsible for superconductivity and superfluidity, Bose-Einstein condensates possess collective amplitude modes. Examples are the amplitude modes of superconductors [Littlewood and Varma(1982)] and cold atomic gases [Altman and Auerbach(2002), Kollath et al.(2007)Kollath, Lauchli, and Altman, Barankov et al.(2004)Barankov, Levitov, and Spivak, Yuzbashyan et al.(2006)Yuzbashyan, Tsyplatyev, and Altshuler, Andreev et al.(2004)Andreev, Gurarie, and Radzihovsky, Barankov and Levitov(2004), Yuzbashyan et al.(2005)Yuzbashyan, Kuznetsov, and Altshuler], in which the density fluctuates between condensed and non-condensed particles, and the formally equivalent Higgs mode of a relativistic scalar condensate [Varma(2002)]. These modes are orthogonal to the phase modes in the order-parameter space, and their excitation is predicted to lead to non-equilibrium states in which the magnitude of the order parameter oscillates [Altman and Auerbach(2002), Kollath et al.(2007)Kollath, Lauchli, and Altman, Barankov et al.(2004)Barankov, Levitov, and Spivak, Yuzbashyan et al.(2006)Yuzbashyan, Tsyplatyev, and Altshuler, Andreev et al.(2004)Andreev, Gurarie, and Radzihovsky, Barankov and Levitov(2004), Yuzbashyan et al.(2005)Yuzbashyan, Kuznetsov, and Altshuler, Eastham and Phillips(2009), Tomadin et al.(2010)].

The strong coupling of excitons and photons in semiconductor microcavities leads to the formation of upper and lower polaritons. Pumping such a microcavity is observed to lead to the formation of Bose-Einstein condensed states of polaritons, characterized by coherent light emission from the structure [Deng et al.(2010)Deng, Haug, and Yamamoto]. This system allows the investigation of physics analogous to that of cold atoms, but in a strongly coupled regime with long range interactions mediated by the photons. The coupling of polaritons to electromagnetic radiation outside the microcavity gives an advantage over cold atoms in that polaritons can be coherently controlled by external pumping, and conversely their dynamics and coherence can be directly observed in the emitted light [Amo et al.(2010)Amo, Sanvitto, and Viña, Deng et al.(2010)Deng, Haug, and Yamamoto]. This has allowed experiments to reveal the collective behavior arising from excitations of the phase mode, including superfluidity [Amo et al.(2009)Amo, Lefrère, Pigeon, Adrados, Ciuti, Carusotto, Houdré, Giacobino, and Bramati], vortex dynamics [Lagoudakis et al.(2011)Lagoudakis, Manni, Pietka, Wouters, Liew, Savona, Kavokin, André, and Deveaud-Plédran, Tosi et al.(2010)Tosi, Marchetti, Sanvitto, Anton, Szymanska, Berceanu, Tejedor, Marrucci, Lemaître, Bloch et al., Sanvitto et al.(2010)Sanvitto, Marchetti, Szymańska, Tosi, Baudisch, Laussy, Krizhanovskii, Skolnick, Marrucci, Lemaître et al.], and the Bogoliubov spectrum [Utsunomiya et al.(2008)Utsunomiya, Tian, Roumpos, Lai, Kumada, Fujisawa, Kuwata-Gonokami, Löffler, Hofling, Forchel et al., Kohnle et al.(2010)Kohnle et al.]. However, the collective behavior associated with the amplitude mode has yet to be considered. Controlled excitation of the amplitude mode of an atomic Fermi gas requires a rapid switch of the magnetic field [Barankov et al.(2004)Barankov, Levitov, and Spivak, Yuzbashyan et al.(2006)Yuzbashyan, Tsyplatyev, and Altshuler, Andreev et al.(2004)Andreev, Gurarie, and

Radzihovsky, Barankov and Levitov(2004), Yuzbashyan et al.(2005)Yuzbashyan, Kuznetsov, and Altshuler], while in superconductors a coexisting charge-density wave is needed [Littlewood and Varma(1982)].

The amplitude mode of the polariton condensate may be identified by considering the Dicke model [Keeling et al.(2005)Keeling, Eastham, Szymanska, and Littlewood, Eastham and Littlewood(2001)], which for weak coupling becomes the BCS model. This limit, with well-known amplitude and phase modes, is adiabatically connected to the strong-coupling limit realized in microcavity experiments, where the collective modes are upper and lower polaritons. Thus the amplitude mode should be identified with the upper polariton. Uniquely, in this system the amplitude mode can be directly driven by resonant excitation, allowing the resulting collective behavior to be studied experimentally. In this paper we predict the collective behavior arising when the amplitude degree-of-freedom of a polariton condensate is manipulated in this way. Whereas driving the phase mode induces superflows, we find that driving the amplitude mode completely destabilizes the condensate, causing the polaritons to spontaneously organize into density-modulated phases (Fig. 17.2). This occurs because the interactions transfer the excess rest mass and interaction energy of the non-equilibrium state into kinetic energy, in a way ruled out for the phase modes by the Landau criterion.

We begin by considering the dynamics of a Dicke model of polariton condensation, because for this model exact solutions for the collective dynamics are available [Barankov and Levitov(2004), Andreev et al.(2004)Andreev, Gurarie, and Radzihovsky, Yuzbashyan et al.(2005)Yuzbashyan, Kuznetsov, and Altshuler, Babelon et al.(2009)Babelon, Cantini, and Doucot]. We show that these solutions, which describe a uniform oscillating condensate, are unstable once excitations at finite wavevector are considered, and systematically identify the instabilities. Our analysis reveals two types (Fig. 17.1): a wave-mixing instability between the lower and upper polariton, and a modulational instability caused by an attractive interaction between upper polaritons. We propose a Ginzburg-Landau theory which captures these instabilities, and provides a realistic model of a microcavity. We use this theory to predict the true steady-states under continuous excitation.

We can systematically establish the amplitude-mode dynamics of a polariton condensate by considering the generalized Dicke model [Keeling et al.(2005)Keeling, Eastham, Szymanska, and Littlewood, Eastham and Littlewood(2001)] :

$$\begin{aligned} \hat{H} = & \sum_k \omega_k \hat{\psi}_k^\dagger \hat{\psi}_k + \sum_i \frac{E}{2} \hat{\sigma}_i^z \\ & + \sum_{i,k} \frac{\Omega_R}{2\sqrt{N}} \left(\hat{\psi}_k^\dagger \hat{\sigma}_i^- e^{-ik \cdot r_i} + \hat{\sigma}_i^+ \hat{\psi}_k e^{ik \cdot r_i} \right). \end{aligned} \quad (17.1)$$

This model describes N localized exciton states with positions r_i , in the limit where exciton-exciton interactions exclude double occupancy. Thus each state may be occupied ($\sigma_i^z = 1$) or unoccupied ($\sigma_i^z = -1$). The excitons are coupled to the two-dimensional microcavity photons with in-plane wavevectors k , annihilation operators $\hat{\psi}_k$, and dispersion relation $\omega_k \approx \omega_0 + |k|^2/2m_{\text{ph}}$ ($\hbar=1$), where $m_{\text{ph}} \approx 10^{-5}m_e$. At this stage we are neglecting important effects such as the dispersion and inhomogeneous broadening of the exciton, the polarization degrees-of-freedom [Shelykh et al.(2005), Shelykh et al.(2006)Shelykh, Rubo, Malpuech, Solnyshkov, and Kavokin], and the finite lifetime of the cavity photons. We consider only the limit of a single exciton energy E , and a single coupling strength parametrized by the Rabi splitting Ω_R . To treat the collective dynamics of (17.1) it is convenient to work with the expectation values $\psi_k = \langle \hat{\psi}_k \rangle / \sqrt{N}$, $P_k = \langle \hat{P}_k \rangle / \sqrt{N} = \frac{1}{N} \sum_i \langle \hat{\sigma}_i^- \rangle e^{-ik \cdot r_i}$, and $D_k = \langle \hat{D}_k \rangle / \sqrt{N} = \frac{1}{N} \sum_i \langle \hat{\sigma}_i^z \rangle e^{-ik \cdot r_i}$. ψ_k and P_k are the macroscopic components of the electric field and polarization at wavevector k , while D_k measures the exciton occupation.

With only a single photon mode the Heisenberg equations for the model (17.1) are integrable, and closed-form solutions for the dynamics of the collective variables are available [Barankov and Levitov(2004), Andreev et al.(2004)Andreev, Gurarie, and Radzihovsky, Yuzbashyan et al.(2005)Yuzbashyan, Kuznetsov, and Altshuler, Babelon et al.(2009)Babelon, Cantini, and Doucot]. These solutions have been studied in the context of quench experiments on atomic Fermi gases, in which the pairing interaction would be rapidly switched, giving an impulsive excitation of the collective amplitude mode (analogous experiments are proposed in light-matter systems [Eastham and Phillips(2009), Tomadin et al.(2010)]). This is predicted to lead to a spatially uniform condensate in which the order parameter oscillates in time, of which an example is [Barankov and Levitov(2004)] :

$$\psi_0 = \phi(t) = \phi_+ \operatorname{dn}(\phi_+ t, \kappa), \quad (17.2)$$

$$P_0 = \rho(t) = 2[-\omega_0 \phi(t) + i\dot{\phi}(t)]/\Omega_R, \quad (17.3)$$

where dn is a Jacobi elliptic function, and the zero of energy is such that ψ_0 is real. The parameters ϕ_+ and κ determine the period T and magnitude of the oscillations, and are known functions of the model parameters and the initial conditions. Note that the oscillation frequency $\Omega = 2\pi/T$ generally differs from Ω_R due to interactions.

Since a microcavity supports a continuum of in-plane modes we must consider the behavior of an oscillating condensate beyond this mean-field approximation. To do this we linearize the equations of motion about the mean-field solution $\psi_0 = \phi(t)$, $P_0 = \rho(t)$ and $D_0 = \Delta(t)$. The fluctuating parts of the collective variables obey

$$\begin{aligned} i\delta\dot{\psi}_k &= \omega_k \delta\psi_k + \Omega_R \delta P_k / 2, \\ i\delta\dot{P}_k &= E \delta P_k - \Omega_R (\Delta \delta\psi_k + \phi \delta D_k) / 2, \\ i\delta\dot{D}_k &= \Omega_R (\rho^* \delta\psi_k + \phi \delta P_{-k}^* - \phi^* \delta P_k - \rho \delta\psi_{-k}^*), \end{aligned} \quad (17.4)$$

where the coefficients are time dependent because the condensate oscillates with angular frequency Ω , which is the gap energy to the occupied amplitude mode. In deriving (17.4) we assume that the wavevectors are much smaller than the inverse of the spacing of the exciton states, so that motional narrowing is effective and the wavevector of the fluctuations is well defined [Brierley and Eastham(2010), Litinskaya and Reineker(2006)].

Eqs. (17.4) have the form $\dot{\Psi} = A(t)\Psi$ where Ψ is a vector of the fluctuating fields and $A(t)$ a time-periodic matrix. Thus Floquet's theorem applies and the solutions are of the form $\Psi = u(t)e^{i\lambda t}$, where $u(t+T) = u(t)$ and the quasi-energy λ is defined up to an integer multiple of Ω . Since u is periodic, the stability of the condensate is determined by the imaginary part of λ .

To illustrate the fluctuation spectrum of an oscillating condensate, we show in Fig. 17.1 the result obtained for the exact solution (17.2,17.3). We measure energy and frequency in units of the Rabi splitting Ω_R , and wavevector in units of $k_0 = \sqrt{2m_{\text{ph}}}\Omega_R$. The spectrum can be understood by considering the limit of small oscillations around an equilibrium condensate, where (17.2) becomes $\phi = \phi_0 + \phi_1 \cos \Omega t$ with $\phi_1 \ll \phi_0$, and we have taken the zero of energy to be the equilibrium chemical potential. In this limit the construction of the spectrum is analogous to that of the nearly free electron model [Ashcroft and Mermin(1976)], with the weak time-periodic component playing the role of the weak periodic potential. To lowest order in the oscillating component, $\phi_1 = 0$, the fluctuation spectrum is that of an equilibrium polariton condensate [Eastham and Littlewood(2001), Keeling et al.(2005)Keeling, Eastham, Szymanska, and Littlewood]. There is a mode starting at zero energy and wavevector, which in the low-density limit is the lower polariton, and in Fig. 17.1 has been renormalized into the linearly-dispersing phase mode. There is also a gapped mode at zero wavevector, which in the low-density limit is the upper polariton [Eastham and Littlewood(2001)], and in Fig. 17.1 is

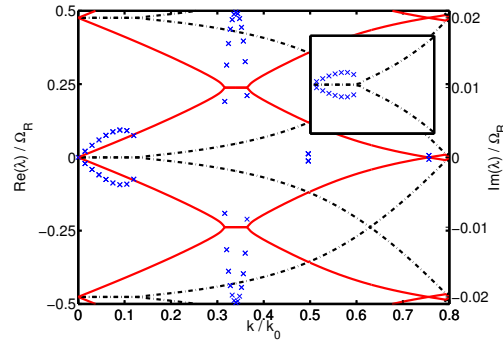


Figure 17.1: (Color online). Quasi-energy spectrum of an oscillating polariton condensate. Lines (left axis) correspond to the real part of the quasi-energy λ , and blue crosses to the non-zero imaginary parts (right axis). Modes with non-zero $\Im\lambda$ are unstable. The line colors indicate modes deriving from the phase mode/lower polariton (red solid), or the amplitude mode/upper polariton (black dashed). Inset: spectrum of a weakly-interacting Bose-Einstein condensate with attractive interactions, qualitatively reproducing the small-wavevector behavior. The parameters are $\omega_0 = -E = 0.4\Omega_R$, $\Delta(0) \approx 0.71$, $\phi(0) \approx 0.44$, implying a renormalized oscillation frequency $\Omega \approx 0.48\Omega_R$.

the collective amplitude mode appearing at the gap frequency Ω [Barankov and Levitov(2006)]. The effect of the oscillations is to fold the spectrum in frequency, and to couple together the resulting levels. Depending on the phase of the coupling this can result in either a level repulsion or attraction. In the latter case the dispersion relation is flattened, and imaginary parts appear for the quasi-energies. This signals an instability of the spatially uniform solution, and an initially exponential growth of modes at finite wavevectors.

The two strongest instabilities in Fig. 17.1 occur near $|k| = |k_1| \approx 0.3k_0$ and $|k| \approx 0$. The first occurs where the positive-energy branch derived from the lower polariton crosses with a replica of the corresponding negative-energy branch. In the low-density limit this occurs only for $E - \omega_0 > 0$. The result is a wave-mixing instability in which an upper and lower polariton from the oscillating condensate at $k = 0$ scatter to a pair of lower polaritons at $\pm k_1$. This is analogous to the instability that drives the microcavity optical parametric oscillator (OPO) [Ciuti et al.(2003)Ciuti, Schwendimann, and Quattropani], with the two $k = 0$ states forming the pump, and the states at $\pm k_1$ the signal and idler. This scattering process has previously been considered as a loss mechanism for incoherent polaritons [Ciuti(2004)]. The creation of phase modes from amplitude oscillations has been considered in the Bose-Hubbard model [Altman and Auerbach(2002)] and a similar instability has recently been found in the BCS model [Dzero et al.(2009)Dzero, Yuzbashyan, and Altshuler].

The second instability in Fig. 17.1 corresponds to the flat dispersion relation in the upper polariton branch near $k = 0$. It occurs because the saturation of the light-matter coupling [Rochat et al.(2000)Rochat, Ciuti, Savona, Piermarocchi, Quattropani, and Schwendimann] reduces the Rabi splitting with increasing excitation. Thus there is an attractive interaction between upper polaritons, and states containing more than a single such excitation are unstable. The resulting form of unstable spectrum is that obtained from the Bogoliubov analysis for a condensate with weak attractive interactions (inset).

The microscopic theory is capable of describing the initial instability, but becomes unwieldy in the nonlinear regime as the unstable modes evolve. To consider the long-term evolution in

a realistic microcavity we study the Ginzburg-Landau theory

$$\begin{aligned}
 i\frac{\partial\psi}{\partial t} &= \left(\omega_0 - \frac{\hbar^2}{2m_{\text{ph}}}\nabla^2\right)\psi + \frac{\Omega_R}{2}(1 - \lambda|P|^2)P \\
 &\quad - i\gamma\psi + \xi + F, \\
 i\frac{\partial P}{\partial t} &= EP + \frac{\Omega_R}{2}(1 - \lambda|P|^2)\psi.
 \end{aligned}
 \tag{17.5}$$

Here $P(x, t)(\psi[x, t])$ represents the macroscopically-occupied exciton (photon) field, which is linearly coupled to the photon (exciton) field to generate polaritons. The nonlinear coupling accounts for the saturation effect. This form can be obtained from (17.1) by representing the exciton operators using the Holstein-Primakoff transformation, and follows directly from the microcavity exciton-photon Hamiltonian [Rochat et al.(2000)Rochat, Ciuti, Savona, Piermarocchi, Quattropani, and Schwendimann] treated in mean-field theory. We neglect the Coulomb interactions between the excitons because they act within each quantum well, and hence are weaker than the saturation nonlinearity when the number of wells is large. This is consistent with the observed redshifting of the upper polariton with density in some microcavities [Kasprzak et al.(2006)Kasprzak, Richard, Kundermann, Baas, Jeambrun, Keeling, Marchetti, Szymanska, Andre, Staehli et al.]. We have introduced a damping constant $\gamma = 0.25 \text{ ps}^{-1}$ to describe decay of the cavity photons, a term F to describe a resonant pump laser, and a noise source ξ to model spontaneous emission noise. We focus on resonant excitation with circularly-polarized light, and hence include only a single polarization of exciton and photon. For numerical work we set $\lambda = 1$, absorbing the nonlinear coupling strength into the definition of density.

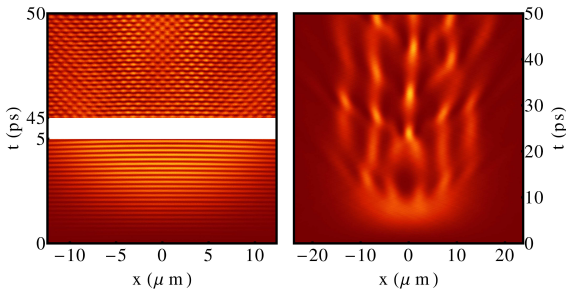


Figure 17.2: (Color online). Photon density from numerical simulations of Eq. (17.5), with resonant pumping of both upper and lower polaritons (left panel), and the upper polariton alone (right panel). The left panel shows the formation of a short-wavelength density modulation due to the wave-mixing instability, and the right the result of the attractive-gas instability. The pump field profile is a Gaussian at $x = 0$ with $\sigma = 10 \mu\text{m}$; $\Omega_R = 20 \text{ meV}$; $E - \omega_0 = 4 \text{ meV}$.

Fig. 17.2 shows the photon densities calculated from (17.5) in one space dimension, with resonant excitation of both the upper and lower polariton (left panel), and the upper polariton alone (right panel). For these parameters pumping both modes initially leads to a uniform induced condensate within the pump spot, in which as in (17.2) the field amplitude oscillates due to intermode beating. However, at later times this uniform state breaks down, and we see the formation of a condensate with a short-wavelength density modulation. We do not see strong signs of thermalization and chaos developing from the instability, as has been suggested for atomic systems [Dzero et al.(2009)Dzero, Yuzbashyan, and Altshuler, Kollath et al.(2007)Kollath, Lauchli, and Altman], presumably due to the presence here of dissipation.

In Fig. 17.2 we also show the behavior when only the upper polariton is pumped. In this case the wave-mixing instability cannot occur, and the dominant instability is due to the attractive interactions. In cold atomic gases, producing a condensate with attractive interactions leads to the Bose supernova [Donley et al.(2001)Donley, Claussen, Cornish, Roberts, Cornell, and

Wieman] where the condensate explodes due to the excess interaction energy of the uniform state. In the microcavity, we instead predict the formation of a large wavelength density modulation. This is because the polaritons can organize in such a way that the excess energy injected by the pump is dissipated, leading to a relatively stable steady-state.

While in the undamped model (17.1) the uniform state is always unstable, the dissipation in (17.5) implies a threshold density below which uniform states are stable. This density n_0 is where the gain due to the interactions exceeds the losses from the modes, and therefore for an interaction strength $g \sim \Omega_R \lambda$ is $gn_0 \sim \gamma$, up to numerical factors which are typically of order one. This is essentially the threshold criterion for the OPO [Ciuti et al.(2003)Ciuti, Schwendimann, and Quattropani], so that the threshold density should be within reach experimentally.

The models we have considered can be extended to include more realistic details. In particular, we have neglected the electron-hole continuum, which can produce resonant damping of the upper polariton, similar to the amplitude-mode damping predicted in atomic systems [Barankov and Levitov(2006), Yuzbashyan and Dzero(2006), Gurarie(2009)]. However, the two-dimensional exciton binding energy in GaAs (CdTe) is 20 (40) meV, so that the upper polariton in Fig. 17.2 is below the continuum, and not too strongly damped by this mechanism.

An interesting extension of our work would be to allow both polarizations of polaritons and excitons in the Ginzburg-Landau theory. In this case there will be additional amplitude modes connected to fluctuations in the degree of polarization of the condensate. It would be interesting to determine whether the polarization oscillations previously seen in OPO simulations [Shelykh et al.(2005)] correspond to these amplitude modes, and to investigate the possibility of nonlinear decay processes similar to those discussed here. It would also be useful to investigate the possibility of attractive interactions between amplitude modes in the BCS model, and hence establish the extent to which the instabilities identified here occur in other systems.

In summary, we have used the Dicke model to show that a uniform condensate in which the amplitude mode is excited is unstable due to (a)an attractive interaction between amplitude modes and (b)scattering between amplitude and phase modes. We have used a Ginzburg-Landau theory to show that these instabilities lead to the formation of spatially inhomogeneous condensates. In a microcavity the amplitude mode corresponds to the upper polariton, and therefore these instabilities can be induced by resonant excitation, leading to features in the real-space density and to bright emission at an angle from the cavity. Our work shows that there is a rich collective behavior associated with condensate amplitude modes, and that microcavities provide a unique opportunity to explore this physics experimentally.

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18 Cavity-induced Eliashberg effect: superconductivity vs charge density wave by Islam, Pini, Flores-Calderón, Piazza

(Denis said that here they get nonequilibrium distribution using Keldysh formalism and evaluation of a diagram, so we can check if our model gives the same outcome? they require linear deviation from thermal distribution of quasiparticles)

Abstract

Recent experiments have shown that non-equilibrium effects can play a key role in cavity-based control of material phases, notably in systems with charge-density-wave order.

Motivated by this, we extend the theory of the Eliashberg effect, originally developed for superconducting phases, to charge-density-wave phases.

Starting from a minimal electronic model where superconductivity and charge-density-wave order are equivalent at equilibrium, we introduce coupling to cavity photons, which are in turn coupled to an environment at a temperature different from the one of the electronic environment.

This drives the system into a non-thermal steady state, which breaks the equivalence between superconductivity and charge-density-wave order.

In the superconducting case, we recover the known behavior: a shift from continuous to discontinuous phase transitions with bistability.

In contrast, the charge-density-wave case displays richer behavior: tuning the cavity frequency induces both continuous and discontinuous transitions, two distinct ordered phases, and a bistable regime ending at a critical point. These findings demonstrate that the scope of cavity-based non-thermal control of quantum materials is broader than at thermal equilibrium, and strongly depends on the targeted phases.

18.1 Theory

Introduction—

Excitation of a material out of equilibrium via laser light has allowed to observe collective phenomena which would have been otherwise prohibited in thermal equilibrium at the given external conditions [De La Torre *et al.*(2021)]. A substantial portion of the work done so far has dealt with light-assisted superconductivity, also due to its technological relevance. This research direction is currently receiving major attention, due to recent experiments observing superconducting behavior close to ambient conditions after applying a laser pulse which excites selected phonon modes that mediate pairing between electrons [Fausti *et al.*(2011), Nicoletti *et al.*(2014), Hu *et al.*(2014), Mitrano *et al.*(2016), Cantaluppi *et al.*(2018), Budden *et al.*(2021), Isoyama *et al.*(2021)]. A different route to the control of superconductivity, implemented experimentally much earlier, is the one that goes under the name of Eliashberg effect (EE) [Eliashberg(1970), ?, Ivlev *et al.*(1973), Wyatt *et al.*(1966), Dayem and Wiegand(1967), Klapwijk *et al.*(1977)]. The latter involves a microwave pulse which directly excites the electrons into a non-equilibrium distribution that favors a larger superconducting gap.

A drawback of both the previous routes is that light pulses are used and thus the interesting non-equilibrium behavior is restricted to a transient. This issue can be solved by confining the light around the material using for instance the two mirrors of a cavity, so that the impact of a single photon is enhanced by reducing the volume of the electromagnetic (EM) modes, which become also separated in frequency [Garcia-Vidal *et al.*(2021), Mivehvar *et al.*(2021), Schlawin *et al.*(2022), Bloch *et al.*(2022)]. Experiments have demonstrated that the material's properties can be altered without laser driving but confining the light [Appugliese *et al.*(2022), Thomas *et al.*(2021), Jarc *et al.*(2023), Thomas *et al.*(2025), Keren *et al.*(2025)], so that only vacuum fluctuations and thermal excitations of photons are present.

Such cavity-material platforms also allow to use light to take the matter out of equilibrium and thus to escape thermal constraints. Since intense light pulses are not needed here, non-thermal stationary states instead of transients can be achieved for an extended cavity-based control of materials [Flores-Calderón *et al.*(2025)]. The EE mentioned above has also theoretically been revisited for cavity-based non-thermal control of superconductors in the stationary state [Curtis *et al.*(2019)].

More recently, the first experimental evidence of the crucial role that non-equilibrium effects can play in the cavity-based control of a material's phase has appeared [Jarc *et al.*(2023), Fassioli *et al.*(2025), Chiriacò(2024)]. This has however not been observed for a superconducting (SC) phase, but for another type of ordered phase, where the system spontaneously breaks translation-invariance by developing a modulation of the electrons' density – a so-called charge-density-wave (CDW) phase. This immediately calls for an extension of the theory of the EE from SC to CDW order.

In this work, we undertake this step. We consider a simple model of electrons, for which SC and CDW order are equivalent at thermal equilibrium. We then add the coupling between

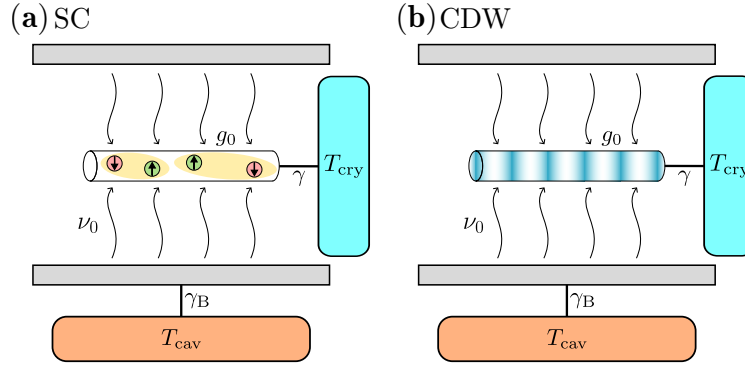


Figure 18.1: Schematics of the setup: a quasi-1D system of electrons featuring (a) superconducting (SC) or (b) charge-density-wave (CDW) order is placed inside a Fabry-Pérot cavity, along a direction parallel to the cavity mirrors. The cavity has a fundamental frequency of ν_0 . The electron current couples to the cavity photons with strength g_0 . Furthermore, the electrons are coupled to a cryostat with temperature T_{cry} , which induces a finite electron damping γ . The cavity photons are also coupled via the mirrors to the electromagnetic (EM) continuum outside, the latter is set at a temperature T_{cav} and also induces a finite photon damping γ_B .

electrons and photons, whereby the latter are confined within a cavity and coupled to the EM modes outside via the cavity mirrors (see Fig. 19.1). This EM environment has a temperature which is assumed to be different from the one of the cryogenic bath coupled to the electrons. The latter are thus in general found in a non-thermal steady state [Flores-Calderón *et al.*(2025)]. We find that this breaks the equivalence between SC and CDW order and leads to a different phenomenology of the EE. For SC, we find the standard EE phenomenology [Eliashberg(1970), ?, Ivlev *et al.*(1973), Curtis *et al.*(2019)] for any finite cavity resonance frequency, leading to a change from a continuous to a discontinuous transition featuring bistable behavior [see Fig. 18.2(a)]. For CDW instead, the EE leads to a much richer phenomenology: By tuning the cavity resonance frequency (via mirror separation) one can switch between three qualitatively different behaviors as a function of temperature [see Fig. 18.2(b)]. Both continuous and discontinuous transitions are found, and two distinct CDW phases appear, sharing their own additional bistable region which ends at a critical point.

Our findings showcase how cavity-based non-thermal control of materials strongly depends on the targeted phase and can be much larger in scope than its equilibrium counterpart, enabling to switch between qualitatively different behaviors by tuning the cavity frequency.

	SC	CDW
$\hat{c}_+(k) / \hat{c}_-(k)$	$\hat{c}_\uparrow(k) / \hat{c}_\downarrow^\dagger(-k)$	$\hat{c}(k + k_F) / \hat{c}(k - k_F)$
ξ_k	$-J \cos(ka)$	$J \sin(ka)$
BZ	$[-\pi/a, \pi/a)$	$[-\pi/2a, \pi/2a)$
$g_{k,k'}^\pm$	$g_0 \sin\left(\frac{ka+k'a}{2}\right)$	$\pm g_0 \cos\left(\frac{ka+k'a}{2}\right)$
η	1	-1

Table 3: Mapping of the Hamiltonian eq. (18.1) for SC and CDW. Rows indicate (from top to bottom) the electron operators, the free electron dispersion, corresponding Brillouin zone (BZ), the light-matter (LM) coupling, and the parameter η that is used below to differentiate between SC and CDW.

Model and approach—

We consider the setup sketched in fig. 19.1 described by the following Hamiltonian ($\hbar$ and k_B are set to unity throughout the manuscript)

$$\begin{aligned}
H &= H_{\text{el}} + H_{\text{ph}} + H_{\text{el-ph}} + H_{\text{el-bath}} + H_{\text{ph-bath}} \\
H_{\text{el}} &= \sum_{k \in \text{BZ}} [\hat{c}_+^\dagger(k) \quad \hat{c}_-^\dagger(k)] \begin{bmatrix} \xi_k & \Delta \\ \Delta^* & -\xi_k \end{bmatrix} \begin{bmatrix} \hat{c}_+(k) \\ \hat{c}_-(k) \end{bmatrix} + \frac{|\Delta|^2}{\kappa} \\
H_{\text{ph}} &= \sum_q \nu_q \left(\hat{a}_q^\dagger \hat{a}_q + \frac{1}{2} \right) \\
H_{\text{el-ph}} &= \sum_{k,k'} \sum_{\alpha=\{+,-\}} g_{k,k'}^\alpha \frac{(\hat{a}_{k-k'} + \hat{a}_{k'-k}^\dagger)}{\sqrt{2\pi\nu_{k-k'}/\nu_0}} \hat{c}_\alpha^\dagger(k) \hat{c}_\alpha(k').
\end{aligned} \tag{18.1}$$

Here, H_{el} is a mean-field Hamiltonian for the electrons, allowing for spontaneous symmetry breaking with order parameter Δ and κ is the strength of the pairing coupling. The electrons are divided into two Nambu components ($\hat{c}_+$ and $\hat{c}_-$) and are coupled through Δ and Δ^* . In the rest of the paper, we will choose the gauge where Δ is real without loss of generality. This simple model can be used to study both superconducting (SC) and charge-density-wave (CDW) transitions with the mapping described in table 3. We consider electrons hopping with strength J along a 1D lattice with lattice constant a , corresponding to the free dispersion ξ_k with bandwidth $2J$, and we assume the band to be half-filled. This can model electrons in quasi-1D nanotubes [Hümmer *et al.*(2016), Jeantet *et al.*(2016)]. Apart from its simplicity, the main reason to choose this model is that, at thermal equilibrium, the SC and CDW order parameters satisfy the same equation (18.4). This allows us to clearly identify the asymmetric out-of-equilibrium behavior of the EE between SC and CDW.

In order to drive the system out of equilibrium, coupling to cavity photons is added. The latter are described by the Hamiltonian H_{ph} , where $\hat{a}_q$ is the photon annihilation operator, and $\nu_q = \sqrt{\nu_0^2 + c^2 q^2}$ is the dispersion, where the momentum q is along the 1D tube, c is the speed of light, and $\nu_0 = \pi c/L$ is the cavity fundamental frequency that can be tuned using the distance between the cavity-mirrors L . The cavity photons couple to the electron current via $H_{\text{el-ph}}$ [Schlawin *et al.*(2019), Rao and Piazza(2023)]. The light-matter (LM) coupling $g_{k,k'}^\alpha$ (with strength g_0) depends on the electronic momenta and Nambu components, and it differs between SC and CDW, as tabulated in table 3. The cavity photons are connected to the EM environment with temperature T_{cav} through the mirrors. This coupling is indicated by

$$\begin{aligned}
H_{\text{ph-bath}} &= \sum_{s,q} \left[\nu_s(q) \left(\hat{b}_{sq}^\dagger \hat{b}_{sq} + \frac{1}{2} \right) \right. \\
&\quad \left. + \frac{t_s^{\text{ph}}(q)}{2\sqrt{\nu_q \nu_s(q)}} (\hat{a}_{-q}^\dagger + \hat{a}_q) (\hat{b}_{s-q}^\dagger + \hat{b}_{sq}) \right],
\end{aligned} \tag{18.2}$$

where we consider an extensive number of modes $\hat{b}_{sq}$ of the external EM environment couple to each cavity photon mode. The bath induces a spectral-width γ_B for the photons. We consider it to be large compared to the LM coupling so that the feedback on photons by the electrons can be neglected and the photons are always at thermal equilibrium at temperature T_{cav} . The electrons are connected to a cryostat with temperature T_{cry} via the Hamiltonian

$$\begin{aligned}
H_{\text{el-bath}} &= \sum_{\alpha,k,s} \left\{ \epsilon_{s\alpha}(k) \hat{f}_{s\alpha k}^\dagger \hat{f}_{s\alpha k} \right. \\
&\quad \left. + t_{s\alpha}(k) \left[\hat{c}_\alpha^\dagger(k) \hat{f}_{s\alpha k} + \hat{f}_{s\alpha k}^\dagger \hat{c}_\alpha(k) \right] \right\}.
\end{aligned} \tag{18.3}$$

Here we consider an extensive number of modes $\hat{f}_{s\alpha k}$ for the cryostat, labeled by s and coupled to each electron. The cryostat provides spectral width γ to the electrons [Flores-Calderón *et al.*(2025)].

Without coupling to the photons ($g_0 = 0$), the electrons are at thermal equilibrium, and the phase transition is described by the gap-equation (see the End Matter and Refs. [Abrikosov *et al.*(1975), Altland and Simons(2010), Tinkham(2004)])

$$\int_{\Delta}^{\infty} dE \Lambda(E) \frac{1 - 2n_0(E)}{E} = \frac{1}{\kappa}, \quad (18.4)$$

where $n_0(\omega) = \frac{1}{\exp(\omega/T_{\text{cry}})+1}$ is the Fermi-Dirac distribution function and $\Lambda(E) = \frac{2}{\pi} \frac{E}{\sqrt{E^2 - \Delta^2}} \frac{\Theta(E - \Delta) \Theta(\sqrt{J^2 + \Delta^2} - E)}{\sqrt{J^2 + \Delta^2} \sqrt{J^2 + \Delta^2 - E^2}}$ is the density of states (DOS) on a lattice for gaped quasi-particles. The gap equation (18.4) is valid for both SC and CDW, which thus behave identically in this thermal equilibrium case, due to the particle-hole symmetry (see the End Matter).

For finite LM coupling, the system in general achieves a non-equilibrium steady state (NESS), due to the presence of two separate baths. So, to describe it, we resort to the Schwinger-Keldysh field theory (SKFT) [Keldysh(1965), ?, Schwinger(1961), Kamenev(2023), Rammer(2007), Sieberer *et al.*(2016)]. Under a weak-coupling approximation and for well-defined fermionic quasi-particles (for which we also neglect the light-induced modification of the spectrum) we arrive at the following out-of-equilibrium gap equation (see the End Matter)

$$\int_{\Delta}^{\infty} dE \Lambda(E) \frac{1 - 2[n_0(E) + \delta n(E)]}{E} = \frac{1}{\kappa}. \quad (18.5)$$

As in the standard description of the EE [Eliashberg(1970), Ivlev *et al.*(1973)], the above approximations lead to the fact that the coupling to light does not modify the structure of the gap equation, but only the electron energy distribution δn entering the latter. The non-equilibrium modification of the distribution reads

$$\delta n(E) = \frac{2g_0^2}{\gamma} \int_{\Delta}^{\infty} dE' \Lambda(E') [\Gamma_{\Delta}^{\eta}(E, E') + \Gamma_{\Delta}^{\eta}(E, -E')], \quad (18.6)$$

where

$$\Gamma_{\Delta}^{\eta}(E, E') = \left(1 + \eta \frac{\Delta^2}{EE'}\right) \text{Im} D_0^R(E - E') H_0(E, E') \quad (18.7)$$

describes the electron-scattering processes mediated by photons with spectral function

$$\text{Im} D_0^R(\omega) = \frac{1}{4\pi} \left[\frac{\gamma_B}{(\omega + \nu_0)^2 + \gamma_B^2} - \frac{\gamma_B}{(\omega - \nu_0)^2 + \gamma_B^2} \right]. \quad (18.8)$$

The scattering processes depend on both the electron and photon distributions, which are contained in the factor

$$\begin{aligned} H_0(\omega, \omega') \\ = [N_{T_{\text{cav}}}(\omega - \omega') - N_{T_{\text{cry}}}(\omega - \omega')] [n_0(\omega) - n_0(\omega')] \end{aligned} \quad (18.9)$$

that keeps track of the difference in temperature between the two baths. Here, $N_T(\omega) = \frac{1}{\exp(\omega/T)-1}$ is the Bose-Einstein distribution function at the temperature T .

For the photon spectral function [see eq. (18.8)] we have considered only zero momentum photons as their effective mass is much less than that of the electrons and hence finite momenta contributions are negligible (forward scattering approximation). This would make the

light-induced scattering negligible unless one includes the effect of elastic scattering on static impurities that are generally present in real materials. This is indeed always done in the description of the EE and allows for scattering processes where momentum is not conserved. In the simplest modeling, one can replace the momentum conserving delta function by a constant $(\Lambda_F/\tau_{\text{el}})^{-1}$ where $\Lambda_F = 2/\pi J$ is the DOS at the Fermi surface and τ_{el} is the elastic scattering rate [Mattis and Bardeen(1958), Curtis *et al.*(2019)]. We further absorb the factor $(\Lambda_F/\tau_{\text{el}})$ in γ . Moreover, one can average the LM coupling on the Fermi Surface and describe the processes in terms of energy transfers only [Curtis *et al.*(2019)].

Note that within our description the whole difference between SC and CDW is condensed in a single sign $\eta = \pm 1$ (see table 3), which enters the non-equilibrium correction to the electron distribution via the scattering function Γ_{Δ}^{η} . The relevant scattering processes are best understood in the Bogoliubov rotated basis where the free electron Hamiltonian becomes diagonal and we have two bands separated by the gap. In this construction, $\Gamma_{\Delta}^{\eta}(E, E')$ corresponds to intraband processes, while $\Gamma_{\Delta}^{\eta}(E, -E')$ to interband processes. Note that for SC (CDW), the intraband (interband) processes have larger amplitudes than the interband (intraband) processes. The intraband processes enhance the gap, while interband processes always suppress the gap. We thus generically expect the EE to enhance the SC gap more than the CDW gap. Furthermore, for $\nu_0 < \Delta$, the interband processes are heavily suppressed, while for $\nu_0 > \Delta$ (but $\nu_0 < 2J$) they become stronger than intraband processes. Hence, the gap enhancement is expected for smaller cavity frequencies.

Results—

At thermal equilibrium without coupling to light, the electronic system shows two phases as temperature is varied. Using eq. (18.4), one can show that at high temperatures the system stays in a metallic phase with $\Delta = 0$. Below a certain temperature $T_{\text{cry}}^{\text{c0}}$ (that depends on κ), the system undergoes a continuous phase transition to an ordered state with a finite gap Δ . This behavior is depicted by the dashed line in the insets of fig. 18.2 and is exactly the same for SC and CDW.

On the other hand, the NESS behavior of the system at $g_0 \neq 0$ is described by eq. (18.5), where more parameters appear. We study the T_{cry} dependence of the order parameter as ν_0 is varied, since these are easily controllable parameters in experiments [Jarc *et al.*(2023)]. In fig. 18.2, we present the phase diagram in the $\nu_0 - T_{\text{cry}}$ plane for SC [panel (a)] and CDW [panel (b)]. Here, we keep $T_{\text{cav}} > T_{\text{cry}}^{\text{c0}}$, which is the case in typical experiments [Jarc *et al.*(2023)]. We show a few representative Δ -vs- T_{cry} curves in the inset, where the ν_0 values are indicated in the main panel by arrows of corresponding colors. In the insets, the solid (dotted) lines represent stable (unstable) finite- Δ solutions of the NESS gap equation eq. (18.5).

In the SC case, as apparent from the inset of fig. 18.2(a), the finite- Δ solution is stable below $T_{\text{cry}} = T_{\text{cry}}^{\text{u}}$ (where the curves have a vertical tangent), while the $\Delta = 0$ solution is stable above $T_{\text{cry}} = T_{\text{cry}}^{\text{c}}$ (where the dotted lines touch zero). For $T_{\text{cry}}^{\text{c}} < T_{\text{cry}} < T_{\text{cry}}^{\text{u}}$, both the finite- Δ and $\Delta = 0$ solutions coexist. So, instead of the thermal continuous transition, in the NESS we have a first-order phase transition between the SC and metallic phase, and in the shaded-green region the two coexist. Furthermore, for small ν_0/J one has $T_{\text{cry}}^{\text{u}} > T_{\text{cry}}^{\text{c0}}$, indicating an increase of the SC transition temperature. This cavity-induced EE behavior for SC order is consistent with the predictions of [Curtis *et al.*(2019)], and qualitatively the same as found for the standard EE induced by driving with coherent light [Eliashberg(1970), ?, Ivlev *et al.*(1973)].

The CDW phase diagram is much richer, as shown in fig. 18.2(b). We find three different phases: a metallic phase with $\Delta = 0$, a CDW with a smaller gap Δ (CDW_L) and a CDW with a larger Δ (CDW_H). In the shaded-green region, metallic and CDW_H phases coexist, while in the shaded-yellow region CDW_L and CDW_H coexist. Note that the latter region ends in a critical point indicated by a black star. The way in which the system transitions between these

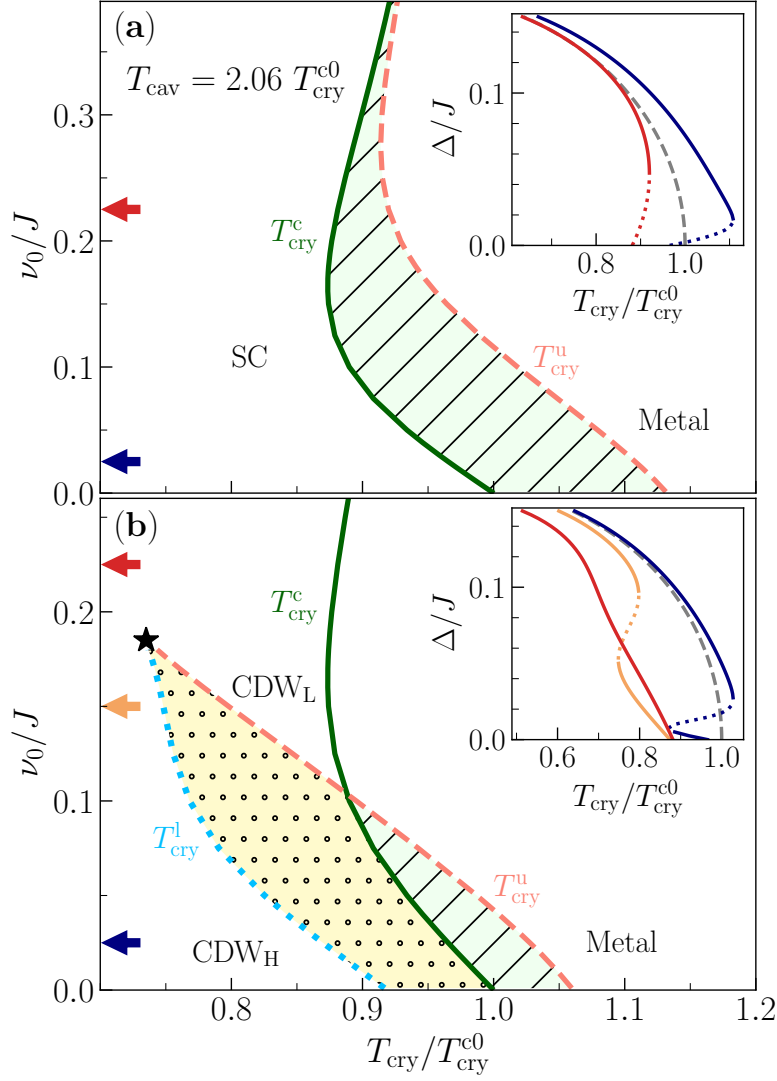


Figure 18.2: NESS phase diagram in the $\nu_0 - T_{\text{cry}}$ plane for the SC (a) and the CDW (b) case. We set $T_{\text{cav}} = 2.06 T_{\text{cry}}^{\text{c0}}$, $\kappa = 0.5 J$, $g_0 = 0.1 J$, $\gamma = 0.01 J$ and $\gamma_B = 0.5 \nu_0$. The shaded areas indicate phase-coexistence regions and the black star a critical point. Insets: Δ vs T_{cry} for different values of ν_0 which are indicated by arrows of the corresponding colors in the main panel. The solid (dotted) part of the curves represents a stable (unstable) solution of the gap equation. The dashed gray equilibrium curve is added for reference.

phases is correspondingly more complex. In the inset of fig. 18.2(b) we show Δ vs T_{cry} for three representative ν_0 values corresponding to qualitatively different types of transitions. Note that, depending on ν_0 , the curves have a right inflection point at $T_{\text{cry}} = T_{\text{cry}}^{\text{u}}$ (corresponding to the dashed light-red line in the main panel) or a left inflection point at $T_{\text{cry}} = T_{\text{cry}}^{\text{l}}$ (corresponding to the dotted light-blue line in the main panel). We find stable finite- Δ solutions for $T_{\text{cry}} < T_{\text{cry}}^{\text{u}}$ (corresponding to CDW_H) and for $T_{\text{cry}}^{\text{l}} < T_{\text{cry}} < T_{\text{cry}}^{\text{c}}$ (corresponding to CDW_L). For instance, for the blue curve in the inset (small ν_0/J), starting from the metallic phase, as T_{cry} is reduced the system first enters a coexistence region with metallic and CDW_H, then a coexistence region with CDW_L and CDW_H, and finally the CDW_H at low T_{cry} . That is, two first-order transitions are encountered in this case. For the orange curve instead (intermediate ν_0/J), the system enters the CDW_L smoothly, then a coexistence region with CDW_L and CDW_H, and finally the CDW_H at low T_{cry} . That is, first a continuous and then a first-order transition. For the red curve (large ν_0/J , which lies above the critical point), we only observe a continuous transition to the a CDW phase, as CDW_H and CDW_L are not anymore separated by a transition but

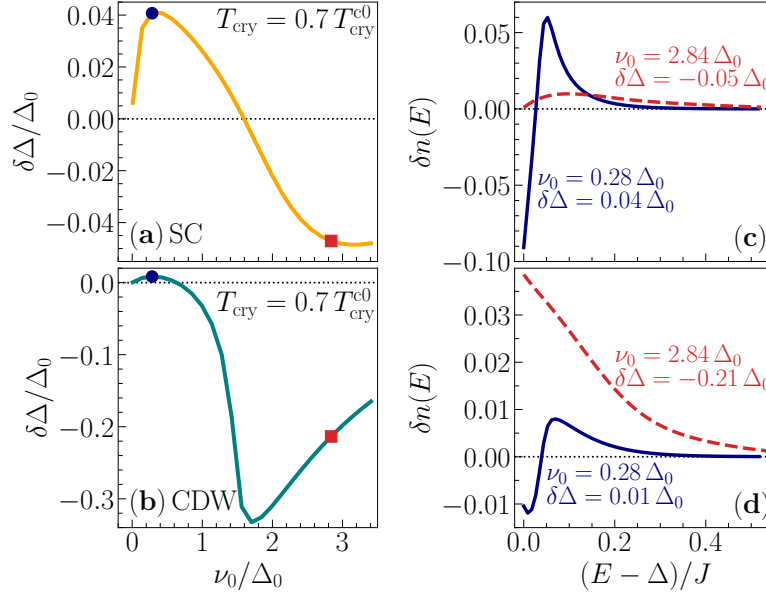


Figure 18.3: Non-thermal modification of the gap $\delta\Delta = \Delta - \Delta_0$ vs ν_0 is shown for SC (a) and CDW (b). Δ_0 is the equilibrium gap value. Modification of the distribution δn vs E is shown for SC (c) and CDW (d) at two representative ν_0 values marked in (a) and (b). We set $T_{\text{cry}} = 0.7 T_{\text{cry}}^{\text{c0}}$ and other parameters are the same as in fig. 18.2.

rather a smooth crossover. Moreover, also for CDW we find $T_{\text{cry}}^{\text{u}} > T_{\text{cry}}^{\text{c0}}$ for small ν_0/J values, indicating an increase of the transition temperature. As expected, the enhancement is smaller than for SC and persists up-to a smaller value of ν_0/J .

A necessary (but not sufficient) condition for the increase of the transition temperature is the enhancement of the gap. In fig. 18.3(a) and (b), we show how the gap Δ is modified from its equilibrium value Δ_0 as ν_0 is varied deep in the ordered phase for SC and CDW, respectively. The gap is enhanced for both at small ν_0/Δ_0 . Consistently with the phase diagram of fig. 18.2, for CDW the enhancement is smaller and persists up to a smaller value of ν_0/Δ_0 . Furthermore, the enhancement/suppression of the gap is related to the sign of δn at energies close to the gap [Eliashberg(1970), ?, Ivlev *et al.*(1973), Curtis *et al.*(2019)]. In fig. 18.3(c) and (d) we show δn as a function of energy for SC and CDW, respectively, at specific cavity frequencies marked in fig. 18.3(a) and (b) with corresponding colors. For the solid blue curves at small ν_0/Δ_0 , $\delta n < 0$ near $E = \Delta$ indicates that the quasiparticles are pushed away from the edge of the band, effectively enhancing the gap. On the other hand, for the dashed red curves at larger ν_0/Δ_0 , the redistribution is not favorable for the gap.

At thermal equilibrium, Ginzburg-Landau (GL) theory is a tool for understanding phase transitions based on the properties of the free energy as a function of the order parameter [Ginzburg and Landau(2009), ?]. In the present model at $g_0 = 0$, an expansion of the free energy in powers of Δ shows that, while the equilibrium free energy contains terms quadratic and quartic in Δ , the cubic term is absent. The transition is continuous, with a gap that initially grows as $\Delta \sim \sqrt{T_{\text{cry}}^{\text{c0}} - T_{\text{cry}}}$ by lowering the temperature below its critical value [Altland and Simons(2010), Tinkham(2004)].

For $g_0 > 0$, a similar expansion yields the NESS free energy (see Supplemental Materials)

[Schmid(1977)]

$$\begin{aligned}\mathcal{F}_{\text{neq}}[\Delta] = & \frac{T_{\text{cry}} - T_{\text{cry}}^c}{2T_{\text{cry}}^c} \Delta^2 + \eta \frac{g_0^2 [N_{T_{\text{cav}}}(\nu_0) - N_{T_{\text{cry}}^c}(\nu_0)]}{3\pi\gamma J T_{\text{cry}}^c} \Delta^3 \\ & + \frac{7\zeta(3)}{32\pi^2} \frac{1}{(T_{\text{cry}}^c)^2} \Delta^4 + \dots, \end{aligned} \quad (18.10)$$

where $\zeta(3) \approx 1.2$ is the Riemann zeta function. In order to obtain an analytical expression, we assumed that $\gamma_B \ll \nu_0, T_{\text{cav}}$. We see that escaping thermal equilibrium allows for a cubic term in the NESS free energy. Such a term has also been found in the standard EE case of a SC driven by coherent light [Ivlev *et al.*(1973), Schmid(1977)]. Here we extended this to the present two-bath cavity-electron setup, both for the SC and CDW case. We see that the cubic term alone encodes the information about the temperature difference between the two baths (through the boson-occupation difference), as well as the information about the difference between SC and CDW (through the sign-factor η). Furthermore, the cubic term implies a linear-in-temperature growth of the order parameter solution as opposed to the square-root behavior at thermal equilibrium:

$$\Delta[T_{\text{cry}}] \approx \frac{\pi\gamma J}{g_0^2} \frac{\eta}{N_{T_{\text{cav}}}(\nu_0) - N_{T_{\text{cry}}^c}(\nu_0)} (T_{\text{cry}} - T_{\text{cry}}^c). \quad (18.11)$$

When $T_{\text{cav}} > T_{\text{cry}}^c$, the solution for SC (CDW) is unstable (stable). The situation is reversed when $T_{\text{cav}} < T_{\text{cry}}^c$ (See the End Matter for a more detailed analysis of the latter situation).

Conclusions—

Motivated by recent experiments demonstrating cavity-based control of quantum materials in non-equilibrium steady states, we extended the theory of the Eliashberg effect from superconductivity to charge-density-wave order. The latter shows a much broader scope for the control of the ordering: by tuning the cavity resonance frequency, both continuous and discontinuous transitions, as well as multiple ordered phases merging at a critical point are selectively accessible. In the future, our analysis shall be extended to include the non-equilibrium fluctuations of the order parameter, which are expected to lead to non-thermal distributions within the Ginzburg-Landau free-energy landscape. Moreover, analog extensions of the EE shall be applied to other types of order, including magnetic phases.

18.1.1 End Matter

NESS gap equation—

The NESS gap equation is derived using SKFT [Keldysh(1965), ?, Schwinger(1961), Kamenev(2023), Rammer(2007), Sieberer *et al.*(2016)]. We start by writing the Keldysh action for the Hamiltonian in eq. (18.1) following the notations in Ref. [Kamenev(2023)]. Note that one needs to treat Δ as a field to arrive at the correct Keldysh action. We perform a Bogoliubov rotation on the Keldysh action so that the free part of the action corresponding to H_{el} becomes diagonal. The dispersions of the two energy bands are given by $\pm E_k$ where $E_k = \sqrt{\Delta^2 + \xi_k^2}$. Under weak coupling approximation for $g_0 \ll J, \gamma, \gamma_B$, the effect of the LM coupling is included perturbatively through its self-energy. Then, we integrate out the electronic degrees of freedom to obtain an effective Keldysh action in terms of the gap field Δ . Finally, we take the classical saddle point of the Keldysh action to get the NESS gap equation. Assuming $\gamma \ll \Delta, \gamma_B, T_{\text{cry}}, T_{\text{cav}}$ we perform a

quasi-particle approximation to arrive at

$$\sum_k \frac{1 - 2[n_0(E_k) + \delta n(k, E_k)]}{E_k} = \frac{1}{\kappa}, \quad (18.12)$$

where the change in the distribution function is obtained through the quantum kinetic equation

$$\begin{aligned} \delta n(k, E_k) = \frac{2}{\gamma} \sum_{k'} \sum_q \int d\Omega (g_{k,k'}^+)^2 & \left[\left(1 + \frac{\xi_k \xi_{k'} + \eta \Delta^2}{E_k E_{k'}} \right) \text{Im} D_0^R(q, \Omega) H_0(E_k, E_{k'}) \delta(\Omega - (E_k - E_{k'})) \right. \\ & \left. + \left(1 - \frac{\xi_k \xi_{k'} + \eta \Delta^2}{E_k E_{k'}} \right) \text{Im} D_0^R(q, \Omega) H_0(E_k, -E_{k'}) \delta(\Omega - (E_k + E_{k'})) \right] \delta(q - (k - k')) \end{aligned} \quad (18.13)$$

See the Supplemental Material for a detailed derivation. In the above expression, the first term describes scattering processes within the same energy band and the second term between the bands. Note the energy and momenta conservations are described by the delta functions. Furthermore, we make a forward scattering approximation $\text{Im} D_0^R(q, \omega) \approx \text{Im} D_0^R(q = 0, \omega) \equiv \text{Im} D_0^R(\omega)$ and consider the effect of static impurities in real materials (see Refs. [Curtis *et al.*(2019), Mattis and Bardeen(1958)]) to replace $\delta(q - (k - k'))$ with a constant, as discussed in the main text. We also average the LM coupling on the Fermi surface [Curtis *et al.*(2019)]. Finally, we convert the momentum sums to energy integration with the help of the density of states for the gaped quasi-particles $\Lambda(E)$ to arrive at eq. (18.5). Note that by putting $g_0 = 0$, we get $\delta n = 0$ and recover the equilibrium gap equation in eq. (18.4).

Equivalence of SC and CDW gap equation at equilibrium—

At thermal equilibrium, the mapping between SC and CDW cases is ensured by the symmetry of the free electron dispersion around the Fermi surface, given by $\epsilon_{k+k_F} = -\epsilon_{k-k_F}$. This is guaranteed since we consider a half-filled band ($k_F = \pi/2a$) with free-electron dispersion $\epsilon_k = -J \cos(ka)$. Note that this is a consequence of the particle-hole symmetry (PHS) of H_{el} under the transformation $\hat{c}_\sigma(k \pm k_F) \rightarrow \hat{c}_\sigma^\dagger(-k \pm k_F)$. However, $H_{\text{el-ph}}$ is not invariant under this transformation and breaks PHS (see the Supplemental Material). Hence, out-of-equilibrium the order parameters for the SC and CDW cases behave differently.

Additional results for cooler photon-bath—

In the main text, we discuss the phase diagram for $T_{\text{cav}} > T_{\text{cry}}^{\text{c0}}$, which is generally the case in experiments [Jarc *et al.*(2023)]. However, for completeness, it is also interesting to explore the opposite case where $T_{\text{cav}} < T_{\text{cry}}^{\text{c0}}$. From eq. (18.11) within GL theory, we can see that in this case the linear growth of the gap solution with $|T_{\text{cry}} - T_{\text{cry}}^{\text{c}}|$ will flip sign for both SC and CDW with respect to the situation described in the main text. The stability of the solution will also change accordingly.

In fig. 18.4, we show the solution of the gap equation for a hotter [fig. 18.4(a)] and a cooler [fig. 18.4(b)] photon-bath. The numerical solution is in excellent agreement with the analytical expression of eq. (18.11) when the gap is small. Further we notice an emergent symmetry between SC and CDW in this case. From eq. (18.11) we infer that the symmetry operation is given by $T_{\text{cav}} \rightarrow T'_{\text{cav}}$ such that $N_{T'_{\text{cav}}}(\nu_0) = 2N_{T_{\text{cry}}^{\text{c}}}(\nu_0) - N_{T_{\text{cav}}}(\nu_0)$ and $\text{SC} \rightarrow \text{CDW}$ i.e. $\eta \rightarrow \eta' = -\eta$. Note that this emergent symmetry is valid only in the small- Δ region where the GL expansion is valid and not for the general NESS gap equation in eq. (18.5).

In fig. 18.5, we show the phase diagram in the $\nu_0 - T_{\text{cry}}$ plane for a cooler photon-bath i.e. $T_{\text{cav}} < T_{\text{cry}}^{\text{c0}}$. For the SC case in fig. 18.5(a), we see that at high temperatures we have a metallic phase with $\Delta = 0$ that exists on the right of the solid green line. On the left of the

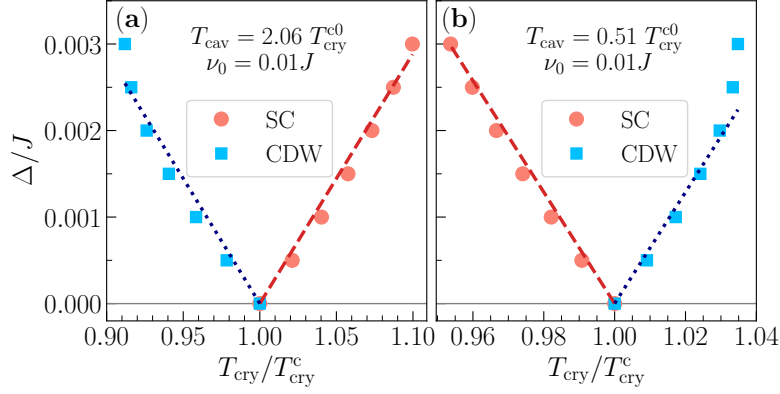


Figure 18.4: Δ vs T_{cry} for (a) hotter photon-bath ($T_{\text{cav}} = 2.06 T_{\text{cry}}^{\text{c0}}$) and (b) cooler photon-bath ($T_{\text{cav}} = 0.51 T_{\text{cry}}^{\text{c0}}$) for small Δ . We set $\nu_0 = 0.01 J$. Other parameters are the same as in fig. 18.2. The markers represent the numerical solution of the gap equation and the lines are obtained using the analytical expression in eq. (18.11).

solid green line, we have ordered phases with finite Δ . We see two kinds of ordered phases: SC_L , with a smaller value of Δ , which exists between the dashed light-red line and the solid green line, and SC_H , with a larger value of Δ , which exists on the left of the dotted light-blue line. In the shaded-yellow region SC_H and SC_L can coexist, similarly to the behavior of CDW in the dotted yellow region of fig. 18.2(b) in the main text. Note that also here the phase-coexistence region ends in a critical point marked by a black star. The phase diagram is constructed from the dependence of Δ on T_{cry} as ν_0 is varied. We show a few representative Δ vs T_{cry} curves in the inset, where the ν_0 values are indicated in the main plot by arrows of the corresponding colors. Here, the solid (dotted) lines represent stable (unstable) finite- Δ solutions of the NESS gap equation in eq. (18.5). Note that the finite- Δ solution is stable below $T_{\text{cry}} = T_{\text{cry}}^{\text{c}}$ (represented by the solid green line in the main plot) where the curves touch zero, while the $\Delta = 0$ solution is stable above $T_{\text{cry}} = T_{\text{cry}}^{\text{c}}$, indicating a second-order continuous transition. For the blue curve, we see two stable branches connected by a dotted line. The branch with smaller Δ exists for $T_{\text{cry}}^{\text{l}} < T_{\text{cry}} < T_{\text{cry}}^{\text{c}}$ while the branch with larger Δ exists for $T_{\text{cry}} < T_{\text{cry}}^{\text{u}}$. Hence for $T_{\text{cry}}^{\text{l}} < T_{\text{cry}} < T_{\text{cry}}^{\text{u}}$, both the finite- Δ solutions exist and this indicates a first-order transition between SC_H and SC_L . Note that in the main plot the dashed light-red line represents $T_{\text{cry}}^{\text{u}}$ and the dotted light-blue line stands for $T_{\text{cry}}^{\text{l}}$, which constitute the boundaries of the phase-coexistence region in the phase diagram. So, depending on ν_0 , we see only a continuous transition or a continuous transition accompanied by a first-order one as T_{cry} is lowered.

On the other hand, for the CDW case in fig. 18.5(b), we always see at least one first-order transition. In this phase diagram, on the right of the dashed-dotted purple line ($T_{\text{cry}} > T_{\text{cry}}^{\text{m}}$), there is the metallic phase. We see again two kinds of ordered phases: CDW_L , with a smaller value of Δ , which exists between the dashed light-red line and the solid green line ($T_{\text{cry}}^{\text{u}} < T_{\text{cry}} < T_{\text{cry}}^{\text{c}}$), and CDW_H , with a larger value of Δ , which exists on the left of the dotted light-blue line ($T_{\text{cry}} < T_{\text{cry}}^{\text{l}}$). In this case, however, there are two phase-coexistence regions: in the shaded-green region ($T_{\text{cry}}^{\text{c}} < T_{\text{cry}} < T_{\text{cry}}^{\text{m}}$), metallic phase and CDW_L coexist, while in the shaded-yellow region ($T_{\text{cry}}^{\text{l}} < T_{\text{cry}} < T_{\text{cry}}^{\text{u}}$), CDW_L and CDW_H coexist. The latter region ends in a critical point marked by a black star. So, at small ν_0 , we can observe two first-order transitions as T_{cry} is varied, while at larger ν_0 we observe only one first-order transition. In the inset we show a few representative Δ vs T_{cry} , from which the phase diagram is constructed. For the red curve, a finite- Δ solution and a $\Delta = 0$ solution coexist between the temperature where the dotted part of the curve touches zero and the rightmost inflection point ($T_{\text{cry}}^{\text{c}} < T_{\text{cry}} < T_{\text{cry}}^{\text{m}}$). This indicates a first-order transition. For $T_{\text{cry}} < T_{\text{cry}}^{\text{c}}$, Δ grows smoothly without any further transition. On

the other hand for the blue curve, in addition to the $\Delta = 0$ and finite Δ coexistence we see another range of temperature between the left inflection point and a right (not the rightmost) inflection point where two finite Δ solutions coexist. This indicates an additional first-order transition.

In conclusion, for a cooler photon-bath the behavior of the gap solution close to $\Delta = 0$ is inverted with respect to the hotter photon bath case due to an emergent symmetry for small Δ . However, for the whole phase diagram which is determined by finite values of Δ , this symmetry is broken and a new structure of phase coexistence regions emerges both for the SC and the CDW cases.

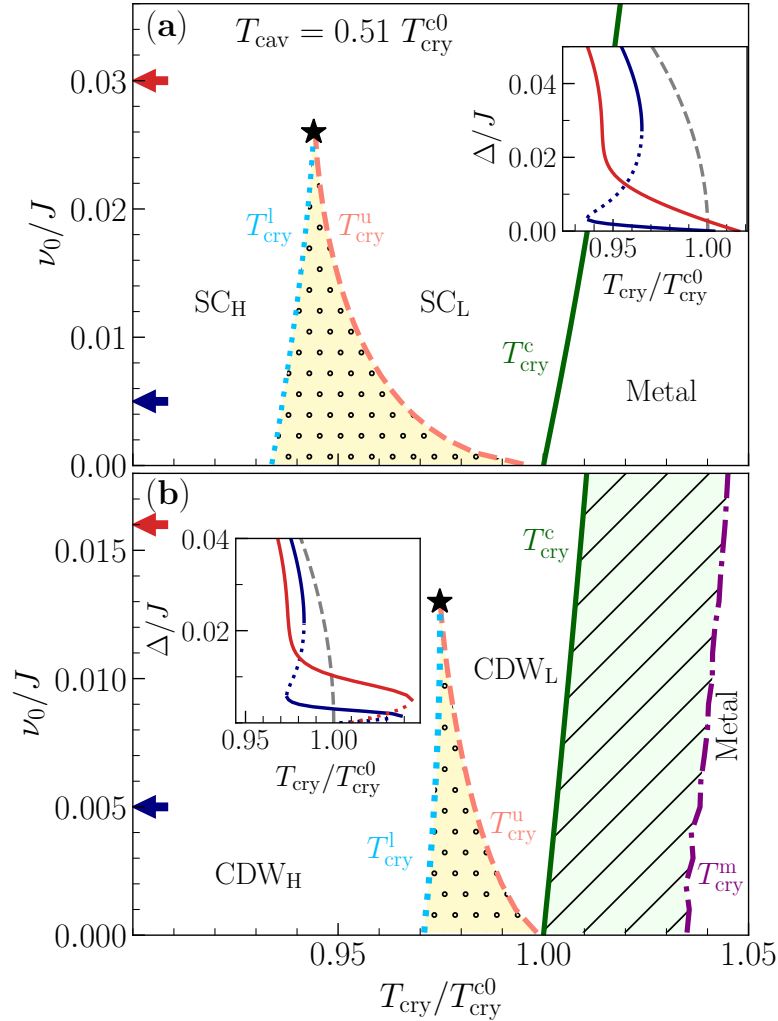


Figure 18.5: NESS phase diagram in the $\nu_0 - T_{\text{cry}}$ plane for the SC (a) and the CDW (b) case for a cooler photon-bath. We set $T_{\text{cav}} = 0.51 T_{\text{cry}}^{\text{c0}}$, $\kappa = 0.5 J$, $g_0 = 0.1 J$, $\gamma = 0.01 J$ and $\gamma_B = 0.5 \nu_0$. The shaded areas indicate phase-coexistence regions and the black stars critical points. Insets: Δ vs T_{cry} for different values of ν_0 which are indicated by arrows of the corresponding colors in the main panel. The solid (dotted) part of the curves represents a stable (unstable) solution of the gap equation. The dashed gray equilibrium curve is added for reference.

18.2 SM: The non-thermal gap equation: a Keldysh field theory derivation

18.2.1 The Model

We consider the system sketched in Fig. 1 in the main text. A system of fermions in a quasi-1D lattice is placed between the parallel mirrors of a Fabry-Perot cavity. Additionally, the fermions are connected to a cryostat and the photons interact with the environment outside through the cavity mirrors. This setup is described by the following Hamiltonian ($\hbar$ and k_B are set to unity throughout the letter)

$$H = (H_{\text{el}} + H_{\text{el-bath}}) + (H_{\text{ph}} + H_{\text{ph-bath}}) + H_{\text{el-ph}} \quad (18.14)$$

We divided the Hamiltonian in three parts: (i) the matter part ($H_{\text{el}} + H_{\text{el-bath}}$), (ii) the light part ($H_{\text{ph}} + H_{\text{ph-bath}}$) and the (iii) the light-matter coupling $H_{\text{el-ph}}$. We describe them separately below.

Matter

We consider a system of fermions in Nambu basis described by the following mean-field Hamiltonian

$$H_{\text{el}} = \sum_k [\hat{c}_+^\dagger(k) \quad \hat{c}_-^\dagger(k)] \begin{bmatrix} \xi_k & \Delta \\ \Delta^* & -\xi_k \end{bmatrix} \begin{bmatrix} \hat{c}_+(k) \\ \hat{c}_-(k) \end{bmatrix} + \frac{|\Delta|^2}{\kappa}, \quad (18.15)$$

where κ is the strength of pairing coupling. In this context it can be electron-phonon coupling or two-particle electron-electron interaction strength. Due to the presence of the off-diagonal terms (Δ), this model can describe a spontaneous symmetry-breaking transition where Δ acts as an order parameter as well as the energy gap. We can describe both the superconducting (SC) and charge-density-wave (CDW) transitions with the following definition.

For CDW (particle-hole channel) :

$$\hat{c}_+(k) = \hat{c}(k + k_F) \quad \text{and} \quad \hat{c}_-(k) = \hat{c}(k - k_F). \quad (18.16)$$

and for SC (particle-particle channel) :

$$\hat{c}_+(k) = \hat{c}_\uparrow(k) \quad \text{and} \quad \hat{c}_-(k) = \hat{c}_\downarrow(-k) \quad (18.17)$$

We consider that the free fermions are hopping on a lattice with lattice constant a . This results in a dispersion $\epsilon(k) = -J \cos(ka)$ in the Brillouin zone $k \in [-\pi/a, \pi/a]$. Here, J determines the bandwidth for the fermions and the energy ranges from $-J$ to $+J$. We also consider a half-filled band so that $\epsilon_F = 0$ and $k_F = \frac{\pi}{2a}$. So we have

$$\xi_k = \begin{cases} \epsilon(k + k_F) = -J \cos(ka + \pi/2) = J \sin(ka) & \text{where } k \in [-\pi/2a, \pi/2a] \quad \text{for CDW} \\ \epsilon(k) = -J \cos(ka) & \text{where } k \in [-\pi/a, \pi/a] \quad \text{for SC.} \end{cases} \quad (18.18)$$

So, in eq. (18.15), the momentum sum represents

$$\sum_k \rightarrow \begin{cases} \int_{-\pi/a}^{\pi/a} \frac{dk}{2\pi} & \text{for SC, and} \\ \int_{-\pi/2a}^{\pi/2a} \frac{dk}{\pi} & \text{for CDW.} \end{cases} \quad (18.19)$$

We have also considered that the fermions are connected to a cryostat with temperature T_{cry} that is described by

$$H_{\text{el-bath}} = \sum_{\alpha, k, s} \left\{ \epsilon_{s\alpha}(k) \hat{f}_{s\alpha k}^\dagger \hat{f}_{s\alpha k} + t_{s\alpha}(k) \left[\hat{c}_\alpha^\dagger(k) \hat{f}_{s\alpha k} + \hat{f}_{s\alpha k}^\dagger \hat{c}_\alpha(k) \right] \right\}. \quad (18.20)$$

Here we consider an extensive number of degrees of freedom $\hat{f}_{s\alpha k}$ for the cryostat, labeled by s and coupled to each electron momentum k and Nambu component α . The cryostat provides the electrons with a spectral width γ [Flores-Calderón *et al.*(2025)]. Note that T_{cry} is a controllable parameter in an experiment, and we will study the behavior of the system as T_{cry} is varied.

Light

The cavity photons are described by

$$H_{\text{ph}} = \sum_q \nu_q \left(\hat{a}_q^\dagger \hat{a}_q + \frac{1}{2} \right) \quad (18.21)$$

The bosonic annihilation and creation operators $\hat{a}_q, \hat{a}_q^\dagger$ describe the photons where $[\hat{a}_q, \hat{a}_{q'}^\dagger] = i\delta(q - q')$. The photons dispersion is given by $\nu_q = \sqrt{\nu_0^2 + c^2 q^2}$, where c is the speed of light and ν_0 is the cavity fundamental frequency given in terms of the distance between the cavity mirrors as $\nu_0 = \pi c/L$.

Cavity photons are also connected to the electromagnetic continuum of the external environment through the cavity mirrors. This coupling is described by

$$H_{\text{ph-bath}} = \sum_{s,q} \left[\nu_s(q) \left(\hat{b}_{sq}^\dagger \hat{b}_{sq} + \frac{1}{2} \right) + \frac{t_s^{\text{ph}}(q)}{2\sqrt{\nu_q \nu_s(q)}} (\hat{a}_{-q}^\dagger + \hat{a}_q) (\hat{b}_{s-q}^\dagger + \hat{b}_{sq}) \right], \quad (18.22)$$

where we consider an extensive number of modes $\hat{b}_{sq}$ of the external EM environment couple to each cavity photon mode. This keeps the photons in thermal equilibrium at temperature T_{cav} . This coupling also provide the photons with a spectral width γ_B [Flores-Calderón *et al.*(2025)].

Light-matter coupling

The fermion current couples to the cavity photons under paramagnetic approximation [Schlawin *et al.*(2019), Rao and Piazza(2023)]. In terms of free fermion operators it is given by

$$H_{\text{el-ph}} = \sum_{k,k'} \sum_{\sigma=\{\uparrow,\downarrow\}} g_0 \sin \left(\frac{ka + k'a}{2} \right) \hat{c}_\sigma^\dagger(k) \hat{c}_\sigma(k') \hat{\phi}(k - k'), \quad (18.23)$$

where $\hat{\phi}(q) = \sqrt{\frac{\nu_0}{\nu_q}} \frac{(\hat{a}_q + \hat{a}_{-q}^\dagger)}{\sqrt{2\pi}}$ and g_0 is the light-matter coupling strength. Now we need to re-write this in terms of the Nambu fermions. For SC,

$$\begin{aligned} H_{\text{el-ph}} &= \sum_{k,k'} g_0 \sin \left(\frac{ka + k'a}{2} \right) \left[\hat{c}_\uparrow^\dagger(k) \hat{c}_\uparrow(k') \hat{\phi}(k - k') + \hat{c}_\downarrow^\dagger(k) \hat{c}_\downarrow(k') \hat{\phi}(k - k') \right] \\ &= \sum_{k,k'} g_0 \sin \left(\frac{ka + k'a}{2} \right) \left[\hat{c}_\uparrow^\dagger(k) \hat{c}_\uparrow(k') \hat{\phi}(k - k') - \hat{c}_\downarrow^\dagger(-k) \hat{c}_\downarrow(-k') \hat{\phi}(k' - k) \right] \\ &= \sum_{k,k'} g_0 \sin \left(\frac{ka + k'a}{2} \right) \left[\hat{c}_+^\dagger(k) \hat{c}_+(k') \hat{\phi}(k - k') - \hat{c}_-(k) \hat{c}_-^\dagger(k') \hat{\phi}(k' - k) \right] \\ &= \sum_{k,k'} g_0 \sin \left(\frac{ka + k'a}{2} \right) \left[\hat{c}_+^\dagger(k) \hat{c}_+(k') \hat{\phi}(k - k') + \hat{c}_-^\dagger(k) \hat{c}_-(k') \hat{\phi}(k - k') \right] \end{aligned} \quad (18.24)$$

For CDW we can ignore the spin as the phenomenology is independent of spin. We have

$$\begin{aligned}
H_{\text{el-ph}} &= \sum_{k,k'} g_0 \left[\sin \left(\frac{ka + k'a}{2} + \frac{\pi}{2} \right) \hat{c}^\dagger(k + k_F) \hat{c}(k' + k_F) + \sin \left(\frac{ka + k'a}{2} - \frac{\pi}{2} \right) \hat{c}^\dagger(k - k_F) \hat{c}(k' - k_F) \right] \hat{\phi}(k - k') \\
&+ \sum_{k,k'} g_0 \sin \left(\frac{ka + k'a}{2} \right) \left[\hat{c}^\dagger(k + k_F) \hat{c}(k' - k_F) \hat{\phi}(k - k' + 2k_F) - \hat{c}^\dagger(k - k_F) \hat{c}(k' + k_F) \hat{\phi}(k - k' - 2k_F) \right] \\
&= \sum_{k,k'} g_0 \cos \left(\frac{ka + k'a}{2} \right) \left[\hat{c}_+^\dagger(k) \hat{c}_+(k') \hat{\phi}(k - k') - \hat{c}_-^\dagger(k) \hat{c}_-(k') \hat{\phi}(k - k') \right] \\
&+ \sum_{k,k'} g_0 \sin \left(\frac{ka + k'a}{2} \right) \left[\hat{c}_+^\dagger(k) \hat{c}_-(k') \hat{\phi}(k - k' + 2k_F) - \hat{c}_-^\dagger(k) \hat{c}_+(k') \hat{\phi}(k - k' - 2k_F) \right] \\
&\approx \sum_{k,k'} g_0 \cos \left(\frac{ka + k'a}{2} \right) \left[\hat{c}_+^\dagger(k) \hat{c}_+(k') \hat{\phi}(k - k') - \hat{c}_-^\dagger(k) \hat{c}_-(k') \hat{\phi}(k - k') \right]
\end{aligned} \tag{18.25}$$

Note that we can neglect the terms that couple $\hat{c}_+$ and $\hat{c}_-$ as the photon has to carry large momenta while we will see later that photons can carry only negligible momentum.

We can summarize both cases in the compact notation

$$H_{\text{el-ph}} = \sum_{k,k'} \sum_{\alpha=\{+,-\}} g_{k,k'}^\alpha \hat{c}_\alpha^\dagger(k) \hat{c}_\alpha(k') \hat{\phi}(k - k'). \tag{18.26}$$

Here the light-matter couplings are given by

$$g_{k,k'}^+ = \eta g_{k,k'}^- = g_{k,k'} = \begin{cases} g_0 \sin \left(\frac{ka+k'a}{2} + k_F a \right) = g_0 \cos \left(\frac{ka+k'a}{2} \right) & \text{for CDW} \\ g_0 \sin \left(\frac{ka+k'a}{2} \right) & \text{for SC,} \end{cases} \tag{18.27}$$

where

$$\eta = \begin{cases} -1 & \text{for CDW} \\ +1 & \text{for SC.} \end{cases} \tag{18.28}$$

For zero light-matter coupling ($g_0 = 0$), fermions at thermal equilibrium with the cryostat and have Fermi-Dirac distribution $n_0(\omega) = \frac{1}{\exp(\omega/T_{\text{cry}})+1}$. We define the distribution factor $F_0(\omega) = 1 - 2n_0(\omega) = \tanh \left(\frac{\omega}{2T_{\text{cry}}} \right)$. Similarly the photons are at thermal equilibrium with the environment and have distribution $N_{T_{\text{cav}}}$, where we define Bose-Einstein distribution $N_T = \frac{1}{\exp(\omega/T)-1}$. We also define $B_T(\omega) = 1 + 2N_T(\omega) = \coth \left(\frac{\omega}{2T} \right)$ for future purpose.

18.2.2 Keldysh action and the gap equation

We follow the standard procedure described in Ref. [Kamenev(2023)] to write the Keldysh action for the Hamiltonian in Eq. (18.15) and it is given by (in Nambu basis for electrons)

$$\begin{aligned}
S_N = & \int d\omega \sum_k \begin{bmatrix} \bar{c}_{+1}(k, \omega) & \bar{c}_{-1}(k, \omega) & \bar{c}_{+2}(k, \omega) & \bar{c}_{-2}(k, \omega) \end{bmatrix} [G_{N0}^{-1}(\Delta_{cl}, \Delta_q)] \begin{bmatrix} c_{+1}(k, \omega) \\ c_{-1}(k, \omega) \\ c_{+2}(k, \omega) \\ c_{-2}(k, \omega) \end{bmatrix} \\
& - \frac{1}{\kappa} [\Delta_{cl}^* \Delta_q + \Delta_q^* \Delta_{cl}] + S_L + S_{LM}, \text{ where} \\
S_L = & \int \frac{d\Omega}{2\pi} \sum_q L \begin{bmatrix} \phi_{cl}(q, \Omega) & \phi_q(q, \Omega) \end{bmatrix} \begin{bmatrix} 0 & (\Omega - i\gamma_B)^2 - \nu_q^2 \\ (\Omega + i\gamma_B)^2 - \nu_q^2 & 2i\gamma_B \Omega B_{T_{cav}}(\Omega) \end{bmatrix} \begin{bmatrix} \phi_{cl}(q, \Omega) \\ \phi_q(q, \Omega) \end{bmatrix} \text{ and} \\
S_{LM} = & \int \frac{d\omega}{2\pi} \frac{d\omega'}{2\pi} \sum_{\alpha=\{+,-\}} \sum_{k,k'} g_{k,k'}^\alpha \{ \phi_{cl}(k-k', \omega-\omega') [\bar{c}_{\alpha 1}(k, \omega) c_{\alpha 1}(k', \omega') + \bar{c}_{\alpha 2}(k, \omega) c_{\alpha 2}(k', \omega')] \\
& + \phi_q(k-k', \omega-\omega') [\bar{c}_{\alpha 1}(k, \omega) c_{\alpha 2}(k', \omega') + \bar{c}_{\alpha 2}(k, \omega) c_{\alpha 1}(k', \omega')] \}.
\end{aligned} \tag{18.29}$$

Here, $\{+, -\}$ are the Nambu indices and $\{1, 2\}$ are the Larkin-Ovchinnikov indices. The “free” fermionic inverse Green’s function G^{-1} is given by

$$[G_{N0}^{-1}(\Delta_{cl}, \Delta_q)] = \begin{bmatrix} \omega - \xi_k + i\gamma & -\Delta_{cl} & [G_0^{-1}]_{++}^K & -\Delta_q + [G_0^{-1}]_{+-}^K \\ -\Delta_{cl}^* & \omega + \xi_k + i\gamma & -\Delta_q^* + [G_0^{-1}]_{-+}^K & [G_0^{-1}]_{--}^K \\ 0 & -\Delta_q & \omega - \xi_k - i\gamma & -\Delta_{cl} \\ -\Delta_q^* & 0 & -\Delta_{cl}^* & \omega + \xi_k - i\gamma \end{bmatrix} \tag{18.30}$$

Note that we have assumed the static limit (zero frequency for the bosonic Δ fields). The distribution function that enters $[G^{-1}]_{\alpha\beta}^K$ and they can be non-thermal in general. It is useful to write this 4×4 matrix in terms of 2×2 blocks as

$$G_{N0}^{-1}(\Delta_{cl}, \Delta_q) = \begin{bmatrix} G_{N0}^{R-1} & G_{N0}^{K-1} + G_{N0}^{q-1} \\ G_{N0}^{q-1} & G_{N0}^{A-1} \end{bmatrix} \tag{18.31}$$

where we have in the Nambu basis

$$\begin{aligned}
G_{N0}^{R-1} &= \begin{bmatrix} \omega - \xi_k + i\gamma & -\Delta_{cl} \\ -\Delta_{cl}^* & \omega + \xi_k + i\gamma \end{bmatrix} & G_{N0}^{K-1} &= \begin{bmatrix} [G_0^{-1}]_{++}^K & [G_0^{-1}]_{+-}^K \\ [G_0^{-1}]_{-+}^K & [G_0^{-1}]_{--}^K \end{bmatrix} \\
G_{N0}^{q-1} &= \begin{bmatrix} 0 & -\Delta_q \\ -\Delta_q^* & 0 \end{bmatrix} & G_{N0}^{A-1} &= \begin{bmatrix} \omega - \xi_k - i\gamma & -\Delta_{cl} \\ -\Delta_{cl}^* & \omega + \xi_k - i\gamma \end{bmatrix}
\end{aligned} \tag{18.32}$$

At this point we do the Bogoliubov transformation given by (generalized for the additional Keldysh indices)

$$\begin{bmatrix} \psi_{p1}(k, \omega) \\ \psi_{m1}(k, \omega) \\ \psi_{p2}(k, \omega) \\ \psi_{m2}(k, \omega) \end{bmatrix} = \mathcal{U}(\Delta_{cl}, k) \begin{bmatrix} c_{+1}(k, \omega) \\ c_{-1}(k, \omega) \\ c_{+2}(k, \omega) \\ c_{-2}(k, \omega) \end{bmatrix} \tag{18.33}$$

where $\{p, m\}$ are Bogoliubov indices. We have

$$\mathcal{U} = \begin{bmatrix} U & 0_{2 \times 2} \\ 0_{2 \times 2} & U \end{bmatrix} \text{ and } U = \begin{bmatrix} u_k & v_k \\ -v_k^* & u_k^* \end{bmatrix} = \begin{bmatrix} \sqrt{\frac{1}{2} \left(1 + \frac{\xi_k}{E_k}\right)} & \frac{\Delta_{cl}}{|\Delta_{cl}|} \sqrt{\frac{1}{2} \left(1 - \frac{\xi_k}{E_k}\right)} \\ -\frac{\Delta_{cl}^*}{|\Delta_{cl}|} \sqrt{\frac{1}{2} \left(1 - \frac{\xi_k}{E_k}\right)} & \sqrt{\frac{1}{2} \left(1 + \frac{\xi_k}{E_k}\right)} \end{bmatrix} \tag{18.34}$$

where

$$E_k = \sqrt{\xi_k^2 + |\Delta_{\text{cl}}|^2}. \quad (18.35)$$

Under this transformation the action becomes

$$S_B = \int d\omega \sum_k \begin{bmatrix} \bar{\psi}_{p1}(k, \omega) & \bar{\psi}_{m1}(k, \omega) & \bar{\psi}_{p2}(k, \omega) & \bar{\psi}_{m2}(k, \omega) \end{bmatrix} [G_0^{-1}(\Delta_{\text{cl}}, \Delta_{\text{q}})] \begin{bmatrix} \psi_{p1}(k, \omega) \\ \psi_{m1}(k, \omega) \\ \psi_{p2}(k, \omega) \\ \psi_{m2}(k, \omega) \end{bmatrix} \\ - \frac{1}{\kappa} [\Delta_{\text{cl}}^* \Delta_{\text{q}} + \Delta_{\text{q}}^* \Delta_{\text{cl}}] + S_L + S_{LM}, \quad (18.36)$$

where

$$G_0^{-1}(\Delta_{\text{cl}}, \Delta_{\text{q}}) = \begin{bmatrix} G_0^{R-1} & G_0^{K-1} + G_0^{q-1} \\ G_0^{q-1} & G_0^{A-1} \end{bmatrix} = \mathcal{U} G_{N0}^{-1} \mathcal{U}^\dagger. \quad (18.37)$$

The 2×2 blocks

$$G_0^{R-1} = U G_{N0}^{R-1} U^\dagger = \begin{bmatrix} \omega - E_k + i\gamma & 0 \\ 0 & \omega + E_k + i\gamma \end{bmatrix} = [G_0^{A-1}]^\dagger, \quad (18.38)$$

$$G_0^{K-1} = U G_{N0}^{K-1} U^\dagger = \begin{bmatrix} 2i\gamma F_0(\omega) & 0 \\ 0 & 2i\gamma F_0(\omega) \end{bmatrix} \quad (18.39)$$

and

$$G_0^{q-1} = U G_{N0}^{q-1} U^\dagger = \begin{bmatrix} -u_k v_k^* \Delta_{\text{q}} - u_k^* v_k \Delta_{\text{q}}^* & -u_k^2 \Delta_{\text{q}} + v_k^2 \Delta_{\text{q}}^* \\ + (v_k^*)^2 \Delta_{\text{q}} - (u_k^*)^2 \Delta_{\text{q}}^* & u_k v_k^* \Delta_{\text{q}} + u_k^* v_k \Delta_{\text{q}}^* \end{bmatrix}. \quad (18.40)$$

The light-matter coupling in this rotated basis

$$S_{LM} = \int \frac{d\omega}{2\pi} \frac{d\omega'}{2\pi} \sum_{\alpha, \beta = \{p, m\}} \sum_{k, k'} \tilde{g}_{k, k'}^{\alpha\beta} \{ \phi_{\text{cl}}(k - k', \omega - \omega') [\bar{\psi}_{\alpha 1}(k, \omega) \psi_{\beta 1}(k', \omega') + \bar{\psi}_{\alpha 2}(k, \omega) \psi_{\beta 2}(k', \omega')] \\ + \phi_{\text{q}}(k - k', \omega - \omega') [\bar{\psi}_{\alpha 1}(k, \omega) \psi_{\beta 2}(k', \omega') + \bar{\psi}_{\alpha 2}(k, \omega) \psi_{\beta 1}(k', \omega')] \} \quad (18.41)$$

where

$$\tilde{g}_{k, k'}^{pp} = g_{k, k'}^+ u_k u_{k'}^* + g_{k, k'}^- v_k v_{k'}^* = \frac{g_{k, k'}}{2} \left[\sqrt{\left(1 + \frac{\xi_k}{E_k}\right) \left(1 + \frac{\xi_{k'}}{E_{k'}}\right)} + \eta \sqrt{\left(1 - \frac{\xi_k}{E_k}\right) \left(1 - \frac{\xi_{k'}}{E_{k'}}\right)} \right] \\ \tilde{g}_{k, k'}^{pm} = -g_{k, k'}^+ u_k v_{k'} + g_{k, k'}^- v_k u_{k'} = -\frac{g_{k, k'}}{2} \frac{\Delta_{\text{cl}}}{|\Delta_{\text{cl}}|} \left[\sqrt{\left(1 + \frac{\xi_k}{E_k}\right) \left(1 - \frac{\xi_{k'}}{E_{k'}}\right)} - \eta \sqrt{\left(1 - \frac{\xi_k}{E_k}\right) \left(1 + \frac{\xi_{k'}}{E_{k'}}\right)} \right] \\ \tilde{g}_{k, k'}^{mp} = -g_{k, k'}^+ v_k^* u_{k'}^* + g_{k, k'}^- u_k^* v_{k'}^* = -\frac{g_{k, k'}}{2} \frac{\Delta_{\text{cl}}^*}{|\Delta_{\text{cl}}|} \left[\sqrt{\left(1 - \frac{\xi_k}{E_k}\right) \left(1 + \frac{\xi_{k'}}{E_{k'}}\right)} - \eta \sqrt{\left(1 + \frac{\xi_k}{E_k}\right) \left(1 - \frac{\xi_{k'}}{E_{k'}}\right)} \right] \\ \tilde{g}_{k, k'}^{mm} = g_{k, k'}^+ v_k^* v_{k'} + g_{k, k'}^- u_k^* u_{k'} = \frac{g_{k, k'}}{2} \left[\sqrt{\left(1 - \frac{\xi_k}{E_k}\right) \left(1 - \frac{\xi_{k'}}{E_{k'}}\right)} + \eta \sqrt{\left(1 + \frac{\xi_k}{E_k}\right) \left(1 + \frac{\xi_{k'}}{E_{k'}}\right)} \right]. \quad (18.42)$$

Now, we will treat the light-matter coupling in a mean-field way. This will induce self energy to the fermions of the form

$$\Sigma = \begin{bmatrix} \Sigma^R & \Sigma^K \\ \Sigma^q & \Sigma^A \end{bmatrix} \quad (18.43)$$

where $\Sigma^{R/A/K/q}$ is 2×2 matrix in the Bogoliubov basis. So, the action becomes

$$S_B \approx \int d\omega \sum_k [\bar{\psi}_{p1}(k, \omega) \quad \bar{\psi}_{m1}(k, \omega) \quad \bar{\psi}_{p2}(k, \omega) \quad \bar{\psi}_{m2}(k, \omega)] [G^{-1}(\Delta_{cl}, \Delta_q)] \begin{bmatrix} \psi_{p1}(k, \omega) \\ \psi_{m1}(k, \omega) \\ \psi_{p2}(k, \omega) \\ \psi_{m2}(k, \omega) \end{bmatrix} - \frac{1}{\kappa} [\Delta_{cl}^* \Delta_q + \Delta_q^* \Delta_{cl}] + S_L, \quad (18.44)$$

where

$$G^{-1}(\Delta_{cl}, \Delta_q) = G_0^{-1}(\Delta_{cl}, \Delta_q) - \Sigma \quad (18.45)$$

Integrating out the fermions one gets the effective action

$$\tilde{S}_B = -i \text{Tr} [\ln \{-iG^{-1}(\Delta_{cl}, \Delta_q)\}] - \frac{1}{\kappa} [\Delta_{cl}^* \Delta_q + \Delta_q^* \Delta_{cl}] + S_L \quad (18.46)$$

Classical saddle point equations for this action can be found out by setting

$$\left. \frac{\delta \tilde{S}_B}{\delta \Delta_q} \right|_{\Delta_q=0} = 0 \quad \text{and} \quad \left. \frac{\delta \tilde{S}_B}{\delta \Delta_q^*} \right|_{\Delta_q=0} = 0. \quad (18.47)$$

So, we get

$$-i \text{Tr} \left[iG(\Delta_{cl}, \Delta_q = 0) \frac{\delta \{-iG^{-1}(\Delta_{cl}, \Delta_q)\}}{\delta \Delta_q} \right]_{\Delta_q=0} - \frac{1}{\kappa} \Delta_{cl}^* = 0. \quad (18.48)$$

Note that it is okay to consider only one of the equations as they are Hermitian conjugate of each other and at the end produce the same equation. To evaluate these equations we need $G(\Delta_{cl}, \Delta_q = 0)$, which can be found out by inverting

$$G^{-1}(\Delta_{cl}, \Delta_q = 0) = \begin{bmatrix} G^{R-1} & [G^{-1}]^K \\ 0 & G^{A-1} \end{bmatrix} = \begin{bmatrix} G_0^{R-1} - \Sigma^R(\Delta_{cl}, 0) & [G_0^{-1}]^K - \Sigma^K(\Delta_{cl}, 0) \\ 0 & G_0^{A-1} - \Sigma^A(\Delta_{cl}, 0) \end{bmatrix}, \quad (18.49)$$

where $G_0^{R-1} = \begin{bmatrix} \omega - E_k + i\gamma & 0 \\ 0 & \omega + E_k + i\gamma \end{bmatrix} = [G_0^{A-1}(\Delta_{cl})]^\dagger$.

By inverting the free part we get,

$$G_0(\Delta_{cl}) = [G_0(\Delta_{cl}, \Delta_q = 0)] = \begin{bmatrix} G_0^R(\Delta_{cl}) & G_0^K(\Delta_{cl}) \\ 0 & G_0^A(\Delta_{cl}) \end{bmatrix} \quad (18.50)$$

where

$$G_0^R = [G_0^A]^\dagger = \begin{bmatrix} G_{0pp}^R & G_{0pm}^R \\ G_{0mp}^R & G_{0mm}^R \end{bmatrix} = \begin{bmatrix} \frac{1}{\omega - E_k + i\gamma} & 0 \\ 0 & \frac{1}{\omega + E_k + i\gamma} \end{bmatrix}, \quad (18.51)$$

and

$$G_0^K = [G_0^R - G_0^A]F_0 = \begin{bmatrix} G_{0pp}^K & G_{0pm}^K \\ G_{0mp}^K & G_{0mm}^K \end{bmatrix} = \begin{bmatrix} 2i \text{Im} G_{0pp}^R F_0 & 0 \\ 0 & 2i \text{Im} G_{0mm}^R F_0 \end{bmatrix}. \quad (18.52)$$

In general we will have

$$G(\Delta_{cl}, \Delta_q = 0) = \begin{bmatrix} G^R(\Delta_{cl}) & G^K(\Delta_{cl}) \\ 0 & G^A(\Delta_{cl}) \end{bmatrix} = \begin{bmatrix} G_0^R + G_0^R \Sigma^R G_0^R & G^K \\ 0 & G_0^A + G_0^A \Sigma^A G_0^A \end{bmatrix}, \quad (18.53)$$

For the Keldysh Green's function we use the standard steady-state ansatz

$$\begin{aligned}
 G^K &= \begin{bmatrix} G_{pp}^K & G_{pm}^K \\ G_{mp}^K & G_{mm}^K \end{bmatrix} = G^R F - F G^A = \begin{bmatrix} G_{pp}^R & G_{pm}^R \\ G_{mp}^R & G_{mm}^R \end{bmatrix} \begin{bmatrix} F_{pp} & F_{pm} \\ F_{mp} & F_{mm} \end{bmatrix} - \begin{bmatrix} F_{pp} & F_{pm} \\ F_{mp} & F_{mm} \end{bmatrix} \begin{bmatrix} G_{pp}^A & G_{pm}^A \\ G_{mp}^A & G_{mm}^A \end{bmatrix} \\
 &= \begin{bmatrix} (G_{pp}^R - G_{pp}^A)F_{pp} + G_{pm}^R F_{mp} - F_{pm} G_{mp}^A & G_{pm}^R F_{mm} - F_{pp} G_{pm}^A + (G_{pp}^R - G_{mm}^A)F_{pm} \\ G_{mp}^R F_{pp} - F_{mm} G_{mp}^A + (G_{mm}^R - G_{pp}^A)F_{mp} & (G_{mm}^R - G_{mm}^A)F_{mm} + G_{mp}^R F_{pm} - F_{mp} G_{pm}^A \end{bmatrix}
 \end{aligned} \quad (18.54)$$

Now using the similar notation of breaking up the 4×4 matrix into 2×2 blocks, we get

$$\frac{\delta \{-iG^{-1}(\Delta_{\text{cl}}, \Delta_q)\}}{\delta \Delta_q} \Big|_{\Delta_q=0} = -i \begin{bmatrix} 0 & \frac{\delta G_0^{q-1}}{\delta \Delta_q} \Big|_{\Delta_q=0} \\ \frac{\delta G_0^{q-1}}{\delta \Delta_q} \Big|_{\Delta_q=0} & 0 \end{bmatrix} + i \begin{bmatrix} \frac{\delta \Sigma^R}{\delta \Delta_q} \Big|_{\Delta_q=0} & \frac{\delta \Sigma^K}{\delta \Delta_q} \Big|_{\Delta_q=0} \\ \frac{\delta \Sigma^q}{\delta \Delta_q} \Big|_{\Delta_q=0} & \frac{\delta \Sigma^A}{\delta \Delta_q} \Big|_{\Delta_q=0} \end{bmatrix} \quad (18.55)$$

where

$$\begin{aligned}
 \frac{\delta G_0^{q-1}}{\delta \Delta_q} \Big|_{\Delta_q=0} &= \begin{bmatrix} -u_k v_k^* & -u_k^2 \\ (v_k^*)^2 & u_k v_k^* \end{bmatrix} = \begin{bmatrix} -\frac{\Delta_{\text{cl}}^*}{2E_k} & -\frac{1}{2} \left(1 + \frac{\xi_k}{E_k}\right) \\ \frac{(\Delta_{\text{cl}}^*)^2}{2|\Delta_{\text{cl}}|^2} \left(1 - \frac{\xi_k}{E_k}\right) & \frac{\Delta_{\text{cl}}^*}{2E_k} \end{bmatrix} \quad \text{and} \\
 \frac{\delta G_0^{q-1}}{\delta \Delta_q^*} \Big|_{\Delta_q=0} &= \begin{bmatrix} -u_k^* v_k & v_k^2 \\ -(u_k^*)^2 & u_k^* v_k \end{bmatrix} = \begin{bmatrix} -\frac{\Delta_{\text{cl}}}{2E_k} & \frac{(\Delta_{\text{cl}})^2}{2|\Delta_{\text{cl}}|^2} \left(1 - \frac{\xi_k}{E_k}\right) \\ -\frac{1}{2} \left(1 + \frac{\xi_k}{E_k}\right) & \frac{\Delta_{\text{cl}}}{2E_k} \end{bmatrix}
 \end{aligned} \quad (18.56)$$

Note that we have also considered a Σ^q that is absent in the standard Keldysh structure and vanishes when $\Delta_q = 0$.

So, Eq. (18.48) produces

$$-i \text{Tr} \left[G^K \frac{\delta G_0^{q-1}}{\delta \Delta_q} \Big|_{\Delta_q=0} \right] + i \text{Tr} [G^R \bar{\Sigma}^R + G^A \bar{\Sigma}^A + G^K \bar{\Sigma}^q] = \frac{\Delta_{\text{cl}}^*}{\kappa}. \quad (18.57)$$

Here, we have used the short-hand notation $\bar{\Sigma}^{(R/A/K/q)} = \frac{\delta \Sigma^{(R/A/K/q)}}{\delta \Delta_q} \Big|_{\Delta_q=0}$ and $\bar{\Sigma}^{*(R/A/K/q)} = \frac{\delta \Sigma^{*(R/A/K/q)}}{\delta \Delta_q^*} \Big|_{\Delta_q=0}$.

Writing the components of G^K explicitly for the first part we get

$$i \sum_k \int \frac{d\omega}{2\pi} [u_k v_k^* (G_{pp}^K - G_{mm}^K) - (v_k^*)^2 G_{pm}^K + u_k^2 G_{mp}^K] + i \text{Tr} [G^R \bar{\Sigma}^R + G^A \bar{\Sigma}^A + G^K \bar{\Sigma}^q] = \frac{\Delta_{\text{cl}}^*}{\kappa}. \quad (18.58)$$

Using the parametrization of G^K in Eq. (18.54), we get

$$\begin{aligned}
 i \sum_k \int \frac{d\omega}{2\pi} &\{ [u_k v_k^* (G_{pp}^R - G_{pp}^A) + (v_k^*)^2 G_{pm}^A + u_k^2 G_{mp}^R] F_{pp} + [-u_k v_k^* (G_{mm}^R - G_{mm}^A) - (v_k^*)^2 G_{pm}^R - u_k^2 G_{mp}^A] F_{pm} \\
 &+ [-u_k v_k^* (G_{mp}^R + G_{mp}^A) - (v_k^*)^2 (G_{pp}^R - G_{mm}^A)] F_{pm} + [u_k v_k^* (G_{pm}^R + G_{pm}^A) + (u_k)^2 (G_{mm}^R - G_{pp}^A)] F_{pp} \} \\
 &+ i \sum_k \int \frac{d\omega}{2\pi} \sum_{\alpha, \beta=\{p,m\}} \left[G_{\alpha\beta}^R \bar{\Sigma}_{\beta\alpha}^R + G_{\alpha\beta}^A \bar{\Sigma}_{\beta\alpha}^A + \sum_{\lambda=\{p,m\}} (G_{\alpha\lambda}^R F_{\lambda\beta} - F_{\alpha\lambda} G_{\lambda\beta}^A) \bar{\Sigma}_{\beta\alpha}^q \right] = \frac{\Delta_{\text{cl}}^*}{\kappa}.
 \end{aligned} \quad (18.59)$$

This is a general gap equation that does not assume anything about the fermion dispersion (or the Green's functions) and the distribution other than the steady state ansatz.

18.2.3 Equilibrium gap equation: equivalence of SC and CDW

In thermal equilibrium (decoupled from the cavity: $\Sigma = 0$, $G_{pm}^{R/A} = G_{mp}^{R/A} = 0$) we have $F_{pm} = F_{mp} = 0$ and $F_{pp} = F_{mm} = F_0 = \tanh\left(\frac{\omega}{2T}\right)$. So, from eq. (18.59) we get

$$i \sum_{\mathbf{k}} \int \frac{d\omega}{2\pi} u_{\mathbf{k}} v_{\mathbf{k}}^* [2i \text{Im} G_{0pp}^R - 2i \text{Im} G_{0mm}^R] F_0 = \frac{\Delta_{\text{cl}}^*}{\kappa} \quad (18.60)$$

or, explicitly,

$$-2 \sum_{\mathbf{k}} \int \frac{d\omega}{2\pi} \frac{\Delta_{\text{cl}}^*}{2E_{\mathbf{k}}} \left[-\frac{\gamma}{(\omega - E_{\mathbf{k}})^2 + \gamma^2} + \frac{\gamma}{(\omega + E_{\mathbf{k}})^2 + \gamma^2} \right] F_0(\omega) = \frac{\Delta_{\text{cl}}^*}{\kappa}. \quad (18.61)$$

In the $\gamma \rightarrow 0$ limit we get

$$\sum_{\mathbf{k}} \frac{F_0(E_{\mathbf{k}})}{E_{\mathbf{k}}} = \frac{1}{\kappa}. \quad (18.62)$$

This is the standard gap equation at thermal equilibrium.

Let us now show that this gap equation is identical for SC and CDW. We define the notation $\Delta = \Delta_{\text{cl}}$ as this is the physical gap. We have $E_{\mathbf{k}} = \sqrt{\xi_{\mathbf{k}}^2 + |\Delta|^2} = \sqrt{J^2 \cos^2(ka) + |\Delta|^2}$ for SC and $E_{\mathbf{k}} = \sqrt{\xi_{\mathbf{k}}^2 + |\Delta|^2} = \sqrt{J^2 \sin^2(ka) + |\Delta|^2}$ for CDW. Hence the gap equations

$$\int_{-\pi/a}^{\pi/a} \frac{dk}{2\pi} \frac{F_0\left(\sqrt{J^2 \cos^2(ka) + |\Delta|^2}\right)}{\sqrt{J^2 \cos^2(ka) + |\Delta|^2}} = \frac{1}{\kappa} \text{ for SC, and } \int_{-\pi/2a}^{\pi/2a} \frac{dk}{\pi} \frac{F_0\left(\sqrt{J^2 \sin^2(ka) + |\Delta|^2}\right)}{\sqrt{J^2 \sin^2(ka) + |\Delta|^2}} = \frac{1}{\kappa} \text{ for CDW.} \quad (18.63)$$

The SC gap equation can be easily mapped to the CDW gap equation by a change of variables $k \rightarrow k \pm k_F$ where $k_F = \pi/2a$. Note that we can do this because we considered a half-filled free electron band $\epsilon_{\mathbf{k}} = -J \cos(ka)$ that results in the symmetry $\epsilon_{k+k_F} = -\epsilon_{-k+k_F}$. This is a consequence of the symmetry of H_{el} under the particle-hole transformation $\hat{c}_{\sigma}(k \pm k_F) \rightarrow \hat{c}_{\sigma}^{\dagger}(-k \pm k_F)$. In the Nambu structure this transformation translates to $\hat{c}_{\alpha}(k \pm k_F) \rightarrow \hat{c}_{\alpha}^{\dagger}(-k \pm k_F)$ for SC and $\hat{c}_{\alpha}(k) \rightarrow \hat{c}_{\alpha}^{\dagger}(-k)$ for CDW. Let us show that H_{el} is invariant under this

transformation. For SC, we have

$$\begin{aligned}
 H_{\text{el}} &= \int_{-\pi/a}^{\pi/a} \frac{dk}{2\pi} \left[\xi_k \hat{c}_+^\dagger(k) \hat{c}_+(k) - \xi_k \hat{c}_-^\dagger(k) \hat{c}_-(k) + \Delta \hat{c}_+^\dagger(k) \hat{c}_-(k) + \Delta^* \hat{c}_-^\dagger(k) \hat{c}_+(k) \right] \\
 &= \int_{-\pi/2a}^{\pi/2a} \frac{dk}{2\pi} \left[\xi_{k+k_F} \hat{c}_+^\dagger(k+k_F) \hat{c}_+(k+k_F) - \xi_{k+k_F} \hat{c}_-^\dagger(k+k_F) \hat{c}_-(k+k_F) \right. \\
 &\quad \left. + \Delta \hat{c}_+^\dagger(k+k_F) \hat{c}_-(k+k_F) + \Delta^* \hat{c}_-^\dagger(k+k_F) \hat{c}_+(k+k_F) \right] \\
 &\quad + \int_{-\pi/2a}^{\pi/2a} \frac{dk}{2\pi} \left[\xi_{k-k_F} \hat{c}_+^\dagger(k-k_F) \hat{c}_+(k-k_F) - \xi_{k-k_F} \hat{c}_-^\dagger(k-k_F) \hat{c}_-(k-k_F) \right. \\
 &\quad \left. + \Delta \hat{c}_+^\dagger(k-k_F) \hat{c}_-(k-k_F) + \Delta^* \hat{c}_-^\dagger(k-k_F) \hat{c}_+(k-k_F) \right] \\
 &\rightarrow \tilde{H}_{\text{el}} = \int_{-\pi/2a}^{\pi/2a} \frac{dk}{2\pi} \left[\xi_{k+k_F} \hat{c}_+(-k+k_F) \hat{c}_+^\dagger(-k+k_F) - \xi_{k+k_F} \hat{c}_-(-k+k_F) \hat{c}_-^\dagger(-k+k_F) \right. \\
 &\quad \left. + \Delta^* \hat{c}_+(-k+k_F) \hat{c}_-^\dagger(-k+k_F) + \Delta \hat{c}_-(-k+k_F) \hat{c}_+^\dagger(-k+k_F) \right] \\
 &\quad + \int_{-\pi/2a}^{\pi/2a} \frac{dk}{2\pi} \left[\xi_{k-k_F} \hat{c}_+(-k-k_F) \hat{c}_+^\dagger(-k-k_F) - \xi_{k-k_F} \hat{c}_-(-k-k_F) \hat{c}_-^\dagger(-k-k_F) \right. \\
 &\quad \left. + \Delta^* \hat{c}_+(-k-k_F) \hat{c}_-^\dagger(-k-k_F) + \Delta \hat{c}_-(-k-k_F) \hat{c}_+^\dagger(-k-k_F) \right] \\
 &= \int_{-\pi/2a}^{\pi/2a} \frac{dk}{2\pi} \left[\xi_{k+k_F} \hat{c}_+^\dagger(k+k_F) \hat{c}_+(k+k_F) - \xi_{k+k_F} \hat{c}_-^\dagger(k+k_F) \hat{c}_-(k+k_F) \right. \\
 &\quad \left. - \Delta \hat{c}_+^\dagger(k+k_F) \hat{c}_-(k+k_F) - \Delta^* \hat{c}_-^\dagger(k+k_F) \hat{c}_+(k+k_F) \right] \\
 &\quad + \int_{-\pi/2a}^{\pi/2a} \frac{dk}{2\pi} \left[\xi_{k-k_F} \hat{c}_+^\dagger(k-k_F) \hat{c}_+(k-k_F) - \xi_{k-k_F} \hat{c}_-^\dagger(k-k_F) \hat{c}_-(k-k_F) \right. \\
 &\quad \left. - \Delta \hat{c}_+^\dagger(k-k_F) \hat{c}_-(k-k_F) - \Delta^* \hat{c}_-^\dagger(k-k_F) \hat{c}_+(k-k_F) \right] \\
 &= \int_{-\pi/2a}^{\pi/2a} \frac{dk}{2\pi} \left[\xi_{k+k_F} \hat{c}_+^\dagger(k+k_F) \hat{c}_+(k+k_F) - \xi_{k+k_F} \hat{c}_-^\dagger(k+k_F) \hat{c}_-(k+k_F) \right. \\
 &\quad \left. + \Delta \hat{c}_+^\dagger(k+k_F) \hat{c}_-(k+k_F) + \Delta^* \hat{c}_-^\dagger(k+k_F) \hat{c}_+(k+k_F) \right] \\
 &\quad + \int_{-\pi/2a}^{\pi/2a} \frac{dk}{2\pi} \left[\xi_{k-k_F} \hat{c}_+^\dagger(k-k_F) \hat{c}_+(k-k_F) - \xi_{k-k_F} \hat{c}_-^\dagger(k-k_F) \hat{c}_-(k-k_F) \right. \\
 &\quad \left. + \Delta \hat{c}_+^\dagger(k-k_F) \hat{c}_-(k-k_F) + \Delta^* \hat{c}_-^\dagger(k-k_F) \hat{c}_+(k-k_F) \right] = H_{\text{el}}
 \end{aligned} \tag{18.64}$$

Note that we used the fact that $\xi_{-k \pm k_F} = -\xi_{k \pm k_F}$. We also absorbed a minus sign in Δ as it

has a gauge symmetry and the gap equation only depends on $|\Delta|^2$. For CDW we have instead

$$\begin{aligned}
 H_{\text{el}} &= \int_{-\pi/2a}^{\pi/2a} \frac{dk}{\pi} \left[\xi_k \hat{c}_+^\dagger(k) \hat{c}_+(k) - \xi_k \hat{c}_-^\dagger(k) \hat{c}_-(k) + \Delta \hat{c}_+^\dagger(k) \hat{c}_-(k) + \Delta^* \hat{c}_-^\dagger(k) \hat{c}_+(k) \right] \\
 &\rightarrow \tilde{H}_{\text{el}} = \int_{-\pi/2a}^{\pi/2a} \frac{dk}{\pi} \left[\xi_k \hat{c}_+(-k) \hat{c}_+^\dagger(-k) - \xi_k \hat{c}_-(-k) \hat{c}_-^\dagger(-k) + \Delta^* \hat{c}_+(-k) \hat{c}_-^\dagger(-k) + \Delta \hat{c}_-(-k) \hat{c}_+^\dagger(-k) \right] \\
 &= \int_{-\pi/2a}^{\pi/2a} \frac{dk}{\pi} \left[\xi_k \hat{c}_+^\dagger(k) \hat{c}_+(k) - \xi_k \hat{c}_-^\dagger(k) \hat{c}_-(k) - \Delta \hat{c}_+^\dagger(k) \hat{c}_-(k) - \Delta^* \hat{c}_-^\dagger(k) \hat{c}_+(k) \right] \\
 &= \int_{-\pi/2a}^{\pi/2a} \frac{dk}{\pi} \left[\xi_k \hat{c}_+^\dagger(k) \hat{c}_+(k) - \xi_k \hat{c}_-^\dagger(k) \hat{c}_-(k) + \Delta \hat{c}_+^\dagger(k) \hat{c}_-(k) + \Delta^* \hat{c}_-^\dagger(k) \hat{c}_+(k) \right] = H_{\text{el}}
 \end{aligned} \tag{18.65}$$

Here we used the fact that $\xi_{-k} = -\xi_k$. So, under this transformation H_{el} stays invariant.

Now let us see how $H_{\text{el-ph}}$ behaves under this transformation. For SC, we have

$$\begin{aligned}
 H_{\text{el-ph}} &= \sum_{\alpha=\{+,-\}} \int_{-\pi/a}^{\pi/a} \frac{dk}{2\pi} \int_{-\pi/a}^{\pi/a} \frac{dk'}{2\pi} g_{k,k'}^\alpha \hat{c}_\alpha^\dagger(k) \hat{c}_\alpha(k') \hat{\phi}(k-k') \\
 &= \sum_{\alpha=\{+,-\}} \int_{-\pi/2a}^{\pi/2a} \frac{dk}{2\pi} \int_{-\pi/2a}^{\pi/2a} \frac{dk'}{2\pi} \left\{ [g_{k+k_F,k'+k_F}^\alpha \hat{c}_\alpha^\dagger(k+k_F) \hat{c}_\alpha(k'+k_F) + g_{k-k_F,k'-k_F}^\alpha \hat{c}_\alpha^\dagger(k-k_F) \hat{c}_\alpha(k'-k_F)] \hat{\phi}(k-k') \right. \\
 &\quad \left. + g_{k+k_F,k'-k_F}^\alpha \hat{c}_\alpha^\dagger(k+k_F) \hat{c}_\alpha(k'-k_F) \hat{\phi}(k-k'+2k_F) + g_{k-k_F,k'+k_F}^\alpha \hat{c}_\alpha^\dagger(k-k_F) \hat{c}_\alpha(k'+k_F) \hat{\phi}(k-k'-2k_F) \right\} \\
 &\rightarrow \tilde{H}_{\text{el-ph}} = \sum_{\alpha=\{+,-\}} \int_{-\pi/2a}^{\pi/2a} \frac{dk}{2\pi} \int_{-\pi/2a}^{\pi/2a} \frac{dk'}{2\pi} \left\{ g_{k+k_F,k'+k_F}^\alpha \hat{c}_\alpha^\dagger(-k+k_F) \hat{c}_\alpha^\dagger(-k'+k_F) \hat{\phi}(k-k') \right. \\
 &\quad \left. + g_{k-k_F,k'-k_F}^\alpha \hat{c}_\alpha^\dagger(-k-k_F) \hat{c}_\alpha^\dagger(-k'-k_F) \hat{\phi}(k-k') \right. \\
 &\quad \left. + g_{k+k_F,k'-k_F}^\alpha \hat{c}_\alpha^\dagger(-k+k_F) \hat{c}_\alpha^\dagger(-k'-k_F) \hat{\phi}(k-k'+2k_F) \right. \\
 &\quad \left. + g_{k-k_F,k'+k_F}^\alpha \hat{c}_\alpha^\dagger(-k-k_F) \hat{c}_\alpha^\dagger(-k'+k_F) \hat{\phi}(k-k'-2k_F) \right\} \\
 &= \sum_{\alpha=\{+,-\}} \int_{-\pi/2a}^{\pi/2a} \frac{dk}{2\pi} \int_{-\pi/2a}^{\pi/2a} \frac{dk'}{2\pi} \left\{ - [g_{-k'+k_F,-k+k_F}^\alpha \hat{c}_\alpha^\dagger(k+k_F) \hat{c}_\alpha(k'+k_F) + g_{-k'-k_F,-k-k_F}^\alpha \hat{c}_\alpha^\dagger(k-k_F) \hat{c}_\alpha(k'-k_F) \right. \\
 &\quad \left. - g_{-k'+k_F,-k-k_F}^\alpha \hat{c}_\alpha^\dagger(k+k_F) \hat{c}_\alpha(k'-k_F) \hat{\phi}(k-k'-2k_F) - g_{-k'-k_F,-k+k_F}^\alpha \hat{c}_\alpha^\dagger(k-k_F) \hat{c}_\alpha(k'+k_F) \right. \\
 &= \sum_{\alpha=\{+,-\}} \int_{-\pi/2a}^{\pi/2a} \frac{dk}{2\pi} \int_{-\pi/2a}^{\pi/2a} \frac{dk'}{2\pi} \left\{ - [g_{k+k_F,k'+k_F}^\alpha \hat{c}_\alpha^\dagger(k+k_F) \hat{c}_\alpha(k'+k_F) + g_{k'-k_F,k-k_F}^\alpha \hat{c}_\alpha^\dagger(k-k_F) \hat{c}_\alpha(k'-k_F)] \right. \\
 &\quad \left. + g_{k+k_F,k'-k_F}^\alpha \hat{c}_\alpha^\dagger(k+k_F) \hat{c}_\alpha(k'-k_F) \hat{\phi}(k-k'-2k_F) - g_{k-k_F,k'+k_F}^\alpha \hat{c}_\alpha^\dagger(k-k_F) \hat{c}_\alpha(k'+k_F) \hat{\phi}(k-k'+2k_F) \right\} \\
 &\neq H_{\text{el-ph}}.
 \end{aligned} \tag{18.66}$$

Here we have used the fact that $g_{-k+k_F,-k'+k_F}^\alpha = g_{k+k_F,k'+k_F}^\alpha$, $g_{-k-k_F,-k'-k_F}^\alpha = g_{k-k_F,k'-k_F}^\alpha$, $g_{-k+k_F,-k'-k_F}^\alpha = -g_{k+k_F,k'-k_F}^\alpha$ and $g_{-k-k_F,-k'+k_F}^\alpha = -g_{k-k_F,k'+k_F}^\alpha$.

For CDW we have

$$\begin{aligned}
 H_{\text{el-ph}} &= \sum_{\alpha=\{+,-\}} \int_{-\pi/2a}^{\pi/2a} \frac{dk}{\pi} \int_{-\pi/2a}^{\pi/2a} \frac{dk'}{\pi} g_{k,k'}^{\alpha} \hat{c}_{\alpha}^{\dagger}(k) \hat{c}_{\alpha}(k') \hat{\phi}(k-k') \\
 &\rightarrow \tilde{H}_{\text{el-ph}} = \sum_{\alpha=\{+,-\}} \int_{-\pi/2a}^{\pi/2a} \frac{dk}{\pi} \int_{-\pi/2a}^{\pi/2a} \frac{dk'}{\pi} g_{k,k'}^{\alpha} \hat{c}_{\alpha}(-k) \hat{c}_{\alpha}^{\dagger}(-k') \hat{\phi}(k-k') \\
 &= \sum_{\alpha=\{+,-\}} \int_{-\pi/2a}^{\pi/2a} \frac{dk}{\pi} \int_{-\pi/2a}^{\pi/2a} \frac{dk'}{\pi} (-1) g_{-k',-k}^{\alpha} \hat{c}_{\alpha}^{\dagger}(k) \hat{c}_{\alpha}(k') \hat{\phi}(k-k') \\
 &= \sum_{\alpha=\{+,-\}} \int_{-\pi/2a}^{\pi/2a} \frac{dk}{\pi} \int_{-\pi/2a}^{\pi/2a} \frac{dk'}{\pi} (-1) g_{k,k'}^{\alpha} \hat{c}_{\alpha}^{\dagger}(k) \hat{c}_{\alpha}(k') \hat{\phi}(k-k') \\
 &\neq H_{\text{el-ph}}.
 \end{aligned} \tag{18.67}$$

So, $H_{\text{el-ph}}$ is not invariant under the particle-hole transformation. So, SC and CDW cannot be mapped into each other when they are coupled to the cavity photons. Hence, we expect to observe different out-of-equilibrium behavior for them.

18.3 SM: Material in optical cavity

The gap equation in Eq. (18.59) depends on the specifics of the distribution function. We put the material in an optical cavity and use the resultant non-thermal distribution function in the gap equation. In the weak light-matter coupling strength (g_0) we do a perturbative calculation of $\mathcal{O}(g_0^2)$. So, we derive the self-energies of the electrons and the non-thermal distribution using quantum kinetic equations (QKE) due to coupling to the cavity before deriving the non-thermal gap equation.

18.3.1 Quantum Kinetic Equations

From the gap equation in Eq. (18.59), it is clear that we need to calculate the self-energies of the electrons due to the light-matter coupling. In this section we do that before deriving the non-thermal gap equation.

Let's re-derive QKE assuming a full F matrix in the Bogoliubov basis. We have [Kamenev(2023), Flores-Calderón *et al.*(2025)]

$$[G_0^{-1}]^K - \left(G_0^{R^{-1}} \cdot F - F \cdot G_0^{A^{-1}} \right) = \Sigma^K - (\Sigma^R \cdot F - F \cdot \Sigma^A) \tag{18.68}$$

where

$$\begin{aligned}
 \text{L.H.S.} &= [G_0^{-1}]^K - \left(G_0^{R^{-1}} \cdot F - F \cdot G_0^{A^{-1}} \right) \\
 &= \begin{bmatrix} 2i\gamma F_0 & 0 \\ 0 & 2i\gamma F_0 \end{bmatrix} - \left(\begin{bmatrix} \omega - E_k + i\gamma & 0 \\ 0 & \omega + E_k + i\gamma \end{bmatrix} \cdot \begin{bmatrix} F_{pp} & F_{pm} \\ F_{mp} & F_{mm} \end{bmatrix} - h.c. \right) \\
 &= \begin{bmatrix} 2i\gamma(F_0 - F_{pp}) & 2(E_k - i\gamma)F_{pm} \\ -2(E_k + i\gamma)F_{mp} & 2i\gamma(F_0 - F_{mm}) \end{bmatrix}
 \end{aligned} \tag{18.69}$$

and

R.H.S.

$$\begin{aligned}
 &= \Sigma^K - (\Sigma^R \cdot F - F \cdot \Sigma^A) \\
 &= \begin{bmatrix} \Sigma_{pp}^K & \Sigma_{pm}^K \\ \Sigma_{mp}^K & \Sigma_{mm}^K \end{bmatrix} - \left(\begin{bmatrix} \Sigma_{pp}^R & \Sigma_{pm}^R \\ \Sigma_{mp}^R & \Sigma_{mm}^R \end{bmatrix} \cdot \begin{bmatrix} F_{pp} & F_{pm} \\ F_{mp} & F_{mm} \end{bmatrix} - \begin{bmatrix} F_{pp} & F_{pm} \\ F_{mp} & F_{mm} \end{bmatrix} \cdot \begin{bmatrix} \Sigma_{pp}^A & \Sigma_{pm}^A \\ \Sigma_{mp}^A & \Sigma_{mm}^A \end{bmatrix} \right) \\
 &= \begin{bmatrix} \Sigma_{pp}^K - (\Sigma_{pp}^R - \Sigma_{pp}^A)F_{pp} + (F_{pm}\Sigma_{mp}^A - \Sigma_{pm}^R F_{mp}) & \Sigma_{pm}^K + (F_{pp}\Sigma_{pm}^A - \Sigma_{pm}^R F_{mm}) - (\Sigma_{pp}^R - \Sigma_{mm}^A)F_{pm} \\ \Sigma_{mp}^K + (F_{mm}\Sigma_{mp}^A - \Sigma_{mp}^R F_{pp}) - (\Sigma_{mm}^R - \Sigma_{pp}^A)F_{mp} & \Sigma_{mm}^K - (\Sigma_{mm}^R - \Sigma_{mm}^A)F_{mm} + (F_{mp}\Sigma_{pm}^A - \Sigma_{mp}^R F_{pm}) \end{bmatrix}
 \end{aligned} \tag{18.70}$$

We will do a perturbative calculation in the light-matter coupling g and keep terms upto $\mathcal{O}(g^2)$. Equating off-diagonal terms of L.H.S. and R.H.S. we get

$$\begin{aligned}
 F_{pm}(k, \omega) &\approx \frac{\Sigma_{pm}^K - (\Sigma_{pm}^R - \Sigma_{pm}^A)F_0}{2(E_k - i\gamma)} = \frac{\Sigma_{pm}(k, \omega)}{2(E_k - i\gamma)} \\
 F_{mp}(k, \omega) &\approx -\frac{\Sigma_{mp}^K - (\Sigma_{mp}^R - \Sigma_{mp}^A)F_0}{2(E_k + i\gamma)} = -\frac{\Sigma_{mp}(k, \omega)}{2(E_k + i\gamma)}
 \end{aligned} \tag{18.71}$$

From the diagonal terms we get

$$\begin{aligned}
 2i\gamma(F_0 - F_{pp}) &\approx \Sigma_{pp}^K - (\Sigma_{pp}^R - \Sigma_{pp}^A)F_0 = \Sigma_{pp} \quad \text{and} \\
 2i\gamma(F_0 - F_{mm}) &\approx \Sigma_{mm}^K - (\Sigma_{mm}^R - \Sigma_{mm}^A)F_0 = \Sigma_{mm}
 \end{aligned} \tag{18.72}$$

So,

$$\begin{aligned}
 F_{pp}(k, \omega) &= F_0(\omega) - \frac{\Sigma_{pp}(k, \omega)}{2i\gamma} \quad \text{and} \\
 F_{mm}(k, \omega) &= F_0(\omega) - \frac{\Sigma_{mm}(k, \omega)}{2i\gamma}
 \end{aligned} \tag{18.73}$$

Note that we have defined

$$\Sigma_{\alpha\beta}(k, \omega) = \Sigma_{\alpha\beta}^K(k, \omega) - [\Sigma_{\alpha\beta}^R(k, \omega) - \Sigma_{\alpha\beta}^A(k, \omega)]F_0(\omega). \tag{18.74}$$

18.3.2 Self-energies and non-thermal distribution

We need to derive the self-energies to the fermions due to the light-matter coupling. Note that we will derive them at the classical saddle point i.e. mean field approximation. In that case we set $\Delta_q = 0$ and consider only Δ_{cl} . Before deriving them using diagrammatic techniques, note that the photon Green's functions

$$\begin{aligned}
 D_0^R(q, \omega) &= [D_0^A(q, \omega)]^* = \frac{\nu_0}{2\pi} \frac{1}{(\omega + i\gamma_B)^2 - \nu_q^2} = \frac{\nu_0}{4\pi\nu_q} \left[\frac{1}{\omega + i\gamma_B - \nu_q} - \frac{1}{\omega + i\gamma_B + \nu_q} \right], \\
 \text{Im}D_0^R(q, \omega) &= -\frac{\nu_0}{4\pi\nu_q} \left[\frac{\gamma_B}{(\omega - \nu_q)^2 + \gamma_B^2} - \frac{\gamma_B}{(\omega + \nu_q)^2 + \gamma_B^2} \right], \quad \text{and} \\
 D^K(q, \omega) &= 2i\text{Im}D_0^R(q, \omega)B_{T_{\text{cav}}}(\omega)
 \end{aligned} \tag{18.75}$$

In fig. 18.6 we sketch the self-energies which are lowest order in the light-matter coupling $[\mathcal{O}(g_0^2)]$. For these diagrams we followed the notation in Ref. [Kamenev(2023)], i.e. solid lines

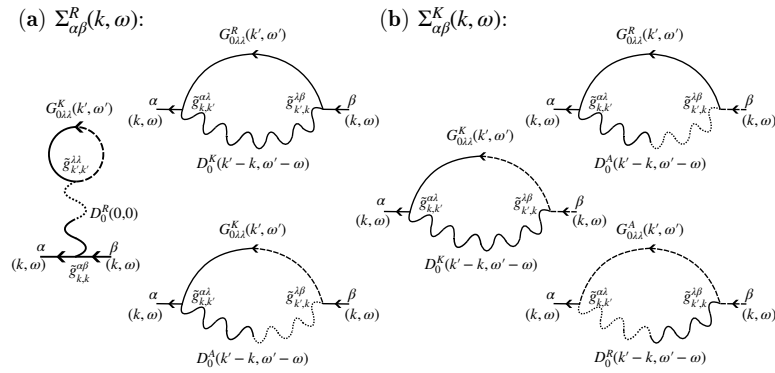


Figure 18.6: Self-energy diagrams of electron due to light-matter coupling for $\Delta_q = 0$. Solid lines represent $\psi_{\alpha 1}(k, \omega)$ and dashed lines represent $\psi_{\alpha 2}(k, \omega)$. Solid curly lines represent $\phi_{\text{cl}}(q, \Omega)$ and dotted curly lines represent $\phi_q(q, \Omega)$. All the diagrams are of $\mathcal{O}(g_0^2)$. (a) Retarded self-energy diagrams are listed. The left-most one is Hartree diagram while the other two are Fock diagrams. (b) Keldysh self-energy diagrams are listed. All of them are Fock diagrams.

represent, $\psi_1/\bar{\psi}_1$, dashed lines represent $\psi_1/\bar{\psi}_1$, solid wavy lines represent ϕ_{cl} and dotted wavy lines represent ϕ_{q} . From fig. 18.6(a), we get the retarded self-energy

$$\begin{aligned}
\Sigma_{\alpha\beta}^R(k, \omega) &= [\Sigma_{\beta\alpha}^A(k, \omega)]^* \\
&= i \sum_{\lambda=\{p,m\}} \sum_{k'} \int \frac{d\omega'}{2\pi} (-1) (i\tilde{g}_{k,k}^{\alpha\beta}) (i\tilde{g}_{k',k'}^{\lambda\lambda}) iG_{0\lambda\lambda}^K(k', \omega') iD_0^R(0, 0) \\
&+ i \sum_{\lambda=\{p,m\}} \sum_{k'} \int \frac{d\omega'}{2\pi} (i\tilde{g}_{k,k'}^{\alpha\lambda}) (i\tilde{g}_{k',k}^{\lambda\beta}) [iG_{0\lambda\lambda}^R(k', \omega') iD_0^K(k' - k, \omega' - \omega) + iG_{0\lambda\lambda}^K(k', \omega') iD_0^A(k' - k, \omega' - \omega)] \\
&= - \sum_{\lambda=\{p,m\}} \sum_{k'} \int \frac{d\omega'}{\pi} \tilde{g}_{k,k'}^{\alpha\lambda} \tilde{g}_{k',k}^{\lambda\beta} [G_{0\lambda\lambda}^R(k', \omega') \text{Im}D_0^R(k' - k, \omega' - \omega) B_{T_{\text{cav}}}(\omega' - \omega) + \text{Im}G_{0\lambda\lambda}^R(k', \omega') D_0^A(k' - k, \omega' - \omega)]
\end{aligned}
\tag{18.76}$$

Note that the Hartree term vanishes. From fig. 18.6(b), we get the Keldysh self-energy

$$\begin{aligned}
& \Sigma_{\alpha\beta}^K(k, \omega) \\
&= i \sum_{\lambda=\{p,m\}} \sum_{k'} \int \frac{d\omega'}{2\pi} (i\tilde{g}_{k,k'}^{\alpha\lambda})(i\tilde{g}_{k',k}^{\lambda\beta}) [iG_{0\lambda\lambda}^R(k', \omega') iD_0^A(k' - k, \omega' - \omega) + iG_{0\lambda\lambda}^A(k', \omega') iD_0^R(k' - k, \omega' - \omega)] \\
&+ i \sum_{\lambda=\{p,m\}} \sum_{k'} \int \frac{d\omega'}{2\pi} (i\tilde{g}_{k,k'}^{\alpha\lambda})(i\tilde{g}_{k',k}^{\lambda\beta}) [iG_{0\lambda\lambda}^K(k', \omega') iD_0^K(k' - k, \omega' - \omega)] \\
&= -2 \sum_{\lambda=\{p,m\}} \sum_{k'} \int \frac{d\omega'}{\pi} \tilde{g}_{k,k'}^{\alpha\lambda} \tilde{g}_{k',k}^{\lambda\beta} \{ \text{Im} G_{0\lambda\lambda}^R(k', \omega') \text{Im} D_0^R(k' - k, \omega' - \omega) [F_0(\omega') B_{T_{\text{cav}}}(\omega' - \omega) - 1] \}.
\end{aligned} \tag{18.77}$$

Hence we get (noting $\tilde{g}_{k,k'}^{\alpha\beta} = [\tilde{g}_{k',k}^{\beta\alpha}]^*$),

$$\begin{aligned} \Sigma_{\alpha\beta}(k, \omega) &= \Sigma_{\alpha\beta}^K(k, \omega) - [\Sigma_{\alpha\beta}^R(k, \omega) - \Sigma_{\alpha\beta}^A(k, \omega)]F_0(\omega) \\ &= -i2 \sum_{\lambda=\{p,m\}} \sum_{k'} \int \frac{d\omega'}{\pi} \tilde{g}_{k,k'}^{\alpha\lambda} \tilde{g}_{k',k}^{\lambda\beta} \text{Im} G_{0\lambda\lambda}^R(k', \omega') \text{Im} D_0^R(k' - k, \omega' - \omega) \tilde{H}_0(\omega, \omega') \end{aligned} \quad (18.78)$$

where the distribution factor

$$\tilde{H}_0(\omega, \omega') = B_{T_{\text{cav}}}(\omega' - \omega) [F_0(\omega') - F_0(\omega)] + F_0(\omega')F_0(\omega) - 1 = [B_{T_{\text{cav}}}(\omega - \omega') - B_{T_{\text{cry}}}(\omega - \omega')] [F_0(\omega) - F_0(\omega')] \quad (18.79)$$

So for the distribution functions we get

$$F_{pp}(k, \omega) = -F_{mm}(k, -\omega) = F_0(\omega) - \frac{1}{2\gamma} \sum_{k'} (g_{k,k'})^2 \left[\left(1 + \frac{\xi_k \xi_{k'} + \eta |\Delta_{cl}|^2}{E_k E_{k'}} \right) \text{Im} D_0^R(k' - k, E_{k'} - \omega) \tilde{H}_0(\omega, E_{k'}) \right. \\ \left. - \left(1 - \frac{\xi_k \xi_{k'} + \eta |\Delta_{cl}|^2}{E_k E_{k'}} \right) \text{Im} D_0^R(k' - k, E_{k'} + \omega) \tilde{H}_0(-\omega, E_{k'}) \right] \quad (18.80)$$

and

$$F_{pm}(k, \omega) = [F_{mp}(k, \omega)]^* \\ = i \frac{\Delta_{cl}}{2(E_k - i\gamma)} \sum_{k'} (g_{k,k'}^+)^2 \left(\frac{\eta \xi_k - \xi_{k'}}{E_k E_{k'}} \right) \left[\text{Im} D_0^R(k' - k, E_{k'} - \omega) \tilde{H}_0(\omega, E_{k'}) + \text{Im} D_0^R(k' - k, E_{k'} + \omega) \tilde{H}_0(-\omega, E_{k'}) \right] \quad (18.81)$$

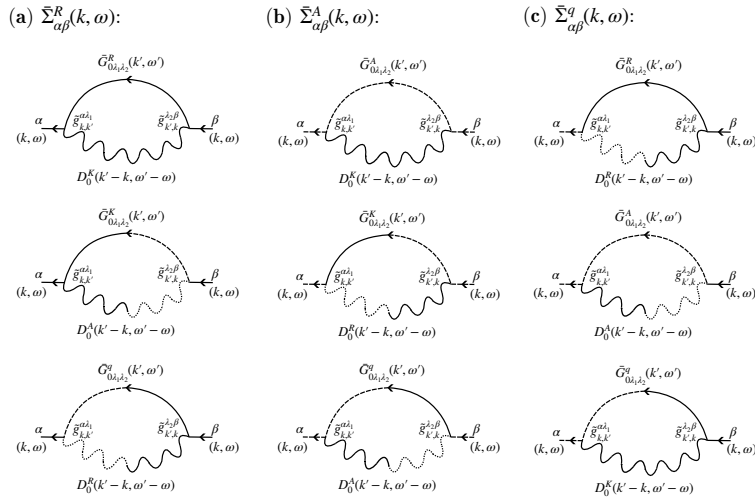


Figure 18.7: Diagrammatic representation of the derivative of the self-energies with respect to Δ_q defined as $\bar{\Sigma}^{R/A/q} = \frac{\partial \Sigma^{R/A/q}}{\partial \Delta_q} \Big|_{\Delta_q=0}$. The derivative directly acts on the electronic Green's

function in the diagrams and they are defined by $\bar{G}^{R/A/K/q} = \frac{\partial G^{R/A/K/q}}{\partial \Delta_q} \Big|_{\Delta_q=0}$.

(a) Diagrams for derivative of the retarded self-energy. (b) Diagrams for derivative of the advanced self-energy.

(c) Diagrams for derivative of the 'quantum' self-energy. Note that while $\Sigma^q = 0$, $\bar{\Sigma}^q$ is finite.

•Variation of self-energies:

In the non-thermal gap equation in eq. (18.59), in addition to the mean field self-energies, we need the variation of them w.r.t. Δ_q and for that we need to derive self-energies linear in Δ_q . We sketch them in fig. 18.7. In the self-energy diagrams we need to consider fermion Green's functions that depends on Δ_q . When we differentiate the self-energy w.r.t. Δ_q , that will apply on the fermion Green's function in the diagrams. So, let us first list them before evaluating the $\bar{\Sigma}$. Note that

$$G_0(\Delta_q) G_0^{-1}(\Delta_q) = \mathbb{I}_{4 \times 4} \implies \frac{\delta G_0(\Delta_q)}{\delta \Delta_q} G_0^{-1}(\Delta_q) + G_0(\Delta_q) \frac{\delta G_0^{-1}(\Delta_q)}{\delta \Delta_q} = 0 \\ \implies \frac{\delta G_0(\Delta_q)}{\delta \Delta_q} \Big|_{\Delta_q=0} = -G_0(0) \frac{\delta G_0^{-1}(\Delta_q)}{\delta \Delta_q} \Big|_{\Delta_q=0} G_0(0) \quad (18.82)$$

Now,

$$G_0 = \begin{bmatrix} G_0^R & G_0^K \\ 0 & G_0^A \end{bmatrix} \quad \text{and} \quad \frac{\delta G_0^{-1}(\Delta_q)}{\delta \Delta_q} \Big|_{\Delta_q=0} = \begin{bmatrix} 0 & \frac{\delta G_0^{q-1}}{\delta \Delta_q} \Big|_{\Delta_q=0} \\ \frac{\delta G_0^{q-1}}{\delta \Delta_q} \Big|_{\Delta_q=0} & 0 \end{bmatrix} \quad (18.83)$$

So,

$$\begin{aligned} \frac{\delta G_0(\Delta_q)}{\delta \Delta_q} \Big|_{\Delta_q=0} &= \begin{bmatrix} \bar{G}_0^R & \bar{G}_0^K \\ \bar{G}_0^q & \bar{G}_0^A \end{bmatrix} = \begin{bmatrix} \frac{\delta G_0^R(\Delta_q)}{\delta \Delta_q} \Big|_{\Delta_q=0} & \frac{\delta G_0^K(\Delta_q)}{\delta \Delta_q} \Big|_{\Delta_q=0} \\ \frac{\delta G_0^q(\Delta_q)}{\delta \Delta_q} \Big|_{\Delta_q=0} & \frac{\delta G_0^A(\Delta_q)}{\delta \Delta_q} \Big|_{\Delta_q=0} \end{bmatrix} \\ &= \begin{bmatrix} -G_0^K \frac{\delta G_0^{q-1}}{\delta \Delta_q} \Big|_{\Delta_q=0} G_0^R & -G_0^R \frac{\delta G_0^{q-1}}{\delta \Delta_q} \Big|_{\Delta_q=0} G_0^A - G_0^K \frac{\delta G_0^{q-1}}{\delta \Delta_q} \Big|_{\Delta_q=0} G_0^K \\ -G_0^A \frac{\delta G_0^{q-1}}{\delta \Delta_q} \Big|_{\Delta_q=0} G_0^R & -G_0^A \frac{\delta G_0^{q-1}}{\delta \Delta_q} \Big|_{\Delta_q=0} G_0^K \end{bmatrix} \end{aligned} \quad (18.84)$$

Hence

$$\begin{aligned} \bar{G}_0^R &= \frac{\delta G_0^R(\Delta_q)}{\delta \Delta_q} \Big|_{\Delta_q=0} = -G_0^K \frac{\delta G_0^{q-1}}{\delta \Delta_q} \Big|_{\Delta_q=0} G_0^R = - \begin{bmatrix} G_{0pp}^K & 0 \\ 0 & G_{0mm}^K \end{bmatrix} \begin{bmatrix} -u_k v_k^* & -u_k^2 \\ (v_k^*)^2 & u_k v_k^* \end{bmatrix} \begin{bmatrix} G_{0pp}^R & 0 \\ 0 & G_{0mm}^R \end{bmatrix} \\ &= \begin{bmatrix} u_k v_k^* G_{0pp}^K G_{0pp}^R & u_k^2 G_{0pp}^K G_{0mm}^R \\ -(v_k^*)^2 G_{0mm}^K G_{0pp}^R & -u_k v_k^* G_{0mm}^K G_{0mm}^R \end{bmatrix} \end{aligned} \quad (18.85)$$

$$\begin{aligned} \bar{G}_0^A &= \frac{\delta G_0^A(\Delta_q)}{\delta \Delta_q} \Big|_{\Delta_q=0} = -G_0^A \frac{\delta G_0^{q-1}}{\delta \Delta_q} \Big|_{\Delta_q=0} G_0^K = - \begin{bmatrix} G_{0pp}^A & 0 \\ 0 & G_{0mm}^A \end{bmatrix} \begin{bmatrix} -u_k v_k^* & -u_k^2 \\ (v_k^*)^2 & u_k v_k^* \end{bmatrix} \begin{bmatrix} G_{0pp}^K & 0 \\ 0 & G_{0mm}^K \end{bmatrix} \\ &= \begin{bmatrix} u_k v_k^* G_{0pp}^A G_{0pp}^K & u_k^2 G_{0pp}^A G_{0mm}^K \\ -(v_k^*)^2 G_{0mm}^A G_{0pp}^K & -u_k v_k^* G_{0mm}^A G_{0mm}^K \end{bmatrix} \end{aligned} \quad (18.86)$$

$$\begin{aligned} \bar{G}_0^q &= \frac{\delta G_0^q(\Delta_q)}{\delta \Delta_q} \Big|_{\Delta_q=0} = -G_0^A \frac{\delta G_0^{q-1}}{\delta \Delta_q} \Big|_{\Delta_q=0} G_0^R = - \begin{bmatrix} G_{0pp}^A & 0 \\ 0 & G_{0mm}^A \end{bmatrix} \begin{bmatrix} -u_k v_k^* & -u_k^2 \\ (v_k^*)^2 & u_k v_k^* \end{bmatrix} \begin{bmatrix} G_{0pp}^R & 0 \\ 0 & G_{0mm}^R \end{bmatrix} \\ &= \begin{bmatrix} u_k v_k^* G_{0pp}^A G_{0pp}^R & u_k^2 G_{0pp}^A G_{0mm}^R \\ -(v_k^*)^2 G_{0mm}^A G_{0pp}^R & -u_k v_k^* G_{0mm}^A G_{0mm}^R \end{bmatrix} \end{aligned} \quad (18.87)$$

$$\begin{aligned} \bar{G}_0^K &= \frac{\delta G_0^K(\Delta_q)}{\delta \Delta_q} \Big|_{\Delta_q=0} = -G_0^R \frac{\delta G_0^{q-1}}{\delta \Delta_q} \Big|_{\Delta_q=0} G_0^A - G_0^K \frac{\delta G_0^{q-1}}{\delta \Delta_q} \Big|_{\Delta_q=0} G_0^K \\ &= - \begin{bmatrix} G_{0pp}^R & 0 \\ 0 & G_{0mm}^R \end{bmatrix} \begin{bmatrix} -u_k v_k^* & -u_k^2 \\ (v_k^*)^2 & u_k v_k^* \end{bmatrix} \begin{bmatrix} G_{0pp}^A & 0 \\ 0 & G_{0mm}^A \end{bmatrix} - \begin{bmatrix} G_{0pp}^K & 0 \\ 0 & G_{0mm}^K \end{bmatrix} \begin{bmatrix} -u_k v_k^* & -u_k^2 \\ (v_k^*)^2 & u_k v_k^* \end{bmatrix} \begin{bmatrix} G_{0pp}^K & 0 \\ 0 & G_{0mm}^K \end{bmatrix} \\ &= \begin{bmatrix} u_k v_k^* (G_{0pp}^R G_{0pp}^A + G_{0pp}^K G_{0pp}^K) & u_k^2 (G_{0pp}^R G_{0mm}^A + G_{0pp}^K G_{0mm}^K) \\ -(v_k^*)^2 (G_{0mm}^R G_{0pp}^A + G_{0mm}^K G_{0pp}^K) & -u_k v_k^* (G_{0mm}^R G_{0mm}^A + G_{0mm}^K G_{0mm}^K) \end{bmatrix}. \end{aligned} \quad (18.88)$$

Evaluating the diagrams in fig. 18.7 we get

$$\begin{aligned}
& \bar{\Sigma}_{\alpha\beta}^R(k, \omega) \\
&= i \sum_{\lambda_1, \lambda_2=\{p, m\}} \sum_{k'} \int \frac{d\omega'}{2\pi} \tilde{g}_{k, k'}^{\alpha\lambda_1} \tilde{g}_{k', k}^{\lambda_2\beta} [\bar{G}_{0\lambda_1\lambda_2}^R(k', \omega') D_0^K(k' - k, \omega' - \omega) + \bar{G}_{0\lambda_1\lambda_2}^K(k', \omega') D_0^A(k' - k, \omega' - \omega) \\
&\quad + \bar{G}_{0\lambda_1\lambda_2}^q(k', \omega') D_0^R(k' - k, \omega' - \omega)] \\
&= i \sum_{k'} \int \frac{d\omega'}{2\pi} \tilde{g}_{k, k'}^{\alpha p} \tilde{g}_{k', k}^{p\beta} [\bar{G}_{0pp}^R D_0^K + \bar{G}_{0pp}^K D_0^A + \bar{G}_{0pp}^q D_0^R] + i \sum_{k'} \int \frac{d\omega'}{2\pi} \tilde{g}_{k, k'}^{\alpha p} \tilde{g}_{k', k}^{m\beta} [\bar{G}_{0pm}^R D_0^K + \bar{G}_{0pm}^K D_0^A + \bar{G}_{0pm}^q D_0^R] \\
&\quad + i \sum_{k'} \int \frac{d\omega'}{2\pi} \tilde{g}_{k, k'}^{\alpha m} \tilde{g}_{k', k}^{p\beta} [\bar{G}_{0mp}^R D_0^K + \bar{G}_{0mp}^K D_0^A + \bar{G}_{0mp}^q D_0^R] + i \sum_{k'} \int \frac{d\omega'}{2\pi} \tilde{g}_{k, k'}^{\alpha m} \tilde{g}_{k', k}^{m\beta} [\bar{G}_{0mm}^R D_0^K + \bar{G}_{0mm}^K D_0^A + \bar{G}_{0mm}^q D_0^R] \\
&= i \sum_{k'} \int \frac{d\omega'}{2\pi} \frac{\Delta_{cl}^*}{2E_{k'}} \tilde{g}_{k, k'}^{\alpha p} \tilde{g}_{k', k}^{p\beta} [G_{0pp}^K G_{0pp}^R D_0^K + (G_{0pp}^R G_{0pp}^A + G_{0pp}^K G_{0pp}^K) D_0^A + G_{0pp}^A G_{0pp}^R D_0^R] \\
&\quad + i \sum_{k'} \int \frac{d\omega'}{2\pi} \frac{1}{2} \left(1 + \frac{\xi_{k'}}{E_{k'}}\right) \tilde{g}_{k, k'}^{\alpha p} \tilde{g}_{k', k}^{m\beta} [G_{0pp}^K G_{0mm}^R D_0^K + (G_{0pp}^R G_{0mm}^A + G_{0pp}^K G_{0mm}^K) D_0^A + G_{0pp}^A G_{0mm}^R D_0^R] \\
&\quad - i \sum_{k'} \int \frac{d\omega'}{2\pi} \frac{\Delta_{cl}^*}{2\Delta_{cl}} \left(1 - \frac{\xi_{k'}}{E_{k'}}\right) \tilde{g}_{k, k'}^{\alpha m} \tilde{g}_{k', k}^{p\beta} [G_{0mm}^K G_{0pp}^R D_0^K + (G_{0mm}^R G_{0pp}^A + G_{0mm}^K G_{0pp}^K) D_0^A + G_{0mm}^A G_{0pp}^R D_0^R] \\
&\quad - i \sum_{k'} \int \frac{d\omega'}{2\pi} \frac{\Delta_{cl}^*}{2E_{k'}} \tilde{g}_{k, k'}^{\alpha m} \tilde{g}_{k', k}^{m\beta} [G_{0mm}^K G_{0mm}^R D_0^K + (G_{0mm}^R G_{0mm}^A + G_{0mm}^K G_{0mm}^K) D_0^A + G_{0mm}^A G_{0mm}^R D_0^R].
\end{aligned} \tag{18.89}$$

Note that for brevity we have omitted the momentum and energy arguments of the Green's functions. It is easily understood that all the fermionic Green's functions have the argument (k', ω') and the photonic Green's function carry $(k' - k, \omega' - \omega)$ as the argument. Similarly we also get

$$\begin{aligned}
& \bar{\Sigma}_{\alpha\beta}^A(k, \omega) \\
&= i \sum_{\lambda_1, \lambda_2=\{p, m\}} \sum_{k'} \int \frac{d\omega'}{2\pi} \tilde{g}_{k, k'}^{\alpha\lambda_1} \tilde{g}_{k', k}^{\lambda_2\beta} [\bar{G}_{0\lambda_1\lambda_2}^A(k', \omega') D_0^K(k' - k, \omega' - \omega) + \bar{G}_{0\lambda_1\lambda_2}^K(k', \omega') D_0^R(k' - k, \omega' - \omega) \\
&\quad + \bar{G}_{0\lambda_1\lambda_2}^q(k', \omega') D_0^A(k' - k, \omega' - \omega)] \\
&= i \sum_{k'} \int \frac{d\omega'}{2\pi} \tilde{g}_{k, k'}^{\alpha p} \tilde{g}_{k', k}^{p\beta} [\bar{G}_{0pp}^A D_0^K + \bar{G}_{0pp}^K D_0^R + \bar{G}_{0pp}^q D_0^A] + i \sum_{k'} \int \frac{d\omega'}{2\pi} \tilde{g}_{k, k'}^{\alpha p} \tilde{g}_{k', k}^{m\beta} [\bar{G}_{0pm}^A D_0^K + \bar{G}_{0pm}^K D_0^R + \bar{G}_{0pm}^q D_0^A] \\
&\quad + i \sum_{k'} \int \frac{d\omega'}{2\pi} \tilde{g}_{k, k'}^{\alpha m} \tilde{g}_{k', k}^{p\beta} [\bar{G}_{0mp}^A D_0^K + \bar{G}_{0mp}^K D_0^R + \bar{G}_{0mp}^q D_0^A] + i \sum_{k'} \int \frac{d\omega'}{2\pi} \tilde{g}_{k, k'}^{\alpha m} \tilde{g}_{k', k}^{m\beta} [\bar{G}_{0mm}^A D_0^K + \bar{G}_{0mm}^K D_0^R + \bar{G}_{0mm}^q D_0^A] \\
&= i \sum_{k'} \int \frac{d\omega'}{2\pi} \frac{\Delta_{cl}^*}{2E_{k'}} \tilde{g}_{k, k'}^{\alpha p} \tilde{g}_{k', k}^{p\beta} [G_{0pp}^A G_{0pp}^K D_0^K + (G_{0pp}^R G_{0pp}^A + G_{0pp}^K G_{0pp}^K) D_0^R + G_{0pp}^A G_{0pp}^R D_0^A] \\
&\quad + i \sum_{k'} \int \frac{d\omega'}{2\pi} \frac{1}{2} \left(1 + \frac{\xi_{k'}}{E_{k'}}\right) \tilde{g}_{k, k'}^{\alpha p} \tilde{g}_{k', k}^{m\beta} [G_{0pp}^A G_{0mm}^K D_0^K + (G_{0pp}^R G_{0mm}^A + G_{0pp}^K G_{0mm}^K) D_0^R + G_{0pp}^A G_{0mm}^R D_0^A] \\
&\quad - i \sum_{k'} \int \frac{d\omega'}{2\pi} \frac{\Delta_{cl}^*}{2\Delta_{cl}} \left(1 - \frac{\xi_{k'}}{E_{k'}}\right) \tilde{g}_{k, k'}^{\alpha m} \tilde{g}_{k', k}^{p\beta} [G_{0mm}^A G_{0pp}^K D_0^K + (G_{0mm}^R G_{0pp}^A + G_{0mm}^K G_{0pp}^K) D_0^R + G_{0mm}^A G_{0pp}^R D_0^A] \\
&\quad - i \sum_{k'} \int \frac{d\omega'}{2\pi} \frac{\Delta_{cl}^*}{2E_{k'}} \tilde{g}_{k, k'}^{\alpha m} \tilde{g}_{k', k}^{m\beta} [G_{0mm}^A G_{0mm}^K D_0^K + (G_{0mm}^R G_{0mm}^A + G_{0mm}^K G_{0mm}^K) D_0^R + G_{0mm}^A G_{0mm}^R D_0^A].
\end{aligned} \tag{18.90}$$

and

$$\begin{aligned}
 & \bar{\Sigma}_{\alpha\beta}^q(k, \omega) \\
 &= i \sum_{\lambda_1, \lambda_2=\{p,m\}} \sum_{k'} \int \frac{d\omega'}{2\pi} \tilde{g}_{k,k'}^{\alpha\lambda_1} \tilde{g}_{k',k}^{\lambda_2\beta} \\
 & \times [\bar{G}_{0\lambda_1\lambda_2}^q(k', \omega') D_0^K(k' - k, \omega' - \omega) + \bar{G}_{0\lambda_1\lambda_2}^R(k', \omega') D_0^R(k' - k, \omega' - \omega) + \bar{G}_{0\lambda_1\lambda_2}^A(k', \omega') D_0^A(k' - k, \omega' - \omega)] \\
 &= i \sum_{k'} \int \frac{d\omega'}{2\pi} \left\{ \tilde{g}_{k,k'}^{\alpha p} \tilde{g}_{k',k}^{p\beta} [\bar{G}_{0pp}^q D_0^K + \bar{G}_{0pp}^R D_0^R + \bar{G}_{0pp}^A D_0^A] + \tilde{g}_{k,k'}^{\alpha p} \tilde{g}_{k',k}^{m\beta} [\bar{G}_{0pm}^q D_0^K + \bar{G}_{0pm}^R D_0^R + \bar{G}_{0pm}^A D_0^A] \right. \\
 & \quad \left. + \tilde{g}_{k,k'}^{\alpha m} \tilde{g}_{k',k}^{p\beta} [\bar{G}_{0mp}^q D_0^K + \bar{G}_{0mp}^R D_0^R + \bar{G}_{0mp}^A D_0^A] + \tilde{g}_{k,k'}^{\alpha m} \tilde{g}_{k',k}^{m\beta} [\bar{G}_{0mm}^q D_0^K + \bar{G}_{0mm}^R D_0^R + \bar{G}_{0mm}^A D_0^A] \right\} \\
 &= i \sum_{k'} \int \frac{d\omega'}{2\pi} \frac{\Delta_{cl}^*}{2E_{k'}} \tilde{g}_{k,k'}^{\alpha p} \tilde{g}_{k',k}^{p\beta} [G_{0pp}^A G_{0pp}^R D_0^K + G_{0pp}^K G_{0pp}^R D_0^R + G_{0pp}^A G_{0pp}^K D_0^A] \\
 & + i \sum_{k'} \int \frac{d\omega'}{2\pi} \frac{1}{2} \left(1 + \frac{\xi_{k'}}{E_{k'}} \right) \tilde{g}_{k,k'}^{\alpha p} \tilde{g}_{k',k}^{m\beta} [G_{0pp}^A G_{0mm}^R D_0^K + G_{0pp}^K G_{0mm}^R D_0^R + G_{0pp}^A G_{0mm}^K D_0^A] \\
 & - i \sum_{k'} \int \frac{d\omega'}{2\pi} \frac{\Delta_{cl}^*}{2\Delta_{cl}} \left(1 - \frac{\xi_{k'}}{E_{k'}} \right) \tilde{g}_{k,k'}^{\alpha m} \tilde{g}_{k',k}^{p\beta} [G_{0mm}^A G_{0pp}^R D_0^K + G_{0mm}^K G_{0pp}^R D_0^R + G_{0mm}^A G_{0pp}^K D_0^A] \\
 & - i \sum_{k'} \int \frac{d\omega'}{2\pi} \frac{\Delta_{cl}^*}{2E_{k'}} \tilde{g}_{k,k'}^{\alpha m} \tilde{g}_{k',k}^{m\beta} [G_{0mm}^A G_{0mm}^R D_0^K + G_{0mm}^K G_{0mm}^R D_0^R + G_{0mm}^A G_{0mm}^K D_0^A] .
 \end{aligned} \tag{18.91}$$

18.3.3 The non-thermal gap equation for material in cavity

We want to calculate the perturbative correction to the gap equation in the small g_0 limit. In this limit, the gap equation in eq. (18.59) can be written as

$$\begin{aligned}
 & i \sum_k \int \frac{d\omega}{2\pi} \left\{ u_k v_k^* [(G_{0pp}^R - G_{0pp}^A) - (G_{0mm}^R - G_{0mm}^A)] F_0 \right. \\
 & \quad + u_k v_k^* [(G_{0pp}^R - G_{0pp}^A) \delta F_{pp} - (G_{0mm}^R - G_{0mm}^A) \delta F_{mm}] \\
 & \quad + [u_k v_k^* (\delta G_{pp}^R - \delta G_{pp}^A) - u_k v_k^* (\delta G_{mm}^R - \delta G_{mm}^A) - (v_k^*)^2 (\delta G_{pm}^R - \delta G_{pm}^A) + u_k^2 (\delta G_{mp}^R - \delta G_{mp}^A)] F_0 \\
 & \quad \left. + [-(v_k^*)^2 (G_{0pp}^R - G_{0pp}^A) F_{pm} + u_k^2 (G_{0mm}^R - G_{0pp}^A) F_{mp}] \right\} \\
 & + i \sum_k \int \frac{d\omega}{2\pi} \sum_{\alpha=\{p,m\}} [G_{0\alpha\alpha}^R \bar{\Sigma}_{\alpha\alpha}^R + G_{0\alpha\alpha}^A \bar{\Sigma}_{\alpha\alpha}^A + (G_{0\alpha\alpha}^R - G_{0\alpha\alpha}^A) F_0 \bar{\Sigma}_{\alpha\alpha}^q] = \frac{\Delta_{cl}^*}{\kappa}
 \end{aligned}$$

where $\delta G_{\alpha\beta}^{R/A} = G_{0\alpha\alpha}^{R/A} \Sigma_{\alpha\beta}^{R/A} G_{0\beta\beta}^{R/A}$ is the perturbative correction to the Green's functions. Noting that

$$u_k = \sqrt{\frac{1}{2} \left(1 + \frac{\xi_k}{E_k} \right)} \quad \text{and} \quad v_k = \frac{\Delta_{cl}}{|\Delta_{cl}|} \sqrt{\frac{1}{2} \left(1 - \frac{\xi_k}{E_k} \right)} \tag{18.92}$$

we get

$$\begin{aligned}
 & \sum_k \left\{ \frac{F_0(E_k)}{E_k} + \left[\frac{\delta F_{pp}(k, E_k) - \delta F_{mm}(k, -E_k)}{2E_k} \right] \right. \\
 & \quad \left. + i \int \frac{d\omega}{2\pi} \left[-\frac{1}{2} \left(1 - \frac{\xi_k}{E_k} \right) \frac{(G_{0pp}^R - G_{0mm}^A)}{\Delta_{cl}} F_{pm} + \frac{1}{2} \left(1 + \frac{\xi_k}{E_k} \right) \frac{(G_{0mm}^R - G_{0pp}^A)}{\Delta_{cl}^*} F_{mp} \right] \right\} \\
 & + i \int \frac{d\omega}{2\pi} \left[\frac{(\delta G_{pp}^R - \delta G_{pp}^A) - (\delta G_{mm}^R - \delta G_{mm}^A)}{2E_k} \right. \\
 & \quad \left. - \frac{1}{2} \left(1 - \frac{\xi_k}{E_k} \right) \frac{(\delta G_{pm}^R - \delta G_{pm}^A)}{\Delta_{cl}} + \frac{1}{2} \left(1 + \frac{\xi_k}{E_k} \right) \frac{(\delta G_{mp}^R - \delta G_{mp}^A)}{\Delta_{cl}^*} \right] F_0 \\
 & + i \sum_k \int \frac{d\omega}{2\pi} \sum_{\alpha=\{p,m\}} \frac{1}{\Delta_{cl}^*} [G_{0\alpha\alpha}^R \bar{\Sigma}_{\alpha\alpha}^R + G_{0\alpha\alpha}^A \bar{\Sigma}_{\alpha\alpha}^A + (G_{0\alpha\alpha}^R - G_{0\alpha\alpha}^A) F_0 \bar{\Sigma}_{\alpha\alpha}^q] = \frac{1}{\kappa}
 \end{aligned}$$

From now on we write $\Delta = \Delta_{cl}$ as this is the physically relevant one. Furthermore, we choose the gauge where Δ is real without loss of generality. We make a quasi-particle approximation i.e. $\gamma \ll T_{cry}, T_{cav}, \Delta$ and get for the gap equation

$$\sum_k \frac{F_0(E_k) + \delta F(k, E_k)}{E_k} = \frac{1}{\kappa}, \quad (18.93)$$

where

$$\begin{aligned}
 \delta F(k, E_k) = & -\frac{1}{\gamma} \sum_{k'} \sum_q \int d\Omega (g_{k,k'}^+)^2 \left[\left(1 + \frac{\xi_k \xi_{k'} + \eta \Delta^2}{E_k E_{k'}} \right) \text{Im} D_0^R(q, \Omega) \tilde{H}_0(E_k, E_k + \Omega) \delta(\Omega - (E_{k'} - E_k)) \right. \\
 & \left. - \left(1 - \frac{\xi_k \xi_{k'} + \eta \Delta^2}{E_k E_{k'}} \right) \text{Im} D_0^R(q, \Omega) \tilde{H}_0(-E_k, -E_k + \Omega) \delta(\Omega - (E_{k'} + E_k)) \right] \delta
 \end{aligned} \quad (18.94)$$

In the above equation, the delta functions represent energy and momentum conservations. However, in real materials, the presence of static impurities causes the momentum to be not conserved. Hence, the corresponding delta function can be replaced by a constant $(\Lambda_F/\tau_{el})^{-1}$, where $\Lambda_F = \frac{2}{\pi J}$ is the density of states at the Fermi surface and τ_{el} is the elastic scattering rate [Curtis *et al.*(2019), Mattis and Bardeen(1958)]. Absorbing the factor (Λ_F/τ_{el}) in γ , we get

$$\begin{aligned}
 \delta F(k, E_k) = & -\frac{1}{\gamma} \sum_{k'} \sum_q (g_{k,k'}^+)^2 \left[\left(1 + \frac{\xi_k \xi_{k'} + \eta \Delta^2}{E_k E_{k'}} \right) \text{Im} D_0^R(q, E_{k'} - E_k) \tilde{H}_0(E_k, E_{k'}) \right. \\
 & \left. - \left(1 - \frac{\xi_k \xi_{k'} + \eta \Delta^2}{E_k E_{k'}} \right) \text{Im} D_0^R(q, E_{k'} + E_k) \tilde{H}_0(-E_k, E_{k'}) \right].
 \end{aligned} \quad (18.95)$$

Furthermore we convert the momentum sums to energy integration with the help of a density of states for the gapped quasi-particles $\Lambda(E) = \frac{2}{\pi} \frac{E}{\sqrt{E^2 - \Delta^2} \sqrt{J^2 + \Delta^2 - E^2}}$ and use Fermi surface averaged values for explicitly momentum dependent quantities [Curtis *et al.*(2019)]. Also note the speed of light is much larger than the Fermi velocity, hence it suffices to consider only the $q = 0$ photons. We define the shorthand notation $\text{Im} D_0^R(\omega) = \text{Im} D_0^R(0, \omega)$. Incorporating all

of these we arrive at

$$\int_{\Delta}^{\sqrt{J^2+\Delta^2}} dE \Lambda(E) \frac{F_0(E) + \delta F(E)}{E} = \frac{1}{\kappa}, \text{ where}$$

$$\delta F(E) = -\frac{g_0^2}{\gamma} \int_{\Delta}^{\sqrt{J^2+\Delta^2}} dE' \Lambda(E') \left[\left(1 + \frac{\eta \Delta^2}{EE'}\right) \text{Im} D_0^R(E' - E) \tilde{H}_0(E, E') - \left(1 - \frac{\eta \Delta^2}{EE'}\right) \text{Im} D_0^R(E' + E) \tilde{H}_0(E, E') \right] \quad (18.96)$$

We can also re-write the gap equation in terms of the Fermi and Bose functions as

$$\int_{\Delta}^{\sqrt{J^2+\Delta^2}} dE \Lambda(E) \frac{1 - 2[n_0(E) + \delta n(E)]}{E} = \frac{1}{\kappa}, \text{ where}$$

$$\delta n(E) = \frac{2g_0^2}{\gamma} \int_{\Delta}^{\sqrt{J^2+\Delta^2}} dE' \Lambda(E') \left[\left(1 + \frac{\eta \Delta^2}{EE'}\right) \text{Im} D_0^R(E - E') H_0(E, E') + \left(1 - \frac{\eta \Delta^2}{EE'}\right) \text{Im} D_0^R(E + E') H_0(E, E') \right]$$

$$H_0(\omega, \omega') = [N_{T_{\text{cav}}}(\omega - \omega') - N_{T_{\text{cry}}}(\omega - \omega')] [n_0(\omega) - n_0(\omega')]. \quad (18.97)$$

18.4 SM: Ginzburg-Landau theory of the phase transition

We start with the full gap equation expressed in terms of the energy E

$$\int_{\Delta}^{\sqrt{J^2+\Delta^2}} dE \Lambda(E) \frac{F_0(E) + \delta F(E)}{E} = \frac{1}{\kappa}. \quad (18.98)$$

with

$$\Lambda(E) = \frac{2}{\pi} \frac{E}{\sqrt{E^2 - \Delta^2} \sqrt{J^2 + \Delta^2 - E^2}}. \quad (18.99)$$

Set $E^2 = x^2 + \Delta^2$. Then $E = \Delta \Rightarrow x = 0$, $E = \sqrt{J^2 + \Delta^2} \Rightarrow x = J$, and $E dE = x dx$:

$$\frac{2}{\pi} \int_0^J \frac{dx}{\sqrt{J^2 - x^2} \sqrt{\Delta^2 + x^2}} \left[F_0(\sqrt{x^2 + \Delta^2}) + \delta F(\sqrt{x^2 + \Delta^2}) \right] = \frac{1}{\kappa}. \quad (18.100)$$

We now approximate as in equilibrium the metallic density of states $\frac{2}{\pi} \frac{1}{\sqrt{J^2 - x^2}}$ as a constant with main contribution from the fermi energy $\varepsilon \simeq \varepsilon_F$ or here $x = 0$, we denote it $D(\varepsilon_F) = D_F = \frac{2}{\pi J}$.

$$\delta F(\sqrt{x^2 + \Delta^2}) = -\frac{g_0^2}{\gamma} \int_0^J d\tilde{x} D_F \left\{ \left(1 + \eta \frac{\Delta^2}{E(x) E(\tilde{x})}\right) \text{Im} D_0^R(E(\tilde{x}) - E(x)) \tilde{H}_0(E(x), E(\tilde{x})) \right. \\ \left. - \left(1 - \eta \frac{\Delta^2}{E(x) E(\tilde{x})}\right) \text{Im} D_0^R(E(\tilde{x}) + E(x)) \tilde{H}_0(-E(x), E(\tilde{x})) \right\}, \quad (18.101)$$

with $E(x) = \sqrt{x^2 + \Delta^2}$. We specialize now sharp photons, meaning $\gamma_B \rightarrow 0$ allows us to express the imaginary parts as delta functions.

$$\text{Im} D_0^R(\omega) \Big|_{\gamma_B \rightarrow 0} = -\frac{1}{4} \left[\delta(\omega - \nu_0) - \delta(\omega + \nu_0) \right]. \quad (18.102)$$

We now want to relate the condition for the non-zero value of the delta function in E to the condition for $\tilde{x}$.

$$\text{Im} D_0^R(E(x) \pm E(\tilde{x})) = -\frac{1}{4} \left[\delta(E(x) \pm E(\tilde{x}) - \nu_0) - \delta(E(x) \pm E(\tilde{x}) + \nu_0) \right]. \quad (18.103)$$

Define $f_s(x, \tilde{x}) = E(x) + pE(\tilde{x}) + s\nu_0$, ($s, p = \pm 1$). Zeros for the tilde variables obey $\tilde{x}^2 + \Delta^2 = (p\sqrt{x^2 + \Delta^2} + s\nu_0)^2$ and $\partial f_s / \partial \tilde{x} = p\tilde{x} / E(\tilde{x})$. Hence

$$\delta(E(x) + pE(\tilde{x}) - \nu_0) = \frac{E(\tilde{x})}{\tilde{x}} \delta(\tilde{x} - \tilde{x}_{sp}),$$

Where we have defined

$$\tilde{x}_{sp} = \sqrt{(\sqrt{x^2 + \Delta^2} + ps\nu_0)^2 - \Delta^2} \quad (18.104)$$

Let us now analyze what the delta functions mean. Physically when we have $\tilde{E} - E - \nu_0 = 0$ it can be interpreted as a cavity photon being absorbed such that if before the quasiparticle had energy E it will then have energy $\tilde{E} + \nu_0$. Similarly for $\tilde{E} - E + \nu_0 = 0$ it will emit a photon and thus reduce its energy to $E - \nu_0$. Now since a superconductor or CDW state has a gap and the energies are only positive this can only occur if $E - \nu_0 > 0$. Since the energies are positive we see that $\tilde{E} + E + \nu_0 = 0$ can never occur for positive cavity frequency. Moreover, the process with $\tilde{E} + E - \nu_0 = 0$ can occur only for $\nu_0 - E > 0$. Going back to the integral in energy we can see this from the integral being expressed as $\int_0^{\nu_0} dE$ with vanishing width. We may therefor focus only on the process with $\tilde{E} - E - \nu_0 = 0$ and $\tilde{E} - E + \nu_0 = 0$ and $\tilde{E} + E - \nu_0 = 0$. The distribution will then become:

$$\begin{aligned} \delta F(\sqrt{x^2 + \Delta^2}) = \frac{g_0^2 D_F}{4\gamma} \int_0^J d\tilde{x} & \left[\left(1 + \eta \frac{\Delta^2}{E(x) E(\tilde{x})}\right) \frac{E(\tilde{x})}{\tilde{x}} \tilde{H}_0(E(x), E(x) + \nu_0) \delta(\tilde{x} - \tilde{x}_+) \right. \\ & - \left(1 + \eta \frac{\Delta^2}{E(x) E(\tilde{x})}\right) \frac{E(\tilde{x})}{\tilde{x}} \tilde{H}_0(E(x), E(x) - \nu_0) \delta(\tilde{x} - \tilde{x}_-) \Theta(E(x) - \nu_0) \Big] \\ & - \left(1 - \eta \frac{\Delta^2}{E(x) E(\tilde{x})}\right) \frac{E(\tilde{x})}{\tilde{x}} \tilde{H}_0(-E(x), \nu_0 - E(x)) \delta(\tilde{x} - \tilde{x}_-) \Theta(\nu_0 - E(x)) \Big], \end{aligned} \quad (18.105)$$

Simplifying we have

$$\begin{aligned} \delta F(\sqrt{x^2 + \Delta^2}) = \frac{g_0^2 D_F}{4\gamma} & \left[\left(E(x) + \nu_0 + \eta \frac{\Delta^2}{E(x)}\right) \frac{1}{\tilde{x}_+} \tilde{H}_0(E(x), E(x) + \nu_0) \right. \\ & - \left(E(x) - \nu_0 + \eta \frac{\Delta^2}{E(x)}\right) \frac{1}{\tilde{x}_-} \tilde{H}_0(E(x), E(x) - \nu_0) \Theta(E(x) - \nu_0) \Big] \\ & - \left(\nu_0 - E(x) - \eta \frac{\Delta^2}{E(x)}\right) \frac{1}{\tilde{x}_-} \tilde{H}_0(-E(x), \nu_0 - E(x)) \Theta(\nu_0 - E(x)) \Big] \end{aligned} \quad (18.106)$$

We now write

$$\begin{aligned} \int dE \nu(E) \frac{\delta F(E)}{E} = \frac{g_0^2 D_F^2}{4\gamma} \int_0^J dx & \left[\left(1 + \frac{\nu_0}{E(x)} + \eta \frac{\Delta^2}{E^2(x)}\right) \frac{1}{\tilde{x}_+} \tilde{H}_0(E(x), E(x) + \nu_0) \right. \\ & - \left(1 - \frac{\nu_0}{E(x)} + \eta \frac{\Delta^2}{E^2(x)}\right) \frac{1}{\tilde{x}_-} \tilde{H}_0(E(x), E(x) - \nu_0) \Theta(E(x) - \nu_0) \Big] \\ & - \left(\frac{\nu_0}{E(x)} - 1 - \eta \frac{\Delta^2}{E^2(x)}\right) \frac{1}{\tilde{x}_-} \tilde{H}_0(-E(x), \nu_0 - E(x)) \Theta(\nu_0 - E(x)) \Big], \end{aligned} \quad (18.107)$$

Using now that $\tilde{H}_0(\omega, \omega') = (F_0(\omega') - F_0(\omega))(B_{T_{\text{cav}}}(\omega' - \omega) - B_{T_{\text{cry}}}(\omega' - \omega))$ we can write

$$\begin{aligned}\tilde{H}_0(E(x), E(x) + \nu_0) &= (F_0(E(x) + \nu_0) - F_0(E(x)))(B_{T_{\text{cav}}}(\nu_0) - B_{T_{\text{cry}}}(\nu_0)) \\ &= \tilde{B}(\nu_0)(F_0(E(x) + \nu_0) - F_0(E(x)))\end{aligned}\quad (18.108)$$

We defined $\tilde{B}(\nu_0) = B_{T_{\text{cav}}}(\nu_0) - B_{T_{\text{cry}}}(\nu_0)$. Similarly we have:

$$\begin{aligned}\tilde{H}_0(E(x), E(x) - \nu_0) &= (F_0(E(x) - \nu_0) - F_0(E(x)))(B_{T_{\text{cav}}}(-\nu_0) - B_{T_{\text{cry}}}(-\nu_0)) \\ &= \tilde{B}(\nu_0)(F_0(E(x)) - F_0(E(x) - \nu_0))\end{aligned}\quad (18.109)$$

and also

$$\begin{aligned}\tilde{H}_0(-E(x), \nu_0 - E(x)) &= (F_0(\nu_0 - E(x)) - F_0(-E(x)))(B_{T_{\text{cav}}}(\nu_0) - B_{T_{\text{cry}}}(\nu_0)) \\ &= \tilde{B}(\nu_0)(F_0(E(x)) - F_0(E(x) - \nu_0))\end{aligned}\quad (18.110)$$

Approximate now a small gap $\Delta \ll T_{\text{cry}}$ so that $F_0(E/2T_{\text{cry}}) = \tanh(E/2T_{\text{cry}}) \simeq E/2T_{\text{cry}}$

$$F_0(E(x) + \nu_0) - F_0(E(x)) \simeq \frac{\nu_0}{2T_{\text{cry}}}, \quad F_0(E) - F_0(E - \nu_0) \simeq \frac{\nu_0}{2T_{\text{cry}}}. \quad (18.111)$$

Therefore we have then

$$\begin{aligned}\int dE \nu(E) \frac{\delta F(E)}{E} &= \frac{g_0^2 D_F^2 \nu_0 \tilde{B}(\nu_0)}{8\gamma T_{\text{cry}}} \int_0^J dx \left[\left(1 + \frac{\nu_0}{E(x)} + \eta \frac{\Delta^2}{E^2(x)}\right) \frac{1}{\tilde{x}_+} \right. \\ &\quad \left. - \left(1 - \frac{\nu_0}{E(x)} + \eta \frac{\Delta^2}{E^2(x)}\right) \frac{1}{\tilde{x}_-} \Theta(E(x) - \nu_0) \right] \\ &\quad + \left(1 - \frac{\nu_0}{E(x)} + \eta \frac{\Delta^2}{E^2(x)}\right) \frac{1}{\tilde{x}_-} \Theta(\nu_0 - E(x)) \Big],\end{aligned}\quad (18.112)$$

Let us now assume that Δ is small such that we can approximate

$$\tilde{x}_+ = \sqrt{(\sqrt{x^2 + \Delta^2} + \nu_0)^2 - \Delta^2} = \sqrt{x^2 + \Delta^2 + \nu_0^2 + 2\sqrt{x^2 + \Delta^2}\nu_0 - \Delta^2} \approx \sqrt{x^2 + \nu_0^2 + 2\sqrt{x^2}\nu_0} = |x + \nu_0| \quad (18.113)$$

similarly $\tilde{x}_- \approx |x - \nu_0|$. The integrals we must calculate are then

$$\begin{aligned}\int dE \nu(E) \frac{\delta F(E)}{E} &\approx \frac{g_0^2 D_F^2 \nu_0 \tilde{B}(\nu_0)}{8\gamma T_{\text{cry}}} \int_0^J dx \left[\left(1 + \frac{\nu_0}{\sqrt{x^2 + \Delta^2}} + \eta \frac{\Delta^2}{x^2 + \Delta^2}\right) \frac{1}{x + \nu_0} \right. \\ &\quad \left. - \left(1 - \frac{\nu_0}{\sqrt{x^2 + \Delta^2}} + \eta \frac{\Delta^2}{x^2 + \Delta^2}\right) \frac{1}{|x - \nu_0|} \Theta(E(x) - \nu_0) \right] \\ &\quad + \left(1 - \frac{\nu_0}{\sqrt{x^2 + \Delta^2}} + \eta \frac{\Delta^2}{x^2 + \Delta^2}\right) \frac{1}{|x - \nu_0|} \Theta(\nu_0 - E(x)) \Big],\end{aligned}\quad (18.114)$$

The first integral of the form $\int dx 1/(x \pm \nu_0)$ will only give logarithmic corrections independent of Δ which we do not care about for the gap equation expansion. The second terms with $\int dx \nu_0/(E(x))$ are second order in the small cavity frequency $\nu_0 \ll J, T_{\text{cry}}$, since there is already a ν_0 multiplying everything. We therefore need to only worry about the terms with the

energy squared in the denominator. We want to find the lowest order in Δ . We notice that the integrand of

$$\int_0^J dx \frac{\Delta^2}{x^2 + \Delta^2} \frac{1}{|x - \nu_0|} \Theta(E(x) - \nu_0) \quad (18.115)$$

has a divergence at $x = \nu_0$ which will give the main contribution for $\nu_0 \gg \Delta$ and a coefficient in front of $\frac{\Delta^2}{\nu_0^2 + \Delta^2}$ which has as a lowest order expansion in the gap only quadratic terms. There is no contribution for small x since the Heaviside forbids so. Nevertheless, looking at the other integral we notice that

$$\int_0^J dx \frac{\Delta^2}{x^2 + \Delta^2} \frac{1}{|x - \nu_0|} \Theta(\nu_0 - E(x)) \quad (18.116)$$

has an integrand also with a divergent term but in this case because of the Heaviside function we are allowed to have x small since $\nu_0 - \Delta > 0$, thus for small x we may consider the contribution to the integral for small x up to the smallest scale Δ

$$\int_0^\Delta dx \frac{\Delta^2}{x^2 + \Delta^2} \frac{1}{|x - \nu_0|} \approx \int_0^\Delta dx \frac{\Delta^2}{\Delta^2} \frac{1}{|\nu_0|} = \frac{\Delta}{\nu_0} \quad (18.117)$$

A similar reasoning can be applied to the integral with $1/(x + \nu_0)$, the remaining part of the integral gives logarithmic contributions or quadratic in the gap. Thus keeping only the lowest order contributions that have a gap dependence, neglecting constants, we get:

$$\int dE \nu(E) \frac{\delta F(E)}{E} \approx \eta \frac{g_0^2 D_F^2 \tilde{B}(\nu_0)}{4\gamma T_{cry}} \Delta + \mathcal{O}(\Delta^2) \quad (18.118)$$

We now fit this result together with the gap equation and the Ginzburg-Landau analysis, let us define

$$I(\Delta, T_{cry}) \equiv \int_\Delta^{\sqrt{J^2 + \Delta^2}} dE \Lambda(E) \frac{F_0(E) + \delta F(E)}{E} = \frac{1}{\kappa}.$$

For $T_{cry} = T_{cry}^c$ (non-eq. critical temperature) one has $I(\Delta=0, T_{cry}^c) = 1/\kappa$. Expand about $(\Delta, T_{cry}) = (0, T_{cry}^c)$:

$$I(\Delta, T_{cry}) = I(0, T_{cry}^c) + \left. \frac{\partial I}{\partial T_{cry}} \right|_{\Delta=0} (T_{cry} - T_{cry}^c) + \left. \frac{\partial I}{\partial \Delta} \right|_{\Delta=0} \Delta + \frac{1}{2} \left. \frac{\partial^2 I}{\partial \Delta^2} \right|_{\Delta=0} \Delta^2 + \dots \quad (18.119)$$

The gap equation becomes

$$\left. \frac{\partial I}{\partial T_{cry}} \right|_{\substack{\Delta=0 \\ T_{cry}=T_{cry}^c}} (T_{cry} - T_{cry}^c) + \left. \frac{\partial I}{\partial \Delta} \right|_{\Delta=0} \Delta + \frac{1}{2} \left. \frac{\partial^2 I}{\partial \Delta^2} \right|_{\Delta=0} \Delta^2 = 0$$

has no linear-in- Δ contribution from equilibrium terms (they start at $\propto \Delta^2$). In fact, $\left. \frac{1}{2} \frac{\partial^2 I^{eq}}{\partial \Delta^2} \right|_{\Delta=0} = -\frac{7D_F \zeta(3)}{8\pi^2 (T_{cry}^c)^2}$ while $\left. \frac{\partial I}{\partial T_{cry}} \right|_{\Delta=0, T_{cry}=T_{cry}^c} = \frac{-D_F}{T_{cry}^c}$ so that the gap equation becomes for $T_{cry} \approx T_{cry}^c$ and $\Delta \approx 0$:

$$\Delta \left(\frac{T_{cry}^c - T_{cry}}{T_{cry}^c} - \frac{7\zeta(3)}{8\pi^2} \frac{\Delta^2}{(T_{cry}^c)^2} + \eta \frac{g_0^2 D_F (B_{T_{cav}}(\nu_0) - B_{T_{cry}^c}(\nu_0))}{4\gamma T_{cry}^c} \Delta + \dots \right) = 0. \quad (18.120)$$

The Landau functional can be obtained by integrating this and noting a minus sign is present

$$\mathcal{F}_{\text{neq}}[\Delta] = \frac{T_{\text{cry}} - T_{\text{cry}}^c}{2T_{\text{cry}}^c} \Delta^2 + \eta \frac{g_0^2 D_F(B_{T_{\text{cry}}^c}(\nu_0) - B_{T_{\text{cav}}}(\nu_0))}{12\gamma T_{\text{cry}}^c} \Delta^3 + \frac{7\zeta(3)}{8\pi^2} \frac{1}{4(T_{\text{cry}}^c)^2} \Delta^4 + \dots \quad (18.121)$$

Noting that $D_F = 2/\pi J$ we get

$$\mathcal{F}_{\text{neq}}[\Delta] = \frac{T_{\text{cry}} - T_{\text{cry}}^c}{2T_{\text{cry}}^c} \Delta^2 + \eta \frac{g_0^2 (N_{T_{\text{cry}}^c}(\nu_0) - N_{T_{\text{cav}}}(\nu_0))}{3\pi\gamma T_{\text{cry}}^c} \Delta^3 + \frac{7\zeta(3)}{8\pi^2} \frac{1}{4(T_{\text{cry}}^c)^2} \Delta^4 + \dots \quad (18.122)$$

The solution of the gap equation can be approximated as

$$\Delta[T_{\text{cry}}] \approx \frac{\pi\gamma J}{g_0^2} \frac{\eta}{N_{T_{\text{cav}}}(\nu_0) - N_{T_{\text{cry}}^c}(\nu_0)} (T_{\text{cry}} - T_{\text{cry}}^c). \quad (18.123)$$

Note that it grows linearly with $|T_{\text{cry}} - T_{\text{cry}}^c|$ and has opposite slopes for SC and CDW (determined by η). This is in contrast to equilibrium where the gap grows as $\sqrt{T_{\text{cry}}^c - T_{\text{cry}}}$.

19 Non-thermal pairing glue of electrons in the steady state by Pini, Johansen, Piazza

Abstract

The study of mechanisms for enhancing superconductivity has been a central topic in condensed matter physics due to the combination of fundamental and technological interests. One promising route is to exploit non-equilibrium effects in the steady state. Efforts in this direction have so far focused on enhancing the pairing mechanism known from thermal equilibrium through modified distributions for the electrons or the bosons mediating the electron-electron interaction. In this work, we identify an additional pairing mechanism that is active only outside thermal equilibrium. By generalizing Eliashberg theory to non-equilibrium steady states using the Keldysh formalism, we derive a set of Eliashberg equations that capture the effect of this genuinely non-thermal pairing glue even in the weak-coupling regime. We discuss two examples where this mechanism has a major impact. First, in a temperature-bias setup, we find that superconductivity is enhanced when the boson mediator is colder than the electrons. Second, we find that an incoherent drive of the boson mediator at energies much greater than the temperature pushes the system far from thermal equilibrium but leaves the critical coupling essentially unchanged, owing to a competition between electron heating and the enhancement of pairing by the non-thermal glue.

19.1 Introduction

In recent years, a series of experiments on transient light-induced superconductivity [1, 2, Hu *et al.*(2014), Mitrano *et al.*(2016), Cantaluppi *et al.*(2018), Budden *et al.*(2021), Isoyama *et al.*(2021), 8] has reignited the interest in discovering non-thermal mechanisms to enhance or induce superconductivity, either in a transient or a non-equilibrium steady state (NESS). Focusing on systems in the steady state, the idea of using photons confined within a cavity and coupled to quantum materials [65, 12, 49] appears as a promising direction, as the confinement-enhanced light-matter coupling allows to avoid intense driving, still appreciably affecting the material. In particular, experiments have indicated that this might be used to modify superconducting pairing [10, 11]. The simplest model for a light-induced non-thermal steady-state

of matter that enhances superconductivity was first considered in the context of the Eliashberg *effect* [Eliashberg(1970), ?, 19, 21, 20, Klapwijk *et al.*(1977)], recently explored also in cavity settings [30, 41]. The idea is to directly drive the electrons into a non-thermal distribution that is more prone to electron-pair formation. An alternative route is to instead focus on the boson mediating electron-electron interactions that lead to pairing and superconductivity, either by directly employing cavity photons as mediators [Schlawin *et al.*(2019), 31, 37, 47, 39], or by coupling them to a degree of freedom in the material (phonon, surface plasmon, etc.) [44, 45, Eckhardt *et al.*(2024), 27]. The latter option, albeit with intense laser light (instead of cavity light) and thus confined to a transient, is also the one employed by the experiments mentioned at the beginning.

When considering all the above scenarios, we find that the attention has so far focused on enhancing the “glue” that is responsible for pairing also at thermal equilibrium. Indeed, even in the cases where non-equilibrium effects are directly considered, the main approach consists of replacing the thermal distributions of the electrons [Eliashberg(1970), 19, 21, 20, Klapwijk *et al.*(1977), 30, 41] or of the boson mediators [Eckhardt *et al.*(2024), 39] in the gap equation with their non-equilibrium counterparts. This amounts to modifying the pairing mechanism that is active already at thermal equilibrium. In this work, we identify instead an additional pairing mechanism that is active only when boson mediators and electrons are not in mutual thermal equilibrium. This mechanism manifests as a new independent equation that couples to the regular gap equation. Out of equilibrium, the extended system of equations has to be solved simultaneously, giving rise to a genuinely non-thermal pairing mechanism that *cannot* be described by a gap equation that differs from the thermal equilibrium one just by the functional form of energy distributions or by the value of the coupling constants.

A good starting point to describe a system of boson mediators and superconducting electrons is Eliashberg *theory* of phonon-mediated superconductivity [14, 15] (for recent reviews, see [23, 24]). This theory improves on BCS theory by including retardation effects (i.e. a frequency dependence) in the electron-electron interaction due to the dynamics of the boson mediator. At thermal equilibrium, these retardation effects are typically important in intermediate and strong-coupling regimes [57, 58], but introduce only minor corrections to BCS theory in the weak-coupling regime [59, 60, 61]. In contrast, we find that, outside of thermal equilibrium, retardation effects are essential also in the weak-coupling regime, as they capture important non-thermal contributions in the gap equation that are missing in BCS theory.

As a first step, we generalize Eliashberg theory to the case of a time-translation invariant NESS by employing the Keldysh diagrammatic formalism [70, 71]. Differently from previous dynamical implementations, which considered momentum-averaged models of transient [25, Sentef *et al.*(2016), 33] or quasistationary Floquet-driven [29, Babadi *et al.*(2017)] superconductors, we also go beyond momentum-averaging, which turns out to be necessary to generate the new non-thermal contribution to the pairing. As a specific implementation, we focus on the case of a perfect forward-scattering electron-electron interaction where the boson mediator does not transfer any momentum. For this implementation, our approach is analytically tractable in the weak-coupling regime. This type of interaction is also experimentally relevant, as it is used to describe for example photon-mediated interactions in driven cavity materials [31, 37] and phonon-mediated interactions in FeSe thin films on dielectric substrates [42, 43, 44]. While we find that this model is particularly suited for highlighting non-thermal pairing effects, we remark that the non-equilibrium pairing mechanism that we identified is not limited to the perfect forward-scattering case.

Applying our formalism to this model, we compute both analytically and numerically the steady-state properties at the superconducting transition for two different examples: (i) a temperature-bias setup with bosons and electrons coupled to two different thermal baths (similarly to Ref. [40]) and (ii) a setup where the bosons are incoherently driven at energies much

greater than the temperature (similarly to Ref. [Eckhardt *et al.*(2024)]).

In both setups, the non-thermal pairing mechanism significantly affects the physics of the superconducting phase transition. Specifically, the critical coupling g_c in the weak-coupling regime can be expressed in terms of three self-consistently determined parameters: the effective electron temperature T_e , the electron spectral width at the Fermi surface (FS) η , and the non-thermal pairing parameter α_{nth} . The first two describe modifications of the electron distribution and spectrum, respectively. The last parameter describes instead a genuinely non-thermal effect, as it arises from the breaking of the thermal fluctuation-dissipation relations in the anomalous self-energy. It is non-zero only outside of thermal equilibrium and its sign determines whether it contributes to enhance or suppress electron pairing with respect to the thermal case.

The paper is organized as follows. In section 19.2 we summarize the key results of the paper and verify their validity for the two examples. section 19.3 introduces the model Hamiltonian. section 19.4 discusses the structure of the NESS-Eliashberg equations. In section 19.5, expressions for the spectral gap and the critical coupling in the NESS are derived. section 19.6 discusses in more detail the two considered examples, and, finally, section 19.7 contains our conclusions. Additional details on the derivation and formal structure of the theory, as well as details on the methods to solve the NESS-Eliashberg equations, are given in sections 19.8.2 to 19.8.12, which are summarized at the end of the main text.

Throughout this paper, we will work in units where the Boltzmann constant k_B and the reduced Planck constant $\hbar$ are equal to unity.

19.2 Key results

In this Section, we present our key result: an analytical expression for the critical coupling of the superconducting phase transition in the NESS, which explicitly contains an additional contribution from the genuinely non-thermal pairing mechanism, as illustrated in Fig. 19.1(a). We then briefly discuss the main predictions and the underlying physical picture for two different examples, whereby we also benchmark our analytics with the numerical solution of the NESS Eliashberg equations.

19.2.1 Superconducting critical point in a NESS at weak coupling

Consider an interacting system of bosons and electrons in a NESS that is generated by linearly coupling them to two different environments, as sketched in Fig. 19.1(b). The bosons have a spectrum with a characteristic frequency ω_0 , spectral width κ and mediate an attractive forward-scattering interaction for the electrons with coupling g (see section 19.3 for more details on the model). We consider a regime where the coupling between bosons and electrons is small compared to ω_0 and κ , such that the feedback of the electrons on the bosons is negligible (see section 19.8.6).

The key result of our paper is that the critical coupling g_c of the superconducting phase transition in the NESS is modified by a genuinely non-thermal contribution. This becomes evident in the weak-coupling regime with well-defined electronic quasiparticles, where the critical coupling can be approximated as (see section 19.5.2)

$$g_c \approx \left[\frac{1}{g_{c,\text{th}}(T_e)} - \frac{|D_0^R| \alpha_{\text{nth}}}{2\eta} \right]^{-1}, \quad (19.1)$$

where

$$g_{c,\text{th}}(T_e) = \frac{2T_e}{|D_0^R|} \quad (19.2)$$

is the weak-coupling expression for the critical coupling at thermal equilibrium and $D_0^R = -\frac{\omega_0}{2(\omega_0^2 + \kappa^2)}$. In these expressions, η and T_e denote the spectral width (or inverse damping) and effective temperature of the electrons, respectively, both defined in the vicinity of the FS. The non-thermal pairing parameter α_{nth} appears only away from thermal equilibrium, resulting in $g_c \neq g_{c,\text{th}}$ whenever $\alpha_{\text{nth}} \neq 0$. Having $\alpha_{\text{nth}} \neq 0$, therefore, signifies that the non-thermal mechanism illustrated in Fig. 19.1(a) is activated. These three quantities generally depend on the NESS, which we show by considering two protocols that realizes different NESSs. For these two examples we have derived analytical weak-coupling expressions describing the NESS's properties (see section 19.6). In general, they need to be self-consistently determined by the NESS-Eliashberg equations for the normal (Σ_ω) and anomalous (Δ_ω) self-energies (see section 19.4).

The non-thermal pairing parameter is extracted from the Keldysh component of the anomalous self-energy Δ_ω evaluated at the FS ($\omega = 0$):

$$\alpha_{\text{nth}} = \frac{\text{Im} \Delta_{\omega=0}^K}{\Delta_{\omega=0}^R}, \quad (19.3)$$

where the functional dependence on ω is indicated by the subscript. At the phase transition the anomalous self-energies become infinitesimal. However, the ratio of the Keldysh component Δ_ω^K to the retarded component, i.e. the spectral gap, $\Delta_{\omega=0}^R$, is finite at and below the phase transition. In thermal equilibrium, Δ_ω^K is fixed by the thermal fluctuation-dissipation relation (FDR) $\Delta_\omega^K = \tanh(\omega/2T)(\Delta_\omega^R - \Delta_\omega^A)$ (see section 19.8.3) and becomes zero at $\omega = 0$, implying $\alpha_{\text{nth}} = 0$ and $g_c = g_{c,\text{th}}$, as anticipated.

As soon as the system is brought outside of thermal equilibrium, thermal FDRs are broken and α_{nth} becomes continuously non-zero, directly influencing the critical coupling through eq. (19.1). Its sign determines if it contributes to enhance ($\alpha_{\text{nth}} < 0 \Rightarrow g_c < g_{c,\text{th}}$) or decrease ($\alpha_{\text{nth}} > 0 \Rightarrow g_c > g_{c,\text{th}}$) the pairing with respect to thermal equilibrium at the temperature T_e .

The magnitude of the non-thermal contribution to g_c in eq. (19.1) is inversely proportional to the electron spectral width, given by

$$\eta = -\text{Im} \Sigma_{\omega=0}^R. \quad (19.4)$$

In absence of other broadening effects, the inclusion of the normal self-energy Σ_ω in the theory is essential to obtain a meaningful result for the non-thermal contribution.

Finally, an effective temperature T_e can be defined from the low-energy occupation of the electrons even in a NESS. This is self-consistently determined as

$$T_e = \lim_{\omega \rightarrow 0} \frac{1}{2\partial_\omega F_\omega^e} = \lim_{\omega \rightarrow 0} \left[\partial_\omega \frac{\text{Im} \Sigma_\omega^K}{\text{Im} \Sigma_\omega^R} \right]^{-1}, \quad (19.5)$$

i.e. from the slope of the electron distribution function F_ω^e around the FS. Estimating this temperature allows us to capture thermal effects related to cooling or heating of the electrons.

Before turning to the specific examples, let us stress a property of our newly introduced pairing mechanism that demonstrated its genuinely non-thermal nature. It is known from Eliashberg theory at thermal equilibrium, that in the weak-coupling limit the usual BCS result has to be obtained, apart from minor corrections [56, 60, 61]. In our eq. (19.1), $g_{c,\text{th}}$ indeed corresponds to the BCS result in a forward-scattering case [42]. However, in our NESS, we find that the additional pairing mechanism containing α_{nth} remains finite in the weak-coupling limit. This immediately tells us that it is not smoothly connected to any thermal phenomena, because the latter are fully contained in the BCS expression.

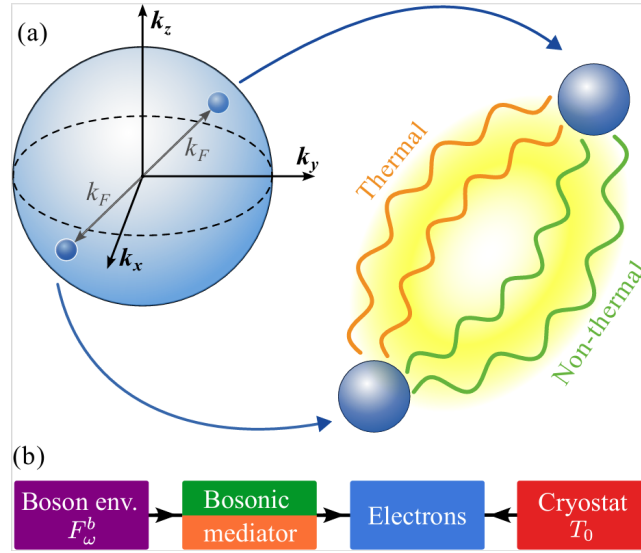


Figure 19.1: (a) Illustration of the coexistence of both thermal and non-thermal pairing mechanisms between two electrons at opposite sides of the FS in the NESS. Both mechanisms originate from interactions mediated by the bosonic mode. (b) Sketch of the generic setup used to consider both thermal and NESS superconductivity. It consists of a system composed of electrons which interact through a bosonic mediator. The system is coupled to two different environments: one coupled to the bosons, defined by the distribution F_ω^b , and a second coupled directly to the electrons. The latter is a thermal bath at a fixed temperature T_0 and therefore referred to as the “cryostat”.

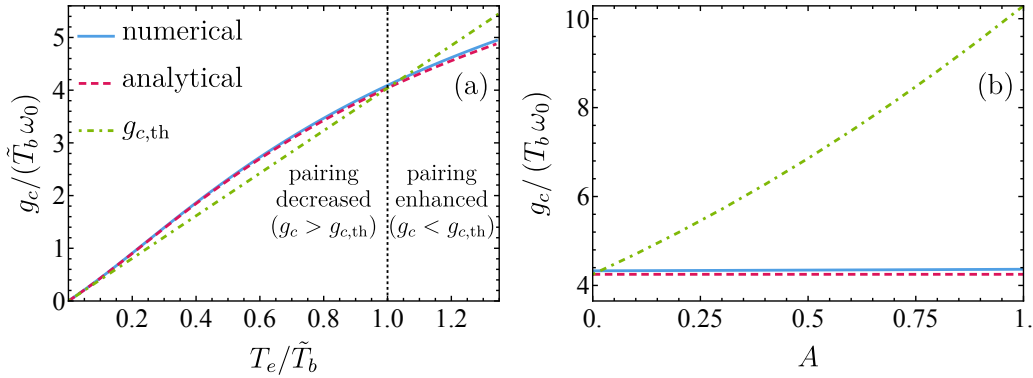


Figure 19.2: Critical coupling g_c for the two different example settings for NESS superconductivity. The “numerical” (blue solid) lines are the numerically converged self-consistent solutions of eq. (19.26) in the NESS. The “analytical” (red dashed) lines correspond to the self-consistent analytical solutions in the weak-coupling regime. The “ $g_{c,th}$ ” lines (green dot-dashed) correspond to the thermal equilibrium weak-coupling approximation given by eq. (19.2) evaluated at T_e . (a) Critical coupling g_c in the temperature-bias setting as a function of the effective electron temperature T_e at fixed boson temperature $T_b = \tilde{T}_b = 10^{-3}\omega_0$. The two curves intersect when $T_e = \tilde{T}_b$ in the NESS (vertical black dotted line). The analytical weak-coupling solution is calculated via eq. (19.46). The bath-induced spectral widths for bosons and electrons are $\kappa/\omega_0 = 0.1$ and $\eta_0/\omega_0 = 10^{-6}$, respectively. (b) Critical coupling g_c in the example with incoherent driving of the bosons as a function of the driving strength A . The boson spectral width is $\kappa/\omega_0 = 0.25$ and the thermal (low energy) part of the boson distribution is described by the temperature $T_b/\omega_0 = 10^{-3}$. The incoherent drive [parametrized in eq. (19.48)] is centered at the boson frequency ω_0 and has a Gaussian envelope with a width $v/\omega_0 = 0.05$.

19.2.2 NESS examples

To test the validity of eq. (19.1) and get a better understanding of the non-thermal effects, we have carefully analyzed two qualitatively different scenarios for generating NESSs.

Temperature bias – In the first example we consider a situation where the system is subjected to a temperature bias, i.e. the boson environment and the cryostat in Fig. 19.1(b) are two thermal baths with temperatures T_b and T_0 , respectively. We assume that the coupling of the bosons to their bath is strong enough for them to be treated as thermal at temperature T_b . The effective temperature T_e of the electrons generically ends up between the two external bath temperatures. Within this setting, we can either generate a NESS by keeping T_b fixed to a constant value and vary T_0 , or remain in thermal equilibrium under simultaneous variation of the baths' temperatures $T_b = T_0$. Such a setting can be realized, for example, in the case of cavity implementations, where the electrons are in contact with a cryostat, while the cavity photons are in contact with an external electromagnetic environment, as theoretically considered in Refs. [30, 40, 41] and experimentally realized in Ref. [9].

Within this setting, weak-coupling analytical self-consistent solutions for η , T_e and α_{nth} can be found (see section 19.6.1). In particular, in the NESS close to thermal equilibrium ($T_0 \approx T_e \approx T_b$) we find that

$$\alpha_{\text{nth}} \approx \mathcal{C} \frac{T_b - T_e}{T_e}, \quad (19.6)$$

where $\mathcal{C}$ is a positive constant that depends on T_b and the spectrum of both bosons and electrons. Following the discussion of eq. (19.1), we see that the pairing is non-thermally enhanced ($\alpha_{\text{nth}} < 0$) for bosons cooler than electrons ($T_b < T_e$) and decreased ($\alpha_{\text{nth}} > 0$) for hotter bosons ($T_b > T_e$).

To highlight this result, Fig. 19.2(a) shows the numerical result for the critical coupling g_c in the NESS against the effective electron temperature T_e obtained by keeping the temperature of the bosonic bath fixed to $T_b = \tilde{T}_b = 10^{-3}\omega_0$ and varying the cryostat temperature T_0 , compared to the thermal equilibrium result where the temperatures $T_b = T_e = T_0$ are all varied together. Note that, by plotting g_c against T_e instead of against T_0 , only genuinely non-thermal effects lead to deviations from the thermal equilibrium result, as “thermal-like” equilibration effects are already absorbed in the value of $T_e \neq T_0$. In Fig. 19.2(a), we observe genuinely non-thermal deviations throughout the whole temperature range, with the pairing in the NESS being non-thermally enhanced for $T_b < T_e$ and decreased for $T_b > T_e$, in line with the result sketched in eq. (19.6).

This example clearly shows that one can separate two-types of non-equilibrium effects: 1) “thermal-like” effects, whereby objects that were defined already at thermal equilibrium still exist in the NESS but with a modified value, and 2) “genuinely non-thermal” effects, whereby new objects that did not exist at thermal equilibrium appear. In the present case, the NESS modification of the electron temperature and damping belong to category 1), while α_{nth} belongs to 2), and so does in particular its effect on the critical coupling, corresponding to the difference between the green dot-dashed and the blue curve in Fig. 19.2(a).

Combining the weak-coupling analytical self-consistent solutions for η , T_e and α_{nth} with eq. (19.1), we also find a closed analytical expression for the critical coupling in the NESS [see eq. (19.46)]. This analytical solution is also plotted in Fig. 19.2(a) and agrees very well with the numerical result, confirming the validity of eq. (19.1) to capture the main contributions to the critical coupling.

More detailed results and discussions, including numerical and analytical results for η , T_e and α_{nth} , are given in section 19.6.1.

Incoherent driving of bosons – As a complementary example, we consider decoupling the cryostat from the electrons and instead generate a NESS by driving bosons at a temperature T_b incoherently around their resonance frequency ω_0 with a driving strength A and a width v

[see the parametrization in eq. (19.48)]. Such protocol should be possible within the setting of cavity-mediated superconductivity, where the cavity [31] can be driven with a high level of control. Incoherent driving of the bosons in superconductors was considered in [Eckhardt *et al.*(2024)], but their model was fundamentally different and they did not include non-thermal effects at a self-consistent level.

In this case, the physics is different from the temperature-bias case, as the drive ($\omega \approx \omega_0$) and thermal ($\omega \approx 0$) effects are energetically well separated. By increasing the drive strength the electrons heat up [see eq. (19.54)] resulting in an increase of the thermal part of the critical coupling $g_{c,\text{th}}$ and making it harder to reach a superconducting state. This is seen in Fig. 19.2(b) where both g_c and $g_{c,\text{th}}$ are plotted as a function of the drive strength. Taking into account the genuinely non-thermal effects, one observes that g_c does not follow the same behavior as $g_{c,\text{th}}$ but is instead essentially unperturbed by the drive

$$g_c \approx g_{c,\text{th}}(T_b) = g_c(A = 0). \quad (19.7)$$

The stability of g_c against incoherent driving can be understood from the closed-form expressions for T_e , η , and α_{nth} (see section 19.6.2), combined with eq. (19.1). Using the same language as for the discussion of the previous example, we can describe the physics of the present example as follows. While the “thermal-like” effects increase T_e and η , and thus also g_c , the “genuinely non-thermal” mechanism α_{nth} enhances the bosons’ ability to pair electrons in equal measure, resulting in a net cancellation of the effects of the incoherent drive. Additionally, we are able to show that this result is independent of the exact protocol for the incoherent drive, as long as the energy separation and weak-coupling criteria are satisfied. The incoherent driving of the bosons at energies much larger than T_e pushes the system into a highly non-thermal state while only weakly affecting the critical coupling.

In summary, we find that our approximation in eq. (19.1) works well in the weak-coupling regime. In the two qualitatively different protocols we have analyzed, we find that the genuinely non-thermal pairing mechanism that we propose here can have a substantial impact. Furthermore, our approach allows for a deep understanding of the weak-coupling regime, as we can derive analytical expression for the relevant quantities. Our result highlight the genuinely non-thermal pairing as an additional tuning knob for controlling the superconducting threshold.

19.3 Model

Having presented our key results, we now turn to a more detailed discussion of the underlying model. To investigate superconductivity in a non-equilibrium steady state (NESS), we consider a system coupled to external environments, as illustrated in Fig. 19.1(b). The total Hamiltonian reads

$$\hat{H}_{\text{tot}} = \hat{H}_{\text{sys}} + \hat{H}_{\text{env}}, \quad (19.8)$$

where $\hat{H}_{\text{sys}}$ describes an interacting electron–boson system and $\hat{H}_{\text{env}}$ accounts for the environments and their coupling to the system.

The system Hamiltonian can be decomposed as

$$\hat{H}_{\text{sys}} = \hat{H}_{e,0} + \hat{H}_{b,0} + \hat{H}_{e-b}, \quad (19.9)$$

where

$$\hat{H}_{e,0} = \sum_{\mathbf{k}} \sum_{\sigma=\uparrow,\downarrow} \xi_{\mathbf{k}} \hat{c}_{\sigma,\mathbf{k}}^\dagger \hat{c}_{\sigma,\mathbf{k}}, \quad (19.10)$$

is the Hamiltonian for free electrons with mass m and chemical potential μ , such that their dispersion is $\xi_{\mathbf{k}} = \mathbf{k}^2/2m - \mu$. We focus on a system with a single boson mode with frequency ω_0 giving the bare boson Hamiltonian the form

$$\hat{H}_{b,0} = \omega_0 \hat{a}^\dagger \hat{a}. \quad (19.11)$$

The electron-boson interaction is considered to be a Yukawa-like density coupling,

$$\hat{H}_{e-b} = 2g_0 \sum_{\mathbf{k}, \sigma} \hat{c}_{\sigma, \mathbf{k}}^\dagger \hat{c}_{\sigma, \mathbf{k}} (\hat{a}^\dagger + \hat{a}), \quad (19.12)$$

where g_0 is the interaction strength. This form of coupling realizes a perfect forward-scattering scenario: bosons do not transfer momentum to the electrons. Such a model is relevant for photon–electron coupling in cavity-driven materials [31, 37] and for phonon-mediated interactions in monolayer FeSe on SrTiO₃ substrates [42, 43], which has also been considered in cavity-driven contexts [44]. More generally, this model allows us to investigate non-thermal effects in a relatively simple yet realistic setting. Extensions to more general electron–boson interactions [Schlawin *et al.*(2019), Eckhardt *et al.*(2024)] are conceptually straightforward. We note that the main result in eq. (19.1) depends on the momentum dependence of the interaction. We expect smoother interactions to decrease the non-thermal contribution to g_c by reducing electronic coherence (see section 19.8.12).

To generate a NESS, we assume linear coupling between the system and the external environments (see sections 19.8.7 and 19.8.8). The linearity allows the environmental degrees of freedom to be integrated out exactly, yielding renormalized spectra and distribution functions for both electrons and bosons [see eqs. (19.27) to (19.30) in section 19.4]. When the environments are not mutually thermalized, the system settles into a NESS.

For the bosons, we focus on the regime in which the electron–boson coupling g_0 is weak compared to the boson–environment coupling. In this case, the back-action of the electrons on the bosons is negligible (see section 19.8.6). For the electrons, we consider the opposite regime: their coupling to the cryostat is weak relative to g_0 . In this limit, the properties of the electrons in the interacting NESS can differ substantially from those determined solely by the cryostat.

19.4 NESS-Eliashberg equations

In this section, we present the structure of the NESS-Eliashberg equations. A detailed derivation, based on a real-time path-integral formalism, is provided in section 19.8.2. We focus on the two-point correlation functions, which contain information about both the spectral properties and the particles distribution. Since the NESS is non-thermal, neither the spectrum nor the distribution of the electrons are known a priori, and both must be determined self-consistently. To this end, the electron propagators acquire a 2×2 matrix structure [71]

$$\mathcal{G}_{x, x'} = \begin{pmatrix} \mathcal{G}^K & \mathcal{G}^R \\ \mathcal{G}^A & 0 \end{pmatrix}_{x, x'}, \quad (19.13)$$

where $x = (\mathbf{r}, t)$ denotes the space-time coordinate. We assume a spin-balanced system, such that the propagators for both spin-components are identical. The retarded propagator describes the linear response of the system to single-particle excitations. For the electrons, it is defined via the expectation value of the anti-commutator

$$\mathcal{G}_{x, x'}^R = -i\theta_{t-t'} \left\langle \left\{ \hat{c}_{\sigma, x}, \hat{c}_{\sigma, x'}^\dagger \right\} \right\rangle, \quad (19.14)$$

where $\theta_{t-t'}$ is the Heaviside step function. The advanced propagator is related to the retarded propagator through the relation $\mathcal{G}_{x, x'}^A = (\mathcal{G}_{x', x}^R)^*$, with the asterisk denoting complex conjugation. Assuming a state that is invariant under translations in both space and time, the space-time dependence of $\mathcal{G}_{x, x'}$ can be diagonalized by transforming to momentum and frequency space. Additionally, the perfect forward scattering in eq. (19.12), decouples different momentum modes, allowing us to focus solely on the Fermi momentum $|\mathbf{k}| = k_F$, which is the

first mode to develop an instability in the pairing channel [72]. Consequently, we will discuss the physics at k_F for the remainder of paper. A more general momentum-resolved description is provided in section 19.8.2.

The retarded propagator provides direct access to the steady-state electron spectral function $\mathcal{A}_\omega$, which encodes the density of single-particle excitations, through the propagator's imaginary part

$$\mathcal{A}_\omega = i (\mathcal{G}_\omega^R - \mathcal{G}_\omega^A) = -2 \text{Im } \mathcal{G}_\omega^R. \quad (19.15)$$

While the retarded propagator only contains information about the spectrum, the Keldysh propagator contains information about how this spectrum is occupied. It is defined via the expectation value of the commutator

$$\mathcal{G}_{x,x'}^K = -i \left\langle \left[\hat{c}_{\sigma,x}, \hat{c}_{\sigma,x'}^\dagger \right] \right\rangle. \quad (19.16)$$

To make the connection to the occupation explicit, the steady-state Keldysh propagator in frequency space at k_F can be reparameterized in terms of the spectral function

$$\mathcal{G}_\omega^K = -i F_\omega^e \mathcal{A}_\omega, \quad (19.17)$$

where the distribution function F_ω^e fully characterizes the occupation of the spectrum. In thermal equilibrium, it is fixed by the thermal FDRs, taking the familiar form $F_\omega^e = \tanh(\omega/2T)$.

In the superconducting phase the bosonic mediator leads to two types of interaction channels. One is the exchange channel, which gives rise to the normal self-energy (Σ), the other is the pairing channel, which generates the anomalous self-energy (Δ). Out of equilibrium both self-energies acquire a 2×2 matrix structure

$$\Sigma_\omega = \begin{pmatrix} 0 & \Sigma^A \\ \Sigma^R & \Sigma^K \end{pmatrix}_\omega, \quad \Delta_\omega = \begin{pmatrix} 0 & \Delta^A \\ \Delta^R & \Delta^K \end{pmatrix}_\omega, \quad (19.18)$$

where the retarded and advanced components are connected by complex conjugation². Using the retarded component of the normal self-energy, the Fermi momentum can be defined self-consistently as

$$k_F = \sqrt{2m(\mu - \text{Re } \Sigma_{\mathbf{k}=\mathbf{k}_F, \omega=0}^R)}. \quad (19.19)$$

For convenience it is useful to define the normal-state propagator (i.e. the normal propagator without the anomalous self-energy) as

$$G_\omega = [(\omega - \xi_{\mathbf{k}_F})\sigma_x - \Sigma_\omega]^{-1} = (\omega\sigma_x - \Sigma_\omega + \text{Re } \Sigma_0)^{-1}, \quad (19.20)$$

where σ_x is the first Pauli-matrix. In the second equality we used $\xi_{\mathbf{k}_F} = k_F^2/2m - \mu$ and eq. (19.19) to simplify the expression. As the system enters the superconducting phase, the anomalous self-energy becomes finite and the normal electron propagator becomes

$$\mathcal{G}_\omega = (G_\omega^{-1} + \Delta_\omega G_{-\omega}^T \bar{\Delta}_\omega^T)^{-1}, \quad (19.21)$$

where (see section 19.8.4)

$$\bar{\Delta}_\omega = \begin{pmatrix} 0 & \bar{\Delta}^A \\ \bar{\Delta}^R & \bar{\Delta}^K \end{pmatrix}_\omega = \begin{pmatrix} 0 & (\Delta^A)^* \\ (\Delta^R)^* & -(\Delta^K)^* \end{pmatrix}_\omega, \quad (19.22)$$

²The conjugation relation between anomalous retarded (Δ_ω^R) and advanced (Δ_ω^A) self-energies is true only at $|\mathbf{k}| = k_F$ and for the gauge choice $\Delta_{\omega=0}^R \in \mathbb{R}$. This is discussed in detail in section 19.8.4.

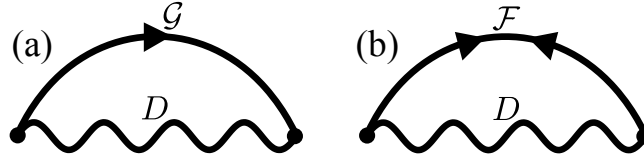


Figure 19.3: Structure of (a) normal and (b) anomalous self-energy diagrams. The wavy line represents the boson propagator, while the straight line with a single arrow (two opposing arrows) denote the normal (anomalous) electron propagator.

resulting in the retarded component of the normal propagator taking the form

$$\mathcal{G}_\omega^R = \frac{1}{(G_\omega^R)^{-1} + G_{-\omega}^A (\Delta_\omega^A)^* \Delta_\omega^R}. \quad (19.23)$$

Considering electrons at k_F allows us to fix a gauge in which $(\Delta_\omega^A)^* = \Delta_\omega^R$ (cf. section 19.8.4). As follows from eq. (19.15), the electron spectrum is fully determined by the retarded propagator. Combined with eq. (19.23), this implies that the spectral gap does not explicitly depend on Δ_ω^K . In thermal equilibrium and the weak-coupling regime, Δ_ω^R reduces to the conventional BCS gap. More generally, weak coupling implies that Δ_ω^R varies slowly in the vicinity of the FS. Under this assumption, the energy gap in the electron spectrum is set by Δ_0^R , which is purely real in the chosen gauge ³.

The spectral gap arises in the superconducting phase due to the condensation of Cooper pairs, giving rise to finite anomalous propagators

$$\begin{aligned} \mathcal{F}_{x,x'}^R &= -i\theta_{t-t'} \langle \{\hat{c}_{\uparrow,x}, \hat{c}_{\downarrow,x'}\} \rangle, \\ \mathcal{F}_{x,x'}^A &= i\theta_{t'-t} \langle \{\hat{c}_{\uparrow,x}, \hat{c}_{\downarrow,x'}\} \rangle, \\ \mathcal{F}_{x,x'}^K &= -i \langle [\hat{c}_{\uparrow,x}, \hat{c}_{\downarrow,x'}] \rangle, \end{aligned} \quad (19.24)$$

which can be directly computed from the normal and anomalous self-energies

$$\mathcal{F}_\omega = G_\omega \Delta_\omega \mathcal{G}_{-\omega}^T = \mathcal{G}_\omega \Delta_\omega G_{-\omega}^T, \quad (19.25)$$

with a 2×2 matrix structure analogous to eq. (19.13). Within the path-integral formulation outlined in section 19.8.2, the boson-mediated interaction gives rise to normal and anomalous self-energies of a one-loop form. Their diagrammatic representation is shown in fig. 19.3, while the corresponding NESS-Eliashberg equations read

$$\begin{aligned} \Sigma_\omega^R &= ig \int_\epsilon (D_{\omega-\epsilon}^R \mathcal{G}_\epsilon^K + D_{\omega-\epsilon}^K \mathcal{G}_\epsilon^R), \\ \Sigma_\omega^K &= ig \int_\epsilon (D_{\omega-\epsilon}^R \mathcal{G}_\epsilon^R + D_{\omega-\epsilon}^A \mathcal{G}_\epsilon^A + D_{\omega-\epsilon}^K \mathcal{G}_\epsilon^K), \\ \Delta_\omega^R &= ig \int_\epsilon (D_{\omega-\epsilon}^R \mathcal{F}_\epsilon^K + D_{\omega-\epsilon}^K \mathcal{F}_\epsilon^R), \\ \Delta_\omega^K &= ig \int_\epsilon (D_{\omega-\epsilon}^R \mathcal{F}_\epsilon^R + D_{\omega-\epsilon}^A \mathcal{F}_\epsilon^A + D_{\omega-\epsilon}^K \mathcal{F}_\epsilon^K), \end{aligned} \quad (19.26)$$

where we have introduced the notation $\int_\epsilon = \int_{-\infty}^\infty \frac{d\epsilon}{2\pi}$. The effective electron-electron interaction strength is $g = 8g_0^2 \sqrt{1 + \kappa^2/\omega_0^2}$ and $D_\omega^{R/A/K}$ is the propagator for the real quadrature of the boson. In thermal equilibrium, the Keldysh components Σ_ω^K and Δ_ω^K satisfy FDRs analogous

³Note that in principle the spectral gap should be self-consistently evaluated at the gap edge via the equation $\Delta_0 = \text{Re} \Delta_{\omega=\Delta_0}^R$ [61], but $\Delta_0 \ll \omega_0, \kappa$ in weak-coupling regime, so that $\Delta_0 = \Delta_{\omega \approx 0}^R$ is a good approximation.

to those of the electron propagator, and Eqs. (19.26) become fully equivalent to the real-frequency analytic continuation of the Eliashberg equations in Matsubara frequencies, as shown in section 19.8.3.

In Eqs. (19.26), no equations for the boson self-energies have been included. This is justified by the bosons being much more strongly coupled to their environment than they are to the electrons, such that the back-action of the electrons can be neglected (see section 19.8.6 for justification). Within this approximation, only the environment-induced self-energies have to be included in the boson propagators. As shown in section 19.8.7, including these results in the boson propagators taking the form

$$\begin{aligned} D_{\omega}^{R(A)} &= \frac{1}{2} \frac{\omega_0}{(\omega \pm i\kappa)^2 - \omega_0^2}, \\ D_{\omega}^K &= 2iF_{\omega}^b \operatorname{Im} D_{\omega}^R, \end{aligned} \quad (19.27)$$

where κ is an environment-induced spectral width and the renormalization of the boson energy has been reabsorbed into ω_0 . The bosonic distribution function, $F_{\omega}^b = n_{\omega}^E - n_{-\omega}^E$, is fully determined by the odd part of the bosonic environment's occupation function (n_{ω}^E). For instance, if the boson environment is a thermal bath at a temperature T_b , the boson distribution function becomes

$$F_{T_b, \omega}^{b, \text{th}} = \coth \frac{\omega}{2T_b}. \quad (19.28)$$

Analogously, we include the possibility of weakly coupling the electrons to an external environment, which we model as a cryostat. The cryostat drives the electron distribution toward a thermal one at temperature T_0 ⁴. As shown in section 19.8.8, the thermalization effect of the cryostat can be captured by adding the following cryostat-induced contributions to the normal self-energy in eq. (19.20)

$$\begin{aligned} \Sigma_{\omega}^{\text{cryo}, R} &= -i\eta_0, \\ \Sigma_{\omega}^{\text{cryo}, K} &= -2i\eta_0 F_{\omega}^{\text{cryo}}, \end{aligned} \quad (19.29)$$

where $F_{\omega}^{\text{cryo}} = \tanh(\omega/2T_0)$ and η_0 is the spectral broadening of the electrons induced by the cryostat, reflecting the strength of the cryostat-electron coupling. The effect of the cryostat on the electron is thus implemented through the substitutions

$$\begin{aligned} \Sigma_{\omega}^R &\rightarrow \Sigma_{\omega}^R + \Sigma_{\omega}^{\text{cryo}, R}, \\ \Sigma_{\omega}^K &\rightarrow \Sigma_{\omega}^K + \Sigma_{\omega}^{\text{cryo}, K}. \end{aligned} \quad (19.30)$$

19.5 Non-thermal contributions to the spectral gap and the superconducting phase transition

In this section, we will highlight the genuinely non-thermal effects contained in Eqs. (19.26). We will first sketch how they arise in the spectral gap within the superconducting phase, and then concentrate on properties at phase transition, deriving our main result in eq. (19.1).

19.5.1 Spectral gap

As discussed in the previous section, the gap that opens in the electron spectrum is given by Δ_0^R in the weak-coupling regime. This spectral gap is self-consistently determined from the Δ_{ω}^R

⁴We use a linear coupling to a fermionic bath to describe this process. Such a simplified description avoids the additional complexity of a bosonic bath, and since we are anyway focusing on a weakly-coupled cryostat the precise modeling details have minimal impact on the observed physics.

equation in Eqs. (19.26), evaluated at $\omega = 0$. Consider now a regime where the spectral width η of electrons at the FS is small compared to the characteristic energy scales of the bosons (ω_0 , κ) and the effective electron temperature T_e . In this regime, the electron propagators are sharply peaked near the FS, whereas the boson propagators vary slowly with frequency. This qualitative difference allows one to approximate the integrals in the equation for Δ_0^R by evaluating the boson propagators at $\omega = 0$ and bringing them outside of the integral, obtaining

$$\Delta_0^R = igD_0^R \int_{\epsilon} \mathcal{F}_{\epsilon}^K + igD_0^K \int_{\epsilon} \mathcal{F}_{\epsilon}^R. \quad (19.31)$$

The second term vanishes because the anomalous retarded propagator $\mathcal{F}^R$ contains the expectation value of an equal-time anti-commutator

$$\int_{\epsilon} \mathcal{F}_{\epsilon}^R = \mathcal{F}_{\tau=0}^R = -i\theta_{\tau=0} \langle \{\hat{c}_{\uparrow, \mathbf{k}_F, t}, \hat{c}_{\downarrow, -\mathbf{k}_F, t}\} \rangle = 0, \quad (19.32)$$

where the relative-time variable $\tau = t - t'$ has been introduced.

In the weak-coupling regime, the spectral gap therefore simplifies to

$$\Delta_0^R = igD_0^R \mathcal{F}_{\tau=0}^K. \quad (19.33)$$

By Fourier transforming the anomalous Keldysh propagator defined in eq. (19.24), one identifies $\mathcal{F}_{\tau=0}^K$ as $2i$ times the Cooper-pair expectation value $\langle \hat{c}_{\uparrow, \mathbf{k}_F, t} \hat{c}_{\downarrow, -\mathbf{k}_F, t} \rangle$. Since they are linearly connected, either Δ_0^R or $\langle \hat{c}_{\uparrow, \mathbf{k}_F, t} \hat{c}_{\downarrow, -\mathbf{k}_F, t} \rangle$ can thus serve as an order parameter; throughout this work, we adopt Δ_0^R .

Although our main results concern the phase transition, one can show that the spectral gap inside the superconducting phase also acquires a genuinely non-thermal correction. This can be seen by explicitly evaluating $\mathcal{F}_{\tau=0}^K = \int_{\epsilon} \mathcal{F}_{\epsilon}^K$ in eq. (19.31) using eq. (19.25), leading to (cf. section 19.8.5)

$$\begin{aligned} \mathcal{F}_{\tau=0}^K = \int_{\epsilon} & \left[2iF_{\epsilon}^{\Sigma} \text{Im} \mathcal{F}_{\epsilon}^R - 2i(F_{\epsilon}^{\Delta} - F_{\epsilon}^{\Sigma}) \right. \\ & \left. \times |\mathcal{G}_{\epsilon}^R|^2 (1 - |G_{\epsilon}^R|^2 |\Delta_{\epsilon}^R|^2) \text{Im} \Delta_{\epsilon}^R \right]. \end{aligned} \quad (19.34)$$

Here the Keldysh components of the normal (Σ_{ϵ}^K) and anomalous (Δ_{ϵ}^K) self-energies have been parametrized by generalized distributions F_{ϵ}^{Δ} and F_{ϵ}^{Σ} , analogously to the parametrization of the electron propagator in eq. (19.17) [cf. eqs. (19.190) and (19.191)]. In thermal equilibrium, both generalized distributions reduce to $\tanh(\epsilon/2T)$, and only the first term in the integrand of eq. (19.34) contributes. In a NESS, however, F_{ϵ}^{Δ} and F_{ϵ}^{Σ} generally differ, giving rise to a genuinely non-thermal correction to the spectral gap in eq. (19.33) through the second term in eq. (19.34).

19.5.2 Critical coupling

We now consider how the genuinely non-thermal effects modify the onset of superconductivity. Building on the previous section, where we showed that the spectral properties of the electrons acquire non-thermal corrections in superconducting phase, we now analyze how these corrections enter the phase transition itself. At the critical transition point ($g = g_c$), the anomalous self-energy Δ_{ω} becomes infinitesimal. Consequently, in Eqs. (19.26) one can replace $\mathcal{G}_{\omega} \rightarrow G_{\omega}$, which implies $F_{\omega}^{\Sigma} \rightarrow F_{\omega}^e$ (cf. section 19.8.5). This renders $\mathcal{F}_{\omega}$ in eq. (19.25), and therefore the anomalous self-energy equations in (19.26), linear in Δ_{ω} . The expression for $\mathcal{F}_{\tau=0}^K$ in eq. (19.34) then simplifies to

$$\begin{aligned} \mathcal{F}_{\tau=0}^K = - \int_{\epsilon} & \left\{ 2iF_{\epsilon}^e \left\{ \text{Im} \left[(G_{\epsilon}^R)^2 \Delta_{\epsilon}^R \right] \right. \right. \\ & \left. \left. + |G_{\epsilon}^R|^2 \text{Im} \Delta_{\epsilon}^R \right\} + |G_{\epsilon}^R|^2 \Delta_{\epsilon}^K \right\}, \end{aligned} \quad (19.35)$$

Again exploiting that the electron spectrum remains sharply peaked around the FS, whereas the anomalous components Δ_ϵ^R and Δ_ϵ^K vary slowly, $\mathcal{F}_{\tau=0}^K$ can be approximated as

$$\mathcal{F}_{\tau=0}^K \approx 2i\mathcal{I}_K \Delta_0^R + i\mathcal{I}_R \text{Im} \Delta_0^K, \quad (19.36)$$

where

$$\begin{aligned} \mathcal{I}_K &= - \int_\epsilon F_\epsilon^e \text{Im} \left[(G_\epsilon^R)^2 \right], \\ \mathcal{I}_R &= - \int_\epsilon |G_\epsilon^R|^2. \end{aligned} \quad (19.37)$$

Furthermore, in the weak-coupling regime, the retarded electron propagator takes the approximate low-energy form

$$G_\epsilon^R \approx \frac{1}{\omega + i\eta}, \quad (19.38)$$

with spectral width defined in eq. (19.4). The electron distribution is likewise approximated by

$$F_\epsilon^e \approx \frac{\epsilon}{2T_e}, \quad (19.39)$$

with the effective electron temperature defined in eq. (19.5). These approximations allow a closed-form solution of the integrals in eq. (19.37), yielding $\mathcal{I}_R = -1/2\eta$ and $\mathcal{I}_K = 1/4T_e$. Substituting back into $\mathcal{F}_{\tau=0}^K$ and using the result in eq. (19.33) leads to the critical coupling

$$g_c = \left[\frac{|D_0^R|}{2} \left(\frac{1}{T_e} - \frac{\alpha_{\text{nth}}}{\eta} \right) \right]^{-1}, \quad (19.40)$$

where $D_0^R = -\frac{\omega_0}{2(\omega_0^2 + \kappa^2)}$ and $\alpha_{\text{nth}} = \text{Im}(\Delta_0^K) / \Delta_0^R$. This is our main result presented in eq. (19.1).

19.6 Non-thermal steady states: Two examples

The expression for g_c derived in eq. (19.40), provides a compact analytic measure of how genuinely non-thermal effects reshape the superconducting instability. Yet the size and qualitative character of these corrections depend sensitively on the structure of the NESS. In order to get an understanding of how important the correction is and to connect the analytic weak-coupling estimates to fully self-consistent solutions, we considered two concrete examples in section 19.2.2. In this section we will examine these two examples in more detail. We discuss the relevant physical quantities at the phase transition obtained both from self-consistent numerical solutions of Eqs. (19.26) and analytically in the weak-coupling regime. The numerical iterative method employed to solve Eqs. (19.26) is described in section 19.8.9.

19.6.1 Temperature bias

As a first concrete realization of a NESS, we consider a finite temperature bias between the two external environments in Fig. 19.1(b). For the results in the NESS, we keep the boson temperature T_b fixed at $\tilde{T}_b$ and vary the cryostat temperature T_0 , thereby tuning the degree of non-thermality. Thermal equilibrium is restored when the bias is removed ($T_b = T_0$), so the corresponding results are instead obtained by varying the two temperatures simultaneously. All the results shown in this section are computed using the same parameters as in Fig. 19.2(a), with a sufficiently small $\tilde{T}_b$ and a sufficiently large κ to stay within the weak-coupling regime and justify neglecting the back-action of the electrons on the bosons. The derivations of the analytical expressions for the various physical quantities can be found in section 19.8.10.

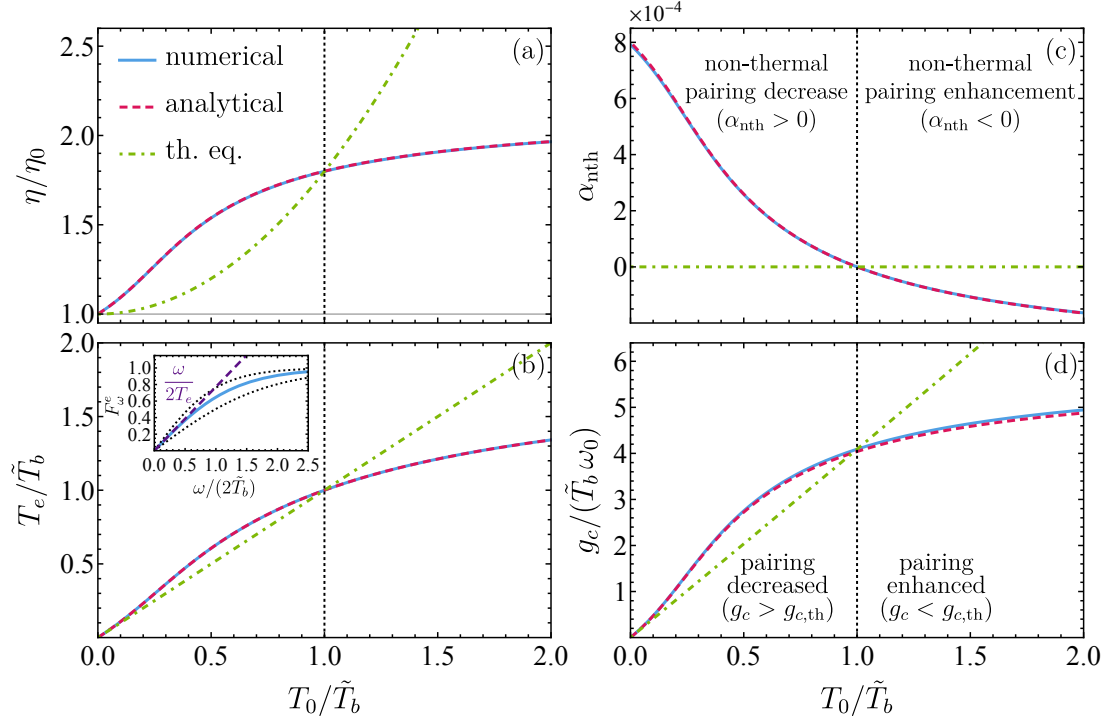


Figure 19.4: Physical quantities appearing in eq. (19.1) for the temperature-bias example as a function of the cryostat temperature T_0 : (a) electron spectral width, (b) effective electron temperature, (c) non-thermal pairing parameter, (d) critical coupling. The NESS is realized by keeping $T_b = \tilde{T}_b$ fixed and varying T_0 . The input parameters are identical to those in Fig. 19.2(a). Blue solid lines are numerical results in the NESS, red dashed lines are the corresponding weak-coupling analytical expressions given in eqs. (19.43) to (19.46), green dot-dashed lines are numerical results in thermal equilibrium ($T_b = T_0$ along the curve). The vertical black dotted line ($T_0 = \tilde{T}_b$) separates the regions of decreased ($T_0 < \tilde{T}_b$) and enhanced ($T_0 > \tilde{T}_b$) pairing. Inset of (b): Numerical estimation of T_e by fitting the electron distribution F_ω^e (blue solid line) with $\omega/2T_e$ (violet dashed line) for the point at $T_0/\tilde{T}_b = 1.8$. The black dotted lines correspond to the Fermi distribution at temperatures T_b (above) and T_0 (below).

In section 19.2 we showed that genuinely non-thermal effects can enhance pairing when the bosons are cooler than the electrons. To disentangle the underlying mechanisms, we now examine separately the behavior of the electron spectral width η , the effective electron temperature T_e and the genuinely non-thermal pairing parameter α_{nth} , clarifying how each quantity influences the critical coupling g_c via eq. (19.1). For each of these physical quantities we find a weak-coupling analytical expression which, when combined with eq. (19.1), yields a closed analytical expression for g_c expressed solely in terms of the external parameters (T_0 , T_b , η_0 , ω_0 and κ).

Before considering the analytical approximations for the physical quantities, we point out that many of these can be expressed in terms of the constant

$$\gamma = -\frac{\text{Im } D_0^K}{2T_b} = \frac{2\omega_0\kappa}{(\kappa^2 + \omega_0^2)^2}, \quad (19.41)$$

which has the units of inverse energy squared (see section 19.8.10). This quantity governs, for instance, the first-order broadening of the electron spectrum due to the interaction with bosons at T_b

$$\eta^{(1)} = g_c \gamma T_b. \quad (19.42)$$

Including the cryostat contribution, the total spectral width becomes

$$\eta \approx \eta_0 + g_c \gamma T_b. \quad (19.43)$$

To confirm the validity of the analytical approximation in eq. (19.43), we compare it to the numerical value obtained from $\eta = \eta_0 - \text{Im } \Sigma_0^R$ in Fig. 19.4(a). In the figure one sees that the T_0 dependence of η obtained numerically excellently agrees with the analytical result. The thermal equilibrium spectral width is also shown for comparison. Interestingly, we see that for $T_0 > T_b$ (cooler bosons) the electrons have a smaller spectral width than in thermal equilibrium, which boosts the genuinely non-thermal enhancement of pairing in this region due to its $\sim \alpha_{\text{nth}}/\eta$ dependence in eq. (19.1).

Having established the behavior of the spectral width, we now consider the effective electron temperature T_e . Its weak-coupling behavior is captured by

$$T_e \approx \frac{\eta_0 + g_c \gamma T_b}{\eta_0 + g_c \gamma T_0} T_0, \quad (19.44)$$

from which the thermal equilibrium result $T_e = T_0$ immediately follows when $T_b = T_0$. In the NESS, however, T_e acquires a non-trivial dependence on the self-consistently determined g_c , via eq. (19.1), and the temperature bias. Numerically, T_e can be extracted by the slope of F_ω^e around the FS as shown in eq. (19.5). This corresponds to fitting the electron distribution with $\omega/2T_e$ close to the FS [see inset of Fig. 19.4(b)]. The main panel of Fig. 19.4(b) shows that these numerical results again agree extremely well with the analytical expression. As a consistency check, it is observed that T_e always lies between the temperatures of the two environments, T_0 and T_b .

The final self-consistent ingredient required to determine g_c is the genuinely non-thermal pairing parameter α_{nth} , defined in eq. (19.3). In the weak-coupling regime, its analytical approximation is found to be

$$\alpha_{\text{nth}} \approx \frac{g_c \gamma}{1 + \frac{g_c \gamma T_b}{\eta}} \frac{T_b - T_e}{T_e} \approx \frac{g_c \gamma \eta_0}{\eta_0 + 2g_c \gamma T_b} \frac{T_b - T_0}{T_0}, \quad (19.45)$$

from which one can directly extract the constant $\mathcal{C}$ in eq. (19.6). In thermal equilibrium, $T_b = T_e = T_0$, which immediately implies $\alpha_{\text{nth}} = 0$. By contrast, in the NESS, α_{nth} becomes

negative for $T_0 > T_e > T_b$, i.e. when the bosons are cooler than the cryostat, as already discussed in section 19.2.2. Note that the sign-changing factor in eq. (19.45) exhibits the same functional dependence whether expressed in terms of T_e or T_0 . This follows directly from the monotonic dependence of T_e on T_0 , shown in Fig. 19.4(b), together with the fact that $T_e = T_0$ when $T_0 = T_b$.

In Fig. 19.4(c) we plot α_{nth} as a function of T_0 , comparing the numerical solution from eq. (19.3) with the analytical expression in eq. (19.45). The two approaches show excellent agreement over the considered temperature range. Small deviations appear only at low temperatures, where $T_e \sim \eta$ and the assumption $\eta \ll T_e$ underlying the derivation of eq. (19.45) ceases to be valid. As discussed in section 19.2, the sign of α_{nth} determines whether the genuinely non-thermal contribution in eq. (19.1) enhances ($\alpha_{\text{nth}} < 0$) or suppresses ($\alpha_{\text{nth}} > 0$) pairing relative to the thermal-like contribution. Consistent with this interpretation, Fig. 19.4(c) shows that pairing enhancement occurs when the boson are cooler than the cryostat ($T_0 > T_b$), in agreement with the behavior observed in Fig. 19.2(a).

In Fig. 19.2(a), the critical coupling g_c was presented as a function of the effective temperature T_e to emphasize non-thermal effects. For completeness, Fig. 19.4(d) displays the same data as a function of the cryostat temperature T_0 , which is the independent parameter. This representation makes clear that both the thermal-like renormalization of the temperature $T_e < T_0$ and the genuinely non-thermal contribution ($\alpha_{\text{nth}} < 0$) act constructively, leading to an enhanced pairing instability when the bosons are cooled below the cryostat temperature.

The analytical weak-coupling expression for g_c , shown in Fig. 19.2(a) and Fig. 19.4(d), is obtained by combining eq. (19.1) with the analytical solutions for η , T_e and α_{nth} given in eqs. (19.43) to (19.45). This leads to

$$g_c = \frac{2T_b}{|D_0^R|} + \frac{1}{2} \left(\frac{2T_b}{|D_0^R|} + \frac{\eta_0}{2\gamma T_0} \right) \times \left(\sqrt{1 + \frac{4\eta_0(T_0 - T_b)}{\gamma T_0 |D_0^R|} \left(\frac{2T_b}{|D_0^R|} + \frac{\eta_0}{2\gamma T_0} \right)^{-2}} - 1 \right). \quad (19.46)$$

At thermal equilibrium ($T_b = T_e = T_0$), the second term vanishes and one recovers the thermal result $g_c = 2T_e/|D_0^R|$ from eq. (19.2). As seen in Fig. 19.2(a) and Fig. 19.4(d), the weak-coupling approximation in eq. (19.46) reproduces the numerical results very well, thereby confirming the validity of the key result in eq. (19.1)⁵.

To summarize, we find that pairing is enhanced by cooling the boson mediating electron-electron interactions below the cryostat temperature. This enhancement arises from the combined action of a thermal-like reduction of the electronic temperature T_e , and a genuinely non-thermal contribution encoded in the finite value of α_{nth} .

19.6.2 Incoherent driving of bosons

In the previous example, a NESS was generated by a finite temperature bias. An alternative route to a NESS is to directly couple the bosons to a non-thermal environment, for instance by incoherently driving the bosons on top of their thermal background. With the amplitude of the incoherent drive acting as a tuning parameter for the degree of non-thermality, we can simplify the description by decoupling the cryostat in Fig. 19.1(b). Within our formalism, this corresponds to setting $\eta_0 = 0$ in eq. (19.43), such that the steady-state properties of the electrons are entirely determined by the interactions with the bosons.

⁵We remark that the small discrepancy of the weak-coupling solution at larger T_0 comes mostly from the weak-coupling approximation of the thermal contribution $g_{c,\text{th}}$ of eq. (19.2), rather than from the non-thermal contribution in eq. (19.1). This is also discussed at the end of section 19.6.2.

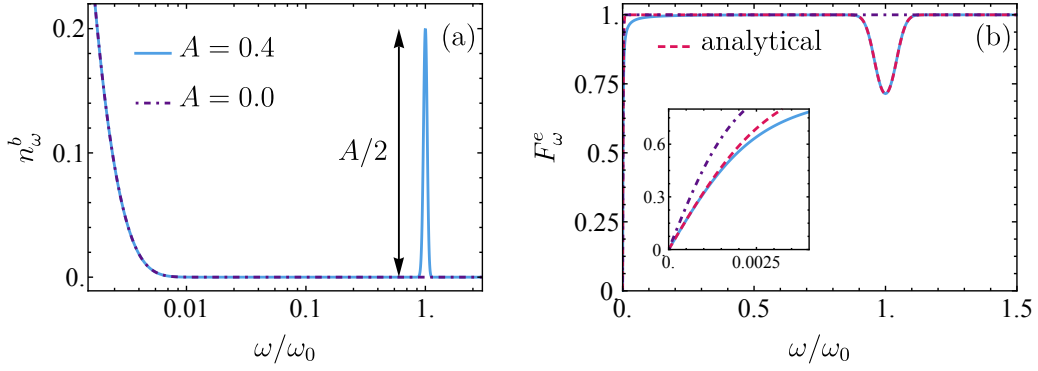


Figure 19.5: (a) Boson occupation function $n_\omega^b = (F_\omega^b - 1)/2$ with F_ω^b given by eq. (19.47) and (b) numerically computed electron distribution function F_ω^e when the bosons are incoherently driven with strength A . The $A = 0.0$ (blue solid) lines represent the undriven/thermal case, while the $A = 0.4$ (violet dot-dashed) lines denote the distributions for a finite drive strength. The additional “analytical” (red dashed) curve corresponds to the weak-coupling approximation in eq. (19.50) for $A = 0.4$, with the effective electron temperature T_e determined via eq. (19.54). The inset shows a magnification of the behavior close to the FS, highlighting its thermal character.

As in the previous section, we separately analyze the electron spectral width η , the effective electron temperature T_e and the genuinely non-thermal pairing parameter α_{nth} as a function of the control parameter, here being the drive strength. For all three quantities, analytical expressions can be derived in the weak-coupling regime. Substituting these results into eq. (19.1), we obtain a closed analytical expression for g_c . Details of the weak-coupling derivations are provided in section 19.8.11.

The incoherent drive is modeled by augmenting the bosonic distribution with an additional non-thermal term (see section 19.8.7)

$$F_\omega^b = F_{T_b, \omega}^{b, \text{th}} + F_\omega^{b, d}, \quad (19.47)$$

where $F_{T_b, \omega}^{b, \text{th}}$ is the thermal distribution defined in eq. (19.28). We parametrize the incoherent drive by a Gaussian

$$F_\omega^{b, d} = A \left[\exp \left(-\frac{(\omega - \omega_d)^2}{v^2} \right) - \exp \left(-\frac{(\omega + \omega_d)^2}{v^2} \right) \right], \quad (19.48)$$

where A and v control the strength and width of the incoherent drive, respectively, and the peak is centered at frequency ω_d . To maximize the electronic response, we choose $\omega_d = \omega_0$, i.e., resonant with the bosonic mode.

To remain within the weak-coupling regime, we consider a low-temperature $T_b = 10^{-3}\omega_0$. Together with a large bosonic spectral width $\kappa = 0.25\omega_0$, this justifies neglecting the back-action of the electrons on the bosons. The width of the drive is fixed to $v = 0.05\omega_0$, which is much smaller than the driving frequency, ensuring that the thermal and driven contributions to F_ω^b are well separated in frequency. These parameters coincide with those used in Fig. 19.2(b).

The resulting bosonic occupation function for finite drive strength is shown in Fig. 19.5(a), together with the undriven thermal occupation function. For $A = 0$, the bosons are purely thermal, and the entire system thermalizes to the temperature T_b . This thermalization is a direct consequence of the decoupling from the cryostat, which leaves the boson as the sole source for determining the electron distribution in the steady state.

In the steady state, there exists a contribution to F^e that is independent of the electron-boson coupling

$$F_{\omega}^{e,(0)} = \frac{1}{F_{\omega}^b} = \frac{1}{F_{T_b,\omega}^{b,\text{th}} + F_{\omega}^{b,d}}. \quad (19.49)$$

Although the time required to reach this contribution diverges for $g_c \rightarrow 0$, it is always there in the steady state. For $A = 0$, this expression reduces to $F_{\omega}^e = F_{\omega}^{e,(0)} = 1/\coth(\omega/2T_b) = \tanh(\omega/2T_b)$, as expected in thermal equilibrium.

For finite drive strength ($A \neq 0$), the electronic distribution acquires a non-thermal component, related to $1/F_{\omega}^{b,d}$, that is non-perturbative in the coupling. The numerical result for F_{ω}^e is shown in Fig. 19.5(b). In the weak-coupling regime, the behavior of the electron distribution near ω_d closely mirrors the inverse behavior of the driven bosonic distribution shown in Fig. 19.5(a). By contrast, as highlighted in the inset of Fig. 19.5(b), the distribution near the FS deviates from the zeroth-order result in eq. (19.49). This deviation reflects the combined effect of interactions and driving, which leads to an increase of the effective electron temperature T_e . To capture this behavior, we refine our weak-coupling approximation for the electron distribution (19.49) by replacing the boson temperature T_b with the effective electron temperature T_e

$$F_{\omega}^e \approx \frac{1}{F_{T_e,\omega}^{b,\text{th}} + F_{\omega}^{b,d}}. \quad (19.50)$$

As demonstrated in Fig. 19.5(b), this ansatz provides a good description of the electron distribution both near the drive frequency ω_d and close to the FS.

To obtain a reliable self-consistent approximation for the effective electronic temperature T_e , it is first necessary to understand how incoherent driving modifies the electronic spectrum and, in particular, the spectral width η at the FS. The spectral width extracted from the numerical determination of $\text{Im } \Sigma_0^R$ from Eqs. (19.26) is shown in Fig. 19.6(a). Strong incoherent driving of the bosons is seen to significantly increase the electronic spectral width compared to the undriven case. This demonstrates that incoherent driving of the bosons substantially modifies the thermal-like properties of the electrons and, as evident from eq. (19.1), an increased spectral width reduces the effectiveness of the genuinely non-thermal contribution to the critical coupling.

To understand the mechanism behind the increase of η , we now consider its weak-coupling approximation. In contrast to the electronic distribution, modifications to the spectrum start appearing at linear order in g_c . Specifically, the first-order contribution to the imaginary part of the retarded self-energy takes the form

$$\text{Im } \Sigma_{\omega}^{R,(1)} = \frac{g_c \text{Im } D_{\omega}^K}{2}, \quad (19.51)$$

where D_{ω}^K is given in eq. (19.27) and contains the driven bosonic distribution F_{ω}^b defined in eq. (19.47). Since we consider $T_b \ll \omega_d$, the incoherent drive is energetically well-separated from the low-energy thermal sector. This separation allows us to decompose $\text{Im } \Sigma_{\omega}^{R,(1)}$ into two physically distinct contributions. In the vicinity of the FS, $\text{Im } \Sigma_{\omega}^{R,(1)}$ is governed entirely by the thermal background at temperature T_b , adding $g_c \gamma T_b$ to η . This contribution is identical to the thermal broadening obtained in the temperature-bias example [cf. eq. (19.42)]. At energies close to the drive frequency ω_d , however, the incoherent drive creates spectral weight that was not present in the temperature-bias example. Owing to the convolution structure of the self-energies in Eqs. (19.26), this spectral weight makes the electrons at the FS in the NESS sensitive to the behavior of the electron distributions near ω_d . As a result an additional contribution arises in η , such that its weak-coupling approximation becomes

$$\eta \approx g_c (\gamma T_b + g_c I_{\eta}), \quad (19.52)$$

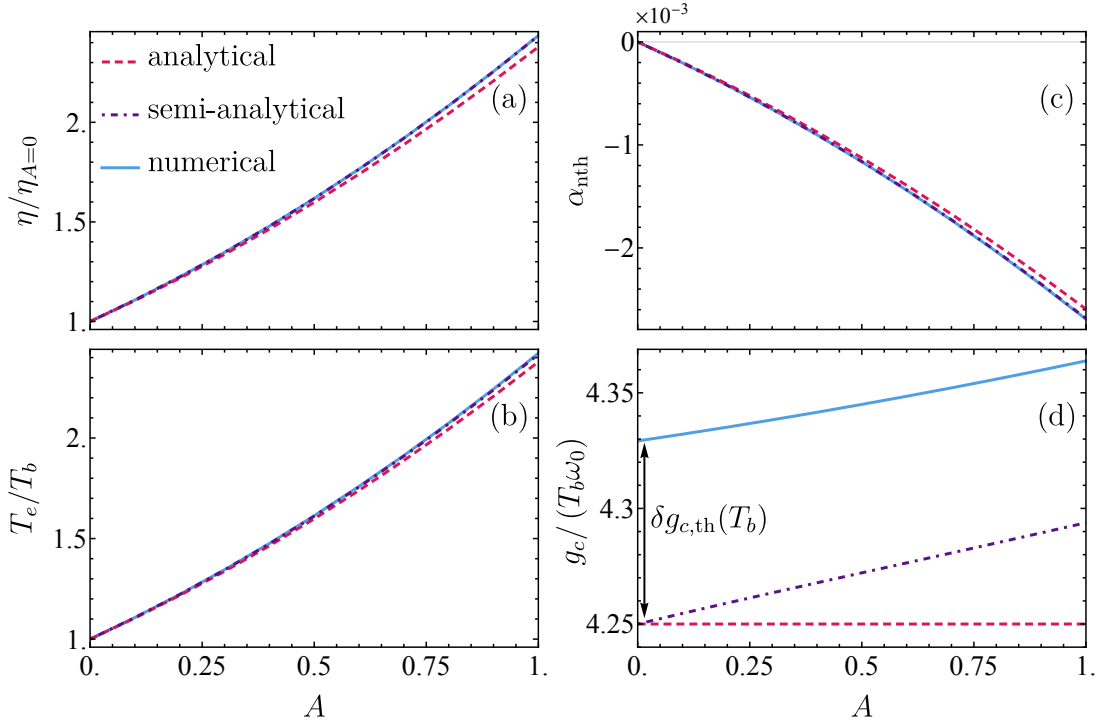


Figure 19.6: Physical quantities appearing in eq. (19.1) for the setting with incoherent driving of the bosons as a function of the driving strength: (a) electron spectral width, (b) effective electron temperature, (c) non-thermal pairing parameter, (d) critical coupling [zoom of the plot in Fig. 19.2(b)]. Curves denoted with “numerical” (blue solid) are obtained via the self-consistent numerical solution of eq. (19.26), curves denoted with “analytical” (red dashed) are the corresponding weak-coupling approximations obtained via eqs. (19.52) to (19.56) and curves denoted with “semi-analytical” (violet dot-dashed) are computed with the weak-coupling approximations eqs. (19.1) and (19.52) to (19.55), but with the numerically determined g_c . The difference between the numerical and analytical thermal equilibrium critical coupling is denoted as $\delta g_{c,\text{th}}(T_b)$ in panel (d). The input parameters are identical to those used in Fig. 19.2(b).

where the effect of the drive is encoded in

$$I_\eta = 2 \int_\epsilon \left[2 \operatorname{sign}(\epsilon) F_\epsilon^{b,d} + (F_\epsilon^{b,d})^2 \right] \frac{(\operatorname{Im} D_\epsilon^R)^2}{\epsilon^2}, \quad (19.53)$$

which is strictly positive. Anticipating the closed analytical expression for g_c in eq. (19.56), the approximation for η in eq. (19.52) is compared to the numerical result in Fig. 19.6(a). We find a good qualitative agreement over the full range of drive strengths, with the analytical expression slightly underestimating η at larger A . This underestimate can be mitigated by inserting the numerically determined value of g_c into eq. (19.52), as shown by the semi-analytical curve in Fig. 19.6(a). This observation confirms that the functional form of eq. (19.52) correctly captures the dominant mechanism underlying the increase of η .

Having established a weak-coupling description of the modification of the electron spectrum via eq. (19.51), we now turn to the effective electron temperature T_e . As already indicated in Fig. 19.5(b) and shown explicitly in Fig. 19.6(b), T_e increases monotonically with the drive strength A . This heating arises due to the interacting nature of the system and leads to an increase of the thermal contribution $g_{c,\text{th}}$ defined in eq. (19.2). To derive a weak-coupling approximation for T_e , we note that, since the bosons near the FS remain thermal with a temperature T_b , the heating of the electrons must originate from the spectral weight in $\operatorname{Im} G_\omega^R$ at energies $\omega \sim \omega_d$. This spectral weight is generated by $\operatorname{Im} \Sigma_\omega^{R,(1)}$ in eq. (19.51), highlighting the essential role of the spectral modifications in determining T_e . Exploiting the energy separation between the thermal and driven sectors, one finds the estimate

$$T_e \approx \frac{\eta}{g_c \gamma} = T_b \left(1 + \frac{g_c I_\eta}{\gamma T_b} \right). \quad (19.54)$$

As for η , using the analytical approximation (19.56) for g_c in eq. (19.54) leads to a slight underestimate of T_e observed in Fig. 19.6(b), whereas substituting the numerical value of g_c yields excellent agreement. eq. (19.54) makes explicit that the heating of the electrons is governed by the same quantity I_η that controls the increase of the spectral width.

With the electronic properties near the FS under control, the final ingredient required to estimate the critical coupling in eq. (19.1) is the genuinely non-thermal pairing parameter α_{nth} . Within a weak-coupling treatment, its evaluation requires an appropriate ansatz for the frequency dependence of the anomalous self-energy near ω_d . At lowest order, this dependence is well approximated by that of the bosonic propagators (cf. fig. 19.7), leading to

$$\alpha_{\text{nth}} \approx \frac{g_c}{1 + \frac{g_c \gamma T_b}{\eta}} \left[\gamma \left(\frac{T_b}{T_e} - 1 \right) - \frac{2I_\eta}{|D_0^R|} \right]. \quad (19.55)$$

This weak-coupling approximation and the numerical result for α_{nth} are compared in Fig. 19.6(c). As for η and T_e , the analytical approximation slightly underestimates the magnitude of α_{nth} , an effect that is again remedied by inserting the numerically determined value of g_c , as done in the semi-analytical curve.

At thermal equilibrium ($A = 0$), α_{nth} vanishes identically, since $T_e = T_b$ and $I_\eta = 0$ as a consequence of $F_\omega^{b,d}|_{A=0} = 0$. The first term in the parenthesis of eq. (19.55) is identical to the temperature-bias expression for α_{nth} in eq. (19.45). Because the incoherent drive increases T_e , this term leads to a negative contribution to α_{nth} , corresponding to an enhancement of the genuinely non-thermal pairing effect. The second term in eq. (19.55), is also negative and proportional to I_η , which is positive and increases with the drive strength. Remarkably, this shows that the same mechanism responsible for the increase of T_e and η also enhances the genuinely non-thermal pairing.

Having obtained analytical expressions for η , T_e and α_{nth} as functions of g_c , we can now substitute them into the weak-coupling approximation for g_c in eq. (19.1) and solve for g_c . This yields

$$g_c \approx g_{c,\text{th}}(T_b), \quad (19.56)$$

indicating that, to leading order, the critical coupling is essentially unaffected by the incoherent drive. A detailed comparison with the numerical result in Fig. 19.6(d) shows that the analytical approximation slightly underestimates the thermal value, leading to an offset $\delta g_{c,\text{th}}(T_b) = g_{c,A=0} - g_{c,\text{th}}(T_b) \approx 0.02 g_{c,\text{th}}(T_b)$. By using the “semi-analytical” results for η , T_e and α_{nth} in eq. (19.1) the offset is unaffected, but the monotonic increase with A , observed for the numerical result in Fig. 19.6(d), is captured. Including high-order corrections in $g_{c,\text{th}}(T_e)$ would reduce the offset $\delta g_{c,\text{th}}(T_b)$ and capture the small increase of g_c with A , but we do not pursue this extension here, since the correction would be anyway subleading [$\delta g_{c,\text{th}}(T_b) \ll g_{c,\text{th}}(T_b)$]⁶. It is nevertheless notable that the dominant discrepancy arises entirely from the thermal contribution $g_{c,\text{th}}$, while the non-thermal contributions are captured with high accuracy.

In summary, when the incoherent drive is well separated from low-energy thermal modes, the critical coupling is only weakly sensitive to the specific form of the drive distribution, rendering this result applicable to a broad class of incoherent driving protocols. In this regime, the electrons near the FS heat up and additional occupation near the drive frequency is generated, yet these effects are largely compensated by the enhanced genuinely non-thermal pairing. As a consequence, the critical coupling is almost unperturbed, despite the system being driven far from thermal equilibrium and exhibiting a substantially increased electron temperature.

19.7 Conclusions

We have developed a non-thermal extension of Eliashberg theory to describe superconductivity in steady states where the bosonic pairing mediator and the electrons are not in mutual thermal equilibrium. Within this framework, we derived an analytical expression for the critical coupling that depends on three self-consistently determined quantities: the effective electron temperature, the quasiparticle spectral width, and a genuinely non-thermal pairing contribution. Applying this formalism to two representative non-thermal examples in the weak-coupling regime, we obtained self-consistent analytic approximations for these quantities and found good agreement with fully numerical solutions. These results highlight the analytically tractable nature of the theory and demonstrated that, away from thermal-equilibrium, the retarded nature of the boson-mediated interaction is essential even in the weak-coupling regime.

More specifically, we showed that bosons colder than the electrons enhance pairing due to a combination of a thermal-like cooling and genuinely non-thermal enhancement of the pairing strength. By contrast, incoherent driving of the bosons at energies much larger than the temperature leaves the critical coupling nearly unchanged. In this case the heating of the electrons is largely compensated by an enhancement of the genuinely non-thermal pairing contribution. Nevertheless, the incoherent drive pushes the system into a strongly non-thermal steady state, which is simultaneously hotter and more strongly paired than the corresponding thermal equilibrium state.

Our work demonstrates that non-thermal states of the boson mediators provide a new mechanism to modify superconductivity and motivate further exploration of regimes beyond weak coupling, steady states, finite-momentum interactions, and dynamics inside the superconducting phase.

⁶More refined weak-coupling treatments at thermal equilibrium have been developed by accounting for the frequency dependence of Δ_ω^R arising from the bosonic propagator (cf. fig. 19.7) in eq. (19.26) [42, 60, 61].

19.8 SM

19.8.1 Summary of the appendices

The appendices are organized as follows. section 19.8.2 details the derivation of NESS-Eliashberg equations using a real-time Keldysh path-integral formalism. section 19.8.3 shows how the thermal equilibrium results are recovered by the NESS-Eliashberg equations and how they relate to Matsubara formalism. section 19.8.4 clarifies the causal structure of anomalous and normal propagators at the mean-field level in Keldysh formalism and how the choice of the gauge for the anomalous self-energy can lead to simplifications. section 19.8.5 presents useful expressions for the retarded and Keldysh components of the normal and anomalous propagators, both in the superconducting phase and at phase transition. section 19.8.6 discusses why the feedback of the electrons on the bosons can be neglected in the weak-coupling regime. sections 19.8.7 and 19.8.8 respectively presents how to describe the effect of the boson environment and the cryostat. section 19.8.9 illustrates the numerical method employed to solve the NESS-Eliashberg equations at the phase transition. In sections 19.8.10 and 19.8.11 weak-coupling analytical expressions for relevant physical quantities are derived for the temperature-bias and the incoherently driven settings, respectively. Finally, in section 19.8.12 the implementation of the NESS-Eliashberg equations for smooth momentum-dependent electron-electron interactions and the subsequent momentum averaging within the quasi-classical approximation are discussed.

19.8.2 Path-integral derivation of NESS-Eliashberg equations

The first step in deriving the NESS-Eliashberg equations, is to construct a steady-state real-time partition function based on the Hamiltonian (19.9). This can be done using the Schwinger-Keldysh path-integral formulation [71, 37]. After having integrated out the P -quadrature of the bosonic field and having included the boson environment self-energy (see section 19.8.7), the partition function for the Bose-Fermi system becomes

$$\mathcal{Z} = \prod_{\sigma} \int \mathcal{D}[\bar{\psi}_{\sigma}, \psi_{\sigma}] \int \mathcal{D}[X] \times \exp \left\{ i \left(S_{e,0}[\bar{\psi}, \psi] + S_{b,0}[X] + S_{e-b}[\bar{\psi}, \psi, X] \right) \right\}, \quad (19.57)$$

where

$$S_{e,0}[\bar{\psi}, \psi] = \int_{x,x'} \sum_{\sigma} \begin{pmatrix} \bar{\psi}_{\sigma}^c \\ \bar{\psi}_{\sigma}^q \end{pmatrix}_x^T \begin{pmatrix} 0 & [G_0^{-1}]^A \\ [G_0^{-1}]^R & [G_0^{-1}]^K \end{pmatrix}_{x-x'} \begin{pmatrix} \psi_{\sigma}^c \\ \psi_{\sigma}^q \end{pmatrix}_{x'} \quad (19.58)$$

is the free action for the electrons written in the space-time coordinates $x = (\mathbf{r}, t)$ with the shorthand notation $\int_x = \sum_{\mathbf{r}} \int dt$, while

$$S_{b,0}[X] = \frac{1}{2} \int_{t,t'} \begin{pmatrix} X^c \\ X^q \end{pmatrix}_t^T \begin{pmatrix} 0 & [D^{-1}]^A \\ [D^{-1}]^R & [D^{-1}]^K \end{pmatrix}_{t-t'} \begin{pmatrix} X^c \\ X^q \end{pmatrix}_{t'} \quad (19.59)$$

is the free action for the real bosons, and

$$S_{e-b}[\bar{\psi}, \psi, X] = -2 \left[4(1 + \kappa^2/\omega_0^2) \right]^{1/4} g_0 \times \int_x \sum_{\sigma} \begin{pmatrix} X^c & X^q \end{pmatrix}_t \begin{pmatrix} \bar{\psi}_{\sigma}^c \psi_{\sigma}^q + \bar{\psi}_{\sigma}^q \psi_{\sigma}^c \\ \bar{\psi}_{\sigma}^c \psi_{\sigma}^c + \bar{\psi}_{\sigma}^q \psi_{\sigma}^q \end{pmatrix}_x \quad (19.60)$$

is the action for the electron-boson interaction vertex. Here $\bar{\psi}_{\sigma,x}$ and $\psi_{\sigma,x}$ are the Grassmann fermionic fields associated with the electronic operators $\hat{c}_{\sigma\mathbf{r}}^\dagger$ and $\hat{c}_{\sigma\mathbf{r}}$, and $X(t) = [a^*(t) + a(t)] [4(1 + \kappa^2/\omega_0^2)]^{-1/4}$ ⁷ is the real quadrature of the bosonic field, where the complex fields $a^*(t)$, $a(t)$ are associated to the bosonic operators $\hat{a}^\dagger$, $\hat{a}$. The classical (c) and quantum (q) components of the fields are related to the fields on the forward (+) and backward (−) time contours through the Keldysh rotation [71]

$$\begin{aligned}\psi_{\sigma,x}^{(c/q)} &= (\psi_{\sigma,x}^+ \pm \psi_{\sigma,x}^-) / \sqrt{2}, \\ \bar{\psi}_{\sigma,x}^{(c/q)} &= (\bar{\psi}_{\sigma,x}^+ \pm \bar{\psi}_{\sigma,x}^-) / \sqrt{2}\end{aligned}\tag{19.61}$$

for the fermionic fields and

$$X_t^{(c/q)} = (X_t^+ \pm X_t^-) / 2\tag{19.62}$$

for the real bosonic field. Note that we do not employ the Larkin-Ovchinnikov rotation for fermions, but instead use the same transformations as for the bosons. Consequently, the fermionic and bosonic propagators will have the same Keldysh structure. Inverting the matrix structures in eqs. (19.58) and (19.59), one obtains a free electron propagator with the retarded, advanced and Keldysh (RAK) matrix structure

$$[G_0]_{x-x'} = \begin{pmatrix} [G_0]_{x-x'}^K & [G_0]_{x-x'}^R \\ [G_0]_{x-x'}^A & 0 \end{pmatrix},\tag{19.63}$$

and a real boson propagator with the RAK matrix structure

$$D_{t-t'} = \begin{pmatrix} D_{t-t'}^K & D_{t-t'}^R \\ D_{t-t'}^A & 0 \end{pmatrix}.\tag{19.64}$$

The Fourier transformed boson propagators are defined in eq. (19.27), while the Fourier transformed free electron propagators are given by

$$\begin{aligned}[G_0]_{\mathbf{k},\omega}^{R(A)} &= \frac{1}{\omega - \xi_{\mathbf{k}} \pm i0^+}, \\ [G_0]_{\mathbf{k},\omega}^K &= 2iF_{0,\omega} \text{Im} [G_0]_{\mathbf{k},\omega}^R,\end{aligned}\tag{19.65}$$

with $F_{0,\omega}$ being the initial distribution function of the electrons⁸.

Derivation of NESS-Eliashberg equations

To derive the NESS-Eliashberg equations, we generalize the approach introduced for the thermal equilibrium case in Ref. [51] to a NESS. The steps of the procedure are the following: (i) we integrate out the bosons exactly, obtaining a purely fermionic action with an effective boson-mediated interaction vertex, (ii) we perform a double Hubbard-Stratonovich (HS) transformation in the Cooper and exchange channels, introducing the four-component auxiliary fields Δ and $\mathcal{S}_\sigma$, which are respectively related to the anomalous and normal self-energies, (iii) we reorganize the electron propagator in a Nambu structure and integrate out the electron fields, obtaining an effective action for the auxiliary fields, and finally (iv) we perform a saddle-point (or mean-field) approximation, obtaining the NESS-Eliashberg equations for the auxiliary fields as the stationary path of their effective action.

⁷The scaling constant is chosen to have a particularly simple form of the propagator, as shown in eq. (19.235).

⁸Note that the initial distribution of the electrons will actually never enter the steady-state results, since the memory of it will be lost due to the interaction with the cryostat and/or the bosons.

Integrating out the bosons and deriving the effective interaction for the electrons

By performing the Gaussian functional integral of the X -field in eq. (19.57), one finds an effective action S_e for the electron fields $(\bar{\psi}, \psi)$ (see for example the supplemental material of Ref. [37]). The action $S_e = S_{e,0} + S_{\text{int}}$ includes an effective boson-mediated interaction term S_{int} which can be expressed as

$$\begin{aligned}
S_{\text{int}}[\bar{\psi}, \psi] = & -\frac{g}{2} \sum_{\sigma, \sigma'} \int_{x, x'} \\
& \left[D_{t-t'}^R (\bar{\psi}_{\sigma, x}^c \bar{\psi}_{\sigma', x'}^c \psi_{\sigma', x'}^c \psi_{\sigma, x}^c + \bar{\psi}_{\sigma, x}^c \bar{\psi}_{\sigma', x'}^q \psi_{\sigma', x'}^q \psi_{\sigma, x}^q \right. \\
& \quad + \bar{\psi}_{\sigma, x}^q \bar{\psi}_{\sigma', x'}^c \psi_{\sigma', x'}^c \psi_{\sigma, x}^c + \bar{\psi}_{\sigma, x}^q \bar{\psi}_{\sigma', x'}^q \psi_{\sigma', x'}^q \psi_{\sigma, x}^c) \\
& + D_{t-t'}^A (\bar{\psi}_{\sigma, x}^c \bar{\psi}_{\sigma', x'}^c \psi_{\sigma', x'}^q \psi_{\sigma, x}^c + \bar{\psi}_{\sigma, x}^c \bar{\psi}_{\sigma', x'}^q \psi_{\sigma', x'}^c \psi_{\sigma, x}^c \\
& \quad + \bar{\psi}_{\sigma, x}^q \bar{\psi}_{\sigma', x'}^c \psi_{\sigma', x'}^q \psi_{\sigma, x}^q + \bar{\psi}_{\sigma, x}^q \bar{\psi}_{\sigma', x'}^q \psi_{\sigma', x'}^c \psi_{\sigma, x}^q) \\
& + D_{t-t'}^K (\bar{\psi}_{\sigma, x}^c \bar{\psi}_{\sigma', x'}^c \psi_{\sigma', x'}^q \psi_{\sigma, x}^q + \bar{\psi}_{\sigma, x}^c \bar{\psi}_{\sigma', x'}^q \psi_{\sigma', x'}^c \psi_{\sigma, x}^q \\
& \quad \left. + \bar{\psi}_{\sigma, x}^q \bar{\psi}_{\sigma', x'}^c \psi_{\sigma', x'}^q \psi_{\sigma, x}^c + \bar{\psi}_{\sigma, x}^q \bar{\psi}_{\sigma', x'}^q \psi_{\sigma', x'}^c \psi_{\sigma, x}^c) \right],
\end{aligned} \tag{19.66}$$

where we defined the coupling $g = 8g_0^2 \sqrt{1 + \kappa^2/\omega_0^2}$. We can now separate eq. (19.66) in two terms, which either describe interactions between electrons with the same or opposite spin:

$$S_{\text{int}}[\bar{\psi}, \psi] = \sum_{\sigma} S_{\text{int}}^{\sigma\sigma}[\bar{\psi}, \psi] + S_{\text{int}}^{\uparrow\downarrow}[\bar{\psi}, \psi]. \tag{19.67}$$

The opposite-spin interaction term of eq. (19.67) can be reorganized in the matrix form

$$S_{\text{int}}^{\uparrow\downarrow}[\bar{\psi}, \psi] = -g \int_{x, x'} (\bar{\rho}_{x, x'}^{\uparrow\downarrow})^T \mathcal{D}_{t-t'} \rho_{x, x'}^{\uparrow\downarrow}, \tag{19.68}$$

where we defined the (symmetric) 4×4 matrix

$$\mathcal{D}_{t-t'} = \begin{pmatrix} 0 & D_{t-t'}^A & D_{t-t'}^R & D_{t-t'}^K \\ D_{t-t'}^A & 0 & D_{t-t'}^K & D_{t-t'}^R \\ D_{t-t'}^R & D_{t-t'}^K & 0 & D_{t-t'}^A \\ D_{t-t'}^K & D_{t-t'}^R & D_{t-t'}^A & 0 \end{pmatrix} \tag{19.69}$$

and the four-dimensional electron bilinears

$$\rho_{x, x'}^{\uparrow\downarrow} = \begin{pmatrix} \psi_{\downarrow, x'}^c \psi_{\uparrow, x}^c \\ \psi_{\downarrow, x'}^q \psi_{\uparrow, x}^c \\ \psi_{\downarrow, x'}^c \psi_{\uparrow, x}^q \\ \psi_{\downarrow, x'}^q \psi_{\uparrow, x}^q \end{pmatrix}, \quad \bar{\rho}_{x, x'}^{\uparrow\downarrow} = \begin{pmatrix} \bar{\psi}_{\uparrow, x}^c \bar{\psi}_{\downarrow, x'}^c \\ \bar{\psi}_{\uparrow, x}^c \bar{\psi}_{\downarrow, x'}^q \\ \bar{\psi}_{\uparrow, x}^q \bar{\psi}_{\downarrow, x'}^c \\ \bar{\psi}_{\uparrow, x}^q \bar{\psi}_{\downarrow, x'}^q \end{pmatrix}. \tag{19.70}$$

Analogously, the same-spin interaction term in eq. (19.67) can be written as

$$S_{\text{int}}^{\sigma\sigma}[\bar{\psi}, \psi] = \frac{g}{2} \int_{x, x'} (\bar{\rho}_{x, x'}^{\sigma})^T \mathcal{D}_{t-t'} \rho_{x, x'}^{\sigma}, \tag{19.71}$$

where the electron bilinears are defined as

$$\rho_{x, x'}^{\sigma} = \begin{pmatrix} \bar{\psi}_{\sigma, x'}^c \psi_{\sigma, x}^c \\ \bar{\psi}_{\sigma, x'}^q \psi_{\sigma, x}^c \\ \bar{\psi}_{\sigma, x'}^c \psi_{\sigma, x}^q \\ \bar{\psi}_{\sigma, x'}^q \psi_{\sigma, x}^q \end{pmatrix}, \quad \bar{\rho}_{x, x'}^{\sigma} = \begin{pmatrix} \bar{\psi}_{\sigma, x}^c \psi_{\sigma, x'}^c \\ \bar{\psi}_{\sigma, x}^c \psi_{\sigma, x'}^q \\ \bar{\psi}_{\sigma, x}^q \psi_{\sigma, x'}^c \\ \bar{\psi}_{\sigma, x}^q \psi_{\sigma, x'}^q \end{pmatrix}. \tag{19.72}$$

Before introducing auxiliary fields through the HS transformation, the symmetric matrix (19.69) must be diagonalized

$$g\mathcal{D}_{t-t'} = U^T \Lambda_{t-t'} U, \quad (19.73)$$

where $\Lambda_{t-t'}$ is a diagonal matrix with the entries

$$\begin{aligned} (\lambda_{t-t'})_1 &= g [D_{t-t'}^A - D_{t-t'}^R - D_{t-t'}^K], \\ (\lambda_{t-t'})_2 &= g [-D_{t-t'}^A + D_{t-t'}^R - D_{t-t'}^K], \\ (\lambda_{t-t'})_3 &= g [-D_{t-t'}^A - D_{t-t'}^R + D_{t-t'}^K], \\ (\lambda_{t-t'})_4 &= g [D_{t-t'}^A + D_{t-t'}^R + D_{t-t'}^K], \end{aligned} \quad (19.74)$$

and U is a time-independent orthogonal ($U^{-1} = U^T$) matrix given by

$$U = \frac{1}{2} \begin{pmatrix} -1 & -1 & 1 & 1 \\ -1 & 1 & -1 & 1 \\ 1 & -1 & -1 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}. \quad (19.75)$$

Note that the specific form of $\Lambda_{t-t'}$ and U are not necessary for the following derivation, and are reported here only for completeness. With the transformation (19.73) one can then rewrite the two terms of eq. (19.67) as

$$\begin{aligned} S_{\text{int}}^{\uparrow\downarrow} [\bar{\psi}, \psi] &= - \int_{x,x'} (U \bar{\rho}_{x,x'}^{\uparrow\downarrow})^T \Lambda_{t-t'} (U \rho_{x,x'}^{\uparrow\downarrow}) \\ &= - \int_{x,x'} \sum_n (\tilde{\rho}_{x,x'}^{\uparrow\downarrow})_n (\lambda_{t-t'})_n (\tilde{\rho}_{x,x'}^{\uparrow\downarrow})_n, \end{aligned} \quad (19.76)$$

and

$$\begin{aligned} S_{\text{int}}^{\sigma\sigma} [\bar{\psi}, \psi] &= \frac{1}{2} \int_{x,x'} (U \bar{\rho}_{xx'}^\sigma)^T \Lambda_{t-t'} (U \rho_{xx'}^\sigma) \\ &= \frac{1}{2} \int_{x,x'} \sum_n (\tilde{\rho}_{x,x'}^\sigma)_n (\lambda_{t-t'})_n (\tilde{\rho}_{x,x'}^\sigma)_n, \end{aligned} \quad (19.77)$$

where the transformed electron bilinears have been denoted with the tilde symbol ($\tilde{\rho} = U\rho$).

Hubbard-Stratonovich transformation in the Cooper and exchange channels

To make the quartic interaction terms tractable one performs a HS transformation [71, 72] in the Cooper channel for the opposite-spin interaction of eq. (19.68) and in the exchange channel for the same-spin interaction of eq. (19.71). This is done by introducing two four-component auxiliary complex fields Δ and $\mathcal{S}^\sigma$ through the Gaussian identities

$$\begin{aligned} 1 &= \prod_n \int \mathcal{D} [\bar{\Delta}_n, \Delta_n] \\ &\quad \times \exp \left\{ \int_{x,x'} \left[i \frac{(\bar{\Delta}_{x,x'})_n (\Delta_{x,x'})_n}{(\lambda_{t-t'})_n} \right] \right\}, \end{aligned} \quad (19.78)$$

and

$$\begin{aligned} 1 &= \prod_n \int \mathcal{D} [\bar{\mathcal{S}}_n^\sigma, \mathcal{S}_n^\sigma] \\ &\quad \times \exp \left\{ \int_{x,x'} \left[-2i \frac{(\bar{\mathcal{S}}_{x,x'}^\sigma)_n (\mathcal{S}_{x,x'}^\sigma)_n}{(\lambda_{t-t'})_n} \right] \right\}. \end{aligned} \quad (19.79)$$

For these relations to be true, the Gaussian functional integrals over the auxiliary fields need to be convergent. Strictly speaking, this is the case only if $\text{Im}(\lambda_{t-t'})_n < 0$ in the first integral and $\text{Im}(\lambda_{t-t'})_n > 0$ in the second integral. However, from eq. (19.74) one sees that in general $\text{Im}(\lambda_{t-t'})_n$ is given by $\pm \text{Im} D_{t-t'}^K$, which is sign-changing as a function of $(t-t')$ ⁹. To mitigate this sign problem we follow a similar approach as the one used in Ref. [53] for the derivation of Eliashberg equations in thermal equilibrium for repulsive interactions.

We start by introducing two “modified” four-component complex fields Γ and ζ^σ satisfying the following Gaussian identities

$$1 = \prod_n \int \mathcal{D}[\Gamma_n^*, \Gamma_n] \times \exp \left\{ \int_{x,x'} \left[i \frac{[(v_{t-t'})_n]^2}{(\lambda_{t-t'})_n} (\tilde{\Gamma}_{x,x'})_n^* (\Gamma_{x,x'})_n \right] \right\}, \quad (19.80)$$

and

$$1 = \prod_n \int \mathcal{D}[\zeta_n^{\sigma*}, \zeta_n^\sigma] \times \exp \left\{ \int_{x,x'} \left[-i \frac{2[(\tilde{v}_{t-t'})_n]^2}{(\lambda_{t-t'})_n} (\zeta_{x,x'}^\sigma)_n^* (\zeta_{x,x'}^\sigma)_n \right] \right\}, \quad (19.81)$$

where the $*$ denotes complex conjugation and we have introduced the factors

$$\begin{aligned} (v_{t-t'})_n &= \frac{1}{2} \{1 - \text{sign}[\text{Im}(\lambda_{t-t'})_n]\} \\ &\quad + \frac{i}{2} \{1 + \text{sign}[\text{Im}(\lambda_{t-t'})_n]\}, \\ (\tilde{v}_{t-t'})_n &= \frac{1}{2} \{1 + \text{sign}[\text{Im}(\lambda_{t-t'})_n]\} \\ &\quad + \frac{i}{2} \{1 - \text{sign}[\text{Im}(\lambda_{t-t'})_n]\}. \end{aligned} \quad (19.82)$$

From the definitions in eq. (19.82) it follows that $[(v_{t-t'})_n]^2 = -\text{sign}[\text{Im}(\lambda_{t-t'})_n]$ and $[(\tilde{v}_{t-t'})_n]^2 = \text{sign}[\text{Im}(\lambda_{t-t'})_n]$, ensuring the convergence of the Gaussian integrals of eqs. (19.80) and (19.81).

We now multiply the interaction term $\exp(iS_{\text{int}}^{\uparrow\downarrow}[\bar{\psi}, \psi])$ in the partition function by the identity (19.80) and perform a change of variable

$$\begin{aligned} (\tilde{\Delta}_{x,x'})_n &= (\Gamma_{x,x'})_n - \frac{(\lambda_{t-t'})_n}{(v_{t-t'})_n} (\tilde{\rho}_{x,x'}^{\uparrow\downarrow})_n, \\ (\bar{\tilde{\Delta}}_{x,x'})_n &= (\Gamma_{x,x'})_n^* - \frac{(\lambda_{t-t'})_n}{(v_{t-t'})_n} (\bar{\rho}_{x,x'}^{\uparrow\downarrow})_n, \end{aligned} \quad (19.83)$$

to obtain

$$\begin{aligned} e^{iS_{\text{int}}^{\uparrow\downarrow}[\bar{\psi}, \psi]} &= \prod_n \int \mathcal{D}[\tilde{\Delta}_n, \bar{\tilde{\Delta}}_n] \\ &\quad \times e^{i(S_{\text{HS},0}^{\uparrow\downarrow}[\bar{\tilde{\Delta}}, \tilde{\Delta}] + S_{\text{HS,int}}^{\uparrow\downarrow}[\bar{\psi}, \psi, \bar{\tilde{\Delta}}, \tilde{\Delta}])}, \end{aligned} \quad (19.84)$$

⁹The symmetry of the real boson forces D_t^R to be real and D_t^K to be imaginary.

where

$$S_{\text{HS},0}^{\uparrow\downarrow} [\tilde{\Delta}, \tilde{\Delta}] = \int_{x,x'} \frac{[(v_{t-t'})_n]^2}{(\lambda_{t-t'})_n} (\tilde{\Delta}_{x,x'})_n (\tilde{\Delta}_{x,x'})_n, \quad (19.85)$$

and

$$\begin{aligned} S_{\text{HS,int}}^{\uparrow\downarrow} [\bar{\psi}, \psi, \tilde{\Delta}, \tilde{\Delta}] &= \int_{x,x'} (v_{t-t'})_n \\ &\times \left[(\tilde{\Delta}_{x,x'})_n (\tilde{\rho}_{x,x'}^{\uparrow\downarrow})_n + (\tilde{\rho}_{x,x'}^{\uparrow\downarrow})_n (\tilde{\Delta}_{x,x'})_n \right]. \end{aligned} \quad (19.86)$$

We note that, as discussed in [53] $\tilde{\Delta}$ and $\tilde{\Delta}$ are not necessarily connected by complex conjugation.

Equivalently, one can multiply the interaction term $\exp(iS_{\text{int}}^{\sigma\sigma}[\bar{\psi}, \psi])$ by the identity (19.81) and perform the change of variables

$$\begin{aligned} \tilde{\mathcal{S}}_n^\sigma(x, x') &= (\zeta_{x,x'}^\sigma)_n - \frac{(\lambda_{t-t'})_n}{2(\tilde{v}_{t-t'})_n} (\tilde{\rho}_{x,x'}^\sigma)_n, \\ (\tilde{\mathcal{S}}_{x,x'}^\sigma)_n &= (\zeta_{x,x'}^\sigma)_n^* - \frac{(\lambda_{t-t'})_n}{2(\tilde{v}_{t-t'})_n} (\tilde{\rho}_{x,x'}^\sigma)_n, \end{aligned} \quad (19.87)$$

to obtain

$$\begin{aligned} e^{iS_{\text{int}}^{\sigma\sigma}[\bar{\psi}, \psi]} &= \prod_n \int \mathcal{D} [\tilde{\mathcal{S}}_n^\sigma, \tilde{\mathcal{S}}_n^\sigma] \\ &\times e^{i(S_{\text{HS},0}^{\sigma\sigma}[\tilde{\mathcal{S}}, \tilde{\mathcal{S}}] + S_{\text{HS,int}}^{\sigma\sigma}[\bar{\psi}, \psi, \tilde{\mathcal{S}}, \tilde{\mathcal{S}}])}, \end{aligned} \quad (19.88)$$

where

$$\begin{aligned} S_{\text{HS},0}^{\sigma\sigma} [\tilde{\mathcal{S}}, \tilde{\mathcal{S}}] &= - \int_{x,x'} \frac{2[(\tilde{v}_{t-t'})_n]^2}{(\lambda_{t-t'})_n} (\tilde{\mathcal{S}}_{x,x'}^\sigma)_n (\tilde{\mathcal{S}}_{x,x'}^\sigma)_n, \end{aligned} \quad (19.89)$$

and

$$\begin{aligned} S_{\text{HS,int}}^{\sigma\sigma} [\bar{\psi}, \psi, \tilde{\mathcal{S}}, \tilde{\mathcal{S}}] &= - \int_{x,x'} (\tilde{v}_{t-t'})_n \\ &\times \left[(\tilde{\mathcal{S}}_{x,x'}^\sigma)_n (\tilde{\rho}_{x,x'}^\sigma)_n + (\tilde{\rho}_{x,x'}^\sigma)_n (\tilde{\mathcal{S}}_{x,x'}^\sigma)_n \right]. \end{aligned} \quad (19.90)$$

The quartic interaction terms (19.76) and (19.77) for the electrons have then been replaced with a linear coupling of electron bilinears ($\tilde{\rho}$) with the auxiliary fields ($\tilde{\Delta}$, $\tilde{\mathcal{S}}$), plus the free quadratic action for the auxiliary fields. We can now use the transformation matrix of eq. (19.75) to express the fermion bilinears $\tilde{\rho}$ back in terms of the bilinears ρ in the original Keldysh basis [eqs. (19.70) and (19.72)] as

$$\tilde{\rho}_n^{\uparrow\downarrow} = \sum_m U_{nm} \rho_m^{\uparrow\downarrow}, \quad \tilde{\rho}_n^{\uparrow\downarrow} = \sum_m U_{nm} \tilde{\rho}_m^{\uparrow\downarrow}, \quad (19.91)$$

and

$$\tilde{\rho}_n^\sigma = \sum_m U_{nm} \rho_m^\sigma, \quad \tilde{\rho}_n^\sigma = \sum_m U_{nm} \tilde{\rho}_m^\sigma. \quad (19.92)$$

Moreover, one can now redefine the auxiliary fields as

$$\Delta_n = \sum_m U_{nm} v_m \tilde{\Delta}_m, \quad \bar{\Delta}_n = \sum_m U_{nm} v_m \tilde{\bar{\Delta}}_m, \quad (19.93)$$

and

$$\mathcal{S}_n^\sigma = \sum_m U_{nm} \tilde{v}_m \tilde{\mathcal{S}}_m^\sigma, \quad \bar{\mathcal{S}}_n^\sigma = \sum_m U_{nm} \tilde{v}_m \tilde{\bar{\mathcal{S}}}_m^\sigma, \quad (19.94)$$

where the space-time indices (x, x') have been dropped for simplicity. The introduction of the factors v_m and $\tilde{v}_m$ in this redefinition allows us to go back to auxiliary fields similar to the ones introduced in the “naive” formulation of the HS transformations in eqs. (19.78) and (19.79), with the advantage that now the Gaussian integrals are convergent, because the measure of the integrals $\int \mathcal{D}[\bar{\Delta}, \Delta]$ and $\int \mathcal{D}[\bar{\mathcal{S}}, \mathcal{S}]$ keeps memory of the rotations of eqs. (19.93) and (19.94). With these transformations plus eq. (19.73), one can then rewrite eq. (19.84) as

$$\begin{aligned} & e^{iS_{\text{int}}^{\uparrow\downarrow}[\bar{\psi}, \psi]} \\ &= \int \mathcal{D}[\bar{\Delta}, \Delta] e^{i(S_{\text{HS},0}^{\uparrow\downarrow}[\bar{\Delta}, \Delta] + S_{\text{HS,int}}^{\uparrow\downarrow}[\bar{\psi}, \psi, \bar{\Delta}, \Delta])} \end{aligned} \quad (19.95)$$

where we used the shorthand notation $\int \mathcal{D}[\bar{\Delta}, \Delta] = \prod_n \int \mathcal{D}[\bar{\Delta}_n, \Delta_n]$ and we defined

$$S_{\text{HS},0}^{\uparrow\downarrow}[\bar{\Delta}, \Delta] = \frac{1}{g} \int_{x,x'} (\bar{\Delta}_{x,x'})^T \mathcal{D}_{t-t'}^{-1} \Delta_{x,x'}, \quad (19.96)$$

and

$$\begin{aligned} & S_{\text{HS,int}}^{\uparrow\downarrow}[\bar{\psi}, \psi, \bar{\Delta}, \Delta] = \\ & \int_{x,x'} [(\bar{\Delta}_{x,x'})^T \rho_{x,x'}^{\uparrow\downarrow} + (\bar{\rho}_{x,x'}^{\uparrow\downarrow})^T \Delta_{x,x'}], \end{aligned} \quad (19.97)$$

where we reorganized the fields Δ_n (and their barred counterparts) in the vector form $\Delta^T = (\Delta_1, \Delta_2, \Delta_3, \Delta_4)$. Equivalently, one can rewrite eq. (19.88) as

$$\begin{aligned} & e^{iS_{\text{int}}^{\sigma\sigma}[\bar{\psi}, \psi]} \\ &= \int \mathcal{D}[\bar{\mathcal{S}}, \mathcal{S}] e^{i(S_{\text{HS},0}^{\sigma\sigma}[\bar{\mathcal{S}}, \mathcal{S}] + S_{\text{HS,int}}^{\sigma\sigma}[\bar{\psi}, \psi, \bar{\mathcal{S}}, \mathcal{S}])} \end{aligned} \quad (19.98)$$

where we used the shorthand notation $\int \mathcal{D}[\bar{\mathcal{S}}, \mathcal{S}] = \prod_n \int \mathcal{D}[\bar{\mathcal{S}}_n^\sigma, \mathcal{S}_n^\sigma]$ and we defined

$$S_{\text{HS},0}^{\sigma\sigma}[\bar{\mathcal{S}}, \mathcal{S}] = -\frac{2}{g} \int_{x,x'} (\bar{\mathcal{S}}_{x,x'}^\sigma)^T \mathcal{D}_{t-t'}^{-1} \mathcal{S}_{x,x'}^\sigma, \quad (19.99)$$

$$\begin{aligned} & S_{\text{HS,int}}^{\sigma\sigma}[\bar{\psi}, \psi, \bar{\mathcal{S}}, \mathcal{S}] = \\ & - \int_{x,x'} [(\bar{\mathcal{S}}_{x,x'}^\sigma)^T \rho_{x,x'}^\sigma + (\bar{\rho}_{x,x'}^\sigma)^T \mathcal{S}_{x,x'}^\sigma], \end{aligned} \quad (19.100)$$

where we reorganized the fields $\mathcal{S}_n^\sigma$ (and their barred counterparts) in the vector form $(\mathcal{S}^\sigma)^T = (\mathcal{S}_1^\sigma, \mathcal{S}_2^\sigma, \mathcal{S}_3^\sigma, \mathcal{S}_4^\sigma)$.

Nambu structure and integrating out the electrons

The next step is integrating out the fermionic fields $(\bar{\psi}, \psi)$ to obtain an effective action for the auxiliary fields $(\bar{\Delta}, \Delta, \bar{\mathcal{S}}, \mathcal{S})$. To this extent, it is convenient to transform the action back into momentum-frequency space $k = (\mathbf{k}, \omega)$. This relies on the system being translationally invariant in space and time, so that the auxiliary fields and the propagators are only functions of the difference $(x - x')$. The resulting action is

$$\begin{aligned} S_{\text{HS},0}^{\uparrow\downarrow}[\bar{\Delta}, \Delta] &= \frac{1}{g} \sum_{k,k'} \bar{\Delta}_k^T \mathcal{D}_{k-k'}^{-1} \Delta_{k'}, \\ S_{\text{HS,int}}^{\uparrow\downarrow}[\bar{\psi}, \psi, \bar{\Delta}, \Delta] &= \frac{1}{g} \sum_k \left[(\bar{\rho}_{k,-k}^{\uparrow\downarrow})^T \Delta_k + (\bar{\Delta}_k)^T \rho_{k,-k}^{\uparrow\downarrow} \right], \end{aligned} \quad (19.101)$$

for the Δ -fields and

$$\begin{aligned} S_{\text{HS},0}^{\sigma\sigma}[\bar{\mathcal{S}}, \mathcal{S}] &= -\frac{2}{g} \sum_{k,k'} (\bar{\mathcal{S}}_k^\sigma)^T \mathcal{D}_{k-k'}^{-1} \mathcal{S}_{k'}, \\ S_{\text{HS,int}}^{\sigma\sigma}[\bar{\psi}, \psi, \bar{\mathcal{S}}, \mathcal{S}] &= -\frac{1}{g} \sum_k \left[(\bar{\rho}_{k,k}^\sigma)^T \mathcal{S}_k^\sigma + (\bar{\mathcal{S}}_k^\sigma)^T \rho_{k,k}^\sigma \right], \end{aligned} \quad (19.102)$$

for the $\mathcal{S}^\sigma$ -fields, where we have used the shorthand notations $\sum_k = \sum_{\mathbf{k}} \int \frac{d\omega}{2\pi}$ and $\mathcal{D}_{k-k'}^{-1} = \mathcal{D}_{\omega-\omega'}^{-1} \delta_{\mathbf{k},\mathbf{k}'}$. Finally, the free part of the electron action in eq. (19.58) can be expressed in the momentum-frequency basis as

$$S_{e,0}[\bar{\psi}, \psi] = \sum_{k,\sigma} \begin{pmatrix} \bar{\psi}_\sigma^c & \bar{\psi}_\sigma^q \end{pmatrix}_k \begin{pmatrix} 0 & [G_0^{-1}]^A \\ [G_0^{-1}]^R & [G_0^{-1}]^K \end{pmatrix}_k \begin{pmatrix} \psi_\sigma^c \\ \psi_\sigma^q \end{pmatrix}_k. \quad (19.103)$$

To integrate out the electrons, it is necessary to reorganize the ψ -dependent terms in a Nambu structure [72], which is achieved by introducing the Keldysh-Nambu spinors

$$\Psi_k = \begin{pmatrix} \psi_{\uparrow k}^c \\ \psi_{\uparrow k}^q \\ \bar{\psi}_{\downarrow -k}^c \\ \bar{\psi}_{\downarrow -k}^q \end{pmatrix}, \quad \bar{\Psi}_k = \begin{pmatrix} \bar{\psi}_{\uparrow k}^c \\ \bar{\psi}_{\uparrow k}^q \\ \psi_{\downarrow -k}^c \\ \psi_{\downarrow -k}^q \end{pmatrix}, \quad (19.104)$$

and reorganize the auxiliary field vectors in the 2×2 matrix forms

$$\Delta_k = \begin{pmatrix} (\Delta_k)_1 \\ (\Delta_k)_2 \\ (\Delta_k)_3 \\ (\Delta_k)_4 \end{pmatrix} \rightarrow \mathbf{\Delta}_k = \begin{pmatrix} (\Delta_k)_1 & (\Delta_k)_2 \\ (\Delta_k)_3 & (\Delta_k)_4 \end{pmatrix}, \quad (19.105)$$

and

$$\mathcal{S}_k^\sigma = \begin{pmatrix} (\mathcal{S}_k^\sigma)_1 \\ (\mathcal{S}_k^\sigma)_2 \\ (\mathcal{S}_k^\sigma)_3 \\ (\mathcal{S}_k^\sigma)_4 \end{pmatrix} \rightarrow \mathbb{S}_k^\sigma = \begin{pmatrix} (\mathcal{S}_k^\sigma)_1 & (\mathcal{S}_k^\sigma)_2 \\ (\mathcal{S}_k^\sigma)_3 & (\mathcal{S}_k^\sigma)_4 \end{pmatrix}, \quad (19.106)$$

and equivalently for the barred fields. The three ψ -dependent terms can then be written in the form

$$\begin{aligned} S_{e,\text{HS}}[\bar{\psi}, \psi, \bar{\Delta}, \Delta, \bar{\mathcal{S}}, \mathcal{S}] &\equiv S_{e,0}[\bar{\psi}, \psi] + S_{\text{HS,int}}^{\uparrow\downarrow}[\bar{\psi}, \psi, \bar{\Delta}, \Delta] \\ &+ \sum_{\sigma} S_{\text{HS,int}}^{\sigma\sigma}[\bar{\psi}, \psi, \bar{\mathcal{S}}, \mathcal{S}] = \sum_k \bar{\Psi}_k^T \mathbb{G}_k^{-1} \Psi_k, \end{aligned} \quad (19.107)$$

with the inverse Keldysh-Nambu propagator defined as the following block matrix

$$\mathbb{G}_k^{-1} = \begin{pmatrix} G_k^{\uparrow -1} & \Delta_k \\ \bar{\Delta}_k^T & -G_{-k}^{\downarrow -T} \end{pmatrix}, \quad (19.108)$$

where we used the shorthand notation

$$G_k^\sigma = ([G_0^{-1}]_k - \Sigma_k^\sigma)^{-1} \quad (19.109)$$

for the normal-state propagator and we have defined

$$\Sigma_k^\sigma = \mathbb{S}_k^\sigma + \bar{\mathbb{S}}_k^{\sigma T}, \quad (19.110)$$

anticipating that this symmetrized combination of the auxiliary fields $(\bar{\mathbb{S}}_k^\sigma, \mathbb{S}_k^\sigma)$ will assume the role of the normal electron self-energy. With this interpretation of Σ_k^σ , eq. (19.109) simply becomes the Dyson equation for the normal-state propagator. From the structure of eq. (19.108), one sees that $-\Delta_k$ can instead be interpreted as an anomalous self-energy.

Proceeding with the Gaussian integration of the electron fields $(\bar{\Psi}, \Psi)$, one obtains

$$\begin{aligned} e^{iS_{\text{HS}}[\bar{\Delta}, \Delta, \bar{\mathcal{S}}, \mathcal{S}]} &= \int \mathcal{D}[\bar{\psi}, \psi] e^{iS_{e, \text{HS}}[\bar{\psi}, \psi, \bar{\Delta}, \Delta, \bar{\mathcal{S}}, \mathcal{S}]} \\ &= \det(-i\mathbb{G}^{-1}) = e^{\ln[\det(-i\mathbb{G}^{-1})]} = e^{\text{Tr}[\ln(-i\mathbb{G}^{-1})]}, \end{aligned} \quad (19.111)$$

where the trace operator (Tr) sums both over the diagonal terms in the Keldysh-Nambu structure and sums/integrates over $k = (\mathbf{k}, \omega)$. The partition function then reads

$$\begin{aligned} \mathcal{Z} &= \int \mathcal{D}[\bar{\Delta}, \Delta] \prod_\sigma \int \mathcal{D}[\bar{\mathcal{S}}^\sigma, \mathcal{S}^\sigma] \\ &\times e^{i(S_{\text{HS}, 0}^{\uparrow\downarrow}[\bar{\Delta}, \Delta] + \sum_\sigma S_{\text{HS}, 0}^{\sigma\sigma}[\bar{\mathcal{S}}, \mathcal{S}] + S_{\text{HS}}[\bar{\Delta}, \Delta, \bar{\mathcal{S}}, \mathcal{S}])}. \end{aligned} \quad (19.112)$$

Saddle-point equations for the auxiliary fields

The final step of our derivation is to perform a saddle-point (or mean-field) approximation, i.e. approximating our auxiliary fields with their non-fluctuating values that extremize the action in eq. (19.112). This is done by setting to zero the functional derivatives of the action in eq. (19.112) with respect to the auxiliary fields.

Let us start with the saddle-point equation for the normal self-energy $\Sigma_k^\sigma = \mathbb{S}_k^\sigma + \bar{\mathbb{S}}_k^{\sigma T}$. As the system has balanced spin populations ($\mu = \mu_\uparrow = \mu_\downarrow$), one has $\Sigma_k^\uparrow = \Sigma_k^\downarrow$ making it sufficient to only derive the equations for $\Sigma_k^\downarrow$.

Taking the functional derivative of the free action $S_{\text{HS}, 0}^{\uparrow\downarrow}$ defined in eq. (19.102) with respect to the fields $\bar{\mathcal{S}}_n^\downarrow$ and $\mathcal{S}_n^\downarrow$, one obtains

$$\begin{aligned} \frac{\delta S_{\text{HS}, 0}^{\uparrow\downarrow}[\bar{\mathcal{S}}, \mathcal{S}]}{\delta(\bar{\mathcal{S}}_k^\downarrow)_n} &= -\frac{2}{g} \sum_{k'} \sum_m (\mathcal{D}_{k-k'}^{-1})_{nm} (\mathcal{S}_{k'}^\downarrow)_m, \\ \frac{\delta S_{\text{HS}, 0}^{\uparrow\downarrow}[\bar{\mathcal{S}}, \mathcal{S}]}{\delta(\mathcal{S}_k^\downarrow)_n} &= -\frac{2}{g} \sum_{k'} \sum_m (\bar{\mathcal{S}}_{k'}^\downarrow)_m (\mathcal{D}_{k'-k}^{-1})_{mn}. \end{aligned} \quad (19.113)$$

The functional derivatives of S_{HS} defined in eq. (19.111) are instead given by

$$\begin{aligned} \frac{\delta S_{\text{HS}}[\bar{\Delta}, \Delta, \bar{\mathcal{S}}, \mathcal{S}]}{\delta(\bar{\mathcal{S}}_k^\downarrow)_n} &= -i \frac{\delta \text{Tr}[\ln(-i\mathbb{G}_{k'}^{-1})]}{\delta(\bar{\mathcal{S}}_k^\downarrow)_n} = -i \text{Tr} \left[\mathbb{G}_{k'} \frac{\delta \mathbb{G}_{k'}^{-1}}{\delta(\bar{\mathcal{S}}_k^\downarrow)_n} \right], \end{aligned} \quad (19.114)$$

and

$$\begin{aligned} & \frac{\delta S_{\text{HS}} [\bar{\Delta}, \Delta, \bar{\mathcal{S}}, \mathcal{S}]}{\delta (\mathcal{S}_k^\downarrow)_n} \\ &= -i \frac{\delta \text{Tr} [\ln(-i\mathbb{G}_{k'}^{-1})]}{\delta (\mathcal{S}_k^\downarrow)_n} = -i \text{Tr} \left[\mathbb{G}_{k'} \frac{\delta \mathbb{G}_{k'}^{-1}}{\delta (\mathcal{S}_k^\downarrow)_n} \right]. \end{aligned} \quad (19.115)$$

To further proceed, one needs then to compute the full Keldysh-Nambu propagator $\mathbb{G}_k$ by inverting the matrix $\mathbb{G}_k^{-1}$ defined in eq. (19.108). By employing the Schur complement formula for the inversion of a block matrix [75], one obtains

$$\mathbb{G}_k = \begin{pmatrix} \mathcal{G}_k^\uparrow & \mathcal{F}_k \\ \bar{\mathcal{F}}_k & \bar{\mathcal{G}}_k^\downarrow \end{pmatrix} \quad (19.116)$$

where $\mathcal{G}$ and $\mathcal{F}$ are respectively the normal and anomalous electron propagators in the superconducting phase, and they are given by

$$\begin{aligned} \mathcal{G}_k^\uparrow &= (G_k^{\uparrow-1} + \Delta_k G_{-k}^\downarrow \bar{\Delta}_k^T)^{-1}, \\ \bar{\mathcal{G}}_k^\downarrow &= -(G_{-k}^{\downarrow-T} + \bar{\Delta}_k^T G_k^\uparrow \Delta_k)^{-1}, \\ \mathcal{F}_k &= G_k^\uparrow \Delta_k (G_{-k}^{\downarrow-T} + \bar{\Delta}_k^T G_k^\uparrow \Delta_k)^{-1} = -G_k^\uparrow \Delta_k \bar{\mathcal{G}}_k^\downarrow, \\ \bar{\mathcal{F}}_k &= (G_{-k}^{\downarrow-T} + \bar{\Delta}_k^T G_k^\uparrow \Delta_k)^{-1} \bar{\Delta}_k^T G_k^\uparrow = -\bar{\mathcal{G}}_k^\downarrow \bar{\Delta}_k^T G_k^\uparrow. \end{aligned} \quad (19.117)$$

At the mean-field level, it can be shown (see section 19.8.4) that the upper and lower matrix blocks are connected by the relations $\bar{\mathcal{F}}_k = -\sigma_z \mathcal{F}_k^\dagger \sigma_z$ (σ_z being the third Pauli matrix) and $\bar{\mathcal{G}}_k^\downarrow = -\mathcal{G}_{-k}^{\downarrow T}$.

The functional derivatives of $\mathbb{G}_k^{-1}$ are instead given by

$$\begin{aligned} \frac{\delta \mathbb{G}_{k'}^{-1}}{\delta (\bar{\mathcal{S}}_k^\downarrow)_n} &= \begin{pmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbb{I}_n \delta_{k+k'} \end{pmatrix}, \\ \frac{\delta \mathbb{G}_{k'}^{-1}}{\delta (\mathcal{S}_k^\downarrow)_n} &= \begin{pmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbb{I}_n^T \delta_{k+k'} \end{pmatrix}, \end{aligned} \quad (19.118)$$

where $\delta_{k+k'} = \delta_{\mathbf{k}, -\mathbf{k}'} \delta_{\omega+\omega'}$ and the matrix $\mathbb{I}_n$ is defined as

$$\mathbb{I}_n = \begin{pmatrix} \delta_{n,1} & \delta_{n,2} \\ \delta_{n,3} & \delta_{n,4} \end{pmatrix}, \quad (19.119)$$

i.e. a matrix with a 1 in the position n and the rest of the elements zero. Plugging eqs. (19.116) and (19.118) in eq. (19.115), one obtains

$$\begin{aligned} \frac{\delta S_{\text{HS}} [\bar{\Delta}, \Delta, \bar{\mathcal{S}}, \mathcal{S}]}{\delta (\bar{\mathcal{S}}_k^\downarrow)_n} &= i \text{Tr} \left(\mathcal{G}_{-k'}^{\downarrow T} \mathbb{I}_n \delta_{k+k'} \right) = i (\mathcal{G}_k^\downarrow)_n, \\ \frac{\delta S_{\text{HS}} [\bar{\Delta}, \Delta, \bar{\mathcal{S}}, \mathcal{S}]}{\delta (\mathcal{S}_k^\downarrow)_n} &= i \text{Tr} \left(\mathcal{G}_{-k'}^{\downarrow T} \mathbb{I}_n^T \delta_{k+k'} \right) = i (\mathcal{G}_k^{\downarrow T})_n, \end{aligned} \quad (19.120)$$

where the n -th element of the matrix $\mathcal{G}_k^\downarrow = (G_k^{\uparrow-1} + \Delta_k G_{-k}^\downarrow \bar{\Delta}_k^T)^{-1}$ is defined as in the matrix structure of eq. (19.119). The saddle-point equations for $\bar{\mathcal{S}}_n^\downarrow$ and $\mathcal{S}_n^\downarrow$ are then

$$\begin{aligned} \frac{2}{g} \sum_{k'} \sum_m (\mathcal{D}_{k-k'}^{-1})_{nm} (\mathcal{S}_{k'}^\downarrow)_m &= i (\mathcal{G}_k^\downarrow)_n, \\ \frac{2}{g} \sum_{k'} \sum_m (\bar{\mathcal{S}}_{k'}^\downarrow)_m (\mathcal{D}_{k'-k}^{-1})_{mn} &= i (\mathcal{G}_k^{\downarrow T})_n. \end{aligned} \quad (19.121)$$

Taking the transpose of the second equation and summing the two equation together, one obtains the saddle-point equations for the self-energy $\Sigma_k^\downarrow = \mathbb{S}_k^\downarrow + \bar{\mathbb{S}}_k^{\downarrow T}$:

$$\frac{1}{g} \sum_{k'} \sum_m (\mathcal{D}_{k-k'}^{-1})_{nm} (\Sigma_{k'}^\downarrow)_m = i(\mathcal{G}_k^\downarrow)_n. \quad (19.122)$$

Finally, using the inversion relation for the matrix $\mathcal{D}_{k-k'}$

$$\sum_{k'} \sum_{n'} (\mathcal{D}_{k-k'})_{nn'} (\mathcal{D}_{k'-k''}^{-1})_{n'm} = \delta_{nm} \delta_{k-k''}, \quad (19.123)$$

and dropping the spin-indices, the saddle-point equations for the normal self-energy become

$$\Sigma_n(k) = ig \sum_{k'} \sum_m (\mathcal{D}_{k-k'})_{nm} (\mathcal{G}_{k'})_m, \quad (19.124)$$

where the structure of a Fock self-energy is now evident [cf. Fig. 19.3(a)].

Due to the convolution structure of eq. (19.124), Σ_k inherits the causal structure of $\mathcal{D}_{k-k'}$ and $\mathcal{G}_k$ [see eq. (19.183)], resulting in $\Sigma_1 = 0$. The remaining matrix elements can be identified as the advanced ($\Sigma_2 \rightarrow \Sigma^A$), retarded ($\Sigma_3 \rightarrow \Sigma^R$) and Keldysh ($\Sigma_4 \rightarrow \Sigma^K$) components of the normal self-energy. The matrix Σ_k of eq. (19.110) is then simplified to the form

$$\Sigma_k = \begin{pmatrix} 0 & \Sigma_k^A \\ \Sigma_k^R & \Sigma_k^K \end{pmatrix}. \quad (19.125)$$

Using this RAK notation, and recalling the definition for $(\mathcal{D}_{k-k'})_{nm} = (\mathcal{D}_{\omega-\omega'})_{nm} \delta_{\mathbf{k}, \mathbf{k}'}$ as the Fourier transform of eq. (19.69), eq. (19.124) becomes

$$\begin{pmatrix} \Sigma_{\mathbf{k}, \omega}^A \\ \Sigma_{\mathbf{k}, \omega}^R \\ \Sigma_{\mathbf{k}, \omega}^K \end{pmatrix} = ig \int_{\omega'} \begin{pmatrix} D^A \mathcal{G}^K + D^K \mathcal{G}^A \\ D^R \mathcal{G}^K + D^K \mathcal{G}^R \\ D^K \mathcal{G}^K + D^R \mathcal{G}^R + D^A \mathcal{G}^A \end{pmatrix}, \quad (19.126)$$

where we used the shorthand notations $\int_{\omega'} = \int \frac{d\omega'}{2\pi}$ and $D^\alpha \mathcal{G}^\beta = D_{\omega-\omega'}^\alpha \mathcal{G}_{\mathbf{k}, \omega'}^\beta$.

Next, we consider the saddle-point equation for the Δ -fields. Taking the functional derivative of the free action term $S_{\text{HS},0}^{\uparrow\downarrow}$ defined in eq. (19.101) with respect to $\bar{\Delta}_n$, one obtains

$$\frac{\delta S_{\text{HS},0}^{\uparrow\downarrow} [\bar{\Delta}, \Delta]}{\delta \bar{\Delta}_n(k)} = \frac{1}{g} \sum_{k'} \sum_m (\mathcal{D}_{k-k'}^{-1})_{nm} (\Delta_{k'})_m, \quad (19.127)$$

The functional derivative of S_{HS} defined in eq. (19.111) is instead given by

$$\begin{aligned} & \frac{\delta S_{\text{HS}} [\bar{\Delta}, \Delta, \bar{\mathcal{S}}, \mathcal{S}]}{\delta (\bar{\Delta}_k)_n} \\ &= -i \frac{\delta \text{Tr} [\ln (-i \mathbb{G}_{k'}^{-1})]}{\delta (\bar{\Delta}_k)_n} = -i \text{Tr} \left[\mathbb{G}_{k'} \frac{\delta \mathbb{G}_{k'}^{-1}}{\delta (\bar{\Delta}_k)_n} \right], \end{aligned} \quad (19.128)$$

where $\mathbb{G}_k$ is defined in eq. (19.116), while the functional derivative of $\mathbb{G}_k^{-1}$ is given by

$$\frac{\delta \mathbb{G}_{k'}^{-1}}{\delta (\bar{\Delta}_k)_n} = \begin{pmatrix} \mathbf{0} & \mathbf{0} \\ \mathbb{I}_n^T \delta_{k-k'} & \mathbf{0} \end{pmatrix}, \quad (19.129)$$

where we used the same notations of eq. (19.118). Plugging eqs. (19.116) and (19.129) in eq. (19.128) one obtains then

$$\frac{\delta S_{\text{HS}} [\bar{\Delta}, \Delta, \bar{\mathcal{S}}, \mathcal{S}]}{\delta (\bar{\Delta}_k)_n} = -i \text{Tr} (\mathcal{F}_{k'} \mathbb{I}_n^T \delta_{k-k'}) = -i (\mathcal{F}_k)_n, \quad (19.130)$$

where the n -th element of the matrix $\mathcal{F}_k = G_k^\dagger \Delta_k \mathcal{G}_{-k}^{\dagger T}$ is defined as in the matrix structure of eq. (19.119). The saddle-point equations for the anomalous self-energy are then obtained by setting the sum of eqs. (19.127) and (19.130) to zero:

$$\frac{1}{g} \sum_{k'} \sum_m (\mathcal{D}_{k-k'}^{-1})_{nm} (\Delta_{k'})_m = i (\mathcal{F}_k)_n. \quad (19.131)$$

Using again the inversion relation for the matrix $\mathcal{D}_{k-k'}$ of eq. (19.123) one obtains the equations for the anomalous self-energy

$$(\Delta_k)_n = ig \sum_{k'} \sum_m (\mathcal{D}_{k-k'})_{nm} (\mathcal{F}_{k'})_m, \quad (19.132)$$

Analogously to the normal self-energy of eq. (19.125), the mean-field structure of Δ inherits the causal structure of the boson propagator and $\mathcal{F}$ [see eq. (19.170)]. As a result the component Δ_1 of the anomalous self-energy is zero, and the remaining matrix elements can be identified as the advanced ($\Delta_2 \rightarrow \Delta^A$), retarded ($\Delta_3 \rightarrow \Delta^R$) and Keldysh ($\Delta_4 \rightarrow \Delta^K$) components of the anomalous self-energy. The matrix Δ_k of eq. (19.105) then takes the form

$$\Delta_k = \begin{pmatrix} 0 & \Delta_k^A \\ \Delta_k^R & \Delta_k^K \end{pmatrix}. \quad (19.133)$$

Using this RAK notation for Δ_k and $\mathcal{F}_k$, eq. (19.132) becomes

$$\begin{pmatrix} \Delta_{k,\omega}^A \\ \Delta_{k,\omega}^R \\ \Delta_{k,\omega}^K \end{pmatrix} = ig \int_{\omega'} \begin{pmatrix} D^A \mathcal{F}^K + D^K \mathcal{F}^A \\ D^R \mathcal{F}^K + D^K \mathcal{F}^R \\ D^K \mathcal{F}^K + D^R \mathcal{F}^R + D^A \mathcal{F}^A \end{pmatrix}, \quad (19.134)$$

or, by dropping the spin indices and writing the structure of $\mathcal{F}_k = -G_k \Delta_k \bar{\mathcal{G}}_k = G_k \Delta_k \mathcal{G}_{-k}^T$ explicitly:

$$\begin{pmatrix} \Delta_{k,\omega}^A \\ \Delta_{k,\omega}^R \\ \Delta_{k,\omega}^K \end{pmatrix} = ig \int_{\omega'} \begin{pmatrix} D^K G^A \mathcal{G}_{-}^R + D^A G^K \mathcal{G}_{-}^R & D^A G^R \mathcal{G}_{-}^K & D^A G^R \mathcal{G}_{-}^R \\ D^R G^K \mathcal{G}_{-}^R & D^K G^R \mathcal{G}_{-}^A + D^R G^R \mathcal{G}_{-}^K & D^R G^R \mathcal{G}_{-}^R \\ D^A G^A \mathcal{G}_{-}^R + D^K G^K \mathcal{G}_{-}^R & D^R G^R \mathcal{G}_{-}^A + D^K G^R \mathcal{G}_{-}^K & D^K G^R \mathcal{G}_{-}^R \end{pmatrix} \begin{pmatrix} \Delta_{k,\omega'}^A \\ \Delta_{k,\omega'}^R \\ \Delta_{k,\omega'}^K \end{pmatrix}, \quad (19.135)$$

where we used the shorthand notations $D^\alpha \mathcal{F}^\beta = D_{\omega-\omega'}^\alpha \mathcal{F}_{k,\omega'}^\beta$ and $D^\alpha G^\beta \mathcal{G}_{-}^\gamma = D_{\omega-\omega'}^\alpha G_{k,\omega'}^\beta \mathcal{G}_{-k,-\omega'}^\gamma$.

The NESS-Eliashberg eqs. (19.126) and (19.135), together with the Dyson equation for the normal-state propagator G of eq. (19.109) and the definition of the full normal propagator $\mathcal{G}$ in eq. (19.117), form a close set of equations. In the normal phase or at the phase transition, one can simply replace $\mathcal{G} \rightarrow G$ in eqs. (19.126) and (19.135). After this replacement, eq. (19.135) leads to the linearized equations for the anomalous self-energy, which admit a non-zero solution only at the phase transition. Both eqs. (19.126) and (19.135) can equivalently be rewritten in terms of distributions and retarded propagators, as shown in section 19.8.3.

At thermal equilibrium, the Keldysh components Σ^K and Δ^K satisfy thermal fluctuation-dissipation relations and eqs. (19.126) and (19.135) are largely simplified, as discussed in section 19.8.3. In the same appendix, it is also shown that these simplified thermal equilibrium equations are completely equivalent to the real-frequency analytic continuation of Eliashberg equations in Matsubara frequencies.

19.8.3 Connection to thermal equilibrium

In this appendix, we show how the NESS-Eliashberg eqs. (19.126) and (19.135) are connected to thermal equilibrium by expressing them in terms of distributions and relating them to the Eliashberg equations in the Matsubara formalism. For simplicity, we will work here with the linearized equations at the phase transition and at $|\mathbf{k}| = k_F$, dropping the parametric momentum dependence of propagators and self-energies.

NESS-Eliashberg equations in terms of distributions and thermal equilibrium condition

Our starting point is the linearized version ($\mathcal{G} \rightarrow G$) of the self-energy equations (19.126) at $|\mathbf{k}| = k_F$. In the steady state, one can express the Keldysh components of the electron and boson propagator in terms of their distributions i.e.

$$G_\omega^K = 2iF_\omega^e \text{Im } G_\omega^R, \quad (19.136)$$

$$D_\omega^K = 2iF_\omega^b \text{Im } D_\omega^R. \quad (19.137)$$

Combining eq. (19.136) with the Keldysh component of the Dyson equation, i.e. $G_\omega^K = G_\omega^R \Sigma_\omega^K G_\omega^A$, one also has

$$\Sigma_\omega^K = 2iF_\omega^e \text{Im } \Sigma_\omega^R, \quad (19.138)$$

so, at the phase transition, $F_\omega^\Sigma = F_\omega^e$. Replacing eqs. (19.136) and (19.137) in the equation for Σ_ω^R , one obtains

$$\begin{aligned} \Sigma_\omega^R = & -2g_c \int_\epsilon \left[D_{\omega-\epsilon}^R F_\epsilon^e \text{Im } G_\epsilon^R \right. \\ & \left. + F_{\omega-\epsilon}^b G_\epsilon^R \text{Im } D_{\omega-\epsilon}^R \right]. \end{aligned} \quad (19.139)$$

Similarly for Σ_ω^K one finds

$$\Sigma_\omega^K = -4ig_c \int_\epsilon (F_{\omega-\epsilon}^b F_\epsilon^e + 1) \text{Im } D_{\omega-\epsilon}^R \text{Im } G_\epsilon^R. \quad (19.140)$$

Replacing this expression and the imaginary part of eq. (19.139) in eq. (19.138), one finally obtains the kinetic equation for the distribution F_ω^e

$$\begin{aligned} & \int_\epsilon [F_{\omega-\epsilon}^b F_\epsilon^e + 1 - F_\omega^e (F_\epsilon^e + F_{\omega-\epsilon}^b)] \\ & \times \text{Im } (D_{\omega-\epsilon}^R) \text{Im } (G_\epsilon^R) = 0. \end{aligned} \quad (19.141)$$

Let us consider now the linearized equations for the anomalous self-energy (19.135) at $|\mathbf{k}| = k_F$. First of all, one can express the Keldysh component of the anomalous self-energy in terms of its generalized distribution defined through

$$\Delta_\omega^K = 2iF_\omega^\Delta \text{Im } \Delta_\omega^R. \quad (19.142)$$

Replacing this expression together with eqs. (19.136) and (19.137) in the equation for Δ_ω^R and using the particle-hole symmetry at $|\mathbf{k}| = k_F$, i.e. $G_{-\omega}^A = -G_\omega^R$ and $F_{-\omega}^e = -F_\omega^e$, one obtains

$$\begin{aligned} \Delta_\omega^R = & 2g_c \int_\epsilon \left\{ F_{\omega-\epsilon}^b (G_\epsilon^R)^2 \Delta_\epsilon^R \text{Im } D_{\omega-\epsilon}^R \right. \\ & + D_{\omega-\epsilon}^R \left\{ F_\epsilon^e \text{Im } [(G_\epsilon^R)^2 \Delta_\epsilon^R] \right. \\ & \left. \left. + (F_\epsilon^\Delta - F_\epsilon^e) |G_\epsilon^R|^2 \text{Im } \Delta_\epsilon^R \right\} \right\}. \end{aligned} \quad (19.143)$$

Considering now the equation for Δ_ω^K , one can rewrite it by adding the zero integral

$$0 = -ig_c \int_\epsilon (D_{\omega-\epsilon}^R G_\epsilon^A G_{-\epsilon}^R \Delta_\epsilon^A + D_{\omega-\epsilon}^A G_\epsilon^R G_{-\epsilon}^A \Delta_\epsilon^R) \quad (19.144)$$

to its r.h.s. Using again eqs. (19.136), (19.137) and (19.142) one obtains

$$\begin{aligned} F_\omega^\Delta \text{Im } \Delta_\omega^R &= 2g_c \int_\epsilon \text{Im} (D_{\omega-\epsilon}^R) \\ &\times \left\{ (F_{\omega-\epsilon}^b F_\epsilon^e + 1) \text{Im} \left[(G_\epsilon^R)^2 \Delta_\epsilon^R \right] \right. \\ &\left. + F_{\omega-\epsilon}^b (F_\epsilon^\Delta - F_\epsilon^e) |G_\epsilon^R|^2 \text{Im } \Delta_\epsilon^R \right\}. \end{aligned} \quad (19.145)$$

Replacing the imaginary part of eq. (19.143) on the l.h.s. of this equation and bringing everything on the r.h.s., one finally arrives at the equation for F_ω^Δ

$$\begin{aligned} 0 &= \int_\epsilon \text{Im}(D_{\omega-\epsilon}^R) \left\{ [F_{\omega-\epsilon}^b F_\epsilon^e + 1 \right. \\ &\quad \left. - F_\omega^\Delta (F_\epsilon^e + F_{\omega-\epsilon}^b)] \text{Im} \left[(G_\epsilon^R)^2 \Delta_\epsilon^R \right] \right. \\ &\quad \left. + (F_{\omega-\epsilon}^b - F_\omega^\Delta) (F_\epsilon^\Delta - F_\epsilon^e) |G_\epsilon^R|^2 \text{Im } \Delta_\epsilon^R \right\}. \end{aligned} \quad (19.146)$$

eqs. (19.139), (19.141), (19.143) and (19.146) form a set of self-consistent equations equivalent to eqs. (19.126) and (19.135).

Assuming that bosons and electrons are at thermal equilibrium at the temperature T , one can show that eqs. (19.141) and (19.146) are self-consistently solved by the thermal distributions¹⁰

$$F_\omega^b = \coth\left(\frac{\omega}{2T}\right), \quad F_\omega^e = F_\omega^\Delta = \tanh\left(\frac{\omega}{2T}\right). \quad (19.147)$$

This can be proved by replacing eq. (19.147) in eqs. (19.141) and (19.146) and showing that the terms in the square brackets vanish because of the relation [71]

$$\begin{aligned} 1 + \coth\left(\frac{\omega - \epsilon}{2T}\right) \tanh\left(\frac{\epsilon}{2T}\right) \\ = \tanh\left(\frac{\omega}{2T}\right) \left[\tanh\left(\frac{\epsilon}{2T}\right) + \coth\left(\frac{\omega - \epsilon}{2T}\right) \right], \end{aligned} \quad (19.148)$$

and that the last term in eq. (19.146) vanishes because $F_\omega^\Delta = F_\omega^e$. Note that, for the same reason, also the last term on the r.h.s. of eq. (19.143) becomes zero.

At thermal equilibrium, one has to self-consistently solve only the eqs. (19.139) and (19.143) for Σ_ω^R and Δ_ω^R , as the distributions are already fixed by the thermal fluctuation-dissipation relations in eq. (19.147).

Analytic continuation of Matsubara Eliashberg equations at thermal equilibrium

The Eliashberg eqs. (19.139) and (19.143) for Σ_ω^R and Δ_ω^R at thermal equilibrium could be equivalently obtained as the analytic continuation to real frequencies of Eliashberg equations in Matsubara formalism, which is the one typically employed in thermal equilibrium calculations.

¹⁰These distributions correspond to the thermal Bose-Einstein $n_\omega^b = (F_\omega^b - 1)/2 = (e^{\omega/T} - 1)^{-1}$ and Fermi-Dirac $n_\omega^e = (1 - F_\omega^e)/2 = (e^{\omega/T} + 1)^{-1}$ occupations for bosons and electrons, respectively.

Within this formalism, the frequency dependence of propagators and self-energies is expressed in terms of discrete fermionic Matsubara frequencies $\omega_n = (2n+1)\pi T$ (n integer) on the imaginary frequency axis, rather than in terms of real frequencies ω [73]. The Eliashberg equations in Matsubara frequencies for our model can be derived following Ref. [51] and in the linearized case at $|\mathbf{k}| = k_F$ read ¹¹

$$\Sigma_{i\omega_n} = -2g_c T \sum_{\omega_m} D_{i\omega_n - i\omega_m} G_{i\omega_m}, \quad (19.149)$$

$$\Delta_{i\omega_n} = -2g_c T \sum_{\omega_m} D_{i\omega_n - i\omega_m} G_{i\omega_m} G_{-i\omega_m} \Delta_{i\omega_m}, \quad (19.150)$$

where the dressed fermion propagator is defined through the Dyson equation

$$G_{\mathbf{k}, i\omega_n} = \frac{1}{i\omega_n - \xi_{\mathbf{k}} - \Sigma_{\mathbf{k}, i\omega_n}}, \quad (19.151)$$

and the boson propagator is

$$D_{i\Omega_n} = \frac{1}{2} \frac{\omega_0}{[i\Omega_n + i\kappa \operatorname{sgn} \Omega_n]^2 - \omega_0^2}, \quad (19.152)$$

with $\Omega_n = 2\pi nT$ (n integer) is a bosonic Matsubara frequency. Note that the $\operatorname{sgn} \Omega_n$ factor in eq. (19.152) is necessary to ensure the proper causality structure $\Pi_{-i\Omega_n}^{\text{env}} = [\Pi_{i\Omega_n}^{\text{env}}]^*$ of the boson environment self-energy in Matsubara formalism.

eqs. (19.149) and (19.150) can be analytically continued exactly from the positive Matsubara axis ($\omega_n > 0$) to frequencies $\omega + i0^+$ just above the real axis to obtain the Eliashberg equations for the retarded components Δ_{ω}^R and Σ_{ω}^R [55, 58]. To do so, we express the propagators (19.152) and (19.151) through their spectral representations [73]

$$\begin{aligned} D_{i\Omega_n} &= -2 \int_{\Omega} \frac{\operatorname{Im} D_{\Omega}^R}{i\Omega_n - \Omega} \\ &= -2 \int_{\Omega} \operatorname{Im} D_{\Omega}^R \frac{\Omega}{(i\Omega_n)^2 - \Omega^2}, \end{aligned} \quad (19.153)$$

where the symmetry $\operatorname{Im} D_{-\Omega}^R = -\operatorname{Im} D_{\Omega}^R$ has been used in the second line, and

$$G_{i\omega_n} = -2 \int_{\epsilon} \frac{\operatorname{Im} G_{\epsilon}^R}{i\omega_n - \epsilon}. \quad (19.154)$$

Moreover, one can also express the function $G_{i\omega_m} G_{-i\omega_m} \Delta_{i\omega_m}$ in eq. (19.150) through its spectral representation

$$G_{i\omega_m} G_{-i\omega_m} \Delta_{i\omega_m} = -2 \int_{\epsilon} \frac{\operatorname{Im} (G_{\epsilon}^R G_{-\epsilon}^A \Delta_{\epsilon}^R)}{i\omega_n - \epsilon}. \quad (19.155)$$

We can now replace these spectral representations in eqs. (19.149) and (19.150), obtaining

$$\begin{aligned} \Sigma_{i\omega_n} &= -8g_c \int_{\Omega} \int_{\epsilon} \operatorname{Im} D_{\Omega}^R \operatorname{Im} G_{\epsilon}^R \\ &\quad \times T \sum_{\omega_m} \frac{\Omega}{(i\omega_n - i\omega_m)^2 - \Omega^2} \frac{1}{i\omega_m - \epsilon}, \end{aligned} \quad (19.156)$$

¹¹The factor of 2 in eqs. (19.149) and (19.150) comes from a different definition of the free boson Hamiltonian in eq. (19.11) with respect to Ref. [51].

and

$$\begin{aligned} \Delta_{i\omega_n} &= -8g_c \int_{\Omega} \int_{\epsilon} \text{Im } D_{\Omega}^R \text{Im } (G_{\epsilon}^R G_{-\epsilon}^A \Delta_{\epsilon}^R) \\ &\times T \sum_{\omega_m} \frac{\Omega}{(i\omega_n - i\omega_m)^2 - \Omega^2} \frac{1}{i\omega_m - \epsilon}. \end{aligned} \quad (19.157)$$

The Matsubara frequency sum in both equations can be computed by transforming it in a contour integration [73]

$$\begin{aligned} &T \sum_{\omega_m} \frac{\Omega}{(i\omega_n - i\omega_m)^2 - \Omega^2} \frac{1}{i\omega_m - \epsilon} \\ &= -\frac{1}{2} \tanh\left(\frac{\epsilon}{2T}\right) \frac{\Omega}{(\epsilon - i\omega_n)^2 - \Omega^2} - \frac{1}{4} \coth\left(\frac{\Omega}{2T}\right) \\ &\times \left(\frac{1}{i\omega_n - \Omega - \epsilon} + \frac{1}{i\omega_n + \Omega - \epsilon} \right), \end{aligned} \quad (19.158)$$

where we have used the property $\tanh(i\omega_n - z) = -\coth(z)$ to obtain the coth in the last term. Note that this last replacement is essential to obtain the correct causal structure of the functions when the analytic continuation is performed, i.e. to obtain functions that are bound and analytic in the upper complex plane [55]. We can now perform the analytic continuation $i\omega_n \rightarrow \omega + i0^+$ of the external Matsubara frequency in the previous expressions, obtaining

$$\begin{aligned} \Sigma_{\omega}^R &= 8g_c \int_{\Omega} \int_{\epsilon} \text{Im } D_{\Omega}^R \text{Im } G_{\epsilon}^R \\ &\times \left[\frac{1}{2} \tanh\left(\frac{\epsilon}{2T}\right) \frac{\Omega}{(\omega - \epsilon + i0^+)^2 - \Omega^2} + \frac{1}{4} \coth\left(\frac{\Omega}{2T}\right) \right. \\ &\times \left. \left(\frac{1}{\omega - \Omega - \epsilon + i0^+} + \frac{1}{\omega + \Omega - \epsilon + i0^+} \right) \right], \end{aligned} \quad (19.159)$$

and

$$\begin{aligned} \Delta_{\omega}^R &= 8g_c \int_{\Omega} \int_{\epsilon} \text{Im } D_{\Omega}^R \text{Im } (G_{\epsilon}^R G_{-\epsilon}^A \Delta_{\epsilon}^R) \\ &\left[\frac{1}{2} \tanh\left(\frac{\epsilon}{2T}\right) \frac{\Omega}{(\omega - \epsilon + i0^+)^2 - \Omega^2} + \frac{1}{4} \coth\left(\frac{\Omega}{2T}\right) \right. \\ &\times \left. \left(\frac{1}{\omega - \Omega - \epsilon + i0^+} + \frac{1}{\omega + \Omega - \epsilon + i0^+} \right) \right]. \end{aligned} \quad (19.160)$$

Performing the change of variable $\Omega \rightarrow -\Omega$ in the second term in round brackets, one can replace

$$\begin{aligned} &\left(\frac{1}{\omega - \Omega - \epsilon + i0^+} + \frac{1}{\omega + \Omega - \epsilon + i0^+} \right) \\ &\rightarrow \frac{2}{\omega - \Omega - \epsilon + i0^+}, \end{aligned} \quad (19.161)$$

as $\text{Im}(D_{\Omega}^R) \coth\left(\frac{\Omega}{2T}\right)$ is symmetric in Ω . Moreover, one can identify in the previous expressions the spectral representations for the retarded functions [73]

$$D_{\omega-\epsilon}^R = -2 \int_{\Omega} \text{Im } D_{\Omega}^R \frac{\Omega}{(\omega - \epsilon + i0^+)^2 - \Omega^2}, \quad (19.162)$$

$$G_{\omega+\Omega}^R = -2 \int_{\epsilon} \frac{\text{Im } G_{\epsilon}^R}{\omega + \Omega - \epsilon + i0^+}, \quad (19.163)$$

and

$$G_{\omega+\Omega}^R G_{-\omega-\Omega}^A \Delta_{\omega+\Omega}^R = -2 \int_{\epsilon} \frac{\text{Im} (G_{\epsilon}^R G_{-\epsilon}^A \Delta_{\epsilon}^R)}{\omega + \Omega - \epsilon + i0^+}, \quad (19.164)$$

to obtain

$$\begin{aligned} \Sigma_{\omega}^R &= -2g_c \int_{\epsilon} \tanh\left(\frac{\epsilon}{2T}\right) D_{\omega-\epsilon}^R \text{Im} G_{\epsilon}^R \\ &\quad - 2g_c \int_{\Omega} \coth\left(\frac{\Omega}{2T}\right) G_{\omega+\Omega}^R \text{Im} D_{\Omega}^R, \end{aligned} \quad (19.165)$$

and

$$\begin{aligned} \Delta_{\omega}^R &= -2g_c \int_{\epsilon} \tanh\left(\frac{\epsilon}{2T}\right) D_{\omega-\epsilon}^R \text{Im} (G_{\epsilon}^R G_{-\epsilon}^A \Delta_{\epsilon}^R) \\ &\quad - 2g_c \int_{\Omega} \coth\left(\frac{\Omega}{2T}\right) G_{\omega+\Omega}^R G_{-\omega-\Omega}^A \Delta_{\omega+\Omega}^R \text{Im} D_{\Omega}^R. \end{aligned} \quad (19.166)$$

Finally, by performing the change of variables $\epsilon = \Omega + \omega$ in the second integrals and using the relation $G_{-\omega}^A = -G_{\omega}^R$, one finds that eqs. (19.165) and (19.166) are equivalent to eqs. (19.139) and (19.143) at thermal equilibrium, i.e. obtained by setting $F_{\omega}^e = F_{\omega}^{\Delta} = \tanh(\frac{\omega}{2T})$ and $F_{\omega}^b = \coth(\frac{\omega}{2T})$.

19.8.4 Causal structure of mean-field theory

In this appendix we will consider the causal structure of our theory at the mean-field level. The mean-field level means that we neglect the quantum fluctuations of the normal and anomalous self-energies and they are therefore described fully by eq. (19.26). The fermionic partition function then collapses to

$$\mathcal{Z} = \int \mathcal{D}[\bar{\psi}, \psi] e^{i \sum_k \bar{\Psi}_k^T \mathbb{G}_k^{-1} \Psi_k}, \quad (19.167)$$

with the Keldysh-Nambu spinors defined in eq. (19.104) and the inverse Keldysh-Nambu propagator defined in eq. (19.108). We can now derive the casual structure of the sixteen elements in $\mathbb{G}$ by relating them to two-point correlation functions of the electrons. Due to the quadratic nature of the mean-field action, this can be straightforwardly obtained by adding linear source fields and taking functional derivatives thereof [71].

Anomalous propagators and self-energies

The eight different anomalous terms in $\mathbb{G}$ are connected to the fermionic expectation values

$$\begin{aligned} \mathcal{F}_k^{(1)} &= -i \langle \psi_{\uparrow,k}^c \psi_{\downarrow,-k}^c \rangle, & \bar{\mathcal{F}}_k^{(1)} &= -i \langle \bar{\psi}_{\downarrow,-k}^c \bar{\psi}_{\uparrow,k}^c \rangle, \\ \mathcal{F}_k^{(2)} &= -i \langle \psi_{\uparrow,k}^c \psi_{\downarrow,-k}^q \rangle, & \bar{\mathcal{F}}_k^{(2)} &= -i \langle \bar{\psi}_{\downarrow,-k}^c \bar{\psi}_{\uparrow,k}^q \rangle, \\ \mathcal{F}_k^{(3)} &= -i \langle \psi_{\uparrow,k}^q \psi_{\downarrow,-k}^c \rangle, & \bar{\mathcal{F}}_k^{(3)} &= -i \langle \bar{\psi}_{\downarrow,-k}^q \bar{\psi}_{\uparrow,k}^c \rangle, \\ \mathcal{F}_k^{(4)} &= -i \langle \psi_{\uparrow,k}^q \psi_{\downarrow,-k}^q \rangle, & \bar{\mathcal{F}}_k^{(4)} &= -i \langle \bar{\psi}_{\downarrow,-k}^q \bar{\psi}_{\uparrow,k}^q \rangle, \end{aligned}$$

where the numbering follows the structure in eq. (19.105). To find the explicit causal structure of these objects we map the fields from the Keldysh contour back into time-ordered operators. To this extent, the first step is to Fourier transform to real-space. To ensure consistency with the definition used for the $\bar{\Delta}$ fields, we define the transformation of $\bar{\mathcal{F}}$ with the same sign as

$\bar{\Delta}$. Due to the temporal and spatial translational invariance of the system, the real-space form depends only on the difference $x - x'$ and assumes the form

$$\mathcal{F}_{x-x'}^{(n)} = -i\langle\psi_{\uparrow,x}^{\alpha_n}\psi_{\downarrow,x'}^{\beta_n}\rangle, \quad \bar{\mathcal{F}}_{x-x'}^{(n)} = -i\langle\bar{\psi}_{\downarrow,x'}^{\alpha_n}\bar{\psi}_{\uparrow,x}^{\beta_n}\rangle, \quad (19.168)$$

where $\alpha_n, \beta_n \in \{c, q\}$. The next steps for mapping back to operators is to undo the Keldysh rotation (19.61) and using the appropriate contour ordering, followed by writing out the time-ordering [70]. To keep the presentation self-contained, this procedure is shown for one of the expectation values

$$\begin{aligned} \langle\psi_{\uparrow,x}^c\psi_{\downarrow,x'}^q\rangle &= \frac{1}{2}\langle(\psi_{\uparrow,x}^+ + \psi_{\uparrow,x}^-)(\psi_{\downarrow,x'}^+ - \psi_{\downarrow,x'}^-)\rangle, \\ &= \frac{1}{2}\left(\langle\psi_{\uparrow,x}^+\psi_{\downarrow,x'}^+\rangle - \langle\psi_{\uparrow,x}^-\psi_{\downarrow,x'}^-\rangle \right. \\ &\quad \left. + \langle\psi_{\uparrow,x}^-\psi_{\downarrow,x'}^+\rangle - \langle\psi_{\uparrow,x}^+\psi_{\downarrow,x'}^-\rangle\right), \\ &= \frac{1}{2}\left(\mathcal{T}\langle\hat{c}_{\uparrow,x}\hat{c}_{\downarrow,x'}\rangle - \mathcal{T}^-\langle\hat{c}_{\uparrow,x}\hat{c}_{\downarrow,x'}\rangle \right. \\ &\quad \left. + \langle\hat{c}_{\uparrow,x}\hat{c}_{\downarrow,x'}\rangle + \langle\hat{c}_{\downarrow,x'}\hat{c}_{\uparrow,x}\rangle\right), \\ &= \frac{1}{2}\left[\theta_{t-t'}\left(\langle\hat{c}_{\uparrow,x}\hat{c}_{\downarrow,x'}\rangle + \langle\hat{c}_{\downarrow,x'}\hat{c}_{\uparrow,x}\rangle \right. \right. \\ &\quad \left. \left. + \langle\hat{c}_{\uparrow,x}\hat{c}_{\downarrow,x'}\rangle + \langle\hat{c}_{\downarrow,x'}\hat{c}_{\uparrow,x}\rangle\right) \right. \\ &\quad \left. - \theta_{t'-t}\left(\langle\hat{c}_{\downarrow,x'}\hat{c}_{\uparrow,x}\rangle + \langle\hat{c}_{\uparrow,x}\hat{c}_{\downarrow,x'}\rangle \right. \right. \\ &\quad \left. \left. - \langle\hat{c}_{\uparrow,x}\hat{c}_{\downarrow,x'}\rangle - \langle\hat{c}_{\downarrow,x'}\hat{c}_{\uparrow,x}\rangle\right)\right], \\ &= \theta_{t-t'}\langle\{\hat{c}_{\uparrow,x}, \hat{c}_{\downarrow,x'}\}\rangle = i\mathcal{F}_{x-x'}^R, \end{aligned} \quad (19.169)$$

where $\{\cdot, \cdot\}$ is the anticommutator and the hat has been used to highlight the operator nature. In the first line the Keldysh rotation has been undone. To get from the second to the third line we have used the contour ordering to write the fields as operators. This is done by knowing that all fields on the $+$ ($-$) branch must be (anti-)time-ordered [71], otherwise the fields on the $+$ branch must be evaluated before the ones on the $-$ branch. Going to the fourth line, the time-ordering have been written explicitly, which highlights the cancellations leading to the final result. As a last step the result have been identified with the retarded anomalous propagators defined in eq. (19.24). Following the same procedure for the other seven expectation

values, the causal structure of all the different anomalous propagators is given by

$$\begin{aligned}
 \mathcal{F}_{x-x'}^{(1)} &= -i \langle [\hat{c}_{\uparrow,x}, \hat{c}_{\downarrow,x'}] \rangle = \mathcal{F}_{x-x'}^K, \\
 \mathcal{F}_{x-x'}^{(2)} &= -i\theta_{t-t'} \langle \{\hat{c}_{\uparrow,x}, \hat{c}_{\downarrow,x'}\} \rangle = \mathcal{F}_{x-x'}^R, \\
 \mathcal{F}_{x-x'}^{(3)} &= i\theta_{t'-t} \langle \{\hat{c}_{\uparrow,x}, \hat{c}_{\downarrow,x'}\} \rangle = \mathcal{F}_{x-x'}^A, \\
 \mathcal{F}_{x-x'}^{(4)} &= 0, \\
 \bar{\mathcal{F}}_{x-x'}^{(1)} &= -i \left\langle \left[\hat{c}_{\downarrow,x'}^\dagger, \hat{c}_{\uparrow,x}^\dagger \right] \right\rangle = -(\mathcal{F}_{x-x'}^K)^*, \\
 \bar{\mathcal{F}}_{x-x'}^{(2)} &= -i\theta_{t'-t} \left\langle \left\{ \hat{c}_{\downarrow,x'}^\dagger, \hat{c}_{\uparrow,x}^\dagger \right\} \right\rangle = (\mathcal{F}_{x-x'}^A)^*, \\
 \bar{\mathcal{F}}_{x-x'}^{(3)} &= i\theta_{t-t'} \left\langle \left\{ \hat{c}_{\downarrow,x'}^\dagger, \hat{c}_{\uparrow,x}^\dagger \right\} \right\rangle = (\mathcal{F}_{x-x'}^R)^*, \\
 \bar{\mathcal{F}}_{x-x'}^{(4)} &= 0.
 \end{aligned} \tag{19.170}$$

Transforming back to energy-momentum space, the relation between the two off-diagonal 2×2 blocks in $\mathbb{G}$ is thus found to be

$$\bar{\mathcal{F}}_k = -\sigma_z \mathcal{F}_k^\dagger \sigma_z. \tag{19.171}$$

With this relation we can connect Δ to $\bar{\Delta}$. To do this we follow the same procedure used for deriving the equations for Δ to derive the equations for $\bar{\Delta}$. The result of the almost identical derivations is

$$\begin{pmatrix} \bar{\Delta}_{\mathbf{k},\omega}^A \\ \bar{\Delta}_{\mathbf{k},\omega}^R \\ \bar{\Delta}_{\mathbf{k},\omega}^K \end{pmatrix} = ig \int_{\omega'} \begin{pmatrix} \mathcal{F}^{A*} D^K - \mathcal{F}^{K*} D^A \\ \mathcal{F}^{R*} D^K - \mathcal{F}^{K*} D^R \\ \mathcal{F}^{R*} D^R + \mathcal{F}^{A*} D^A - \mathcal{F}^{K*} D^K \end{pmatrix}, \tag{19.172}$$

where we have used the notation $\mathcal{F}^{\alpha*} D^\beta = (\mathcal{F}_{\mathbf{k},\omega'}^\alpha)^* D_{\omega'-\omega}^\beta$ and used the relation between $\mathcal{F}$ and $\bar{\mathcal{F}}$ in eq. (19.171). As the D 's are propagators of a real field, they obey the symmetries

$$D_{-\omega}^R = D_\omega^A \text{ and } D_{-\omega}^K = D_\omega^K. \tag{19.173}$$

Using these relations and comparing to eq. (19.134) one finds the connection between Δ and $\bar{\Delta}$

$$\bar{\Delta}_k^R = (\Delta_k^R)^*, \quad \bar{\Delta}_k^A = (\Delta_k^A)^*, \quad \bar{\Delta}_k^K = -(\Delta_k^K)^*. \tag{19.174}$$

As observed for the anomalous propagators in eq. (19.171) we see that $\bar{\Delta}$ is connected to Δ through complex conjugation, but with an additional change of sign for the Keldysh component.

From eq. (19.134) and eq. (19.172), it follows directly that if

$$\mathcal{F}_k^R = (\mathcal{F}_k^A)^* \text{ and } \mathcal{F}_k^K = -(\mathcal{F}_k^K)^*, \tag{19.175}$$

then it is also true that

$$\Delta_k^R = (\Delta_k^A)^* \text{ and } \Delta_k^K = -(\Delta_k^K)^*, \tag{19.176}$$

and vice versa. However, due to anomalous nature of $\mathcal{F}$, the causal connections in eq. (19.175) are not generically obeyed. To satisfy the first equality in eq. (19.175), one has always the gauge freedom to choose a global phase for $\mathcal{F}_k^R$ (or equivalently for Δ_k^R) such that the retarded and advanced propagators are connected through complex conjugation. In thermal equilibrium, $\mathcal{F}^K$ is then connected to $\mathcal{F}_k^R$ through thermal FDRs [similarly to eq. (19.142) for Δ^K], so its anti-Hermiticity shown in the second equality of eq. (19.175) is also straightforwardly verified. When the system is not in thermal equilibrium, instead, $\mathcal{F}^K$ is not generically anti-Hermitian. We can show this by expressing it via eq. (19.25) in the form

$$\begin{aligned}
 \mathcal{F}_k^K &= G_k^R \Delta_k^K \mathcal{G}_{-k}^R + G_k^K \Delta_k^A \mathcal{G}_{-k}^R + G_k^R \Delta_k^R \mathcal{G}_{-k}^K \\
 &= \mathcal{G}_k^R \Delta_k^K G_{-k}^R + \mathcal{G}_k^K \Delta_k^A G_{-k}^R + \mathcal{G}_k^R \Delta_k^R G_{-k}^K.
 \end{aligned} \tag{19.177}$$

Assuming the causal connections of $\Delta^{R,A,K}$ in eq. (19.176) to be true, and assuming that $\mathcal{G}^{R,A,K}$ (and equivalently $G^{R,A,K}$) follow the standard causal connections $\mathcal{G}_k^R = (\mathcal{G}_k^A)^*$ and $\mathcal{G}_k^K = -(\mathcal{G}_k^K)^*$ [71], the conjugate of eq. (19.177) gives

$$(\mathcal{F}_k^K)^* = -\mathcal{G}_k^A \Delta_k^K G_{-k}^A - \mathcal{G}_k^K \Delta_k^R G_{-k}^A - \mathcal{G}_k^A \Delta_k^A G_{-k}^K. \quad (19.178)$$

We see then that $(\mathcal{F}_k^K)^* = -\mathcal{F}_k^K$ does not hold in general. It holds, however, if we fix the momentum to $|\mathbf{k}| = k_F$. The proof is based on the fact that, due to the momentum structure of our interaction eq. (19.12), the propagators $\mathcal{G}^{R,A,K}$ (and equivalently $G^{R,A,K}$) satisfy the particle-hole symmetry

$$\mathcal{G}_{-\mathbf{k}_F, -\omega}^R = -\mathcal{G}_{\mathbf{k}_F, \omega}^A \quad \text{and} \quad \mathcal{G}_{-\mathbf{k}_F, \omega}^K = -\mathcal{G}_{\mathbf{k}_F, \omega}^K. \quad (19.179)$$

Using this symmetry one finds

$$\begin{aligned} (\mathcal{F}_{\mathbf{k}_F, \omega}^K)^* &= -\mathcal{G}_{-\mathbf{k}_F, \omega}^R \Delta_{\mathbf{k}_F, \omega}^K G_{\mathbf{k}_F, \omega}^R - \mathcal{G}_{-\mathbf{k}_F, -\omega}^K \Delta_{\mathbf{k}_F, \omega}^R G_{\mathbf{k}_F, \omega}^R \\ &\quad - \mathcal{G}_{-\mathbf{k}_F, -\omega}^R \Delta_{\mathbf{k}_F, \omega}^A G_{\mathbf{k}_F, \omega}^K, \end{aligned} \quad (19.180)$$

which is equivalent to $-\mathcal{F}_{\mathbf{k}_F, \omega}^K$ in eq. (19.177). So we find that, even when the system is in a NESS, as long as it obeys particle-hole symmetry as in eq. (19.179), one can always choose a gauge where $\mathcal{F}^K$ is anti-Hermitian, i.e. purely imaginary in momentum-frequency space. In this case $\Delta^{R,A,K}$ inherit the causal structure of $\mathcal{F}^{R,A,K}$.

Normal propagators

Let us consider now the eight different normal terms in $\mathbb{G}$. For this purpose, we reintroduce the spin index $\sigma = \uparrow, \downarrow$ to properly identify them. Similarly to the anomalous terms, we find that they are connected to the fermionic expectation values

$$\begin{aligned} \mathcal{G}_k^{\uparrow(1)} &= -i \langle \psi_{\uparrow, k}^c \bar{\psi}_{\uparrow, k}^c \rangle, & \bar{\mathcal{G}}_k^{\downarrow(1)} &= -i \langle \bar{\psi}_{\downarrow, -k}^c \psi_{\downarrow, -k}^c \rangle, \\ \mathcal{G}_k^{\uparrow(2)} &= -i \langle \psi_{\uparrow, k}^c \bar{\psi}_{\uparrow, k}^q \rangle, & \bar{\mathcal{G}}_k^{\downarrow(2)} &= -i \langle \bar{\psi}_{\downarrow, -k}^c \psi_{\downarrow, -k}^q \rangle, \\ \mathcal{G}_k^{\uparrow(3)} &= -i \langle \psi_{\uparrow, k}^q \bar{\psi}_{\uparrow, k}^c \rangle, & \bar{\mathcal{G}}_k^{\downarrow(3)} &= -i \langle \bar{\psi}_{\downarrow, -k}^q \psi_{\downarrow, -k}^c \rangle, \\ \mathcal{G}_k^{\uparrow(4)} &= -i \langle \psi_{\uparrow, k}^q \bar{\psi}_{\uparrow, k}^q \rangle, & \bar{\mathcal{G}}_k^{\downarrow(4)} &= -i \langle \bar{\psi}_{\downarrow, -k}^q \psi_{\downarrow, -k}^q \rangle. \end{aligned}$$

Using the anticommutation relations of fermionic fields, it is immediately evident that the relation between the 2×2 diagonal blocks in $\mathbb{G}$ is

$$\bar{\mathcal{G}}_k^\sigma = -\mathcal{G}_{-k}^{\sigma T}. \quad (19.181)$$

We can then just work on the $\mathcal{G}_k^\sigma$ block, as the properties of the $\bar{\mathcal{G}}_k^\sigma$ block immediately follow. To find the causal structure of $\mathcal{G}_k^\sigma$, we start by Fourier transforming it to real space, obtaining

$$\mathcal{G}_{x-x'}^{\sigma(n)} = -i \langle \psi_{\sigma, x}^{\alpha_n} \bar{\psi}_{\sigma, x'}^{\beta_n} \rangle. \quad (19.182)$$

We can now map the fields from the Keldysh contour back into time-ordered operators with a procedure analogous to eq. (19.169) and obtain

$$\begin{aligned} \mathcal{G}_{x-x'}^{\sigma(1)} &= -i \langle [\hat{c}_{\sigma, x}, \hat{c}_{\sigma, x'}^\dagger] \rangle = \mathcal{G}_{x-x'}^{\sigma K}, \\ \mathcal{G}_{x-x'}^{\sigma(2)} &= -i \theta_{t-t'} \langle \{ \hat{c}_{\sigma, x}, \hat{c}_{\sigma, x'}^\dagger \} \rangle = \mathcal{G}_{x-x'}^{\sigma R}, \\ \mathcal{G}_k^{\sigma(3)} &= i \theta_{t'-t} \langle \{ \hat{c}_{\sigma, x}, \hat{c}_{\sigma, x'}^\dagger \} \rangle = \mathcal{G}_{x-x'}^{\sigma A}, \\ \mathcal{G}_{x-x'}^{\sigma(4)} &= 0. \end{aligned} \quad (19.183)$$

This result can then be used to derive the causal structure of the normal self-energy, as discussed above eq. (19.125).

19.8.5 RAK components of electron propagators

In this appendix we present some useful expressions for the retarded and Keldysh components of the normal ($\mathcal{G}$) and anomalous ($\mathcal{F}$) electron propagators written in terms of the normal-state propagator (G) and the anomalous self-energy (Δ).

Within the superconducting phase, the electron propagators are given by eq. (19.117):

$$\mathcal{G}_k = (G_k^{-1} + \Delta_k G_{-k}^T \bar{\Delta}_k^T)^{-1}, \quad (19.184)$$

$$\mathcal{F}_k = G_k \Delta_k \mathcal{G}_{-k}^T = \mathcal{G}_k \Delta_k G_{-k}^T, \quad (19.185)$$

where $G_k = [(\omega - \xi_k)\sigma_x - \Sigma_k]^{-1}$ [cf. eq. (19.20)]. By expliciting the RAK matrix structures as in eqs. (19.13), (19.18) and (19.22), one can explicitly compute the RAK components of the normal and anomalous propagator, obtaining

$$\mathcal{G}_k^R = \frac{G_k^R}{1 + G_k^R G_{-k}^A \bar{\Delta}_k^A \Delta_k^R}, \quad (19.186)$$

$$\mathcal{F}_k^R = \frac{G_k^R G_{-k}^A \Delta_k^R}{1 + G_k^R G_{-k}^A \bar{\Delta}_k^A \Delta_k^R}, \quad (19.187)$$

for the retarded components and

$$\mathcal{G}_k^K = \frac{G_k^K - G_k^A G_k^R (G_{-k}^R \bar{\Delta}_k^R \Delta_k^K + G_{-k}^A \bar{\Delta}_k^K \Delta_k^R + G_{-k}^K \bar{\Delta}_k^R \Delta_k^R)}{(1 + G_k^A G_{-k}^R \bar{\Delta}_k^R \Delta_k^A)(1 + G_k^R G_{-k}^A \bar{\Delta}_k^A \Delta_k^R)}, \quad (19.188)$$

$$\mathcal{F}_k^K = \frac{G_{-k}^R G_k^K \Delta_k^A + G_{-k}^K G_k^R \Delta_k^R + G_{-k}^R G_k^K \Delta_k^K - G_{-k}^R G_k^A G_k^R G_{-k}^A \Delta_k^A \bar{\Delta}_k^K \Delta_k^R}{(1 + G_k^A G_{-k}^R \bar{\Delta}_k^R \Delta_k^A)(1 + G_k^R G_{-k}^A \bar{\Delta}_k^A \Delta_k^R)}, \quad (19.189)$$

for the Keldysh components. We remind that $\bar{\Delta}_\omega$ and Δ_ω are related by complex conjugation relations as in eq. (19.174).

To derive the expression for $\mathcal{F}_\omega^K$ used in of eq. (19.34), one can set $|\mathbf{k}| = k_F$ in eq. (19.189) (dropping the momentum dependence) and express Δ_ω^K and Σ_ω^K in terms of distributions

$$\Sigma_\omega^K = 2iF_\omega^\Sigma \text{Im } \Sigma_\omega^R, \quad (19.190)$$

$$\Delta_\omega^K = 2iF_\omega^\Delta \text{Im } \Delta_\omega^R. \quad (19.191)$$

It is easy to show through eq. (19.20) that F_ω^Σ is also the distribution function for the normal-state propagator G , that is

$$G_\omega^K = 2iF_\omega^\Sigma \text{Im } G_\omega^R. \quad (19.192)$$

On the other hand, in the superconducting phase the electron distribution F_ω^e for the full normal propagator $\mathcal{G}$, defined through the parametrization $\mathcal{G}_\omega^K = 2iF_\omega^e \text{Im } \mathcal{G}_\omega^R$, is in general different from F_ω^Σ . Being at $|\mathbf{k}| = k_F$, one can further use the particle-hole symmetry $G_{-\omega}^R = -G_\omega^A$ [cf. eq. (19.179)] and the relation $(\Delta_\omega^R)^* = \Delta_\omega^A$ [cf. (19.176)] to obtain

$$\begin{aligned} \mathcal{F}_\omega^K = 2iF_\omega^\Sigma & \frac{-\text{Im} \left[(G_\omega^R)^2 \Delta_\omega^R \right] - |G_\omega^R|^4 |\Delta_\omega^R|^2 \text{Im } \Delta_\omega^R}{|1 - (G_\omega^R)^2 (\Delta_\omega^R)^2|^2} \\ & - 2i(F_\omega^\Delta - F_\omega^\Sigma) \frac{|G_\omega^R|^2 \text{Im} (\Delta_\omega^R) + |G_\omega^R|^4 |\Delta_\omega^R|^2 \text{Im} (\Delta_\omega^R)}{|1 - (G_\omega^R)^2 (\Delta_\omega^R)^2|^2}, \end{aligned} \quad (19.193)$$

where we have separated a “thermal-like” contribution proportional to F_ω^Σ and a “non-thermal” contribution proportional $F_\omega^\Delta - F_\omega^\Sigma$. From this expression it is directly evident that $\mathcal{F}_\omega^K$ is

purely imaginary. Further rewriting of this expression in terms of eqs. (19.186) and (19.187) at $|\mathbf{k}| = k_F$ finally results in the expression

$$\begin{aligned} \mathcal{F}_\omega^K &= 2iF_\omega^\Sigma \operatorname{Im}(\mathcal{F}_\omega^R) - 2i(F_\omega^\Delta - F_\omega^\Sigma) \operatorname{Im} \Delta_\omega^R \\ &\times |\mathcal{G}_\omega^R|^2 \left(1 - |G_\omega^R|^2 |\Delta_\omega^R|^2\right), \end{aligned} \quad (19.194)$$

which corresponds to the integrand of eq. (19.34).

At the superconducting phase transition, one has that $\mathcal{G}_\omega \rightarrow G_\omega$ and $F_\omega^\Sigma \rightarrow F_\omega^e$, and this allows to simplify eqs. (19.187) and (19.189) for the anomalous propagator as

$$\mathcal{F}_k^R = G_k^R G_{-k}^A \Delta_k^R, \quad (19.195)$$

for the retarded component and

$$\mathcal{F}_k^K = G_{-k}^R G_k^K \Delta_k^A + G_{-k}^K G_k^R \Delta_k^R + G_{-k}^R G_k^R \Delta_k^K, \quad (19.196)$$

for the Keldysh component. Setting again $|\mathbf{k}| = k_F$ one can use the parametrization eq. (19.192) with $F_\omega^\Sigma \rightarrow F_\omega^e$ and the particle-hole symmetry to rewrite eq. (19.196) as

$$\begin{aligned} \mathcal{F}_\omega^K &= -2iF_\omega^e \left\{ \operatorname{Im} \left[(G_\omega^R)^2 \Delta_\omega^R \right] + |G_\omega^R|^2 \operatorname{Im} \Delta_\omega^R \right\} \\ &+ |G_\omega^R|^2 \Delta_\omega^K, \end{aligned} \quad (19.197)$$

which corresponds to the integrand of eq. (19.35).

19.8.6 Electron back-action onto the bosons

In this appendix, we give a justification of why there is always a weak-coupling regime where the boson self-energy $\Pi(\omega)$ due to electron-boson interaction can be neglected. For simplicity, we suppose here to be at the critical coupling g_c of the phase transition, so that we can consider only normal-state propagators G for the electrons.

Starting from the partition function for the boson-electron system of eq. (19.57), one can keep only the one-loop contributions to the boson self-energy to obtain [71]

$$\mathcal{Z} = \int \mathcal{D}[X] e^{i(S_{b,0}[X] + S_{b,\text{int}}[X])}, \quad (19.198)$$

where $S_{b,0}[X]$ is given by eq. (19.59) and

$$\begin{aligned} S_{b,\text{int}}[X] &= -\frac{1}{2} \int_{t,t'} \begin{pmatrix} X^c \\ X^q \end{pmatrix}_t^T \begin{pmatrix} 0 & \Pi^A \\ \Pi^R & \Pi^K \end{pmatrix}_{t-t'} \begin{pmatrix} X^c \\ X^q \end{pmatrix}_{t'}. \end{aligned} \quad (19.199)$$

In the previous expression, the boson self-energy RAK components are given by

$$\Pi_{t-t'}^{R(A)} = 4ig_c \sum_{\mathbf{r}} \left(G_+^{R(A)} G_-^K + G_+^K G_-^{A(R)} \right), \quad (19.200)$$

$$\Pi_{t-t'}^K = 4ig_c \sum_{\mathbf{r}} \left(G_+^K G_-^K + G_+^A G_-^R + G_+^R G_-^A \right), \quad (19.201)$$

with the shorthand notation $G_+^\alpha G_-^\beta = G_{\mathbf{r},t-t'}^\alpha G_{\mathbf{r},t'-t}^\beta$. To have an estimate of the magnitude of this self-energy, we concentrate on the retarded component. In momentum-frequency space, it reads

$$\Pi_\omega^R = 4ig \int_{\mathbf{k}'} \left(G_{\mathbf{k}',\omega+\omega'}^R G_{\mathbf{k}',\omega'}^K + G_{\mathbf{k}',\omega+\omega'}^K G_{\mathbf{k}',\omega'}^A \right), \quad (19.202)$$

where $G_{\mathbf{k},\omega}$ are the electronic propagators, and we used the shorthand continuum notation $\int_{\mathbf{k}'} = V_d \int d\mathbf{k}' / (2\pi)^d \int d\omega' / 2\pi$, where V_d is the volume of the system in d dimensions. Note that Π_{ω}^R does not have momentum dependence because in our model the bosons carry zero momentum. To further simplify the calculation, we perform a quasi-particle approximation on the electronic propagators as $G_{\omega}^R \approx 1/(\omega + i\eta)$. Within this approximation, one has

$$\Pi_{\omega}^R = 16g_c\eta \int_{\mathbf{k}'} \frac{\omega' F_{\omega'+\xi_{\mathbf{k}'}}^e}{[\omega'^2 + (\omega + i\eta)^2] (\omega'^2 + \eta^2)}. \quad (19.203)$$

Evaluating this self-energy on-shell, i.e. at the frequency of the bosonic mode $\omega = \omega_0$, performing the change of variables $\tilde{\varepsilon} = \mathbf{k}^2/2m\omega_0$ and $\tilde{\omega} = \omega'/\omega_0$ and assuming $\eta \ll \omega_0$, one obtains

$$\Pi_{\omega_0}^R = g_c\eta \frac{C_d}{\omega_0^2} \left(\frac{\omega_0}{\omega_V} \right)^{d/2}, \quad (19.204)$$

where we defined the energy scale $\omega_V = 1/(2mV_d^{2/d})$ and the dimensionless constant

$$C_d = \frac{8\Omega_d}{(2\pi)^{d+1}} \int_0^\infty d\tilde{\varepsilon} \tilde{\varepsilon}^{\frac{d}{2}-1} \times \text{P.V.} \int_{-\infty}^\infty d\tilde{\omega} \frac{F_{(\tilde{\omega}+\tilde{\varepsilon})\omega_0-\mu}^e}{\tilde{\omega} (\tilde{\omega}^2 + 1)}, \quad (19.205)$$

with Ω_d the solid angle in d dimensions and P.V. denotes that the integral is taken in the Cauchy principal value sense. We find that at lowest order for small g_c and η , $\Pi_{\omega_0}^R$ scales as $g_c\eta/\omega_0^2$. This has to be compared to the real part of the on-shell inverse retarded propagator $\text{Re} \{ [D^{-1}]_{\omega_0}^R \} = 2\kappa^2/\omega_0$. This means that, for finite κ and ω_0 , there is always a small- g_c or small- η regime where the photon self-energy can be neglected. While the value of the constants ω_V and C_d depends on the specific model parameters, we at least made sure that the condition $g_c\eta \ll \kappa^2\omega_0$ is always satisfied in our numerical results.

19.8.7 Boson propagator and environment-induced self-energy

In this appendix, we first show how, starting from the action for a complex boson field, one can integrate away the $\mathcal{P}$ -quadrature to obtain the boson propagator for the $\mathcal{X}$ -quadrature. We then show how to derive the environment-induced self-energy for the system bosons in the case of a linear coupling between the $\mathcal{X}$ -quadrature of the environment and the X -quadrature of the system boson.

$\mathcal{P}$ -quadrature integration and boson propagator for the $\mathcal{X}$ -quadrature

Let us start by showing how the $\mathcal{P}$ -quadrature of a boson field can be integrated away. This procedure can be applied to both the multi-mode environment bosons and the single-mode system bosons. The required structure for this procedure is that the bosonic field only interacts with other fields through its $\mathcal{X}$ -quadrature. The generic starting point is therefore a Hamiltonian of the form

$$\hat{H} = \hat{H}_{\text{sys}} + \sum_s \left(\nu_s \hat{b}_s^\dagger \hat{b}_s + \frac{t_s}{\sqrt{2}} \hat{\mathcal{X}}_s \hat{\mathcal{O}} \right), \quad (19.206)$$

where $\hat{b}_s^\dagger$ is the bosonic creation operator for the bare mode s with energy ν_s and

$$\hat{\mathcal{X}}_s = \frac{1}{\sqrt{2Y_b}} (\hat{b}_s + \hat{b}_s^\dagger) \quad (19.207)$$

is the real quadrature operator, with Y_b being a constant that is chosen based on convenience. The operators $\hat{H}_{\text{sys}}$ and $\hat{\mathcal{O}}$ do not include any additional $\hat{b}_s$, $\hat{b}_s^\dagger$ operators. Lastly, t_s is the coupling strength between the $\mathcal{X}_s$ -quadrature and the $\hat{\mathcal{O}}$ operator. The $\mathcal{P}_s$ -quadrature of the boson is defined as

$$\hat{\mathcal{P}}_s = \frac{1}{i\sqrt{2Y_b}} (\hat{b}_s - \hat{b}_s^\dagger). \quad (19.208)$$

This quadrature only appears in the bare quadratic part of the Hamiltonian and can therefore be integrated away. The bare part of the environment action is given by [71]

$$S_0 [b_s^*, b_s] = \int_{\omega} \sum_s \begin{pmatrix} b_s^c \\ b_s^q \end{pmatrix}_{\omega}^{\dagger} \mathcal{M}_{s,\omega} \begin{pmatrix} b_s^c \\ b_s^q \end{pmatrix}_{\omega}, \quad (19.209)$$

where $b_s^{c/q}$ are the classical and quantum components of the complex bosonic fields resulting from translating the term that is quadratic in $\hat{b}$ -operators in the Hamiltonian eq. (19.206) to the $\pm$ time contour, and then performing the Keldysh rotation $b_s^{c/q} = (b_s^+ \pm b_s^-)/\sqrt{2}$, analogously to eqs. (19.61) and (19.62). The matrix $\mathcal{M}_{s,\omega}$ is the inverse complex boson propagator, and is given by

$$\mathcal{M}_{s,\omega} = \begin{pmatrix} 0 & \omega - \nu_s - i0^+ \\ \omega - \nu_s + i0^+ & i2F_{\omega}0^+ \end{pmatrix}, \quad (19.210)$$

with 0^+ being a positive infinitesimal and F_{ω} being the distribution function of the bosons (assumed here to be s -independent, like in the thermal case). The coupling to the $\hat{\mathcal{O}}$ operator in eq. (19.206) gives rise to a term in the action of the form

$$S_c [\mathcal{X}, \mathcal{O}] = - \int_{\omega} \sum_s t_s \begin{pmatrix} \bar{\mathcal{X}}_s^c \\ \bar{\mathcal{X}}_s^q \end{pmatrix}_{-\omega}^T \begin{pmatrix} \mathcal{O}_1 \\ \mathcal{O}_2 \end{pmatrix}_{\omega}, \quad (19.211)$$

where $\mathcal{O}_{1,2}$ are two objects that depend on the explicit form of the operator $\hat{\mathcal{O}}$. Inverting eqs. (19.207) and (19.208), the $b_{\omega}^{c/q}$ and $\bar{b}_{\omega}^{c/q}$ fields can be written in terms of the $\mathcal{X}^{c/q}$ and $\mathcal{P}^{c/q}$ quadratures as

$$\begin{aligned} b_{\omega}^{c/q} &= \sqrt{Y_b} (\mathcal{X}_{s,\omega}^{c/q} + i\mathcal{P}_{s,\omega}^{c/q}), \\ (b_{\omega}^{c/q})^* &= \sqrt{Y_b} (\mathcal{X}_{s,-\omega}^{c/q} - i\mathcal{P}_{s,-\omega}^{c/q}). \end{aligned} \quad (19.212)$$

As this is nothing but a linear transformation, it is straightforward to rewrite S_0 in terms of the quadratures. Since both quadratures are real fields, one can rewrite the action into the following symmetric form

$$\begin{aligned} S_0 [\mathcal{X}, \mathcal{P}] &= \int_{\omega} \sum_{s;\alpha,\beta=c,q} \left(\frac{1}{2} \mathcal{X}_{s,-\omega}^{\alpha} (\mathcal{M}_{s,\omega}^S)_{\alpha,\beta} \mathcal{X}_{s,\omega}^{\beta} \right. \\ &\quad \left. + \frac{1}{2} \mathcal{P}_{s,-\omega}^{\alpha} (\mathcal{M}_{s,\omega}^S)_{\alpha,\beta} \mathcal{P}_{s,\omega}^{\beta} - i \mathcal{P}_{s,-\omega}^{\alpha} (\mathcal{M}_{s,\omega}^A)_{\alpha,\beta} \mathcal{X}_{s,\omega}^{\beta} \right), \end{aligned} \quad (19.213)$$

where the symmetric and anti-symmetric inverse propagators are defined as

$$\begin{aligned} \mathcal{M}_{s,\omega}^S &= Y_b (\mathcal{M}_{s,\omega} + \mathcal{M}_{s,-\omega}^T), \\ \mathcal{M}_{s,\omega}^A &= Y_b (\mathcal{M}_{s,\omega} - \mathcal{M}_{s,-\omega}^T). \end{aligned} \quad (19.214)$$

The functional integrals over the $\mathcal{P}$ -fields can now be performed

$$\begin{aligned} \mathcal{Z}_b &= \int \mathcal{D} [\mathcal{P}, \mathcal{X}] \exp [i (S_0 [\mathcal{X}, \mathcal{P}] + S_c [\mathcal{X}, \mathcal{O}_A])] \\ &= \int \mathcal{D} [\mathcal{X}] \exp [i (S_{\mathcal{X},0} [\mathcal{X}] + S_c [\mathcal{X}, \mathcal{O}_A])], \end{aligned} \quad (19.215)$$

where the $\mathcal{X}$ -quadrature action is found to be

$$S_{\mathcal{X},0}[\mathcal{X}] = \int_{\omega} \sum_{s;\alpha,\beta=c,q} \frac{1}{2} \mathcal{X}_{s,-\omega}^{\alpha} (\mathcal{Q}_{s,\omega})_{\alpha,\beta} \mathcal{X}_{s,\omega}^{\beta}, \quad (19.216)$$

and the bare inverse $\mathcal{X}$ propagator is given by

$$\begin{aligned} \mathcal{Q}_{s,\omega} &= \mathcal{M}_{s,\omega}^S + [\mathcal{M}_{s,-\omega}^A]^T [\mathcal{M}_{s,\omega}^S]^{-1} \mathcal{M}_{s,-\omega}^A \\ &= \begin{pmatrix} 0 & Q_{s,\omega}^A \\ Q_{s,\omega}^R & Q_{s,\omega}^K \end{pmatrix}, \end{aligned} \quad (19.217)$$

where the elements are given by

$$\begin{aligned} Q_{s,\omega}^R &= (Q_{s,\omega}^A)^* = 2Y_b \frac{(\omega + i0^+)^2 - \nu_s^2}{\nu_s}, \\ Q_{s,\omega}^K &= \frac{2Y_b i0^+ F_{e,\omega}}{\nu_s^2} [\omega^2 + \nu_s^2 + (0^+)^2] + \frac{4Y_b i0^+ \omega F_{o,\omega}}{\nu_s}. \end{aligned} \quad (19.218)$$

Here the even and odd parts of the boson distribution function have been separated out such that

$$\begin{aligned} F_{e,\omega} &= F_{\omega} + F_{-\omega}, \\ F_{o,\omega} &= F_{\omega} - F_{-\omega}. \end{aligned} \quad (19.219)$$

Environment-induced self-energy

We consider now a linear coupling between the $\mathcal{X}$ -quadrature of the environment bosons and the X -quadrature of the system bosons with bare frequency $\tilde{\omega}_0$, resulting in $\hat{\mathcal{O}} = (\hat{a}^\dagger + \hat{a})/\sqrt{2Y_a} = \hat{X}$. The linear nature of the coupling means that the Gaussian functional integrals over the environment $\mathcal{X}_s$ -fields can be directly performed

$$\int \mathcal{D}[\mathcal{X}] \exp[i(S_{\mathcal{X},0}[\mathcal{X}] + S_c[\mathcal{X}, X])] = \exp(iS_{X,\text{env}}[X]), \quad (19.220)$$

with

$$S_{X,\text{env}}[X] = -\frac{1}{2} \int_{\omega} \sum_{\alpha,\beta=c,q} X_{-\omega}^{\alpha} (\mathbf{\Pi}_{\omega}^{\text{env}})_{\alpha,\beta} X_{\omega}^{\beta}, \quad (19.221)$$

being the resulting action describing the effect of the environment on the system bosons. The environment-induced self-energy for the system bosons is then given by

$$\begin{aligned} \mathbf{\Pi}_{\omega}^{\text{env}} &= \begin{pmatrix} 0 & \Pi_{\omega}^{\text{env},A} \\ \Pi_{\omega}^{\text{env},R} & \Pi_{\omega}^{\text{env},K} \end{pmatrix}, \\ &= \sum_s t_s^2 \sigma_x [\mathcal{Q}_{s,\omega}]^{-1} \sigma_x, \\ &= \sum_s t_s^2 \begin{pmatrix} 0 & C_{s,\omega}^A \\ C_{s,\omega}^R & C_{s,\omega}^K \end{pmatrix}. \end{aligned} \quad (19.222)$$

Inverting $\mathcal{Q}_{s,\omega}$ leads to the following RAK components of the environment boson propagators

$$\begin{aligned} C_{s,\omega}^R &= (C_{s,\omega}^A)^* = \frac{1}{2Y_b} \frac{\nu_s}{(\omega + i0^+)^2 - \nu_s^2}, \\ C_{s,\omega}^K &= \frac{-i2\nu_s \omega 0^+ F_{\mathcal{X},s,\omega}^{\text{env}}}{Y_b |(\omega + i0^+)^2 - \nu_s^2|^2}, \end{aligned} \quad (19.223)$$

where the environment $\mathcal{X}$ -quadrature distribution function has been defined as

$$F_{\mathcal{X},s,\omega}^{\text{env}} = \frac{F_{o,\omega}^{\text{env}}}{2} + F_{e,\omega}^{\text{env}} \frac{\omega^2 + \nu_s^2 + (0^+)^2}{4\omega\nu_s}. \quad (19.224)$$

As this distribution function is odd in ω , the Keldysh component $C_{s,\omega}^K$ is even by construction, which is a consequence of the formulation in terms of real fields [71].

To make further progress, the environment energies ν_s and couplings t_s must be specified. This is conveniently done by introducing the spectral density

$$J_\omega = \sum_s \frac{t_s^2 A_{s,\omega}}{2\pi}, \quad (19.225)$$

where $A_{s,\omega} = -2\text{Im} C_{s,\omega}^R$ is the environment spectral function. As all the ω -dependence of J_ω originates from $A_{s,\omega}$ and given the expression for $C_{s,\omega}^R$ in eq. (19.223), J_ω is antisymmetric in ω . Using the spectral density as a weight, the discrete sum over the environment energies in $\Pi_\omega^{\text{env},R}$ can be written as an integral

$$\Pi_\omega^{\text{env},R} = \int_\epsilon \frac{2\pi\epsilon J_\epsilon}{(\omega + i0^+)^2 - \epsilon^2}. \quad (19.226)$$

If J_ω is analytic on the real axis and $\lim_{|\omega| \rightarrow \infty} |J_\omega| = \text{const.}$, then the integral can be evaluated directly by infinitesimally shifting the integration contour into the complex plane. The result is

$$\begin{aligned} \Pi_\omega^{\text{env},R} &= -\frac{i\pi}{2} (J_\omega - J_{-\omega}) \\ &\quad - \text{P.V.} \int_\epsilon \frac{2\pi\epsilon J_\epsilon}{(\epsilon - \omega)(\epsilon + \omega)}. \end{aligned} \quad (19.227)$$

For $\Pi_\omega^{\text{env},K}$ one notices that, as J_ω is an antisymmetric function, then $C_{s,\omega}^K = -iF_{o,\omega}^{\text{env}} A_{s,\omega}/2$. Using the definition (19.225) of J_ω , the Keldysh component of the self-energy takes the form

$$\Pi_\omega^{\text{env},K} = -i\pi J_\omega F_{o,\omega}^{\text{env}}. \quad (19.228)$$

The odd distribution function can be rewritten in terms of the environment occupation $n_\omega^{\text{env}} = (F_\omega^{\text{env}} - 1)/2$ as

$$F_{o,\omega}^{\text{env}} = 2(n_\omega^{\text{env}} - n_{-\omega}^{\text{env}}). \quad (19.229)$$

For the spectral density we consider the simplest choice that satisfies the criteria described above

$$J_\omega = \mathcal{N} \frac{\kappa\omega}{\pi} \exp \left[-\left(\frac{\omega}{\omega_c} \right)^2 \right], \quad (19.230)$$

where $\mathcal{N}$ is a constant with the units of the inverse of an energy that we will choose later and ω_c is a cutoff frequency that ensures the convergence of eq. (19.227). We consider the regime where the cutoff (ω_c) is by far the largest energy scale of the system, such that the system is unaffected by the cutoff. In this regime the real part of the self-energy becomes a constant (on the energy scale of the system) and can therefore be absorbed into the bare frequency. Under this approximation the effective retarded self-energy simplifies to

$$\Pi_\omega^{\text{env},R} = -i\mathcal{N}\omega\kappa, \quad (19.231)$$

and the Keldysh self-energy takes the form

$$\Pi_\omega^{\text{env},K} = -i\mathcal{N}\omega\kappa F_{o,\omega}. \quad (19.232)$$

Due to the quadratic nature of the action (19.221) in the X -fields, one can immediately combine it with the bare action for the single-mode system bosons, which has a form analogous to eq. (19.216) with the replacements $\mathcal{X}_s \rightarrow X$ and $\nu_s \rightarrow \tilde{\omega}_0$, and without the sum over s . This results in the total quadratic action for the X -fields

$$\begin{aligned} S_{b,0}[X] &= \frac{1}{2} \int_{\omega} \begin{pmatrix} X^c \\ X^q \end{pmatrix}_{-\omega}^T \begin{pmatrix} 0 & [D^{-1}]^A \\ [D^{-1}]^R & [D^{-1}]^K \end{pmatrix}_{\omega} \begin{pmatrix} X^c \\ X^q \end{pmatrix}_{\omega} \end{aligned} \quad (19.233)$$

with the inverse retarded propagator given by

$$\begin{aligned} [D^{-1}]_{\omega}^R &= (D_{\omega}^R)^{-1} = 2Y_a \frac{\omega^2 - \tilde{\omega}_0^2}{\tilde{\omega}_0} - \Pi_{\omega}^{\text{env},R}, \\ &= \frac{2Y_a}{\tilde{\omega}_0} \left(\omega^2 - \tilde{\omega}_0^2 + i \frac{\mathcal{N} \tilde{\omega}_0}{2Y_a} \omega \kappa \right), \end{aligned} \quad (19.234)$$

the advanced inverse propagator given by the complex conjugate of eq. (19.234), and the Keldysh inverse propagator given by $[D^{-1}]^K = -\Pi_{\omega}^{\text{env},K}$, where the Keldysh self-energy is defined in eq. (19.222). To have a form of the boson propagator that resembles that of thermal equilibrium, we scale the system quadrature $Y_a = \tilde{\omega}_0/\omega_0$ and chose the constant in the spectral density to be $\mathcal{N} = 4/\omega_0$, where $\omega_0 = \sqrt{\tilde{\omega}_0^2 - \kappa^2}$ is the renormalized energy scale of the boson including the bath coupling. With these choices, the retarded boson propagator obtained by inverting the matrix structure in eq. (19.233) takes the form

$$D_{\omega}^R = \frac{1}{2} \frac{\omega_0}{(\omega + i\kappa)^2 - \omega_0^2}, \quad (19.235)$$

while the Keldysh propagator is given by

$$D_{\omega}^K = |D_{\omega}^R|^2 \Pi_{\omega}^{\text{env},K} = F_{\omega}^b (D_{\omega}^R - D_{\omega}^A). \quad (19.236)$$

Inserting eqs. (19.232) and (19.235) one identifies the boson distribution function as

$$F_{\omega}^b = \frac{F_{o,\omega}^{\text{env}}}{2} = n_{\omega}^{\text{env}} - n_{-\omega}^{\text{env}}. \quad (19.237)$$

For a thermally occupied environment one has $n_{\omega}^{\text{env}} = (e^{\omega/T} - 1)^{-1}$, which leads to the expected result of $F_{\omega}^b = \coth(\omega/2T)$. More generally, we find that, through a linear quadrature coupling, the distribution of the system bosons F_{ω}^b can be directly tuned by engineering the occupation function of the environment n_{ω}^{env} .

19.8.8 Cryostat-induced self-energy

In this appendix, we show how to derive the cryostat-induced self-energy for the electrons shown in eq. (19.29). We model the cryostat as a thermal bath of electrons at temperature T_0 that is linearly coupled to the system electrons. This is achieved by considering a Hamiltonian of the form

$$\begin{aligned} \hat{H}_{\text{cryo}} &= \sum_{s,\sigma,\mathbf{k}} \epsilon_{s,\sigma,\mathbf{k}} \hat{f}_{s,\sigma,\mathbf{k}}^{\dagger} \hat{f}_{s,\sigma,\mathbf{k}} \\ &+ \sum_{s,\sigma,\mathbf{k}} \left(t_{s,\sigma,\mathbf{k}} \hat{c}_{\sigma,\mathbf{k}}^{\dagger} \hat{f}_{s,\sigma,\mathbf{k}} + \text{h.c.} \right), \end{aligned} \quad (19.238)$$

where $\hat{f}_{s,\sigma,\mathbf{k}}^\dagger$ creates a cryostat electron in the mode s with spin σ and momentum $\mathbf{k}$, while $\hat{c}_{\sigma,\mathbf{k}}^\dagger$ creates a system electron with spin σ and momentum $\mathbf{k}$ as in eq. (19.10), $\epsilon_{s,\sigma,\mathbf{k}}$ is the dispersion of the bath electrons and $t_{s,\sigma,\mathbf{k}}$ is the coupling strength between cryostat electrons and the system electrons. We note that, by coupling the electrons at same momentum and spin, this interaction does not generate scattering in momentum but only an energetic redistribution of the electrons.

From the Hamiltonian (19.238) we can write down the part of the partition function that involves the cryostat

$$\mathcal{Z}_{\text{cryo}} = \int \mathcal{D}[\bar{f}, f] e^{iS_{\text{cryo}}[\psi, \bar{\psi}, f, \bar{f}]}, \quad (19.239)$$

where f (ψ) and $\bar{f}$ ($\bar{\psi}$) are the Grassmann fields related to the cryostat (system) electron annihilation and creation operators. The action has the form

$$\begin{aligned} S_{\text{cryo}}[\psi, \bar{\psi}, f, \bar{f}] = \sum_{\mathbf{k}, s, \sigma} \int_{\omega} & \left[\left(\begin{array}{c} \bar{f}_{s,\sigma,\mathbf{k}}^c \\ \bar{f}_{s,\sigma,\mathbf{k}}^q \end{array} \right)^T_{\omega} [(G^f)^{-1}]_{s,\sigma,\mathbf{k},\omega} \cdot \left(\begin{array}{c} f_{s,\sigma,\mathbf{k}}^c \\ f_{s,\sigma,\mathbf{k}}^q \end{array} \right)_{\omega} \right. \\ & \left. + t_{s,\sigma,\mathbf{k}} \left(\begin{array}{c} \bar{f}_{s,\sigma,\mathbf{k}}^c \\ \bar{f}_{s,\sigma,\mathbf{k}}^q \end{array} \right)^T_{\omega} \sigma_x \left(\begin{array}{c} \psi_{\sigma,\mathbf{k}}^c \\ \psi_{\sigma,\mathbf{k}}^q \end{array} \right)_{\omega} + \text{h.c.} \right]. \end{aligned} \quad (19.240)$$

The cryostat electron propagator assumes the matrix form

$$G_{s,\sigma,\mathbf{k},\omega}^f = \begin{pmatrix} G_{s,\sigma,\mathbf{k},\omega}^{f,K} & G_{s,\sigma,\mathbf{k},\omega}^{f,R} \\ G_{s,\sigma,\mathbf{k},\omega}^{f,A} & 0 \end{pmatrix}_{s,\sigma,\mathbf{k},\omega}. \quad (19.241)$$

As the feedback of the system on the cryostat can be neglected, the retarded propagator is simply given by the non-interacting form

$$G_{s,\sigma,\mathbf{k},\omega}^{f,R} = \frac{1}{\omega - \epsilon_{s,\sigma,\mathbf{k}} + i0^+}, \quad (19.242)$$

the advanced propagator is given by its complex conjugate, and the Keldysh propagator takes the form

$$\begin{aligned} G_{s,\sigma,\mathbf{k},\omega}^{f,K} &= F_{\omega}^{\text{cryo}} \left(G_{s,\sigma,\mathbf{k},\omega}^{f,R} - G_{s,\sigma,\mathbf{k},\omega}^{f,A} \right), \\ &= 2iF_{\omega}^{\text{cryo}} \text{Im}G_{s,\sigma,\mathbf{k},\omega}^{\text{cryo},R}, \end{aligned} \quad (19.243)$$

where $F_{\omega}^{\text{cryo}} = \tanh(\omega/2T_0)$. As the action (19.240) is quadratic, the Gaussian functional integral over the cryostat fields can be calculated exactly. The resulting contribution to the action of the system electrons is quadratic in the ψ ($\bar{\psi}$) fields and has the matrix form

$$S_{e-\text{cryo}}[\bar{\psi}, \psi] = - \sum_{\sigma,\mathbf{k}} \int_{\omega} \left(\begin{array}{c} \bar{\psi}_{\sigma,\mathbf{k}}^c \\ \bar{\psi}_{\sigma,\mathbf{k}}^q \end{array} \right)^T_{\omega} \Sigma_{\sigma,\mathbf{k},\omega}^{\text{cryo}} \left(\begin{array}{c} \psi_{\sigma,\mathbf{k}}^c \\ \psi_{\sigma,\mathbf{k}}^q \end{array} \right)_{\omega}, \quad (19.244)$$

where the cryostat-induced self-energy Σ^{cryo} has a similar causal structure as eq. (19.18). The retarded component of the self-energy has the form

$$\Sigma_{\sigma,\mathbf{k},\omega}^{\text{cryo},R} = \sum_s |t_{s,\sigma,\mathbf{k}}|^2 G_{s,\sigma,\mathbf{k},\omega}^{f,R} = \sum_s \frac{|t_{s,\sigma,\mathbf{k}}|^2}{\omega - \epsilon_{s,\sigma,\mathbf{k}} + i0^+}, \quad (19.245)$$

while the Keldysh self-energy is given by

$$\begin{aligned}\Sigma_{\sigma,\mathbf{k},\omega}^{\text{cryo},K} &= \sum_s |t_{s,\sigma,\mathbf{k}}|^2 G_{s,\sigma,\mathbf{k},\omega}^{f,K}, \\ &= -2\pi i \sum_s |t_{s,\sigma,\mathbf{k}}|^2 F_{\omega}^{\text{cryo}} \delta(\omega - \epsilon_{s,\sigma,\mathbf{k}}),\end{aligned}\tag{19.246}$$

where the infinitesimal linewidth (0^+) of the cryostat spectrum has been used to write the cryostat spectral function as a δ -function. Following the same logic as in section 19.8.7, one can introduce the cryostat spectral density as

$$J_{\sigma,\mathbf{k},\omega}^{\text{cryo}} = \sum_s |t_{s,\sigma,\mathbf{k}}|^2 \delta(\omega - \epsilon_{s,\sigma,\mathbf{k}}).\tag{19.247}$$

Using this spectral density, the cryostat-induced self-energy takes a form similar to the one induced by the boson environment in eqs. (19.227) and (19.228):

$$\begin{aligned}\Sigma_{\sigma,\mathbf{k},\omega}^{\text{cryo},R} &= -i\pi J_{\sigma,\mathbf{k},\omega}^{\text{cryo}} - \text{P.V.} \int_{\epsilon} \frac{J_{\sigma,\mathbf{k},\epsilon}^{\text{cryo}}}{\epsilon - \omega}, \\ \Sigma_{\sigma,\mathbf{k},\omega}^{\text{cryo},K} &= -2\pi i J_{\sigma,\mathbf{k},\omega}^{\text{cryo}} F_{\omega}^{\text{cryo}}.\end{aligned}\tag{19.248}$$

Differently from the spectral function for the boson propagator, the cryostat spectral function is positive [73]. Considering a simple spectral density of the form

$$J_{\sigma,\mathbf{k},\omega}^{\text{cryo}} = \frac{\eta_0}{\pi} \exp\left(-\frac{\omega^2}{\omega_c^2}\right),\tag{19.249}$$

and considering the limit where ω_c is a very large energy scale, one finds that real part of the retarded self-energy just gives rise to a constant shift which we absorb into the chemical potential. Instead, the imaginary part gives rise to a spectral width of the system electrons

$$\Sigma_{\sigma,\mathbf{k},\omega}^{\text{cryo},R} = -i\eta_0,\tag{19.250}$$

whereas the Keldysh self-energy has form

$$\Sigma_{\sigma,\mathbf{k},\omega}^{\text{cryo},K} = -2i\eta_0 F_{\omega}^{\text{cryo}}.\tag{19.251}$$

These self-energies are added to the boson-induced normal self-energies, as shown in eq. (19.30). Without the boson self-energies, the electrons are redistributed according to the cryostat distributions, such that electron distribution function in eq. (19.17) becomes

$$F_{\omega}^e = F_{\omega}^{\text{cryo}},\tag{19.252}$$

and thus thermalizes with the cryostat.

19.8.9 Numerical method to solve the NESS-Eliashberg equations at the phase transition

In this appendix, we show the numerical method employed to self-consistently solve the NESS-Eliashberg eqs. (19.126) and (19.135) at the phase transition. The critical coupling g_c is defined as the smallest coupling at which these equations admit a non-zero solution for $\Delta_{\mathbf{k},\omega}^R$. As in section 19.4, we set here the parametric momentum dependence to the Fermi momentum $|\mathbf{k}| = k_F$, as this is the first momentum channel to become unstable towards superconductivity,

dropping the explicit momentum dependence and obtaining eq. (19.26). In the calculation, the Fermi momentum is self-consistently defined through eq. (19.19), i.e. it is the momentum at which the real part of the quasi-particle pole of the retarded normal Green's function $G_{\mathbf{k},\omega}^R$ is zero.

To find the critical coupling g_c , the most naive approach would be solve the non-thermal Eliashberg equations iteratively and, coming from couplings $g > g_c$, find the coupling for which the order parameter Δ_ω^R becomes zero. However, this procedure often has the problem that the iterative algorithm gets very slow close to the phase transition, requiring an extrapolation for $\Delta_\omega^R \rightarrow 0$. A better alternative is to work with the linearized Eliashberg equations, obtained by assuming Δ_ω^R to be infinitesimal and neglecting all the terms that are non-linear in Δ_ω^R [23]. In eq. (19.26), this linearization procedure corresponds to replacing $\mathcal{G} \rightarrow G$ everywhere. One can then formally express the linearized NESS-Eliashberg equations for the fields $\Delta_\omega = (\Delta_\omega^A, \Delta_\omega^R, \Delta_\omega^K)$ and $\Sigma_\omega = (\Sigma_\omega^A, \Sigma_\omega^R, \Sigma_\omega^K)$ as

$$\Sigma_\omega = g_c \int_{\omega'} \Xi_{\omega,\omega'} [\Sigma_{\omega'}], \quad (19.253)$$

$$\Delta_\omega = g_c \int_{\omega'} \mathcal{K}_{\omega,\omega'} \Delta_{\omega'}, \quad (19.254)$$

where the kernels $\mathcal{K}_{\omega,\omega'}$ and $\Xi_{\omega,\omega'}$ are respectively a 3×3 matrix and a 3-dimensional vector in the RAK components with a continuous dependence on frequency variables (ω, ω') [see eq. (19.26) for their definitions]. The critical coupling g_c is then the coupling for which a non-zero Δ_ω solution of eq. (19.254) appears, which corresponds to require

$$\det [\mathbf{1}_3 \delta(\omega - \omega') - g_c \mathcal{K}_{\omega,\omega'}] = 0, \quad (19.255)$$

where $\mathbf{1}_3$ is the 3×3 identity matrix, and the determinant is computed both in discrete RAK space and in continuum frequency space. This equation for g_c , often called “Thouless criterion” [13], can be equivalently found from by looking for a divergence of the pairing susceptibility coming from the normal phase, as was attempted in Ref. [37]. We remark however that in Ref. [37] some important contributions in the RAK structure of the kernel were neglected, which are instead fully included here.

Direct solution of eq. (19.255) is however still quite troublesome, because it requires to discretize frequency space in a way to properly capture the relevant features of the kernel $\mathcal{K}_{\omega,\omega'}$ and calculate a determinant of a very large matrix. This procedure is typically much more straightforward in thermal equilibrium, where eq. (19.255) can be expressed on the naturally discretized imaginary Matsubara frequency axis, and a relatively small number of Matsubara frequencies is needed to fully describe the kernel (of the order of 1000 for our model). Here we employ an alternative method, originally proposed by McMillan in Ref. [54] to solve real-frequency linearized Eliashberg equations at thermal equilibrium, which is based on an iterative algorithm applied directly to eqs. (19.253) and (19.254), bypassing the calculation of the determinant of eq. (19.255). The iterative procedure is the following:

1. Start from a guess for the coupling g_c and the anomalous self-energy Δ_ω .
2. Evaluate the normal self-energy Σ_ω through eq. (19.253) and the corresponding normal-state Green's function G_ω through eq. (19.20).
3. Evaluate the frequency integral on the r.h.s. of eq. (19.254) and then update the coupling g_c so that the updated Δ_ω is rescaled to have $\Delta_{\omega=0}^R = 1$.
4. Repeat from step 2 until convergence on the coupling g_c and the functions Δ_ω and Σ_ω is reached.

Imposing $\Delta_{\omega=0}^R = 1$ in step 3 is equivalent to work in terms of the renormalized order parameter $\Delta_{\omega}/\Delta_{\omega=0}^R$, which remains finite also at the phase transition. This iterative method allows then to converge both on the coupling g_c and on the real-frequency order parameter Δ_{ω} at the same time¹². The most computationally expensive part of the algorithm is the calculation of the frequency integrals in the Δ_{ω} and Σ_{ω} equations, which can be performed directly in frequency space or as a product in time space, by exploiting the convolution form of the integral. Depending on the used algorithm, one can take advantage of the fact that the self-energies inherit symmetry properties from the real bosons and the particle-hole symmetry of the electrons

$$\begin{aligned}\Sigma_{\omega}^R &= -(\Sigma_{-\omega}^R)^*, \\ \Sigma_{\omega}^K &= -\Sigma_{-\omega}^K, \\ \Delta_{\omega}^R &= (\Delta_{-\omega}^R)^*, \\ \Delta_{\omega}^K &= \Delta_{-\omega}^K.\end{aligned}\tag{19.256}$$

Sometimes it was necessary to partially slow down the iterations in order to converge, as it is often the case in the solution of iterative integral equations (see for example Refs. [67, 68, 69]).

19.8.10 Weak-coupling approximations in the temperature-bias example

In this appendix we derive weak-coupling analytical expressions for the temperature-bias example. We first find expressions for the electron spectral width η , the effective temperature T_e and the non-thermal pairing parameter α_{nth} . Secondly, we use these results to derive a closed expression for the critical coupling g_c . In this setup, the bosons are strongly coupled to a bath with temperature T_b and the electrons are weakly coupled to cryostat at the temperature T_0 . These couplings can be modeled via the self-energies $\Pi_{\omega}^{\text{env}}$ (see section 19.8.7) and $\Sigma_{\omega}^{\text{cryo}}$ (see section 19.8.8), respectively. As mentioned in eq. (19.30), the latter self-energy can be introduced in our NESS-Eliashberg equations with the substitution $\Sigma_{\omega} \rightarrow \Sigma_{\omega} + \Sigma_{\omega}^{\text{cryo}}$. Since we are interested in physical quantities at the phase transition, we work here with the linearized NESS-Eliashberg equations at $|\mathbf{k}| = k_F$.

Electron spectral width η

The electron spectral width η is given by the negative imaginary part of the retarded self-energy evaluated at the FS $\omega = 0$. The coupling to the cryostat gives rise to a g_c -independent contribution $\eta_0 = -\text{Im} \Sigma_0^{\text{cryo},R}$, while the interaction with the bosons contributes to the spectral width through $-\text{Im} \Sigma_0^R$. The boson-induced broadening can be computed by taking the imaginary part of the linearized version of eq. (19.26) for Σ_{ω}^R :

$$\text{Im} \Sigma_{\omega}^R = g_c \int_{\epsilon} \text{Re} (D_{\omega-\epsilon}^R G_{\epsilon}^K + D_{\omega-\epsilon}^K G_{\epsilon}^R). \tag{19.257}$$

To perform a weak-coupling approximation of eq. (19.257), one replaces the electron propagators with their quasi-particle approximation $G_{\epsilon}^R \approx 1/(\epsilon + i\eta)$ and $G_{\epsilon}^K = 2iF_{\epsilon}^e \text{Im}(G_{\epsilon}^R) \approx -F_{\epsilon}^e(2i\eta)/(\epsilon^2 + \eta^2)$. Assuming that the electron quasiparticles are long-lived ($\eta \ll T_e, T_b, \omega_0$), the first term in parenthesis can be neglected since $G_{\epsilon}^K \approx -2i\pi F_{\epsilon}^e \delta_{\epsilon}$ and $F_{\epsilon=0}^e = 0$. Using the

¹²We remark that, at thermal equilibrium, McMillian's algorithm [54] is not the best method to obtain the real-frequency order parameter $\Delta_{\omega}^R/\Delta_{\omega=0}^R$ at the phase transition. A more advanced method, which first exploits the numerical advantage of the Matsubara frequency axis and then analytically continues the result to the real axis with a numerically exact procedure, has been developed by Marsiglio and Carbotte [55, 57, 61]. The latter method, however, cannot be applied to our non-thermal case as it still relies on Matsubara frequencies.

approximation $\text{Im } G_\epsilon^R \approx -\pi\delta_\epsilon$, the remaining term $\text{Re} (D_{\omega-\epsilon}^K G_\epsilon^R) = -\text{Im } D_{\omega-\epsilon}^K \text{Im } G_\epsilon^R$ gives the first-order contribution

$$\text{Im } \Sigma_\omega^{R,(1)} = \frac{g_c \text{Im } D_\omega^K}{2}. \quad (19.258)$$

Setting $\omega = 0$ in eq. (19.258), one obtains the first-order contribution for the electron spectral width

$$\begin{aligned} \eta^{(1)} &= -\text{Im } \Sigma_0^{R,(1)} = -\frac{g_c \text{Im } D_0^K}{2} \\ &= 2g_c \frac{\omega_0 \kappa T_b}{(\kappa^2 + \omega_0^2)^2} = g_c \gamma T_b, \end{aligned} \quad (19.259)$$

where we defined the constant

$$\gamma = \frac{2\omega_0 \kappa}{(\kappa^2 + \omega_0^2)^2}, \quad (19.260)$$

which has the units of inverse energy squared, making $g_c \gamma$ a dimensionless quantity. The spectral width of the FS at the leading order in g_c is therefore given by

$$\eta \approx \eta_0 + \eta^{(1)} = \eta_0 + g_c \gamma T_b. \quad (19.261)$$

Effective electron temperature T_e

Let us now consider the effective electron temperature T_e , as defined in eqs. (19.5) and (19.39). First the substitution $\Sigma_\omega \rightarrow \Sigma_\omega + \Sigma_\omega^{\text{cryo}}$ is used in eq. (19.138), obtaining

$$F_\omega^e = F_\omega^{\text{cryo}} - \frac{1}{2i\eta_0} (\Sigma_\omega^K - 2iF_\omega^e \text{Im } \Sigma_\omega^R). \quad (19.262)$$

Using the result in eq. (19.140) and the imaginary part of eq. (19.139) in eq. (19.262), one obtains the following kinetic equation

$$\begin{aligned} F_\omega^e &= F_\omega^{\text{cryo}} + \frac{2g_c}{\eta_0} \int_\epsilon \left[F_{\omega-\epsilon}^b F_\epsilon^e + 1 \right. \\ &\quad \left. - F_\omega^e (F_\epsilon^e + F_{\omega-\epsilon}^b) \right] \text{Im } D_{\omega-\epsilon}^R \text{Im } G_\epsilon^R, \end{aligned} \quad (19.263)$$

which corresponds to eq. (19.141) specialized to the temperature-bias setup. We can now perform the weak-coupling approximation by replacing the electron retarded propagator with its quasiparticle approximation $G_\omega^R \approx 1/(\omega + i\eta)$ with η given by eq. (19.261), and consider an external frequency $\omega \rightarrow 0$ in order to linearize the electron and cryostat distributions as $F_\omega^e \approx \omega/2T_e$ and $F_\omega^{\text{cryo}} \approx \omega/2T_0$, respectively. We obtain then

$$\begin{aligned} \frac{\omega}{2T_e} &= \frac{\omega}{2T_0} + \frac{2g_c}{\eta_0} \int_\epsilon \left[F_{\omega-\epsilon}^b F_\epsilon^e + 1 \right. \\ &\quad \left. - \frac{\omega}{2T_e} (F_\epsilon^e + F_{\omega-\epsilon}^b) \right] \text{Im } D_{\omega-\epsilon}^R \text{Im } G_\epsilon^R, \end{aligned} \quad (19.264)$$

In the weak-coupling regime $\eta \ll T_0, T_b, \omega_0$ such that $\text{Im } G_\epsilon^R \approx -\pi\delta_\epsilon$, which makes it possible to evaluate the integral over ϵ , obtaining

$$\frac{1}{2T_e} = \frac{1}{2T_0} + \frac{g_c \gamma}{2\eta_0} \left(1 - \frac{T_b}{T_e} \right), \quad (19.265)$$

where we have used the explicit form for D_ω^R from eq. (19.27), simplified a common ω factor and then set $\omega \rightarrow 0$ and expanded $F_{-\epsilon}^b \approx -2T_b/\epsilon$ before the integration. Solving for T_e , one finally obtains

$$T_e \approx \frac{\eta_0 + g_c \gamma T_b}{\eta_0 + g_c \gamma T_0} T_0. \quad (19.266)$$

Non-thermal pairing parameter α_{nth}

Let us now consider the linearized equation for Δ_ω^K at $|\mathbf{k}| = k_F$ and express the Keldysh component of the electron and boson propagators in terms of distributions as done in eq. (19.145), which with a slight rewriting takes the form

$$\begin{aligned} \Delta_\omega^K &= 2g_c \int_\epsilon \text{Im } D_{\omega-\epsilon}^R \\ &\times \left\{ 2i (F_{\omega-\epsilon}^b F_\epsilon^e + 1) \text{Im} \left[(G_\epsilon^R)^2 \Delta_\epsilon^R \right] \right. \\ &\left. + F_{\omega-\epsilon}^b (\Delta_\epsilon^K - 2i F_\epsilon^e \text{Im } \Delta_\epsilon^R) |G_\epsilon^R|^2 \right\}. \end{aligned} \quad (19.267)$$

Taking the imaginary part of this equation, dividing it by Δ_ω^R and considering the static limit $\omega \rightarrow 0$, one finds an equation for the non-thermal pairing parameter α_{nth} defined in eq. (19.3):

$$\begin{aligned} \alpha_{\text{nth}} &= 2g_c \int_\epsilon \text{Im } D_\epsilon^R \left\{ 2 (F_\epsilon^b F_\epsilon^e - 1) \text{Im} \left[(G_\epsilon^R)^2 \frac{\Delta_\epsilon^R}{\Delta_0^R} \right] \right. \\ &\left. + F_\epsilon^b \left[\frac{\text{Im } \Delta_\epsilon^K}{\Delta_0^R} - 2F_\epsilon^e \frac{\text{Im } \Delta_\epsilon^R}{\Delta_0^R} \right] |G_\epsilon^R|^2 \right\}, \end{aligned} \quad (19.268)$$

where the odd nature of the Boson spectral function and distribution function has been used. To compute the integral in eq. (19.268) we use the weak-coupling approximation $G_\epsilon^R \approx 1/(\epsilon + i\eta)$, with η given by eq. (19.261), together with a further assumption that $\eta \ll T_e, T_b, \omega_0$. Noticing that the functions $(G_\epsilon^R)^2 = 1/(\epsilon + i\eta)^2$ and $|G_\epsilon^R|^2 = 1/(\epsilon^2 + \eta^2)$ are strongly peaked close to $\epsilon = 0$, it is possible to approximate all the other multiplying function with their static limit, obtaining

$$\begin{aligned} \alpha_{\text{nth}} &\approx g_c \gamma \left\{ \left(\frac{T_b}{T_e} - 1 \right) \int_\epsilon \frac{4\eta\epsilon^2}{(\epsilon^2 + \eta^2)^2} \right. \\ &\left. + \frac{T_b}{\eta} \int_\epsilon \left[\frac{\epsilon}{T_e} \frac{\text{Im } \Delta_\epsilon^R}{\Delta_0^R} - \alpha_{\text{nth}} \right] \frac{2\eta}{\epsilon^2 + \eta^2} \right\}. \end{aligned} \quad (19.269)$$

As $\epsilon \text{Im } \Delta_\epsilon^R / \Delta_0^R \rightarrow 0$ for $\epsilon \rightarrow 0$, the first term in the square brackets can be neglected. The remaining frequency integrals are $\int_\epsilon \frac{4\eta\epsilon^2}{(\epsilon^2 + \eta^2)^2} = 1$ and $\int_\epsilon \frac{2\eta}{\epsilon^2 + \eta^2} = 1$, leading to

$$\alpha_{\text{nth}} \approx g_c \gamma \left[\left(\frac{T_b}{T_e} - 1 \right) - \frac{T_b \alpha_{\text{nth}}}{\eta} \right]. \quad (19.270)$$

One can finally isolate α_{nth} and find the g_c -dependent expression

$$\alpha_{\text{nth}} \approx \frac{g_c \gamma \left(\frac{T_b}{T_e} - 1 \right)}{1 + \frac{g_c \gamma T_b}{\eta}} = \frac{g_c \gamma \eta_0 (T_b - T_0)}{T_0 (\eta_0 + 2g_c \gamma T_b)}, \quad (19.271)$$

where in the last equality eqs. (19.261) and (19.266) was used to simplify the expression. Moreover, by expanding the first equality in eq. (19.271) close to thermal equilibrium ($T_b \approx T_e \approx T_0$), one obtains the expression

$$\alpha_{\text{nth}} \approx \mathcal{C} \frac{T_b - T_e}{T_e}, \quad (19.272)$$

where $\mathcal{C} = \frac{g_{c,\text{th}} \gamma (\eta_0 + g_{c,\text{th}} \gamma T_b)}{\eta_0 + 2g_{c,\text{th}} \gamma T_b}$ is a positive constant, with $g_{c,\text{th}}$ given by eq. (19.2). This is the expression used in eq. (19.6).

Closed-form expression for the critical coupling g_c

Having derived expressions for η in eq. (19.261), T_e in eq. (19.266) and α_{nth} in eq. (19.271), these can now be substituted into the expression for the critical coupling of eq. (19.1). After some straightforward but tedious algebra, one finds a quadratic equation for g_c with the following (positive) solution

$$g_c = \frac{2T_b}{|D_0^R|} + \frac{1}{2} \left(\frac{2T_b}{|D_0^R|} + \frac{\eta_0}{2\gamma T_0} \right) \times \left(\sqrt{1 + \frac{4\eta_0(T_0 - T_b)}{\gamma T_0 |D_0^R|} \left(\frac{2T_b}{|D_0^R|} + \frac{\eta_0}{2\gamma T_0} \right)^{-2}} - 1 \right), \quad (19.273)$$

which is closed-form expression for g_c depending only on the external parameters η_0 , κ , ω_0 , T_0 and T_b .

19.8.11 Weak-coupling approximations in the incoherent drive example

Similarly to section 19.8.10, in this appendix we derive the weak-coupling analytical expressions for the electron spectral width η , the effective temperature T_e , the non-thermal pairing parameter α_{nth} and the critical coupling g_c for the incoherent drive example. In this setup, the bosons are strongly coupled to an environment with a thermal background at temperature T_b and a incoherent drive at the frequency $\omega_d \gg T_b$, as described by the distribution in eq. (19.47). The electrons are instead decoupled from the cryostat ($\Sigma^{\text{cryo}} = 0$), so that their distribution F_ω^e is entirely determined by their coupling to the bosons.

Zeroth-order electron distribution

Since there is no cryostat, the electron distribution has to be self-consistently determined even for arbitrarily weak coupling. The distribution is determined by eq. (19.141). To zeroth order in g_c , the electrons have a vanishing linewidth such that $\text{Im } G_\epsilon^{R,(0)} = -\pi\delta_\epsilon$. The resulting equation for the distribution then takes the form

$$[F_\omega^b F_0^{e,(0)} + 1 - F_\omega^{e,(0)}(F_0^{e,(0)} + F_\omega^b)] \text{Im } D_\omega^R = 0. \quad (19.274)$$

At the FS $F_0^e = 0$, which removes two terms in eq. (19.274). Since $\text{Im } D_\omega^R \neq 0$, which is guaranteed by the coupling to the external environment, the zeroth-order contribution to the electron distribution takes the form

$$F_\omega^{e,(0)} = \frac{1}{F_\omega^b}. \quad (19.275)$$

This result resembles that of thermal equilibrium in eq. (19.147), but in this case it is true only at the zeroth order in g_c , rather than being self-consistently verified at all orders like at thermal equilibrium [cf. eqs. (19.146) to (19.148)].

Electron spectral width η

In the incoherent driving example, the spectral width of the electrons is solely due to the interaction with the bosons at the FS, i.e. $\eta = -\text{Im } \Sigma_0^R$. The first-order contribution to the normal self-energy can be calculated as the temperature-bias example, obtaining [cf. eq. (19.258)]

$$\text{Im } \Sigma_\omega^{R,(1)} = \frac{g_c \text{Im } D_\omega^K}{2}. \quad (19.276)$$

Consequently, the electron spectral width at the FS, to first order in g_c , is $\eta \approx \eta^{(1)} = g_c \gamma T_b$ found in eq. (19.259). As $\omega_d \gg T_b$, there is a large energy separation between the incoherently driven and thermal occupations, and for this reason the incoherent drive only starts contributing to η at second order in g_c . However, since we are in the weak-coupling regime and at the phase transition, it follows that $g_c \sim T_b$, making $\eta^{(1)}$ comparable in magnitude to the second order term. It is therefore essential to also compute the second order term to capture the physics under incoherent driving at $\omega_d \gg T_b$. This can be done by considering again the full frequency integral defining $\text{Im } \Sigma_\omega^R$ in eq. (19.257), setting $\omega = 0$ and estimating the additional contribution from the frequency region around $\epsilon = \omega_d$. To account for this contribution, we insert $\text{Im } \Sigma_\epsilon^{R,(1)}$ in the electron propagator

$$G_\epsilon^{R,(1)} = (\omega - i \text{Im } \Sigma_\epsilon^{R,(1)})^{-1}. \quad (19.277)$$

Note that, in the weak-coupling regime, $\text{Re } \Sigma_\epsilon^{R,(1)}$ can be neglected as the particle-hole symmetry forces it to vanish at the FS, while near ω_d its magnitude has to be compared to ω_d itself and is therefore negligible. The reason that it is sufficient to include eq. (19.276) to capture effects of the drive, is the convolution nature of eq. (19.257) for the retarded self-energy. Using eq. (19.277) inside eq. (19.257) makes the self-energy sensitive to the behavior of the electron distribution near ω_d , where $\text{Im } \Sigma_\omega^{R,(1)}$ is peaked. For this reason, it is also necessary to include the zeroth-order result for the distribution found in eq. (19.275). Within these approximations η takes the form

$$\begin{aligned} \eta &= -\text{Im } \Sigma_0^R \\ &\approx -g_c \int_\epsilon \text{Re} (2i D_{-\epsilon}^R F_\epsilon^{e,(0)} \text{Im } G_\epsilon^{R,(1)} \\ &\quad + 2i G_\epsilon^{R,(1)} F_{-\epsilon}^b \text{Im } D_{-\epsilon}^R) \\ &= -2g_c \int_\epsilon \left(\frac{1}{F_\epsilon^b} - F_\epsilon^b \right) \text{Im } D_\epsilon^R \text{Im } G_\epsilon^{R,(1)} \\ &= -2g_c \int_\epsilon \left[\left(\frac{1}{F_\epsilon^{b,\text{th}} + F_\epsilon^{b,d}} - F_\epsilon^{b,\text{th}} - F_\epsilon^{b,d} \right) \right. \\ &\quad \left. \times \text{Im } D_\epsilon^R \text{Im } G_\epsilon^{R,(1)} \right], \end{aligned} \quad (19.278)$$

where the expression for the boson distribution in eq. (19.47) and the symmetries of the real retarded boson propagator have been used. The large energy separation of the incoherent drive from the thermal part means that, for energies where $F_\epsilon^{b,d}$ is finite, $F_\epsilon^{b,\text{th}}$ looks constant, whereas near the FS $F_\epsilon^{b,d}$ vanishes. This allows one to divide the integral in eq. (19.278) in two contributions: a contribution $\eta_<$ from frequencies around the FS ($\epsilon \approx 0$) and a contribution η_d from frequencies around the drive frequency ($\epsilon \approx \omega_d$). The first contribution can be identified as

$$\eta_< = -2g_c \int_\epsilon \left(\frac{1}{F_\epsilon^{b,\text{th}}} - F_\epsilon^{b,\text{th}} \right) \text{Im } D_\epsilon^R \text{Im } G_\epsilon^{R,(1)}. \quad (19.279)$$

As $F_\epsilon^{b,\text{th}}$ goes to unity for $\epsilon \gg T_b$, the integrand vanishes in this region and therefore only captures behavior near the FS. In the frequency region around the FS, $\text{Im } G_\epsilon^{R,(1)}$ can also be replaced with $\text{Im } G_\epsilon^{R,(0)} = -\pi \delta_\epsilon$, such that one indeed recovers $\eta^< = \eta^{(1)}$.

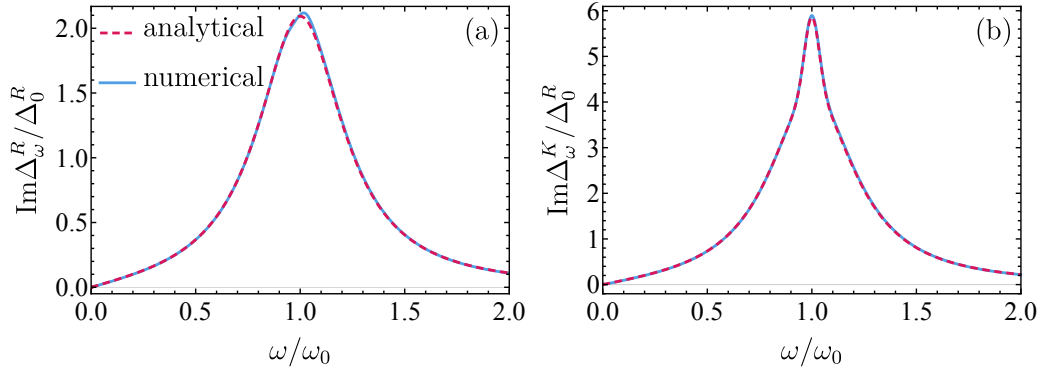


Figure 19.7: Imaginary part of the (a) retarded and (b) Keldysh components of the anomalous self-energies for incoherent driving of the bosons with strength $A = 0.4$. The converged self-consistent numerical solution of eq. (19.26) is labeled “numerical” while the approximation in eq. (19.299) is denoted as “analytical”. The other parameters are set to the ones used in Fig. 19.2(b).

Considering now the contribution from the frequency region around ω_d , we find

$$\eta_d = -2g_c \int_{\epsilon} \left[\left(\frac{1}{\text{sign}(\epsilon) + F_{\epsilon}^{b,d}} - \text{sign}(\epsilon) - F_{\epsilon}^{b,d} \right) \times \text{Im } D_{\epsilon}^R \text{Im } G_{\epsilon}^{R,(1)} \right]. \quad (19.280)$$

Using eqs. (19.276) and (19.277), $\text{Im } G_{\epsilon}^{R,(1)}$ in this integral can be expressed as

$$\text{Im } G_{\epsilon}^{R,(1)} = \frac{2g_c \text{Im } D_{\epsilon}^K}{4\epsilon^2 + (g_c \text{Im } D_{\epsilon}^K)^2}. \quad (19.281)$$

Moreover, in weak-coupling regime and in the region where $\epsilon \sim \omega_d \gg g_c$, the g_c -dependent term in the denominator of eq. (19.281) can be neglected. With this approximation, eq. (19.280) becomes

$$\eta_d \approx g_c^2 I_{\eta}, \quad (19.282)$$

with the coupling-independent constant

$$\begin{aligned} I_{\eta} &= \int_0^{\infty} \frac{d\epsilon}{\pi} \left(1 + F_{\epsilon}^{b,d} - \frac{1}{1 + F_{\epsilon}^{b,d}} \right) \frac{\text{Im } D_{\epsilon}^R \text{Im } D_{\epsilon}^K}{\epsilon^2} \\ &= 2 \int_0^{\infty} \frac{d\epsilon}{\pi} \left[2F_{\epsilon}^{b,d} + (F_{\epsilon}^{b,d})^2 \right] \frac{(\text{Im } D_{\epsilon}^R)^2}{\epsilon^2}, \end{aligned} \quad (19.283)$$

which, given the expression eq. (19.48) for $F_{\epsilon}^{b,d}$, vanishes for $A = 0$ as expected. The total electron spectral width in the weak-coupling regime is then given by the sum of the two contributions

$$\eta \approx \eta^{(1)} + \eta_d = g_c (\gamma T_b + g_c I_{\eta}). \quad (19.284)$$

Effective temperature T_e

From the zeroth-order result in eq. (19.275) it is clear that the electrons inherit the thermal background of the bosons. As discussed in eqs. (19.5) and (19.39), the slope of the distribution

near the FS determines the effective temperature T_e . Although the drive acts at high energies, interactions propagate its influence to lower energies, leading to a drive-induced modification of T_e . To capture this effect we consider the derivative of the defining equation for the distribution function in eq. (19.141) at the FS

$$\begin{aligned} \partial_\omega \int_\epsilon [F_{\omega-\epsilon}^b F_\epsilon^e + 1 - F_\omega^e (F_\epsilon^e + F_{\omega-\epsilon}^b)] \\ \times \text{Im } D_{\omega-\epsilon}^R \text{Im } G_\epsilon^R \Big|_{\omega=0} = 0. \end{aligned} \quad (19.285)$$

Mathematically, the interactions propagate the effects of the drive down to the FS due to the convolution structure of eq. (19.285) and the contribution of the frequency integral around the drive frequency ω_d . The finite value of κ makes $\text{Im } D_\omega^R$ a slowly changing function in the weak-coupling regime, such that its variation can be neglected compared to that of the thermal contribution near the FS. Additionally, the change of electron temperature is going to be a small change on the integrated distributions in eq. (19.285) that depend only on ϵ (as g_c is small). It is thus only in $F_\omega^e \approx \omega/2T_e$ where the effective electron temperature must be included. One can now follow the same approach as for η , namely, using eqs. (19.275) and (19.277) instead of the fully self-consistent electron propagators. With these approximations eq. (19.285) takes the form

$$\begin{aligned} \int_\epsilon \left[\frac{\partial_\omega F_{\omega-\epsilon}^b \Big|_{\omega=0}}{F_\epsilon^b} - F_0^e \partial_\omega F_{\omega-\epsilon}^b \Big|_{\omega=0} \right. \\ \left. - \frac{1}{2T_e} \left(\frac{1}{F_\epsilon^b} - F_\epsilon^b \right) \right] \text{Im } D_\epsilon^R \text{Im } G_\epsilon^{R,(1)} \approx 0. \end{aligned} \quad (19.286)$$

As F_ϵ^e is anti-symmetric around the FS, the second term vanishes and the last term can immediately be identified as being proportional to η in eq. (19.278), giving the result

$$I_0 + \frac{\eta}{4T_e g_c} = 0, \quad (19.287)$$

with

$$I_0 = \int_\epsilon \frac{\partial_\omega F_{\omega-\epsilon}^b}{F_\epsilon^b} \text{Im } D_\epsilon^R \text{Im } G_\epsilon^{R,(1)} \Big|_{\omega=0}. \quad (19.288)$$

The region where $F_{\omega-\epsilon}^b$ has the largest gradient is for $\epsilon \lesssim T_b$, where it diverges as $2T_b/\epsilon$. The large energy separation between the two terms in F_ω^b thus allows one to neglect the large energy contribution to I_0 and to approximate it as

$$\begin{aligned} I_0 &\approx \int_{-\lambda}^\lambda \frac{d\epsilon}{2\pi} \frac{\partial_\omega \coth\left(\frac{\omega-\epsilon}{2T_b}\right)}{\coth\left(\frac{\epsilon}{2T_b}\right)} \text{Im } D_\epsilon^R \text{Im } G_\epsilon^{R,(1)} \Big|_{\omega=0} \\ &= -\frac{1}{T_b} \int_{-\lambda}^\lambda \frac{d\epsilon}{2\pi} \text{csch}\left(\frac{\epsilon}{T_b}\right) \text{Im } D_\epsilon^R \text{Im } G_\epsilon^{R,(1)}, \end{aligned} \quad (19.289)$$

where λ is a suitably chosen cutoff $T_b \ll \lambda \ll \omega_d$. Being in the weak-coupling regime and only considering $\epsilon \leq \lambda$, the broadening $\eta^{(1)}$ is so small that it can again be approximated as infinitesimal, meaning that one can replace $\text{Im } G_\epsilon^{R,(1)} \rightarrow \text{Im } G_\epsilon^{R,(0)} = -\pi\delta_\epsilon$. As the linear behavior of $\text{Im } D_\epsilon^R$ cancels the divergence in $\text{csch}(\epsilon/T_b)$, the integral is finite, giving the following approximate closed-form expression for I_0

$$I_0 \approx -\frac{\omega_0 \kappa}{2(\kappa^2 + \omega_0^2)^2} = -\frac{\gamma}{4}. \quad (19.290)$$

Inserting this in eq. (19.287), the approximation for the effective electron temperature can be written as

$$T_e \approx \frac{\eta}{g_c \gamma} = \frac{T_b \eta}{\eta^{(1)}} = T_b \left(1 + \frac{g_c I_\eta}{\gamma T_b} \right). \quad (19.291)$$

Combining this result with eq. (19.275), the electron distribution can be approximated beyond the zeroth order as the inverse of the boson distribution, but with the thermal background at the temperature T_e

$$F_\epsilon^e \approx \frac{1}{F_{T_e, \omega}^{b, \text{th}} + F_\omega^{b, d}}. \quad (19.292)$$

Non-thermal pairing parameter α_{nth}

Having consistently included the effect of drive in both the spectral width and the effective temperature, one can use these to compute α_{nth} . One starts by inserting the approximation (19.292) for the electron distribution into the eq. (19.268) for α_{nth} and gets to the integral

$$\alpha_{\text{nth}} = 2g_c \int_\epsilon \text{Im } D_\epsilon^R \left\{ 2 \left(\frac{F_{T_b, \epsilon}^{b, \text{th}} + F_\epsilon^{b, d}}{F_{T_e, \omega}^{b, \text{th}} + F_\omega^{b, d}} - 1 \right) \text{Im} \left[(G_\epsilon^R)^2 \frac{\Delta_\epsilon^R}{\Delta_0^R} \right] + (F_{T_b, \epsilon}^{b, \text{th}} + F_\epsilon^{b, d}) \left[\frac{\text{Im } \Delta_\epsilon^K}{\Delta_0^R} - \frac{2 \text{Im } \Delta_\epsilon^R}{(F_{T_e, \epsilon}^{b, \text{th}} + F_\epsilon^{b, d}) \Delta_0^R} \right] \right\} |G_\epsilon^R| \quad (19.293)$$

Using the energy separation between $F_{T_e, \omega}^{b, \text{th}}$ and $F_\omega^{b, d}$, together with the approximated behavior of the electron propagator near the FS, one can define a low-energy contribution to α_{nth}

$$\alpha_{<} = 2g_c \int_\epsilon \text{Im } D_\epsilon^R \left\{ 2 \left(\frac{F_{T_b, \epsilon}^{b, \text{th}}}{F_{T_e, \epsilon}^{b, \text{th}}} - 1 \right) \text{Im} \left[(G_\epsilon^{R, (2)})^2 \frac{\Delta_\epsilon^R}{\Delta_0^R} \right] + F_{T_b, \epsilon}^{b, \text{th}} \left(\frac{\text{Im } \Delta_\epsilon^K}{\Delta_0^R} - \frac{2}{F_{T_b, \epsilon}^{b, \text{th}}} \frac{\text{Im } \Delta_\epsilon^R}{\Delta_0^R} \right) |G_\epsilon^{R, (2)}|^2 \right\}, \quad (19.294)$$

where $G_\omega^{R, (2)} = (\epsilon + i\eta)^{-1}$ ensures that the integrand vanishes for large energy, allowing us to extend the integral to the full energy axis. Similarly to what was found for $\eta_{<}$ in eq. (19.279), one finds that $\alpha_{<}$ has a functional form analogous to the temperature-bias one in eq. (19.270):

$$\alpha_{<} \approx g_c \gamma \left[\left(\frac{T_b}{T_e} - 1 \right) - \frac{T_b \alpha_{\text{nth}}}{\eta} \right]. \quad (19.295)$$

The contribution to the non-thermal pairing parameter due to the incoherent drive is instead captured by the region of the integral around $\epsilon \approx \omega_d$

$$\begin{aligned} \alpha_d &= 2g_c \int_\epsilon \text{Im } D_\epsilon^R \left[2 \left(\frac{\text{sign}(\epsilon) + F_\epsilon^{b, d}}{\text{sign}(\epsilon) + F_\omega^{b, d}} - 1 \right) \text{Im} \left[(G_\epsilon^R)^2 \frac{\Delta_\epsilon^R}{\Delta_0^R} \right] + (\text{sign}(\epsilon) + F_\epsilon^{b, d}) \left(\frac{\text{Im } \Delta_\epsilon^K}{\Delta_0^R} - \frac{2 \text{Im } \Delta_\epsilon^R}{(\text{sign}(\epsilon) + F_\epsilon^{b, d}) \Delta_0^R} \right) \right] |G_\epsilon^R| \\ &= 2g_c \int_\epsilon \text{Im } D_\epsilon^R \left[(\text{sign}(\epsilon) + F_\epsilon^{b, d}) \frac{\text{Im } \Delta_\epsilon^K}{\Delta_0^R} - 2 \frac{\text{Im } \Delta_\epsilon^R}{\Delta_0^R} \right] |G_\epsilon^R|^2, \end{aligned} \quad (19.296)$$

To compute this contribution, one needs approximations for the behavior of the retarded and Keldysh anomalous self-energies near ω_d . These approximation must be obtained from their one-loop definition in Eqs. (19.26). To this end, the crudest approximation one can perform is that the anomalous propagators $\mathcal{F}$ in Eqs. (19.26) are strongly peaked at the FS, so that the bosonic propagators can be taken out of the convolution similarly to what was done in

section 19.5. For the retarded anomalous self-energy, this gives

$$\begin{aligned}\Delta_\omega^R &= ig \int_\epsilon (D_{\omega-\epsilon}^R \mathcal{F}_\epsilon^K + D_{\omega-\epsilon}^K \mathcal{F}_\epsilon^R) \\ &\approx ig \left(D_\omega^R \int_\epsilon \mathcal{F}_\epsilon^K + D_\omega^K \int_\epsilon \mathcal{F}_\epsilon^R \right) \\ &= ig D_\omega^R \int_\epsilon \mathcal{F}_\epsilon^K.\end{aligned}\tag{19.297}$$

where the equal-time limit in eq. (19.32) was used in the last line. Equivalently, the behavior of the anomalous Keldysh self-energy around $\omega = \omega_d$ can be approximated as

$$\begin{aligned}\Delta_\omega^K &= ig \int_\epsilon (D_{\omega-\epsilon}^R \mathcal{F}_\epsilon^R + D_{\omega-\epsilon}^A \mathcal{F}_\epsilon^A + D_{\omega-\epsilon}^K \mathcal{F}_\epsilon^K) \\ &\approx ig \left(D_\omega^R \int_\epsilon \mathcal{F}_\epsilon^R + D_\omega^A \int_\epsilon \mathcal{F}_\epsilon^A + D_\omega^K \int_\epsilon \mathcal{F}_\epsilon^K \right) \\ &= ig D_\omega^K \int_\epsilon \mathcal{F}_\epsilon^K.\end{aligned}\tag{19.298}$$

Using eqs. (19.297) and (19.298) we find the approximations

$$\begin{aligned}\left. \frac{\text{Im } \Delta_\omega^R}{\Delta_0^R} \right|_{\omega \approx \omega_d} &\approx \frac{\text{Im } D_\omega^R}{D_0^R}, \\ \left. \frac{\text{Im } \Delta_\omega^K}{\Delta_0^K} \right|_{\omega \approx \omega_d} &\approx \frac{\text{Im } D_\omega^K}{D_0^K} = \frac{2(\text{sign}(\epsilon) + F_\epsilon^{b,d}) \text{Im } D_\omega^R}{D_0^R}.\end{aligned}\tag{19.299}$$

These approximations are compared with the numerical solutions in fig. 19.7, confirming that they work quite well in the weak-coupling regime¹³. As the approximations (19.299) are used under an integral and multiplied by smooth functions, small deviations do not have a significant impact on the value of the integral.

Inserting the approximations (19.299) into eq. (19.296), together with the approximation that the $\omega \approx \omega_d$ behavior of $|G_\epsilon^R|^2 \approx \epsilon^{-2}$, one obtains an approximation for α_d of the form

$$\begin{aligned}\alpha_d &\approx \frac{4g_c}{D_0^R} \int_\epsilon \frac{(\text{Im } D_\epsilon^R)^2}{\epsilon^2} \left[(\text{sign}(\epsilon) + F_\epsilon^{b,d})^2 - 1 \right] \\ &= \frac{4g_c}{D_0^R} \int_\epsilon \frac{(\text{Im } D_\epsilon^R)^2}{\epsilon^2} \left[2 \text{sign}(\epsilon) F_\epsilon^{b,d} + (F_\epsilon^{b,d})^2 \right] \\ &= \frac{2g_c}{D_0^R} I_\eta,\end{aligned}\tag{19.300}$$

where in the last line eq. (19.283) was used to rewrite the result. Collecting the results of eqs. (19.295) and (19.300) one can write the full weak-coupling approximation for α_{nth}

$$\begin{aligned}\alpha_{\text{nth}} &\approx \alpha_< + \alpha_d, \\ &\approx g_c \gamma \left[\left(\frac{T_b}{T_e} - 1 \right) - \frac{T_b \alpha_{\text{nth}}}{\eta} \right] + \frac{2g_c}{D_0^R} I_\eta \\ &= \frac{g_c}{1 + \frac{g_c \gamma T_b}{\eta}} \left[\gamma \left(\frac{T_b}{T_e} - 1 \right) + \frac{2I_\eta}{D_0^R} \right].\end{aligned}\tag{19.301}$$

¹³We note that a similar approximation for Δ_ω^R is also found at zeroth order in the weak-coupling limit of standard Eliashberg theory at thermal equilibrium, see Ref. [61].

Closed-form expression for the critical coupling g_c

Having derived expressions for η in eq. (19.284), T_e in eq. (19.291) and α_{nth} in eq. (19.301), these can now be substituted into the expression for the critical coupling in eq. (19.1) to obtain

$$\begin{aligned}
 g_c &= \left[\frac{D_0^R}{2} \left(\frac{\alpha_{\text{nth}}}{\eta} - \frac{1}{T_e} \right) \right]^{-1} \\
 &\approx \frac{2(g_c I_\eta + \gamma T_b)}{D_0^R \left[\frac{\frac{2I_\eta}{D_0^R} + \gamma \left(\frac{T_b \gamma}{g_c I_\eta + \gamma T_b} - 1 \right)}{1 + \frac{T_b \gamma}{g_c I_\eta + \gamma T_b}} - \gamma \right]} \\
 &= \frac{2(g_c I_\eta + \gamma T_b)}{D_0^R \frac{2(I_\eta - D_0^R \gamma)(g_c I_\eta + \gamma T_b)}{D_0^R (g_c I_\eta + 2\gamma T_b)}} \\
 &= \frac{g_c I_\eta + 2\gamma T_b}{I_\eta - D_0^R \gamma},
 \end{aligned} \tag{19.302}$$

Isolating g_c , one arrives at the self-consistent analytical approximation

$$g_c \approx -\frac{2T_b}{D_0^R} = g_{c,\text{th}}(T_b). \tag{19.303}$$

19.8.12 Quasi-classical approximation and momentum averaging

In most common implementations of Eliashberg theory for phonon-mediated superconductivity, the phonon-electron interaction is not forward-scattering as in eq. (19.12), but rather has a smooth momentum dependence. Considering a more generic momentum-dependent interaction, the linearized version of the NESS-Eliashberg equation for Δ^R in eq. (19.133) would generalize to the form

$$\begin{aligned}
 \Delta_{\mathbf{k},\omega}^R &= i \int_{\mathbf{k}'} \int_{\omega'} g_{\mathbf{k},\mathbf{k}'} \left[D^R G^K G_-^R \Delta_{\mathbf{k}',\omega'}^A \right. \\
 &\quad + (D^K G^R G_-^A + D^R G^R G_-^R) \Delta_{\mathbf{k}',\omega'}^R \\
 &\quad \left. + D^K G^R G_-^R \Delta_{\mathbf{k}',\omega'}^K \right],
 \end{aligned} \tag{19.304}$$

where $D^\alpha G^\beta G_-^\gamma = D_{\mathbf{k}-\mathbf{k}',\omega-\omega'}^\alpha G_{\mathbf{k}',\omega'}^\beta G_{-\mathbf{k}',-\omega'}^\gamma$ and appropriate dimensionality constants have been reabsorbed into the coupling $g_{\mathbf{k},\mathbf{k}'}$.

To simplify the calculation, the momentum dependence of the coupling $g_{\mathbf{k},\mathbf{k}'}$ and of the boson propagators $D_{\mathbf{k}-\mathbf{k}',\omega-\omega'}^\alpha$ is often dropped, and this in turn makes the anomalous self-energy momentum independent, i.e. $\Delta_{\mathbf{k},\omega} \equiv \Delta_\omega$ [23]. This corresponds to the so-called quasi-classical approximation [77], which is usually applicable in the regime $T_c \ll \omega_0 \ll \mu$, where T_c is the critical temperature, ω_0 is a typical phonon energy scale and μ is the chemical potential [23, 76]. Within this approximation, the only momentum dependent quantities in eq. (19.304) are the electron propagators G . One can then perform the momentum integral by rewriting it as an integral over the energy

$$\int_{\mathbf{k}'} G_{\mathbf{k}',\omega'}^\alpha G_{-\mathbf{k}',-\omega'}^\beta = \int_{-\varepsilon_{\min}}^{\varepsilon_{\max}} d\varepsilon \nu(\varepsilon) G_{\varepsilon,\omega'}^\alpha G_{\varepsilon,-\omega'}^\beta, \tag{19.305}$$

where $\nu(\varepsilon)$ is the density of states at the energy ε , and $\varepsilon_{\min}$ and $\varepsilon_{\max}$ are some energy cutoffs of the electronic bands. The integral is typically estimated by approximating the density of states

$\nu(\varepsilon)$ with its value at the FS ν_F , by assuming particle-hole symmetry ($\varepsilon_{\min} = -\varepsilon_{\max}$) and by taking the limit for $\varepsilon_{\max} \rightarrow \infty$ [23]. The final integral to be computed is of the form

$$\int_{-\infty}^{\infty} d\varepsilon G_{\varepsilon, \omega'}^{\alpha} G_{\varepsilon, -\omega'}^{\beta}. \quad (19.306)$$

Focusing on the combination $G^R G_-^R$, which appears in eq. (19.133) multiplying Δ^K , and replacing the electron propagator with its quasi-particle approximation $G_{\varepsilon, \omega}^R = 1/(\omega - \varepsilon + i\eta)$, the integral becomes

$$\int_{-\infty}^{\infty} d\varepsilon G_{\varepsilon, \omega'}^R G_{\varepsilon, -\omega'}^R = \int_{-\infty}^{\infty} d\varepsilon \frac{1}{(\varepsilon - i\eta)^2 - \omega'^2} = 0. \quad (19.307)$$

This implies that, within the quasi-classical approximation, Δ^R and Δ^K are essentially decoupled at the phase transition and any non-thermal contribution to the critical coupling coming from Δ^K , like the α_{nth} term in eq. (19.1), is going to be suppressed. On the other hand, for any model where the momentum structure of the coupling function $g_{\mathbf{k}, \mathbf{k}'} D_{\mathbf{k}-\mathbf{k}', \omega-\omega'}^R$ (and subsequently of the order parameter) is important, like in our forward-scattering interaction model of eq. (19.12), Δ^R and Δ^K remain coupled and a non-thermal contribution from Δ^K always appears.

20 Nonequilibrium hysteretic phase transitions in periodically light-driven superconductors by Zhang, Takasan, Tsuji

(it is unknown, what are the baths, with which temperature?)
(also, nothing is said about the number of baths)

Abstract

We find nonequilibrium phase transitions accompanied by multiple (nested) hysteresis behaviors in superconductors coupled to baths (how many baths?) under a time-periodic light driving.

The transitions are demonstrated with a full phase diagram in the domain of the driving amplitude and frequency by means of the Floquet many-body theory.

In the weak driving regime with a frequency smaller than half of the superconducting gap, excited quasiparticles are accumulated at the far edges of the bands (how is it?) , realizing a distribution reminiscent of the Eliashberg effect (what is it?) , which suddenly becomes unstable (how is it?) in the strong driving regime due to multi-photon-assisted tunneling across the gap mediated by the in-gap Floquet sidebands. (nothing is clear, what is going on?)

We also show that superconductivity is enhanced in the weak driving regime without effective cooling, which is attributed to the modulation of the spectrum due to Floquet sidebands.

20.1 Theory

Introduction.

Driving quantum phases of matter with light has been a promising route to control electronic states of materials in a nonthermal way [Aoki *et al.*(2014), Oka and Kitamura(2019), de la Torre *et al.*(2021), Murakami *et al.*(2023)] (why do we need to control them and how?? yes, we can

put with light electrons in a nonthermal distribution, so what?) , which may provide various possibilities of phase transitions(which phase transitions?) that are otherwise elusive to access within thermal equilibrium.

This is of particular interest in the context of superconductivity, since thermal phase transitions to superconducting states have been limited, to date, to low temperatures at ambient pressure (and it was not a problem at all, of course except ideas of creating high-Tc superconductors, but it is irrelevant to this topics) . In fact, recent progress of intense laser techniques has allowed one to monitor real-time dynamics of superconducting states in a fast timescale (see, e.g., [Giannetti *et al.*(2016), Shimano and Tsuji(2020), Demsar(2020)] (read these papers!)) and observe light-induced superconducting-like states [Fausti *et al.*(2011), Kaiser *et al.*(2014), Hu *et al.*(2014), Mitrano *et al.*(2016), Niwa *et al.*(2019), Zhang *et al.*(2020), Buzzi *et al.*(2020), Budden *et al.*(2021), Isoyama *et al.*(2021), Katsumi *et al.*(2023), Fava *et al.*(2024), Nishida *et al.*(2024)](read these papers!) , which may provide an opportunity of finding novel nonequilibrium phase transitions (which nonequilibrium phase transitions, for example?) in light-driven superconductors.

During irradiation of multi-cycle pump light, the system is expected to approach a Floquet state [Shirley(1965), Sambe(1973), Tsuji(2024)] (i.e., a time-periodic steady state) (what is it?) that may accommodate external-field-dressed quasiparticles, which have been observed as Floquet-Bloch bands in time-resolved angle-resolved photoemission spectroscopy (tr-ARPES) experiments [Wang *et al.*(2013), Mahmood *et al.*(2016), Ito *et al.*(2023), Zhou *et al.*(2023), Merboldt *et al.*(2024), Choi *et al.*(2024)].

Signatures of Floquet states (what are they?) have also been glimpsed in transport [McIver *et al.*(2020), Liu *et al.*(2025)], transient absorption spectroscopy [Sie *et al.*(2015), Kobayashi *et al.*(2023)], nonlinear optical response [Shan *et al.*(2021)], and tunneling spectroscopy [Park *et al.*(2022), Haxell *et al.*(2023)] measurements.

When the frequency Ω of the driving field is much larger than any other relevant energy scales, the system is well described by an effectively renormalized Hamiltonian, leading to the Floquet engineering [Bukov *et al.*(2015), Oka and Kitamura(2019), Weitenberg and Simonet(2021)]. (what is the Floquet engineering?)

This effect can be used, for example, to enhance superconductivity by suppressing hopping amplitudes [Ido *et al.*(2017), Dasari and Eckstein(2018)] (how is it physically? sounds like they are just different concepts?) .

On the other hand, it is more complicated to understand the behavior of superconducting states at low-frequency regimes. (why? why at all nothing is written about Gorkov's and other workers about it? didn't they describe these systems already?)

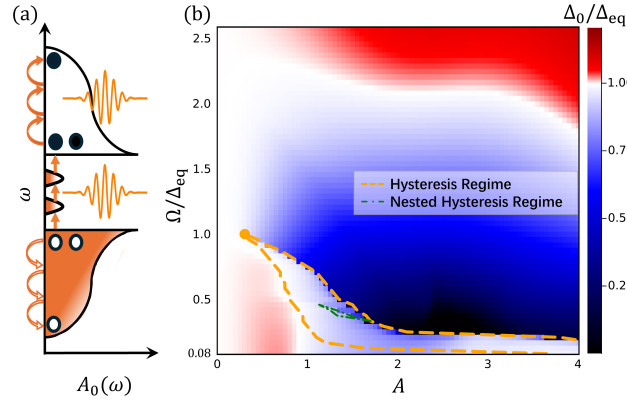


Figure 20.1: (a) Schematic picture of quasiparticles (holes) in superconductors driven by light shown by filled (empty) circles (and what is going on here? some strange orange arrows...) . Intraband scattering leads to an effective cooling of the system (Eliashberg effect), while there can occur multi-photon-assisted tunneling across the gap through Floquet sidebands (small peaks inside the gap) (I don't know any of these effects, what are they?) . (b) Phase diagram of a BCS superconductor coupled to baths under a time-periodic light driving in the domain of the driving amplitude A and frequency Ω . The color code plots the time-averaged superconducting order parameter Δ_0 rescaled by the equilibrium one Δ_{eq} . The area surrounded by the orange (green) curve indicates the (nested) hysteresis regime. The phase diagram is computed with $U = 6.0$, $\Gamma = 0.1$, and $T = 0.2$, which gives $\Delta_{\text{eq}} \approx 2.4$. (see the text to understand what is meant here?)

When Ω becomes much smaller than the superconducting gap 2Δ , thermally excited quasiparticles/holes are kicked away from the gap region (see Fig. 20.1(a)), resulting in an effective cooling of the quasiparticle distribution that may enhance superconductivity. This is known as the Eliashberg effect [Eliashberg(1970), Ivlev and Eliashberg(1971), Demsar(2020), Ivlev *et al.*(1973)], which has been observed in early microwave driving experiments [Wyatt *et al.*(1966), Dayem and Wiegand(1967)]. While this mechanism is plausible, the effect of the modulation of the spectral function, which is beyond the quasithermal description [de la Torre *et al.*(2021)] but will be important for large driving amplitudes with sub-gap frequencies (not too far smaller than 2Δ), has not been considered before. Furthermore, the realized steady-state distribution of quasiparticles might be unstable: The kicked quasiparticles (holes) are accumulated at the far edge of the upper (lower) band, which causes significant population inversion (Fig. 20.1(a)). It has not been clear how the quasiparticle distribution collapses if one increases the driving amplitude beyond the perturbative regime, and whether there exists a nonequilibrium phase transition in between the weak- and strong-driving regimes.

In this Letter, we revisit the problem of light-driven superconductors at sub-gap frequencies, and address the issue of the ultimate fate of the Eliashberg-type distribution of quasiparticles.

To this end, we solve nonequilibrium steady states (NESS) for a model of light-driven BCS superconductors coupled to baths (how many baths?) by means of the Floquet Green's function method [Faisal(1989), Martinez(2003), Tsuji *et al.*(2008), Tsuji *et al.*(2009), Aoki *et al.*(2014)] (read these papers, I don't know any of them? Never heard about Floquet Green's function method) , which can take account of both the effects of nonequilibrium distributions and modulated band structures.

The obtained phase diagram is shown in Fig. 20.1(b). We find multiple (nested) hysteretic phase transitions between the weak and strong driving regimes, which indicate characteristic features of the instability of the Eliashberg-type distribution (what is the Eliashberg-type distribution?) that collapses due to multi-photon-assisted tunneling through in-gap Floquet

sidebands (Fig. 20.1(a)). Moreover, in the weak driving regime we show that superconductivity can be enhanced without effective cooling, which is attributed to the light-induced modulation of the spectrum due to Floquet sidebands (for other mechanisms of light-induced enhancement of superconductivity, see, e.g., Refs. [Tsuji *et al.*(2011), Denny *et al.*(2015), Sentef *et al.*(2016), Okamoto *et al.*(2016), Komnik and Thorwart(2016), Kennes *et al.*(2017), Sentef(2017), Babadi *et al.*(2017), Murakami *et al.*(2017), Coulthard *et al.*(2017), Mazza and Georges(2017), Ido *et al.*(2017), Nava *et al.*(2018), Werner *et al.*(2018), Fabrizio(2018), Dasari and Eckstein(2018), Kaneko *et al.*(2019), Bittner *et al.*(2019), Werner *et al.*(2019a), Werner *et al.*(2019b), Ray *et al.*(2023), Eckhardt *et al.*(2024), Ray and Werner(2024), Wang *et al.*(2024)]). (I'll read some of them, since I don't know anything about light-induced enhancement of superconductivity)

Model and Formalism.

We consider a BCS superconductor on a square lattice illuminated by external continuous-wave light, whose Hamiltonian is given by

$$H_{\text{BCS}}(t) = \sum_{\mathbf{k}, \sigma} \varepsilon_{\mathbf{k} - \frac{e}{\hbar} \mathbf{A}(t)} c_{\mathbf{k}\sigma}^\dagger c_{\mathbf{k}\sigma} - \frac{U}{N} \sum_{\mathbf{k}, \mathbf{p}} c_{\mathbf{k}\uparrow}^\dagger c_{-\mathbf{k}\downarrow}^\dagger c_{-\mathbf{p}\downarrow} c_{\mathbf{p}\uparrow}, \quad (20.1)$$

where $c_{\mathbf{k}\sigma}$ and $c_{\mathbf{k}\sigma}^\dagger$ stand for the electron annihilation and creation operators with momentum $\mathbf{k}$ and spin $\sigma \in \{\uparrow, \downarrow\}$, $U(> 0)$ is the pairing strength between electrons, and N is the number of lattice sites.

The band dispersion is given by $\varepsilon_{\mathbf{k}} = -2t_{\text{hop}} [\cos(k_1 a) + \cos(k_2 a)]$, in which the vector potential $\mathbf{A}(t) = A \cos(\Omega t) \mathbf{e}_p$ is introduced via the Peierls substitution ($\mathbf{k} \rightarrow \mathbf{k} - \frac{e}{\hbar} \mathbf{A}(t)$, e is the electric charge) [Peierls(1933)] (I don't think that it is related to Peierls substitution, but I'll check. this is just a standard way to introduce it) .

The system is set at half filling with the chemical potential $\mu = 0$. We take the polarization along the diagonal direction, $\mathbf{e}_p = \frac{1}{\sqrt{2}}(1, 1)$, and A and Ω represent the driving amplitude and frequency, respectively. To study the superconducting order, we define the gap function $\Delta(t) = -\frac{U}{N} \sum_{\mathbf{k}} \langle c_{-\mathbf{k}\downarrow}(t) c_{\mathbf{k}\uparrow}(t) \rangle$ to decouple the interaction term. We use the units with $\hbar = 1$, $e = 1$, $t_{\text{hop}} = 1$, and the lattice constant $a = 1$.

We assume that the superconductor is coupled to heat baths, and that the system arrives at a time-periodic steady state in the long-time limit where energy injected from the external driving field is balanced with dissipation. For a model of heat baths, we employ the free-fermion bath [Büttiker(1985), Tsuji *et al.*(2009)] (important papers, read them!) with temperature T and the system-bath coupling Γ .

The bath's degrees of freedom can be analytically integrated out, and the resulting effect of the bath can be incorporated in the self-energy, whose retarded component is given by $\Sigma_{\mathbf{k}}^R(\omega) = -i\Gamma$, and the lesser component is given by the fluctuation-dissipation theorem, $\Sigma_{\mathbf{k}}^<(\omega) = -2if_T^{\text{FD}}(\omega)\text{Im}\Sigma_{\mathbf{k}}^R(\omega)$. (is FDT about it?? I haven't seen it in terms of Σ , where is it derived?)

Here, $f_T^{\text{FD}}(\omega)$ is the Fermi-Dirac distribution at temperature T .

In this setup (are we doing some steps?) , there is only an energy exchange without net particle current between the baths and the superconductor (i.e., nonequilibrium superconductivity in an energy mode [Demsar(2020), Tinkham(2004)]) (why no exchange of particles?) .

This is to be distinguished from the previous study [Yang *et al.*(2021)] where superconductors are driven by the chemical potential change (i.e., charge mode [Demsar(2020), Tinkham(2004)]). (also, it is interesting to read it)

The single-particle Green's function is then derived from the Dyson-Keldysh equation [Jauho *et al.*(1994), Haug *et al.*(2008), Eckstein(2018)] (read the papers!) extended to the Nambu space,

$$(G_{\mathbf{k}}^R)^{-1} = (G_{0\mathbf{k}}^R)^{-1} - \Sigma_{\mathbf{k}}^R, \quad (20.2)$$

$$G_{\mathbf{k}}^< = G_{\mathbf{k}}^R * \Sigma_{\mathbf{k}}^< * G_{\mathbf{k}}^A, \quad (20.3)$$

where $G_{0\mathbf{k}}^R$ is the noninteracting retarded Green's function, and $G_{\mathbf{k}}^R$, $G_{\mathbf{k}}^A$, and $G_{\mathbf{k}}^<$ are the full retarded, advanced, and lesser Green's functions, respectively.

Each Green's function takes the form of the matrix in terms of the Floquet and Nambu spaces (Floquet spaces?), and "*" is the matrix multiplication in their product space (I don't think that there is such thing as "product space", it is just a GF of the system) (then in the SM they write that it is also a convolution, so what is it really? see the derivation of it somewhere). The self-consistent equation for the order parameter is given by

$$\Delta(t) = i \frac{U}{2N} \sum_{\mathbf{k}} \text{Tr} (\tau_1 G_{\mathbf{k}}^<(t, t)). \quad (20.4)$$

(how to derive it in this form, through lesser GF??) (there is cutoff assumed)

Here, τ_1 is the Pauli matrix acting on the Nambu space.

In this way, we can take account of arbitrary higher-order harmonics of the gap oscillations self-consistently (is there some meaning of this statement? of course gap is evaluated by the gap equation, but does it meant that there are oscillations in it because of it? I don't think that authors understood this statement).

In practice, we put a cutoff for the Floquet matrix size, for which we have confirmed the convergence of the results numerically. (which cutoff?)

The details of the formalism are explained in Supplemental Material (SM) [foo()]. (I'll read it later)

Results.

In Fig. 20.1(b), we plot the time-averaged (not a steady state) superconducting order parameter Δ_0 rescaled by the equilibrium one Δ_{eq} (shown by the color code) (at which temperatures?) for different amplitudes and frequencies of the driving field.

We first comment on the results in the high-frequency regime briefly. When the frequency Ω is greater than the other energy scales of the system, the steady state is well described by the effective Hamiltonian in the high-frequency expansion [Bukov *et al.*(2015), Eckardt and Anisimovas(2015), Mikami *et al.*(2016), Oka and Kitamura(2019)], the leading term of which is given by the time-averaged one,

$$H^{\text{eff}} = \sum_{\mathbf{k}, \sigma} J_0(A) \varepsilon_{\mathbf{k}} c_{\mathbf{k}\sigma}^\dagger c_{\mathbf{k}\sigma} - \frac{U}{N} \sum_{\mathbf{k}, \mathbf{p}} c_{\mathbf{k}\uparrow}^\dagger c_{-\mathbf{k}\downarrow}^\dagger c_{-\mathbf{p}\downarrow} c_{-\mathbf{p}\uparrow} + O(1/\Omega).$$

Here the kinetic term is renormalized by the zeroth-order Bessel function $J_0(A)$. (why is it so?? read these papers!)

Since $|J_0(A)| < 1$ for any nonzero amplitudes A , this renormalization factor suppresses the hopping amplitude and the bandwidth, leading to dynamical localization when $J_0(A) = 0$ [Dunlap and Kenkre(1986)]. In superconductors, this suppression of the hopping relatively enhances the pairing interaction (with the effective interaction $U_{\text{eff}} = U/|J(A)|$), thus enhancing superconductivity [Ido *et al.*(2017), Dasari and Eckstein(2018)] (see the upper red area of Fig. 20.1(b)). For numerical details, we refer to SM [foo()].

In the low-frequency ($\Omega < \Delta_{\text{eq}}$) and weak-field regime, we can observe another enhanced area in the lower left part of Fig.20.1(b). Such an enhancement of superconductivity with the low-frequency drive is reminiscent of the Eliashberg effect [Wyatt *et al.*(1966), Ivlev *et al.*(1973), Dayem and Wiegand(1967), Eliashberg(1970), Ivlev and Eliashberg(1971), Demsar(2020)] (read, what is it???) , most of the previous arguments of which have been based on the gap equation,

$$\int \frac{d\omega}{\omega} A(\omega)(1 - 2f(\omega)) = \frac{1}{U}, \quad (20.5)$$

with the spectral function $A(\omega)$ and distribution $f(\omega)$.

In the conventional argument, one assumes that the effect of the external field only appears as a perturbation to the Fermi-Dirac distribution, $f(\omega) = f_T^{\text{FD}}(\omega) + \delta f(\omega)$, with the spectral function being fixed to the equilibrium one ($A(\omega) = A_{\text{eq}}(\omega)$), and that the gap equation (20.5) remains to be valid. The perturbation $\delta f(\omega)$ can be separately evaluated by the quantum Boltzmann equation [Chang and Scalapino(1977), de Visser *et al.*(2014), Curtis *et al.*(2019)]. Due to the weight $1/\omega$ in Eq. (20.5), $\delta f(\omega)$ that is negative at lower $\omega (> 0)$ and positive at higher ω is favored for enhancing superconductivity. One can roughly attribute this effect to a shift of an effective temperature defined by $\frac{\delta T}{T} \equiv \int_{-\infty}^{+\infty} \frac{\delta f(\omega)}{\omega} d\omega$. Enhancement of superconductivity may then be expected if δT becomes negative (i.e., effective cooling).

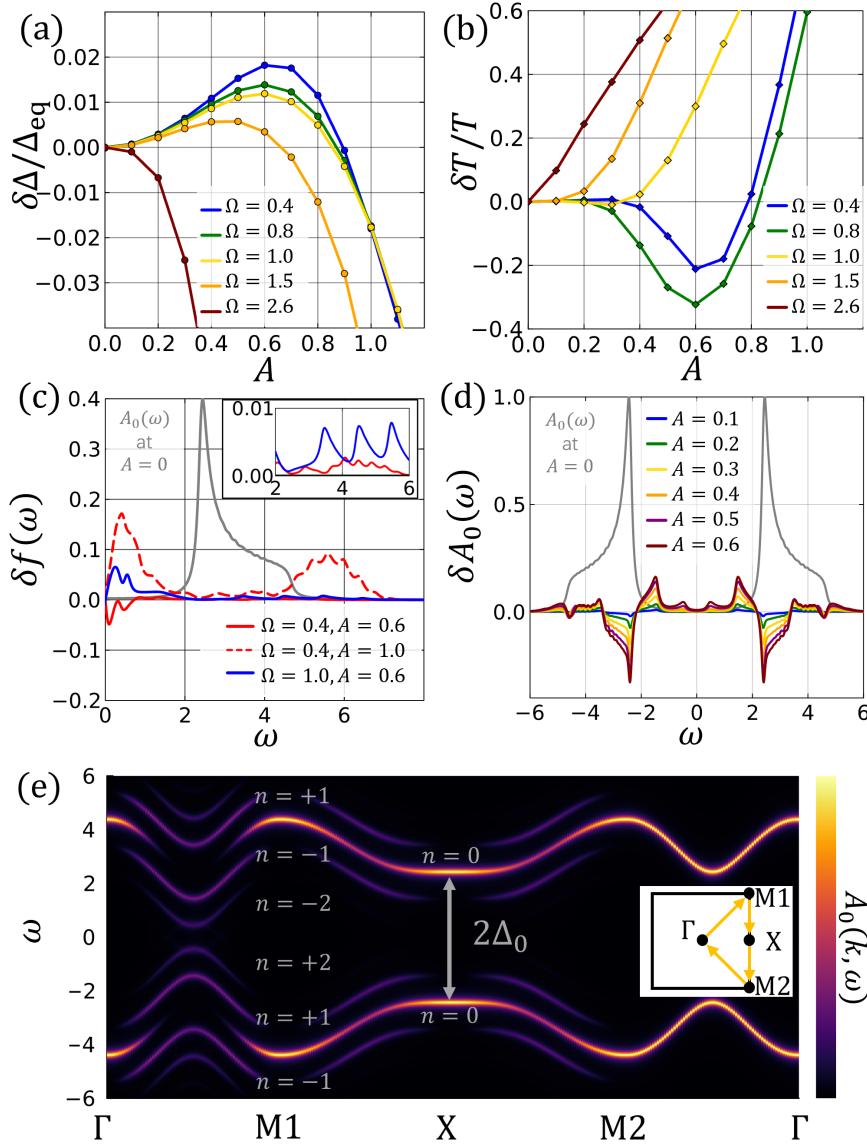


Figure 20.2: Relative changes of (a) the superconducting order parameter $\delta\Delta/\Delta_{\text{eq}}(\approx 2.4)$ and (b) the effective temperature $\delta T/T$ as a function of the driving amplitude A . (c) Changes of the effective distribution $\delta f(\omega)$. The inset shows an enlarged view of $\delta f(\omega)$. The gray solid line shows the time-averaged spectral function $A_0(\omega)$ for $A = 0$, whose maximum is normalized to 0.4. (d) Changes of the spectral function $\delta A_0(\omega)$, which has been normalized to set the maximum of $A_0(\omega)$ at $A = 0$ to 1.0. (e) Momentum-dependent spectral function $A_0(\mathbf{k}, \omega)$ for $\Omega = 1.0$ and $A = 0.6$. The number n marks the Floquet sidebands. The inset shows the path in the Brillouin zone. The other parameters are $U = 6.0$, $\Gamma = 0.1$, and $T = 0.2$.

In our formalism, Eq. (20.5) is not strictly valid due to the strong driving and the coupling to the bath. Instead, we compute the nonequilibrium distribution, the superconducting order parameter, and the spectral function fully self-consistently through Eqs. (20.10)-(20.4). We define the distribution change as $\delta f(\omega) \equiv A_0(\omega)/n_0(\omega) - f_T^{\text{FD}}(\omega)$, where $A_0(\omega)$ and $n_0(\omega)$ are the time-averaged spectral function and occupation function given by the Floquet retarded and lesser Green's functions, respectively [foo()].

In Fig. 20.2(a)-(b), we present the relative change in the time-averaged order parameter $\delta\Delta$ and effective temperature δT across different driving frequencies. The results demonstrate that superconductivity is indeed enhanced for weak A and $\Omega < \Delta_{\text{eq}}(\approx 2.4)$, and such enhancement smoothly turns into suppression as A increases. Figure 20.2(c) provides a detailed view of $\delta f(\omega)$.

Specifically, for $\Omega = 0.4$ and $A = 0.6$, where superconductivity is most strongly enhanced at that frequency, $\delta f(\omega)$ is indeed negative around $\omega = 0$, resulting in negative δT in Fig. 20.2(b). In contrast, at $\Omega = 0.4$ and $A = 1.0$, where suppression of superconductivity happens, $\delta f(\omega)$ becomes entirely positive at $\omega > 0$, thus resulting in positive δT . Although this distribution indicates effective heating, it remains to be of the Eliashberg type in the sense that quasiparticles are accumulation at the far edges of the bands (Fig. 20.2(c)).

A discrepancy between the sign of $\delta\Delta$ and δT arises for $\Omega = 1.0$ and $A = 0.6$, where $\delta f(\omega)$ at $\omega > 0$ and consequently δT are both positive, despite the strongest enhancement of superconductivity occurs. In particular, we do not observe significant negative values of $\delta f(\omega)$ around the position of the coherence peak $\omega \approx \Delta_0$ (see the inset of Fig. 20.2(c)), which is in stark contrast to those obtained from the kinetic equations [de Visser *et al.*(2014), Curtis *et al.*(2019)]. This indicates that enhancement of superconductivity can happen even without effective cooling of quasiparticles, and suggests another mechanism for such enhancement.

We attribute this mechanism to the modulation of the spectral function due to light driving. Let us denote the change of the spectral function by $\delta A_0(\omega)$, which becomes positive around $\omega \sim \Delta_0 - \Omega$, and negative around $\omega \sim \Delta_0$ (Fig. 20.2(d)). These structures roughly correspond to the Floquet sidebands, making replicas of the coherence peak, which can be clearly observed in the momentum-resolved spectral function (Fig. 20.2(e)). If we assume that the distribution remains to be unchanged from $f_{\text{FD}}^T(\omega)$, then $\delta A_0(\omega)$ multiplied by a positive factor $\frac{1-2f_{\text{FD}}^T(\omega)}{\omega}$ peaked at $\omega = 0$ will give a positive contribution to the left-hand side of Eq. (20.5). In order to keep the left-hand side of Eq. (20.5) constant, the coherence peak should be shifted to larger ω , thus enhancing the gap Δ_0 . In actual situations, both the distribution and spectral functions are modulated by periodic driving, and the competition between the two effects will determine the overall change of the superconducting order parameter.

So far, we have demonstrated how the Eliashberg-type enhancement of superconductivity breaks down as the driving strength increases. One might expect the superconducting order to diminish down to zero smoothly. However, this transition is not smooth: In Fig. 20.3(a), we plot Δ_0 as a function of A for different light frequencies Ω . As one can see, there are two branches in the Δ_0 - A curves: The upper branches marked with upper triangles are computed from small to large A , i.e., the initial value of Δ_0 for the self-consistency loop is taken to be the converged value of Δ_0 at smaller A . The lower branches marked with lower triangles are computed in the opposite direction. This discontinuous transition is clearly a non-perturbative effect with respect to A , which cannot be captured by a naive perturbation theory.

When Ω is smaller than Δ_{eq} (≈ 2.4), Δ_0 in the upper branches shows a drastic decline at certain driving amplitudes after the Eliashberg-type enhancement, and merges into the lower branches. The lower branches show similar sudden jumps at different driving amplitudes. For a fixed Ω , the two branches form a hysteresis loop of the superconducting order parameter, indicating a first-order-like phase transition. The hysteresis regions shrink with increasing light frequencies, and completely disappear when the frequency becomes larger than Δ_{eq} (see the curves of $\Omega = 2.5$ in Fig. 20.3(a)). The disappearance of the hysteresis loops at $\Omega \simeq \Delta_{\text{eq}}$ suggests that the two-photon-assisted tunneling triggers the transition. With careful observation, one can see in Fig. 20.3(a) that for each frequency the drastic decline of the two branches happens right before cutting the horizontal dashed line of $\Delta_0 = \Omega$, which meets the two-photon absorption condition since $2\Delta_0$ is the gap size of the superconducting NESS.

One interesting observation here is that such a first-order phase transition can be nested and triggered by higher-order processes, implying a hierarchical structure of nonequilibrium phases with multi-stability. For instance, the upper branch of the loop with $\Omega = 1.0$ in Fig. 20.3(a)

has another steep decline around $A = 1.3$, marked with the red arrow. If we sweep A from smaller to larger values and vice versa within the upper branch around $A = 1.3$, we find a nested hysteresis loop (Fig. 20.3(b)). This secondary hysteresis loop is intersected by the horizontal line of $\Delta_0 = 2.0$ (Fig. 20.3(b)), indicating that the transition is induced by a four-photon-assisted excitation process that requires $2\Delta_0 = 4\Omega$.

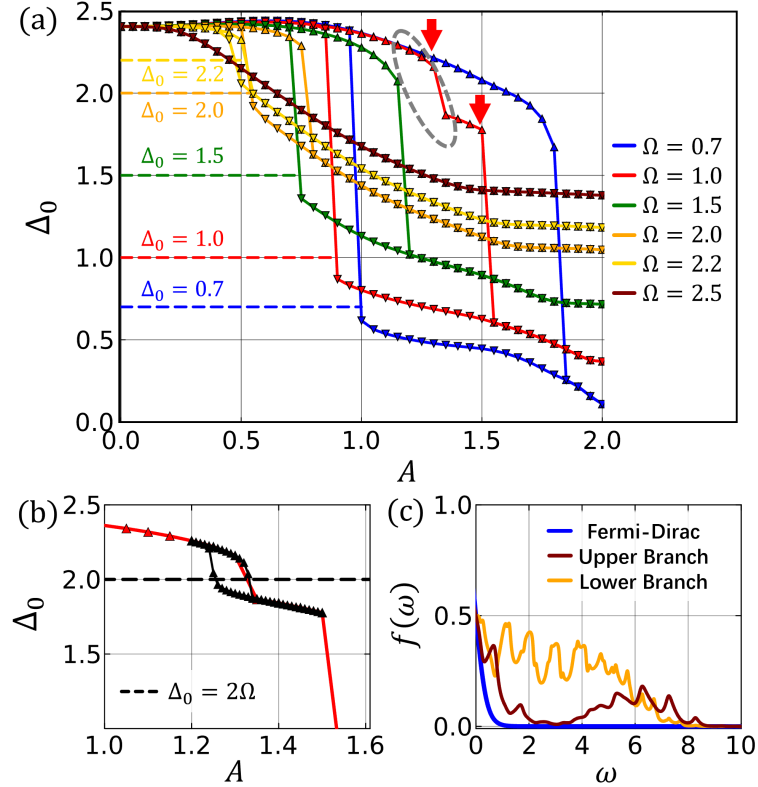


Figure 20.3: (a) Hysteresis behavior of Δ_0 as a function of A with different driving frequencies. The upper (lower) branches correspond to Δ_0 computed from those with smaller (larger) A . The horizontal dashed lines indicate $\Delta_0 = \Omega$. The red arrows show sudden jumps of Δ_0 for $\Omega = 1.0$. (b) Enlarged view of Δ_0 in the region marked by the gray dashed curve in (a). The black curves with triangle markers show the nested hysteresis behavior of Δ_0 for $\Omega = 1.0$. The black dashed horizontal line shows $\Delta_0 = 2\Omega = 2.0$. (c) Effective distributions for $\Omega = 1.0$ and $A = 1.2$ corresponding to the upper (dark red) and lower (orange) branches of the hysteresis loop. The blue curve is the Fermi-Dirac distribution with $T = 0.2$ for reference. The other parameters are $U = 6.0$, $\Gamma = 0.1$, and $T = 0.2$, which gives $\Delta_{\text{eq}} \approx 2.4$.

To understand the nature of each state in the two main branches in the hysteresis loop, we plot the corresponding NESS distribution functions for $A = 1.0$ and $\Omega = 1.0$ in Fig. 20.3(c). The distribution of the upper branch is essentially of the Eliashberg-type, where excited quasiparticles are accumulated at the far band edges, showing a significant population inversion within the bands, while the distribution of the lower branch shows the breakdown of such population inversion accompanied by the massive excitation of quasiparticles. This type of a sudden collapse of the distribution is similar to the recently discussed quantum avalanche effect [Han *et al.*(2023)] assisted by multi-photon processes through in-gap states in the context of dielectric breakdown of correlated insulators.

In the phase diagram (Fig. 20.1(b)), the hysteresis region appears in the area surrounded by the orange dashed line with the endpoint at $A \approx 0.3$ and $\Omega \approx \Delta_{\text{eq}}$. One can smoothly connect the small A and large A phases by going around the endpoint on a higher frequency side. This connectivity may come from the dissipative bath, which smoothens the coherence peaks and

leaves tiny but nonzero density of states within the superconducting gap. Additionally, the phase diagram does not show any trace of single-photon absorptions. This is because we are focusing on the clean limit of the superconductors. We expect the effect of impurities will further enrich the phase diagram in the frequency regime $\Omega \approx 2\Delta_{\text{eq}}$, which we leave as a future problem.

Conclusion.

In this Letter, we have studied the Floquet steady states of superconductors coupled to heat baths under the persistent and periodic light driving by the Floquet Green's function method. When the driving frequency Ω is smaller than Δ_{eq} , we find that enhancement of superconductivity can indeed occur even without effective cooling, which is attributed to the modulation of the spectrum in the Floquet states. We further demonstrated nonequilibrium superconducting phase transitions accompanied by multiple (nested) hysteresis behaviors, which are induced by multi-photon-assisted tunnelings. This process tells us how the Eliashberg-type enhancement of superconductivity breaks down in the strong driving field regime.

We conclude by proposing several potential experimental approaches. In solid-state systems, one can directly observe the modifications of the energy spectrum in Floquet states of superconductors by the time-resolved ARPES with terahertz pump pulses [Reimann *et al.*(2018), Ito *et al.*(2023), Boschini *et al.*(2024)], which would be limited in terms of the energy resolution and pump intensities but will be further developed in the near future. One can also use the transient absorption, optical nonlinearity, and tunneling spectroscopies with continuous microwave irradiation to probe the Floquet steady states and the hysteresis behaviors. Another platform is cold-atom systems, where gauge fields, dissipation, and fermionic condensates can be effectively simulated [Regal *et al.*(2004), Diehl *et al.*(2008), Senaratne *et al.*(2018), Eckardt(2017), Weitenberg and Simonet(2021)].

20.2 Floquet many-body theory for BCS superconductors

20.2.1 Keldysh Formalism

We give the definition of the lesser Green's function, the retarded Green's function, and the advanced Green's function in the real-time domain:

$$G_{\mathbf{k},\alpha\beta}^<(t, t') = +i \left\langle \psi_{\mathbf{k}\beta}^\dagger(t') \psi_{\mathbf{k}\alpha}(t) \right\rangle, \quad (20.6)$$

$$G_{\mathbf{k},\alpha\beta}^R(t, t') = -i\theta(t - t') \left\langle \{ \psi_{\mathbf{k}\alpha}(t), \psi_{\mathbf{k}\beta}^\dagger(t') \} \right\rangle, \quad (20.7)$$

$$G_{\mathbf{k},\alpha\beta}^A(t, t') = +i\theta(t' - t) \left\langle \{ \psi_{\mathbf{k}\alpha}(t), \psi_{\mathbf{k}\beta}^\dagger(t') \} \right\rangle. \quad (20.8)$$

$\psi_{\mathbf{k}\alpha}^\dagger$ and $\psi_{\mathbf{k}\alpha}$ denotes the fermionic creation and annihilation operator with momentum $\mathbf{k}$ and internal degree of freedom α . In the context of superconductors, the fermionic operators ψ and $\psi^\dagger$ stand for the Nambu spinor. $\langle \dots \rangle$ stands for taking the average with respect to the initial density matrix ρ_0 . $\theta(t - t')$ is the step function, and the $\{ \dots, \dots \}$ is the anti-commutator.

The retarded Green's function includes information on the energy spectrum (real part) and the lifetime (imaginary part) of the quasiparticles, while the lesser Green's function contains information on the occupation of quasiparticles. In equilibrium systems, they are connected through the Fluctuation-Dissipation theorem (DFT) [Haug *et al.*(2008)] (read [Haug *et al.*(2008)], since I don't know where it is explained in another place), which says:

$$G_{\mathbf{k}}^<(\omega) = -2i \cdot f_T^{\text{FD}}(\omega) \cdot \text{Im}G_{\mathbf{k}}^R(\omega), \quad (20.9)$$

where ω comes from the Fourier transformation on $(t - t')$ and $f_T^{\text{FD}}(\omega) = \frac{1}{e^{\omega/k_B T} + 1}$ is the Fermi-Dirac distribution at temperature T . In non-equilibrium interacting systems, the relation above is usually invalid, and we have to solve $G_{\mathbf{k},\alpha\beta}^<(t, t')$ and $G_{\mathbf{k},\alpha\beta}^R(t, t')$ respectively.

Denoting non-interacting and interacting Green's functions with $G_{0\mathbf{k},\alpha\beta}^<(t, t')$ and $G_{0\mathbf{k},\alpha\beta}^R(t, t')$ respectively, we have the Dyson-Keldysh equations [Jauho *et al.*(1994)] for non-equilibrium interacting systems:

$$G_{\mathbf{k}}^R = G_{0\mathbf{k}}^R + G_{0\mathbf{k}}^R * \Sigma_{\mathbf{k}}^R * G_{\mathbf{k}}^R, \quad (20.10)$$

$$G_{\mathbf{k}}^< = G_{\mathbf{k}}^R * \Sigma_{\mathbf{k}}^< * G_{\mathbf{k}}^A, \quad (20.11)$$

where the multiplication stands for the abbreviation of time convolution and matrix index contraction. For example,

$$(A * B)_{\alpha\beta}(t, t') \equiv \sum_{\gamma} \int d\bar{t} A_{\alpha\gamma}(t, \bar{t}) B_{\gamma\beta}(\bar{t}, t'). \quad (20.12)$$

For the BCS model used in the main text,

$$H_{\text{BCS}}(t) = \sum_{\mathbf{k},\sigma} \varepsilon_{\mathbf{k}-e\mathbf{A}(t)} c_{\mathbf{k}\sigma}^\dagger c_{\mathbf{k}\sigma} - \frac{U}{N} \sum_{\mathbf{k},\mathbf{p}} c_{\mathbf{k}\uparrow}^\dagger c_{-\mathbf{k}\downarrow}^\dagger c_{-\mathbf{p}\downarrow} c_{\mathbf{p}\uparrow}, \quad (20.13)$$

we adopt the standard mean-field approximation by introducing the superconducting order parameter

$$\Delta(t) \equiv -\frac{U}{N} \sum_{\mathbf{k}} \langle c_{-\mathbf{k}\downarrow}(t) c_{\mathbf{k}\uparrow}(t) \rangle = i \frac{U}{2N} \sum_{\mathbf{k}} \text{Tr} (\tau_1 G_{\mathbf{k}}^<(t, t)). \quad (20.14)$$

The assumption that $\Delta(t)$ is real has been included in the second equation. The mean-field Hamiltonian can then be reshaped in a matrix form with the Nambu spinor:

$$H_{\text{MF}}(t) = \sum_{\mathbf{k}} \psi_{\mathbf{k}}^\dagger H_{0\mathbf{k}}(t) \psi_{\mathbf{k}} + \frac{N}{U} \Delta^\dagger(t) \Delta(t), \quad (20.15)$$

$$H_{0\mathbf{k}}(t) = \begin{pmatrix} \varepsilon_{\mathbf{k}-e\mathbf{A}(t)} & \Delta(t) \\ \Delta(t)^\dagger & -\varepsilon_{-\mathbf{k}-e\mathbf{A}(t)} \end{pmatrix}. \quad (20.16)$$

In the rest of the text, we drop the unimportant second term in Eq.(20.15). The mean-field Hamiltonian is then formally non-interacting, and the corresponding Green's function is denoted by $G_{0\mathbf{k}}^{R/<}$.

20.2.2 Heat Bath

(this is a very unusual approach to bath, see original papers to understand, what is going on here???)

In the main text, we adopted a free fermion bath model for each site of the superconductor, whose Hamiltonian is simply

$$H_{\text{bath}} = \sum_{i,\delta} \phi_{i\delta}^\dagger \varepsilon_{i\delta}^{\text{bath}} \phi_{i\delta}. \quad (20.17)$$

$\phi_{i\delta}$ is the fermionic operator of the bath at the lattice site i and δ denotes the internal degree of freedom.

We then assume a Büttiker's model, where the bath couples to the electrons through

$$H_{\text{coup}} = \sum_{i\delta\alpha} \left(\psi_{i\alpha}^\dagger (V_i)_{\alpha\delta} \phi_{i\delta} + \phi_{i\delta}^\dagger (V_i^\dagger)_{\delta\alpha} \psi_{i\alpha} \right). \quad (20.18)$$

By analytically integrating out the bath degrees of freedom (how to do it???) , coupling to the baths can be addressed with self-energy

$$\Sigma_{i,\alpha\beta}^{R/<} = \sum_{\delta\gamma} \left[(V_i)_{\alpha\delta} \cdot G_{i,\delta\gamma}^{\text{bath},R/<} \cdot (V_i^\dagger)_{\gamma\beta} \right], \quad (20.19)$$

where $G_{i,\delta\gamma}^{\text{bath},R/<}$ is the Green's function of the heat bath.

The baths have a considerably large number of degrees of freedom so they can be treated as is in equilibrium and have a definite temperature T . As is assumed in many previous studies [Tsuji *et al.*(2009), Aoki *et al.*(2014)], the chemical potential of the bath μ^{bath} can be well adjusted so that $\text{Re } G_{\mathbf{k},\delta\gamma}^{\text{bath},R}(\omega) = 0$, which ensures no net particle flow between the bath and the superconductors. The bath is further assumed to have a constant density of states and doesn't mix up the degrees of freedom of the Nambu spinors. Therefore, we can neglect the frequency dependence of the imaginary part of the self-energy function and denote it with a coupling strength constant Γ , that is

$$\Sigma_{\mathbf{k}}^R(\omega) = -i\Gamma \cdot \mathbf{1}, \quad (20.20)$$

the lesser component of the self-energy function is then given according to the fluctuation-dissipation theorem

$$\Sigma_{\mathbf{k}}^<(\omega) = 2if_T^{\text{FD}}(\omega)\Gamma \cdot \mathbf{1}. \quad (20.21)$$

Note that $\mathbf{1}$ is the 2×2 unit matrix in the Nambu space.

20.2.3 Floquet Green's function

For a periodic non-equilibrium many-body system, its discrete time-translation symmetry ensures the Green's functions satisfy $G_{\mathbf{k}}^{R/<}(t, t') = G_{\mathbf{k}}^{R/<}(t + \mathcal{T}, t' + \mathcal{T})$, where $\mathcal{T}$ is the periodic. In terms of $t_{\text{dif}} \equiv t - t'$ and $t_{\text{ave}} \equiv (t + t')/2$, we may perform the following transformation to derive the Floquet Green's function [Tsuji *et al.*(2008)]:

$$G_{\mathbf{k},mn}^{R/<}(\omega) \equiv \frac{1}{\mathcal{T}} \int_0^{\mathcal{T}} dt_{\text{ave}} e^{i(m-n)\Omega t_{\text{ave}}} \int_{-\infty}^{\infty} dt_{\text{dif}} e^{i(\omega + (m+n)\Omega/2)t_{\text{dif}}} G_{\mathbf{k}}^{R/<}(t, t'), \quad (20.22)$$

where m, n are the Floquet indices with integer values and ω is the frequency (energy) that takes value from the Floquet Brillouin Zone $(-\Omega/2, \Omega/2]$ (add references). The transformation above is essentially a continuous Fourier transformation for t_{dif} and a discrete one for t_{ave} . In this sense, the transformation Eq.(20.22) keeps the multiplication structure in the real-time domain, while the multiplication rule now reads [Tsuji *et al.*(2008)]:

$$(A * B)_{\alpha\beta,mn}(\omega) \equiv \sum_{\gamma,p} A_{\alpha\gamma,mp}(\omega) B_{\gamma\beta,pn}(\omega), \quad (20.23)$$

which is formally the multiplication rule for the direct product matrices with Nambu indices α, β, γ and the Floquet indices m, n, p .

In the matrix form, computing the inverse of Green's function becomes straightforward. The equation of motion of the non-interacting retarded Green's function $G_{0\mathbf{k}}^R(t, t')$, which reads

$$(i\partial_t - H_{0\mathbf{k}}(t))G_{0\mathbf{k}}^R(t, t') = \delta(t - t') \cdot \mathbf{1}, \quad (20.24)$$

can be addressed in terms of the Floquet Green's function, which reads

$$\sum_p [(\omega + m\Omega)\delta_{mp} - H_{0\mathbf{k},mp}] G_{0\mathbf{k},pn}^R(\omega) = \delta_{mn} \cdot \mathbf{1}, \quad (20.25)$$

by performing the transformation Eq.(20.22) on both sides of Eq.(20.24). Note that $H_{0\mathbf{k}}(t)G_{0\mathbf{k}}^R(t, t') = H_{0\mathbf{k}}(t)\delta(t - t') * G_{0\mathbf{k}}^R(\bar{t}, t')$, so $H_{0\mathbf{k},mn} = (1/\mathcal{T}) \int_{\mathcal{T}/2}^{\mathcal{T}/2} dt e^{i(m-n)\Omega t} H_{0\mathbf{k}}(t)$, where $\mathcal{T} = 2\pi/\Omega$ is the period. Then, the inverse of the retarded Floquet Green's function reads

$$(G_{0\mathbf{k}}^{R-1})_{mn}(\omega) = (\omega + m\Omega + i\eta)\delta_{mn} \cdot \mathbf{1} - H_{0\mathbf{k},mn}, \quad (20.26)$$

where the infinitesimal η originates from the causal nature of the retarded Green's function. It is then convenient to express the Dyson equation with the inverse of the Floquet Green's functions, that is

$$G_{\mathbf{k}}^{R-1} = G_{0\mathbf{k}}^{R-1} - \Sigma_{\mathbf{k}}^R. \quad (20.27)$$

Note that the self-energy of the bath with Floquet indices reads:

$$\Sigma_{\mathbf{k},mn}^R(\omega) = -i\Gamma \cdot \delta_{mn} \mathbf{1}, \quad (20.28)$$

$$\Sigma_{\mathbf{k},mn}^<(\omega) = 2if_T^{\text{FD}}(\omega + m\Omega)\Gamma \cdot \delta_{mn} \mathbf{1}, \quad (20.29)$$

where ω lies in the Floquet BZ: $(-\Omega/2, \Omega/2]$.

In numerical computations, we truncate the Floquet indices in numerical calculation with $|m|, |n| < N$, which allows us to solve the Dyson-Keldysh equations with finite-sized matrix inverse and multiplication. The error from the truncation is analytically well-bounded [Rudner and Lindner(2020)]. In this study, the truncation is practically always set to meet the condition of $N\Omega > W$, where W is the bandwidth read from the spectral function, to minimize the error. The truncation N becomes very large in the low-frequency regime, which is the main difficulty in the numerical computation.

We comment that no coupling to the baths ($\Gamma = 0$) seemingly results in an illness of the Eq.(20.11), that is because a finite Γ wipes out the information of the initial state. Such illness can be effectively cured by introducing $\Sigma_{\mathbf{k},mn}^<(\omega) = 2if(\omega + m\Omega)\delta_{mn} \cdot \eta$, where $\eta \rightarrow 0+$ and $f(\omega)$ is the distribution of the initial state that can be nonthermal in principle [Eckstein(2018)]. However, it is well known that such a isolated system will eventually be heated to a featureless state, which can be technically reflected by the increasing numerical instability when Γ approaches zero.

20.2.4 Self-consistent Equation for the Order Parameter

The order parameter needs to be self-consistently determined by Eq.(20.14). We express Eq.(20.14) in terms of the Floquet Green's function. Noting that

$$\begin{aligned} G_{\mathbf{k}}^<(t, t) &= \lim_{\delta t \rightarrow 0} G_{\mathbf{k}}^<(t, t - \delta t) \\ &= \frac{1}{2\pi} \sum_{mn} e^{-(m-n)\Omega t} \int_{-\Omega/2}^{\Omega/2} d\omega G_{\mathbf{k},mn}^<(\omega), \end{aligned} \quad (20.30)$$

and $\Delta_{mn} = \Delta_{m-n} \equiv (1/\mathcal{T}) \int_{\mathcal{T}/2}^{\mathcal{T}/2} dt e^{i(m-n)\Omega t} \Delta(t)$, we have

$$\Delta_{mn} = i \frac{U}{2\pi N} \sum_{\mathbf{k}} \sum_{p,q}^{p-q=m-n} \int_{-\Omega/2}^{+\Omega/2} d\omega G_{\mathbf{k},pq}^<(\omega). \quad (20.31)$$

Now, Eq.(20.11),(20.27),(20.31) form the self-consistent equations for the order parameter. We solve them by iteration. We start with an initial order parameter Δ_{mn}^{ini} , which is assumed to be static, i.e. $\Delta_0^{\text{ini}} \neq 0$ and $\Delta_{mn}^{\text{ini}} = 0$ with $m \neq n$.

Substituting Δ_{mn}^{ini} into the Hamiltonian $H_{0\mathbf{k},mn}$ and further into Eq.(20.26), we have the non-interacting retarded Green's function $G_{0\mathbf{k},mn}^R(\omega)$. Taking the inverse of the right-hand side of the Dyson equation Eq.(20.27), we further derive the interacting retarded Green's function $G_{\mathbf{k},mn}^R(\omega)$. Note that the advanced Green's function satisfies $G_{\mathbf{k},mn}^A(\omega) = (G_{\mathbf{k}}^R)_{mn}^\dagger(\omega)$, the lesser Green's function $G_{\mathbf{k},mn}^<(\omega)$ can then be derived from the Keldysh equation Eq.(20.11). Next, we use the right-hand side of Eq.(20.31) to update the order parameter. The procedure above is repeated until we have a converged result for Δ_{mn} .

The convergence for the superconducting states with $\Delta_0 \neq 0$ is typically quick, while the convergence for the normal states is not. We practically have a maximum time of iteration for the normal state, which makes their Δ_0 tiny but nonzero.

In the main text, we focus mainly on the zeroth order component Δ_0 , the time-averaged order parameter, as the indicator of superconductivity in the non-equilibrium steady states (NESS). We also compute the time-averaged spectral function $A_0(\omega)$ and occupation function $n_0(\omega)$ for quasiparticles.

$$A_0(\omega) = -\frac{1}{\pi N} \sum_{\mathbf{k}} \text{Im} [\text{Tr} (G_{\mathbf{k},mm}^R(\omega - m\Omega))] , \quad (20.32)$$

$$n_0(\omega) = \frac{1}{2\pi N i} \sum_{\mathbf{k}} \text{Tr} (G_{\mathbf{k},mm}^<(\omega - m\Omega)) . \quad (20.33)$$

The trace in the formulas above acts on Nambu indices, and the Floquet index m is determined by the condition that $(\omega - m\Omega)$ falls inside the Floquet BZ $(-\Omega/2, \Omega/2]$. In the main text, we introduced the effective distribution function for the NESS:

$$f^{\text{eff}}(\omega) \equiv \frac{n_0(\omega)}{A_0(\omega)}. \quad (20.34)$$

The definition is inspired by the DFT Eq.(20.9), and the effective distribution reduces to Fermi-Dirac distribution $f_T^{\text{FD}}(\omega)$ if the system stays in equilibrium states, like the situation when the light driving is off ($A = 0$).

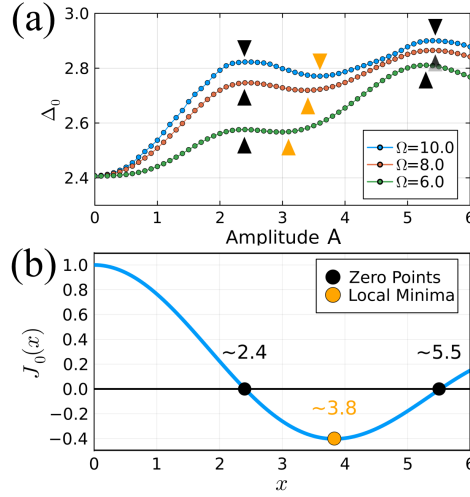


Figure 20.4: (a) The time-averaged order parameter Δ_0 as a function of A , computed with $\Omega = 6.0, 8.0, 10.0$ respectively. The black triangles mark the local maximum of Δ_0 and the yellow triangles mark the local minimum of Δ_0 . The results are computed with $U = 6.0$, $\Gamma = 0.1$, $T = 0.2$ and the truncation of the Floquet indices $N = 10$. (b) The zeroth order Bessel function $J_0(x)$. The black circles mark its two zero points at 2.4 and 5.5 approximately. The yellow circle marks its local minimum at 3.8 approximately.

20.3 Numerical Results in the High-Frequency Regime

In the main text, we stated that when the frequency Ω is greater than the other energy scales of the system, the steady state is then well described by the effective Hamiltonian in the high-frequency expansion [Oka and Kitamura(2019)], the leading term of which is given as

$$H^{\text{eff}} = \sum_{\mathbf{k}, \sigma} J_0(A) \varepsilon_{\mathbf{k}} c_{\mathbf{k}\sigma}^\dagger c_{\mathbf{k}\sigma} - \frac{U}{N} \sum_{\mathbf{k}, \mathbf{p}} c_{\mathbf{k}\uparrow}^\dagger c_{-\mathbf{k}\downarrow}^\dagger c_{-\mathbf{p}\downarrow} c_{-\mathbf{p}\uparrow} + O(1/\Omega). \quad (20.35)$$

When the frequency Ω is even larger than the band width, the lacking of scattering channels in the current model suppresses the heating inside the band, the distribution $f^{\text{eff}}(\omega)$ thus remains the Fermi-Dirac one in the range of the energy $(-\Omega, \Omega)$.

We plot the relation between Δ_0 and A in Fig.20.4(a). We can see the non-monotonic increase of Δ_0 with respect to A . The local maximum of Δ_0 agrees with the zero points of $J_0(A)$ at 2.4 and 5.5. With the frequency Ω increasing, and the local minimum of Δ_0 approaches the extreme points of $J_0(A)$ of 3.8. This shows that Δ_0 follows the effective pairing strength $U^{\text{eff}} = U/|J_0(A)|$ with satisfactory.

One problem of realizing such enhancement with high-frequency driving is apparently the heating from other scattering channels. Recently, it is proposed to utilize the cavity photons instead, which can realize the similar dynamical localization as well as suppressing the heating.

Part XVI

Appendix

.1 Appendix

.1.1 Literature

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.2 Bibliography from Old, Fundamental Articles

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From Stability of nonequilibrium Fermi distributions with respect to Cooper pairing by Aronov, Gurevich

Comments:

²⁾ We have learned that D. A. Kirzhnits and Yu. V. Kopayev are also studying a similar problem.

³⁾ A similar formalism has also been used by Elesin [¹⁶] in order to solve the problem of a semiconductor in the field of a strong electromagnetic wave.

⁴⁾ In this representation the factors $\mathbf{u}_p$ and $\mathbf{v}_p$ are determined by other expressions (see [¹¹]). We take this opportunity to mention that in [¹¹] in expression (2.18) for the collision operator the last two terms were erroneously omitted. Taking these terms into account leads to a change of the numerical coefficients in the final formulas, namely, it leads to the replacement in formula (43) of the coefficient associated with Δ/kT by $1/2 + \pi/4$ and it also leads to the addition of the term $(1 - \ln 2)/2$ in the expression for $\ln C$.

⁵⁾ This implies, as one can easily see, a "negative temperature" of the superconducting excitations.

⁶⁾ This state has subsequently been studied in detail experimentally. [²⁰]

⁷⁾ Equation (4.3), which we have derived, differs slightly from the corresponding equation in [¹²].

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